GEK-83947B

Instructions

Micro Power System Stabilizer With Generic Interface

Micro Power System Stabilizer

With Generic Interface

GEK-83947B

(Supersedes GEK-83947A)

Original Issue Date: May 1988
Revision A: July 1989
Revision B: December 1991

These instructions do not purport to cover all details or variations in equipment, nor to provide for every possible contingency to be met during installation, operation, and maintenance. Should further information be desired or should particular problems arise **that** *are not covered sufficiently for the purchaser's purpose, the matter should be referred to GE Drive Systems, Salem, Virginia, U.S.A.*

© 1988, 1989, 1991 by General Electric Company, U.S.A.
All rights reserved.

Printed in the United States of America.

Safety Symbol Legend

WARNING — Commands attention to an operating procedure, practice, condition or statement, which, if not strictly observed, could result in personal injury or death.

CAUTION — Commands attention to an operating procedure, practice, condition or statement, which, if not strictly observed, could result in damage to or destruction of equipment.

NOTE — Commands attention to an essential operating or maintenance procedure, condition, or statement that must be highlighted.

WARNING

THIS EQUIPMENT CONTAINS A POTENTIAL HAZARD OF ELECTRICAL SHOCK OR BURN. ONLY PERSONNEL WHO ARE ADEQUATELY TRAINED AND THOROUGHLY FAMILIAR WITH THE EQUIPMENT AND THE INSTRUCTIONS SHOULD INSTALL, OPERATE, OR MAINTAIN THIS EQUIPMENT.

ISOLATION OF TEST EQUIPMENT FROM THE EQUIPMENT UNDER TEST PRESENTS POTENTIAL ELECTRICAL HAZARDS. IF THE TEST EQUIPMENT CANNOT BE GROUNDED TO THE EQUIPMENT UNDER TEST, THE TEST EQUIPMENTS CASE MUST BE SHIELDED TO PREVENT CONTACT BY PERSONNEL.

TO MINIMIZE HAZARD OF ELECTRICAL SHOCK OR BURN, APPROVED GROUNDING PRACTICES AND PROCEDURES MUST BE STRICTLY FOLLOWED.

NOTICE

This document contains proprietary information of General Electric Company, U.S.A. and is furnished to its customer solely to assist that customer in the installation, testing, and/or maintenance of the equipment described. This document shall not be reproduced in whole or in part nor shall its contents be disclosed to any third party without the written approval of GE Drive Systems, 1501 Roanoke Boulevard, Salem, Virginia 24153, U.S.A.

DIRECTO-MATIC℠ is a trademark of General Electric Company, U.S.A.
Hayes@ is a registered trademark of Hayes Microcomputer Products, Inc.
Smartmodem 1200℠ is a trademark of Hayes Microcomputer Products, Inc.

TABLE OF CONTENTS

Chapter/Paragraph Page

1 EQUIPMENT DEFINITION

1-1. INTRODUCTION.. 1-1
1-2. CONTROL FUNCTION (SOFTWARE) ..1- 1
1-3. HARDWARE 1-1
1-3.1. Controller.. 1-1
1-3.2. Data Recorders 1-2
1-3.3. Remote Communication 1-2
1-3.4. Excitation System Interface 1-2
1-3.5. Generic Interface 1-3
1-4. SYSTEM SPECIFICATIONS 1-3

2 INSTALLATION

2-1. INTRODUCTION 2-1
2-2. RECEMNG AND HANDLING.. 2-1
2-2.1. Unpacking..... 2-1
2-2.2. Storage and Protection 2-1
2-3. PARTS LIST 2-1
2-4. LOCATION AND CONNECTIONS.. 2-2
2-4.1. Initial Wiring Checks 2-2

3 OPERATIONAL CHECKS

3-1. INTRODUCTION.. 3-1

SECTION I: OFFLINE OPERATION

3-2. PRE-POWER CHECKS 3-1
3-3. POWER SUPPLY 3-1
3-4. FREQUENCY TRANSDUCER.. 3-1
3-5. UNIT PARAMETERS.. 3-1
3-5.1. Unit Identification 3-2
3-5.2. Tuneup..... 3-2
3-5.3. Event Recorder 3-2
3-5.4. Auto Scaling 3-2
3-5.5. Torsional Filter Enable/Disable 3-2
3-5.6. Auto/Manual 3-2
3-5.7. Saving Parameters 3-2
3-5.8. Optional Under/Overvoltage Detection ..3-3
3-5.9. Phase Rotation.. 3-3
3-6. FREQUENCY RESPONSE TEST.. 3-3
3-7. FAULTS 3-4
3-7.1. Offline Fault..... 3-4
3-7.2. Loss of Phase Fault.. 3-4
3-7.3. Frequency Transducer Ceiling Fault.. .... ..3- 4
3-7.4. μPSS Output Ceiling Fault.. 3-4
3-7.5. μPSS Segment Not Running.. 3-5

Chapter/Paragraph **Page**

SECTION II: ONLINE OPERATION

3-8. INITIAL ONLINE OPERATION.. 3-5
3-8.1. Manual Mode 3-5
3-8.2. Auto Mode 3-5
3-8.3. Tuneup 3-5

4 USING THE MONITOR

4-1. INTRODUCTION 1
4-2. USER MONITOR 4-1
4-2.1. AS – Autoscale Command 4-2
4-2.2. DR – Display Rating Command.. 4-2
4-2.3. FC – Fault List Clear Command.. 4-2
4-2.4. FD – Fault Display Command.. 4-2
4-2.5. GP – Gain and Phase Command.. 4-2
4-2.6. HH – Help Command 4-2
4-2.7. ID – Enter Station and Unit Identification Command 4-3
4-2.8. LC – List Clear Command 4-3
4-2.9. LD – List Display Command.. 4-3
4-2.10. LS – List Status Command.. 4-3
4-2.11. MM – Main Monitor Command.. 4-4
4.212. PC – Password Change Command.. 4-4
4-2.13. SS – Stabilizer Status Command 4-4
4-2.14. SU – Station and Unit Display Command..... .4-4
4-2.15. TD – Time Display Command 4-4
4-2.16. TM – Test Mode Display Command.. 4-4
4-2.16.1. C – CURRENT TEST STATUS.. 4-4
4-2.16.2. D – DIGITAL TEST 4-4
4-2.16.3. E – EXCITATION SIGNATURE TEST ..4- 5
4-2.16.4. S – SINGLE FREQUENCY TEST.. 4-5
4-2.17. TS – Time Set Command.. 4-6
4-2.18. TU – Tune Up Command.. 4-6
4-2.18.1. G – GAIN ADJUSTMENT 4-6
4-2.18.2. T – TIME CONSTANT ADJUSTMENT.. 4-6
4-2.18.3. L – LIMIT ADJUSTMENTS 4-6
4-2.18.4. Q – QUIT 4-8
4-2.19. Sample User Session.. 4-8
4-3. MAIN MONITOR 4-8
4-3.1. EEPROM Saves 4-9
4-3.2. Modifying Variables 4-9

5 **DESCRIPTION** OF OPERATION

5-1. CONTROL (DMC) 5-1
5-1.1. Power-up 5-1

TABLE OF CONTENTS – CONT

Chapter/Paragraph		Page
5-1.2.	Control	7-1
5-1.3.	User Interface	5-2
5-2.	TEST RECORDER	5-2
5-3.	TRANSIENT EVENT RECORDER	5-2
5-4.	GENERIC INTERFACE	5-3
5-4.1.	Current Limit Protection	5-3
5-4.1.1.	CURRENT LIMIT IMPLEMENTATION	5-4
5-4.2.	Transmission Monitor Protection	5-4
5-4.2.1.	TRANSMISSION MONITOR IMPLEMENTATION	5-4
5-4.3.	Setting Up the μPSS and Power Amplifier	5-7
5-4.4.	Power Amplifier Gain Adjustment	5-8
5-4.5.	Adjusting Transmission Monitor Protection	5-8
5-4.6.	Gain Phase Measurement with Generic Interface	5-8
5-4.6.	Set Up The Demodulator	5-8
6	MAINTENANCE AND TROUBLESHOOTING	
6-1.	MAINTENANCE	6-1
6-2.	TROUBLESHOOTING	6-1
6-3.	RENEWAL PARTS..	6-1
A	TERMINAL CHARACTERISTICS	
A-1.	TERMINAL SETTINGS	A-1
A-2.	CABLE CHARACTERISTICS	A-1
B	APPLICABLE PUBLICATIONS	
C	HAYES® MODEM SETTINGS	
C-1.	CONFIGURATION SWITCHES	C-1
D	POWER AMPLIFIER	
D-1.	INTRODUCTION	D-1
E	PRINTED WIRING BOARD/MODULE DRAWINGS	
E-I.	INTRODUCTION	E-1
F	SYSTEM ELEMENTARY DIAGRAM	
F-1.	INTRODUCTION	F-1
G	APPLICABLE SOFTWARE LISTINGS	
G-1.	INTRODUCTION	G-1

LIST OF ILLUSTRATIONS

Number		Page
1-1	Measured Gain and Phase vs. Frequency	1-2
5-1	Typical Event Recorder List	5-2
5-2	Transmission Monitor	5-5
5-3	Three-phase Phasor Diagram of PT Voltages	5-6
5-4	Insertion of Transmission Monitor Signal	5-6
5-5	Removing One Phase From Service	5-6
5-6	Monitor Display of Screen 3	5-7

LIST OF TABLES

Number		Page
2-1	µPSS Parts List	2-2
2-2	Medium Electronics Module Parts List	2-2
4-1	User Monitor Commands	4-1

ABBREVIATIONS

The following abbreviations are used in this instruction manual:

CT	Current transformer
CWFT	Continuous wave frequency transducer
DCPT	Direct current potential transformer/Field voltage transducer
DMC	Distributed Micro Controller
DMCC	Distributed Micro Controller printed wiring board
LCP	Local control panel
LED	Light emitting diode
MEM	Medium electronics module
NFTA	Continuous wave frequency transducer, printed wiring board
μPSS	Micro Power System Stabilizer
PWB	Printed Wiring Board
PT	Potential Transformer

APPENDIX G

APPLICABLE SOFTWARE LISTINGS

G-l. INTRODUCTION

This appendix contains listings of software that the user may find helpful in troubleshooting. This software includes I/O logic and relay ladder diagrams.

CHAPTER 1

EQUIPMENT DEFINITION

1-I. INTRODUCTION

This instruction book describes GE's Micro(computer-based) Power System Stabilizer, or μPSS, with Generic Interface. The μPSS processes generator frequency deviations to provide additional control to existing voltage regulator systems. This composite control improves power system stability under certain conditions.

The μPSS is available in the following forms (groups):

Group	Includes
G5 Modem	Transient Event Recorder
G6 Modem	No Transient Event Recorder
G7 No Modem	No Transient Event Recorder
G8 No Modem	Transient Event Recorder

A modem enables the μPSS to be accessed from a remote location. A Transient Event Recorder digitally records data based on either a voltage excursion or a μPSS output excursion.

1-2. CONTROL FUNCTION (SOFTWARE)

The μPSS introduces a control signal into continuously acting voltage regulator equipment. The signal supplements the conventional voltage regulator control with a control function derived so that upon deviations of generator frequency, excitation becomes controlled as a composite function of voltage and frequency. Under certain conditions the composite function be an aid to power system stability.

The following equation is a close approximation of the transfer function:

$$V(SS)/dF = \frac{1}{1 + s\times 2\times .475/204.2 + (s/204.2)^2} \times \frac{1}{1 + s/255}$$

$$\times \frac{K \times sT1 \times s10}{(1+sT1)(1+s10)} \times \frac{1}{1+s\times 2\times .6977/184.43+(s/184.43)^2}$$

$$\times \frac{\sin(w/120)e120}{w/120} \times \frac{(1+sT2)(1+sT4)}{(1+sT3)(1+sT5)}$$

$$\times \frac{1+s \times 2 \times .03/87.96+(s/87.96)^2}{1+s\times 2\times 0.2/68.55+(s/68.55)^2}$$

$$\times \frac{1 + s \times s/17174}{1 + s \times 2 \times .5059/46.39 + (s46.39)^2}$$

V(SS) = Supplementary signal in percentage of generator terminal voltage.

dF = Delta system frequency in percentage.

K = Product of gains in the frequency transducer and the digital transfer function. The transducer gain is adjustable from 0 to 6, and the digital gain from .04 to 10.

T1 = Washout time constant on continuous wave frequency transducer board (CWFT), NFTA/slot 2J. Settable for 0.5, 3, or 10 seconds, or 0 to eliminate the term $sT1/(1+sT1)$.

T2, T4 = Lead time constants, adjustable from 0.2 to 1.5,

T3, T5 = Lag time constants, adjustable from 0.02 to 0.2.

First line: Transfer function for CWFT.

Second *line:* The first term contains two washout functions that remove undesired dc components. The T1 time constant is in the frequency transduce;, . and the 10 second time constant is in the digital transfer function.
The second term is an anti-aliasing low pass analog filter.

Third Line: The first term is the result of digital sampling. The second term is a double lead-lag with adjustable time constants to provide the required phase lead in the output signal.

Fourth and fifth lines: These two digital filter terms are a biquad band reject filter and an elliptical low pass filter. They are used to prevent stimulation of torsional frequencies by the μPSS. If not required, they can be omitted by a program change at the time of installation.

Figure l-l shows measured gain versus frequency characteristics.

1-3. HARDWARE

1-3.1. Controller. The μPSS includes a Distributed Micro Controller (DMC) from GE's DIRECTOMATIC® Plus family of integrated control equipment. The DMC is a microprocessor-based controller designed specifically for analog type regulating applications. It performs decision-making and arithmetic functions under software control.

The DMC consists of printed wiring boards contructed into a standard electronics module and configured according to the application. In this case, the application is the μPSS.

The DMC provides a monitor interface through which the user can display and modify certain parts of the μPSS software. The user can access this monitor with a computer terminal or a remote modem.

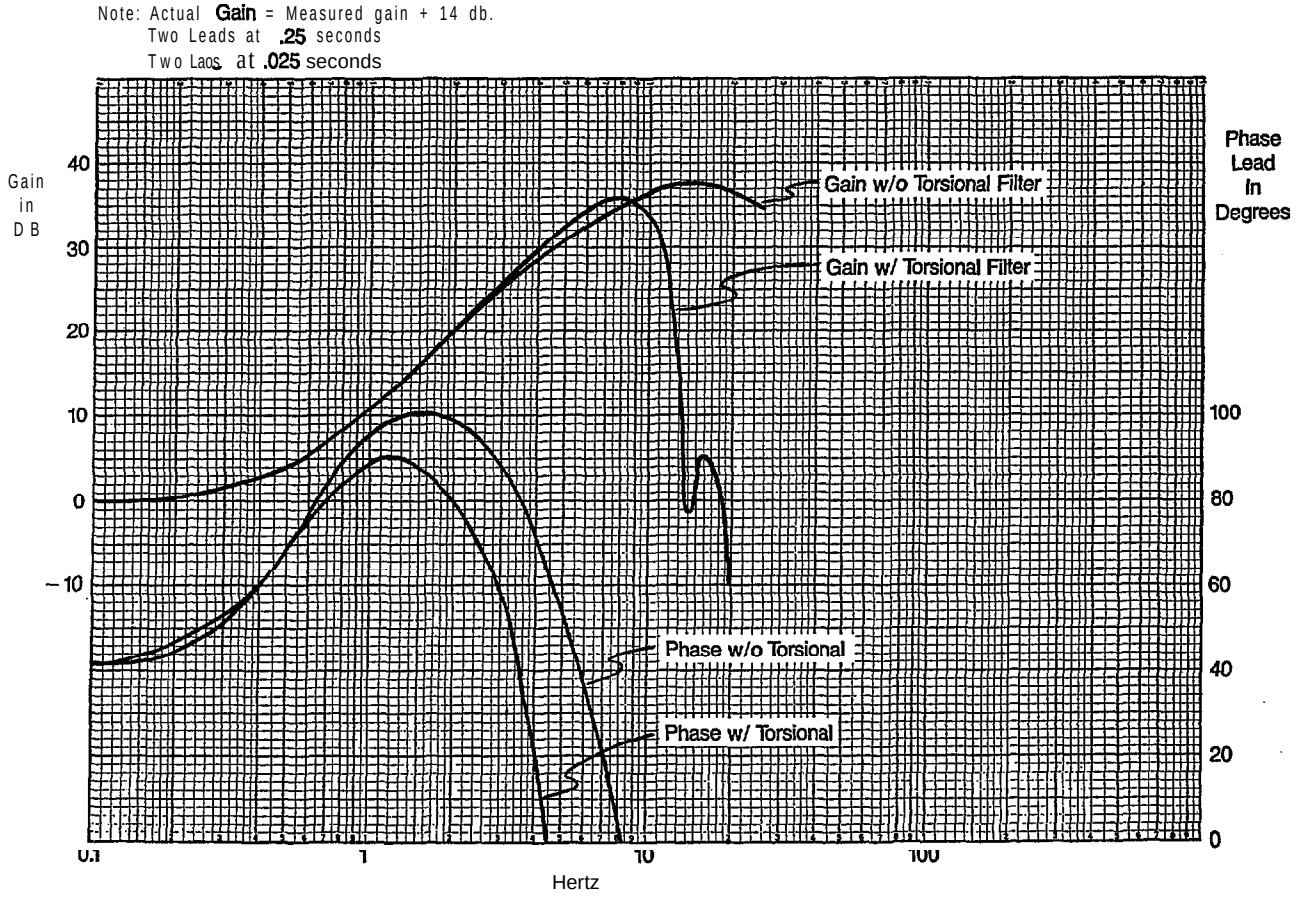


Figure l-l. Measured *Gain* and *Phase vs. Frequency*

In addition to the DMC monitor (Main Monitor), the μPSS provides a "User /Monitor," which enables the user to interface with the μPSS without accessing the Main Monitor. The User Monitor also functions as a protection/buffer because the user must enter a password into the User Monitor to gain access to the Main Monitor. This instruction book contains all the monitor commands needed to set up the μPSS for a particular application, and describes the User Monitor in detail.

As additional system protection, the user must turn a key to access the Main Monitor, perform any function requiring a password, or run any test that may disturb the system.

1-3.2. Data Recorders. The μPSS includes a Test Recorder to record system stability tests. It provides a digital record of date, time, phase volts, phase amperes, watts, vars, μPSS output, frequency, field volts, and the amplitude of the 1/2 Hz sinusoidal test signal. The Test Recorder uses a 20 Hz sample rate, and is not a general purpose high speed digital recorder. The data is recorded for 10.5 seconds.

An optional Transient Event Recorder (μPSS groups G5 and G8) provides a record of four transient events that include pre-transient and post-transient data. Recording is triggered by a preset excursion in average terminal voltage or in the μPSS output.

l-3.3. Remote Communication. An optional modem (μPSS forms G5 and G6) enables user access from a remote location. When needed, the user can perform limited testing from a remote location via this modem.

l-3.4. Excitation System Interface.

CAUTION

Misoperating or misadjusting the μPSS while it is connected to the excitation system can damage the generator. Do not deviate from the operational checks and procedures given in these instructions, without first checking with GE Drive Systems to ensure that they can be safely used.

The μPSS can function with different types of power generation and transmission systems. Via the monitor,

a user can customize μPSS software by changing constants that describe the system. This enables the μPSS to interface with various excitation systems installed within systems that use widely varying parameters.

Nominal constants are installed at the factory; however, they ***do not*** suit many applications. Correct settings for a particular installation can only be obtained through analytical studies or online tests.

*** NOTE ***

It is the customer's responsibility to obtain or approve parameters that are installed in the equipment. GE can assist in performing analytical studies and system tests. Contact the local GE sales office for further information.

GE recommends the "Power System Stabilizer Application Guide" as an aid. The customer can obtain it through the local GE sales representative, or from the GE Systems Development and Engineering Department, Schenectady, NY. Appendix B lists additional references that may help the user in understanding and operating the μPSS and its interfacing system.

l-3.5. **Generic Interface.** For regulators that do not have an input for the normal ± 10 V dc output, this interface is inserted in series with the generator potential transformers (PTs) and the regulator. The output of the μPSS drives the interface in order to modify the ac voltage that the regulator senses.

The generic interface consists of a multiplier printed wiring board and a S-phase transformer isolated output power amplifier. The interface multiplies signals proportional to the generator phase voltages with the μPSS modulation signal. This generates three modulated correction signals, which are fed **to** the three output power amplifiers.

If the modulation signal is zero, the output of the three power amplifiers is also zero. If the modulation signal is greater than zero; the amplifier output adds to the PT voltage in phase, increasing the average sensing voltage. If the modulation signal is less than zero, the amplifier output adds to the PT voltage 180 ° out of phase, decreasing the average sensing voltage.

1-4. SYSTEM SPECIFICATIONS

Power Requirement

115 V ac ±10, 50/60 Hz at 6 A

Power Supply Ratings

BUS	*Volts*	*Tolerance*
P28	28	1 V
P5	5	0.05 v
P15	15.075	0.05 v
N15	-15.075	0.05 v

Type of Operation

Automatic: Configurable to go online automatically when generator reaches a preset power level, and offline at a lower preset level

Manual: Configurable for manual operation, requiring operator to use pushbuttons to put the μPSS online or offline

Analog Signal Inputs

Standard:

From generator *PTs:* 3ϕ, 115 V ac, at 1 VA per phase

From generator CTs: 3ϕ, 5 A, at 1 VA per phase

Event Recorder:

From *field* voltage *DCPT:* Ungrounded lo-O-10 V

Load resistance: 10 kΩ.

Analog Signal Outputs

Standard:

To Voltage *Regulator:* PT Modulation ± 7 V ac

For *customer use:* lo-O-10 V at 2 mA

Optional: As required

Scan Rates

Analog inputs and outputs: 50/60 Hz, dependent on system frequency

Test and Event Recorder:

60 *Hz systems: 20* Hz

50 *Hz systems: 25* Hz

Contact Inputs

Power: From internal 28 V dc

Source: Operator online and offline pushbuttons

Local online and offline pushbuttons, test switch

Generator breaker auxiliary contact

Two spares

Relay Outputs

Power: Contacts rated either 28 V dc, 2 A or 115 V ac, 0.5 A, contacts fused at 3A offline and online

Source: Indicating lamps contacts, internally powered from 28 V dc

Two isolated alarm contacts

One isolated spare contacts

Communications

Via RS-232C serial port

Local: User supplied cable and computer terminal or printer. Refer to Appendix B for terminal and cable requirements.

Remote: (Optional, forms G5 and G6) Hayes Model 1200 Modem at at 300/1200 baud. Requires compatible remote modem and terminal.

Power Amplifier (PA)

Three-phase

Used as interface between μPSS and voltage regulator

Used to modify PT sensing signal

Frequency Transducer

Type: Continuous wave with gain and center frequency null adjustments

Standard Output: 5.0 V per percent frequency change

Adjustable from 0 to 6'V

Spare ***transducer:*** Can be configured for test as a low noise, high stability, 3-phase oscillator

Prewired slot furnished for connection

10-conductor cable, part number DS3815RCA1AL-008X000, required for connecting to normal frequency transducer

Transfer Function

Described in paragraph 1-2

Limits

Output: Adjustable from **1** to 10% change in generator voltage

Ceiling: Adjustable from 1 to 10% change in generator voltage.

Trips μPSS in adjustable time of 1 to 50 seconds

Over/under Voltage: Inhibits μPSS output to exciter for generator voltage deviation which exceeds set level for set time.

Level range: 1 to 20% of generator voltage

Time range: 0.25 to 5 seconds

Transient Event Recorder Trigger

Generator voltage excursion: Settable for 0 ± 20%

μPSS output excursion: Settable for 1 to 5

Fault Detection with Alarm

CWFT Ceiling Fault: Trips at ± 5 V after 2 seconds

μPSS Output Ceiling: Trips at preset level after preset time

Loss of Phase Fault: **Does** not trip

DMC Offline: Does not trip

Test Recorder

Volatile RAM record of a user-initiated test; identical to Transient Event Recorder including station identification and 2 11 scans (lines) of analog data

Scan lines: Each line contains time, date, phase A-B-C voltages and currents, power, vars, μPSS output, frequency deviation, test signal level

No pre-test records in the list

60 *Hz systems:* Line entered **in** record every 50 milliseconds

50 ***Hz*** *systems: Line* entered in record every 40 milliseconds

Forms G7 and G8 (has Transient Event Recorder)

Generator field volts included in data list

Transient Event Recorder

(Optional, forms G7 and G8) Volatile RAM records of events triggered by exceeding preset levels.

Number of event lists: 5

Event trigger: Excursion in average terminal voltage, adjustable from 0 to 20%.

Excursion in μPSS output, adjustable from 1 to 5%

(Excursion defined as change in successive pre-event scans)

Pre-trigger lines: Normally 10 (refer to text)

Event marker: An asterisk preceding a data line

Event list length: 2 11 lines plus station identifier

Line spacing:

60 *Hz systems:* S-line frequency cycles (50 **msec)**

50 Hz systems: Z-line frequency cycles (40 msec)

Data per ***Line:*** Date, time, volts and currents in each phase, total watts and vars, μPSS output, frequency deviation, generator field volts

Quantities expressed in percentage or per unit (PU), except for date and time

Accuracy:

μPSS output and *frequency* deviation: 2%

Voltage and current signals: 3%

Watts and *vars:* 5%

Clock/Calendar

Synchronized to the single-phase input power

Must be reset after a power interruption

Security

Digital program with user-installed constants protected from unauthorized alteration by means of either a key switch in the equipment or by both the key switch and a password

Monitor programs **request** required inputs (password) for access to the various functions

Password can be changed using both the keyswitch and the current password

CHAPTER 2

INSTALLATION

2-1. INTRODUCTION

WARNING

Dangerous voltages are used in this equipment. Be sure to use extreme caution when servicing the equipment.

This equipment contains a potential hazard of electrical shock or burn. Only personnel who are thoroughly familiar with the equipment and the instructions should install, test, or maintain this equipment.

Before installing, testing, or maintaining this equipment, technicians must fully understand and apply standard safety precautions for working with or around high voltages.

The μPSS is normally supplied as a part of the excitation equipment, usually mounted in the regulator or excitation cubicle. However, if it is shipped alone, use the receiving and handling procedures listed below.

2-2. RECEIVING AND HANDLING

GE carefully inspects and packs all equipment before shipping it from the factory. Upon receipt of this equipment, immediately inspect it for missing or damaged items. A packing list included in each case of the equipment itemizes the contents of each package of the shipment. Check the contents of each case against the packing list.

If damage is evident or if there is visual indication of rough handling, immediately file a damage claim with the carrier. Then notify both the transportation company and GE Drive Systems.

2-2.1. Unpacking. It is good practice to not completely unpack the equipment until it has been placed as near as possible to its permanent location. If the equipment has been exposed to low temperatures for an extended period of time, do not unpack it until it has reached room temperature. Otherwise, it sweats. Moisture on certain internal parts can cause electrical failure of insulated windings, especially where high voltages are present.

Use standard unpacking tools, including a nail puller. Carefully remove equipment from its container to avoid damaging or marring the part.

Small parts (such as bolts and screws) are packed in special containers to keep them together, but they may become separated. For this reason, carefully inspect packing material for loose parts before discarding it. Wipe off any particles of packing material or foreign substances that may be lodged in or between the parts.

Z-2.2. Storage and Protection. **If** the equipment is not installed immediately upon receipt, use the following storage procedures to protect it from breakage, corrosion, damage, and deterioration:

1. Unpack the equipment and label it.
2. The packing coverings do not protect the equipment for outdoor storage. Place the equipment under adequate cover, with the following protections:
   - ***a.*** Keep the equipment clean and dry, and protect it from extreme temperature variations, high humidity, salt spray, corrosive materials, dust, and rodents. Avoid sudden changes in temperature and humidity, which cause water condensation.
   - b. Use only breathable (canvas type) covering material – do not use plastic.
3. To prevent moisture from condensing on the equipment, as caused by variations in storage temperature, keep the equipment's internal temperature 5-10 °C higher than outside ambient. This can be done by adding space heaters or by using panel space heaters, when supplied. (Heaters require an external ac power supply.)

CAUTION

Remove all cartons and other miscellaneous material packed inside units before energizing any heaters.

2-3. PARTS LIST

The μPSS is an electronic assembly identified by the manufacturer's catalog (part) number 3S7932LA400G5 thru -G8. This assembly is made up of the components listed in Table 2-1. These components include a medium electronics module, which contains the DMC described in Chapter 1 (paragraph 1-3.1). Table 2-2 is a parts list for this module.

Table 2-2 also lists the location (slot) **where** each board is plugged in the module. For example, location 2B indicates module row 2, slot B.

Table 2-1. μPSS Parts List

Number	Name	Otv.
DS3820PSLB	Power Supply	1
DS3820PCNA	Complex Power Module	1
IC3602A178A	Microsynchronizer Isolation Module	1
Model 1200	Hayes@ Smartmodem (or equivalent)	1
Model 5 18	Encore 3ϕ Power Amplifier	I
*	Medium Electronics Module (MEM)	1

* See Table 2-2 for module parts list.

Table 2-2. Medium Electronics Module Parts List

Board Number	Board Name (Function)	Qty.	Location
DS3800HADA	Address Decoder	1	1B
DS3800HAIC	Analog to Digital Input	1	1F
DS3800HRRB	Relay Output	2	2B, 2D
DS3800HSCC	Contact Input	1	2A
DS3800HXPD	Expander	1	1K
DS3800NDAC	Digital to Analog Output	1	2F
DS3800NFTA	Frequency Transducer (CWFT)	1	2J
DS3800NFTA (opt.)	(Slot wired for spare NFTA)		2K*
DS3800NOAA	Operational Amplifier	1	2E
DS3800NQFA	Quadratic Filter	1	2H
DS3800NGMA	General Purpose Multiplier	2	2C, 2M
DS3800DGPA	DS3800NGMA Daughter Board	2	
DS3800NPTB	Complex Power	1	2L
DS3815DMCC	Microprocessor	1	1L
DS4815PSSA	Memory – RAM & EEPROM	1	1J
DS4815PSSD	Memory – EEPROM	1	1H

* Cable DS3815RCA1AL008X000 required to use this board as a test oscillator.

2-4. LOCATION AND CONNECTIONS

The μPSS is normally furnished as an integral part of the excitation and regulation equipment. In this case, refer to the drawings furnished with the excitation system and the excitation system's instruction manual. They contain the location and connection instructions.

If the μPSS is furnished separately, mount it in a clean, dry location free of excessive vibration. The ambient temperature should not exceed 40 "C.

2-4.1. Initial Wiring Checks. ***There should be no power connected at this point in the installation.***

Check all external wiring connections to the μPSS. Make sure that wire sizes and types are adequate for the requirements. (Refer to the drawings furnished with the excitation equipment.)

Note that the μPSS panel contains a modular telephone jack for directly connecting the (optional) modem. No other termination is provided.

CAUTION

Do not "hi-pot" or megger the µPSS. To do so can damage components. Before hi-potting or meggering any wiring or circuits located externally to the µPSS, the technician must first disconnect them from the µPSS.

Examine each board and the modem to ensure that all jumpers and switches are installed in accordance with the the elementary diagram fin-nished with the equipment.

Make sure that all boards are fully inserted and that all cable plugs are firmly seated in their jacks.

CHAPTER 3

OPERATIONAL CHECKS

3-1. INTRODUCTION

This chapter contains initial offline and online operational checks to ensure that the μPSS is working correctly. These include tuneup and software tests that require use of the monitor.

The initial operation, as well as maintenance and troubleshooting, require a computer terminal to access the User Monitor program in the μPSS software.

Anyone not familiar with the User Monitor program should review and practice using it before conducting the operations in paragraphs 3-5 to the end of the chapter. These checks require knowledge of the program.

To do this, first open the control signal connection between the μPSS and the voltage regulator. Then practice using the program by following procedures in the User Monitor section of Chapter 4. Using the modem, these practice steps can more easily be performed at a remote location within the station or at another site.

Before proceeding with paragraph 3-5, reconnect the output signal lead.

CAUTION

Misoperating or misadjusting the μPSS while it is connected to the excitation system can damage the generator. Do not deviate from the operational checks and procedures given in these instructions, without first checking with GE Drive Systems to ensure that they can be safely used.

SECTION I: OFFLINE OPERATION

3-2. PRE-POWER CHECKS

CAUTION

The μPSS must be out of service or the voltage regulator in the DC Mode when conducting all tests in Section I of this chapter. Failure to do so may damage the generator.

Make sure that the μPSS is out of service or that the voltage regulator is in the DC Mode. Before applying power, connect a computer terminal into the serial jack located on the DMCC.

Appendix G contains relay ladder diagrams that show elements of the OFFLINE/ONLINE control. All relays shown are logical relays, except for output relay PSS-ON. Relays used to control remote indicating lamps are not shown nor are three relays used ***to*** control the various test modes. These are shown in the elementary diagram.

3-3. POWER SUPPLY

Turn on the power to the μPSS by closing the power supply breaker. Check that the two amber lights on the power supply are on. The voltages can be checked at labeled testpoints on the MEM "backsplash" at the top, slightly left of center.

3-4. FREQUENCY TRANSDUCER

The NFTA board, module slot 2J, serves as a continuous wave frequency transducer. With the 3-phase input to the μPSS at rated voltage and frequency, check testpoints VA, VB, VC (on front of the board) with respect to ACOM. The voltages should be 2.1 V ± 0.3 V. Phase rotation ***must*** be A-B-C. Avariable voltage source can be substituted for the normal input from the generator PTs. GE recommends using a spare NFTA as a signal generator.

Check the dc output on TP'7. This voltage should be proportional to the deviation in frequency from rated. If necessary, adjust it to zero using the OFFSET adjustment at the top of the board.

***** NOTE *****

If familiar with the μPSS's User Monitor Program, then proceed with the remaining checks in this chapter. Otherwise, learn to use the monitor at this point, as described in paragraph 3-1.

3-5. UNIT PARAMETERS

CAUTION

The unit parameters entered into the digital programs (such as described in the following paragraphs), are written into RAM. To prevent their loss if a power loss occurs, use the SAVE EEPROM command after entering them. This stores them into EEPROM.

At power-up, the μPSS conducts a number of selftests, which check its hardware and software. After these

checks are complete, hit <RET> (the carriage return) several times to enter the User Monitor.

The software "security system" requires that the user turn a key switch, enter a password, or both.

3-5.1. Unit Identification. Enter the station identification and unit number using the ID command in the User Monitor.

Security: Key Switch in Test

Prompt	Constant
U> ID	Station name (80 characters maximum)
UID>	Unit number (80 characters maximum)

3-5.2. Tuneup. The following parameters must be set using the TU command in the User Monitor. The "Typ." column represents the setting as typically shipped from the factory.

Security: Key Switch In Test, Password

Prompt	Constant	Value	Typ.	Units
TU>G	Gain	____	2	V/V
TU>T	T2. (Lead)	____	2	Seconds
	T3. (Lag)	____	.02	Seconds
	T4. (Lead)	____	2	Seconds
	T5. (Lag)	____	.02	Seconds
TU>L				
L> 1	Upper output limit	____	10	%V(T)
	Lower output limit	____	-10	%V(T)
L> 2	Upper ceiling limit	____	9.5	%V(T)
	Lower ceiling limit	____	-9.5	%V(T)
	Ceiling time delay	____	5	Seconds
L> 3	Overvoltage limit	____	5	%V(T)
	Undervoltage limit	____	-5	%V(T)
	Output inhibit time	____	2	Seconds

3-5.3. Event Recorder,

Security: Key Switch in Test, Password

Prompt	Constant	Value	Typ.	Units
L> 4	Overvoltage event trigger	____	10	%V(T)
	Undervoltage event trigger	____	-10	%V(T)
	PSS output event trigger	____	.05	%V(T)

3-5.4. Auto Scaling.

Security: Key Switch In Test

Prompt	Constant	Value	Typ.
U> AS			
UAS >	Rated generator volts	____	1
UAS >	PT primary volts	____	1
UAS >	PT secondary volts	____	115
UAS >	Rated generatora current	____	.9
UAS >	CT priimary current	____	1
UAS >	CT secondary current	____	5
UAS >	Rated field volts	____	.95
UAS >	DCPT input volts	____	1
UAS >	DCPT output volts	____	5

3-5.5. Torsional Filter Enable/Disable. Enable or disable the torsional filter as required. This must be done throughthe Main Monitor. In general, the filter is enabled for machines other than hydro generators and disabled for hydro generators.

Security: Key Switch in Test, Password

U > MM
PASSWORD
* > M
DISPLAY/MOD
M > M
MODIFY? TORDIS
TORDIS = 1 *(1 = jilter* disabled)
VALUE? 0 *(entering 0 inserts the filter)*
TOR_DIS = 0
M>Q
*>U

3-5.6. Auto/Manual. Set the equipment for automatic or manual operation. (Disabled when the key switch is in Test.)

M > M *(command in Main Monitor)*
MODIFY? PVLT-OPT
PVLT_OPT = 1
VALUE? 0 *(1 = Auto; 0 = Manual)*
PVLT-OPT = 0
M > Q
* >

3-5.7. Saving Parameters. As cautioned at the beginning of these operations, the unit parameters entered in the preceding steps are written into RAM, and are lost if a power interruption occurs. To prevent loss of parameters entered at this point, perform a SAVE EEPROM operation.

*** NOTE ***

If you have made mistakes and are not confident that the parameters entered are correct, it is better to reboot or shut power off and back on, then reenter the values.

```
* > X
XFER SYSTEM
X > S
SAVE IN EEPROM? Y
--- WORKING ---
COMMAND COMPLETE
X > Q
* > U
USER MONITOR
ENTERING THE P.S.S. AND TRANSIENT
EVENT
RECORDERS USER MONITOR TASK
FOR HELP TYPE HH AND RETURN
U >
```

3-5.8. Optional Under/Overvoltage Detection. Set the μPSS output off for a short period of time if the generator deviates significantly from the nominal value.

```
M > M              (command in Main Monitor)
MODIFY? O-V-OPT
VALUE? 1           (1 = enable; 0 = disable)
O-V-OPT = 1
M > Q
* >
```

3-5.9. Phase Rotation . If the system's phase rotation is CBA instead of ABC, the user must rotate the input to the PA connector from the PT and CT signals, so that PA sees the ABC rotation.

This *is* done by switching the phase A and C inputs to the PA connector. Use the Modify command in the Main Monitor to modify the points table per phase rotation CBA.

```
M > M              (command in Main Monitor)
MODIFY? POINTS
POINTS(l) = 12
VALUE? <ret >
M > N
POINTS(S) = 0
VALUE? <ret>
M > N
POINTS(S) = 1
VALUE? 2 <ret >
POINTS(S) = 2
M > N
POINTS(4) = 2
VALUE? 1 <ret>
POINTS(4) = 1
M > N
POINTS(S) = 3
VALUE? <ret >
M > N
POINTS(G) = 4
VALUE? 5 <ret >
POINTS(G) = 5
M > N
POINTS(7) = 5
VALUE? 4 <ret >
POINTS(7) = 4
M > M
MODIFY? K_PH_ROT
K_PH_ROT = 0
VALUE? 1 <ret >
K_PH_ROT = I        (1 enables an alarm that gives a
                    phase rotaion message any time phase
                    A or C trip)
M > Q
* >
```

3-6. FREQUENCY RESPONSE TEST

The initial frequency response test should be run with the μPSS offline. The required S-phase 115 V ac input signal can be from either the normal generator input to the μPSS, or from an independent S-phase source. A third choice, which is recommended, is to configure a spare NFTA in slot 2K to generate S-phase signal level voltages. This requires a 10-conductor cable to connect jacks JB on the two NFTA boards. The JA connector on the NFTA/2J board must be disconnected if this source is used.

```
Security: Key Switch in Test
U> TM
UTM> S
S> 3               (enter amplitude for 1/2 Hz sine wave)
S>                 (1 to modulate system CWFT or 2 to
                    modulate spare CWFT)
```

When the message "SINGLE FREQUENCY TEST COMPLETED" appears enter:

```
S> <ret>
U> GP
```

A message appears giving the gain and phase at 1/2 Hz. Running the tests offline is primarily a check of the functions. These tests are mostly useful in performance checks after the equipment is placed in service.

Initially, more information is desired. Use an external oscillator and phase shift instrument to examine the entire frequency spectrum of interest. (See the references in Appendix B for more complete details for conducting such tests.) The modulating signal should be connected to NOAA/2E TP9 to modulate the system CWFT or NOAA/2E TP7 to modulate the spare CWFT. The input scaling for both signals is 0.2 % delta frequency per volt input, which results in a 1: 1 voltage gain from modulating input to system CWFT output.

The gain measurement is 1/5 of the actual gain of the μPSS. Figure 1-1 (Chapter I) is a gain-fi-equency plot of the μPSS for the parameters listed.

At this point, connect an oscilloscope to TP2 of NOAA/2E and increase the input signal level until the output begins to saturate. Saturation should occur at the maximum and minimum output levels set initially. If necessary, set the input frequency to give the greatest gain, as determined above.

3-7. FAULTS

3-7.1. Offline Fault. The DMC OFFLINE fault indicates that the processor segments have been turned off. It does not occur under normal operating conditions, but is how new PROM sets are shipped from the factory. The processor segments can also be turned off or on from the Main Monitor.

The DMC can be placed online through the Main Monitor.

Security: Key Switch in Test, Password

U > MM

PASSWORD

* > O

ONLINE STAT

O > C *(current and request state)*

NEW STATE? ON *(current and request state)*

O > D (current state of the microcomputer)

O > Q

* > X

XFER SYSTEM

X > S

SAVE IN EEPROM? Y

--- WORKING ---

COMMAND COMPLETE

X > Q

* > U

Use the User Monitor to clear the fault list. The alarm LEDs should be off. All amber and green lights on the front of the boards should be on.

3-7.2. Loss of Phase Fault. While operating with a 3-phase 115 V ac input, disconnect one of the PT input lines on the complex power module. The alarm LED on the Local Control Panel should light after a few seconds time delay. The ONLINE LED should remain on. Reconnect the PT line. Now, examine the fault list.

U > FD

LOSS OF PHASE FAULT

Again, clear the fault list. The alarm LED should go out.

3-7.3. Frequency Transducer Ceiling Fault. This fault is checked continuously as soon as the generator produces a minimum level of power. For initial checkout, the logic variable POFCMD needs to be forced to zero using the Main Monitor.

U > MM

PASSWORD

* > F

FORCE

F > F

FORCE? POFCMD

POFCMD = 1

STATE? 0

POFCMD = OF

F > Q

* > U

Connect a 15-o-15 volt variable dc supply to TP9 on NOAA/2E. Set the signal at 5.2 V. The μPSS should go offline in 2 seconds. The fault list should show:

U > FD

FREQUENCY TRANSDUCER CEILING FAULT

Reduce the signal to zero, reset the alarm, clear the fault list, and place the μPSS back online. Repeat the test using a negative 5.2 V.

Upon completing the test, unforce POFCMD.

U > MM

PASSWORD

* > F

FORCE

F > C

REMOVE ALL? Y

COMMAND COMPLETE

F > Q

* > U

3-7.4. μPSS Output Ceiling Fault. This fault is checked continuously as soon as the generator produces a minimum level of power. For initial checkout, the logic variable POFCMD needs to be forced to zero using the Main Monitor.

T-J > MM

PASSWORD

* > F

FORCE

F > F

FORCE? POFCMD

POFCMD = 1

STATE? 0

POFCMD = OF

F > Q

* > U

Reset the alarm, clear the fault list, and place the μPSS online. Connect the variable dc supply to TP8 of NFTA/2J. Apply a 10 V step. The μPSS should trip after the time delay is set for ceiling time delay, and the alarm LED should light. The fault list should show:

U > FD

PSS OUTPUT CEILING FAULT

If the set time delay exceeds 3 seconds, temporarily set it to slightly less than this value. Repeat the test using -10 V. Reset the time delay if it was changed.

Upon completing the test, unforce POFCMD.

U > MM

PASSWORD

* > F

FORCE

F > C

REMOVE ALL? Y

COMMAND COMPLETE

F > Q

* > U

3-7.5. μPSS Segment Not Running. Turn off the power supply. Remove the cable from the front edge of the NFTA/2J and slide it forward enough to disconnect the connector on the back. Re-apply power. After the DMC runs through its startup checks, examine the fault list.

U > FD

PSS SEGMENT NOT RUNNING

Again, turn off the power supply, reseat the NFTA board, and reconnect the front cable. Turn power back on.

SECTION II: ONLINE OPERATION

3-8. INITIAL ONLINE OPERATION

After completing initial offline testing, place the μPSS online. The voltage regulator should be in the AC Mode and the generator should be operating near rated output.

3-8.1. Manual Mode. For a manually-operated μPSS, online testing should be conducted from the control room. The operator should operate the ON-LINE pushbutton while observing generator voltage, vars, and exciter voltage.

3-8.2. Auto Mode. For a μPSS that operates automatically, online testing requires personnel both at the μPSS and in the control room with a telephone line between them.

To ensure local OFF/ON control, open the 52G generator auxiliary contact. It connects to PSSTB-12, which is normally jumpered to PSSTB-15 fpr Auto operation. Place the μPSS ONLINE with the local pushbutton, while the operator observes generator voltage, vars, and exciter voltage.

3-8.3. **Tuneup.** If operation is satisfactory, the μPSS can now be tuned to the system in accordance with previous planning. Reconnect the 52G contact if it was opened for AUTO operation.

CHAPTER 4

USING THE MONITOR

4-1. INTRODUCTION

As mentioned previously, the μPSS includes a program that enables access to the μPSS software via a computer terminal or modem. The User Monitor is designed for a user to conduct offline and online operational checks, and for entering unit parameters into the digital programs. This chapter describes the User Monitor in detail.

This manual discusses the Main Monitor only in the detail needed for a user to save parameters into EEPROM and to perform minor software changes.

The software "security system" requires that the user turn a key switch, enter a password, or both when using the monitors.

4-2. USER MONITOR

The User Monitor is menu driven, meaning that the user selects a command from a menu or library of available commands. The program responds by either executing the command or by prompting the user for further information required to execute the command. Unknown commands and illegal data entries produce error messages.

After μPSS selftests complete, the user can enter the User Monitor by pressing <ret > (the return key). The following message then appears:

ENTERING THE P.S.S. AND TRANSIENT

EVENT RECORDERS USER MONITOR

FOR HELP TYPE HH AND RETURN

U >

The U> prompt indicates that the User Monitor is online and is ready for a command input. Typing HH <ret > displays the menus of available commands, which vary according to the options purchased. Table 4-l lists these commands in alphabetical order, along with their function and security requirement.

Each Tune Up command causes the User Monitor task to progress to a lower level of processing. If the command is fulfilled immediately upon execution and no additional user input is needed, the monitor returns to its highest level, indicated by the U > prompt. If additional data must be entered, the monitor displays an appropriate lower level prompt, such as "ULD > " for List Display. The prompt, therefore, indicates the current level in the User Monitor. If the operator wants to terminate the current command, hitting <ret > returns the User Monitor to the next highest level.

The following paragraphs describe the User Monitor commands in alphabetical order. An example is also included to show a typical use of each command. Note that these examples, however, do not show error messages or prompts that occur if inputs are not acceptable. Such incorrect inputs are: numbers outside the acceptable range, the Key Switch not in the Test position, or an invalid password. The display's menu-style arrangement should enable even the novice user to properly perform the provided functions. Nonactive functions, such as viewing present limit values, are always possible.

CAUTION

Use care when the Key Switch is in the Test position. That permits changes in the actual control process, whether offline or online.

Table 4-l. User Monitor Commands

Command	Function	[1]Security
As	Auto Scaling of scanned data	KST
DR	Display Ratings	None
FC	Fault Clear	KST
FD	Fault list Display	None
GP	Calculate PSS Gain and Phase	
HH	Help	None
ID	Enter Station and Unit ID	KST
LC (optional)	Clear all Lists except the test list (#6)	None
LD	List Display	None
LS	List Status for all lists	None
MM	Main Monitor	KST, PASS
PC	Password Change	KST, PASS
SS	Current Stabilizer Status	None
s u	Display Station and Unit ID	None
TD	Time Display	None
TM	PSS Test Mode	KST
TS	Time Set	None
TU	Time Set	[2]KST, PASS

[1]KST = Key Switch in Test Position
PASS = Password required

'Except for recorder limits, for which Key Switch can be in Normal position

4-2.1. AS – Autoscale Command . This command elicits data from the user for various equipment ratings, as shown above. The data is used to scale the analog values presented in the recorder lists so that they are displayed in per unit (PU) form. The command can be aborted at any time by hitting <ret> instead of the requested data. An abort results in no scaling changes. To make the scaling changes permanent, an user must perform an EEPROM save of the data (using the Main Monitor).

Security: Key Switch in Test

Example:

U > AS

ENTER THE P.T. TRANSFORMERS PRIMARY RATING IN VOLTS

UAS > 22000

ENTER THE P.T. TRANSFORMERS SECONDARY RATING IN VOLTS

UAS > 120

ENTER THE C.T. TRANSFORMERS PRIMARY RATING IN AMPS

UAS > 12000

ENTER THE C.T. TRANSFORMERS SECONDARY RATING IN AMPS

UAS > 5

ENTER THE GENERATORS LINE POTENTIAL RATING IN VOLTS

UAS > 22000

ENTER THE GENERATORS LINE CURRENT RATING IN AMPS

UAS > 11284

ENTER FIELD DCPT INPUT CALIBRATION IN VOLTS

UAS > 463

ENTER FIELD DCPT OUTPUT CALIBRATION IN VOLTS

UAS > 5

ENTER FIELD DC POTENTIAL RATING IN VOLTS

UAS > 420

AUTO SCALING SUCCESSFULLY COMPLETED

TO SAVE AUTO SCALING RESULTS AN EEPROM WRITE MUST BE PERFORMED

U >

4-2.2. DR – Display Rating Command. This command produces a complete list of faults that are detected. The FD command displays each type that has been detected since the list was last cleared. Entries are not time dated. A fault that occurs, clears, and occurs again is shown only as one entry.

Security: None

Example:

U > DR

(a listing of the scaling data is output)

U >

4-2.3. **FC – Fault List Clear Command.** This command clears the fault list of all entries except those still present in the system. See the FD command for fault display.

Security: Key Switch in Test

Example:

U > FD

POWER SYSTEM STABILIZER FAULT LIST

PSS OUTPUT CEILING FAULT

LOSS OF PHASE FAULT

U > FC

THE FAULT HAS BEEN CLEARED

POWER SYSTEM STABILIZER FAULT LIST

NO FAULTS IN LIST AT PRESENT

4-2.4. FD – Fault Display Command. This command displays the fault list.

Security: None

Example:

U > FD

POWER SYSTEM STABILIZER FAULT LIST

PSS OUTPUT CEILING FAULT

LOSS OF PHASE FAULT

DMC OFFLINE

FREQUENCY TRANSDUCER CEILING FAULT

PSS SEGMENT NOT RUNNING

U >

4-2.5. GP – Gain and Phase Command. This command causes a stabilizer gain and phase calculation to be made on the Test Recorder list using discrete Fourier transform analysis. If the test list is not in *a* complete state, an error message is given. Meaningful results are obtained only if the list was accumulated as the result of running a single frequency test (either system or spare CWFT). A positive value indicates phase lead, and a negative value indicates lag. The Fourier analysis is performed on the list data for the 1/2 Hz sine wave input and the stabilizer output. The displayed gain is the total μPSS gain, including the gain in the CWFT.

Security: None

Example:

U > GP

PSS GAIN = (PSS OUT / .5 HZ SINE) = 7.2

PSS PHASE SHIFT = 7.2

U >

4-2.6. HH – **Help Command.** This command displays in alphabetical order the complete command list. A command denoted with an asterisk may or may not appear on a particular system, depending on the options furnished.

Security: None

Example:

U > HH
AS -- AUTO SCALING OF SCANNED DATE
DR -- DISPLAY RATING
FC -- FAULT CLEAR
FD -- FAULT LIST DISPLAY
GP -- **CALCULATE PSS GAIN AND PHASE**
HH -- HELP
ID -- ENTER STATION AND UNIT ID
LC* -- CLEAR ALL LISTS EXCEPT THE TEST LIST (#6)
LD -- LIST DISPLAY
LS -- LIST STATUS FOR ALL LISTS
MM -- MAIN MONITOR
PC -- PASSWORD CHANGE
SS -- CURRENT STABILIZER STATUS
SU -- DISPLAY STATION AND UNIT ID
TD -- TIME DISPLAY
TM -- PSS TEST MODE
TS -- TIME SET
TU -- TUNE UP
U >

4-2.7. ID -- Enter Station and Unit Identification Command. This command is used to enter the station name and unit for a particular installation. Be ***sure to do an EEPROM write to save the data.***

Security: Key Switch in Test

Example:

U > ID
ENTER STATION (1 TO 80 CHARACTERS)
UID > WORLDS GREATEST POWER CO., ANYTOWN, U.S.A. STATION
ENTER STATION (1 TO 80 CHARACTERS)
UID > UNIT #3
AN EEPROM WRITE MUST BE DONE TO SAVE NEWLY ENTERED DATA
U >

4-2.8. LC -List Clear Command. This command, available only with the optional Transient Event Recorder, clears all recorder lists, except for the Test Recorder list. When the optional Transient Recorder is present, it uses lists 1 to 5 and the Test Recorder uses list 6. When this command executes, data clears from lists 1 to 5, and list 1 immediately begins ***to accumulate*** pre-event data. List 6 is unaffected.

Security: None

Example:

u > LC
THE TRANSIENT EVENT LISTS HAVE BEEN RESET

LIST STATUS		TRIGGERED DATE
01	PRE TRIGGER ACCUMULATION	
02	QUIESCENT	
03	**QUIESCENT**	
04	QUIESCENT	
05	QUIESCENT	
06	**ACCUMULATION**COMPLETE	15 APR 86 11:20:33:64

U >

4-2.9. LD -- List Display Command. **This com-**mand displays the Test Recorder list or a Transient Event Recorder list. The LD command is entered from the highest level of processing in the User Monitor. The User Monitor task responds in one of two ways: (1) If the optional Transient Event Recorder is not present, the Test Recorder List is displayed; or (2) if the Transient Event Recorder is present, the monitor prompts the user to enter the number of the list to be displayed. In this second case, valid list numbers are from 1 to 5 for the Transient Event Recorder lists, or 6 for the always present Test Recorder list. (The Transient Event Recorder lists pre- and post-event data for transient events; the Test Recorder lists only data following the start of a test.)

Hitting <ret > while the list is being displayed aborts the list and returns the operator to the old level.

Figure 5-1 shows the list format for a terminal capable of displaying 132 colunms, which is highly desirable. A terminal with an 80-column display can be used if the wrap-around option is enabled. The μPSS expects an initial baud rate of 1200, which ***can*** be user modified. See Appendix A for a complete listing of the terminal characteristics required to communicate with the μPSS.

Security: None

Example:

U > LD
ENTER NUMBER OF LIST TO BE DISPLAYED
ULD > 1
(See Figure 5-l)

4-2.10. LS -- List Status Command. This command displays the present status of all event lists in the system, including the optional Transient Event Recorder lists and the Test Recorder list. If the optional Transient Event Recorder is present, the LS command displays a response similar to the LD command, where lists 1 to 5 are for the Transient Recorder and list 6 **is for** the Test Recorder. If the Transient Event Recorder is not present, the LS command presents a display similar to the following, in which list 1 is for the Test Recorder.

Security: None

Example:

u > LS

LIST	STATUS	TRIGGERED DATE
01	ACCUMULATION COMPLETE	22 APR 86 08:13:27:53
02	ACCUMULATION COMPLETE	15 APR 86 09:17:28:05
03	PRE TRIGGER ACCUMULATION	
04	QUIESCENT	
05	QUIESCENT	
06	ACCUMULATION COMPLETE	15 APR 86 11:20:33:64

U >

U > LS

LIST	STATUS	TRIGGERED DATE
01	ACCUMULATION COMPLETE	22 APR 86 08:13:27:53

U >

4-2.11. MM – Main Monitor Command. This command is used to enter the DMC's high level monitor. See paragraph 4-3 for details on using this monitor.

4-2.12. PC – Password Change Command. This command is used to change the current password.

Security: Key Switch in Test, Password

Example:

u > PC

ENTER THE CURRENT PASSWORD

> xxxxxx

ENTER THE PROPOSED PASSWORD

> xxxxxx

FOR VERIFICATION PURPOSES AGAIN

ENTER THE PROPOSED PASSWORD

> xxxxxx

AN EEPROM WRITE MUST BE DONE TO SAVE NEWLY ENTERED DATA

U >

4-2.13. SS – Stabilizer Status Command. This command is used to display the current status of μPSS.

Security: None

Example:

u > ss

THE PSS IS CURRENTLY ON

U >

4-2.14. SU -Station and Unit Display Command. This command is used to display the unit and station. It is recommended that any remote user execute this command when first connected, to avoid inadvertently operating on the wrong unit.

Security: None

Example:

u > su

THIS STATION AND UNIT ARE PRESENTLY DEFINED AS THE WORLD'S GREATEST POWER CO., ANYTOWN, U.S.A. STATION

UNIT #3

U >

4-2.15. TD – Time Display Command. This command displays the current time in the μPSS clock calendar.

Security: None

Example:

U > **TD**

CURRENT SYSTEM TIME IS 15-APR-86 11:23:58

U >

4-2.16. TM – Test Mode Display Command. This command is used to perform several different tests to check μPSS operation. The Help command **lists** the options available.

Security: Key Switch in Test

Example:

U > TM

TM > H

(Displays the help menu, which is a list of options accessible in the Test Mode display.)

C -- CURRENT TEST STATUS

D -- DIGITAL TEST

E -- EXCITATION SIGNATURE TEST

H -- HELP

Q -- QUIT

S -- SINGLE FREQUENCY TEST

U >

4-2.16.1. C – **CURRENT TEST STATUS.** This command Displays the current test status with one of the following messages:

NO TEST IN PROGRESS

EXCITATION TEST WITH PSS IN PROGRESS

EXCITATION TEST WITHOUT PSS IN PROGRESS

SINGLE **FREQUENCY** TEST WITH **SYSTEM CWFT IN PROGRESS**

SINGLE FREQUENCY TEST WITH SPARE CWFT **IN** PROGRESS

DIGITAL TEST IN PROGRESS

4-2.16.2. D – DIGITAL TEST. This command runs a jog test that adds a dc voltage step to the output of the frequency transducer. The test is used to examine the μPSS's step response (signature). The step magnitude is adjustable from 0.02 to 0.4% change in generator frequency. Scaling is 5 V/% f%.

Example:

TM > D

THIS JOG TEST INSERTS AN ADJUSTABLE STEP IN THE OUTPUT OF THE FREQUENCY TRANSDUCER. THE MAGNITUDE OF THE STEP IS ADJUSTABLE BETWEEN 0.02 AND 0.4% CHANGE IN GENERATOR FREQUENCY.

1. TURN ON TEST.
2. TURN OFF TEST.
3. ADJUST STEP MAGNITUDE.

TO CHOOSE A FUNCTION, ENTER APPROPRIATE NUMBER (1-3) OR RETURN TO EXIT.

D > 3

PRESENT STEP MAGNITUDE: .02% SPEED CHANGE

NEW STEP PERCENTAGE: .05

D > 1

DIGITAL JOG TEST STARTED

D > 2

DIGITAL JOG TEST TERMINATED

4-2.16.3. E – EXCITATION SIGNATURE TEST. This test is used to study the response of the excitation control to a step input. It injects a step signal of a 5 second duration into the μPSS output. (If necessary, the duration of this test can be changed by changing variable ETST_CNT using the Main Monitor, as described in paragraph 4-3.2.) The normal μPSS output control signal can be either present or disabled when the test is initiated. The step amplitude is expressed in percentage of change of generator terminal voltage. It is adjustable from 0 to 10%.

To start the test, enter 1 or 2. The terminal displays a message indicating the start of the test. Terminal operation is suspended during the 5-second duration of the test.

Example:

TM > E

THIS TEST INJECTS A POSITIVE STEP WITH ADJUSTABLE AMPLITUDE OF LESS THAN 10% INTO THE EXCITATION REGULATOR. THE STEP SIGNAL CAN BE SUPERPOSED WITH THE PSS OUTPUT SIGNAL OR APPLIED ALONE WITH THE PSS DISABLED:

1. START EXCITATION SIGNATURE TEST WITH PSS ENABLED.
2. START EXCITATION SIGNATURE TEST WITHOUT PSS.
3. ADJUST STEP MAGNITUDE.

TO CHOOSE A FUNCTION, ENTER APPROPRIATE NUMBER (l-3) OR RETURN TO EXIT.

E > 3

CURRENT STEP MAGNITUDE: 1.0% OF GENERATOR TERMINAL VOLTAGE

NEW STEP MAGNITUDE: 2

E > 1

EXCITATION CONTROL SYSTEM SIGNATURE TEST WITH PSS STARTED.

(delay until test completion, normally 5 seconds)

EXCITATION CONTROL SYSTEM SIGNATURE TEST WITH PSS COMPLETED.

E >

4-2.16.4. S – SINGLE FREQUENCY TEST. This test performs an overall system check of the μPSS. It modulates the μPSS 60 Hz frequency input with a 1/2 Hz sinewave. Both the 1/2 Hz sinewave and the μPSS output signal are available as analog *signals* on the NDAC board. These can be connected to an external device, such as a strip chart recorder. The external load must be 10 kΩ or more.

There are two ways to run the single frequency test.

1. Generator offline. A spare NFTA board and cable are required to generate the 3-phase 50/60 Hz signals feeding the main NFTA ii-equency transducer. This setup is used mainly during the startup phase. Either or spare or the system NFTA can be modulated to check the input/output relationship.
2. Without a second NFTA. The μPSS must connect to the S-phase power system or another 3-phase 115 V ac source. The NFTA frequency transducer is modulated with a 1/2 Hz sinewave generated in the μPSS's digital portion. This frequency does not stimulate any resonance frequencies and can be run with the generator online without disturbing normal operation.

Example:

TM > S

THIS TEST MODULATES THE PSS INPUT SIGNAL WITH A 1/2 HERTZ SINEWAVE. THE AMPLITUDE IS ADJUSTABLE FROM .01 TO .4% CHANGE IN FREQUENCY. THE TEST ALSO ALLOWS TO CHOOSE THE SYSTEM CWFT CARD OR A SPARE CWFT CONFIGURED AS OSCILLATOR TO GENERATE THE MODULATED SIGNAL.

1. TURN ON TEST USING SYSTEM CWFT.
2. TURN ON TEST USING SPARE CWFT.
3. ADJUST SIGN WAVE AMPLITUDE.

TO CHOOSE A FUNCTION, ENTER APPROPRIATE NUMBER (l-4) OR RETURN TO EXIT.

S > 4

CURRENT SINE WAVE AMPLITUDE: 0.5% CHANGE IN FREQUENCY

NEW SIGN WAVE AMPLITUDE PERCENTAGE: .04

S > 1

SINGLE FREQUENCY TEST STARTED USING SYSTEM CWFT

S > 3

SINGLE FREQUENCY TEST COMPLETED

NO TESTS IN PROGRESS

4-2.17. TS – Time Set Command. This command is used to set the clock-calendar in the μPSS. The time entry format is 2 digits each for the year, month, day, hour, minutes, and seconds, with each separated by a colon. The new time is echoed back to the user.

Security: None

Example:

U > TS

ENTER THE TIME AS YY:MM:DD:HH:MM:SS

UTS > 87:01:22:11:20:15

CURRENT SYSTEM TIME IS 22-JAN-87 11:20:15

U >

4-2.18. TU – Tune Up Command. This command provides the necessary interface to change certain parameters of the μPSS. These parameters include the lead/lag time constants, the gain of the digital filter, and certain limit parameters that ensure proper μPSS operation. The Help command lists the options available.

Security: Key Switch in Test, Password

Example:

U > TU

TU > H

Displays the help menu, which is a list of options accessibb in the Tune Up Mode display. Note that this command differs from the HH command.)

H -- HELP

G -- GAIN ADJUSTMENT

L -- LIMIT ADJUSTMENT

Q -- QUIT

T -- LEAD/LAG TC ADJUSTMENT

4-2.18.1. G – GAIN ADJUSTMENT

CAUTION

Use caution when changing the gain online. Use only small increments of 0.5 maximum to avoid shocking the system.

The digital filter implemented in the μPSS consists of two parts: the tortional rejection filter and a double lead/lag stage.

The tortional rejection filter is a fixed unity gain filter that attenuates the frequency range in which torsional resonances occur. This filter has no adjustments. For application in which torsional rejection is not desired or required, it can be deleted. This leaves a simple high pass (washout) filter.

The second part of the filter is a double lead/lag stage of the form:

$$K \times [(1 + sT2) \times (1 + sT4)]/ [(I + sT3) \times (1 + sT5)]$$

Entering G allows the user to change the gain parameter K. This gain is adjustable in the range of 0.1 to 50. The gain displayed is the product of a gain of 5 in the CWFT and digital changed to allow for the gain margin test. (This test consists of incrementing the gain step-by-step *to* find the critical gain setting.)

Security: Key Switch in Test, Password

Example:

TU > G

THE PSS GAIN CONSTANT IS ADJUSTABLE FROM .1 TO .50

CURRENT GAIN CONSTANT: 5.0

ENTER THE CURRENT PASSWORD

> xxxxxx *(enter password here)*

ENTER NEW GAIN CONSTANT OR RETURN TO EXIT: *5.2*

NEW GAIN CONSTANT WILL BE: 5.2, TYPE "Y" TO EXECUTE: Y

ENTER NEW GAIN CONSTANT OR RETURN TO EXIT: <ret>

TU > *(EEPROM* save *required)*

4-2.18.2. T -TIME CONSTANT ADJUSTMENT. The lead/lag portion of the filter has four adjustable time constants that correspond to the analog representation of the filter in the Laplace domain. They must be set within the constraints listed in the display.

Security: Key Switch in Test, Password

Example:

TU > T

THE FOLLOWING ADJUSTMENTS CAN BE MADE TO THE LEAD/LAG TIME CONSTANTS:

LEAD CONSTANTS T2, T4 ARE ADJUSTABLE FROM *0.20* TO 1.50 SECONDS.

LAG CONSTANTS T3, T5 ARE ADJUSTABLE FROM *0.020* TO 0.20 SECONDS.

CURRENT VALUES: T2 = 0.25, T3 = 0.025, T4 = 0.25, T5 = 0.02

ENTER THE CURRENT PASSWORD

> xxxxxx *(enter password here)*

ENTER NEW TIME CONSTANTS OR RETURN IF NO CHANGE DESIRED

NEW T2: .2

NEW T3: .02

NEW T4: .2

NEW T5: .02

TU > *(μPSS offline save required)*

4-2.18.3. L – LIMIT ADJUSTMENTS. There are four adjustable limit settings, which the monitor lists when the user enters L at the TU > prompt. At the

L > prompt, the user then enters a number (1 – 4) to correspond with the adjustment selected.

Security: Key Switch in Test, Password

*** NOTE ***

No security is provided for the Recorder Limit setting.

Example:

TU > L

THE FOLLOWING LIMIT ADJUSTMENTS CAN BE MADE:

1. PSS OUTPUT LIMIT IS ADJUSTABLE BETWEEN ± 1.0% AND ± 10%.

2. CEILING LIMIT DETECTION IS ADJUSTABLE BETWEEN ± 1 VAND ± 10 V, OVERANADJUSTABLETIME PERIOD OF 1 TO 50 SECONDS.

3. OPTIONAL OVER/UNDERVOLTAGE DETECTION IS ADJUSTABLE BETWEEN ± 1% TO ± 20%. PSS OUTPUT IS FORCED TO 0 FOR AN ADJUSTABLE TIME PERIOD OF .25 SECONDS TO 5 SECONDS.

4. RECORDER EVENT TRIGGER LEVELS ARE ADJUSTABLE FOR GENERATOR VOLTAGE EXCURSIONS FROM 0 TO ±20% AND PSS OUTPUT EXCURSIONS FROM 0.01.

TO CHOOSE A FUNCTION, ENTER APPROPRIATE NUMBER (1 - 4) OR RETURN TO EXIT:

L > *(user enters number from 1 to 4 to select desired limit adjustment)*

1 – μPSS Output Limit Adjustment. This limit determines the maximum effect that the μPSS can have on the main exciter control. The limits can be set independently in the positive and negative directions between 1 an 10 V. This is numerically equivalent to percentage of nominal generator voltage, VT.

Example:

L > 1

CURRENT UPPER OUTPUT LIMIT: 10%, CURRENT LOWER OUTPUT LIMIT: -10%

ENTER THE CURRENT PASSWORD

> xxxxxx *(enter password here)*

ENTER NEW LIMITS DESIRED OR RETURN IF NO CHANGE DESIRED

NEW UPPER OUTPUT LIMIT: 8

NEW LOWER OUTPUT LIMIT: -8

L > (save required)

2 – μPSS Output Ceiling Detection. This feature protects against a sustained dc output of the μPSS. Under normal operating conditions, the μPSS output varies only slightly around zero. Any sustained dc level indicates a system failure. The positive and negative limits are independently adjustable from 1 to 10 V. The time constant over which that dc level has to be maintained is adjustable from 1 to 50 seconds.

Example:

L > 2

CURRENT UPPER OUTPUT LIMIT: 10 V, CURRENT LOWER OUTPUT LIMIT: -10 V

CEILING TIME DELAY: 5.0 SECONDS

ENTER THE CURRENT PASSWORD

> xxxxxx (enter password *here)*

ENTER NEW LIMITS OR RETURN IF NO CHANGE DESIRED

NEW UPPER OUTPUT LIMIT: 8

NEW LOWER OUTPUT LIMIT: -8

NEW TIME DELAY: 2

L > *(save required)*

3 – Under/Overvoltage Detection. This optional feature provides the capability of turning the μPSS output off for a short period if the generator voltage deviates significantly from the nominal value (such as caused by a major fault condition in the system). For some faults, the performance of the overall exciter control can be improved by temporarily disabling the μPSS if a fault condition occurs. The over- and under-voltage limits are independently adjustable between 1 and 20% of nominal generator voltage. Therefore, the μPSS detects whenever over a 50 msec period the generator voltage spread exceeds the limits. The time period for which the μPSS output is zeroed is adjustable from.25 to 5 seconds.

This function is enabled by using the main monitor to modify O_U_OPT to 1.

Example:

L > 3

CURRENT OVERVOLTAGE LIMIT: 5.0%, CURRENT UNDERVOLTAGE LIMIT: -5.0%

VOLTAGE TIME DELAY: 2:00 SECONDS

ENTER THE CURRENT PASSWORD

> xxxxxx *(enter password here)*

ENTER NEW LIMITS DESIRED OR RETURN IF NO CHANGE DESIRED

NEW OVERVOLTAGE LIMIT: 6

NEW UNDERVOLTAGE LIMIT: -6

NEW TIME DELAY: -1

L > (save *required)*

4 – Recorder Event Limit Adjustment. The optional Transient Event Recorder triggers a data record on two different events: either **a** generator voltage excursion or a μPSS output excursion. The event trigger, caused by a generator voltage swing, is adjustable between ± 20% of nominal voltage. The μPSS output event limit is adjustable from 0.01 to 0.05 power unit generator voltage change. (That is, a 5 V μPSS output causes a .05 PU generator voltage change.)

Example:

L > 4

PRESENT OVER VOLTAGE EVENT TRIGGER: 10.0, PRESENT UNDER VOLTAGE EVENT TRIGGER: -10.0

PRESENT PSS OUTPUT EVENT TRIGGER: 0.050 P.U. GENERATOR VOLTAGE

(No security provision, *so no password required)*

ENTER NEW LIMITS OR RETURN IF NO CHANGE DESIRED

NEW OVERVOLTAGE LIMIT: 9

NEW UNDERVOLTAGE LIMIT: -9

NEW PSS OUTPUT LIMIT: .05

L > *(save required)*

4-2.18.4. Q – QUIT. To exit the tuneup display, type Q and <ret > . Hitting <ret > alone also moves the prompt to the next higher level.

Upon exiting the display, a message appears to warn the user that any permanent changes must be saved in EEPROM using the Main Monitor.

4-2.19. Sample User Session. The following is a sample user session for the User Monitor.

ENTERING THE P.S.S. AND TRANSIENT

EVENT RECORDERS USER MONITOR

FOR HELP TYPE HH AND RETURN

U >

LJ > su

THIS STATION AND UNIT ARE PRESENTLY DEFINED AS THE WORLD'S GREATEST POWER CO., ANYTOWN, U.S.A. STATION

UNIT #3

u > LS

LIST	STATUS	TRIGGERED DATE
01	ACCUMULATION COMPLETE	22 APR 86 08:13:27:53
02	ACCUMULATION COMPLETE	15 APR 86 09:17:28:05
03	PRE TRIGGER ACCUMULATION	
04	QUIESCENT	
05	QUIESCENT	
06	ACCUMULATION COMPLETE	15 APR 86 11:20:33:64

U > LD

ENTER NUMBER OF LIST TO BE DISPLAYED

ULD > 1

(See Figure 5-1)

U > LD

ENTER NUMBER OF LIST TO BE DISPLAYED

ULD > 3

SPECIFIED LIST IS EMPTY . COMMAND NOT ALLOWED

u > LC

THE TRANSIENT EVENT LISTS HAVE BEEN RESET

LIST	STATUS	TRIGGERED DATE
01	PRE TRIGGER ACCUMULATION	
02	QUIESCENT	
03	QUIESCENT	
04	QUIESCENT	
05	QUIESCENT	
06	ACCUMULATION COMPLETE	15 APR 86 11:20:33:64

u > SS

THE PSS IS CURRENTLY ON

U > FD

POWER SYSTEM STABILIZER FAULT LIST

PSS OUTPUT CEILING FAULT

LOSS OF PHASE FAULT

DMC OFFLINE

FREQUENCY TRANSDUCER CEILING FAULT

PSS SEGMENT NOT RUNNING

U > FC

THE FAULT HAS BEEN CLEARED

KEYLOCK SWITCH NOT IN TEST MODE

U > TD

CURRENT SYSTEM TIME IS 15-APR-86 11:23:58

U > GP

PSS GAIN = (PSS OUT / .5 HZ SINE) = 7.2

PSS PHASE SHIFT = 5'7.6

U >

4-3. MAIN MONITOR

CAUTION

Take care when using the Main Monitor commands. Certain commands have the potential to disrupt or destroy the system sofhuare, as well as cause undesirable outputs from the μPSS.

The Main Monitor (DMC Monitor) is a powerful tool intended only for diagnostic and tuneup purposes by privileged individuals. This manual lists and describes only Main Monitor commands applicable for μPSS user application. For more detailed information, refer to the DMC User's Manual, GEH-5377, available from GE Drive Systems.

The Main Monitor prompt is * > . Until changed, the password is PLEASE.

Security: Key Switch in Test, Password

Example:

U > MM

ENTER THE CURRENT PASSWORD

u > xxxxxx

*>

4-3.1. EEPROM Saves. To save any changes made to the tuneup constants, password, and such, the user must write them to EEPROM using the Main Monitor. Values not saved in EEPROM remain volatile and are lost if a power interruption occurs. Then when power returns, all such parameters are restored to the last values stored in EEPROM. To perform an EEPROM save, the user must first access the Main Monitor through the User Monitor. If the Key Lock switch is in the Test position and the correct password is known, proceed as in the example, returning to the User Monitor upon completion.

Security: Key Switch in Test, Password

Example:

```
U > MM
ENTER THE CURRENT PASSWORD
U > PLEASE
*> x
XFER SYSTEM
X > S
SAVE IN EEPROM? Y
---WORKING--
COMMAND COMPLETE
X > Q
*> u
USER MONITOR
ENTERING THE PSS AND TRANSIENT
EVENT RECORDERS USER MONITOR TASK
FOR HELP TYPE HH
U >
```

4-3.2. Modifying Variables. If necessary, the user can change the value of system variables. As cautioned, **this feature must only be used by knowledgeable, authorized individuals.**

For example, the duration of the μPSS Excitation Signature Test (described in paragraph 4-2.16.3) can be changed by modifying the value of variable ETST-CNT. Such changes can only be done through the Main Monitor. This variable is scaled such that the value 60 corresponds to 1 second. Therefore, a value 300 corresponds to 5 seconds.

Security: Key Switch in Test, Password

Example:

```
U > MM
ENTER THE CURRENT PASSWORD
u > PASSWORD
*> M
DISPLAY/MOD
M > M
MODIFY? ETST-CNT
ETST-CNT = 300
VALUE? 60
ETST-CNT = 60
M > Q
*> u        (To re-enter the User Monitor)
```

CHAPTER 5

DESCRIPTION OF OPERATION

5-1. CONTROL (DMC)

The purpose of the μPSS is to supply a signal to the excitation control as a function of oscillations in generator frequency. This provides damping of these oscillations. The DMC provides the main control, with protective features to assure proper system operation. Additionally, it includes diagnostic features to evaluate its performance, and troubleshooting diagnostics. The DMC is interrupt driven from generator frequency that significantly suppresses noise from the generator frequency and its harmonics.

5-1.1. Power-up. At power-up, the μPSS's DMC first conducts a series of selftests on memory. Then it initializes the HAIC analog input board to give a 3.5 microsecond switching time delay and a 100 microsecond settling time. (Switching time is the time allowed to switch from one input to the next; settling time is the time waited before the new input is converted from analog to digital.)

The DMC initializes the digital outputs driven by the HRRB relay driver boards, which turns off the μPSS output relay. It also forces the μPSS output to zero for a 1 minute period to ensure proper warmup. Therefore, even if the μPSS output relay is energized, the output remains at zero until the 1 minute time period has passed. This initialization procedure is executed only upon power-up, or if the DMC has been taken internally offline (segment execution is stopped) and then brought online again.

5-1.2. Control. The μPSS executes control on a continuous basis based upon the line frequency interrupt. First it reads the 3-phase voltages, then calculates the average S-phase voltage. It then compares this average with the sample lying 3 scans back (50 msec). If the difference, AV, exceeds an optional over/under-voltage detection limit, the μPSS output is forced to zero for a prescribed time period. This optional feature improves the overall system performance for some applications. During major faults, defined here as large changes in line voltage, forcing the μPSS output to zero sometimes improves the response of the generator control as the μPSS tends to counteract the generator control in those fault situations. The average AV signal also is used **to** trigger the event recorder. If it exceeds the recorder over/under line voltage trigger level an event data line, marked with a preceding asterisk, is recorded followed by 200 lines of post event data associated with the event.

Upon analog-to-digital (A/D) conversion of the input frequency signal, AF (generated by the frequency transducer board), the μPSS calculates its output by applying a two-stage filter algorithm. The first stage consists of a first order high pass filter used to suppress dc and, where required, a torsional rejection filter. The torsional rejection filter is a specialized high order filter that provides low phase shift in the frequency range of interest and a sharp dropoff in the torsional resonance frequency range. (If the rejection of torsional frequencies is not desired, the torsional filter is disabled.) The second stage is an adjustable double lead/lag filter providing the necessary phase lead for the output signal. The gain through the digital filters can be adjusted.

Before the output is applied to the digital-to-analog (D/A) converter, the following conditions have to be met. The output is clamped to limits set by the user. This assures that the output signal cannot completely override the main generator control. An output ceiling detector disables the μPSS if the output voltage exceeds a level set during initial tuneup. The ceiling detection limits cannot be set to exceed the μPSS output clamping values to assure that ceiling detection is not disabled by output clamping. The output signal is also inhibited by the initial one minute timeout or the optional over/under voltage detection. If any tests are in progress, such as the excitation interface test, the calculated output car-be overridden by or superposed with the test signal. This is optional and depends on the test selected. No new output is calculated if the analog AF signal is saturated to avoid windup of internal digital filter states. A prolonged saturated input indicates a failure, so performing the filter calculations is then useless. Another check takes place upon bringing the μPSS online; the command to put the μPSS in service is inhibited if the output to be switched into the system exceeds a fixed value (1 volt). This assures that bringing the μPSS online does not cause a significant system disturbance. Once all these checks are made, the signal is passed on to the NDAC board's D/A converter. The converted analog signal is then buffered, sent through the μPSS output relay, and into the generator excitation control system.

Other diagnostic features include checking the NFTA board for ceiling operation. The DMC reads a frequency signal (biased to zero at 60 Hz) and checks it against a limit value. If the signal exceeds this value for a preset period of time, the μPSS is disabled and an alarm is set to indicate a CWFT failure.

A loss of phase diagnostic generates an alarm 2 seconds after loss of one or two phases feeding into the CWTA (NFTA board). The μPSS is not disabled since one phase alone is sufficient to derive AF.

The test recorder is activated upon initiation of one of the built-in tests. These are a digital test, excitation interface test, and a single frequency test, described in more detail in the command section for the test mode, TM.

The transient event recorder is triggered on either a transient in average phase voltage or a transient in the μPSS output signal. The trigger levels for these events are user adjustable through the monitor Tuneup command.

To minimize the phase shift caused by calculation delays, the μPSS performs a post filter calculation after converting the output signal. This calculation is performed in preparation for the next scan. At this time, the μPSS scans the remaining analog inputs used by the recorder. These include phase currents, watts, vars, field voltage, and test signals.

Upon completing all calculations and diagnostics, the μPSS output segment stops execution, and it waits for the next line frequency interrupt to awaken the segment to execute the next scan.

5-1.3. User Interface. User interface to the μPSS, the Test Recorder and the Transient Event Recorder is via a computer printer or CRT terminal, or a modem connected to jack JB on DMCC/2L in the DMC. The user accesses both the Main Monitor (DMC Monitor) and the User Monitor through the or remote modem.

When the μPSS conducts the selftests at power-up, it produces a number of display messages on the CRT or printer. When these selftests complete, the user can enter the User Monitor or the Main Monitor. Refer to Chapter 4 for a list and definition of all monitor commands, and examples of how to use them.

5-2. TEST RECORDER

The Test Recorder provides RAM storage of 211 scans of all analog data after a selftest is initiated. There is no pre-test data in the list, which consists of 2 11 lines plus a station identifier. Each scan produces one line. The list is number 1 if there is no Transient Event Recorder, or number 6 if there *is a* recorder. The scan rate is 20 Hz for 60 Hz systems and 25 Hz for 50 Hz systems. Figure 5-1 shows a typical test list. This list cannot be cleared, but is overwritten every time a test runs.

5-3. TRANSIENT EVENT RECORDER

A transient event is defined as the excursion of average terminal voltage or μPSS output beyond **a value** pre-

WORLD'S GREATEST POWER CO., STUARTS DRAFT, VA STATION: LOTSAPOWER

WINDMILL NUMBER 115
TRANSIENT EVENT RECORDER LIST NUMBER 1
AT THE TIME OF THE EVENT PSS WAS ON

*DATE TIME"	VOLT-A PERU.	VOLT-B PERU.	VOLT-C PERU.	I-A PERU.	I-B PERU.	I-C PERU.	PWR PERU.	VAR PERU.	PSS-OUT %	PSS-AF %	V-FIELD PER U.
07 DEC 1987 07:30:45.99	0.983	0.984	0.981	0.982	0.904	0.922	0.883	0.064	0.40	1.002	0.75
07 DEC 1987 07:30:46.04	0.983	0.984	0.981	0.982	0.904	0.922	0.883	0.064	0.40	1.002	0.75
07 DEC 1987 07:30:46.09	0.983	0.984	0.981	0.982	0.904	0.922	0.883	0.064	0.40	1.002	0.75
07 DEC 1987 07:30:46.14	0.983	0.984	0.981	0.982	0.904	0.922	0.883	0.064	0.40	1.002	0.75
07 DEC 1987 07:30:46.19	0.983	0.984	0.981	0.982	0.904	0.922	0.883	0.064	0.40	1.002	0.75
07 DEC 1987 07:30:46.24	0.983	0.984	0.981	0.982	0.904	0.922	0.883	0.064	0.40	1.002	0.75
07 DEC 1987 07:30:46.29	0.983	0.984	0.981	0.982	0.904	0.922	0.883	0.064	0.40	1.002	0.75
07 DEC 1987 07:30:46.34	0.983	0.984	0.981	0.982	0.904	0.922	0.883	0.064	0.40	1.002	0.75
07 DEC 1987 07:30:46.39	0.983	0.984	0.981	0.982	0.904	0.922	0.863	0.064	0.40	1.002	0.75
07 DEC 1987 07:30:46.44	0.983	0.984	0.981	0.982	0.904	0.922	0.883	0.064	0.40	1.002	0.75
"07 DEC 1987 07:30:46.49	-0.001	0.000	0.001	0.002	0.000	-0.001	-0.001	-0.003	0.40	1.191	0.0
07 DEC 1987 07:30:46.54	-0.001	0.000	0.001	0.002	0.000	-0.001	-0.001	-0.003	0.40	1.191	0.0
07 DEC 1987 07:30:46.59	-0.001	0.000	0.001	0.002	0.000	-0.001	-0.001	-0.003	0.40	1.191	0.0

continued for a total of 211 records

***Indicates the time that an event occurred.**

Fig-we 5-1. Typical Event Recorder List

defined by the user. The Transient Event Recorder stores data lists for four separate transient events. It contains five lists, but one is always storing pre-event data or post-event data and is not available for examination. Each list includes a specific number of pre-event and post event lines of data. Both the Transient Event and Test Recorder lists store the same analog input quantities and have the same total length.

One list is always in the process of accumulating pre-event data. When an event occurs, a list fills with additional post-event data. The recorder then begins recording pre-event data in the next list. After list 5 fills, list 4 begins recording pre-event data. Lists 4 and 5 toggle, so when all lists are full, lists 4 and 5 alternately fill for subsequent events. This management scheme ensures that the three oldest lists (numbers 1, 2, or 3) and the most recent one (either 4 or 5) are always available for examination until the user clears them. The list that is accumulating pre-event data is not available for examination.

In all lists, the record acquired at the time of the event displayed with a preceding "*". Normally this is the 11 th record. If an event occurs earlier, the asterisk flagged record is displayed closer to the top of the list and the number of post event lines is increased so that the total list length is unchanged. The scan rate is 20 Hz for 60 Hz systems and 25 Hz for 50 Hz systems. A typical list is similar to a test list shown in Figure 5-1.

The status of each list can be examined, and any list not accumulating data can be displayed. The clear command clears all lists. (Refer to paragraph 4.2.8 to review this command.)

It should be noted that the records accumulated in the lists consist of raw count values obtained from the analog to digital converter board in the μPSS module. When a list is displayed, the downloading process outputs scaled values of the raw counts. For some of the scanned values the scaling is fixed but for the rest the scaling is computed from data entered using the Auto Scale command AS. If scaling information is changed between list display commands, the data for which scaling has been changed displays differently.

5-4. GENERIC INTERFACE

The generic interface is shown on elementary diagram sheets 32C, 32CA, 32E, and 32M (see Appendix D). Sheet 32C shows the multipliers, and the input to the power amplifier. Sheet 32CA shows the connection of the power amplifier into the sensing circuit. SW5, SW6 and SW7 *are* CT disconnect switches that should be used to isolate the μPSS and power amplifier from the sensing circuit. Relays 2A, 2B, and 2C are HGA relays used to disconnect the μPSS and power amplifier from the sensing circuit if any protection circuit in the μPSS or power amplifier determines a failure. Sheet 32M shows the multipliers used to demodulate the ac PSS signal used to modulate the input to the voltage regulator. Sheet 32E shows the HRRB (board) contact used to select the demodulated PSS signal, or the output from the μPSS before it is demodulated.

The μPSS has protection that operates on the digital control portion of the stabilizer (see section 4-2.18 Tune-Up Command, and section 3-7, Faults). If any of the control protection schemes determines that the digital output signal is bad, then all three relays 2A, 2B, and 2C drop out. These relays are energized by contacts of the HRRB board in slot 2B, which open on failure. Relays 1A, 1B, and 1C must not be energized for relats 2A, 2B and 2C to be energized. Relays 1A, 1B, and 1C are part of the power amplifier current limit protection.

The generic interface has two types of protection: Current Limit protection and Transmission Monitor protection. Each can remove just the failed power amplifier phase, or all power amplifier phases. How the protection operates is determined by setting certain constants in the μPSS software. Anytime a phase of the power amplifier is removed from service by the protection, an alarm is generated. An alarm message also tells what phase is removed from service, and what function removed the phase from service. Use the FD command in the User Monitor to view the fault message.

The output of each of the power amplifiers is transformer isolated from the PT circuit. The power amplifier's purpose is to vary the sensing voltage to the voltage regulator as controlled by the μPSS. To do this, each power amplifier must sink the current in the PT circuit. If it cannot sink this current, then voltage develops across the winding in the PT circuit, and the voltage regulator thinks that the generator voltage is too low. The voltage regulator then forces the exciter to ceiling. This is not desired.

5-4.1. Current Limit Protection. Current limit protection detects **when** the power amplifier is in current limit, meaning more than 5.8 amps rms line current is in the PT circuit. **A** transistor inside the power amplifier is biased on, and any or all of the relays 1A, 1B, *or* 1C pickup causing the appropriate HGA relay, 2A, 2B or 2C to drop out. The **HSCC** board, which detects voltage logic, monitors the voltage across the internal transistor in the power amplifier. When the μPSS detects that the voltage has decreased below a certain threshold voltage, it opens the appropriate HRRB contacts in slot 2B. This ensures that the appropriate HGA relay has dropped out. **The μPSS must be turned off to reset a tripped phase.**

If all phases of the power amplifier remain in service when current limit is reached, it is likely that a semiconductor in the phase of the power amplifier that reached current limit would fail. This causes the isolation transformer to appear as high impedance. As a result, the exciter remains at its ceiling voltage until a transfer to Manual control occurs (if the excitation has

field overvoltage protection.) By removing the failed phase of the power amplifier from the sensing circuit when there is too much current, the voltage regulator is protected against a subsequent failure in the power amplifier.

Current limit protection also protects the system from failures internal to the Power Amplifier, particularly the Ioss of an internal power supply. Failures of this type cause a dc offset on the power amplifier side of the isolation transformer, and result in the amplifier current exceeding its current limit. This can not be sensed external to the power amplifier because of the isolation transformers.

5-4.1. I. CURRENT LIMIT IMPLEMENTATION. Figure 5-2 shows the power amplifier's output stage, phase C. Phase A and phase B are identical in function. Device Q30 turns on when the current in the positive side of the power amplifier stage exceeds the current limit value. Device U47, a photocoupler, turns on when Q30 turns on. Device Q3 1 turns on when the current in the negative side of the power amplifier stage exceeds the current limit value. Device U48, a photocoupler, turns on when Q31 turns on. Both photocouplers, U47 and U48, are in parallel. An externally mounted relay with a 28 V dc power supply picks up when either or both of the photocouplers are turned on.

A contact of the 28 V relay removes the power amplifier phase from the regulator sensing circuit. The digital PSS also senses that the 28 V relay has picked up and sets alarm messages.

5-4.2. Transmission Monitor Protection. The transmission monitor protection circuit monitors the output of the power amplifier on the power amplifier side of the isolation transformer and determines that the signal is good. Because the μPSS output is typically 0 V, a signal that can be measured at the output must be inserted to the μPSS. This signal must not affect the sensing voltage. Figure 5-3 shows a S-phase phasor diagram of the PT voltages. Figure 5-4 shows the insertion of the transmission monitor signal. The PT line-to-line voltages remain the same, and the voltage regulator does not detect a change in generator voltage because of the transmission monitor signal. The transmission monitor signal is obtained from phase A, and is adjusted typically to be 1% of the sensing voltage. The transmission monitor sensing circuit subtracts out the PSS signal from the total output of the power amplifier, and then rectifies and filters it. This rectified filter signal, VTMX_DC (X for the specific phase), is monitored by the μPSS and used to determine when to take out a failed phase.

The transmission monitor protection also functions to check the gain of the power amplifier when the μPSS is off. It prevents a failed phase from going in service. If already in service, it takes the failed phase out of service.

Figure 5-5 shows that when one phase is removed from service, the transmission monitor, signal causes an unbalance in the sensing voltage. Software allows the user to select the transmission monitor to be turned off when either one or two phases are removed because of failure. As with the current limit protection, the μPSS must be turned off to reset a phase that is tipped by the transmission monitor.

5-4.2.1. TRANSMISSION MONITOR IMPLEMENTATION. On elementary sheet 32F (Appendix F), location 43R is the signal from the processor that controls the transmission monitor. Software calls this signal VTM_DC when the transmission monitor is enabled, and VTM_PRESET when the transmission monitor is disabled for any reason. (VTM_PRESET is typically set to 0, thus the VTM signal shown in Figure 5-3 is 0 if the transmission monitor is shut off.) On sheet 32CA, the dc signal from the processor is input to U4. Multiplier U9 receives the output signal from U4 plus a signal from phase A. U9 then outputs an ac signal in phase with phase A and controlled by the processor. From U9, this ac signal goes through relay contact VTMAC and is input to the power amplifier.

In the power amplifier, the VTM signal is added to each phase output and multiplied by the gain of the power amplifier. The output signal is:

$$PA(output) = (G_{pa})(VTM + PSS) = (G_{pa})(VTM) + (G_{pa})(PSS)$$

Where: G_{pa} = Power Amplifier Gain
VTM = AC Level of Transmission Monitor Signal
PSS = AC Level of PSS Signal

From the PA(output) signal, just the VTM signal is needed. By measuring VTM after is has been through the power amplifier, we can determine that the power amplifier is functioning correctly. To extract the VTM signal, the output of the power amplifier is input to a circuit with a gain adjusted to be $1/G_{pa}$, or K_{tm}. This results in:

$$VTM + PSS = PA(output)(1/G_{pa}) = PA(output)(K_{tm})$$

The μPSS signal is obtained from the power amplifier's input circuit, and then is subtracted from the above. This results in:

$$VTM = PA(output)(K_{tm}) - PSS$$

The VTM signal is rectified and filtered, then input to one of the μPSS A/D converters for signal processing.

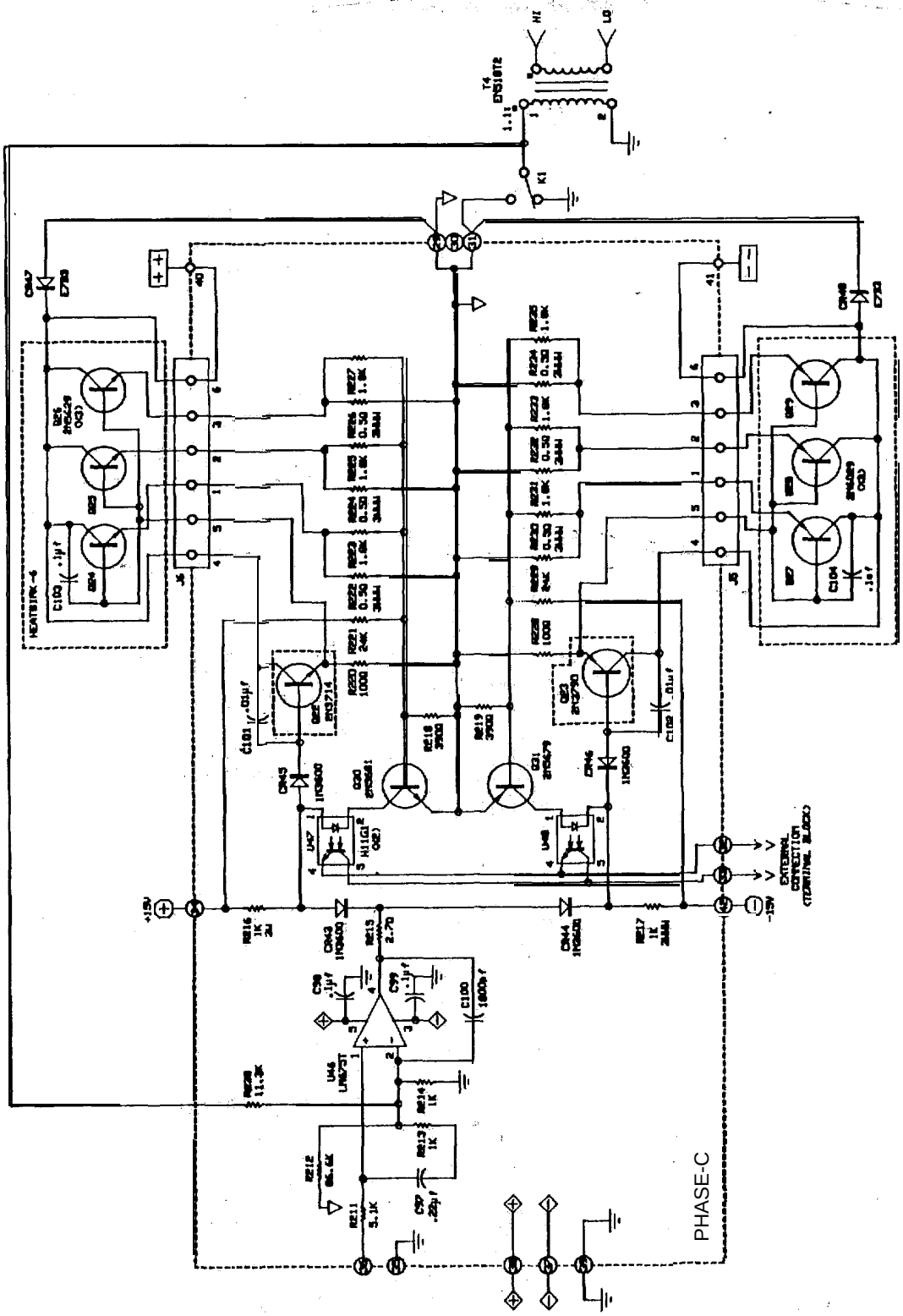


Figure 5-2. Transmission Monitor

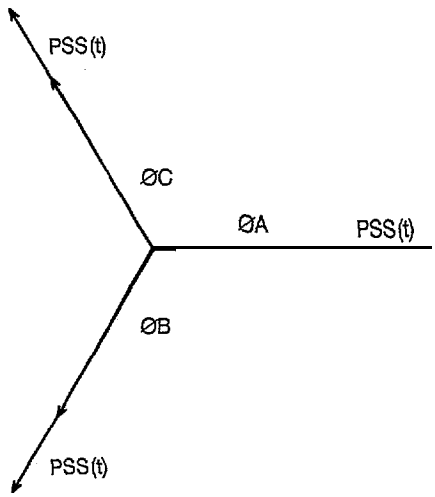


Figure 5-3. Three-phase Phasor Diagram of PT Voltages

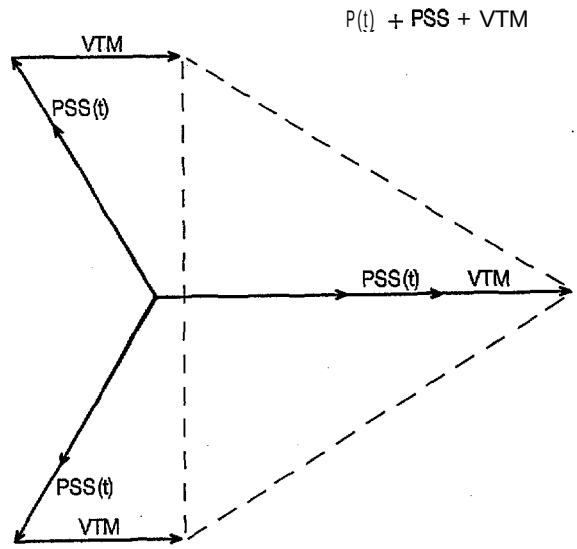


Figure **5-4.** *Insertion* **of Transmission** *Monitor Signal*

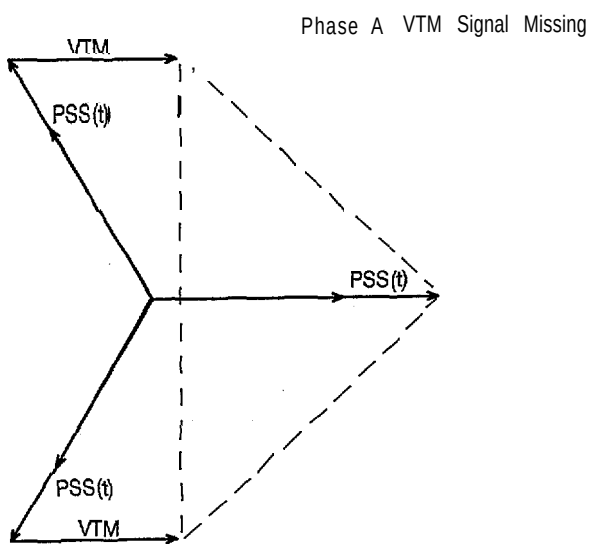


Figure 5-5. Removing One Phase From Service

The μPSS software identifies the outputs of these phases as VTMA_DC, VTMBDC, and VTMCDC.

The μPSS determines if a VTM signal from the power amplifier for phase A is good or bad as follows:

VTMAVE is subtracted from VTMADC, producing signal VTMAERROR. The absolute value of VTMA_ERROR is taken to get ABS-VTMAERROR, which is then compared with VTM-ERROR. If the result is greater than or equal to zero, TMA-_TRIP_EN is set true. TM&TRIP-EN must be true for a time period set by VTMDELAY. If it is, then the output of the delay, TMA_TRIP, is set true. TMATRIP trips the power amplifier's phase A and sets alarm messages for the fault list.

The determination of a good or bad VTM signal from the power amplifier for phase B is done as follows:

VTMAVE is subtracted from VTMBDC, producing signal VTMB_ERROR. The absolute value of VTMB_ERROR is taken to get ABS-VTMB-ERROR, which is then compared with VTM-ERROR. If the result is greater than or equal to zero, TMB-_TRIP_EN is set true. TMB_TRIP_EN must be true for a time period set by VTMDELAY. If it is, then the output of the delay, TMB-TRIP, is set true. TMB-TRIP trips the power amplifier's phase B and sets alarm messages for the fault list.

The determination of a good or bad VTM signal from the power amplifier for phase C is done as follows:

VTMAVE is subtracted from VTMCDC, producing signal VTMC-ERROR. The absolute value of VTMC_ERROR is taken to get ABS_VTMC_ERROR, which is then compared with VTM-ERROR. If the result is greater than or equal to zero, TMC--TRIP-EN is set true. TMC_TRIP_EN must be true for a time period set by VTMDELAY. If it is, then the output of the delay, TMC_TRIP, is set true. TMC-TRIP causes phase C of the power amplifier to trip, and sets alarm messages for the fault list.

Use the Main Monitor's Modify command to display screen #3 to see these variables.

M> R <ret>

SCREEN? 3 < ret > *(buck to modify prompt)*

Screen 3 will look like Figure 5-6, but with different numbers.

Note that in Figure 5-6, variables VTMADC, VTMBDC, and VTMV_DC are all greater than 3.00. The reason is that the gain K_{tm} is not equal to $1/G_{pa}$ If K_{tm} and $1/G_{pa}$ are equal, there is no component from the PSS signal not cancelling out in the transmission monitor circuit. (Refer to paragraphs 5-4.2 and 5-4.4, Transmission Monitor Protection and Amplifier Gain Adjustment .)

5-4.3. Setting Up the μPSS and Power Amplifier.

The software in the μPSS must be set by the user for the desired functionality. The following table is a list of constants that determine power amplifier protection. To set the function, select the Main Monitor and modify the constant using the Modify command. (See paragraph 4-3.2 for an example of using this command.)

K_LPA_ENABLE	Set to 1 to enable power amplifier protection
K_LVTM	Set to 1 to disable transmission monitor protection
K_LCL	Set to 1 to disable CL protection. Must also reconnect 1A, 1B, and 1C
K_TRIP_ONE	Set to 1 cause all power amplifier phases to trip if any one phase trips
K_TRIP_TWO	Set to 1 to cause remaining phase to trip if 2 phases trip
K_PH_ROT	Set to 1 for phase rotation fault. The power system phase rotation is CBA
K_VTM_1	Set to 1 to stop transmission monitor signal with one phase out of service
K_VTM_2	Set to 1 to stop transmission monitor signal with two phases out of service

TRANSMISSION MONITOR I/O

VTM_DC	= 1 .000000	VTM_PRESET	=0.0	VTM_ENABLE	= 1
VTM_COUNT_IN	= 1 .000000	VTM_SCALE	=0.004888	VTMXOUNT	= 3280
VTMAVE	= 5.000000	VTM-ERROR	=4.500000	TMACOUNT	= 11601
VTMADC	= 3.544379	VTMAERROR	= -1.45562	ABS_VTMA_ERR	= 1.455620
TMA_TRI P-EN	= o	TMATRIP	= 0	TMBXOUNT	= 11729
VTMB_DC	= 3.583490	W-MB-ERROR	=-1.41650	ABS_VTMB_ERR	=1.416509
TMB-TRIP	= 0	TMC_COUNT	= 11377	VTMC_DC	=3.475936
VI-MC-ERROR	= -1.52406	ABS_VTMC_ERR	= 1.524063	TMC-TRIP	= 0

Figure 5-6. Monitor Display of Screen 3

K_LPA_ENABLE	Set to I to enable power amplifier protection
VTM_DC	Sets output of VTM signal, 1.0 equals 1% of sensing voltage
VTM_AVE	Sets average value of transmission monitor signal expected by μPSS from power amplifier
VTM-ERROR	Sets amount of error μPSS tolerates in transmission monitor signal from power amplifier
VIM-DELAY	Sets time delay before tripping on transmission monitor failure when conditions are met. The delay is in milliseconds, is typically set to 10,000. It should never be greater than 32'76'7, or negative
VTM_PRESET	Sets output of VTM signal when transmission monitor is off. Should be set to 0 V dc
LPA-TEST	Test function that causes output of the dc transmission monitor signal to be input to multiplier board shown on sheet 32C. This enables calibration or verification of power amplifier gain.

Use the Main Monitor's Modify command and display *screen* 4. No&e that screen 4 must be defined by initially using M > E (for edit).

M> R

SCREEN? 4

CONSTANTS FOR PA PROTECTION

K_LPA_ENABLE = 1	K_TRIP_ONE = 0	K_TRIP_TWO = 0	
K_PH_ROT = 1	K_VTM_1 = 0	K_VTM_2 = 0	
K_LVTM = 0	K_LCL = 0	VI-M-DELAY = 7500	

5-4.4. Power Amplifier Gain Adjustment. Make sure that the CT disconnect switches are opened while testing the μPSS and the power amplifier.

Set K_LPA_ENABLE = 1, K-LVTM = 0, and K_LCL = 0 to enable the power amplifier protection.

The power amplifier should be set up so that 1 V out of the μPSS results in *a* 1% change in terminal voltage. This should be verified by selecting the Main Monitor and modifying LPA_TEST to a value of 1. The PSS must be off and the PA must be enabled. Set VTM--PRESET to 1 .O, and measure the output of the power amplifier. The voltage across the output winding should be:

$$\text{PA(volts)} = \frac{\text{(line-line PT voltage)}}{\sqrt{3}\,(0.01)(\text{VTM_PRESET})}$$

If the PT line-line voltage is 115 V ac rms, and VIM-PRESET is 1 .0, then PA(volts) is 0.664 V ac rms.

Set PA(volts) with accuracy of ± 0.01 V ac rms. However, before making this adjustment, note that if the power amplifier gain (G_{pa}) is adjusted, the gain in transmission monitor circuit (K_{tm}) must also be adjusted.

5-4.5. Adjusting Transmission Monitor Protection

Use the Main Monitor to display screen #3. VTMA-DC, VTMB_DC and VTMC-DC should be 0 ± 0.25. VIM-PRESET should still be 1 .O. Potentiometers R68, R11 1, and R148 are located on the front of the power amplifier. Use these pots to adjust the voltages to 0 ± 0.25. Note that the power amplifier gain must be set as described above before making this transmission monitor adjustment.

Set VTM_PRESET and LPATEST to 0 after setting power amplifier gain and transmission monitor protection. Use the Main Monitor to display screen #3 and verify that TMATRIP, TMB-TRIP and TMC--TRIP are 0. These variables are set to 1 for a transmission monitor trip for any phase. TMADC, TMB_DC and TMC_DC should be about 3 ± 0.25 volts. If they are not, make sure that VTM_DC = 1.0 and make sure that the transmission monitor is enabled (K_LVTM = 0).

Verify that the CT disconnect switches are open. Turn the μPSS on. After a 1.5 second time delay (which is set by TDELAY) the three HGA relays close. After 30 seconds, use the pushbutton on each of the relays 1A, 1B, and 1C. These pushbuttons simulate a current limit condition. Verify that the correct fault list message is displayed as the correct HGA relay drops out. If the current limit fault messages do not appear, check that K_LCL is set to 0. Turn the μPSS off and reset the fault list.

To check transmission monitor fault messages, use the Main Monitor to adjust VTM_DC to 2.5. After the time delay set by VTM_DELAY, the HGA relays should trip. Check the fault list for the transmission monitor message. Turn the μPSS off and reset the fault list.

5-4.6. Gain Phase Measurement with Generic Interface. In addition to power amplifier protection, a demodulator is also provided so that the gain and phase of the μPSS plus the power amplifier can be measured. To do this, the 1/2 Hz single frequency test must be run, then the μPSS gain phase function in the User Monitor is selected. The gain phase function calculates the gain and phase of the μPSS by using FFT on the test list. (These functions are described more in paragraphs 4-2.5 and 4-2.16.4.) Note that if the HRRB contact K6 is closed (see sheet 32E of the elementary), only the output of the μPSS goes into the event recorder test list. If the HRRB *contact* K5 *is* closed, then the demodulated output from the power amplifier goes into the event recorder test list.

5-4.7. Set Up The Demodulator. The input to the μPSS must be rated PT volts. This is because the output of the demodulated signal is not independent of PT voltage. With the μPSS output equal to 1 V dc, the demodulator output should be 1 V dc. To verify this, set VTM_PRESET to -1, turn the μPSS off, and set

PA_TEST to 1. Use a voltmeter on the dc scale. Measure VTP4 (slot 2E), this voltage should be 1 V dc ± .05 V dc. If it is not, adjust VTP4 (slot 2M) with R4 to -0.333 V dc. Adjust VTP3 (slot 2M) with R5 to -0.333 V dc, adjust VTP5 (slot 2M) with R2 to -0.333 V dc. VTP4 (slot 2E) should be 1.0 ±.05 V dc

The demodulator operates linearly up to 4 V dc. Voltages greater than 4 V dc cause nonlinear operation of op amps in the 2G + PSS stage.

Use the 1/2 Hz single frequency test and the Gain Phase function to obtain gain and phase for the PSS + PA (K_LPA_ENABLE = 1) and for just the μPSS (K_LPA_ENABLE = 0). As shipped from the factory, the constants in the μPSS should be: K2, K4 = 0.25; k3, K5 = 0.025; G = 2.0. Using these constants, the phase measurements should agree within ±3.5°, and the amplitude measurement within ± 12%.

When the PSS is initially installed, the one-half Hz single frequency test and the gain phase function should be run - the gain and phase numbers should be recorded. Subsequent μPSS verification procedure can be done by running the single frequency test and using the gain phase function to obtain gain and phase data at one-half Hz. The "recently" obtained" gain and phase data should be compared with the initial values recorded for the gain and phase.

CHAPTER 6

MAINTENANCE AND TROUBLESHOOTING

6-1. MAINTENANCE

The μPSS is static, having no moving parts, except for relays. Therefore, little maintenance should be required.

If high vibration is present, periodically check all screw connections.

Semi-annually, check the null adjustment on the NFTA frequency transducer board, as described in Chapter 5.

Periodically, test the system with the internal 1/2 Hz sinewave test. This can be done remotely without disturbing the system, as described in these instructions.

6-2. TROUBLESHOOTING

The following list contains basic suggestions for troubleshooting, presented only as a guide. The equipment incorporates many diagnostics. The best troubleshooting tools are a thorough familiarity with this instruction book and the equipment's elementary diagram.

CAUTION

To prevent possible component damage, always turn off the power supply before disconnecting or re-inserting boards in the electronic module.

1. If problems occur at startup, carefully check all wiring.
2. Check that all jumpers on boards are installed per the equipment's elementary diagram.
3. If no indicating lamps are on, check that the power supply is energized.
4. Check that all boards are properly inserted and that all cable plugs are firmly seated.
5. Follow the block diagram and check each board for the proper response.
6. If the equipment does not stay in service, connect a computer terminal and check the alarm settings.
7. If the frequency signal from the frequency transducer has a significant dc level, first make sure that the generator is at rated speed, then re-adjust the null.
8. Check for proper grounding of both the exciter and the μPSS system.

6-3. RENEWAL PARTS

When ordering renewal (replacement) parts, the following information should be given:

1. Catalog (part) number stamped on the part, with a complete description of the part, including use and location.

***** NOTE *****

Do not replace board components, but order the entire board. One exception is the factory programmed memory chips mounted in sockets. A gummed label on each replaceable chip contains the part number. These chips can be ordered in sets.

2. Complete nameplate data appearing on the assembly that included the component.
3. If possible, provide data about the original order on which the equipment was first supplied, including all numerical references.

APPENDIX A

TERMINAL CHARACTERISTICS

A-l. TERMINAL SETTINGS

The following list contains the terminal settings required for using the monitor.

Parity	= off
Bits/Character	= 8
Baud Rate	= 1200
[1] Auton XON/XOFF	= On
[2] Wrap	= On
[3] Columns	= 132
[4] Auto Line Feed	= off

[1] = Recommended for CRT display terminals

[2] = Required for SO-column display

[3] = Recommended, but SO ***columns*** will do if wrap is turned on

[4] = If terminal has auto line feed

A-2. CABLE CHARACTERISTICS

The following list contains the cable requirements for a terminal or printer.

Cable with 25-pin female subminiature type D connector is required for DMCC/1K.

PIN	DESIGNATION
1	FRAME GROUND
2	TXD
3	RXD
4	RTS
5	CTS
6	DSR
7	SIGNAL GROUND
8	CXR
20	DTR

APPENDIX B

APPLICABLE PUBLICATIONS

1. "Power System Stabilizer Application Guide" obtainable through your local sales representative from Systems Development and Engineering Department, General Electric Company, Schenectady, NY
2. C. Concordia, F.P. DeMello, "Concepts of Synchronous Machine Stability as Affected by Excitation Control." IEEE Trans. Vol. PAS-88, April 1969, pp. 316-329.
3. E.V. Larsen, D.A. Swann, "Applying Power System Stabilizers, Part I: General Concepts." IEEE Transactions on Power Apparatus and Systems, Vol. PAS-100, No. 6, June 1981, pp 3017-3024.
4. Ibid. "Part II: Performance Objectives and Tuning Concepts." IEEE Transactions on Power Apparatus and Systems, Vol. PAS-100, No. 6, June 1981, pp 3025-3033.
5. Ibid. "Part III: Practical Considerations." IEEE Transactions on Power Apparatus and Systems, Vol. PAS-100, No. 6, June 1981, pp 3034-3046.
6. R.A. Lawson, D.A. Swann, G.F. Wright, "Minimumization of Power System Stabilizer Torsional Interaction." IEEE Transactions.. Vol. PAS-97, January/February 1978, pp. 183-190.
7. F.R. Schleif, R.K. Feeley, W.H. Phillips, R.W. Torluemke, "A Power System Stabilizer Application with Local Mode Cancellation." IEEE Transactions, Vol. PAS-98, May/June 1979, pp. 1054-1060.
8. "Power System Stabilization Via Excitation Control", IEEE Tutorial Text 81 EHO 175-O PWR.
9. M.L. Crenshaw, J.M. Cutler, G.F. Wright, and W.J. Reid, Jr., "Power System Stabilizer Application in a Two Unit Plant - Analytical Studies and Field Tests" IEEE Trans., Vol. PAS 102, No. 2, February 1983, pp. 267-274.
10. Western System Coordinating Council, "Power System Stabilizer Tuning Guide"
11. "DMC User's Manual," Publication GEH-5377, Available from GE Drive Systems, Salem, Virginia.

APPENDIX C

HAYES* MODEM SETTINGS

C-l. CONFIGURATION SW-ITCHES

For the Hayes Smartmodem 1200™ to function correctly, the configuration switches must be set. These **switches** are located behind the Smartmodem's front panel, and are numbered 1 through 10.

Set the switches as listed below.

Switch	Position	Function
1	Down	Ignores DTR
2	Up	Word result codes displayed
3	Down	Result codes sent
4	Up	Characters echoed in command state
5	Up	Auto-answer enabled
6	Down	RS-232C carrier detect on always
7	Up	Single line RJ-11 phone jack
8	Down	Enables commands
9	Up	Compatible with Bell 103/212A modems
10	Up	Modem automatically reset when turned on

APPENDIX D

POWER AMPLIFIER

D-l. INTRODUCTION

This appendix contains the component information on the S-phase power amplifier used in the μPSS. This section is a reproduction of the Encore Electronics component publication for the Model 518 Manual S-Phase Power Amplifier, revision 3/89.

MODEL 518

TEST PROCEDURE

INDEX

TEST EQUIPMENT LIST

TEST PROCEDURE

FULL LOAD TEST CIRCUIT, FIGURE 1

ACTIVE TEST CIRCUIT, FIGURE 2

DATA SHEET

MODEL 518

TEST EQUIPMENT LIST

QTY	MODEL	ITEM	READS	USED FOR
1	WESTON 904	RMS AC VOLTMETER	0-150V	MONITOR LINE VOLTAGE
1	WESTON 528	AC RMS AMMETER	0-15A	MONITOR LINE CURRENT
3	BECKMAN 3010	AC AMMETER	0-10A	MONITOR LOAD CURRENT
3	HITASHI V-302B	DUAL CHANNEL OSCILLOSCOPE	PHASE AND WAVE FORMS	MONITOR INPUTS /OUTPUTS
3	FLUKE 8060A	RMS AC VOLTMETER	0-20V	MONITOR OUTPUT VOLTS
1	ENCORE EN122T2	STEP-DOWN XFMR	0-6V RMS	LINE SYNCRONIZED INPUT SIGNAL
2	GENRAD 15 AMP	0-132V AUTO XFMR	_____	ADJUST AC LINE VOLTAGE
1	ENCORE 518SG	3-PHASE, 60Hz SIG. GEN.	_____	INPUT SIGNAL

MODEL 518

TEST PROCEDURE

(REFER TO TEST CIRCUIT, FIGURE 1)

1. INITIAL TURN-ON: With no input signal or output loads, slowly apply 0 to 115VAC to the amplifier. The AC line current should remain less than 1.0ARMS.

2. GAIN ADJUST RANGE: Apply 7V pk-pk inputs to the amplifier. The pk-pk output voltages shall be adjustable from 1V to 24.5V as the FP trimpot is rotated from its CCW to its CW position on each channel. (A,B,&C)

3. TURN-OFF LOGIC: Turn the power switch off, the amplifier shall remain energized for at least one second after turn-off.

4. TURN-ON LOGIC: Turn the power switch on, no signals shall appear at the outputs for at least 5 seconds.

5. PHASE CONNECTIONS: Check that the' "A" phase input signal on PC-1 appears on the output terminal for PHASE A. Repeat for input signal on PC-2 (Phase B) and PC-3 (Phase C).

6. FULL LOAD TEST: Iterate the gain and the load resistances until all of the outputs are at 12VAC, 5A RMS. Monitor the output wave forms while reducing the input line voltage to 103.5V (-10%). There shall be no evidence of "clipping" on the output waveforms.

Record the AC line current at 126.5V (+10%). The output voltages should remain stable as the line voltage is varied +,-10%. The AC line current shall be less than 6A. Short circuit each output load in turn. The short circuit current for each output shall be less than 10A RMS.

In addition, the current - limit, external closure signals present on the rear panel of the amplifier should be activated.
Test by applying a +12vdc signal through a 1k droping resistor to the + terminal of the phase under test.
(Reference to - terminal, or system ground.)
The voltage at the + terminal will change from +12VDC to approximately zero volts when the overload occurs.

7. ACTIVE LOAD TEST: Reconnect Phase A to the ACTIVE Buck/Boost Test Circuit FIGURE 2 and substitute a low-level line syncronized 60Hz line signal V3 as shown. Channel A shall perform as required by the line signal V3 as shown. Channel A shall perform as required by the procedure outlined on Fig.2, while the other two channels remain as connected for test #6, "FULL LOAD TEST"

Repeat for Phase B and Phase C.

8. LOSS OF LINE VOLTAGE: With the amplifier operating as in Test #7, two channels resistively loaded, and one channel actively loaded in the full "Buck" mode, remove AC line power to the amplifier by removing the line-cord. Repeat several times, and check that the amplifier is still operating normally after power is restored.

9. TRANSMISSION MONITOR SET-UP & TEST:
(Ref. D12971 - A2 PCB)

1. Adjust PSS gain with R153 to minimum, with VTM-0 until step 4.

2. Set VPSS-in to 7V RMS.

3. Use R68 to adjust VTM-DC to zero volts and verify with the oscilloscope that the inputs to R70 and R59 are equal in magnitude but phase-shifted from each other by 180 degrees.

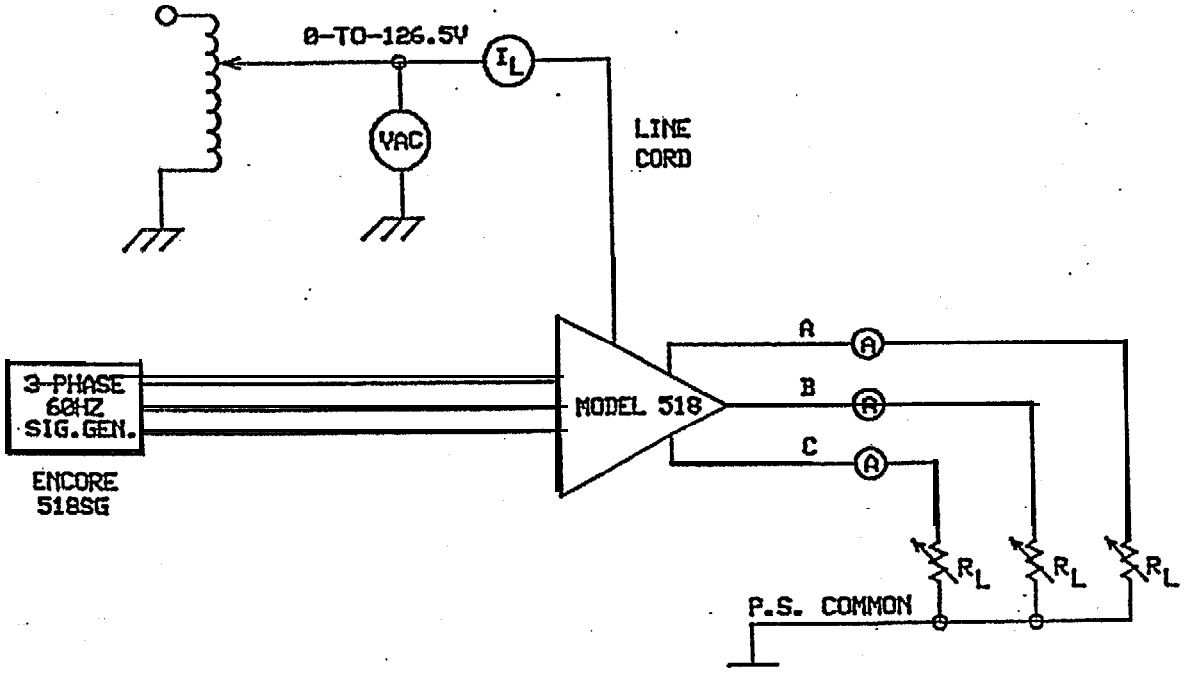
4. Set VTM-AC to 1V RMS.

5. Measure VTM-DC. Should be 7.5VDC with an RMS ripple less than 0.65 volts.

6. Increase the PSS gain to maximum with R153 and repeat steps 2 through 5, but with VPSS reduced to 3.5V RMS in step 2.

7. Measure 2G(PSS + VTM) with PSS gain (R153) at minimum and maximum settings and with PSS-IN at 1VRMS. Monitor for clipping at 2G(PSS + VTM) when adding a 1VRMS VTM-IN signal.

8. Repeat for Phase B and for Phase C.



MODEL 518

RESISTIVE LOAD TEST

FIGURE 1

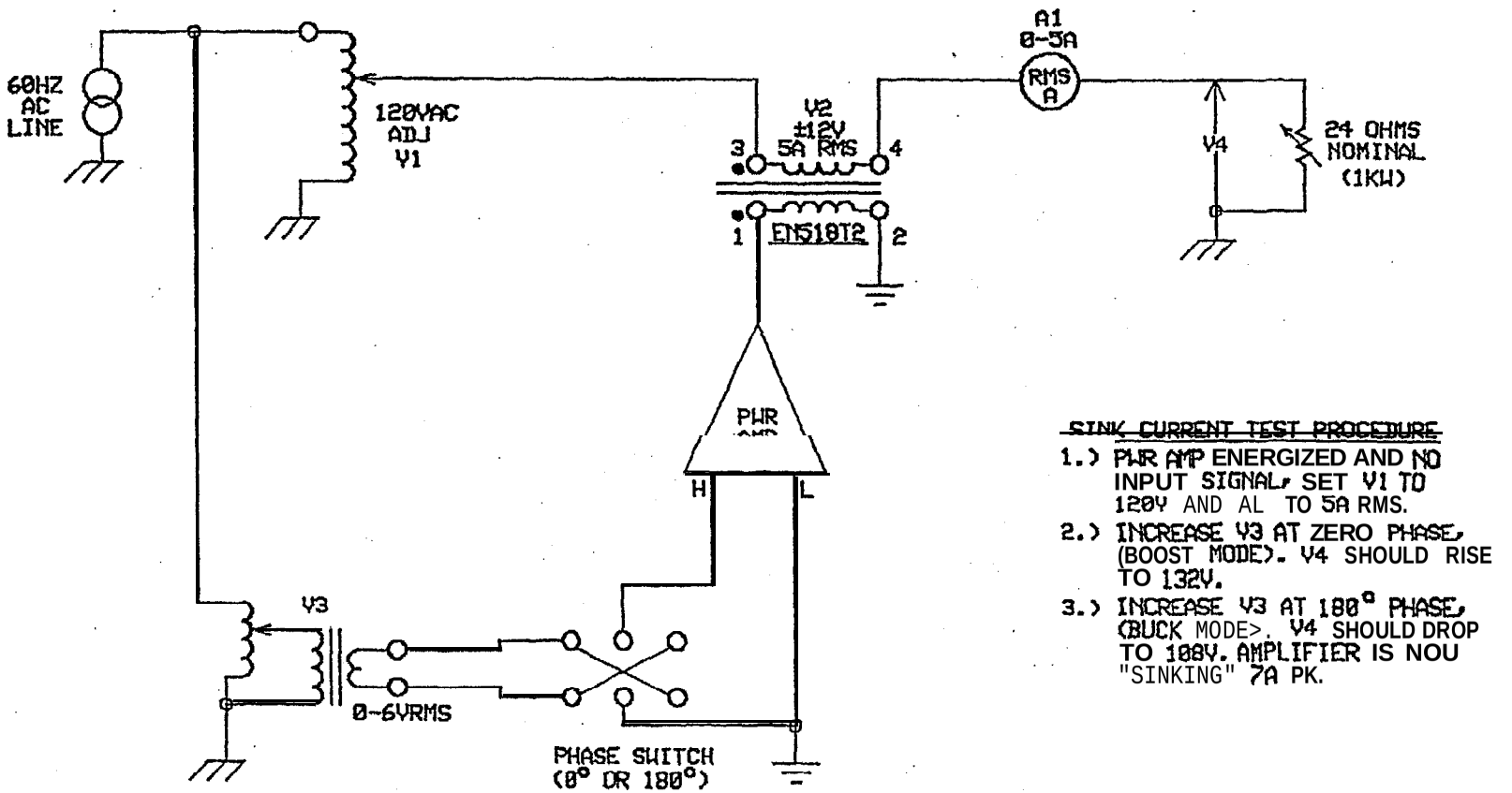


FIGURE 2
MODEL 518 ACTIVE CIRCUIT

MODEL 518
DATA SHEEET

1. INITIAL TURN-ON = I Line @ 115VAC, NL _____ Arms.

2. GAIN ADJUST RANGE:
   Phase "A" Channel: v t ____oV
   Phase "B" Channel: _____v to _____V
   Phase "C" Channel: _____v to _____V

3. TURN-OFF LOGIC: Turn-off Time, Approx: _____ Sec.

4. TURN-ON LOGIC: Turn-on *Time,* Approx: _____ Sec.

5. PHASE CONNECTIONS:

	OK
Phase A	[ ]
Phase B	[
Phase C	[3

6. FULL LOAD TEST:
   No clipping at 103.5V line: OK [ ]
   Line Current at 126.5V line: _____Amps, RMS
   Short Ckt Current A _____Amps, RMS
   B _____Amps, RMS
   B _____Amps, RMS

7. ACTIVE LOAD TEST:

	OK
Buck-Boost on A	[ ]
Buck-Boost on B	[ ]
Buck-Boost on C	[ ]

8. LOSS OF LINE VOLTAGE: No Failure OK [ ]

9. TRANSMISSION MONITOR:

v PSS	VTM-AC	GAIN	VTM-DC		2 (PSS + VTM)
RMM	RMS		VDC	RMS RIPPLE	
7V	0	MIN			---------
7V	1V	MIN			---------
3.5V	0	MIN			-a-------
3.5V	1V	MAX			---------
1.0V	0	MIN	---	----------	
1.0V	0	MAX	---	----------	
1.0V	1V	MAX	---	----------	

MODEL 518

MANUAL

3-PHASE POWER AMPLIFIER

INDEX

- DESCRIPTION
- SPECIFICATION
- INPUT CONNECTOR SIGNAL DEFINITIONS
- TEST PROCEDURE
- TROUBLE SHOOTING PROCEDURE
- RECOMMENDED SPARE PARTS LIST
- POWER SUPPLY & PHASE "A" WIRING DIAGRAM SCHEMATIC D12971
- SCHEMATIC D 13168 3-PHASE POWER AMPLIFIER BOARD
- SCHEMATIC D 13169 3-PHASE POWER OUTPUT BOARD
- OUTLINE DRAWING D12917

REV. 3/89

MODEL 518

DESCRIPTION

The Model 518 is a 3-phase power amplifier, each channel's output is transformer isolated from earth ground and power supply common. Each of the floating outputs is connected in series with a 120V RMS potential transformer and is required to buck or boost the 120V PT by +,-12V RMS while up *to* 5A RMS is flowing through the circuit. A test circuit simulating this mode of operation is shown in Figure 2 of the "TEST PROCEDURE" section.

The amplifier is normally provided with the negative-feedback of each channel sensing the output voltage applied to the primary windings (l-2). of the isolation transformers. This results in a stable signal voltage applied to the primary, while the output voltage at, the isolated secondary winding varies with the IR drops of both the primary and secondary windings. The advantage of this arrangement is that it is relatively easy to stabilize its feedback loop.

The amplifier consists of major sub-sections, as follows: (Reference Schematic D12971)

Power Supply: One common power supply system provides the required DC operating voltages for three amplifier channels. These consist of the following:

One each +,-30VDC at 6A UNREGULATED, referenced to amplifier common. This voltage is used to operate all three transistorized power amplifier stages.

One each +,-15VDC at 0.05A REGULATED, referenced to amplifier common. This voltage is used to operate all three low power op-amp stages.

One each +,-30VDC at 0.15A REGULATED,referenced to amplifier common. This voltage is used to operate all three monolithic power op-amps used to drive the transistorized power stage.

Three each +,-15VDC at .05A REGULATED AND ISOLATED, referenced to the primary sides of each of the three transistorized power stages output transformers.

AMPLIFIERS: Three amplifier channels, one for each input signal. (Phases A,B,and C) Each channel consists of the following:

Three preamplifiers located on PCB A2.

Three power-stage-drivers located on PCB A3.

Three transistorized power output stages located on fan-cooled heatsinks.

Three amplifier output isolation transformers connect the transistorized power output stages to the active loads. Each transformer consists of 1.1 to 1 primary and isolated secondary winding.

CHASSIS (Reference Outline Drawing D 12917)

The chassis accommodates the power supplies and amplifier channels. It has provision for an AC line cord, 3-phase signal input connector, and S-phase output terminal block. It also includes gain-trim potentiometers for each channel, a power UN/OFF switch with indicator, and three fuses. An AC fuse *is* in the line, while there is one DC fuse in each of the +,-30VDC power supply outputs. The purpose of the latter fuses is to minimize thermal damage in the event of a power transistor failure.

TURN-ON/TURN-OFF LOGIC: Since the transistorized power output stages drive active buck-boost loads, it is essential that the amplifier is properly energized anytime the active load is connected. Since the active load is always present, the isolation transformer primaries must be shorted whenever the amplifier is turned off. The short is provided by the NC contacts of a time-delay relay. This allows the amplifier to become fully activated before the relay is energized. When the relay operates it removes the primary short and connects the power amplifier to its isolation transformer primary through a set of NO contacts.

On turn-off a second relay is used to de-energize the "shorting" relay after a brief delay-on-turnoff period. In the event the line voltage to the amplifier is interrupted, a third relay disconnects the coil of the shorting relay and the output transformers are immediately shorted. This protects the amplifier and prevents the introduction of a non-linear impedance into the active load circuit.

PROTECTIVE CIRCUITS:

OVERLOAD CURRENT Limiting and signals for contact closures for each phase are provided when the affected channel's current exceeds safe operating limits,.

TRANSMISSION MONITOR Continually verifies proper operation on the amplifier by injecting an input signal VTM, and then measureing a VTM-DC output. If VTM-DC exceeds its limits, a problem with the amplifier is indicated, and appropriate external relays will take the failed stage out of service.

MODEL 518

SPECIFICATION

EACH CHANNEL

INPUT SIGNAL:	0 to +,-10V
INPUT IMPEDANCE:	100K ohms
GAIN:	.15 to 3.5 with FP trimpots
OUTPUT:	0 to 12V RMS
ACTIVE LOAD:	Amplifier buck/boosts +,-12V RMS, when connected in series with a 120V PT transformer and a 5A RMS load.
OUTPUT ISOLATION:	Output is transfomer isolated to withstand up to 1240VAC, secondary to ground and secondary to amplifier common.
PHASE SHIFT:	<4 degrees between input and output signals at 60Hz. Less than 0.25 degrees change when the 60Hz carrier signal is modulated +,-3Hz.
FAULT PROTECTION:	10A SLO-BLO LINE FUSE 10A FAST BLO POSITIVE AMPLIFIER CKT. 10A FAST BLO NEGATIVE AMPLIFIER CKT.
OVERLOAD PROTECTION:	Output electronically current limited to 10A RMS. Also there is one contact-closure per phase that operates when its channel is in current limit.
TURN ON/TURN OFF PROTECTION:	The amplifier side (primary) of the isolation transformer is disconnected from the amplifier and shorted to PS common whenever the amplifier is de-energized.
TRANSMISSION MONITOR:	Continually verifies operation of the power amplifier by injecting a SIGNAL (VTM) and measuring its DC output level. (VTM-DC)

3-PHASE UNIT SERVICE

AMBIENT TEMPERATURE:	0 degrees C to 50 degrees C operating up to 60 degrees C with derating. -40 degrees C to 80 degrees C shipping and storage.
COOLING:	Internal forced-air
HUMIDITY:	15 to 95% RH, without condensation.

TRANSIT SHOCK:	15G for 11 MS. in any direction.
ALTITUDE:	Up to 3500 feet above sea level, up to 5000 feet with derating.
MTBF:	1000 HRS. with 90% confidence.

CONSTRUCTION: (Reference Outline Drawing D 12917)
Three channels (3-PHASE) and master power supply in one module.

SIZE: 19" W x 8 3/4" H X 15 3/4" D

WEIGHT: 501bs.

CONNECTIONS: INPUT SIGNAL:* Twenty-Six Pin "D" connector. (Ref. TABLE 1, Input Connector Signal Definitions)

* These signals are referenced to PS common.

OUTPUT SIGNALS COMMONS, AND GROUNDS: 12-Pin barrier strip, as marked.

AC LINE: 3-Wire line cord.

POWER: 115V+,-10%, 60Hz, with up to 6A RMS max. line current.

"CURRENT LIMIT" SIGNALS: 6 pin barrier strip, as marked.

TABLE 1

ENCORE ELECTRONICS

MODEL 518

INPUT CONNECTOR SIGNAL DEFINITIONS

PIN#	SIGNAL	ORIGIN
1	PHASE A PSS (+)	80295 - B24
2	GROUND	
3	PHASE B PSS (+)	80295 - B15
4	GROUND	
5	PHASE C PSS (+)	80295 - B6
6	GROUND	
7	VTM - AC	80295 - B25,B16
8	GROUND	- A25, Al6
9	VTM - DC PHASE A	80292 - A21
10	GROUND	
11	VTM - DC PHASE B	80295 - B12
12	GROUND	
13	VTM - DC PHASE C	80295 - B3
14	GROUND	
15	2G(PSS + VTM) PHASE A	80295 - A20
16	GROUND	
17	2G)PSS + VTM) PHASE B	80295 - B11
18	GROUND	
19	2G(PSS + VTM) PHASE C	80295 - B2
20	GROUND	
21	NC	
22	GROUND	
23	NC	
24	GROUND	
25	NC	
26	GROUND	

MODEL 518

RECOMMENDED SPARE PARTS'LIST

QTY	ITEM	MFG	PT.NO.	SOURCE	APPROX. PRICE
1	PREAMPLIFIER BOARD A3	ENCORE		ENCORE	$150.00
1	AMPLIFIER DRIVER BOARD A4	ENCORE		ENCORE	250.00
3	MONOLITHIC POWER OP-AMP	NATIONAL	LM675T	ENCORE*	10.00
1	PNP HEATSINK #1	ENCORE		ENCORE	195.00
1	NPN HEATSINK #2	ENCORE		ENCORE	195.00
3	PNP POWER TRANSISTOR	MOTO	2N6029	ENCORE*	6.00
3	NPN POWER TRANSISTOR	MOTO	2N5629	ENCORE*	5.00
5	10A SLO-BLO LINE FUSE	LITTLEFUSE	313010	ENCORE*	1.50
1 0	10A FAST-BLO LOAD FUSE	LITTLEFUSE	311010	ENCORE*	.75
2	POWER INDICATOR LAMP	SYLVANIA	120PSB	ENCORE*	2.50
1	LINE SENSING RELAY	P&B	KPRA14DG	ENCORE*	30.00
1	DELAY "ON" RELAY	P & B	KRPA14AY	ENCORE	30.00
1	DELAY "OFF" RELAY	P&B	CHB-38-70011	ENCORE	110.00

*CAN ALSO BE PURCHASED FROM ELECTRONICS DISTRIBUTOR.

MODEL 518

TROUBLE SHOOTING PROCEDURE

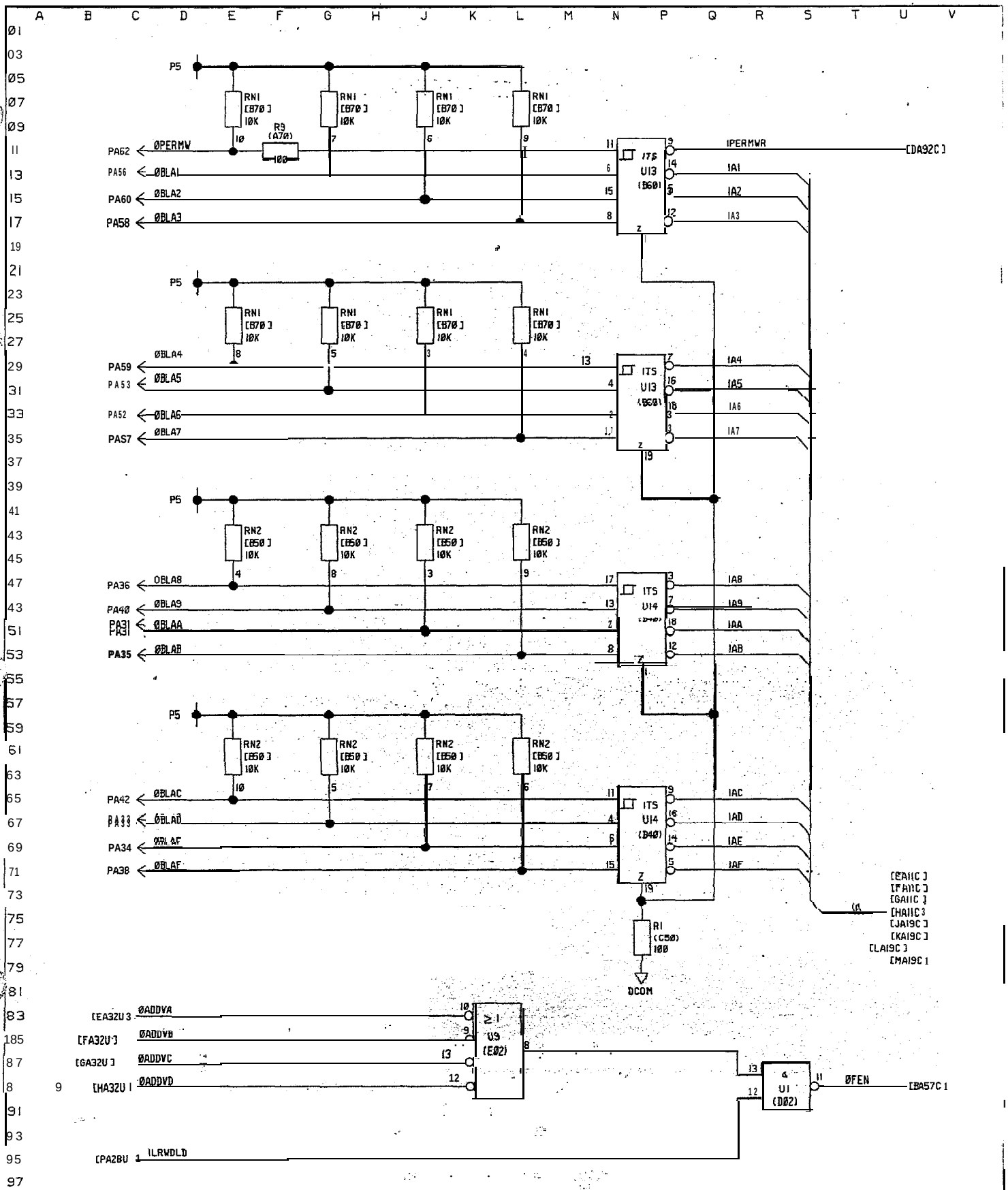
1. Check and replace blown fuses.

2. If line fuse only, unplug heatsinks, and check for failure in unregulated power supplies, starting with the +,-30VDC high power section.

3. Connect line power through a variable transformer. With the power switch on, raise the input line voltage slowly. Excessive line current or a DC component in the power transistor emitter commons indicates a heatsink transistor failure.

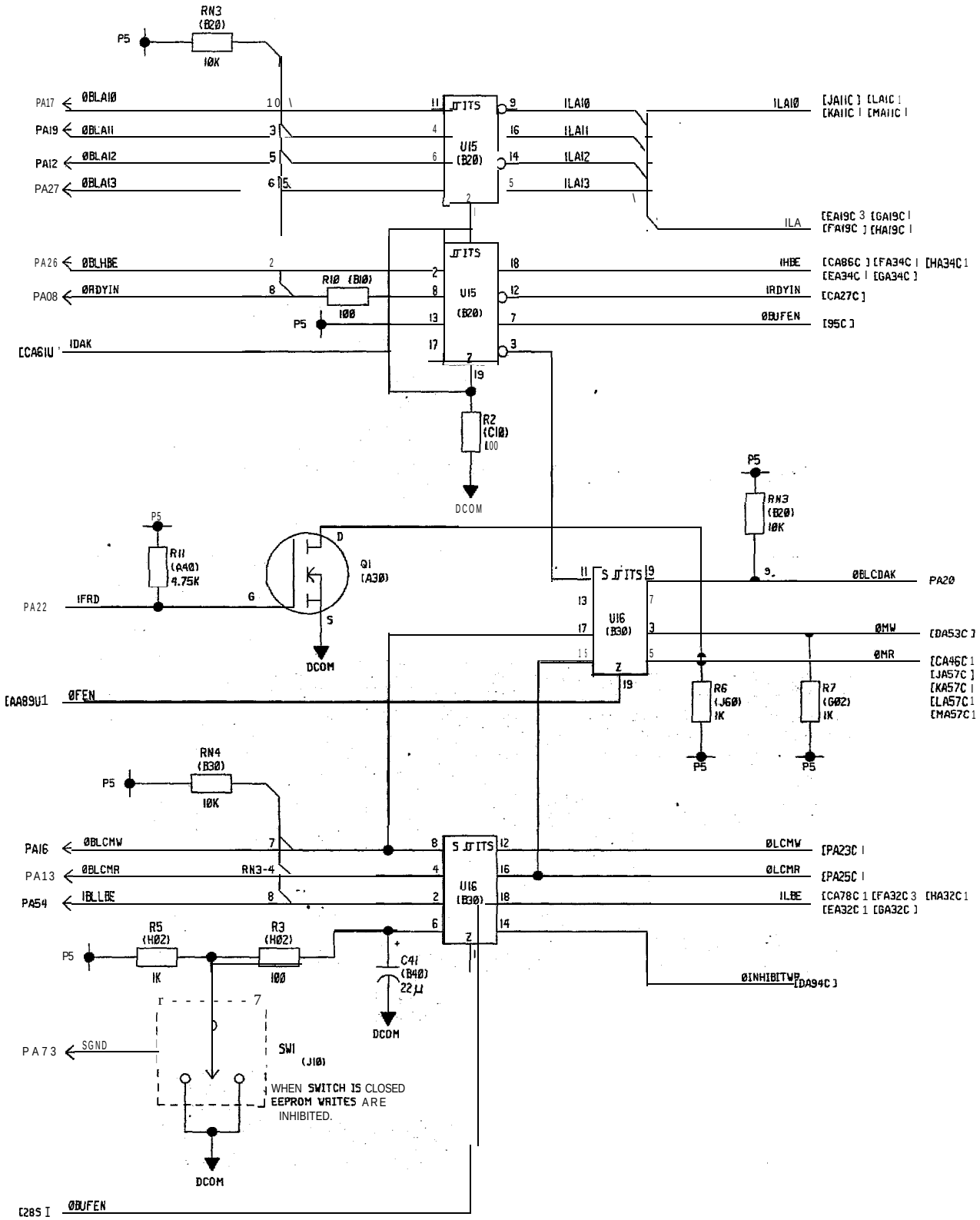
   a. Check power transistors for shorts.
   b. Replace heatsinks that have shorted power transistors, or replace shorted power transistors which can be unplugged from the heatsink.
   c. Replace PC board A2
   d. Replace PC board A3

4. If the unit fails to power-up after the voltage is applied check the logic "turn-on" and "turn-off" circuits and relays. (K1,K2 and X3)



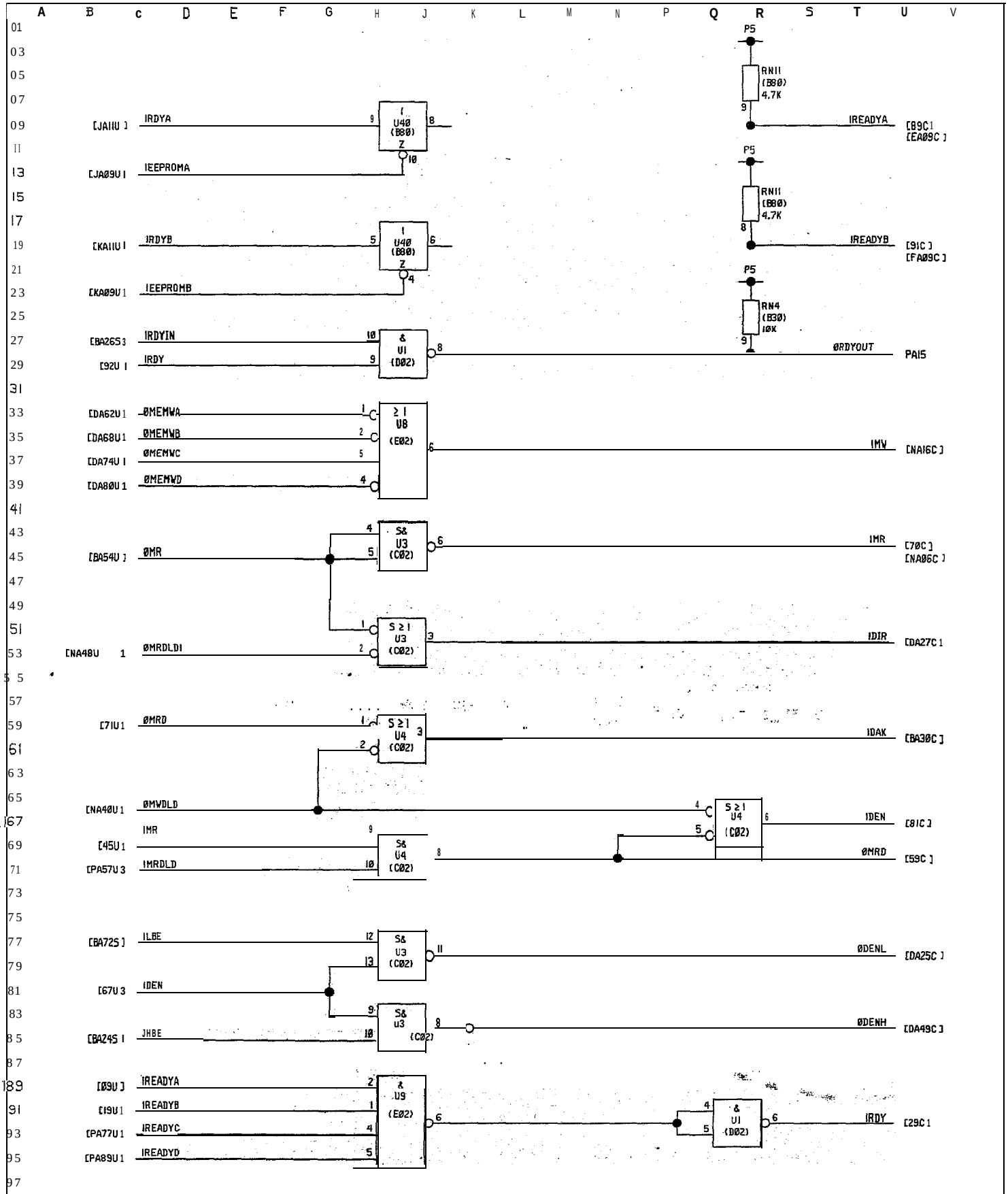
REV. 1	REV.4	REV.5	PRINTS TO DL104 PRI	ENGINEER Roy R. Nanp	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM UNIVERSAL MEMORY B
REV.2	ISSUED 5-8-84			FIRST MADE F	O R	
REV.3	M A D E BY ROY R. NANP			I.C.NO.	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	DS3800HUMB
						CONT.ON SH. 4BA SH.NO. 4AA

CARS



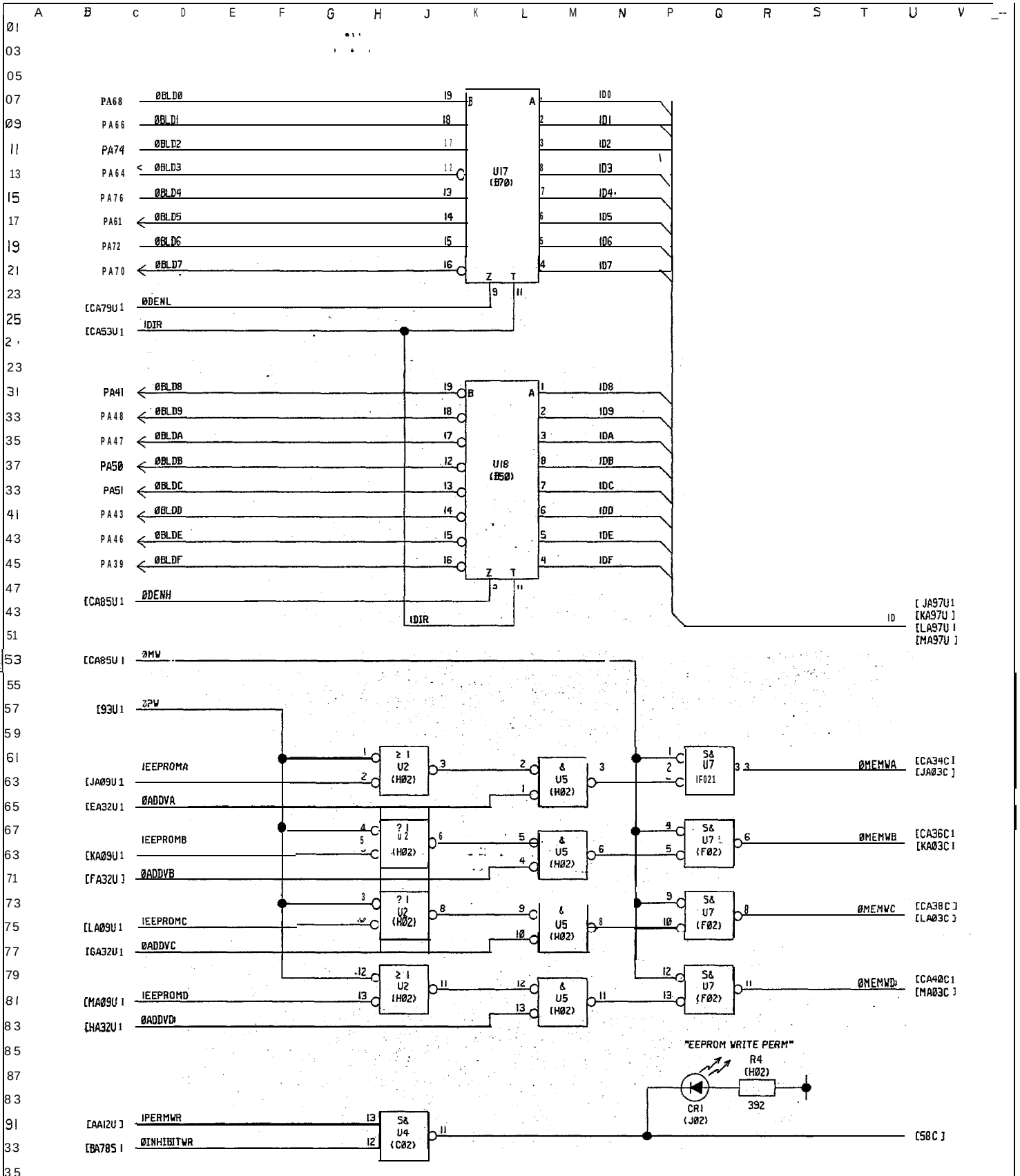
REV.1	REV.4	REV.5	PRINTS TO DL104 PRI	ENGINEER Roy R. Nanz	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM UNIVERSAL MEMORY B
REV.2	ISSUED 5-8-84			FIRST MADE FOR	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	DS3800HU MB
REV.3	MADE BY ROY R. NANZ			I.C.NO.		CONT.ON SH. 4CA SH.NO. 4BA

CARS

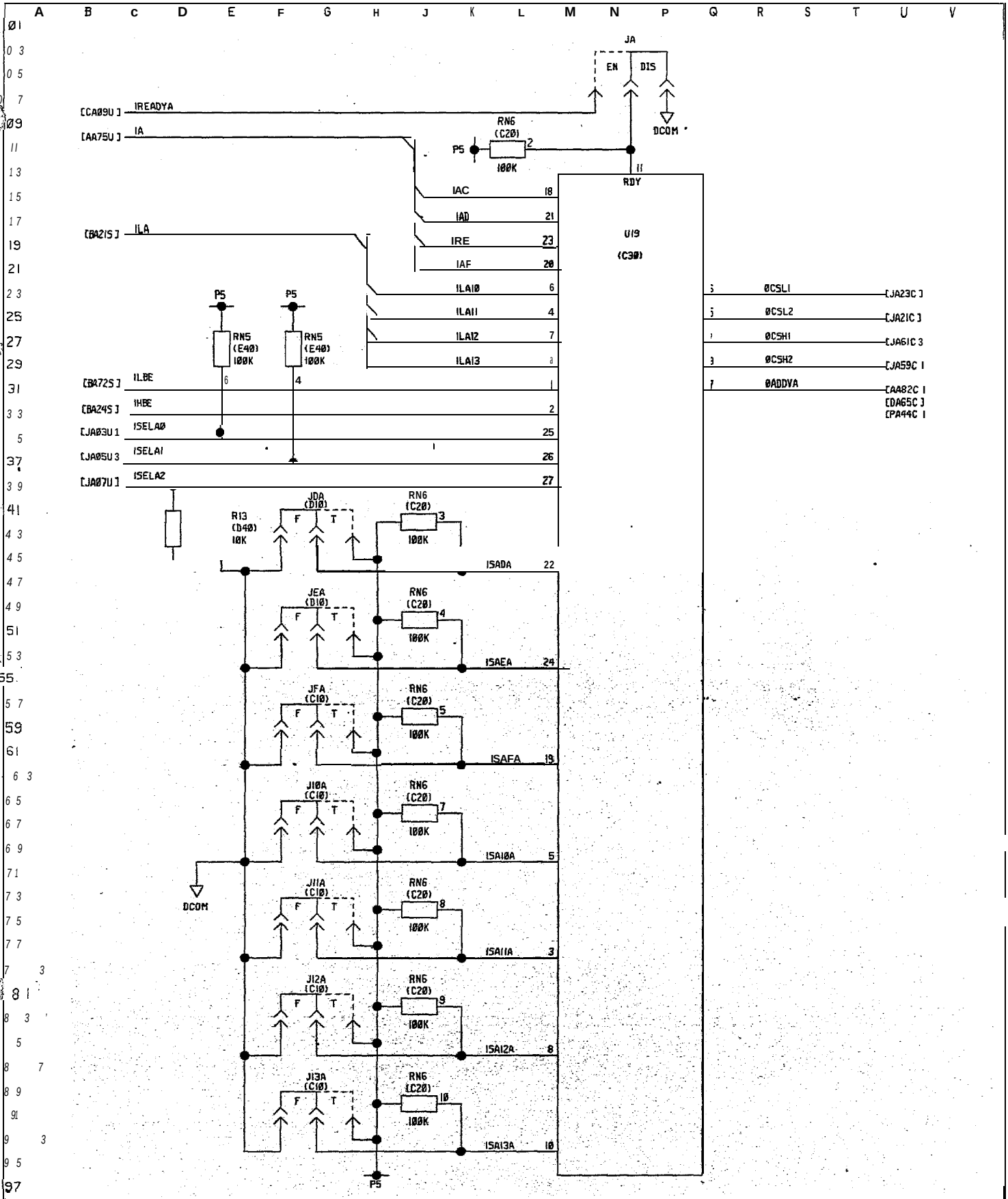


REV.1 OEC WLL 88-05-10	REV.4	REV.5	PRINTS TO DL 104	ENGINEER WL	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM UNIVERSAL MEMORY B
REV.2	ISSUED 5-18-84			FIRST MADE FOR	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	DS3800HUMB
REV.3	MADE BY ROY R. NANZ			I.C.NO.		CONT.ON SH.4DA SH.NO.4CA

CARS

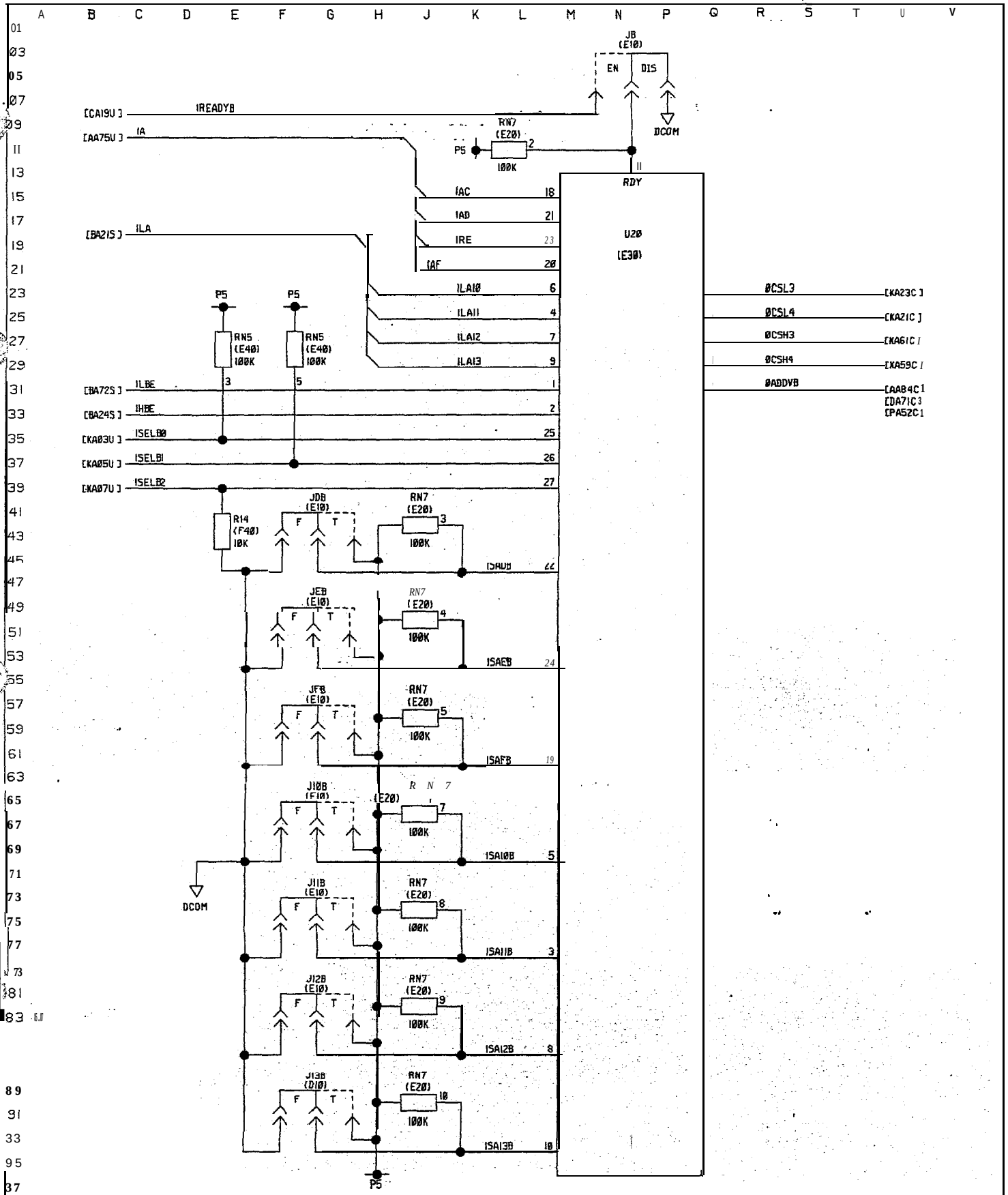


REV.1	REV.4	REV.5	PRINTS TO DL104 PRI	ENGINEER	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM UNIVERSAL MEMORY B
REV.2	ISSUED 5-8-84			FIRST MADE FOR	DRIVE SYSTEMS DEPT. SALEM, VA. U.5.A;	DS3800HUMB
REV.3	MADE BY ROY R. NANZ			I.C.NO.		CONT.ON SH.4EA SH.NO.4DA



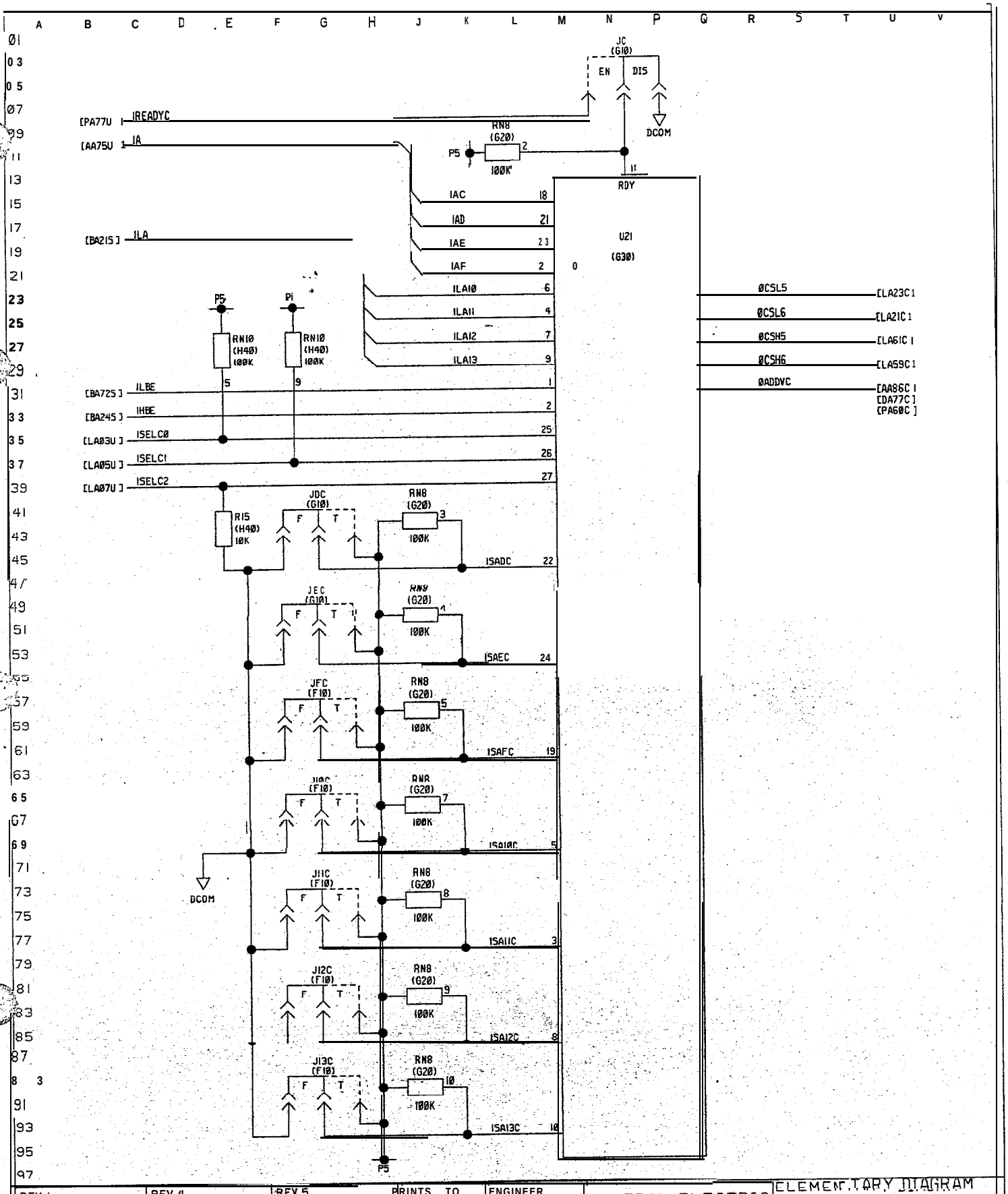
REV.1	REV.4	REV.5 PR	NTS TO DLI04 PRI	ENGINEER	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM UNIVERSAL MEMORY B
REV.2	ISSUED 5- 8-84			FIRST MADE	FOR DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	DS3800HUMB
REV.?	MADE BY ROY R. NANZ			I.C.NO.		CONT.ON SH. 4FA SH.NO.4EA

CAD



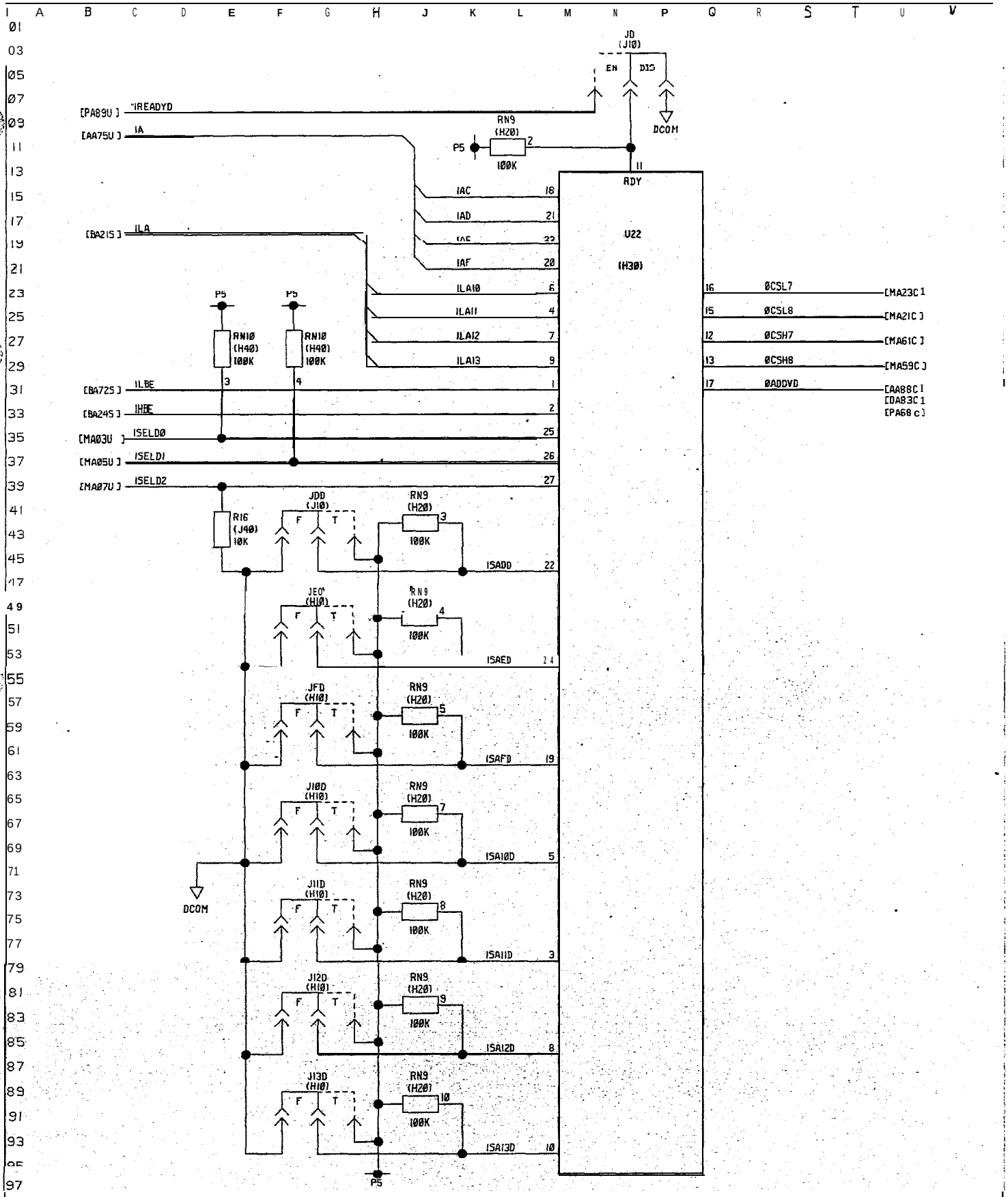
REV.1	REV.4	REV.5	PRINTS TO DL104 PRI	ENGINEER	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM UNIVERSAL MEMORY B
REV.2	ISSUED 5-8-84			FIRST MADE FOR		DS3800HUMB
REV.3	MADE BY ROY R. NANZ			I.C.NO.	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	CONT.ON SH. 4GA SH.NO.4F1

CARS



REV.1	REV.4	REV.5	PRINTS TO DL104 PRI	ENGINEER Roy R. Nanz	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM UNIVERSAL MEMORY B³
REV.2	ISSUED 5-2-84			FIRST MADE F O R		DS3800HUMB
REV.3	MADE BY ROY R. NANZ			I.C.NO.	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	CONT.O N SH. 4HA SH.NO.46A

CARS



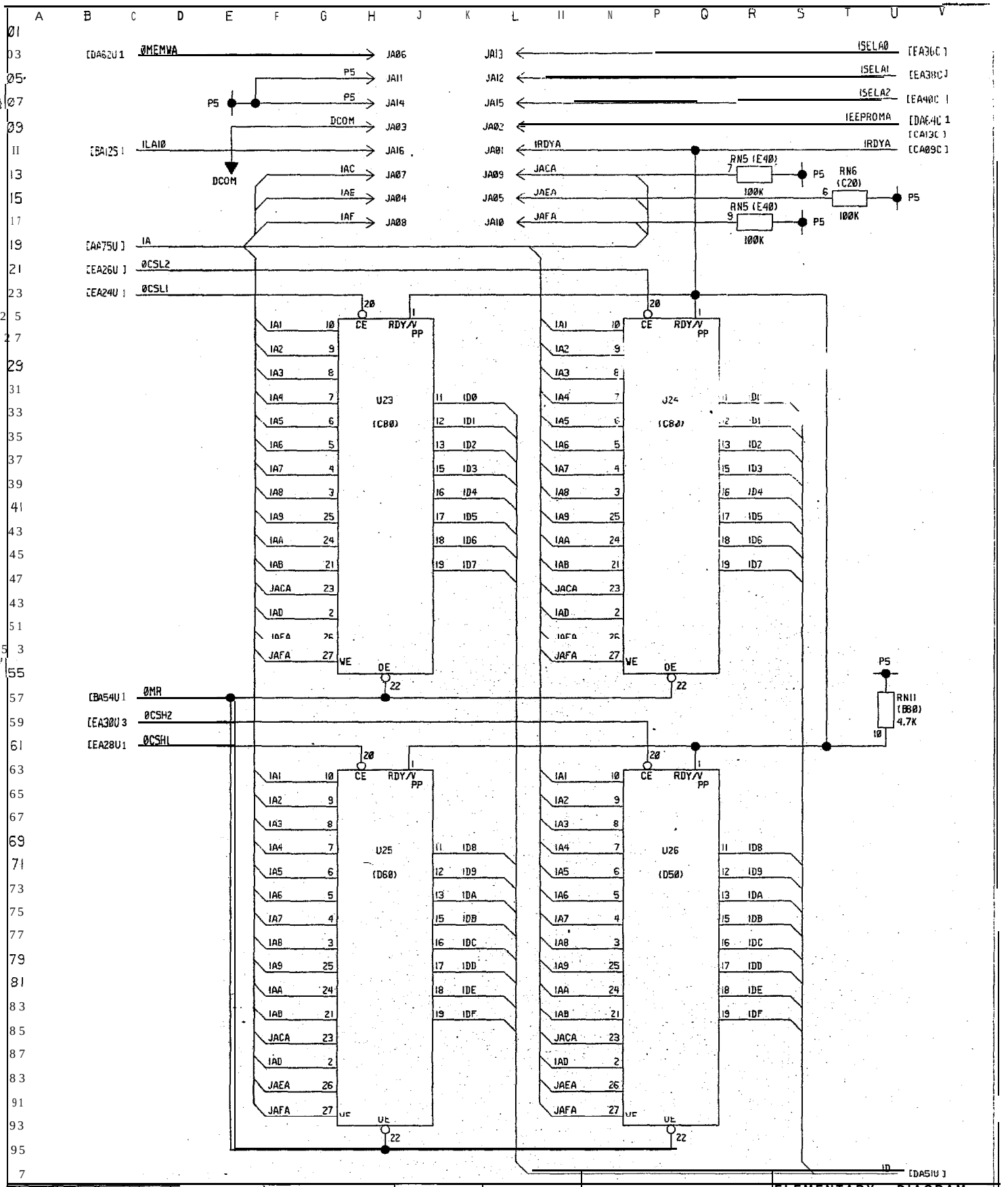
REV. 1	REV.4	REV.5	PRINTS TO DL104 PRI	ENGINEER	GENERAL ELECTRIC COMPANY
REV. 2	ISSUED 5-8-84		FIRST MADE FOR		DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.
REV.3	MADE BY ROY R. NANZ		I.C.NO.		

ELEMENTARY DIAGRAM
UNIVERSAL MEMORY B

DS380ØHUMB

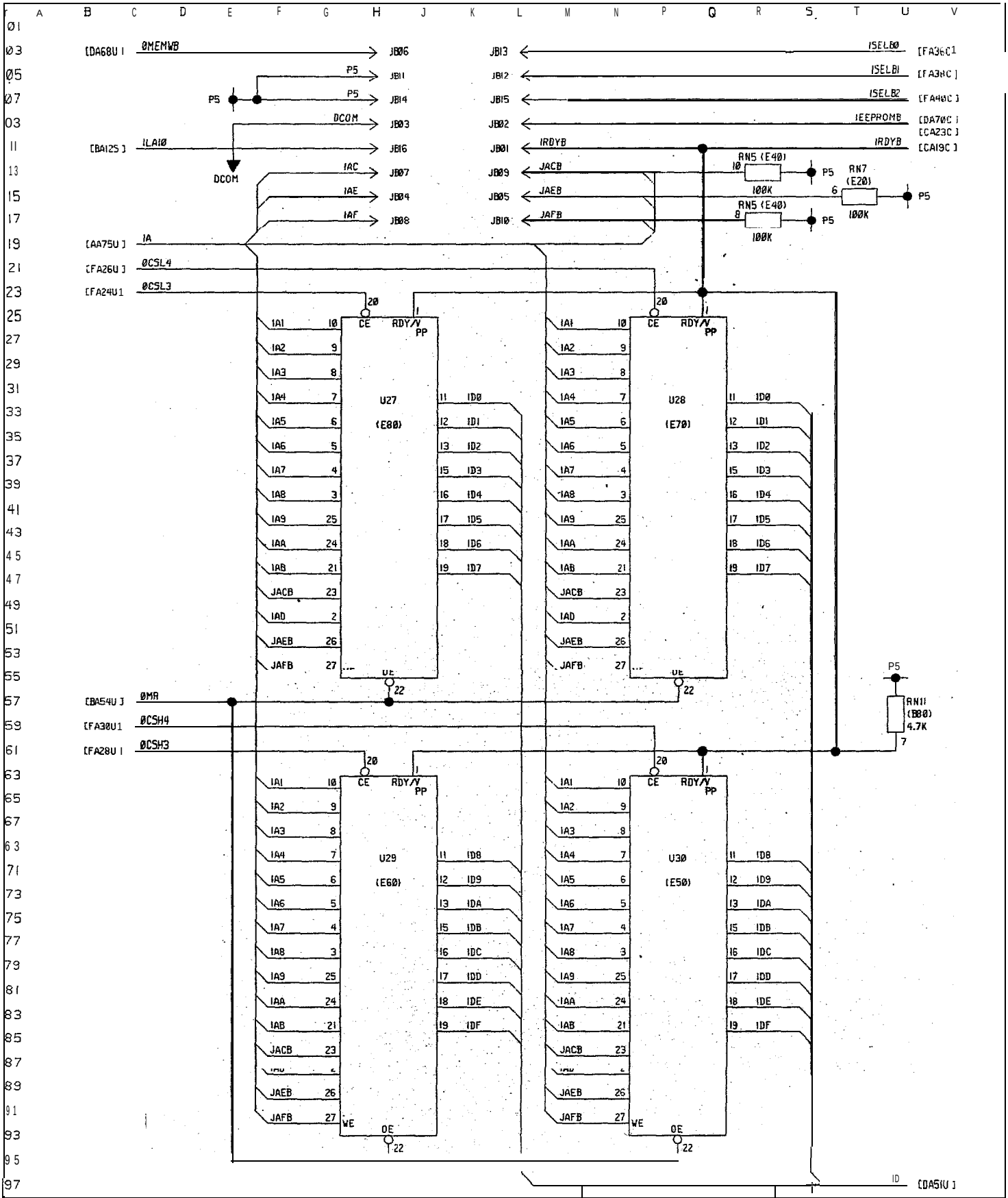
CONT.ON SH. 4JA SH.NO. 4HA

CARS



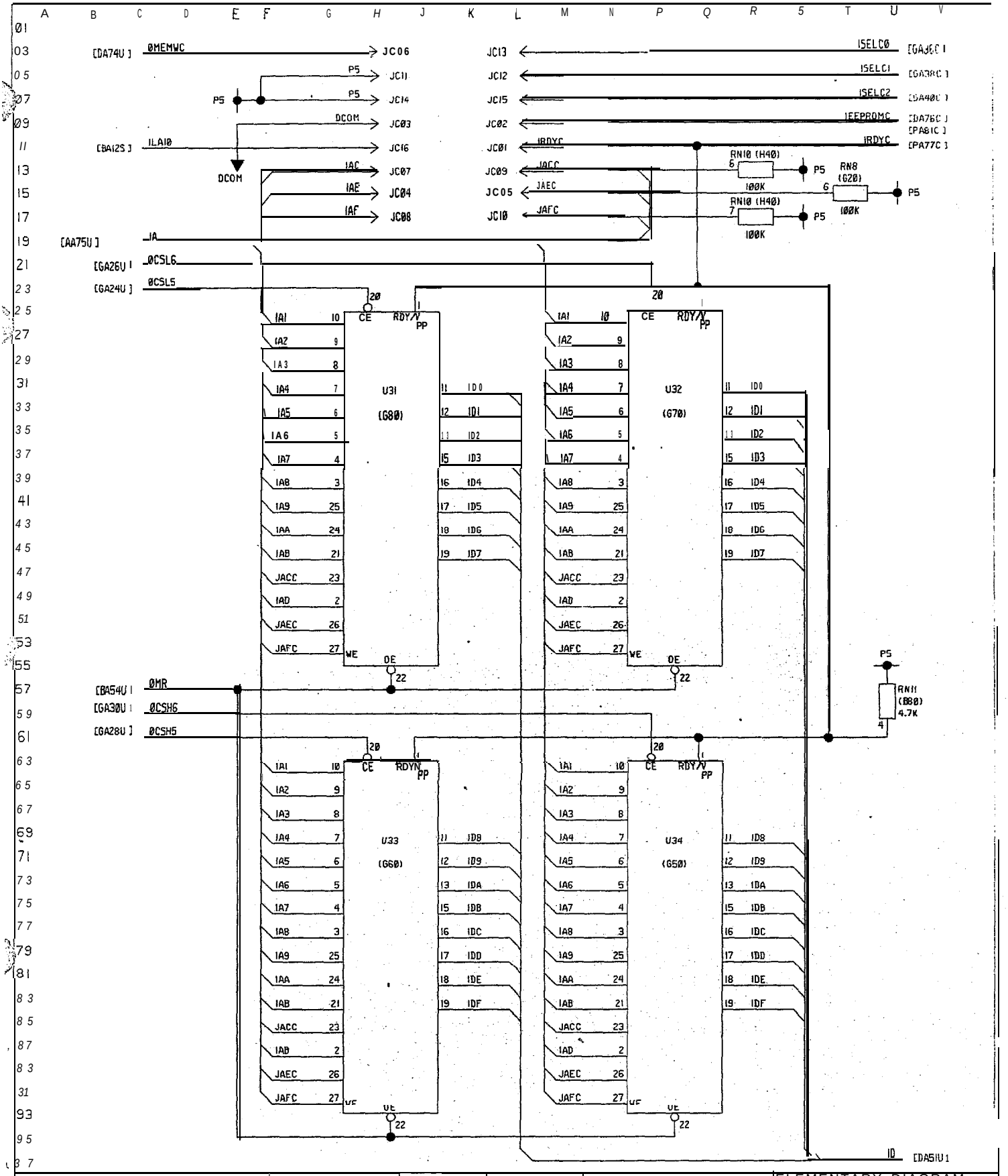
REV.1	REV.4	REV.5	ODTNTS TO DL104 I PRI	ENGINEER	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM UNIVERSAL MEMORY B
REV.2	ISSUED 5-8-84			FIRST MADE FOR	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	DS3800HU MB
REV.3	MADE BY R O Y R . . NANZ			I.C.NO.		CONT.ON SH. 4KA SH.NO. 4JA

CAPS



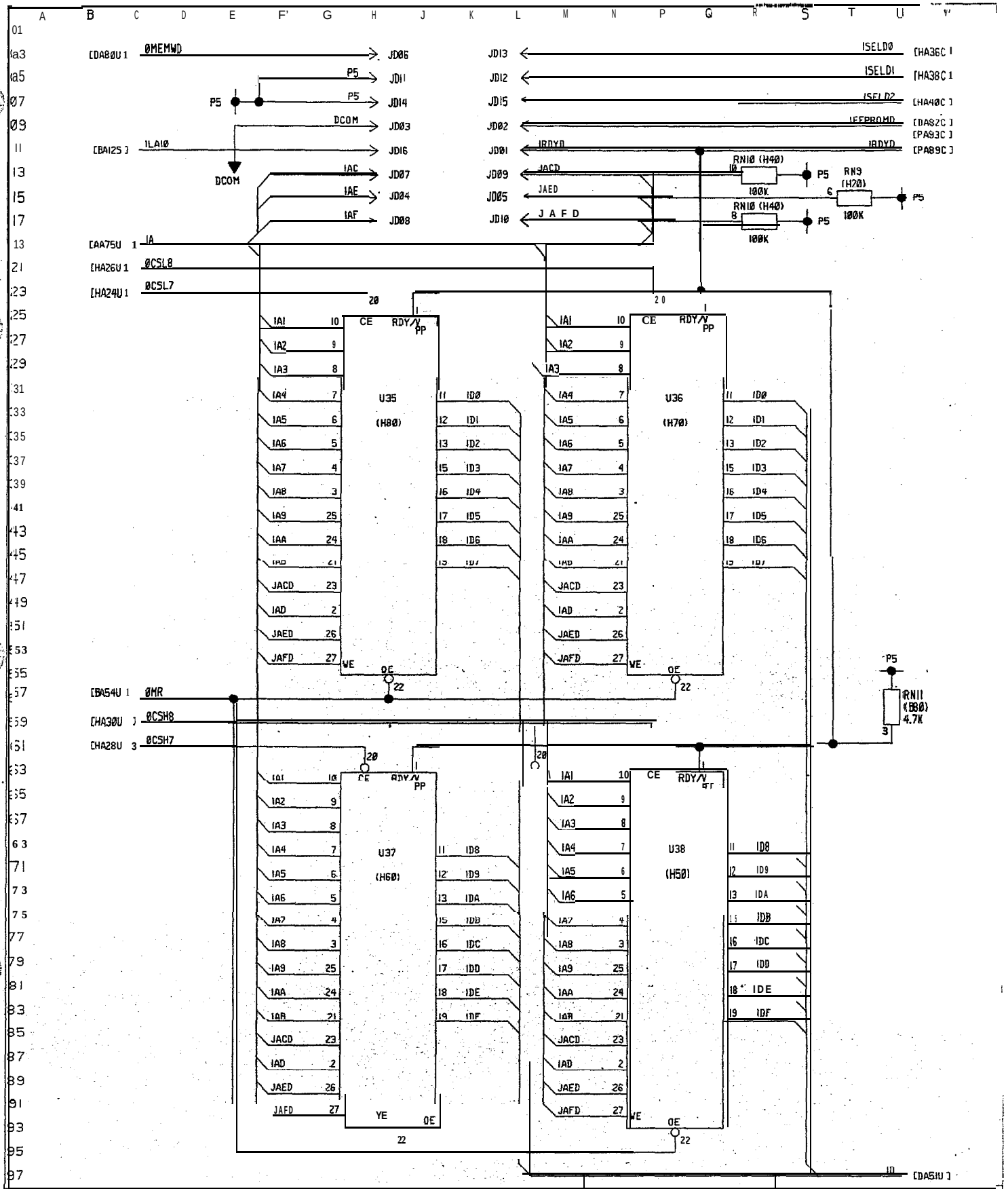
REV.1	REV.4	REV.5	PRINTS TO DL104 PRI	ENGINEER	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM UNIVERSAL MEMORY B
REV.2	ISSUED 5-8-84			FIRST MADE FOR	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	DS38ØØHUMB
REV.3	WADE BY R O Y R . NANZ			I.C.NO.		CONT.ON SH. 4LA SH.NO. 4KA

CARS



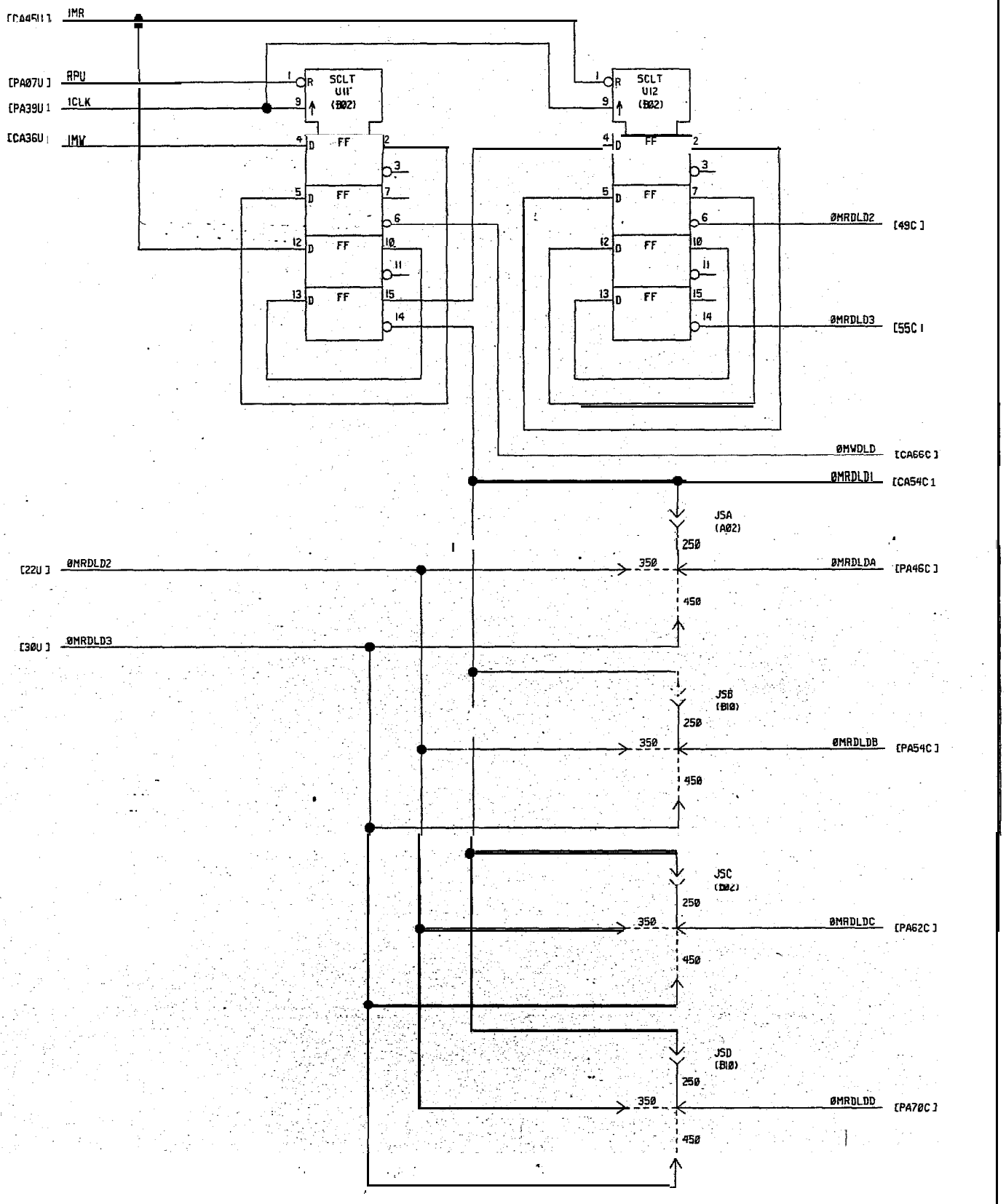
REV.1	REV.4	REV.5	PRINTS TO DL104 PRI	ENGINEER	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM UNIVERSAL MEMORY B .
REV.2	ISSUED 5-8-84			FIRST MADE FOR		DS3800HUMB
REV.3	MADE BY ROY R. NANZ			I.C.NO.	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	CONT.ON SH. 4MA SH.NO.4LA

CARS



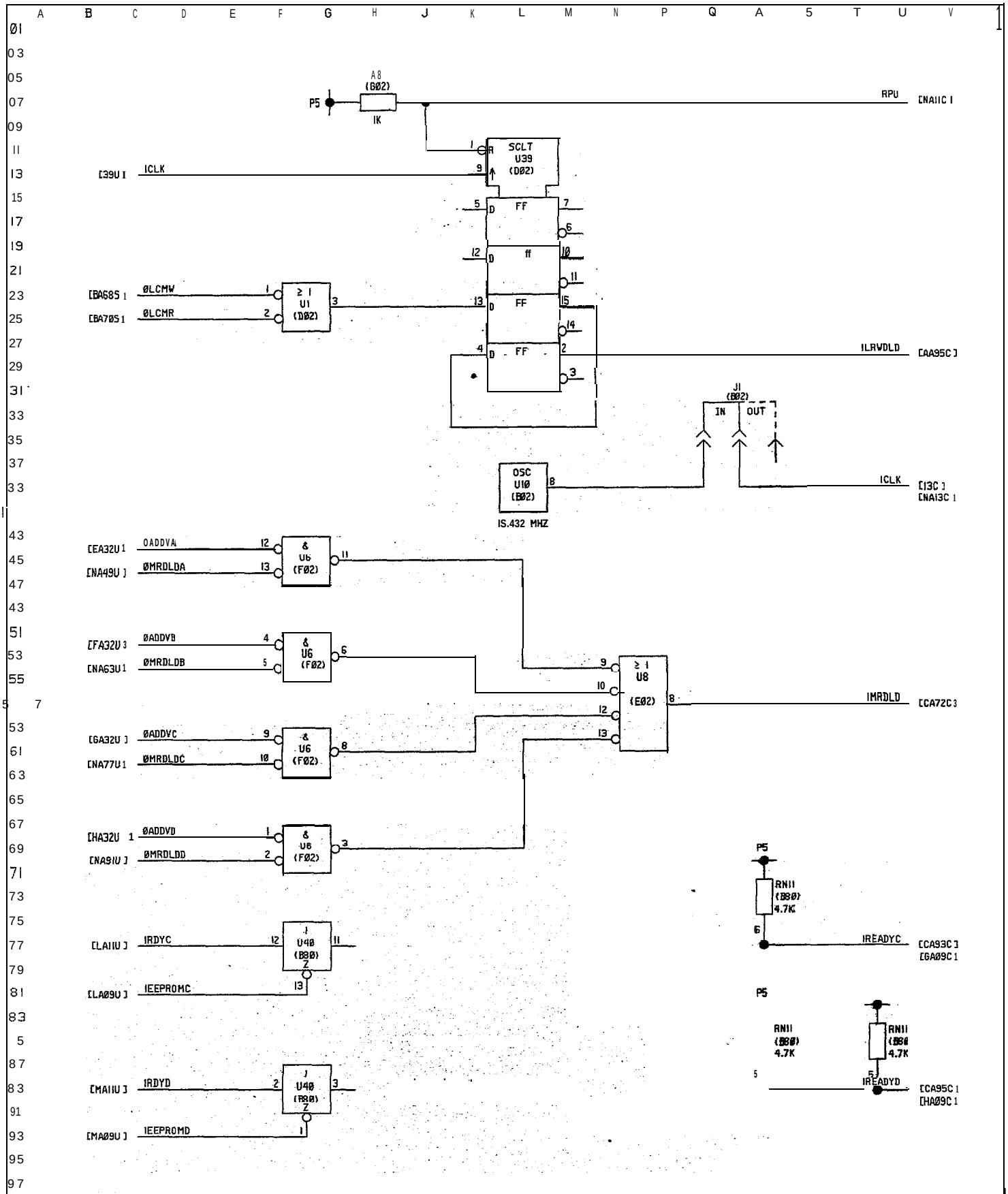
REV.1	REV.4	REV.5	PRINTS TO DL104 PRI	ENGINEER Roy R. Nanz	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM UNIVERSAL MEMORY B
REV. 2	ISSUED S - 6 - 84			FIRST MADE FOR	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	DS380ØHUMB
REV.3	MADE BY ROY R. NANZ			I.C.NO.		CONT.ON SH. 4NA SH.NO. 4MA

CADS



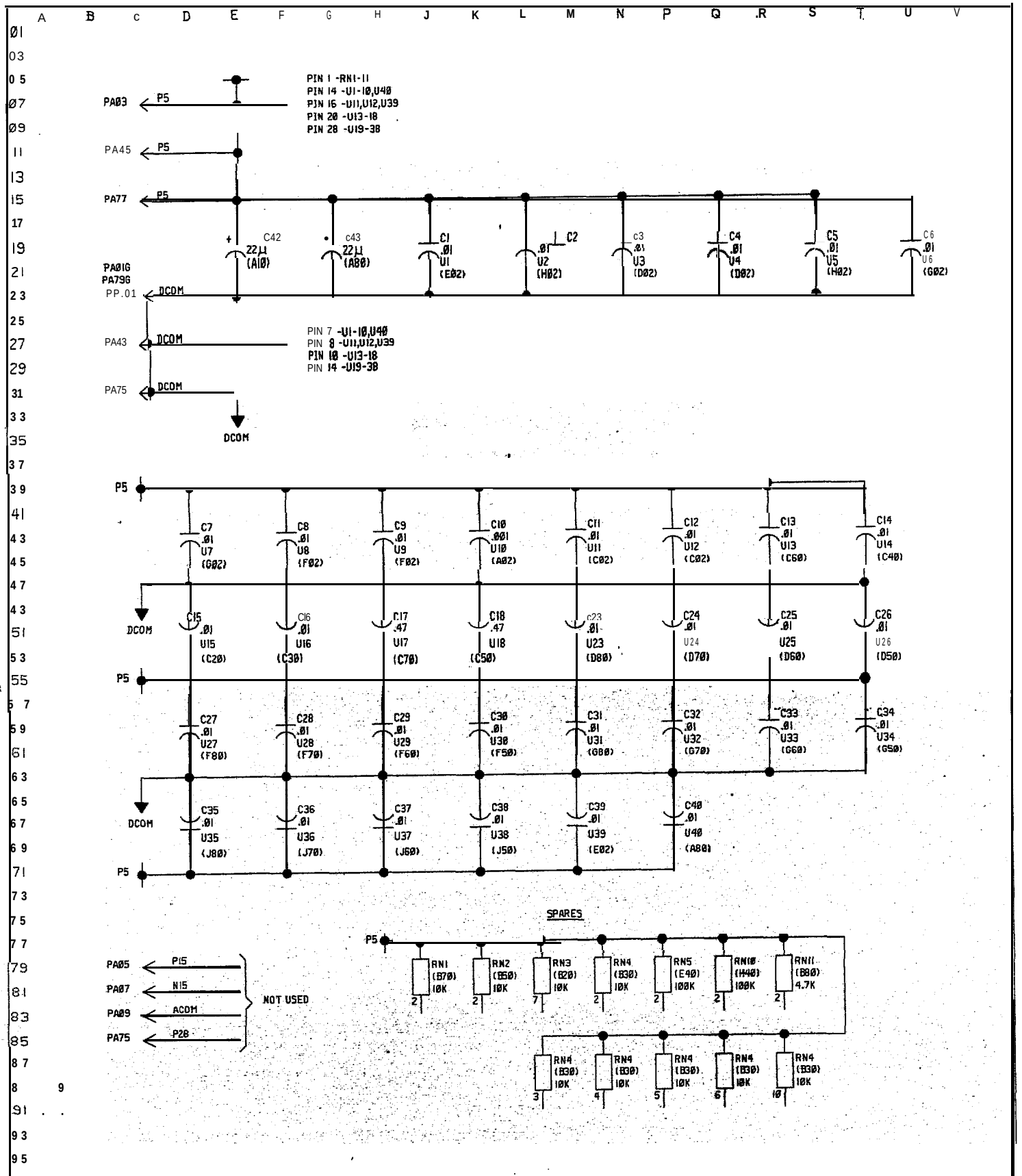
REV.1	REV.4	REV.5	PRINTS TO DL104 PRI	ENGINEER RJR.NANZ	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM UNIVERSAL MEMORY B
REV.2	ISSUED 5-8-84			FIRST MADE FOR		DS3800HUMB
REV.3 R	MADE BY O Y R.NANZ		I.C.NO.		DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	CONT.ON SH.4PA SH.NO.4NA

CARS



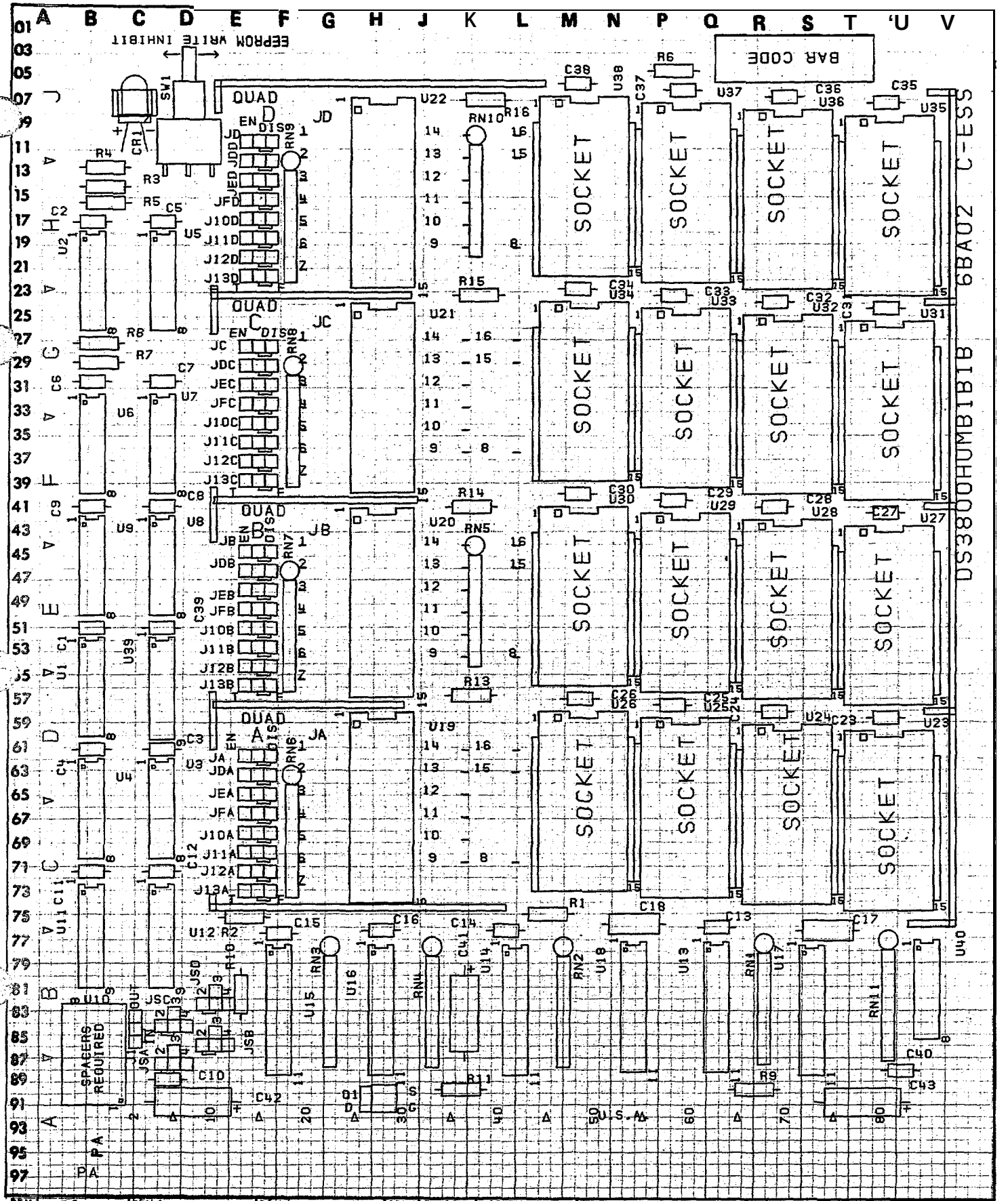
REV.1 OEC WLL 88-05-10	REV.4	REV.5	PRINTS TO DL104	ENGINEER W22	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM UNIVERSAL MEMORY B
REV.2	ISSUED 5-8-84			FIRST MADE FOR	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	DS380ØHUMB
REV.3	MADE BY ROY R. NANZ			I.C.NO.		CONT.ON SH.4RS SH.NO.4PA

CARS



REV.1	REV.4	REV.5 PRINTS TO D L I 0 4	ENGINEER	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM UNIVERSAL MEMORY B
REV.2	ISSUED 5-8-84		FIRST MADE FOR	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	DS38 00HUMB
REV.3	MADE BY ROY R. NANZ		I.C.NO.		CONT.ON SH.45A SH.NO.4RA

CARS



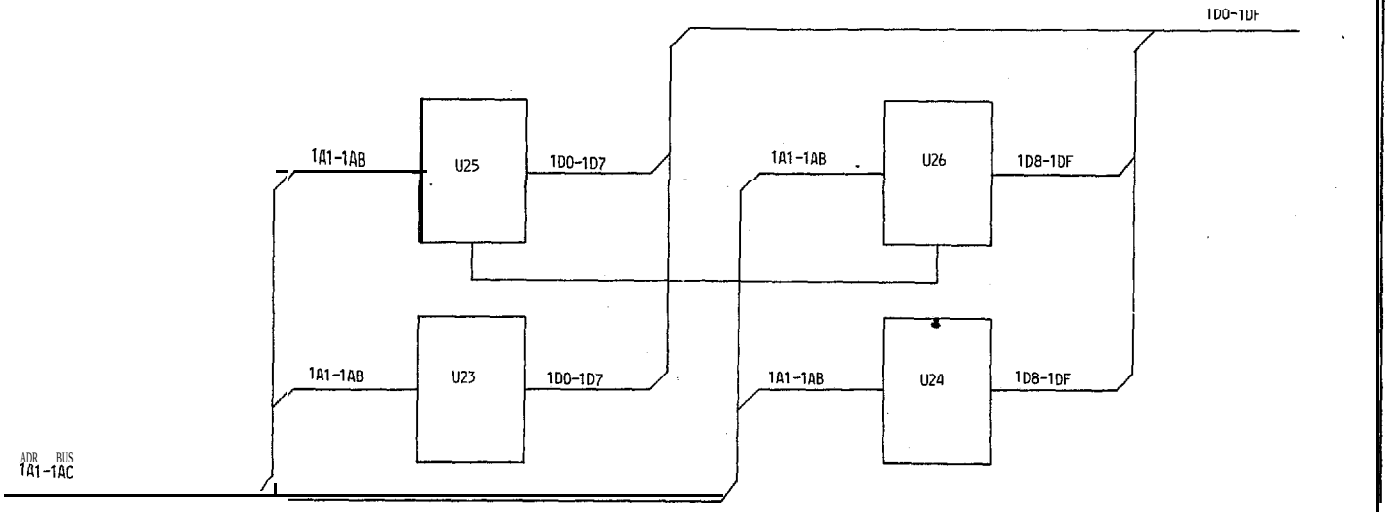
REV.1 DEC WLL 5-10-84	REV.4	REV.5	PRINTS TO DL104	APPROVALS WLL	GENERAL ELECTRIC DRIVE SYSTEMS DEPT. SALEM, VA., U.S.A.
REV.2	ISSUED 5-8-84			FIRST MADE FOR	
REV.3	MADE BY C.W. FITZWATER			LC. NO.	

ELEMENTARY DIAGRAM
UNIVERSAL MEMORY B

DS3800HUMB

CONT. ON SH. FL SH. NO. 4SP

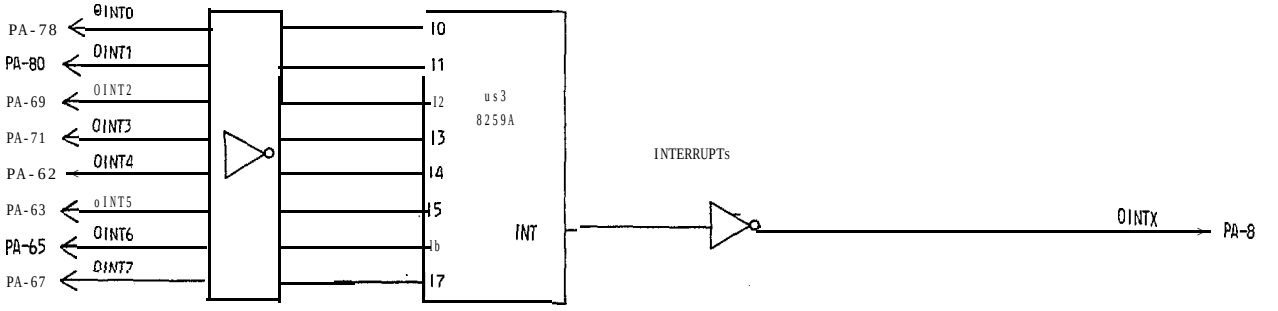
CARS

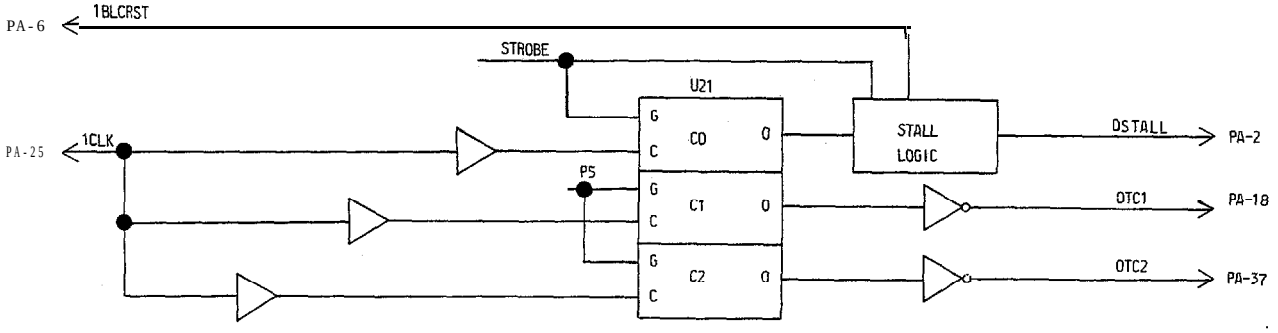


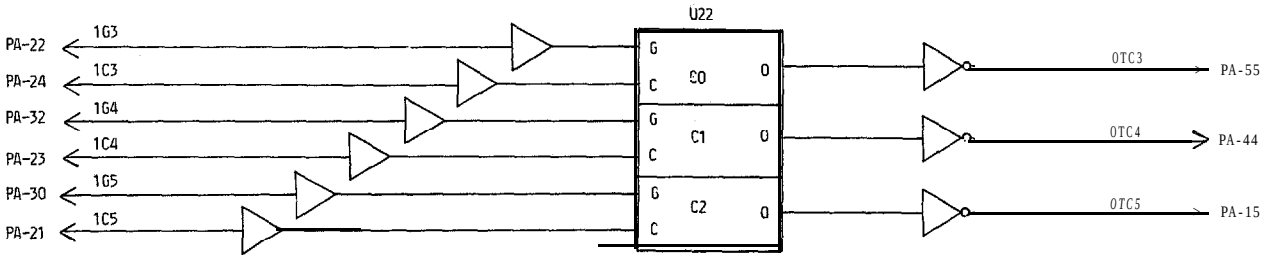
PROM MEMORY

LOW ADR - U23 & U24

HIGH ADR - U25 & U26







U21-22, 8254 COUNTERS

	REV. 4	REV. 5	PRINTS TO PRI DL102	ENGINEER	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.
REV. 2	ISSUED August 9, 1984			MK IV	
REV 3	MADE BY E. DILLON				

ELEMENTARY DIAGRAM

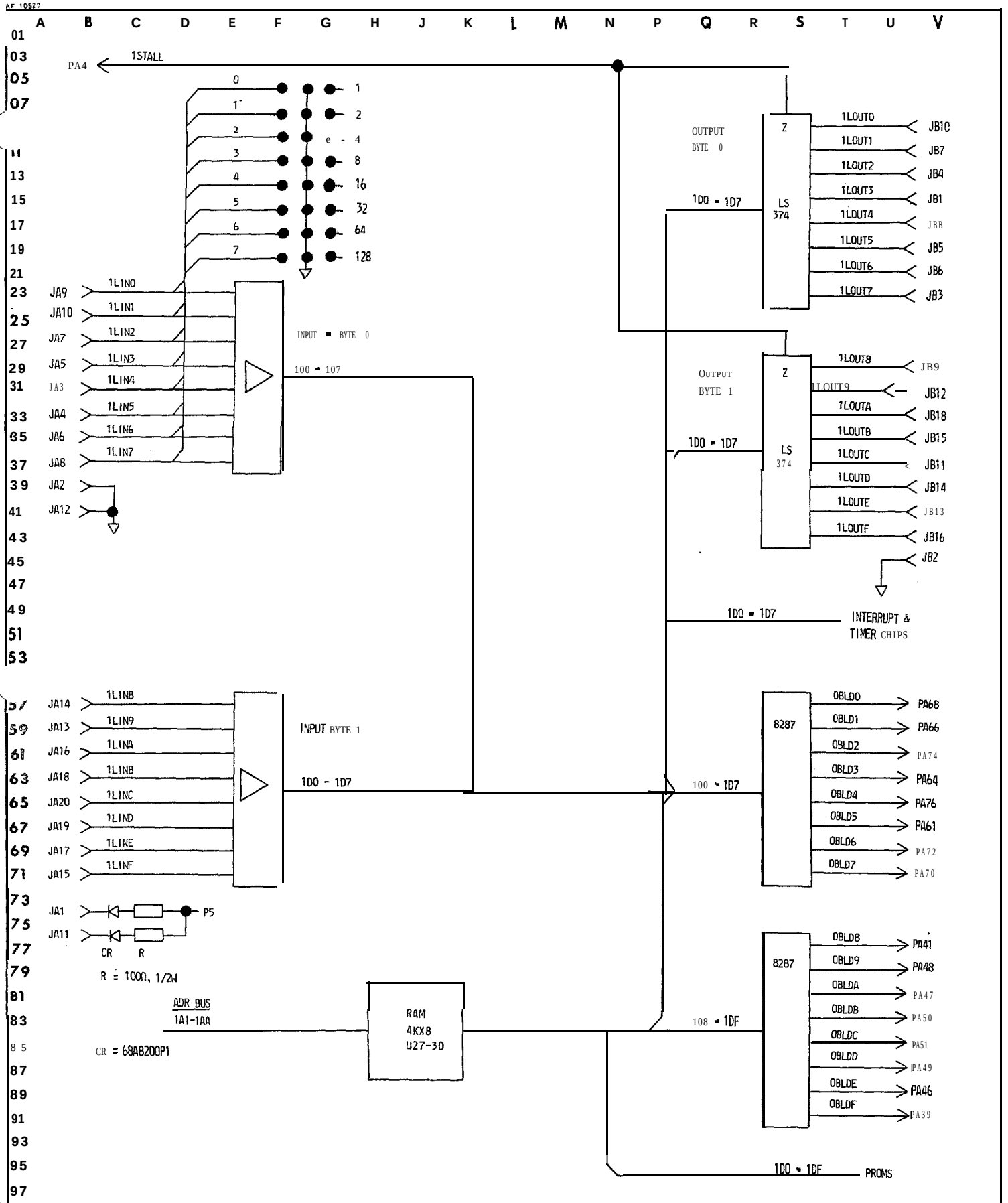
CPU EXPANDER BLOCK DIAGRAM

DS3800HXPD

CONT. ON SH. 2BA SH. NO. 2AA

OSD 0152 (09 76)

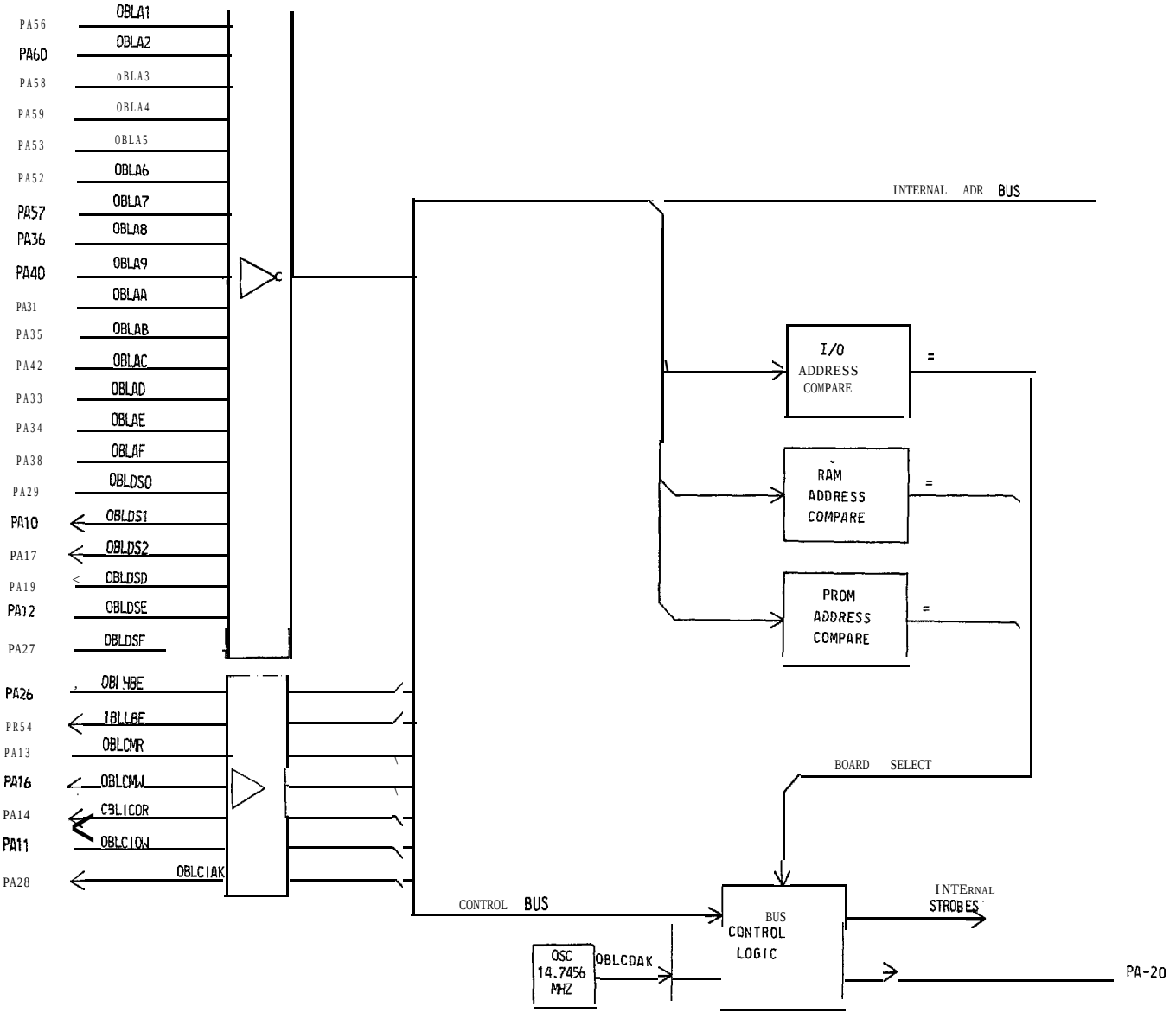
AF 10527



.1	REV. 4	REV. 5	PRINTS TO PRI DL102	ENGINEER	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED August 9, 1984			MK IV	DRIVE SYSTEMS DEPARTMENT	CPU EXPANDER BLOCK DIAGRAM
REV. 3	MADE BY E. DILLON				SALEM, VA. U.S.A.	DS3800HXPD
						CONT. ON SH.2CA SH. NO. 2B1

DSD 0152 (09 78)

CAKS



1	REV. 4	REV.5	PRINTS TO PRI DL102	ENGINEER
REV. 2	ISSUED August 9, 1984			MK IV
REV. 3	MADE BY E. DILLON			

GENERAL@ ELECTRIC
DRIVE SYSTEMS DEPARTMENT
SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM
CPU EXPANDER BLOCK DIAGRAM
DS3800HXPD
CONT. ON SH. 2DA SH. NO. 201

DEFINITION AND FUNCTION

The DS3800HXPD CPU Expander is an L-bus interface board that provides memory and I/O expansion for a microprocessor board. The HXPD contains 4 sockets for Intel* 2764 or 27128 PROMs, and 4 sockets for 2K or 8K RAMs. An on-board PROM functions as the memory address decoder. Berg jumpers select the PROM and RAM segments, starting addresses, and the high byte I/O address. Additionally, the HXPD provides 5 timer/counters, digital I/O, and interrupt expansion.

THEORY OF OPERATION

Refer to the HXPD block diagram, pages 2AA through 2CA.

MEMORY

Programmable Read-Only Memory (PROM)

The HXPD contains sockets for 4 Intel 2764 or 27128 type PROMs for application programming. Berg jumpers must be set according to PROM type used. Berg jumpers on the 6 segment input lines select the memory segment (1, 2, 3, D, E, or F) or no PROM; Berg jumpers for address lines D, E, and F select the starting address.

Refer to Table 1 in the 'APPLICATION DATA' section for Berg jumper settings;

Random Access Memory (RAM)

The HXPD contains 4 sockets for 2K or 8K RAM. Berg jumpers must be set according to amount of RAM used. As with the PROM, Berg jumpers on the 6 segment input lines select the memory segment (1, 2, 3, D, E, or F) or no RAM: Berg jumpers for address lines D, E, and F select the starting address.

Refer to Table 1 in the. "APPLICATION DATA" section for Berg jumper settings.

INPUT/OUTPUT (I/O)

The HXPD provides 16 digital inputs on connector JA and outputs (see Table 3) on JB. Berg jumpers (see Table 1) on input lines 0-7 allow the user to program a constant. If the lines are used as an input from an external source, all jumpers should be placed in the T (true) position.'

A logic high on the 1STALL input (PA4) places all the outputs in a high impedence state. If the 1STALL control function is not used, PA4 should be grounded on the backplane.

* Intel is a registered trademark of the Intel Corporation.

REV.1	REV.4	REV.7	[PRINTS TO DL102 PRI	ENGINEER EW	DESCRIPTION OF OPERATION AND APPLICATION DATA
REV.2	REV.5	ISSUED August 9, 1984		GENERAL ELECTRIC	CPU Expander
REV.3	REV.6	MADE BY E. J. Dillon		COMPANY SALEM,VA	DS3800HXPD CONT.ON SH. 2EA SH.NO. 2DA

Interrupts

The **HXPD** contains an 8259A programmable interrupt controller chip to process 8 interrupt inputs (see Table **2).** Interrupt output is on PA-8. The chip can be used as either a buffered mode slave or by polling.

Timer/counter

The RXPD contains two 8254 programmable interval timer/counter chips that provide 5 general purpose outputs (see Table **2).** An additional output connects to the board's stall timer circuitry, which indicates a board stall **(PA-2).**

I/O Addressing

Figures 1 and 2 show the addressing requirements of the HXPD board;

Figure **1**
HIGH BYTE

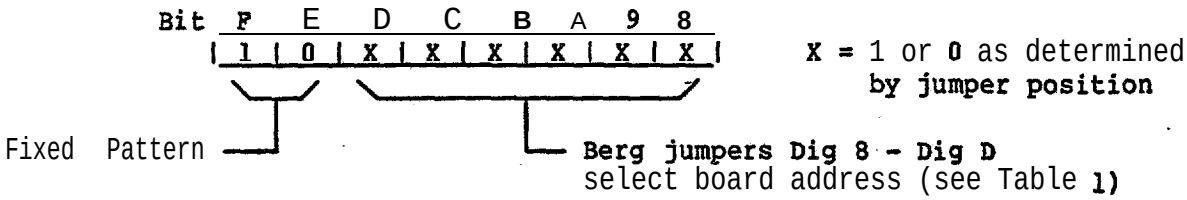


Figure 2
LOW BYTE

Bit	7	6	5	4	3	2	1	0
8259A	X	X	1	0	0	X	*	0
8254 (U21)	X	X	1	0	1	*	*	0
8254 (U22)	X	X	1	1	0	*	*	0
Byte 0 (I/O)	X	X	0	0	0	X	X	0
Byte 1 (I/O)	X	0	0	1	X	X	0	
Reset stall timer	X	X	1	1	1	X	X	0

* = Device address bits
X = Don't *care*

jREV.1	REV.4	jREV.7	PRINTS TO DL102	ENGINEER EQ	DESCRIPTION OF OPERATION AND APPLICATION DATA
REV.2	REV.5	ISSUED August 9, 1984	PRI GENERAL ELECTRIC I		CPU Expander
(REV.3	REV.6	MADE BY E. J. Dillon	COMPANY SALEM,VA		DS3800HXPD CONT.ON SH. 2FA SH.NO. 2EA

POWER REQUIREMENTS

Current: Maximum = 2.1 amps
Typical = 1.5 amps

Heat dissipation: 10.5 watts

BERG JUMPERS

Table 1
BERG JUMPERS

JUMPER NAME	FUNCTION
--se--	PROM size setting: Set "64" for 2764 and "128" for 27128 type
PD PE PF	PROM addressing; set true or false (T/F) on address lines D-F to select starting address
PROM	"P" position when no PROM used: set "0", "1", "2", "D", "E", or "F" to select memory segment
------	RAM size setting; 8K byte position or 2K byte position
J15	RAM size setting; 8K byte position or 2K byte position
RF RE RD	RAM addressing: set T/F on address lines D-F to select starting address
RAM	"R" position when no RAM used: set "0", "1", "2", "D", "E", or "F" to select memory Segment
J17	"RUN" position connects oscillator to board circuitry: "TEST" position disconnects oscillator
DIG 8 DIG 9 DIG A DIG B DIG C DIG D	Set true or false (T or F) on L-bus address lines 8-D to set the I/O device address.
1 2 4 8 16 32 64 128	Connected for a logic 1 on I/O input lines 0-7 respectively.

TEST POINTS

- TP1 - P5 on input line JA1
- TP2 - Output line 3 on JB1

REV.1	REV.4	REV.7	(PRINTS TO DL102 PRI	ENGINEER	DESCRIPTION OF OPERATION AND APPLICATION DATA
REV.2	JREV.5	ISSUED August 9, 1984		GENERAL ELECTRIC	CPU Expander
REV.3	REV.6	MADE BY E. J. Dillon		COMPANY SALEM, VA	DS3800HXPD CONT.ON SH. 2GA SH.NO. 2FA

I/O AND MOUNTING

The HXPD connects electrically to the L-bus through 80-pin backplane connector, PA. Twenty-pin front-edge connectors, JA and JB, provide logic input and output, respectively.

Table 2*
BACKPLANE I/O SIGNALS (L-BUS)

SIGNAL NAME	PIN NO.	DIRECTION	DEFINITION
DCOM	PA1	Input	Power digital common
OSTALL	PA2	Output	Board stall
P5	PA3	Input	Power +5VDC
lSTALL	PA4	Input	Stall
lBLCRST	PA6	Input	Reset
OINTx	PA8	Output	Interrupt output
OBLDS1	PA10	Input	Segment line 1
OBLCIOW	PA11	Input	I/O write
OBLDSE	PA12	Input	Segment line E
OBLCMR	PA13	Input	Memory read
OBLCIOR	PA14	Input	I/O read
OTC5	PA15	Output	Timer/counter output 5
OBLCMW	PA16	Input	Memory write
OBLDS2	PA17	Input	Segment line 2
OTC1	PA18	Output	Timer/counter output 1
OBLDSD	PA19	Input	Segment line D
OBLCDAK	PA20	Output	Device acknowledge
lC5	PA21	Input	Timer clock 5
lG3	PA22	Input	Timer gate 3
lC4	PA23	Input	Timer clock 4
lC3	PA24	Input	Timer clock 3
lCLK	PA25	Input	Timer clock for timer inputs 0, 1, & 2
OBLHBE	PA26	Input	High byte enable
OBLDSF	PA27	Input	Segment line F
OBLCIAK	PA28	Input	I/O acknowledge
OBLDS0	PA29	Input	Segment line 0
lG5	PA30	Input	Timer gate 5
OBLAA	PA31	Input	Address line A
lG4	PA32	Input	Timer gate 4
OBLAD	PA33	Input	Address line D
OBLAE	PA34	Input	Address line E

* Table 2 is continued on the following page.

Table 2 (continued)
BACKPLANE I/O SIGNALS (L-BUS)

SIGNAL NAME	PIN NO.	DIRECTION	DEFINITION
OBLAB	PA35	I Input	Address line B
OBLA8	PA36	Input	Address line 8
OTC2	PA37	Output	Timer/counter output 2
OBLAF	PA38	I Input	Address line F
OBLDF	PA39	Bidir.	I Data line F
OBLA9	PA40	Input	Address line 9
OBLD8	PA41	I Bidir.	Data line 8
OBLAC	PA42	Input	Address line C
DCOM	PA43	Input	Power digital common
OTC4	PA44	Output	I Timer/counter output 4
P5	PA45	I Input	Power +5VDC
OBLDE	PA46	Bidir.	I Data line E
OBLDA	PA47	Bidir.	I Data line A
OBLD9	PA48	Bidir.	I Data line 9
OBLDD	PA49	Bidir.	I Data line D
OBLDB	PA50	I Bidir.	I Data line B
OBLDC	PA51	I Bidir.	I Data line C
OBLAC	PA52	I Input	I Address line 6
OBLA5	PA53	I Input	I Address line 5
IBLLBE	PA54	I Input	I Low byte enable
OTC3	PA55	Output	Timer/counter-output 3
OBLA1	PA56	Input	Address line 1
OBLA7	PA57	I Input	Address line 7
OBLA3	PA58	I Input	Address line 3
OBLA4	PA59	Input	Address line 4
OBLA2	PA60	Input	Address line 2
OBLD5	PA61	I Bidir.	Data line 5
OINT4	PA62	Input	Interrupt 4
OINT5	PA63	Input	Interrupt 5
OBLD3	PA64	Bidir.	Data line 3
OINT6	PA65	I Input	Interrupt 6
OBLD1	PA66	I Bidir.	I Data line 1
OINT7	PA67	Input	I Interrupt 7
OBLDO	PA68	Bidir.	Data line 0
OINTZ	PA69	I Input	Interrupt 2
OBLD7	PA70	I Bidir.	Data line 7
OINT3	PA71	Input	Interrupt 3
OBLD6	PA72	I Bidir.	I Data line 6
OBLD2	PA74	Bidir.	Data line 2
OBLD4	PA76	I Bidir.	Data line 4
OINTO	PA78	Input	Interrupt 0
P5	PA77	Input	Power +5VDC
DCOM	PA79	I Input	I Power digital common
OINT1	PA80	Input	Interrupt 1

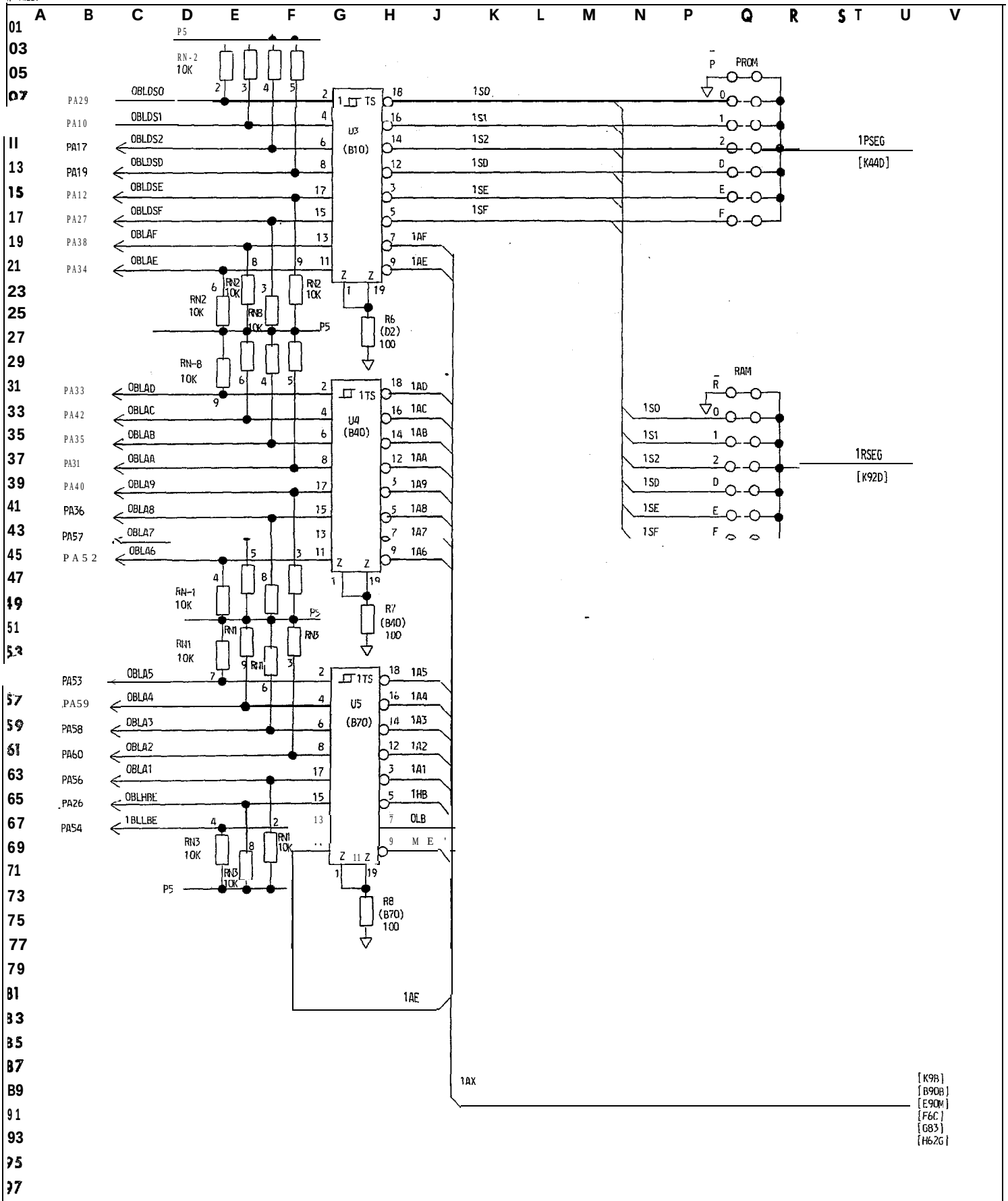
REV.1	(REV.4	REV.7	[PRINTS TO DL102 PRI	(ENGINEER EQ	DESCRIPTION OF OPERATION AND APPLICATION DATA
REV.2	REV.5	ISSUED August 9, 1984		GENERAL ELECTRIC I	CPU Expander
REV.3	REV.6	MADE BY E. J. Dillon		COMPANY SALEM,VA	DS3800HXPD CONT.ON Sfi. 2JA SH.NO. 2HA

Table 3
I/O CONNECTOR SIGNALS

SIGNAL NAME	CONN. PIN #	DIRECTION	DEFINITION
P5	JA1	Output	Power +5VDC
DCOM	JA2	Input	Power digital common
lLIN4	JA3	Input	Input line 4
lLIN5	JA4	Input	Input line 5
lLIN3	JA5	Input	Input line 3
lLIN6	JA6	Input	Input line 6
lLIN2	JA7	Input	Input line 2
lLIN7	JA8	Input	Input line 7
lLIN0	JA9	Input	Input line 0
lLIN1	JA10	Input	Input line 1
P5	JA11	Output	Power +5VDC
DCOM	JA12	Input	Power digital common
lLIN9	JA13	Input	Input line 9
lLIN8	JA14	Input	Input line 8
lLINF	JA15	Input	Input line F
lLINA	JA16	Input	Input line A
lLINE	JA17	Input	Input line E
lLINB	JA18	Input	Input line B
lLIND	JA19	Input	Input line D
lLINC	JA20	Input	Input line C
lLOUT3	JB1	Output	Output line 3
DCOM	JB2	Output	Power digital common
lLOUT4	JB3	Output	Output line 7
lLOUT5	JB4	Output	Output line 2
lLOUT3	JB5	Output	Output line 5
lLOUT6	JB6	Output	Output line 6
lLOUT2	JB7	Output	Output line 1
lLOUT7	JB8	Output	Output line 4
lLOUT0	JB9	Output	Output line 8
lLOUT1	JB10	Output	Output line 0
lLOUTC	JB11	Output	Output line C
lLOUT9	JB12	Output	Output line 9
lLOUT9	JB13	Output	Output line E
lLOUT8	JB14	Output	Output line D
lLOUTF	JB15	Output	Output line B
lLOUTA	JB16	Output	Output line F
lLOUTB	JB18	Output	Output line A

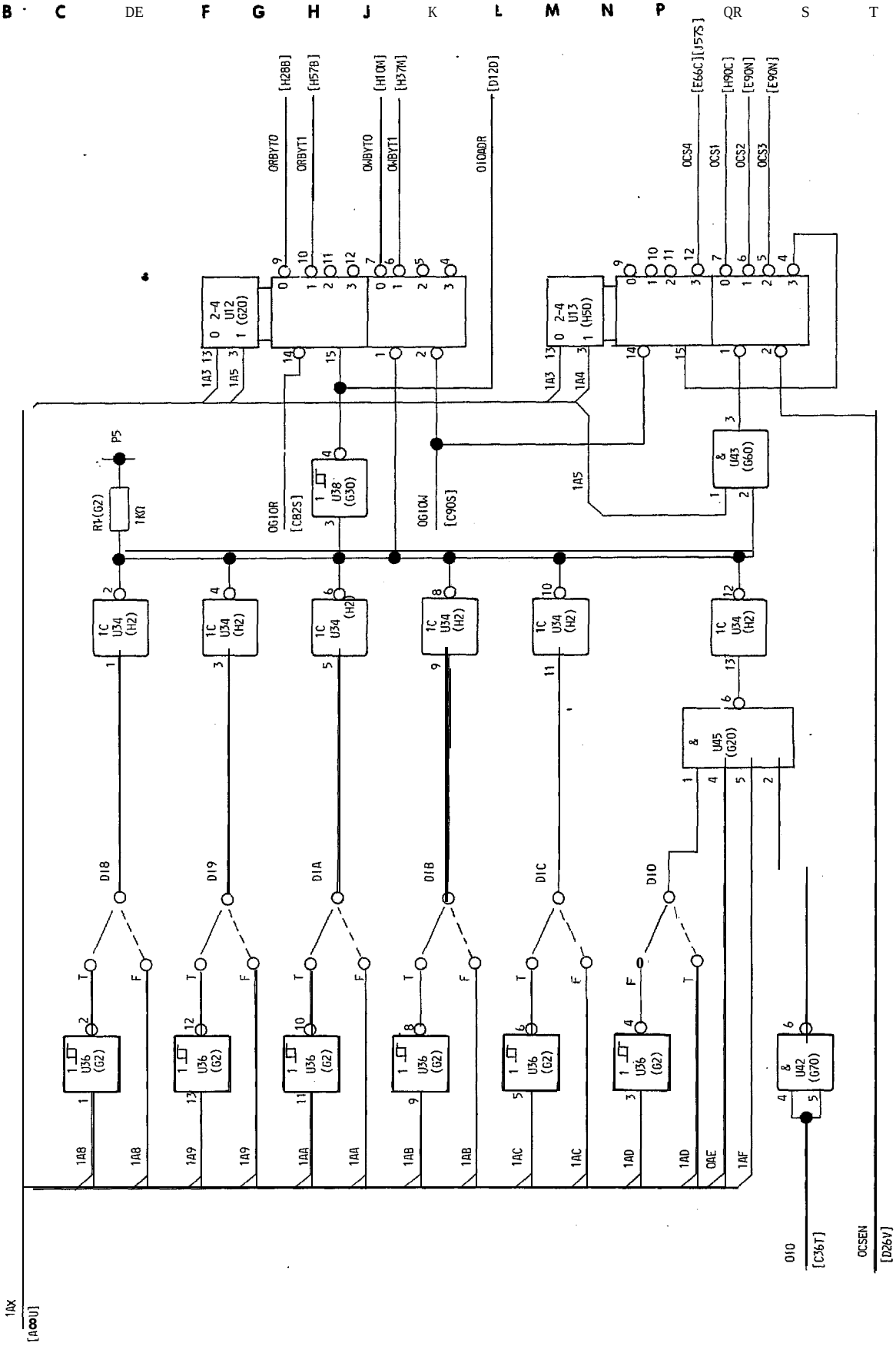
OPERATING AND STORAGE ENVIRONMENT

See GEK-61680.



1	REV. 4	REV. 5	PRINTS TO DL104 PRI	ENGINEER EO	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.
REV. 2	ISSUED 1-21-83				ELEMENTARY DIAGRAM CPU EXPANDER
REV.3	MADE BY E. DILLON				DS3800HXPD CONT. ON SH. 4BA SH. NO. 4AA

AF 10527



.1	REV. 4	REV.5	PRINTS TO DL104 PRI	ENGINEER ED	GENERAL@ ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.
REV.2	ISSUED 1-21-83				
REV. 3	MADE BY E. DILLON				

ELEMENTARY DIAGRAM

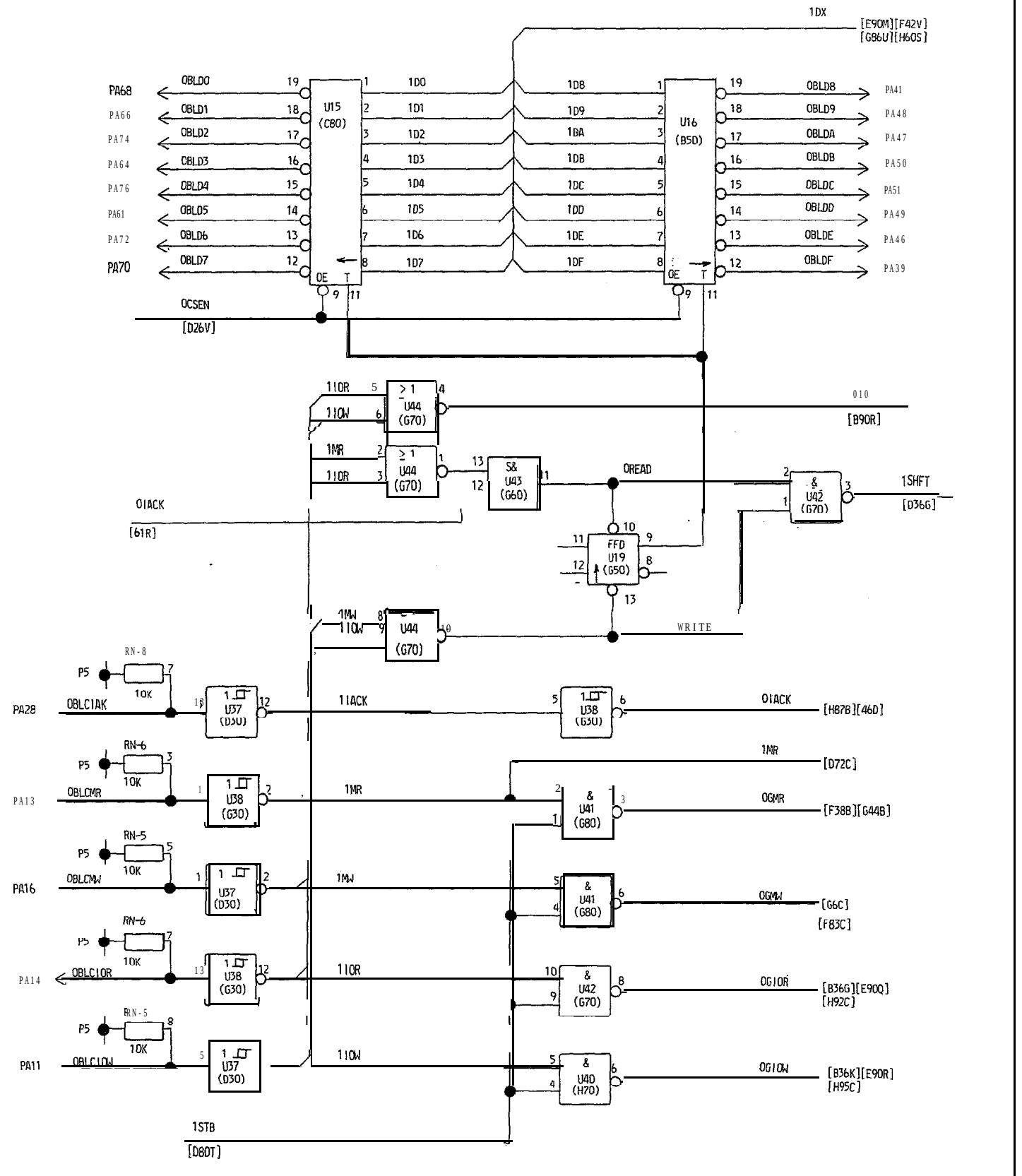
CPU EXPANDER

DS3800HXPD

CONT. ON SH. 4CA SH. NO. 461

DSD 8152 (09 76)

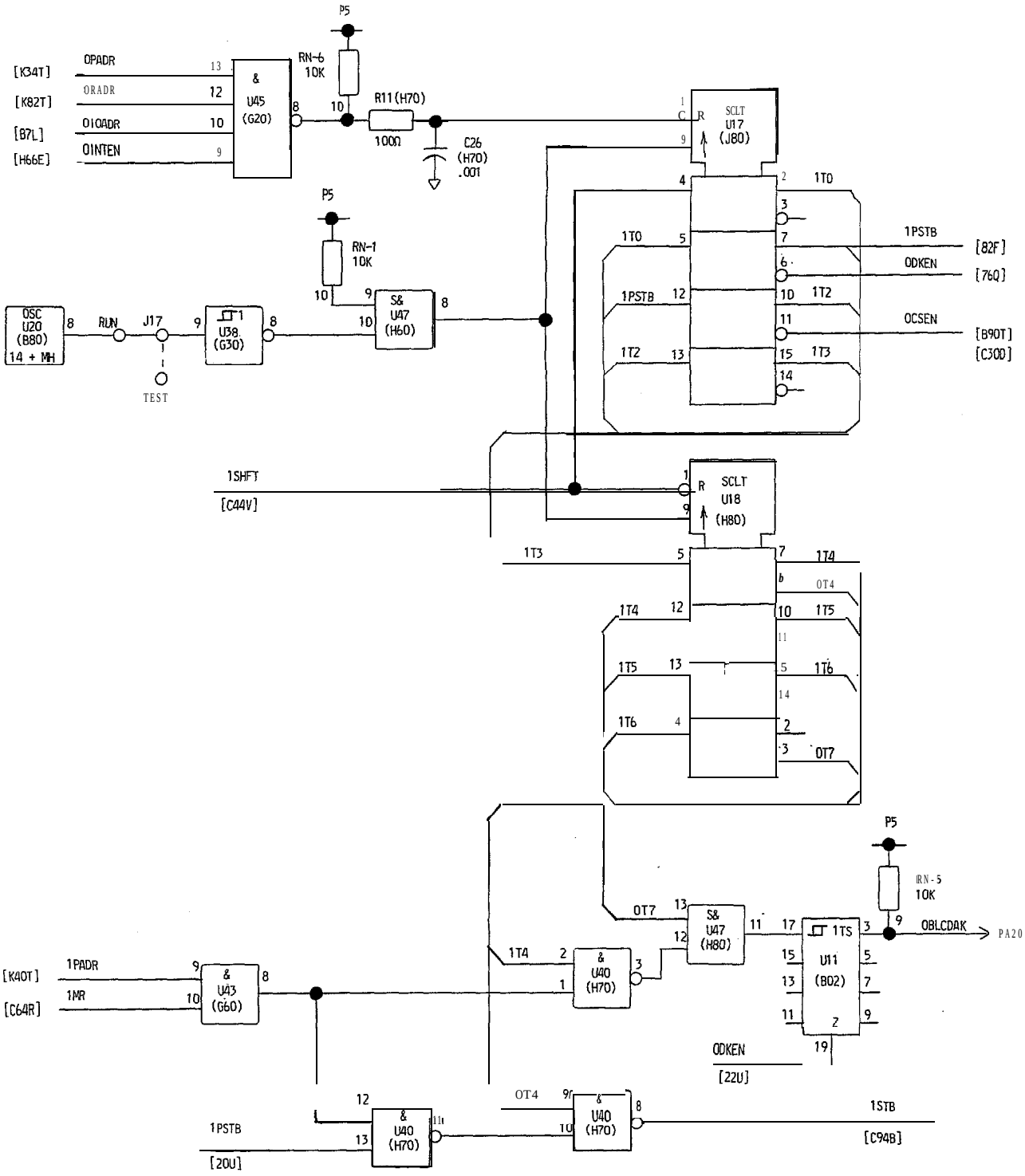
AF 10527



1	REV. 4	REV.5	PRINTS TO DL104 PRI	ENGINEER ED	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM CPU EXPANDER
REV.2	ISSUED 1-21-83					DS3800HXPD
REV. 3	MADE BY E. DILLON					CONT. ON SH. 4DA SH. NO. 4CA

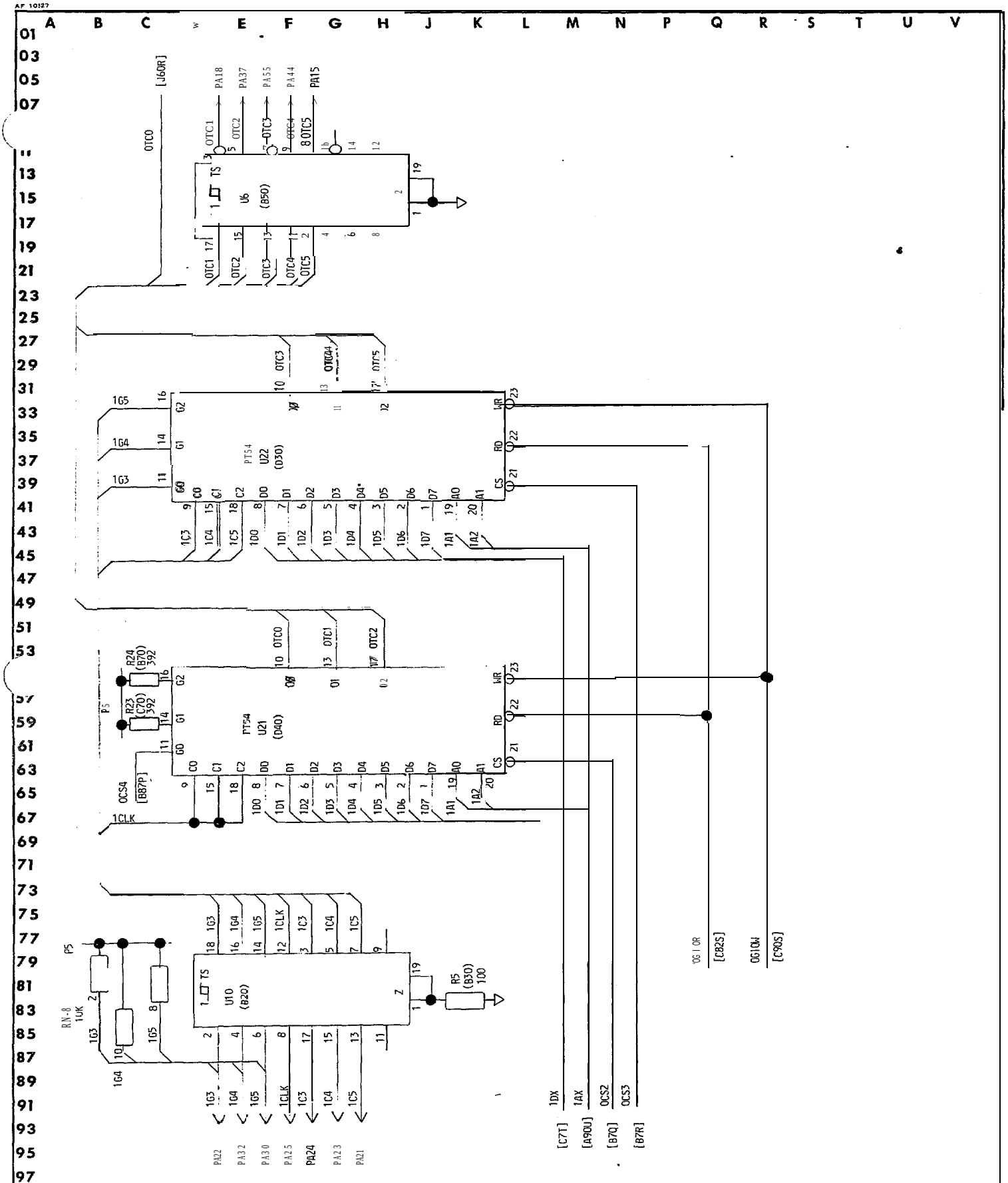
DSD 0152 [09 76]

AF 10527



1	REV. 4	REV. 5	PRINTS TO DL104 PRI	ENGINEER EL	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED 1-21-83				DRIVE SYSTEMS DEPARTMENT	CPU EXPANDER
REV. 3	MADE BY EL E. DILLON				SALEM, VA. U.S.A.	DS3800HXPD
						CONT. ON SHAEA SH. NO. 4DA

DSD 015Z (09-76)



REV. 4	REV. 5	PRINTS TO	ENGINEER
		DL104 PRI	ED

REV. 2	ISSUED
	1-21-83

REV. 3	MADE BY
	E. DILLON

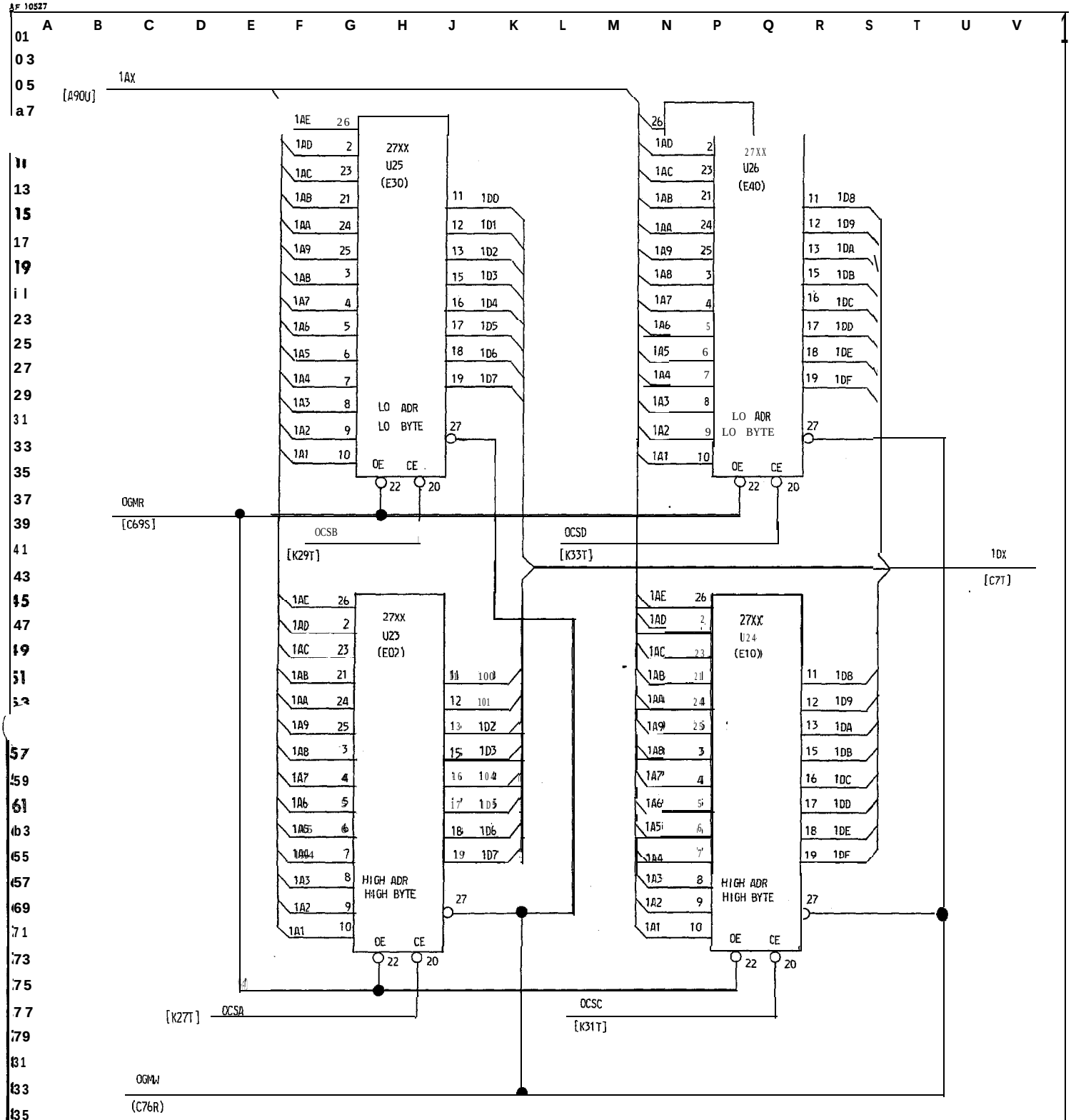
GENERAL ELECTRIC
DRIVE SYSTEMS DEPARTMENT
SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM
CPU EXPANDER

DS3800HXPD

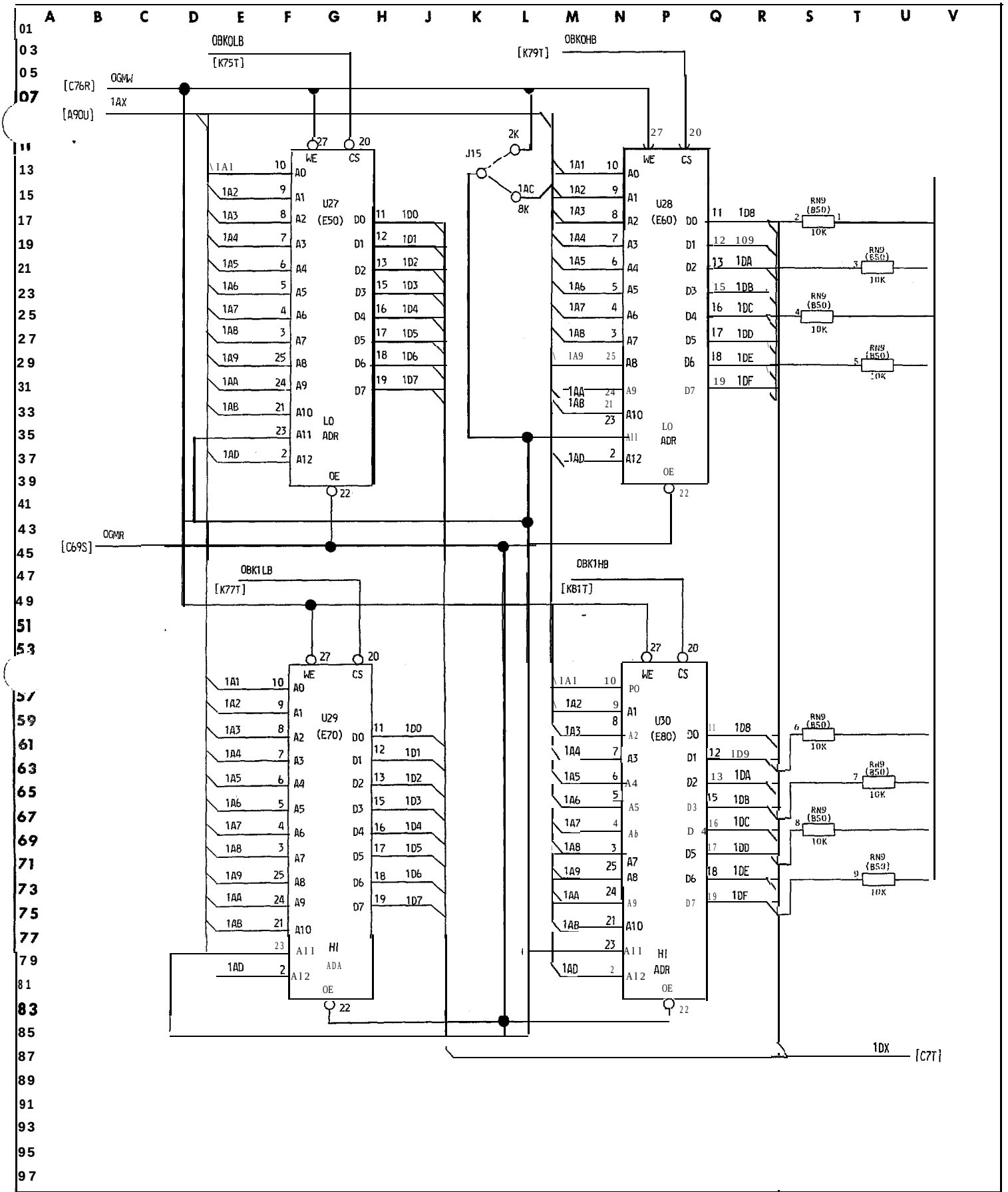
CONT ON SH 4FA SH NO 4FA

AF 10527

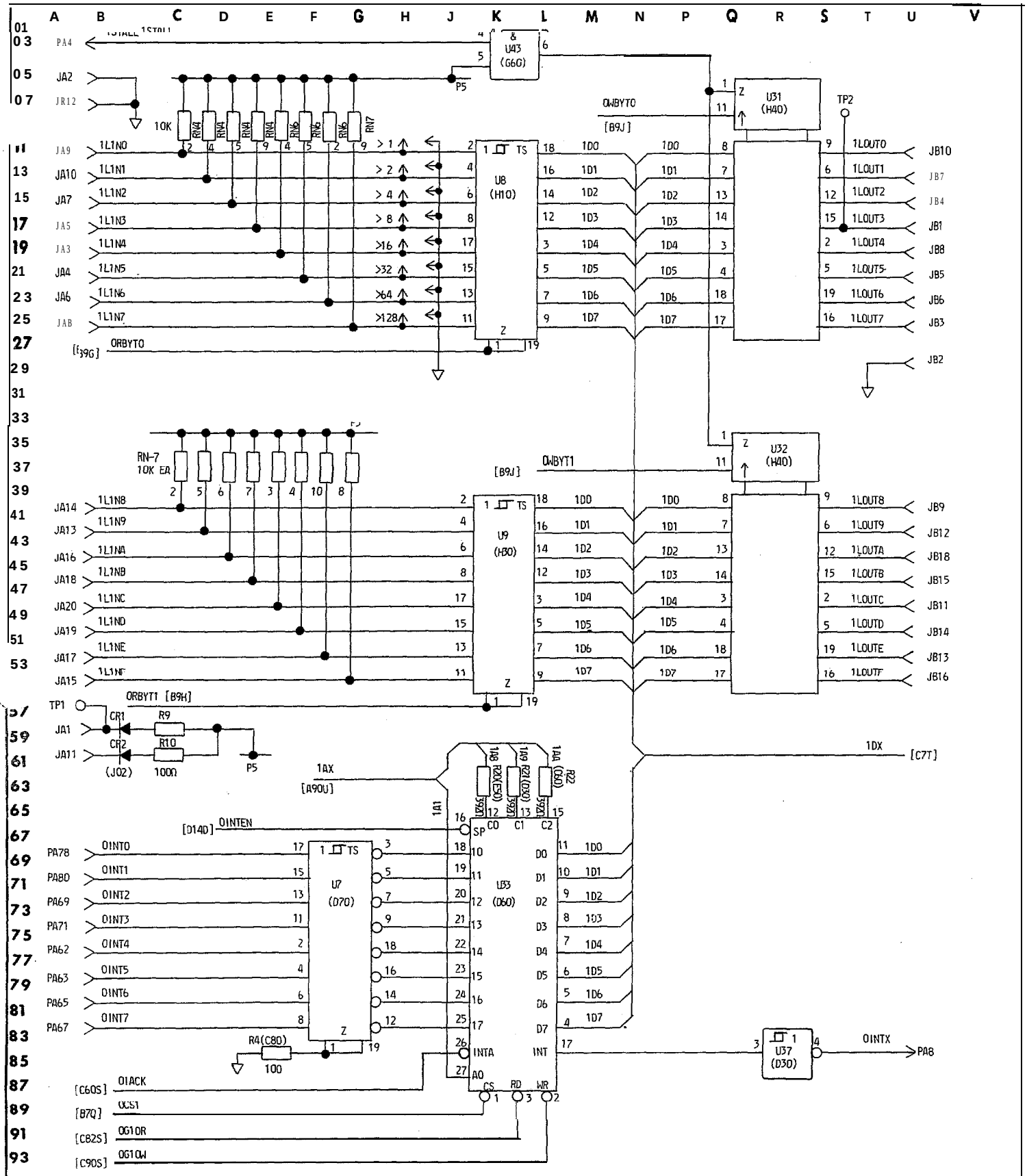


1	REV. 4	REV. 5	PRINTS TO DL104 PRI	ENGINEER ED	GENERAL@ ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED 1-21-83				DRIVE SYSTEMS DEPARTMENT	CPU EXPANDER
REV. 3	MADE BY RL E. DILLON				SALEM, VA. U.S.A.	DS3800HXPD
						CONT. ON SH. 4GA SH. NO. 4FA

DSD 8152 (09 76)



1 DEN 1-89 MAC	REV.4	REV.5	PRINTS TO DL104 PRI	ENGINEER EQ	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.
REV. 2 DEP 5/15/8 3 MAC	ISSUED 1-21-83				ELEMENTARY DIAGRAM CPU EXPANDER DS3800HXPD
REV. 3	MADE BY E. DILLON				CONT. ON SH. 4HA SH. NOW



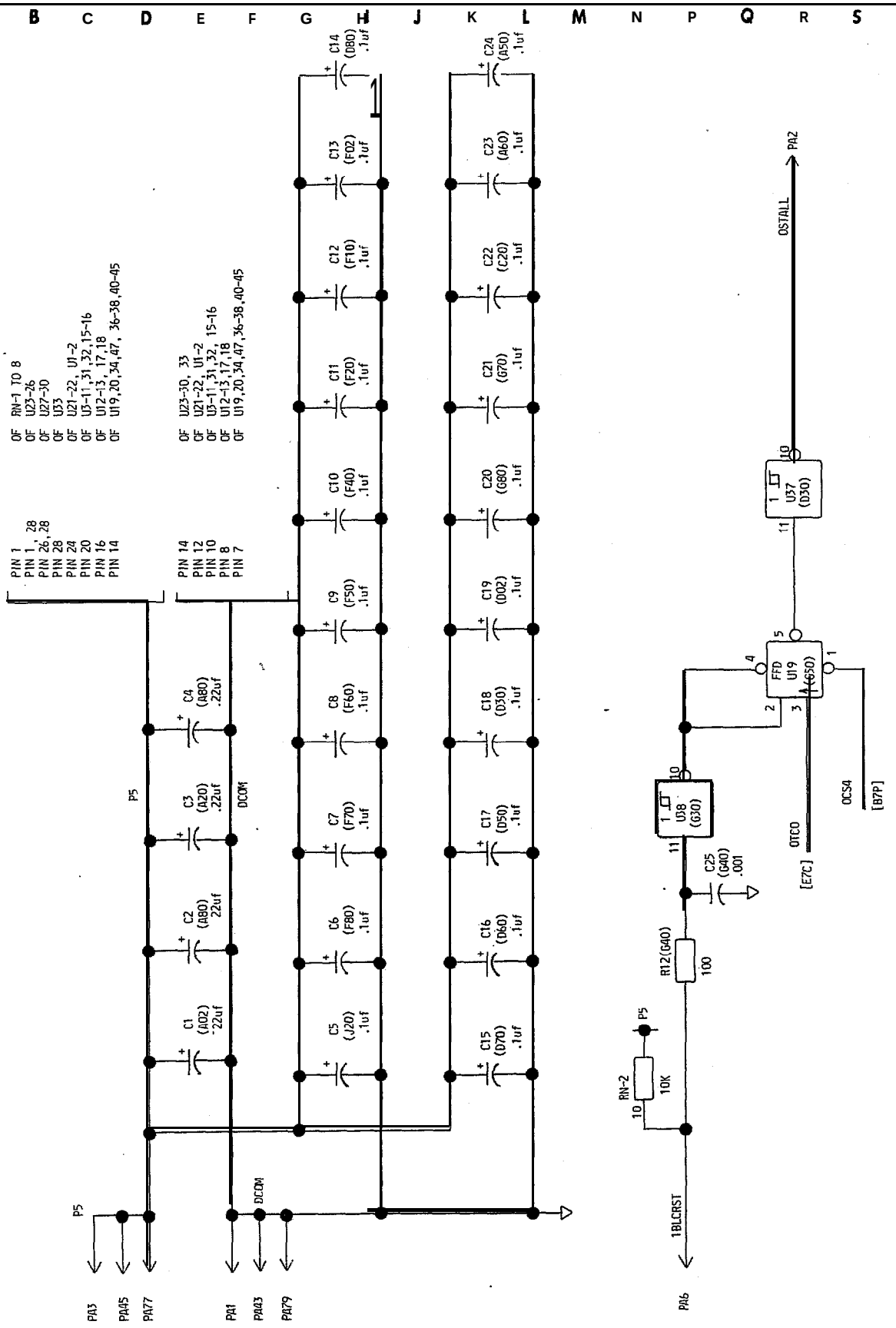
V.1	REV.4	REV. 5	PRINTS TO DL104 PRI	ENGINEER
REV.2	ISSUED 1-21-83			
REV. 3	MADE BY E. DILLON			

GENERAL ELECTRIC
DRIVE SYSTEMS DEPARTMENT
SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM
CPU EXPANDER
DS3800HXPD
CONT. ON SH. 4JA SH. NO 4HA

DSD 0152 (09 76)

F 10527



1	REV. 4	REV. 5	PRINTS TO 2536 DL104 PR1	ENGINEER EO	GENERAL@ ELECTRIC
REV. 2	ISSUED 1-21-83				DRIVE SYSTEMS DEPARTMENT
REV. 3	MADE BY E. DILLON				SALEM, VA. U.S.A.

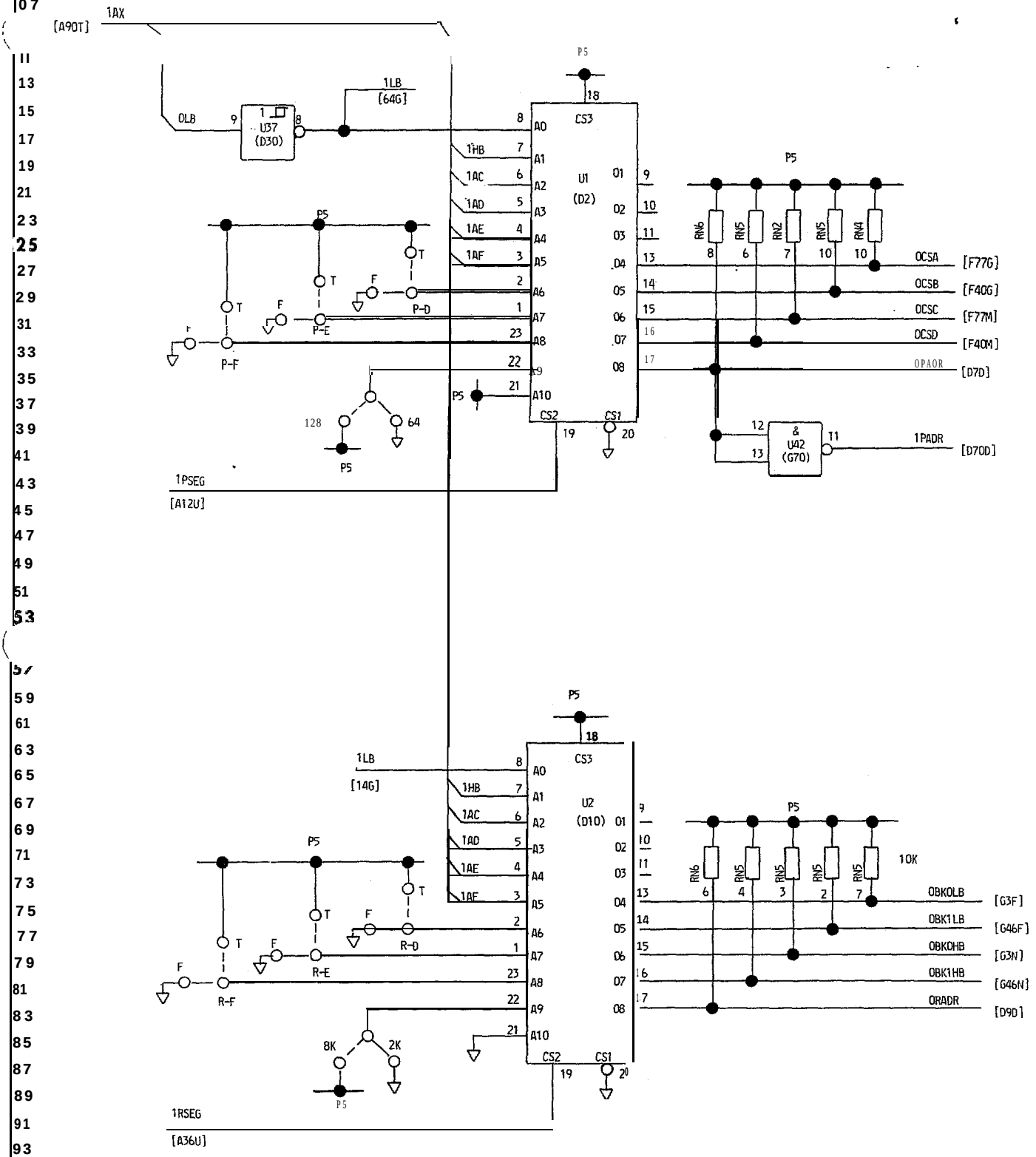
ELEMENTARY DIAGRAM

CPU EXPANDER

DS3800HXPD

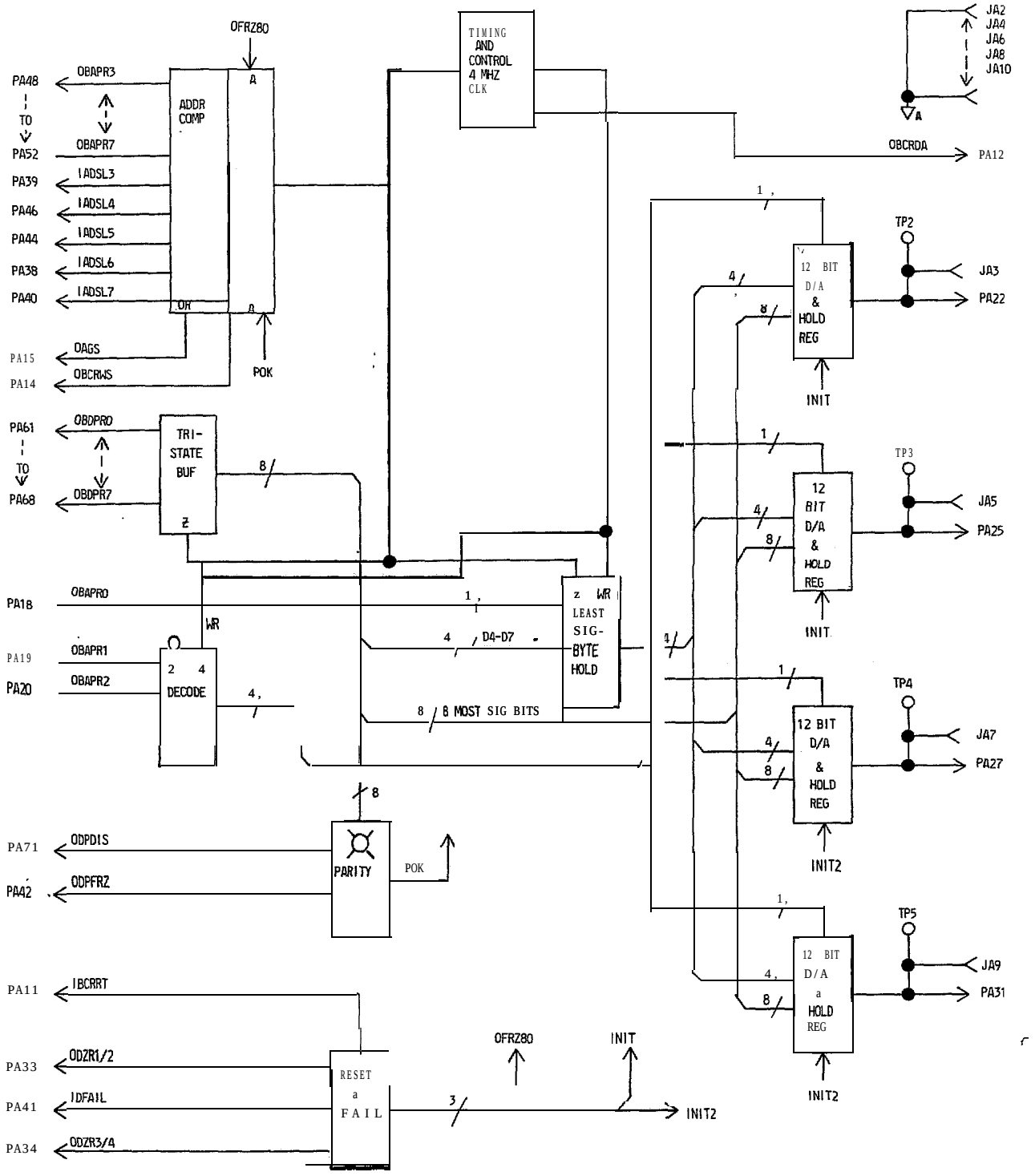
CONT. ON SH. 4KA SH. NO. 4JA

SD 0152 [09 76]



1	REV.4	REV.5	PRINTS TO DL104	ENGINEER EQ	GENERAL@ ELECTRIC	ELEMENTARY DIAGRAM
REV.2	ISSUED 1-21-83				DRIVE SYSTEMS DEPARTMENT	CPU EXPANDER
REV.3	MADE BY E. DILLON				SALEM, VA. U.S.A.	DS3800HXPD
						CONT. ON SH. FL. SH. NO AKA

DSD 0152 (09 16)



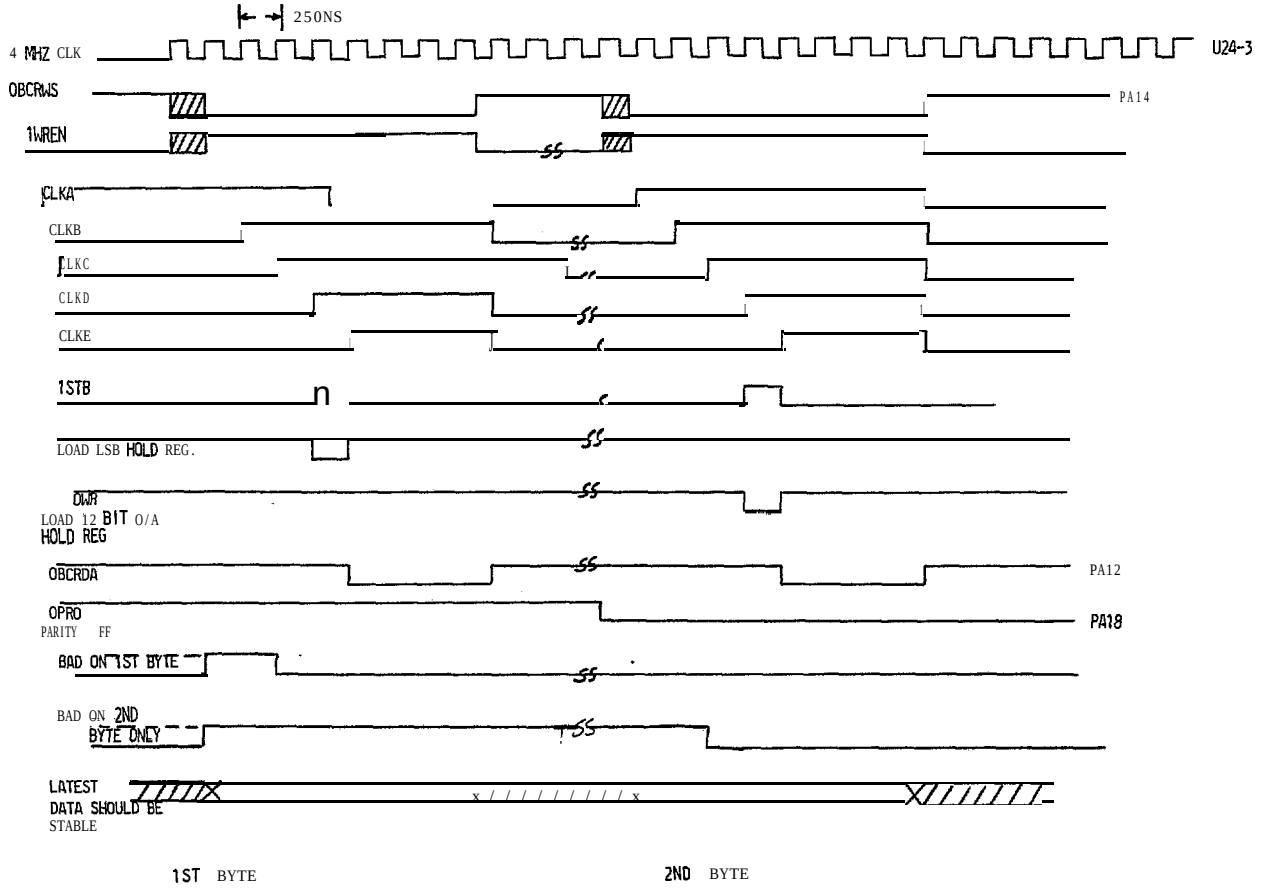
REV. 1	REV. 4	REV.5	PRINTS TO PRI 53-36, TRO DL102 2520	ENGINEER 9/12/77	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.
REV. 2	ISSUED 9/12/77				
REV.3	MADE BY E. BOOTE				

ELEMENTARY DIAGRAM
ANALOG OUTPUT
BLOCK DIAGRAM

D S 3 8 0 O N D A C

CONT. ON SH. 2AB SH. NO. 2AA

DSD 0152 (09-76)



IEV. 1	REV. 4	REV. 5	PRINTS TO PRI 53-3G, TRO DL1 02 2520	ENGINEER MB 9/12/77	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM ANALOG OUTPUT TIMING DIAGRAM
IEV. 2	ISSUED 9/12/77					DS3800NDAC
IEV.3	MADE BY E. BOOTE					CONT. ON SH. 2BA SH. NO. 2AB

DD152 (09-76)

2.0 DEFINITION AND PURPOSE

THIS BOARD IS AN ANALOG OUTPUT BOARD WITH 4 ADDRESSABLE OUTPUTS. EACH OUTPUT WILL BE LOADED BY SENDING TWO CONSECUTIVE BYTES OF DATA, TO THE BOARD. THE 1ST BYTE TO BE THE LEASE SIGNIFICANT 4 BITS REQUIRED BY THE D/A, THE 2ND BYTE TO BE THE 8 MOST SIGNIFICANT BITS REQUIRED BY THE D/A.

OUTPUTS ARE BIPOLAR ±10V MAX.

2.1 THEORY OF OPERATION

REFERENCES ARE TO BLOCK DIAGRAM ON 2AA

2.1.1 ADDRESSING

2.1.1.1 8080 ADDRESSING

8080 CARD SELECT USES DBAPR3-7 ANDED WITH A 1K SELECT LINE ODAA0 (SEE 2AA)

OBAPR3 - OBAPR7 ARE ADDRESS LINES WHICH ARE COMPARED TO SELECT LINES IDASL3-7' THUS WHEN A IDASL LINE IS OPEN THE CORRESPONDING OBAPR LINE MUST BE TRUE ="0" FOR CARD TO BE SELECTED.

ODAA0 IS CONNECTED TO AN 8080, 1K ADDRESS LINE, AND MUST COME TRUE ("0") TO ENABLE THE CARD. SELECTION OF THE 4 OUTPUTS ON EACH CARD IS DONE BY USE OF OBAPR0-2. THESE LINES ARE TAKEN TO A 2 → 4 DECODE [A53E] AND DETERMINE THE 12 BIT HOLD REGISTERS TO BE LOADED. THESE REGISTERS CONSIST OF 3-4 BIT REGISTERS.

2.1.1.2 200 LEVEL ADDRESSING

GROUP SELECT LINES OAGS [A31D] IS OR'D WITH OBAPR3-7 TO ENABLE CARD SELECTION. THUS EITHER 8080 ADDRESS OR GROUP SELECT WILL ENABLE THE CARD. CARD SELECT IS THEN ANDED WITH OBCRWS TO CAUSE WREN [A19H].

REV. 1	REV. 4	REV. 7	PRINTS TO DL102, T... 53-36 2520 ENGINEER PRH 9/12/77	GENERAL@ ELECTRIC	ANALOG OUTPUT
REV. 2	REV. 5	ISSUED 12/14/77		DRIVE SYSTEMS DEPT.	DS3800NDAC
REV. 3	REV. 6	MADE BY E. BOOTE		SALEM. VA. U.S.A.	CONT. ON SH. 2CA SH. NO. 2BA

2.1.2 STROBE GENERATION

IN ORDER TO FREE THE BOARD FROM DEPENOANCE ON WRITE INPUT TIMING A STROBE GENERATOR HAS BEEN USED. [A11L].
THIS GENERATOR IS ENABLED WITH WREN AND IS CLOCKED FROM AN INTERNAL 8MHZ CLOCK, DIVIDED DOWN TO 4 MHZ.
THE TIMING IS SHOWN ON 2AB.
THERE ARE 5 CLOCK OUTPUTS WHICH ARE USED TO TIME DIFFERENT EVENTS.

PARITY FF IS SET OK WHEN CLOCK A RISES AND THE TRI - STATE'DRIVERS ARE ENABLED AT THIS TIME. DATA AND PARITY CHECKING HAVE UNTIL THE RISE OF CLOCK C TO SETTLE. If PARITY IS BAD THE FF IS SET AT CLKE TIME. A FURTHER CLOCK CYCLE IS ALLOWED BEFORE THE STROBE IS. GENERATED. THE TRAILING EDGE OF THE STROBE WILL CLOCK DATA INTO THE LSB LSB HOLD REG [A49M] IF ADDRESS IS ODD OR INTO ONE OF THE 12 BIT HDLD REG'S IF ADDRESS IS EVEN. WHEN THE ADDRESS IS EVEN THE LSB HOLD REG CONTENTS ARE ALSO CLOCKED INTO THE 12 BIT HOLD REG.

OBCRDA IS SET LOW AT THE END OF STROBE TIME AND RISES WHEN WRITE STROBE IS REMOVED.
CLOCK A AND D GO FALSE ON THE 1ST CLOCK RISE AFTER THE WRITE STROBE IS REMOVED. LOADING OF THE 12 BIT HOLD REG'S. ARE THUS BUFFERIEO IN TIME FROM THE START AND END OF THE WRITE STROBE.

2.1.3 OBCRDA IS INHIBITED BY PARITY FF RESET. SINCE PARITY FF IS RESET BY INITIALIZE, OBCRDA IS ALSO INHIBITED DURING INITIALIZE.
PARITY FF IS RESET (BAD) DURING INITIALIZE AND IS SET (GOOD) AT CLKA OF THE 1ST LOAD.

2.1.4 RESET

ON POWER UP OR A HIGH ON IBCRRT THE HOLD REG. ARE RESET TOGETHER WITH THE TIMING GENERATDR. RESET DF THE HOLD REG. PUTS " 1 "
ON THE D/A'S.

2.1.5 EXTERNAL FAIL IDFAIL (PIN 41)

WHEN THE CARD IS USED ON 200 LEVEL THIS PIN IS GROUNEO AND THUSD I SABLED.

WHEN THE CARD IS USED ON 8080 BUS THIS INPUT WILL CAUSE'THE FOLLOWING CONDITIONS DEPENDING ON STRAPPING OF ODZR1/2, ODZR3/4

REFER. TO 2AA

A ODZR1/2 O P E N

ODZR1/2 [A80DC] WILL INHIBIT IDFAIL FROM ENERGIZING INIT1, [A78N] HOWEVER, IDFAIL WILL SET OFRZ80 AND THIS INHIBITS WREN [A19H] PREVENTING WRITING OF DATA INTO THE HOLD REG'S.

B/ ODZR1/2 G R O U N D E D

WHEN IDFAIL GOES HIGH INIT1 IS ENABLED AND THIS RESETS 12 BIT HOLD REG. 1 AND 2 [AA23&39R]. CAUSING " - 1 " ON THE 1 & 2 D/A'S.

C/ ODZR3/4 THIS INPUT-PERFORMS THE SAME FUNCTIONS AS ODZR1/2 EXCEPT VIA INIT2 AND THE OUTPUTS AFFECTED ARE 3 & 4.

REV. 1	REV. 4	REV. 7	PRINTS TO DL102, TRO 53-36 2520	'ENGINEER	GENERAL ELECTRIC DR I VE SYSTEMS DEPT. SALEM, VA. U.S.A.	ANALOG OUTPUT
REV. 2	REV, 5'	ISSUED 2/14/77 PRI				DS 3 8 0 0 N D A C
REV. 3	REV. 6	MADE BY E. 8007-E				CONT. ON SH. 2EA SH. NO. 2DA

2.3 FUNCTIONAL DESCRIPTIONS

2.3.1 INPUTS

2.3.1.1 OBDPR0 - OBDPR7

INPUT DATA WILL BE IN BYTE FORMAT. THE 1ST BYTE TO BE THE LEAST SIGNIFICANT 4 BITS OF DATA. THE 2ND BYTE TO BE THE 8 BITS OF MOST SIGNIFICANT DATA. THE LOWER 4 BITS OF THE LEAST SIGNIFICANT BYTE WILL BE DISSOLVED.

2.3.1.2 OBAPR3 - OBAPR7

8080 ADDRESS BUS FOR CARD SELECT

IDASL3 - IDASL7

SELECT LINES WHICH ARE TIED HIGH TO SELECT THE CORRESPONDING ADDRESS LINE TRUE.

THESE LINES ARE NOT CONN FOR 200 LEVEL WHICH 8080 ADDRESS LINES MUST BE TRUE FOR CARD SELECT ADDRESS. LEFT OPEN FOR 200 LEVEL ADDRESSING.

2.3.1.3 ODAA0

1K ADDRESS SELECT LINE. THIS LINE MAY BE CONNECTED TO ONE OF THE 8080 ADDRESS LINES 8 THRU 13 OR TO AN EXTERNALLY DECODED 8080 ADDRESS LINE. THIS LINE IS NOT CONNECTED FOR 200 LEVEL ADDRESSING.

2.3.1.4 OBAPR2 - OBAPR0

ADDRESS BITS 2 THRU 0 FOR 8080 OR 200 LEVEL ADDRESSING.

2.3.1.5 OAGS

200 LEVEL GROUP SELECT LINE

2.3.1.6 1BCRRT

RESET OR INITIALIZE LINE. THIS INPUT WHEN STROBED HIGH WILL CAUSE THE CARD TO SET ALL OUTPUTS TO ZERO.

2.3.1.7 OBCRWS WRITE STROBE CONNECT TO APPROPRIATE WRITELINE.

2.3.1.8 OBDPRP

PARITY LINE MUST BE SET LOSS BY CONTROLLING BUS WHEN 000 NUMBER OF DATA INPUTS TRUE.
USED ONLY IF PARITY NOT DISABLED.

2.3.1.9 ODPDIS

PARITY DISABLE LINE.
CONNECT TO GROUND TO DISABLE PARITY.

2.3.1.10 DDPFRZ

STRAP LINE TD DETERMINE ACTION IN CASE OF BAD PARITY.
GROUNDING WILL CAUSE ALL OUTPUTS TO REMAIN AT PREVIOUS VALUE
OPEN WILL 'CAUSE ALL OUTPUTS TO ZERO.

2.3.1.11 IDFAIL

8080 BUS LINE GOES HIGH IF COMPUTER FAILS. A HIGH INPUT WILL CAUSE.THE FOLLOWING DEPENDING OF LISTED STRAPPING.

A. ODZR1/2 OPEN

OUTPUTS 1&2 WILL REMAIN AT AS SET BEFORE FAIL.

B. ODZR1/2 GROUNDED

OUTPUTS 1 & 2 WILL GO TO " - 1 " ON FAIL

c. ODZR3/4 OPEN

OUTPUTS 3 & 4 WILL REMAIN AS SET BEFORE FAIL

D. ODZR3/4 GROUNDED

OUTPUTS 3 & 4 WILL GO TO "-1 " ON FAIL

REV. 1	REV. 4	REV. 7	PRINTS TO DL102, TRO 53-36 2528	'ENGINEER 9/12/77	GENERAL ELECTRIC DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.
REV. 2	REV. 5	ISSUED 12/14/72			
REV. 3	REV. 6	MADE BY E. BOOTE			

ANALOG OUTPUT

DS3800NDAC

CONT. ON SH. 2GA SH. NO. 2FA

2.3.2 OUTPUTS

2.3.2. 1 OBCROA

CARD ACKNOWLEDGE GOES LOW WHEN DATA STROBEO INTO OUTPUT REGISTER. IF PARITY ENABLED AND BAD OBCRDA NOT ACTIVATED.

2.3.2.2 AOUT1-AOUT4

ANALOG OUTPUT-S FROM i/A. 5 MA AT + 9.99511 TO -10.000V SHORT CIRCUIT PROTECTED.

2.3.3 PARITY LEO (YEL)

LIT WHEN PARITY GOOD. OUT WHEN PARITY GOES BAD IN EITHER BYTE.

RESET BY NEXT GOOD WRITE.

2.3.4 CARD EDGE TEST LOOPS

TP1	ANALOG GROUND	
TP2	OUTPUT	1
TP3	OUTPUT	2
TP4	OUTPUT	3
TP5	OUTPUT	4

2.4 APPLICATION DATA

THE NOAC MAY BE USED WITH EITHER AN 6080 MICRO. OR ON THE 200 LEVEL BUS.

2.4.1 8080 ADDRESS BUS

ADDRESS BITS OBAPR3 - 7 ARE AVAILABLE FOR THE 8080 BUS. THE CARD ADDRESS IS DETERMINED BY APPROPRIATE IDASL3-7 LINES LEFT OPEN FOR CORRESPONDING OBAPR3-7 LINES REQUIRED TRUE "0".
ODAAO IS A 1K SELECT LINE AND MUST BE TRUE "0" TO SELECT CARD.
MINOR SELECT LINES OBAPR0-2 SELECT APPROPRIATE OUTPUTS 1 THROUGH 4

2.4.2 200 LEVEL ADDRESS BUS

A SINGLE GROUP SELECT LINE OAGS IS USED TO SELECT THE CARD. MINOR SELECT LINES DBAPRD-2 SELECT THE CARD OUTPUTS OUT1 THROUGH OUT4. 2 ADDRESS REQUIRED FOR EACH OUTPUT.

2.4.3 PARITY SELECTION

CONNECT ODPDISTO GROUND IN ORDER TO DISABLE PARITY.

2.4.4 RESET

RESET LINE IBCRRT IS ACTIVE HIGH IT WILL RESET ALL OUTPUTS TO " -1 ".

POWER UP WILL ALSO CAUSE RESET CONDITION.

2.4.5 FAIL (IDFAIL)(8080 ONLY)

ACTIVE HIGH WILL CAUSE THE FOLLOWING DEPENDING ON STRAPPING

A. ODZR1/2 NOT GROUNDED

OUTPUTS 1 & 2 WILL REMAIN AS SET BEFORE FAIL.

B. ODZR1/2 GRDUNDED OUTPUTS 1 & 2 WILL ZERO ON FAIL.

c. ODZR3/4 SAME AS ODZR1/2 EXCEPT OUTPUTS 3 & 4 AFFECTED.

2.6.6 READ WRITE CONTROL

MEMORY MAPPED I/D - 8080, FOR 8080 CONNECT OBCRMW ON HMP1. TO DBERWS ON NDAC.

REV. 1	REV. 4	REV. 7	PRINTS TO PRI DL102, TRO 53-36 2526	ENGINEER 9/12/77	ANALOG OUTPUT
REV. 2	REV. 5	ISSUED 12/14/77		GENERAL ELECTRIC DRIVE SYSTEMS DEPT SALEM, VA. U.S.A.	DS3800NDAC
REV. 3	REV. 6	MADE BY E. BOOTE			CONT. ON SH. 2JA SH. NO. 2HA

I/O ADDRESSING

USE OBCRWS FOR WRITE STROBE

2.4.7 PARITY INDICATION

YEL LEO IS NORMALLY LET. IT WILL GO OUT IF A PARITY OCCURS IN EITHER BYTE OF DATA. THE FIRST BYTE OF A WORD WITH GOOD PARITY WILL RESET THE LAMP.

i.4.B RATING

ACCURACY

± 0.3 0/0 FS FOR 0 TO 70^0C

± 0.2 0/0 FS FOR 15 - 35^0C

SETTLING TIME 4 MICRO SEC TO .01 0/0 F.S.

FOR 20V CHANGE IN OUTPUT

SLEW RATE 20V/MICROSEC

OUTPUT

5MA AT +9.9951V TO -10.000V SHORT CIRCUIT' PROTECTED

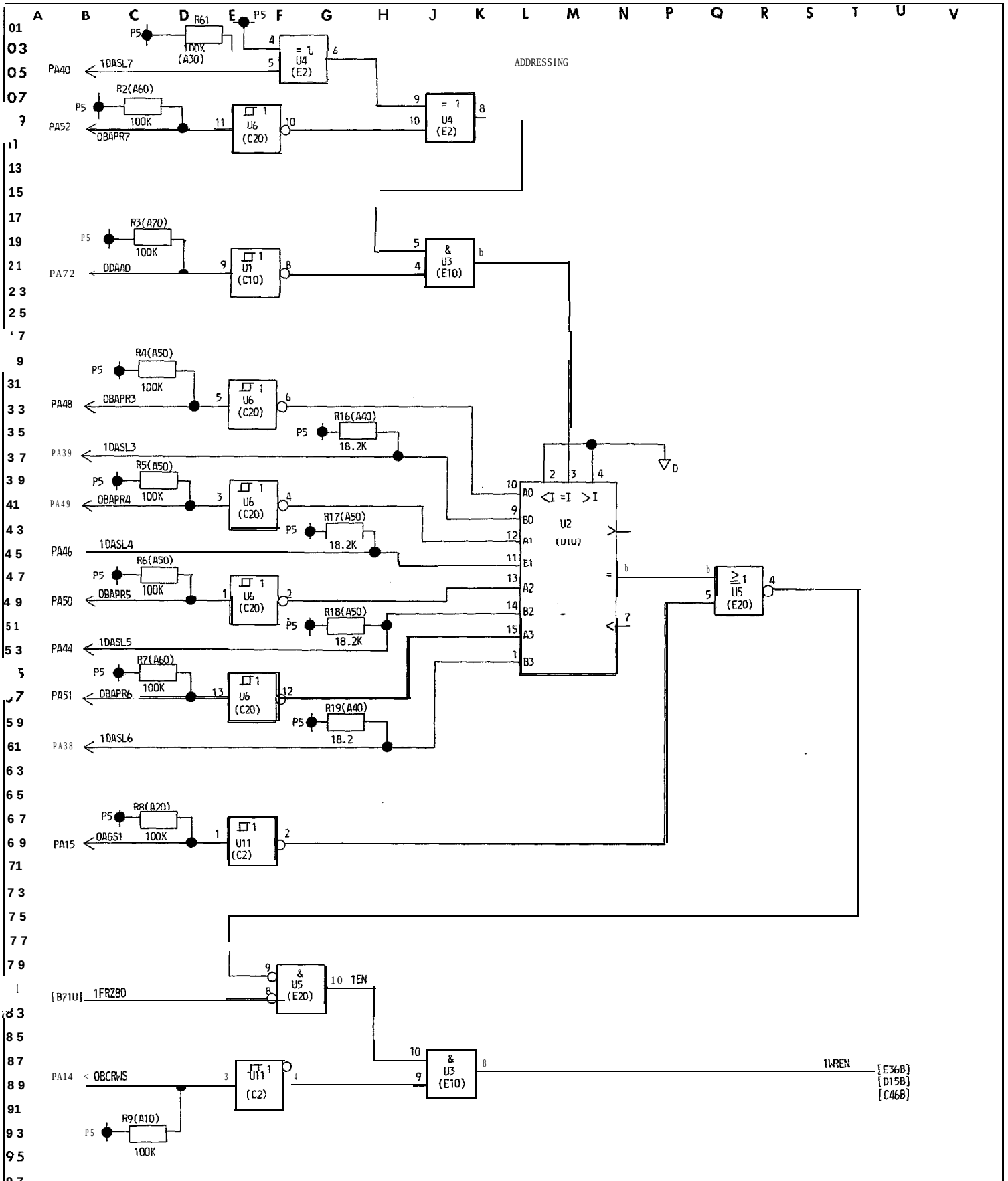
POWER +5V - 0.55A
+15V - 0.16A
-15V - 0.16A

HEAT DISSIPATION 5 WATTS

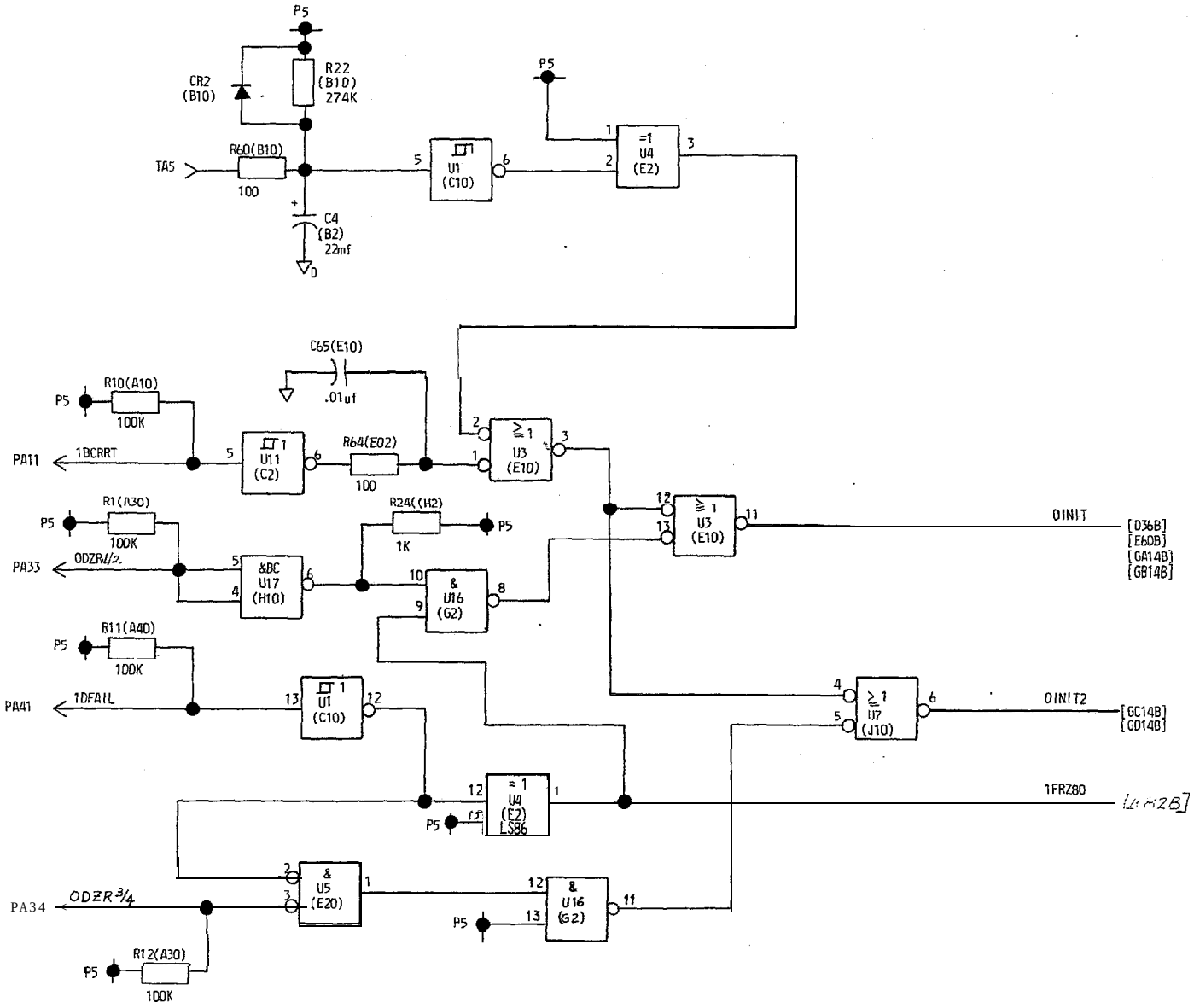
2.4.9 INPUT CODE IN HEX.

OUTPUT VOLTS		CARD INPUT	O/A INPUT	COMPUTER CODE
0.000V	0	FFF	7FF	000
-10.000	-FS	7FF	FFF	800
+9.9951	+FS	800	0DD	7FF
-4. BBMV	-LSB	000	8DD	FFF
+4.88MV	+LSB	FFE	7FE	001

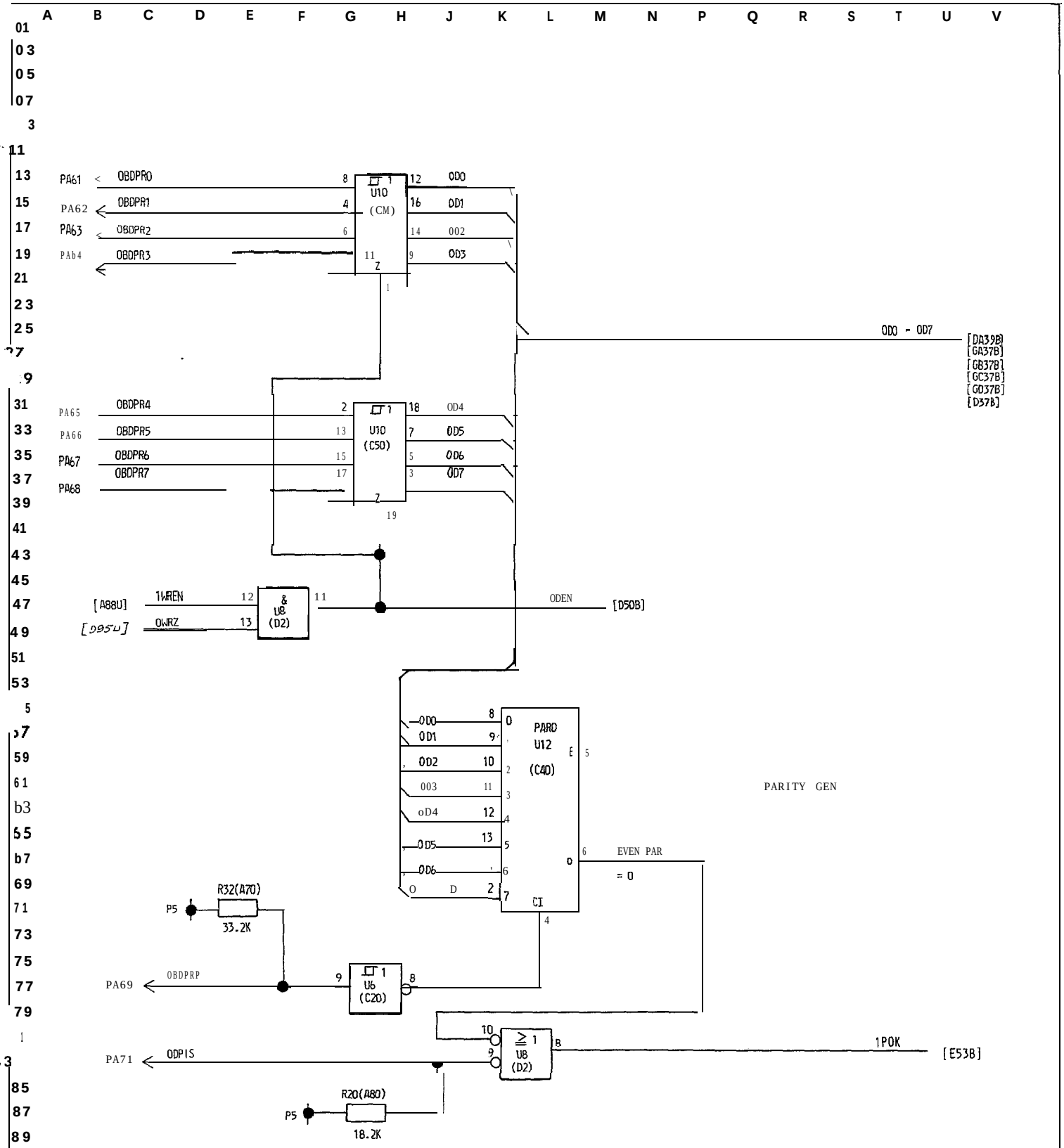
REV. 1 OEE CWS 1/25/79	REV. 4	REV. 7	PRINTS TO PRI DL102, TRO 53-36 2520	ENGINEER MB 9/14/77	GENERAL ELECTRIC DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	ANALOG OUTPUT
REV. 2	REV. 5	ISSUED December 12, 1977				DS 3800NDAC
REV. 3	REV. 6	MADE BY E. BOOTE				CONT. ON SH. F L SH. NO. 2JA



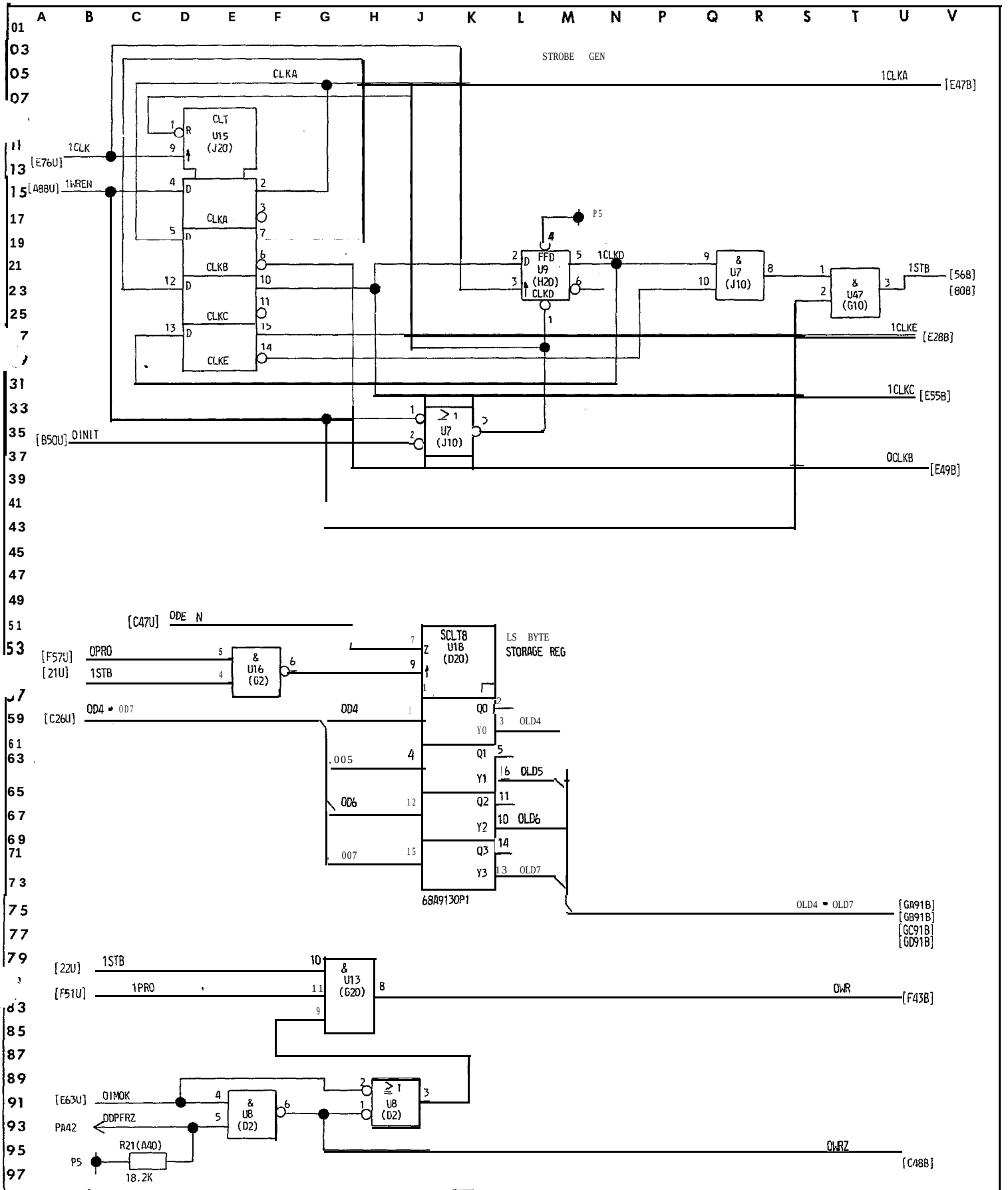
REV. 1	REV 4	REV.5	PRINTS TO PRI 53-36, TRO, DL104	ENGINEER 9/12/77	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED 9/12/77			DOM PLUS	DRIVE SYSTEMS DEPARTMENT	ANALOG OUTPUT
REV.3	MADE BY E. BOOTE				SALEM, VA. U.S.A.	DS380DNDAC
						CONT. ON SH ABA SH. NO. 4AA



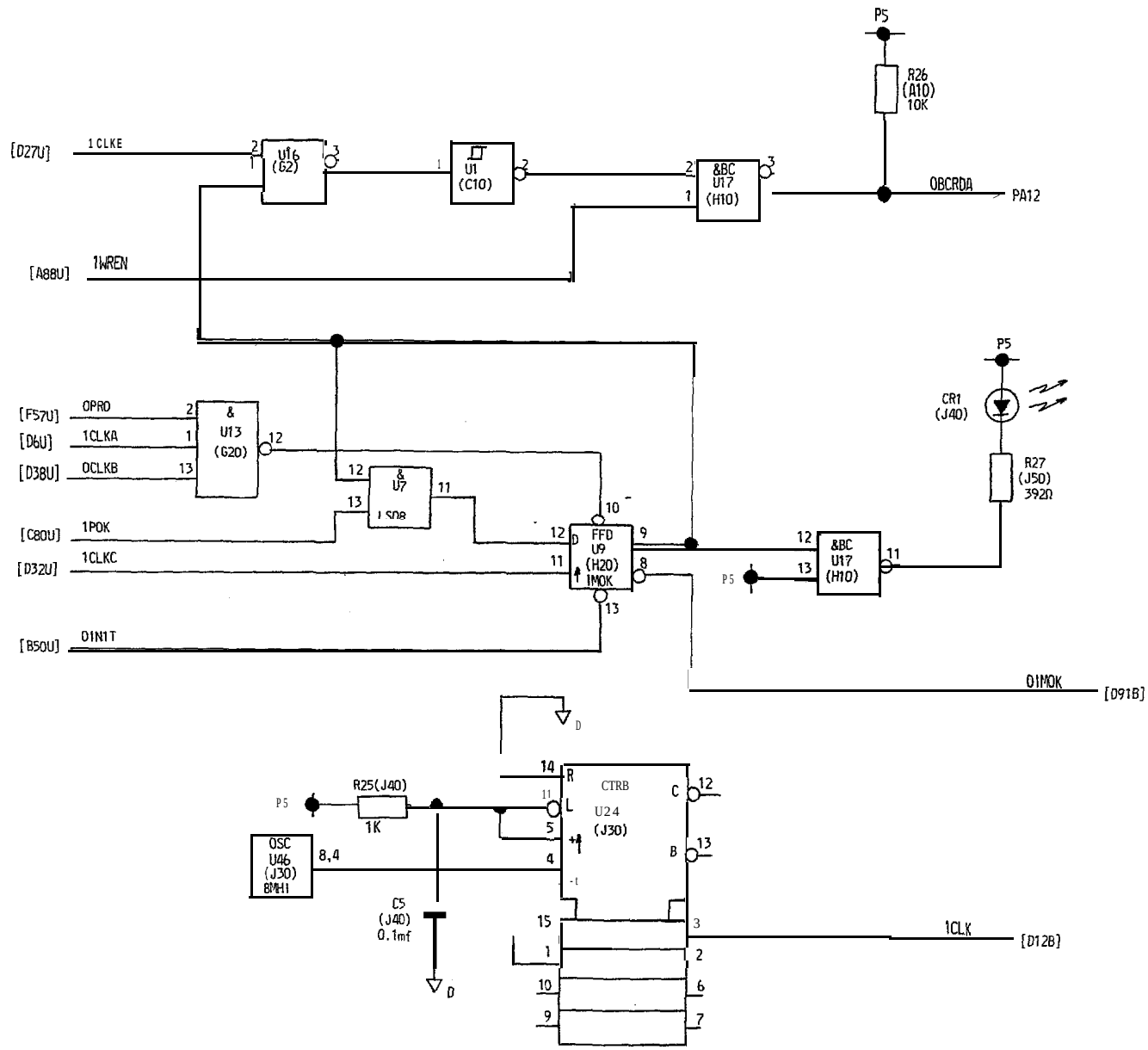
REV. 1 OEB - OEC 1/18/78 CWF	REV. 4	REV. 5	PRINTS TO PRI 53-36, TRO, DL104 2520	ENGINEER AB 9/11/77	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM ANALOG OUTPUT
REV. 2 OEC MAC 3/27/86	ISSUED 9/12/77			DOM PLUS		DS380ONDAC
REV. 3	MADE BY R.F. E. BOOTE					CONT. ON SH. 4CA SH. NO ABA



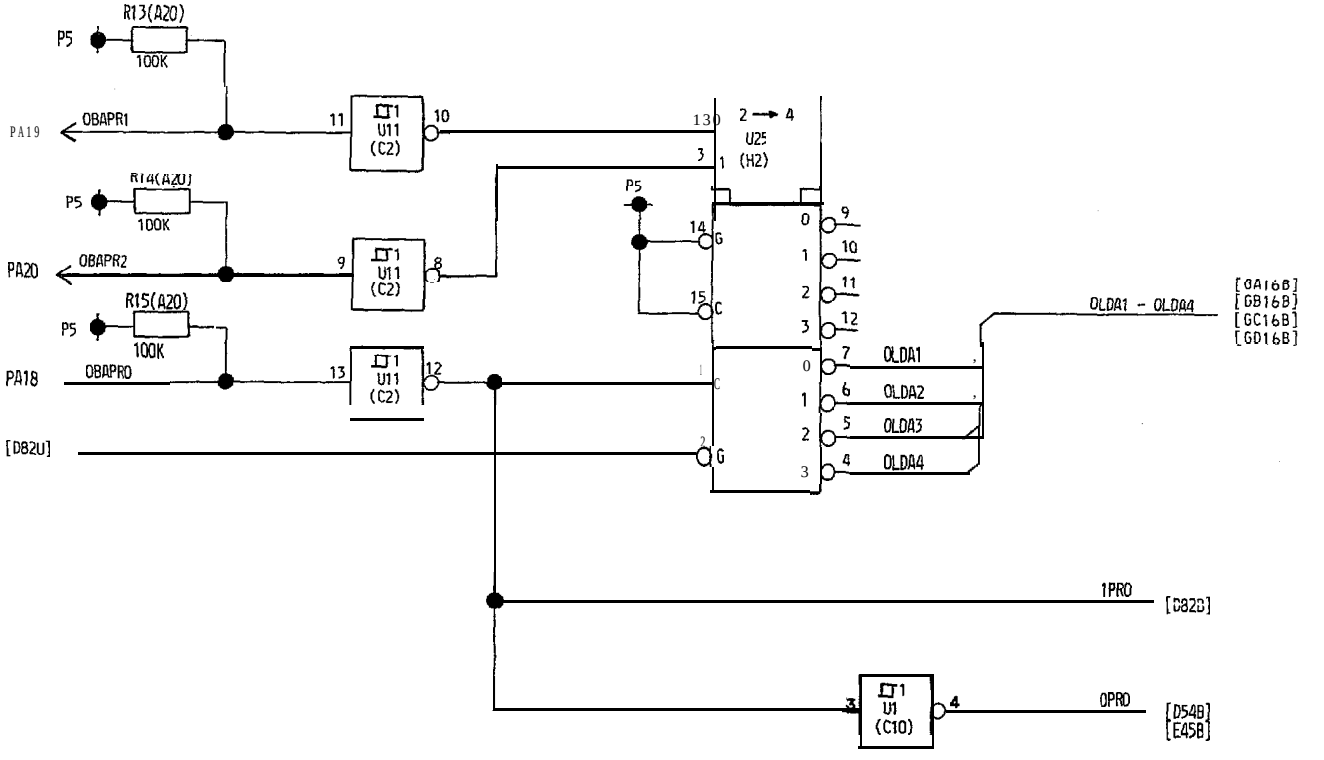
REV.1 OEX MAC 3/24/86	REV.4	REV.5	PRINTS TO 53-36, ~~FRO~~, DL104 2 530	ENGINEER MB 9/12/77	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM ANALOG OUTPUT
REV.2 OEX MAC 3/13/86	ISSUED 9/12/77			DOM PLUS		D S 3 8 0 0 N D A C
REV 3	MADE BY E. BOOTE					CONT. ON SH. 4DA SH. NO. 4CA



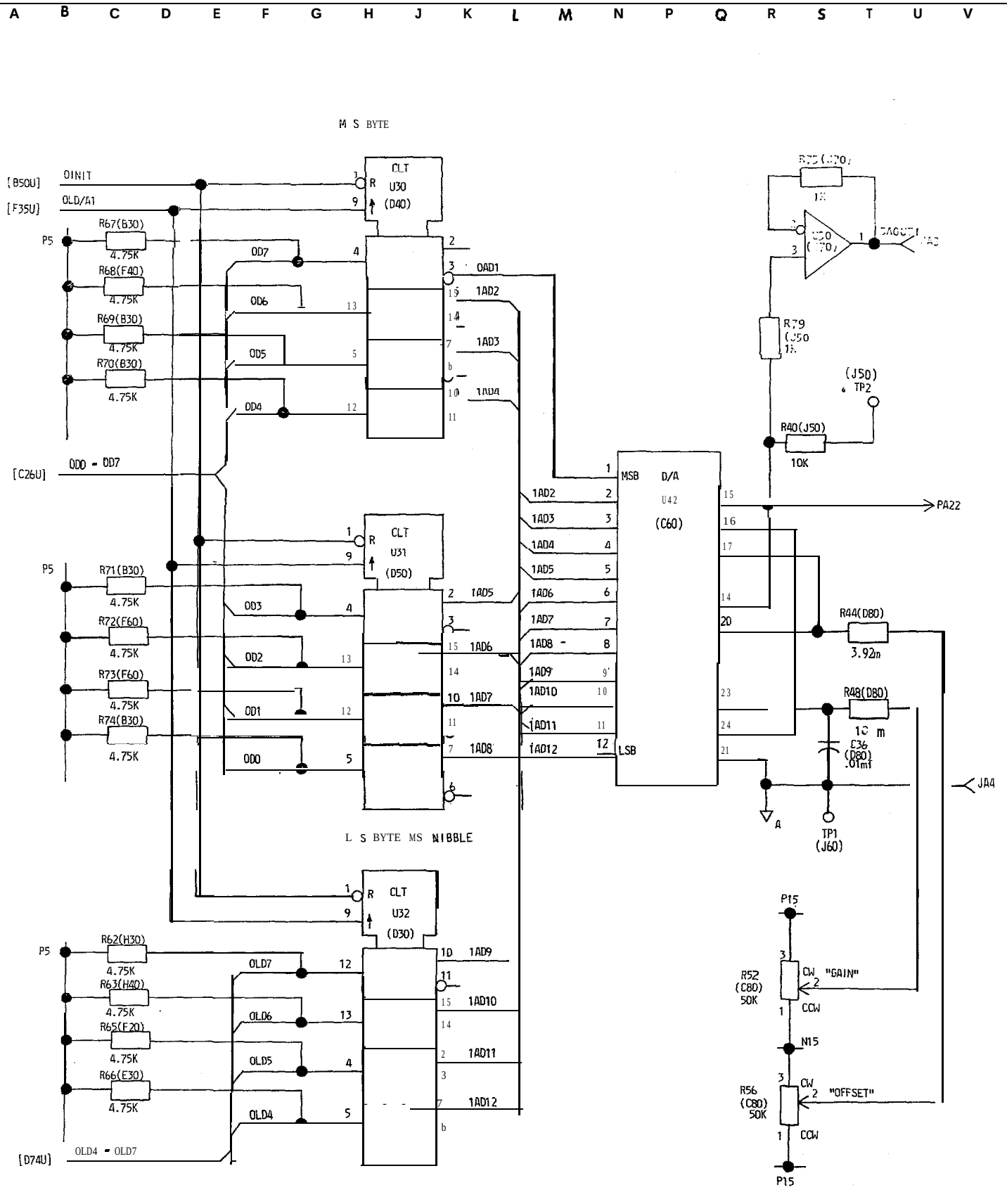
REV. 1 OER 3/24/86	REV. 4	REV. 5	PRINTS TO PRI 53-36, TRG, DL104 J520	ENGINEER JP 9/12/77	ELEMENTARY DIAGRAM
REV. 2 OEX 5/19/86	ISSUED 9-12-77			DOM PLUS	ANALOG OUTPUT
REV. 3	MADE BY E. BOOTE			GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM. VA, U.S.A.	DS3800NDAC CONT. ON SH. 4EA SH. NO. A1



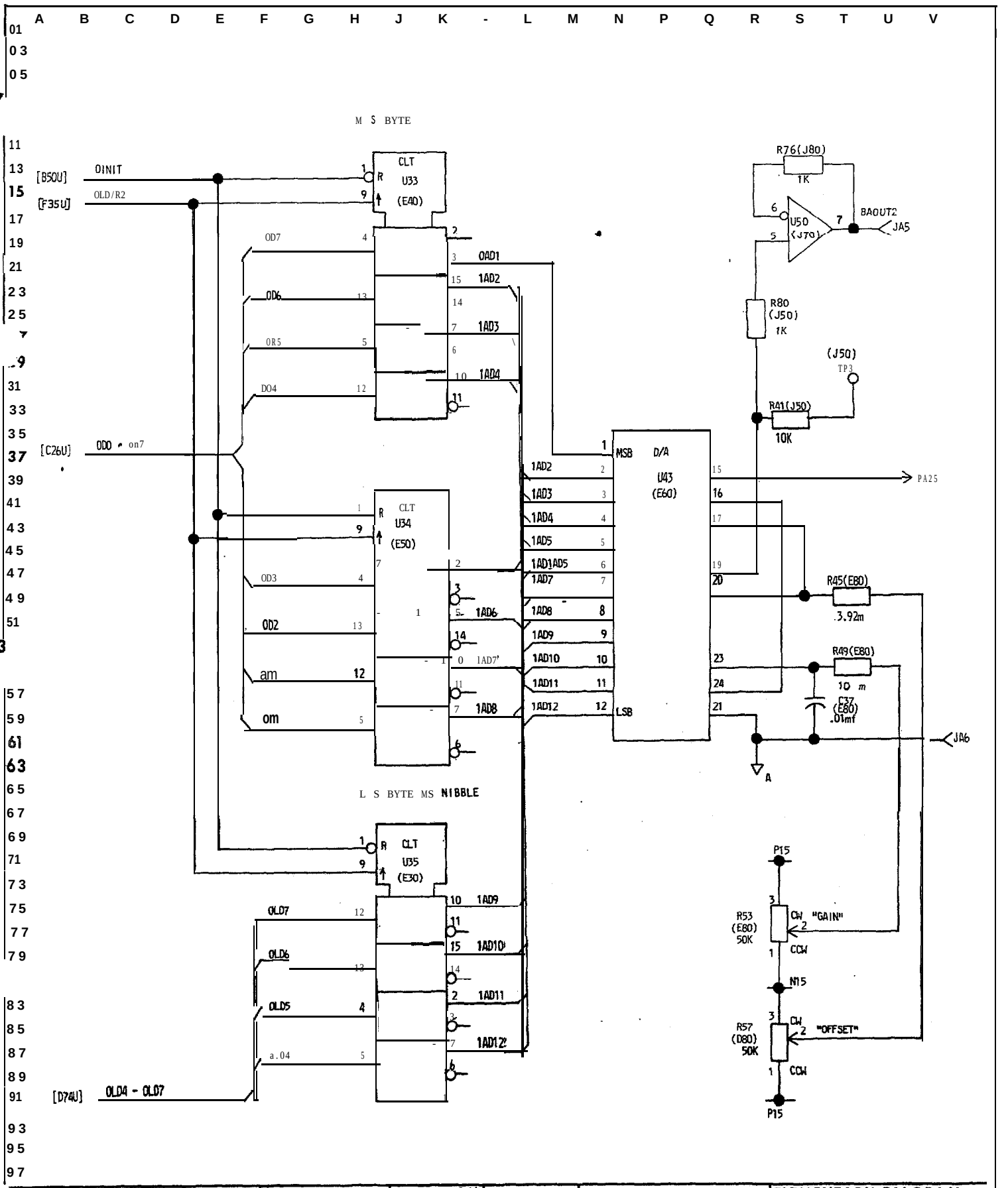
REV. 1 OER REV. 4 MAC 3/24/86	REV.5	PRINTS TO PRI 53-36, FRO, DL104 2520	ENGINEER 9/12/77	GENERAL@ ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM ANALOG OUTPUT	
REV. 2 DEX MAC 5/19/86	ISSUED 9/12/77		DOM PLUS		D S 3 8 0 0 N D A C	
REV.3	MADE BY E. BOOTE				CONT. ON SH. 4FA SH. NO. 4EA	



REV. 1	REV. 4	REV. 5	PRINTS TO 53-36, DL104 PLT TRO 2520	ENGINEER 9/12/77	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM ANALOG OUTPUT
REV. 2	ISSUED 9/12/77			DOM PLUS		D S 3 8 0 0 N D A C
REV. 3	MADE BY E. BOOTE					CONT. ON SH. 4GA SH. NO. 4FA

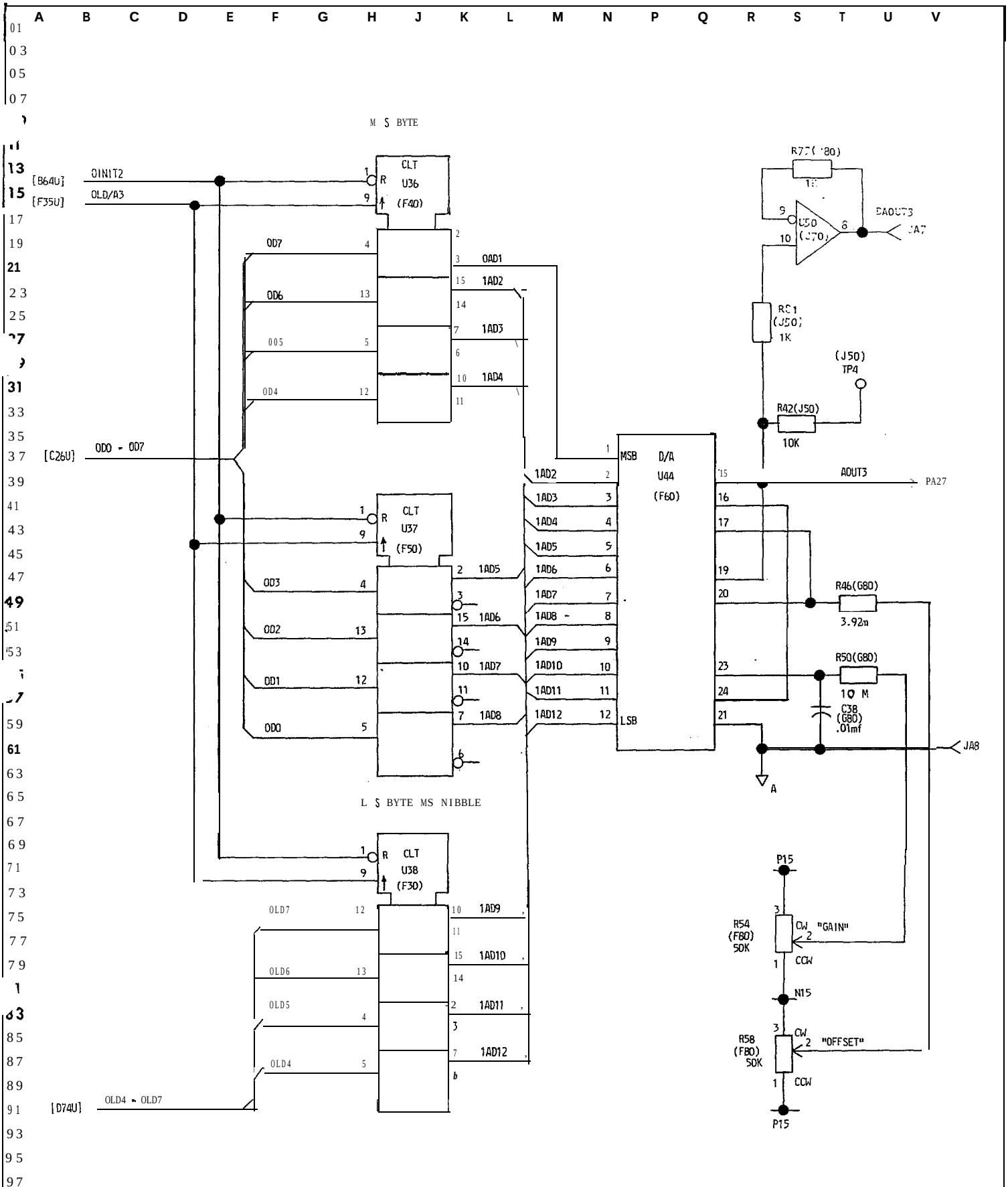


REV 1 NR 1/11 2/24/80	REV. 4	REV.5	PRINTS TO PRI 53-36, TRO; DL104 3536	ENGINEER MB 9/12/77	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM ANALOG OUTPUT
REV.2 DFA DSN 880129	ISSUED 9/12/77			DOM PLUS		D S 3 8 0 0 N D A C
REV. 3	MADE BY E. BOOTE					CONT. ON SH. 4GB SH. NO. 4GA



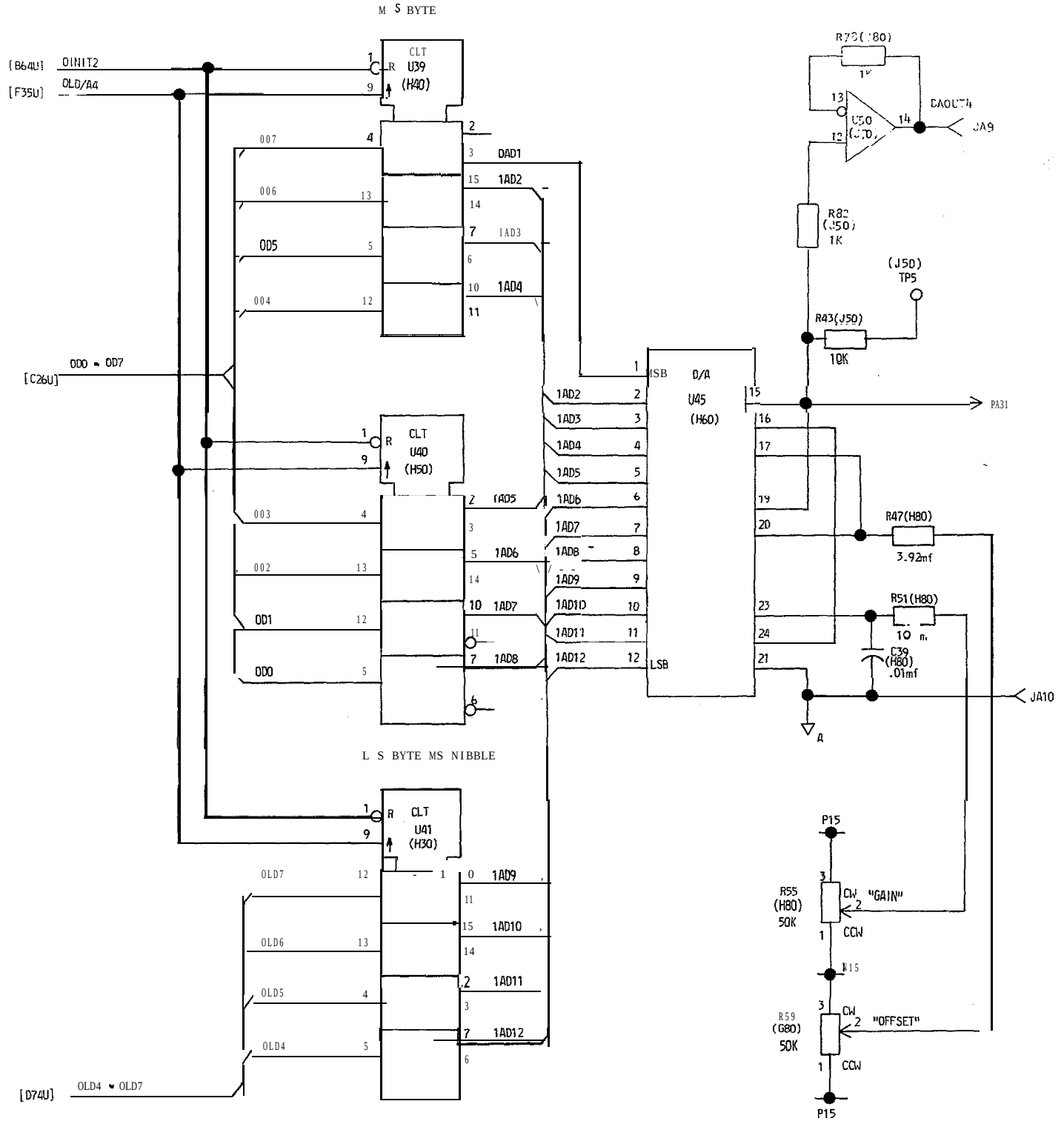
EV.1 OER MAC 3/24/86	REV.4	REV.5	PRINTS TO PRI 53-36, ~~FAB~~, DL104 2620	ENGINEER AB 9/12/77	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM ANALOG OUTPUT
REV.2 OEX MAC 5/19/86	ISSUED 9/12/77			DOM PLUS		D S 3 8 0 0 N D A C
REV.3 OFA DSN 880129	MADE BY E. BOOTE					CONT. ON SH. 4GC SH. NO. 4GE

GSD 0182 (09-76)



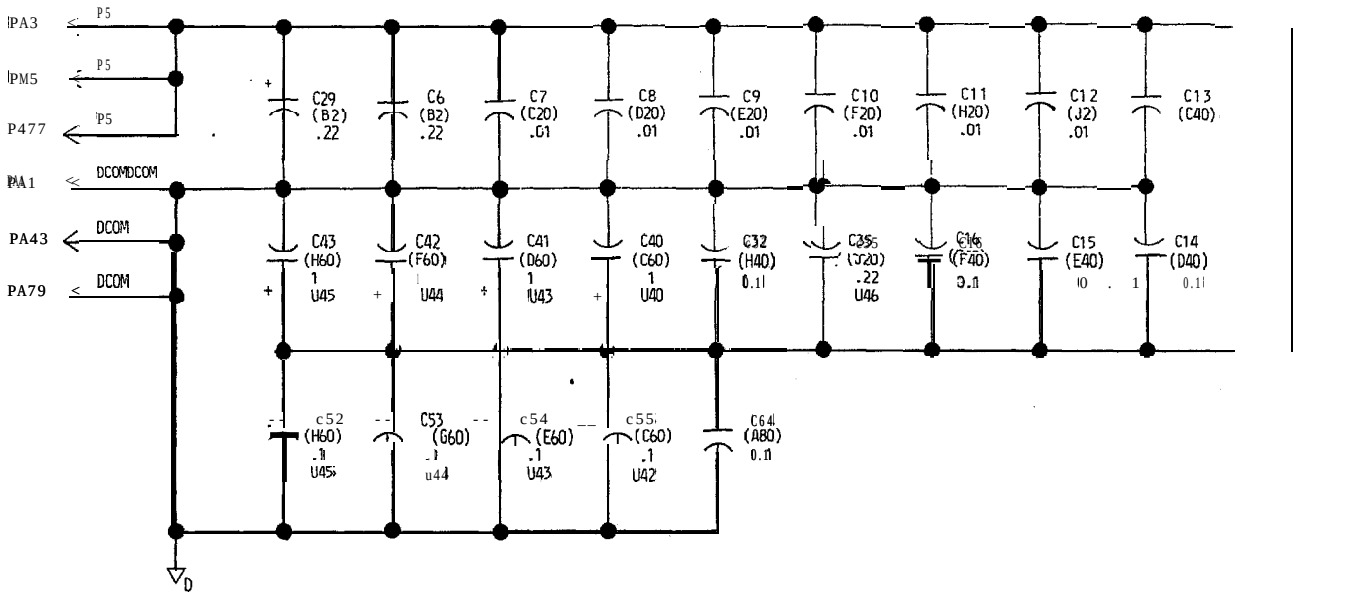
REV.1 OES VA. 3/24/83	REV 4	REV.5	PRINTS TO PRI 53-36, TRO DL104 JSJO	ENGINEER MB 9/12/77	GENERAL@ ELECTRIC	ELEMENTARY DIAGRAM
REV.2 OFA DSN 880129	ISSUED 9/12/77			DOM PLUS	DRIVE SYSTEMS DEPARTMENT	ANALOG OUTPUT
REV 3	MADE BY R.L. E. BOOTE				SALEM, VA. U.S.A.	D S 3 8 0 0 N D A C

CONT. ON SH. 4GD SH. NO. 4GC



REV. 1 JEC 3/27/85	REV. 4	REV. 5	PRINTS TO DRI 53-36, TRO, DL104 76.1	ENGINEER 9/12/77	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM ANALOG OUTPUT
REV. 2 JEA DSN 880129	ISSUED 9/12/77			DOM PLUS		DS 3800NDA
REV. 3	MADE BY L.L. L. BOUTE			-		(CONT. ON SH. 4HA SH. NO. 46

DSU 8152 (09 76)



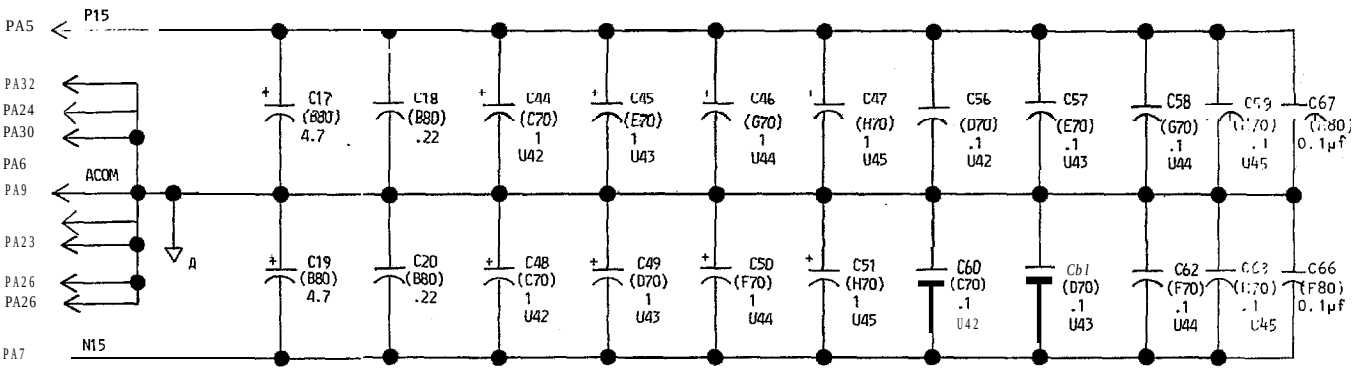
OCOM : PIN 7 OF U1,U3,U4,U5,U6,U7,U8,U9,U11,U12,U13,U16,U17,U46,U47;
PIN 8 OF U2,U15,U18,U24,U25,U30-U41;
PIN ID OF U10

P5 : PIN 13 OF U42-U45; PIN 14 OF U1,U3,U4,U5,U6,U7,U8,U9,U11,U12,U13,U16,U17,U46,U47;
PIN 16 OF U2,U15,U18,U24,U25,U30-U41; PIN 20 OF U10

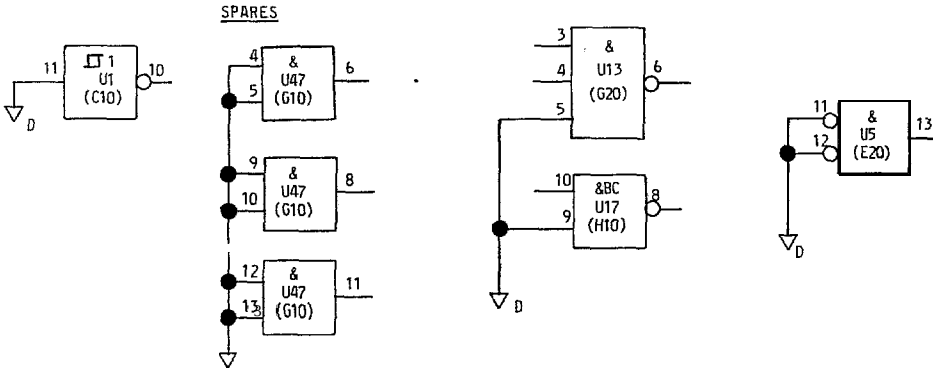
P15: PIN 22 OF U42-U45, PIN 4 OF U50

N15: PIN 14 OF U42-U45, PIN 11 OF U50

ACOM: PIN 21 OF U42-U45

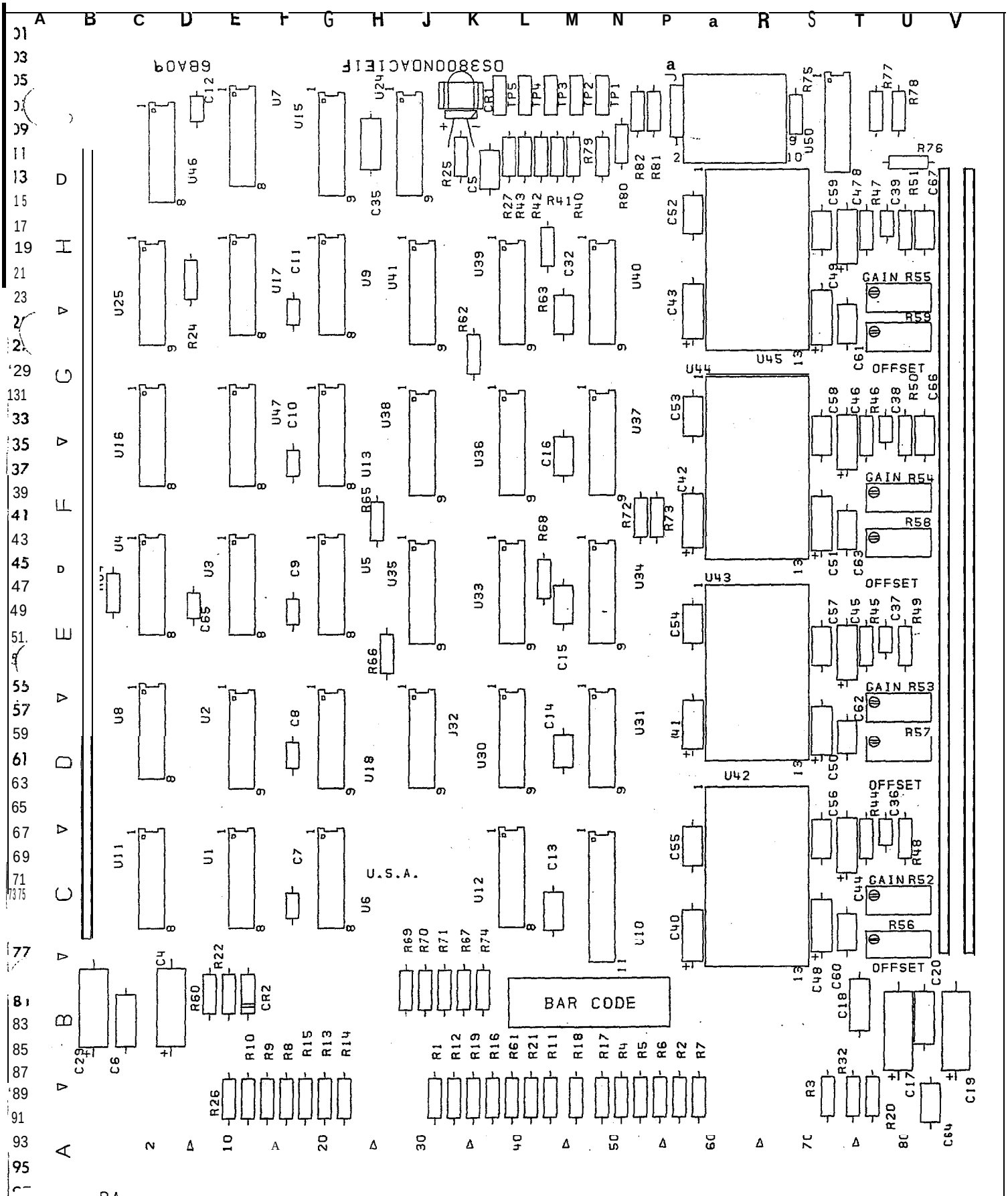


SPARES



REV.1 REH 9/11/81 'ODED BUSSING OF C's	REV.4	REV.5	PRINTS TO DL104	ENGINEER	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.
REV.2 OES MAC 5/29/136	ISSUED 9/12/77			DOM PLUS	
REV 3 OEX MAC 5/19/176 R2	MADE BY E. BOOTE				

ELEMENTARY DIAGRAM
ANALOG OUTPUT
DS380DNDAC
CONT. ON SH.4 JA. SH. NO. 4H

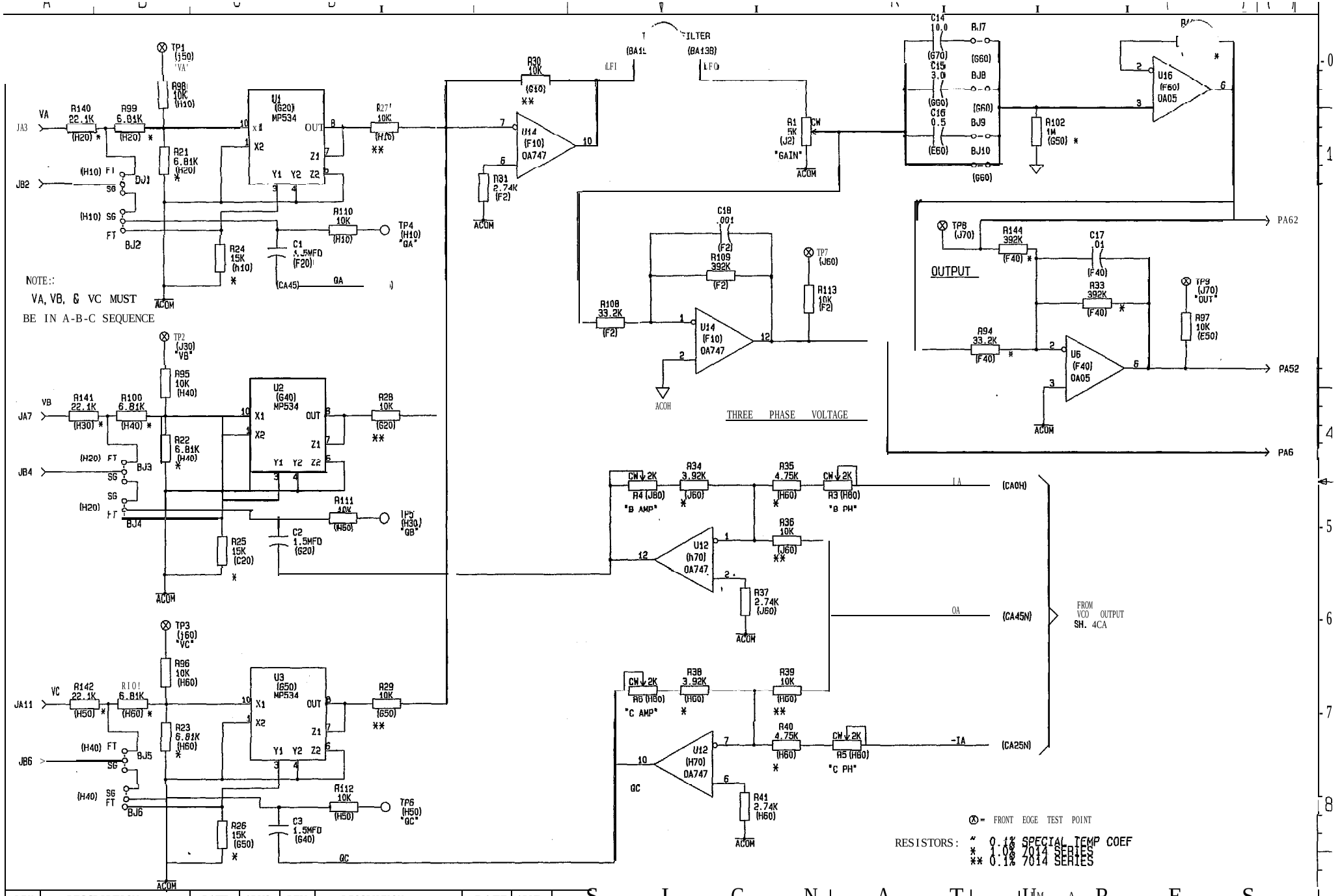


PA

OEK CWF 4/21/79	REV.4 OFA DSN 880129	REV 5	PRINTS TO DL104 PRI 4ME1, 2520	APPROVALS CWF	GENERAL ELECTRIC DRIVE SYSTEMS DEPT. SALEM, VA., U.S.A.;
REV.2 OES RE-ISSUED MAC 3/27/86	FEB MARCH 27, 1986			FIRST MADE FOR	
REV.3 OEX MAC 5/19/86	MADE BY M.A. CONNER			I.C. NO.	

ELEMENTARY DIAGRAM
ANALOG OUTPUT

DS3800NDAC
CONT. ON SH. FL. SH. NO. 4JA



REV	DESCRIPTION	DATE	APVD	REV	DESCRIPTION	DATE	APVD

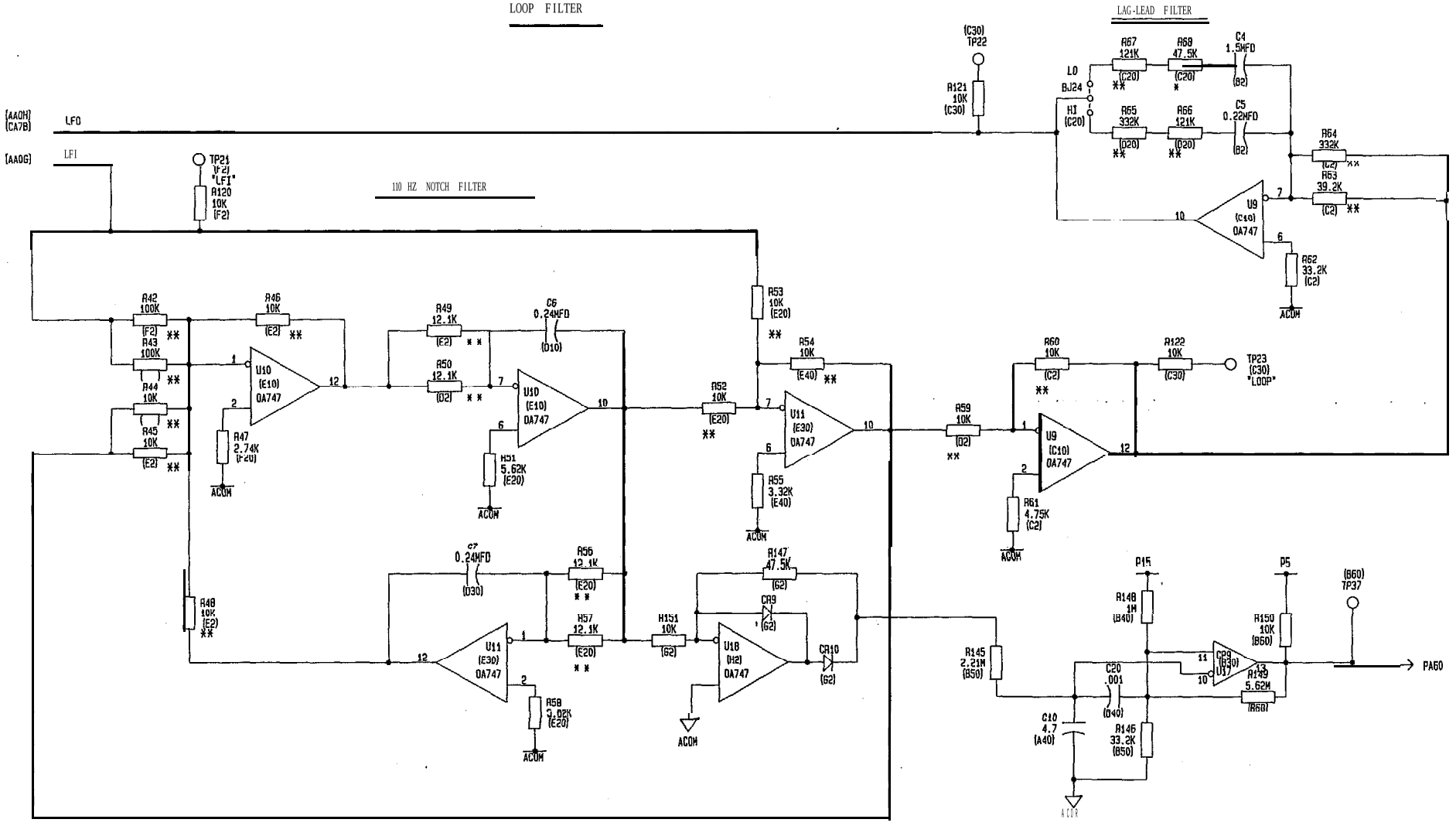
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS OPERATION
SALEM, VIRGINIA

DRAWN		
CHECKED	6/11/86	
ENGRG	G.W.STULTZ	10-16-86
ISSUED	10-16-86	

UM A R CONTINUOUS WAVE FREQUENCY TRANSDUCER 50 / 60 HZ

SIZE	OWG	
C	05380ONFTA	
CONT OF SHEET 4BA	SHEET 4AA	

CARS



REV	DESCRIPTION	DATE	APVO	REV	DESCRIPTION	DATE	APVO
1	CHANGE A146.C10 OEE	3-16-W	GWS				

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS OPERATION
SALEM, VIRGINIA

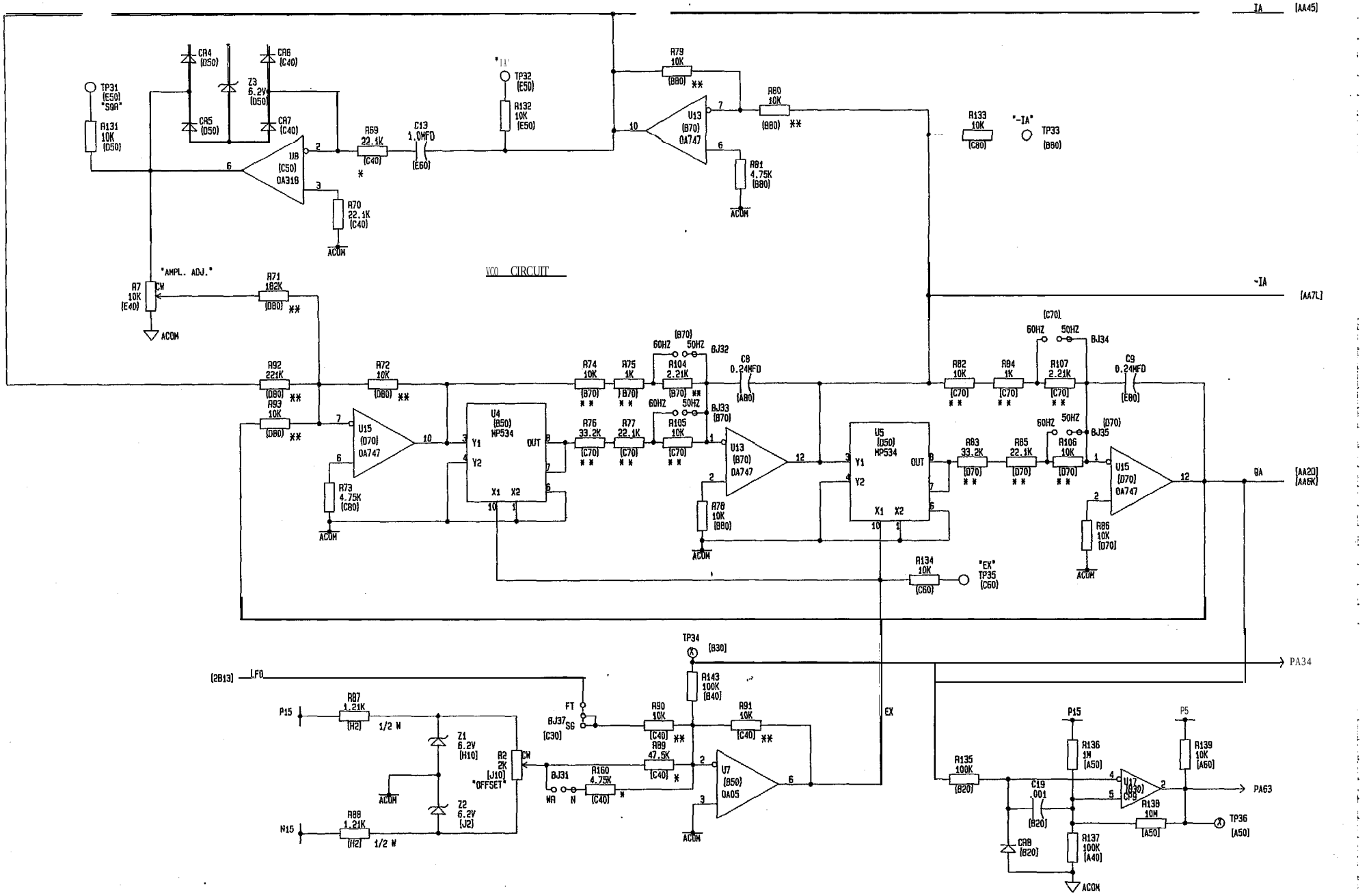
SIGNATURES		DATE
DRAWN	GWS	
CHECKED	6/12/86	
ENGRG	G.W.STULTZ	4/31/89
ISSUED	mk	5/31/89

Re:

ELEMENTARY
CONTINUOUS WAVE FREQUENCY TRANSDUCER 50/60HZ, LOOP FILTER

SIZE c | DL104 | DWG DS3800NFTA

CON ON SHEET 4CA | SHEET 4BA



REV	DESCRIPTION	DATE	APVD	REV	DESCRIPTION	DATE	APVD
1	RES. VALUE CHANGES OEE	12/16/86	GNS				

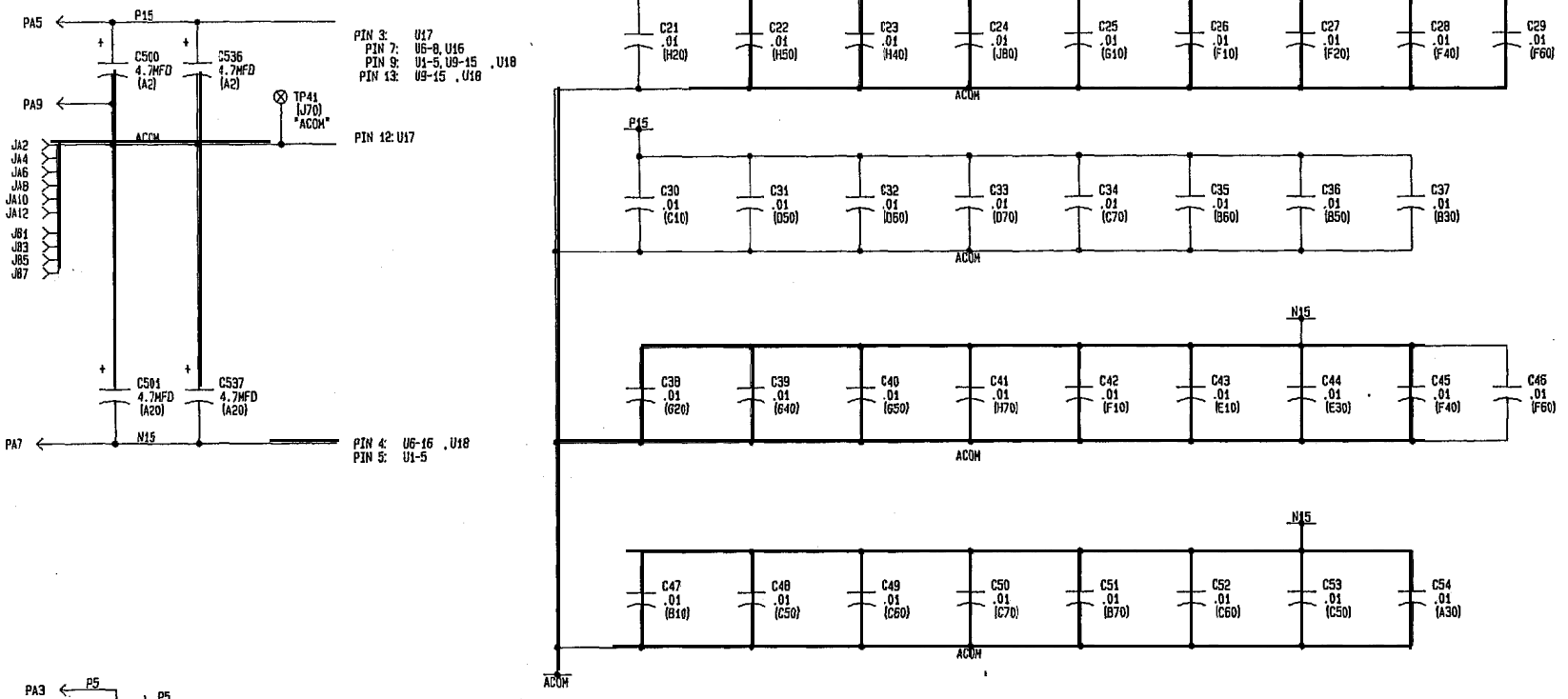
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS OPERATION
SALEM, VIRGINIA

SIGNATURES		DATE ELEMENTARY
DRAWN	GWS	5/31/89
CHECKED	6/12/86	
ENGRG G.W.	5/31/89	
ISSUED	MAC	5/31/89

CONTINUOUS WAVE FREQUENCY TRANSDUCER
50/60 HZ
VOLTAGE CONTROLLED OSCILLATOR

SIZE C | DL104 | DWG DS3800NFTA

CONT ON SHEET 4DA | SHEET 4CA



REV	DESCRIPTION	DATE	APVD	REV	DESCRIPTION	DATE	APVD
1	POWER BUS CHANGE U17 OEE	12/16/86	GWS				

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS OPERATION
SALEM, VIRGINIA

SIGNATURES		DATE
DRAWN	GWS	5/31/89
CHECKED	6/11/86	
ENGRG	G.W.STULTZ	5/31/89
ISSUED	MAC	5/31/89

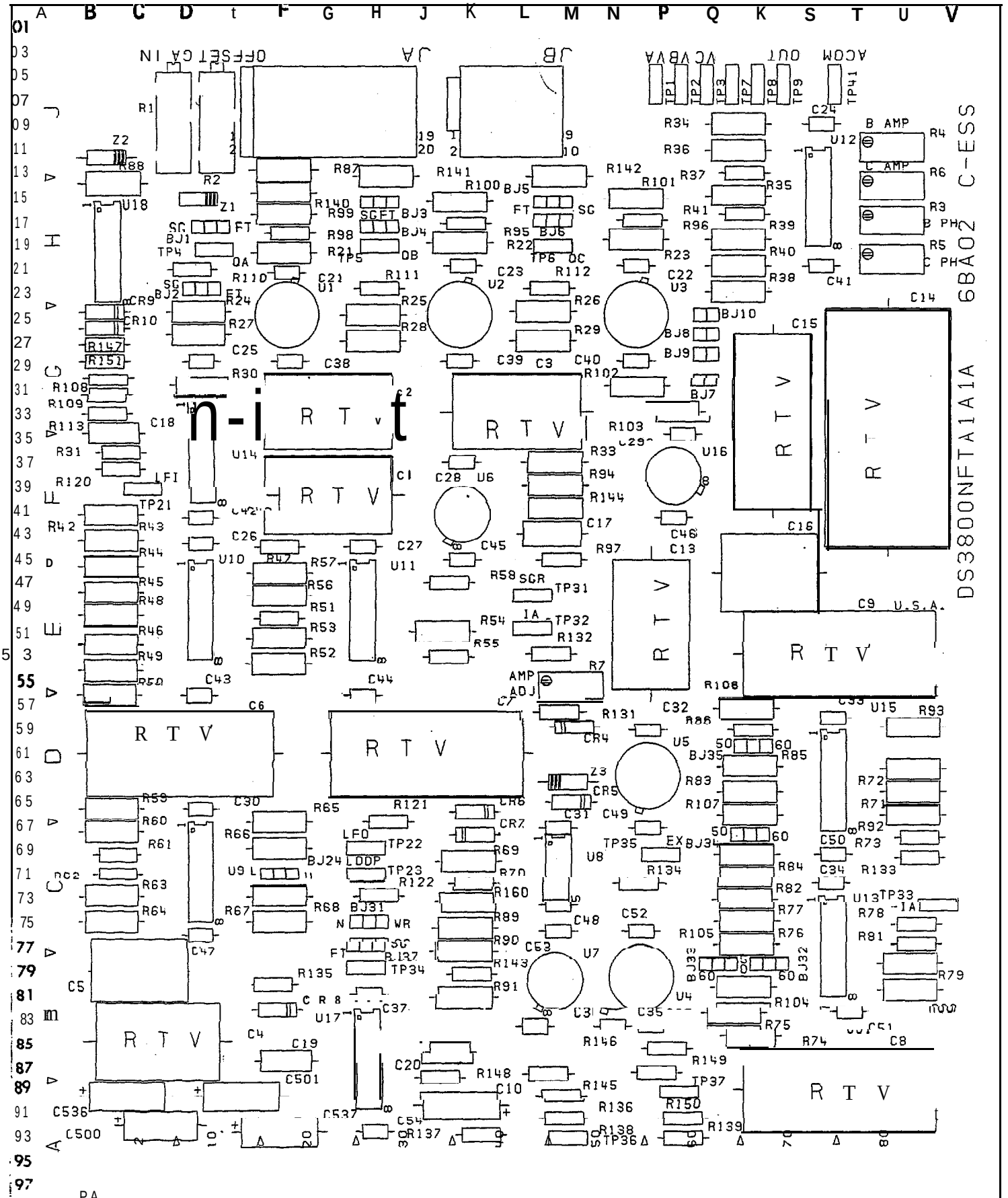
ELEMENTARY

CONTINUOUS WAVE FREQUENCY TRANSDUCER
50 / 60 HZ
POWER SUPPLIES & FILTERING

SIZE C | DL104 | DWG DS3800NFTA

CONT ON SHEET | FL. | SHEET 4DA

MARS



REV 1	REV.4	REV 5	PRINTS TO DL104 4mE1	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV.2	ISSUED Oct. 20, 1986			FIRST MADE FOR	DRIVE SYSTEMS DEPT. SALEM, VA., U.S.A.	
REV 3	MADE BY M.L. REICH			I.C. NO.		DS3800NFTA CONT. ON SH. FL SH. NO. 4 EA

FORM IC 2349 (04-76)

2.1 DEFINITIONS ANB PURPOSE

NGMA IS A GENERAL PURPOSE ANALOG TYPE CARD, PROVIDING:

- 4 VERSITILE OP-AMP CIRCUITS
- 4 FOUR-DUADRANT MULTI PLIERS. GENERAL PURPOSE DAUGHTER BOARD DGPA. SATURATI ON OETECTOR (IMOK)

2.2 THEORY OF OPERATION

THIS SECTION PROVIDED A GENERAL OVERVIEW OF THE OPERATION OF THIS CARD. DETAILS ARE PRESENTED IN SECTION 2.4.

2.2.1 OP-AMP CIRCUITS

EACH OP-AMP CIRCUIT CONSISTS OF A LOWDRIFT INTEGRATED CIRCUIT OP-AMP ARRANGEO TO PERMIT SELECTION OF THE INPUT AND FEEDBACK ELEMENTS, TYPICAL USES INCLUDE

BUFFER
INVERTING AMPLIFIER
INTEGRATOR OR LAG-LEAD COMPENSATION
DIFFERENTIAL INPUT STAGE

2.2.2 MULTIPLIERS

A FOUR-QUADRANT INTEGRATED CIRCUIT IS ARRANGED TO PERMIT USE AS A FOUR-QUADRANT MULTIPLIER OR TWO-QUADRANT DIVIDER. THE CIRCUIT MAY BE CONNECTED AS A SQUARER. CONNECTION AS A SDUARE-ROOT ELEYENT-IS NOT RECOMMENDED BECAUSE OF THE TENDENCY OF THE CIRCUIT TO LATCH-UP UNLESS EXTENSIVE CIRCUITRY IS ADDED TO THE BASIC CONFIGURATION.

2.3 INTERFACE REQUIREMENTS

TYPICALLY, THE ANALOG SIGNALS TO THIS CARD ARE DEVELOPED BY OP-AMP CIRCUITS ON OTHER CARDS. THE SIGNAL LEVEL EXCURSIONS

2.3 INTERFACE REQUIREMENTS (CONT'D)

MUST NOT EXCEED THE ANALOG POWER SUPPLY VOLTAGES (NOMINAL ± 15 VOLTS). IT IS PREFERABLE TO CURRENT LIMIT SIGNALS WHICH APPROACH WITHIN 1.5 VOLTS OF THE SUPPLV VOLTAGES e.g. EXCEED ±13.5 VOLTS WITH ±15 VOLT SUPPLIES, THE SOURCE IMPEDANCE OF SIGNALS OPERATING INTO THE OP-AMP CIRCUITS SHOULD BE INCLUDED IN THE CALCULATION OF INPUT RESISTANCE, CARE SHOULD BE EXERCISED TO DRIVE THE MULTIPLIERS FROM LOW IMPEDANCE SOURCES BECAUSE OF THI LACK OF SPECIFICATION CONTROL OF OFFSET AND BIAS FACTORS OVER THE OPERATING TEMPERATURE RANGE.. THE OUTPUT SIGNAL SWING WILL BE AT ±11 VOLTS @ 1.1 'MA FOR THE OP-AMPS ANO ±11VOLTS @ 6.0 MA FOR THE MULTIPLIERS.

2.4 FUNCTIONAL DESCRIPTIONS

2.4.1 OP-AMP CIRCUITS

OP-AMP CIRCUIT FUNCTION	CKTA	CKTB	CKTC	CKTD
INVERTING INPUT'	AN11	AN5	AN9	AN114
NEW-INVERTING INPUT	AN13	AN117	AN11	AN116
CALIBRATION-INPUT	AN12	AN116	AN10	AN115
FILTERED OUTPUT	A01	A02	A03	A04
FEEDBACK (FROM OUTPUT)	AF2	AF4	AF6	AF8
FEEDBACK(1 .0K TO INPUT)*	AF1	AF3	AF5	AF7

FEEDBACK ELEMENTS SHOULD BE CONNECTED BETWEEN THE TWO LISTED FEEDBACK PA I RS. THE SUMMING JUNCTION INPUT (THROUGH. 1.0K-) IS TERMINATED ON THE DAUGHTER BOARD TO FACILITATE CONNECTION.

THE CALIBRATION INPUT RESISTANCE IS THE SAME AS THE INVERTING INPUT RESISTANCE TO PERMIT CALIBRATION OR CHECK OF THE SYSTEM.. THE CALIBRATION INPUT CAN BE ACCESSED VIA THE CORRESPONDING TEST POINT. THE OUTPUT SIGNAL FROM EACH AMPLIFIER CAN BE MONITORED.

* ROLE OF THESE INPUTS MAY BE REVERSEO.

2.4 FUNCTIONAL DESCRIPTIONS (CONT'D)

2.4.1 OP-AMP CIRCUITS (CONT'D)

AT THE CORRESPONDING TEST POINT.

2.4.2 MULTIPLIERS

THE BASIC EQUATION SOLVED IS

$$V_0 = \frac{(X_1 - X_2)(Y_1 - Y_2)}{10.0} + Z_2$$

IN THE MULTIPLY MODE THE TOTAL ERROR IS LESS THAN ±100MV, FOR $-10V < (X, Y) < +10V$, A NO ZERO IMPEDANCE INPUT SIGNALS..,

IN THE DIVIDE MODE, THE CIRCUIT CONNECTIONS RESULT IN THE BASIC EQUATION BEING MODIFIED TO

$$V_0 = (10.0)\frac{(Z_2 - Z_1)}{(X_1 - X_2)} + Y_1 \quad \text{WHERE } X_1 > X_2$$

FOR THIS CONFIGURATION THE TOTAL ERROR IS LESS THAN:

FOR	ERROR
X=1DV -10V< z < +10V	±75MV
X=1V -10V < Z < +10V	±200MV

(WITH-ZERD IMPEDANCE INPUT SIGNALS)

SELECTION OF THE MODE OF OPERATION HAS BEEN FACILITATED BY THE JUMPERS BJ1-4

THE FILTERED OUTPUT SIGNAL FROY EACH MULTIPLIER CHIP CAN BE MONITORED AT THE APPROPRIATE TEST POINT.

2.4.3 INSTALLATION AND MAINTENANCE AIDS

2.4.3.1 IMOK LIGHT TO OETECT A FAULTY OP-AMP IN SATURATION.

2.4.3.2 TEST POINTS (RING TERMINALS)

TEST POINT	SIGNAL NAME	CIRCUIT		FUNCTION
TP2	AN12	OP-AMP	CKTA	TEST INPUT
TP6	A01	OP-AMP	CKTA	OUTPUT SIGNAL
TP8	AN110	OP-AMP	CKTB	TEST INPUT
TP12	A02	OP-AMP	CKTB	OUTPUT SIGNAL
TP7	AN15	OP-AMP	CKTC	TEST INPUT
TP10	A03	OP-AMP	CKTC	OUTPUT SIGNAL
TP13	AN116	OP-AMP	CKTO	TEST INPUT
TP11	A04	OP-AMP.	CKTD	OUTPUT SIGNAL
TP4	OUT1	MULT#1		OUTPUT SIGNAL
TP3	OUT2	MULT#2		OUTPUT SIGNAL
TP5	OUT3	MULT#3		OUTPUT SIGNAL
TP9	OUT4	MULT#4		OUTPUT SIGNAL
TP1	ACOM			ACOM

2.4.4 POWER AND COOLING REQUIREMENTS

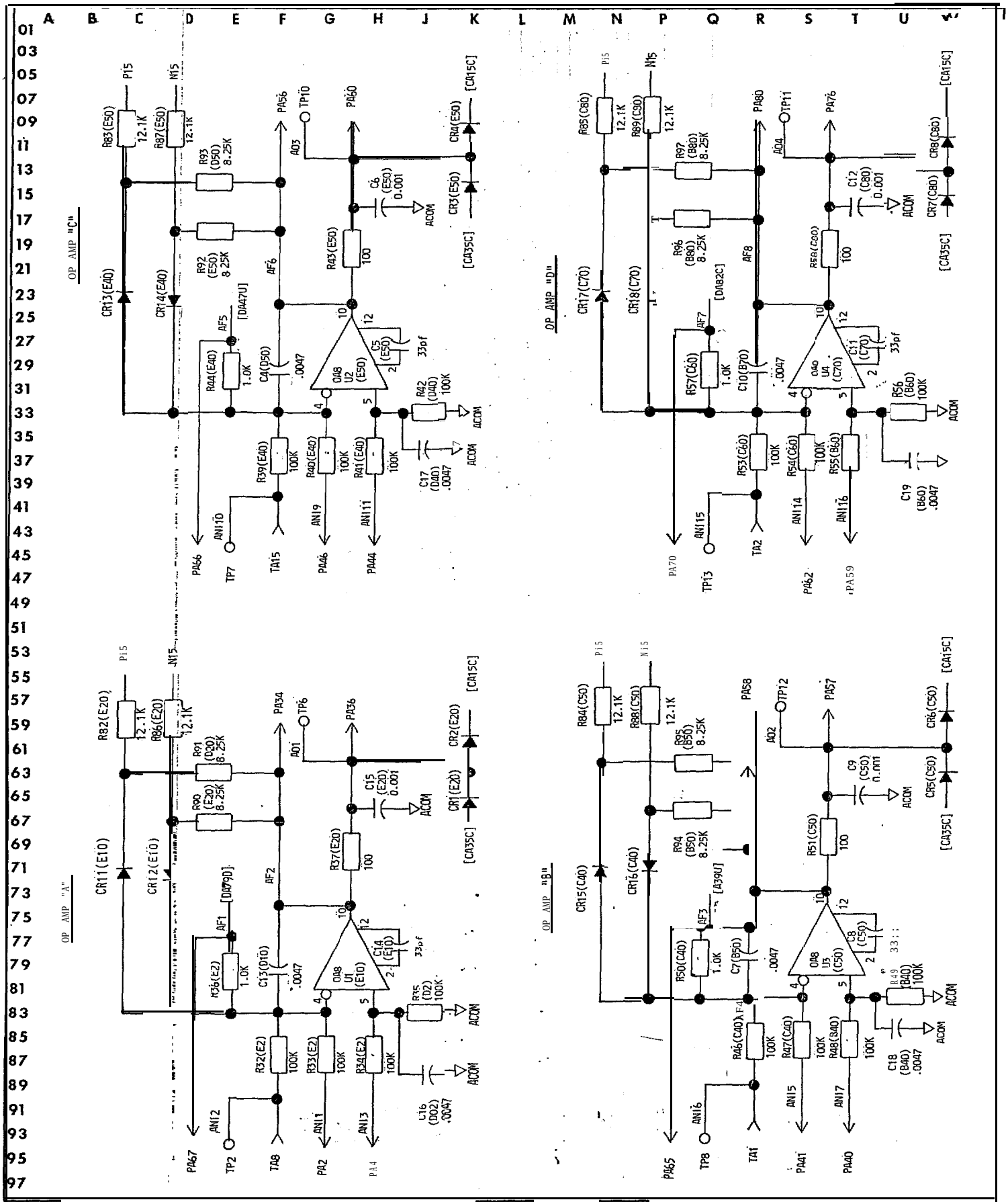
P15 - 15 MA

N15 - 24 MA

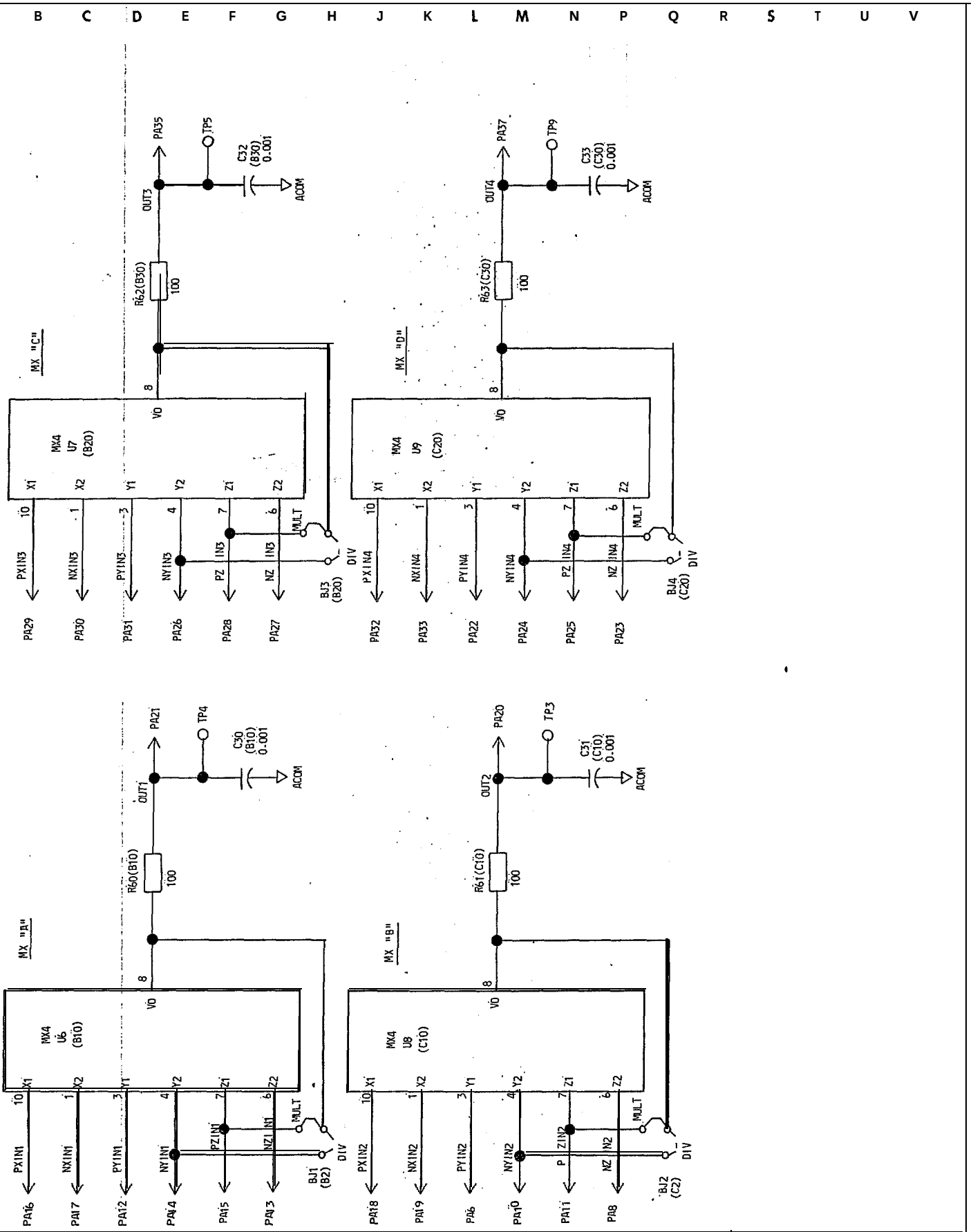
P5, P28 (NONE)

POWER APPROX. 800 MW.

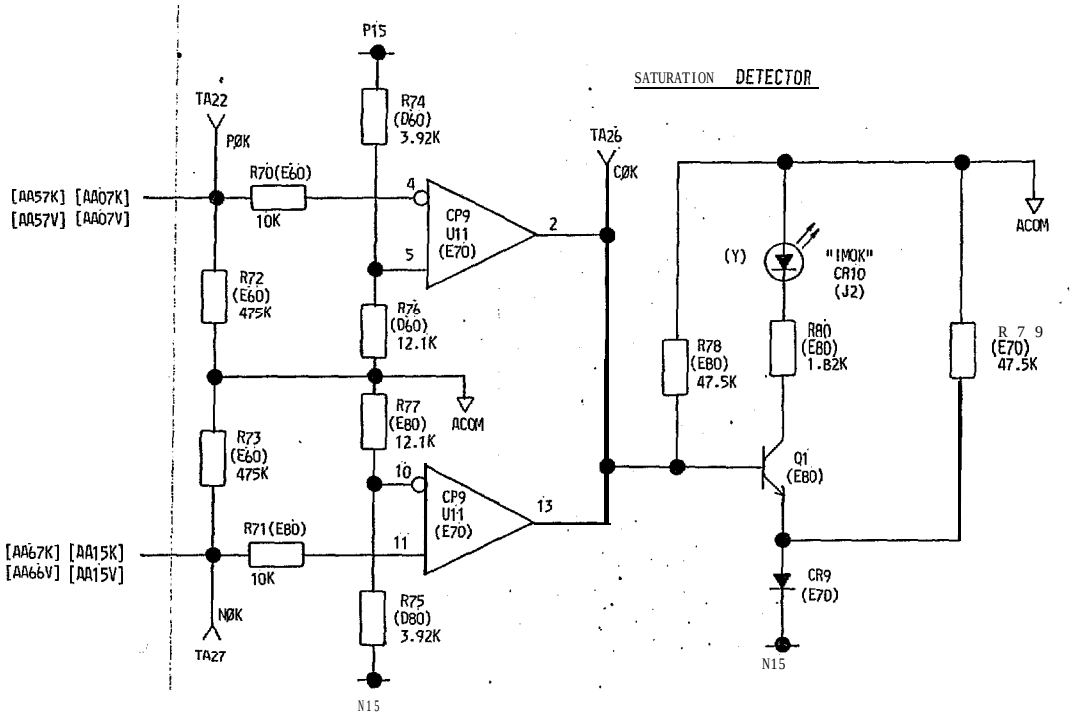
REV. 1	REV. 4	REV. 7	PRINTS TO DL102 PRI	ENGINEER 11/28/78	GENERAL ELECTRIC	GENERAL PURPOSE MULTIPLIER
REV. 2	REV. 5	ISSUED NOV 3 1978			DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	DS3800NGMA CONT. ON SH. FL. SH. NO. 2DA
REV. 3	REV. 6	MADE BY J.H. CUTLER				

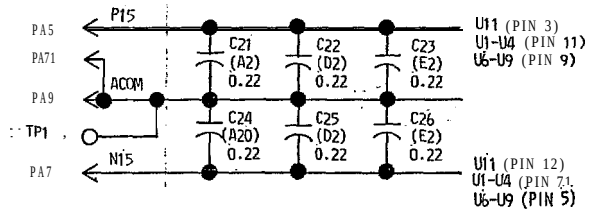


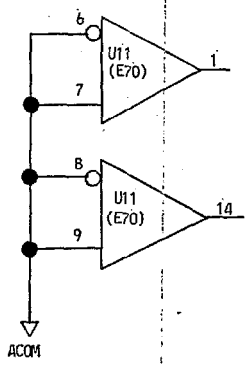
REV. 1	REV. 4	REV. 5	PRINTS TO PRI DL104	ENGINEER	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.
REV. 2	ISSUED Nov. 3, 1978				ELEMENTARY DIAGRAM GENERAL PURPOSE MULTIPLIER
REV. 3	MADE BY C. JORDAN				DS3800NGMA CONT ON SHEET SH. NO.



EV. 1	REV. 4	REV.5	PRINTS TO PRI DL104	ENGINEER	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
EV. 2	ISSUED NOV 3,1978				DRIVE SYSTEMS DEPARTMENT	GENERAL PURPOSE MULTIPLIER
EV. 3	MADE BY C. JORDAN				SALEM, VA. U.S.A.	DS3800NGMA CONT. ON SH. 4CA SH. NO. 4BA

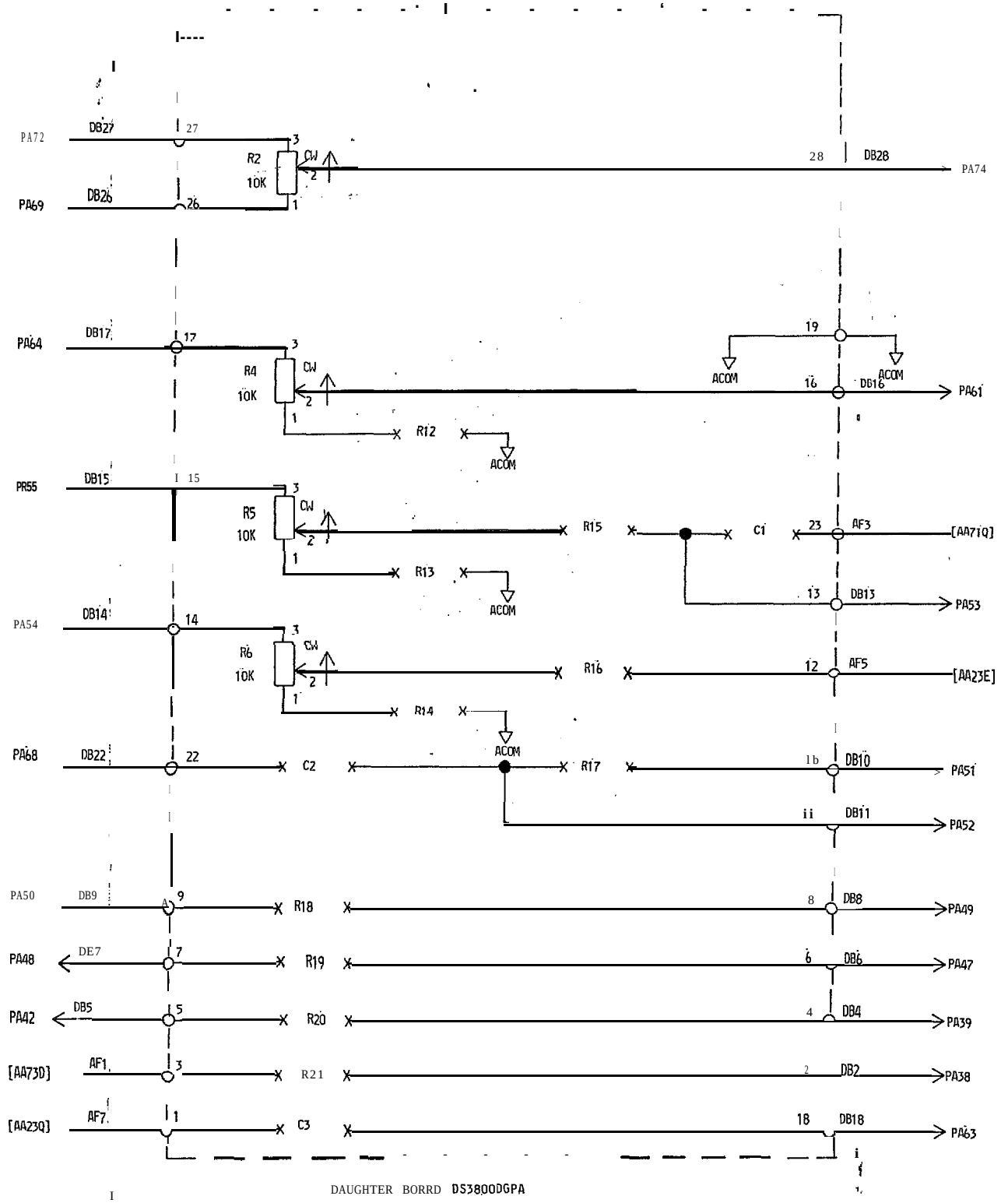




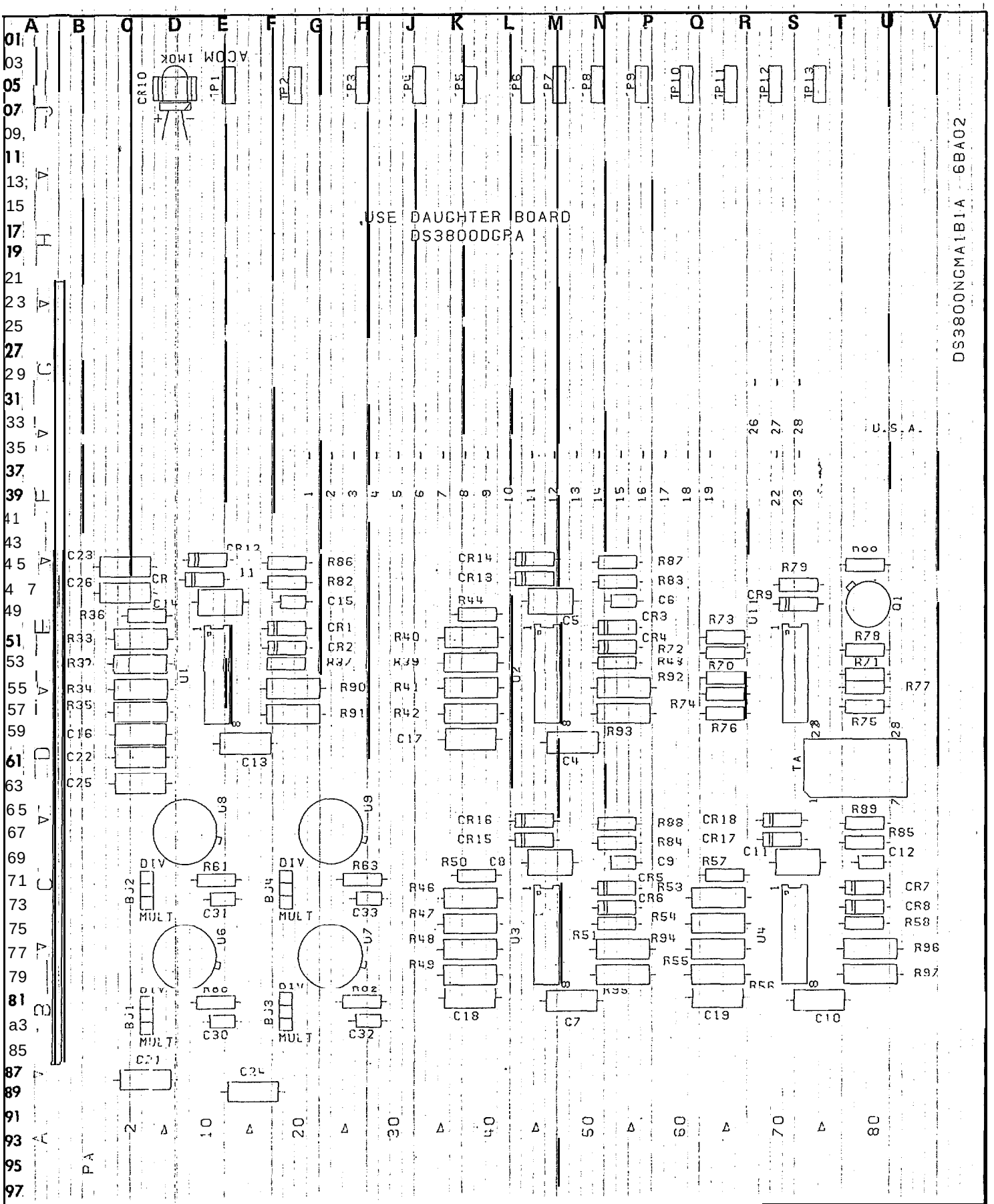


IEV. 1	REV. 4	REV. 5	PRINTS TO PRI DL104	ENGINEER	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
IEV. 2	ISSUED: Nov. 3, 1978				DRIVE SYSTEMS DEPARTMENT	GENERAL PURPOSE MULTIPLIER
EV. 3	MADE BY C. JORDAN				SALEM, VA. U.S.A.	DS3800NGMA
						CONT. ON SH. 40A SH. NO. 4CA

D 0152 (09 76)



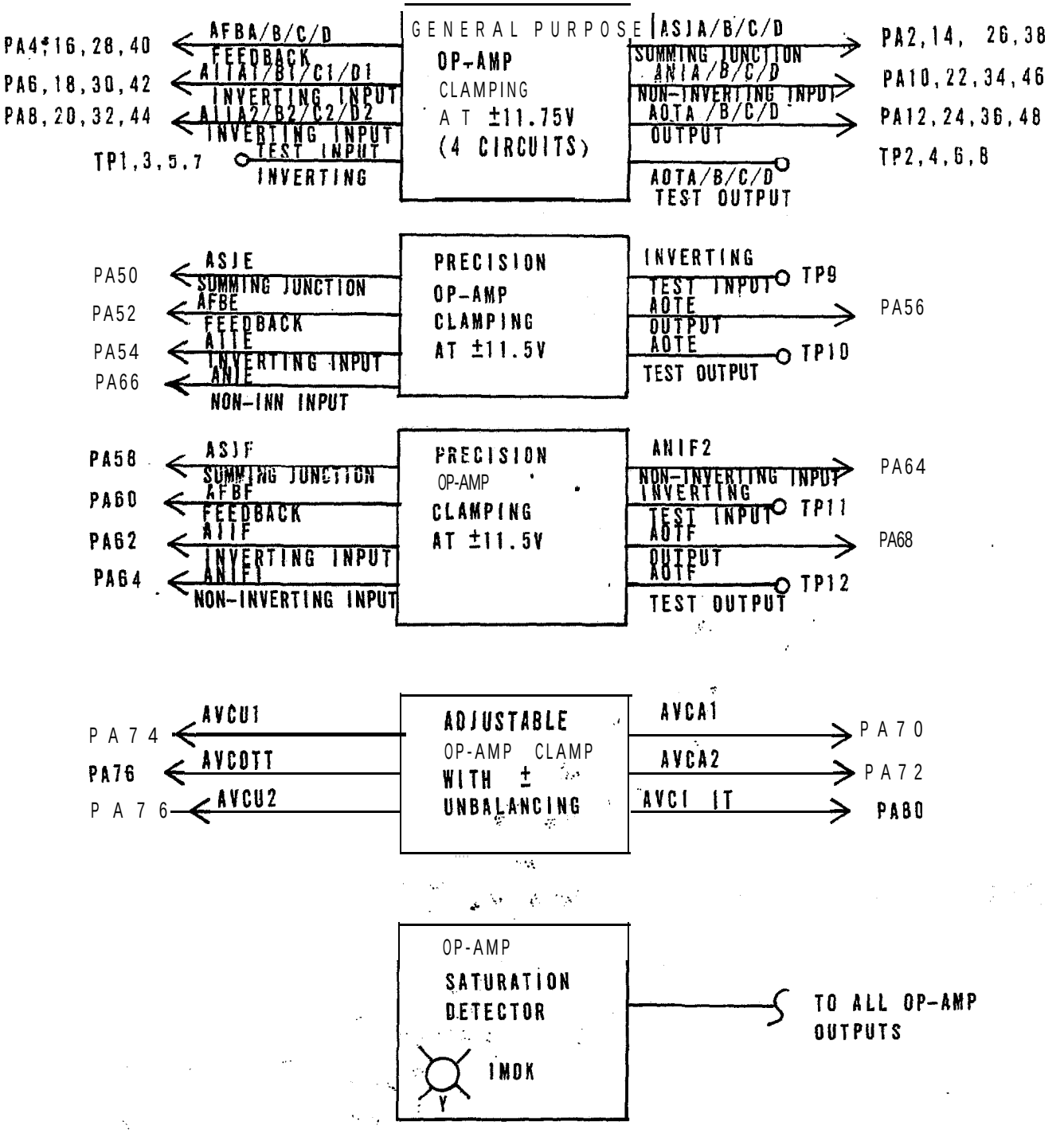
V.1 OEC HQ 6/27/79	REV4 :	REV.5	PRINTS TO PRI DL104	ENGINEER JMCutler 11/6/78	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM GENERAL PURPOSE MULTIPLIER
V.2	ISSUED NOV. 3, 1978					
V.3	MADE BY J. CUTLER					DS3800NGMA CONT. ON SHEA SH. NO 408



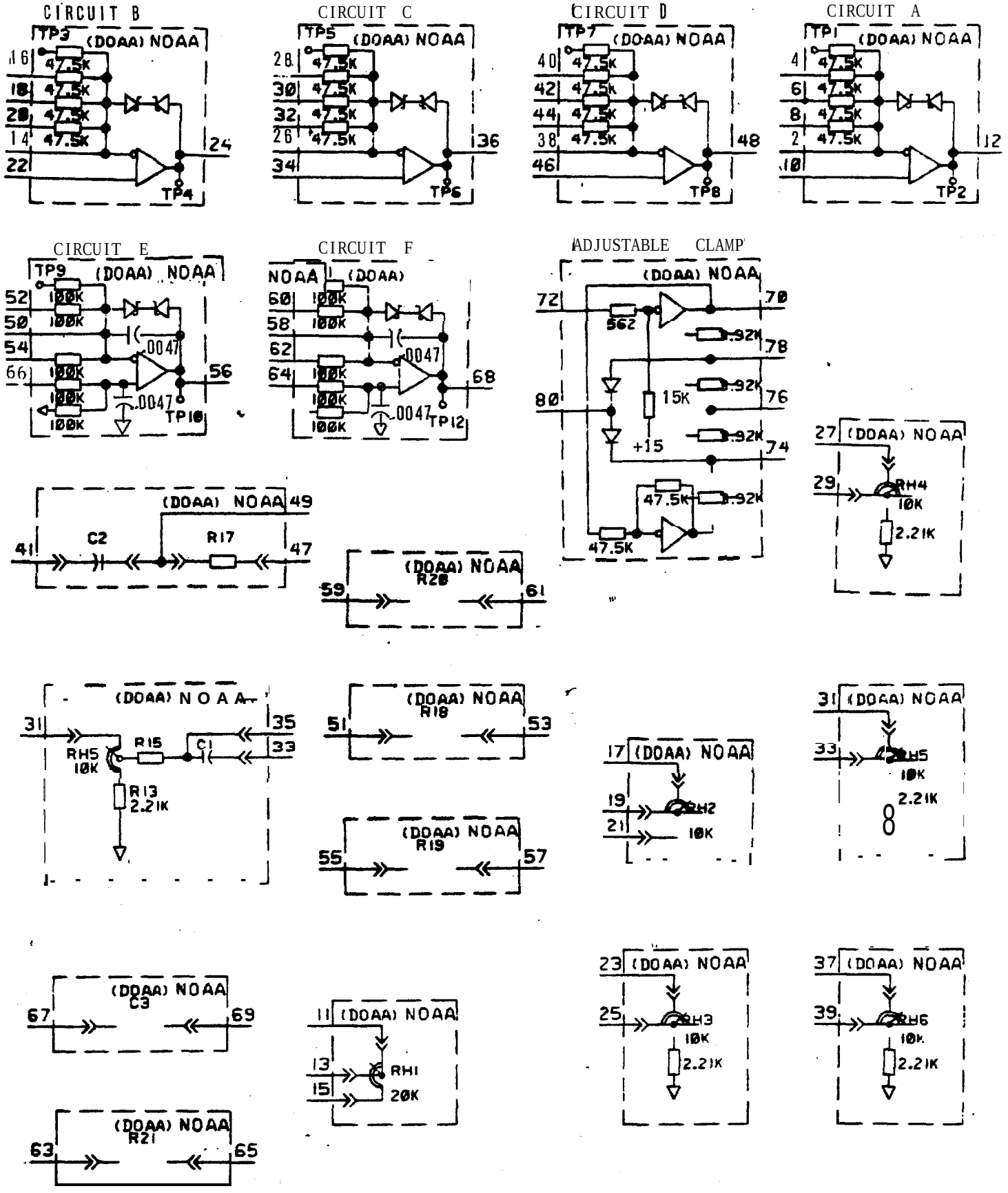
REV.1 OEF MLR	REV.4 5/13/81	REV 5	PRINTS TO DL104 4MEL	APPROVALS MNR	GENERAL ELECTRIC
REV. 2 RE	ISSUED 5-12-81			FIRST MADE FOR	DRIVE SYSTEMS DEPT. SALEM, VA., U.S.A.
REV.3	MADE BY M.L.REICH			I.C. NO.	DS3800NGMA CONT. ON SH. FL SM. NO. 4EA

ELEMENTARY DIAGRAM

NOAA BLOCK DIAGRAM



REV. 1 OEY ALB 5/21/80	REV.4	REV. 7	PRINTS TO DRI DL102	ENGINEER 6-21-79 R.Salibrici	GENERAL ELECTRIC	BLOCK DIAGRAM
REV. 2	REV. 5	ISSUED 8-24-79		FIRST	DRIVE SYSTEMS DEPT	DS3800NOAA
REV. 3	REV. 6	MADE BY R.A.SALIBRICI		MADE FOR DOM +	SALEM, VA. U.S.A.	CONT. ON SH. 2AB SH. NO. 2AA



REV. 1	REV. 4	REV. 7	PRINTS – DL102	ENGINEER 6-21-79 R. Salibrici	GENERAL ELECTRIC
REV. 2	REV. 5	ISSUED 8-24-79			DRIVE SYSTEMS DEPT.
REV. 3	REV. 6	MADE BY R.A.SALIBRICI			SALEM, VA. U.S.A.

DETAILED BLOCK DIAGRAM

DS3800NOAA

CONT. ON SH. 2BA SH. NO. 2AB

2.0 DEFINITION AND PURPOSE

THE NOAA CARD CONSISTS OF SEVERAL DIFFERENT CONFIGURATIONS OF UNCOMMITTED OPERATIONAL AMPLIFIERS. THERE ARE FOUR GENERAL PURPOSE AMPLIFIERS,, AND TWO PRECISION DIFFERENTIAL AMPLIFIERS. ALL

AMPLIFIER OUTPUTS ARE CLAMPED AT HIGH LEVEL SWING, ANO IN ADDITION THE CLAMP LEVEL OF THE FOUR GENERAL PURPOSE AMP-LIFIERS CAN BE DECREASED OR UNBALANCED BY STRAPPING A SPECIAL CIRCUIT INTO THE SELECTED OP-AMP CIRCUIT. AN DP-AMP SATURATION DETECTOR CIRCUIT (IMOK) INDICATES FAULT SATURATION LEVELS ON ALL OP-AMP OUTPUTS.

2.1 THEORY OF OPERATION

REFER TO THE BLOCK DIAGRAM FOR REFERENCE.

2.1.1 GENERAL PURPOSE OP-AMPS

THE CARD CONTAINS FOUR IDENTICAL GENERAL PURPOSE OP-AMP CIRCUITS IDENTIFIED- BY THE LABELS A,B,C AND D, ALL OF WHICH ARE INTEGRATED CIRCUIT DEVICES WITH THE SIGNAL LEADS BROUGHT OUT (EQUIVALENT TO LM747). FOR THIS DISCUSSION THE A LABEL WILL BE USED, BUT THE INFORMATION WILL APPLY TO ALL FOUR . CIRCUITS. ASJA IS THE DP-AMP INVERTING INPUT LEAD, ANIA THE NON-INVERTING INPUT, AND ADTA THE OUTPUT LEAD. THE OUTPUT VOLTAGE *LEVEL OF THE DP-AMP IS CLAMPED TO A MAXIMUM SWING O F ±11.75V±.25V, BEYOND WHICH THE OUTPUT CHANGES VERY LITTLE WITH INCREASING INPUT DRIVE.
ASJA AND ANIA MUST NOT BE DIRECTLY EXPOSED TO VOLTAGES IN EXCESS OF±13.5V, TO PREVENT IRREVERSIBLE DAMAGE TO THE OP-AMPS. THE OUTPUT OF THE AMPLIFIERS IS SHORT - CIRCUIT PROTECTED TD ACOM, P15ORN15, BUT AN EXTENDED SHORT ONONE OR BOTH AMPLIFIERS TO ANY BUS MAY CAUSE THE INTEGRATED CIR-CUIT TD EXCEED SAFE POWER DISSIPATION LIMITS, AND FAIL,' THE OUTPUT CAN DRIVE 10MA OF CURRENT CONTINUOUSLY, WHICH IS EDUIVALENT TO A 1.5KΩ RESISTOR LOAD OVER-THE FULL OUTPUT SWING.

THE OP-AMP INTERNAL CURRENT LIMIT CIRCUITRY MAY CUT IN AT HEAVIER LOADS, INHIBITING ADDITIONAL OUTPUT CURRENT . AND RESULTING IN SIGNAL CLIPPING. AOTA SHOULD NEVER BE EXTERNALLY DRIVEN SINCE IT CAN DAMAGE THE OP-AMPS.

THE NON-INVERTING LEAD, ANIA. MUST BE SHIELDED TO ACOM WHEN BROUGHT OFF THE CARD TO AN EXTERNAL SIGNAL SOURCE, TO PREVENT OSCILLATIONS.

2.1.1 GENERAL PURPOSE OP-AMPS (CONT'D)

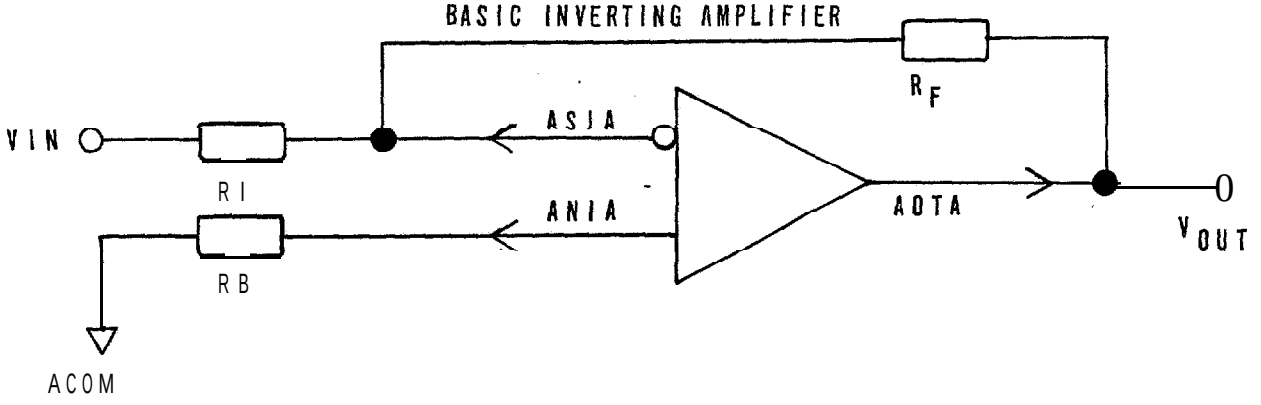
IT IS NOT NECESSARY TO SHIELD WHEN TYING ANIA TO ACOM THROUGH A DAUGHTER-BOARD RESISTOR, SINCE THE BACKPLANE JUMPER WIRE IS SHORT AND DOES NOT LEAVE THE BACKPLANE CARD CONNECTOR. WHEN AN AMPLIFIER IS NOT USED, ANIA SHOULD BE TIED TO ACOM, TO PREVENT THE IMOK CIRCUIT FROM BEING TRIPPED.

A TEST INPUT IS AVAILABLE WHICH CONNECTS TO THE SUMMING JUNCTION, ASIA THROUGH A 47.5KΩ RESISTOR, PROVIDING AN INVERTING I'NPUT, AND A TEST OUTPUT IS PROVIDED WHICH IS THE SAME AS AOTA.

THE FEEDBACK 'RESISTOR (AOTA TO ASIA) SHOULD NOT EXCEED THE PROVIDED 47.5KΩ WITHOUT NEUTRALZING; I.E. CONNECT A SMALL 'CAPACITOR ACROSS THE FEEDBACK RESISTOR, 20pf MINIMUM. THIS WILL PREVENT POSSIBLE OSCILLATION.

SEVERAL AMPLIFIER APPLICATION CONFIGURATIONS WITH ASSOCIATED GAIN EQUATIONS FOLLOW.

BASIC INVERTING AMPLIFIER



$$\frac{V_{OUT}}{V_{IN}} = -\frac{R_F}{R_I}$$

WHERE RI INCLUDES YIN SOURCE RESISTANCE, RS.

$$R_B = \frac{R_F R_I}{R_F + R_I}$$

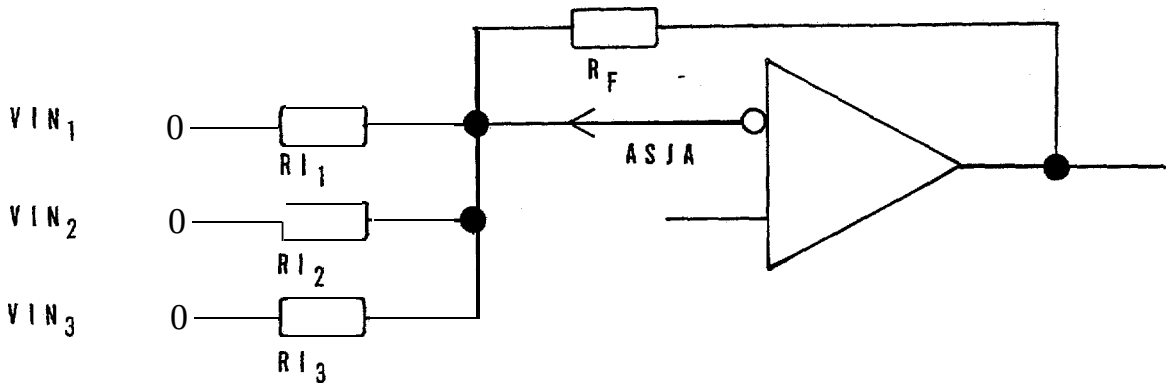
USE TO MINIMIZE THE EFFECTS OF BIAS CURRENTS FROM THE OP-AMPS INPUT LEADS FOR LOW OFFSET.

REV. 1	REV. 4	REV. 7	PRINTS TO DL102 PRI	ENGINEER 6-21-79 R.A.Salibrici	GENERAL ELECTRIC DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	APPLICATION NOTES DS3800NOAA CONT. ON SH. 2DA SH. NO. 2CA
REV. 2	REV. 5	ISSUED 8-24-79				
REV. 3	REV. 6	MADE BY R.A.SALIBRICI				

LEAD CONNECTIONS

AFBA	CONNECTS TO ASJA THROUGH A 47.5KΩ RESISTOR ON THE CARD, AND CAN BE USED AS RI OR RF.
A1IA1	CONNECTS TO ASIA THROUGH A 47.5KΩ RESISTOR ON THE CARD, AND CAN BE USED AS RI OR RF.
A1IA2	CONNECTS TO ASJA THROUGH A 47.5KΩ RESISTOR ON THE CARD, AND CAN BE USED AS RI OR RF.
ANIA	MUST BE EXTERNALLY WIRED TO THE DAUGHTER-BOARD FOR RB WHEN NECESSARY FOR LOW OFFSET VOLTAGE. IF RB IS NOT NEEDED, TIE ANIA DIRECTLY TO ACOM.
ASJA	MAY BE EXTERNALLY WIRED TO THE DAUGHTER-BOARD FOR RF OR RI IF DESIRED.

SUMMING AMPLIFIER



THE SUMMING AMPLIFIER IS WIRED THE SAME AS THE BASIC INVERTING AMPLIFIER EXCEPT THAT MULTIPLE INPUT SIGNALS ARE COUPLED TO THE SUMMING JUNCTION, ASJA, THROUGH INPUT RESISTORS, RI, $-RI_3$.

REV. 1	REV. 4	REV. 7	PRINTS TO DRI DL102	ENGINEER 6-21-79 R. Salibrici	GENERAL ELECTRIC	APPLICATION NOTES
REV. 2	REV. 5	ISSUED 8-24-79			DRIVE SYSTEMS DEPT.	DS3800NOAA
REV. 3	REV. 6	MADE BY R.A.SALIBRICI			SALEM, VA. U.S.A.	CONT. ON SH. 2EA SH. NO. 2DA

DISD 0151 (09-76)

CARS

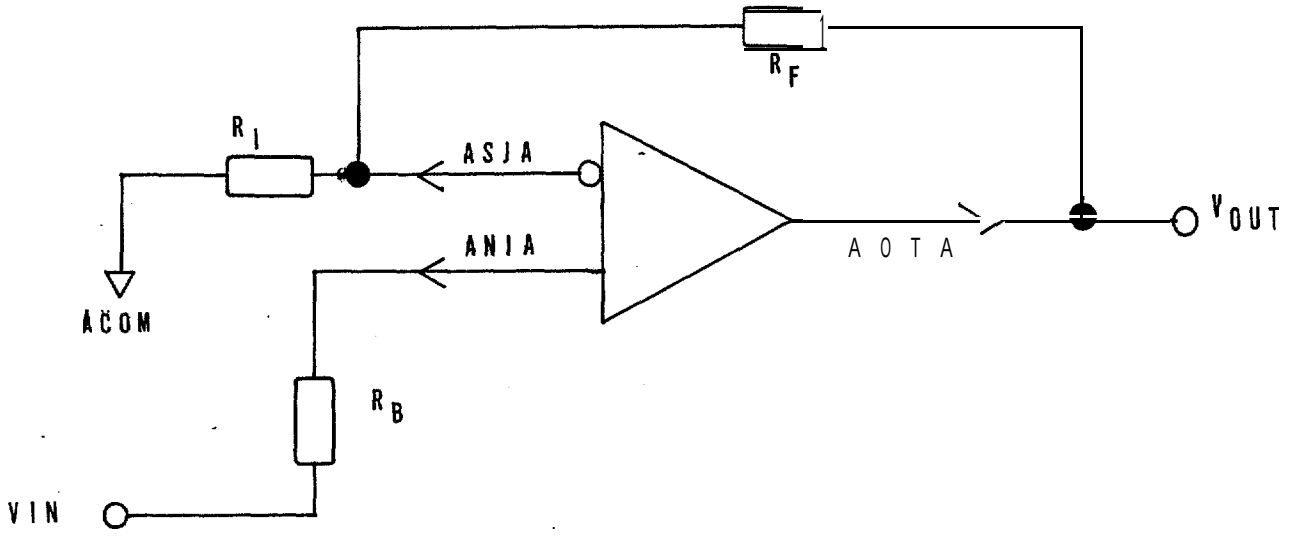
$$V_{OUT} = \left(-VIN_1 \frac{R_F}{RI_1}\right) + \left(-VIN_2 \frac{R_F}{RI_2}\right) + \left(-VIN_3 \frac{R_F}{RI_3}\right)$$

ALL OTHER CONSIDERATIONS ARE THE SAME AS FOR THE BASIC INVERTING AMPLIFIER, WITH $R_B = \frac{RI_\tau \; RF}{RI_\tau + R\,F}$ WHERE RI_τ IS THE PARALLEL RESULTANT VALUE OF $RI_1 RI_2 RI_3$

$$RI_\tau = \frac{1}{1/RI_1 + 1/RI_2 + 1/RI_3}$$

ANY NUMBER OF INPUTS CAN BE SUMMED.

BASIC NON-INVERTING AMPLIFIER



$$\frac{V_{OUT}}{V_{IN}} = \frac{R_F}{R_I} + 1$$

$$R_B = \frac{R_F \; R_I}{R_F + R_I} - R_S$$

USE TO MINIMIZE THE AFFECTS OF BIAS CURRENTS FROM THE BP-AMP INPUT LEADS. RS IS THE V_{IN} SOURCE RESISTANCE.

REV. 1	REV. 4	REV. 7	PRINTS TO PRI DL102	ENGINEER 8-21-79 R Salibrici	GENERAL ELECTRIC	APPLICATION NOTES
REV. 2	REV. 5	ISSUED 8-24-79			DRIVE SYSTEMS DEPT.	DS3800N0AA
REV. 3	REV. 6	MADE BY R.A.SALIBRICI			SALEM, VA. U.S.A.	CONT. ON SH. 2FA SH. NO. 2EA

DSD 0151 (09-76)

CARS

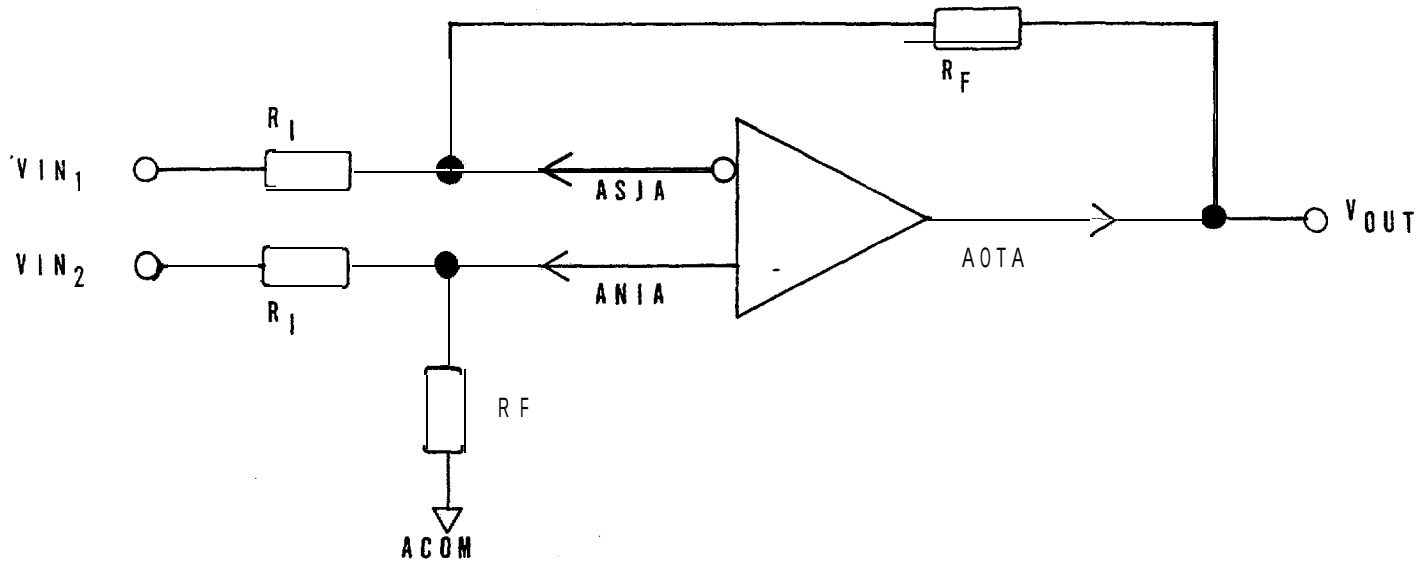
LEAD CONNECTIONS

AFBA, A1IA1, A1IA2 — CONNECT TO ASJA THROUGH A 47.5Kf-I RESISTOR ON THE CARD, AND CAN BE USED AS RI OR RF.

ASJA — MAY BE EXTERNALLY WIRED TO THE DAUGHTER BOARD FOR RF OR RI IF DESIRED.

ANIA — MUST BE EXTERNALLY WIRED TO THE DAUGHTER BOARD FOR RB WHEN NECESSARY FOR LOW OFFSET VOLTAGE. IF RB IS NOT NEEDED, TIE ANIA DIRECTLY TO THE INPUT.

BASIC DIFFERENTIAL AMPLIFIER



$$V_{OUT} = (V_{IN_2} - V_{IN_1}) \frac{RF}{R_I}$$

WHERE RI INCLUDES VIN SOURCE RESISTANCE, WHICH SHOULD BE THE SAME FOR BOTH.

REV. 1	REV. 4	REV. 7	PRINTS TO DL102	ENGINEER 6-21-79 R. Salibrici	GENERAL@ ELECTRIC DRIVE SYSTEMS DEPT.	APPLICATION NOTES
REV. 2	REV. 5	ISSUED 8-24-79				DS3800NOAA
REV. 3	REV. 6	MADE BY R.A.SALIBRICI			SALEM, VA. U.S.A.	CONT. ON SH. 2GA SH. NO. 2FA

DSD 0151 (09-76)

CARS

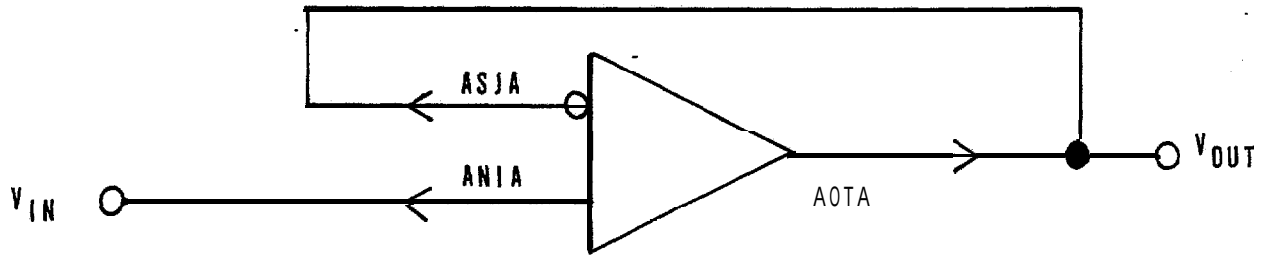
BASIC DIFFERENTIAL AMPLIFIER (CONT'D)

THE DIFFERENTIAL AMPLIFIER REJECTS COMMON MODE INPUT ($VIN_1 = VIN_2$, OR $VIN_2 - VIN_1 = 0$) AND AMPLIFIES DIFFERENTIAL INPUT ($VIN_1 \neq VIN_2$, OR $VIN_2 VIN_1 \neq 0$) BETTER MATCHING BETWEEN THE TWO SETS OF RESISTORS IMPROVES COMMON MODE REJECTION, AND MINIMIZES OFFSETS DUE TO BIAS CURRENTS ON THE OP-AMP INPUT LEADS.

LEAD CONNECTIONS

AFBA, A1IA1, A1IA2	CONNECTS TO ASJA THROUGH A 47.5KΩ RESISTOR ON THE CARD, AND CAN BE USED AS RF OR RI.
ASJA	MAY BE EXTERNALLY WIRED TO THE DAUGHTER-BOARD FOR RF OR RI IF DESIRED.
ANIA	MUST BE EXTERNALLY WIRED TO THE DAUGHTER-BOARD FOR RI AND RF.

BASIC VOLTAGE FOLLOWER



$$V_{OUT} = V_{IN}$$

IF THE V_{IN} SIGNAL SOURCE HAS SIGNIFICANT RESISTANCE TO GROUND, ASJA MAY BE TIED TO AOTA THROUGH A MATCHING RESISTANCE TO NEUTRALIZE BIAS CURRENTS ON THE INPUT LEADS FOR LOW OFFSET VOLTAGE.

REV. 1	REV. 4	REV. 7	PRINTS TO PRI DL102	ENGINEER 8-21-79 R. Salibrici	GENERAL ELECTRIC	APPLICATION NOTES
REV. 2	REV. 5	ISSUED 8-24-79			DRIVE SYSTEMS DEPT.	DS3800NOAA
REV. 3	REV. 6	MADE BY R.A.SALIBRICI			SALEM, VA. U.S.A.	CONT. ON SH. 2HA SH. NO. 2GA

DSD 0151 (09-76)

CARS

2.1.2 PRECISION INVERTING OP-AWP

THE PRECISION DIFFERENTIAL DP-AMP CIRCUITS CAN BE CONSIDERED THE SAME AS A GENERAL PURPOSE OP-AMP, AND IMPLEMENTED PER THAT DISCUSSION (2.1.1) WITH **THE FOLLOWING EXCEPTIONS :**

1) THESE AMPLIFIERS ARE LABELED E AND F.

2) THE NON-INVERTING OP-AMP LEAD IS TIED TO ACOM ON THE CARD THROUGH 100KΩ AND ALSO BROUGHT OUT THROUGH ANOTHER 100K AS ANIE,F.

3) THE EQUIVALENT OP-AMP INTEGRATED CIRCUIT DEVICE IS THE LM308A.

4) THE OUTPUT VOLTAGE CLAMP LEVEL IS ±11.5V±0.25V.

5) THE OUTPUT CAN DRIVE 4MA OF CURRENT CONTINUOUSLY, OR A 2.74KΩ RESISTOR LOAD OVER A ±11.0V SWING.

6) THE TEST INVERTING INPUTS ARE TIED TO ASJE,F THROUGH A 100KΩ RESISTOR; TP9 AND TP11).

7) AFBE AND AIIE ARE TIED TO ASJE THROUGH 100K RESISTORS.

8) THIS AMPLIFIER HAS LOWER BIAS CURRENTS ON THE OP-AMP INPUT LEADS, LOWER OFFSET VOLTAGE, LOWER DRIFT WITH TEMPERATURE, AND HIGHER ACCURACY THAN THE GENERAL PURPOSE OP-AMPS.

9) WITH INPUTS ON AIIE(F) AND ANIE(F), AND AFBE (F) TIED TO AOTE (F), THE AMPLIFIERS -HAVE A DIFFERENTIAL GAIN OF ONE.

10) WHEN USED AS A SINGLE INPUT AMPLIFIER, ANIE(F) OR AIIE(F) CANBE DRIVEN INDIVIDUALLY WITH THE OTHER LEAD TIED TO GROUND. REMEMBER THAT IN THIS MODE, VOLTAGE SIGNALS APPLIED TO ANIE ARE DIVIDED IN HALF BY THE TWO 100KΩ RESISTORS WIRED TO THE NON-INVERTHNG INPUT OF THE OP-AMP, ONE OF WHICH GOES TO GROUND.

11) WHEN USED IN THE DIFFERENTIAL MODE THE CLAMPING ACTION ON THE OUTPUT WILL BE UNBALANCED, BUT THE TOTAL SWING WILL STILL BE 2 TIMES (11.5V±.25V).

12) THE FEEDBACK RESISTOR (A O T E T O ASJE) I s ALREADY NEUTRALIZED ON THE CARD BY A 0.0047uf, AND THE 100KΩ TO GROUND ON THE NONINVERTING INPUT IS ALSO BYPASSED By .0047uf.

REV. 1	REV. 4	REV. 7	PRINTS TO DL102	ENGINEER 8-21-79 R Salibrici	GENERAL ELECTRIC
REV. 2	REV. 5	ISSUED 8-24-79			DRIVE SYSTEMS DEPT.
REV. 3	REV. 6	MADE BY R.A.SALIBRICI			SALEM, VA. U.S.A.

APPLICATION NOTES

DS3800N0AA

CONT. ON SH. 2JA SH. NO. 2HA

2. T.3 ADJUSTABLE OP-AMP CLAMP CIRCUIT

THIS CIRCUIT PROVIDES THE CAPABILITY TO VARY THE OP-AMP OUTPUT SWING CLAMP LEVEL ON THE GENERAL PURPOSE AMPLIFIERS DISCUSSED IN SECTION 2.1.1 BY CONNECTING TWO LEADS OF THIS CIRCUIT TO THE APPROPRIATE POINTS ON THE DESIRED OP-AMP CIRCUIT. IN ADDITION, THE ± SYMMETRY OF THE CLAMP (OUTPUT SWING IS NORMALLY CLAMPED. AT THE SAME MAGNITUDE OF POSITIVE AND NEGATIVE VOLTAGES) CAN BE UNBALANCED BY CONNECTING RESISTANCE, RU INTO THIS CIRCUIT.

TO OPERATE, CONNECT A RESISTOR R BETWEEN AVCA1 AND AVCA2, WHICH ESTABLISHES THE CLAMP LEVEL, TIE AVCOTT TO THE OUTPUT OF THE DESIRED OP-AMP (AOTA,B,C,D) AND TIE AVCIIT TO THE SUMMING JUNCTION OF THE SAME OP-AMP (ASJA,B,C,D). THE NEW CLAMP LEVEL FOR THAT OP-AMP WILL BE [R (KΩ) + 0.562 + 1.4] VOLTS, ±.25V, WHERE R (KΩ) IS THE VALUE IN KILOHMS OF THE RESISTOR BETWEEN AVCA1 AND AVCA2.

THE CLAMPING ACTION WILL STILL BE SYMETRICAL,POSITIVE AND NEGATIVE, AND NOTE THAT THE CLAMP LEVEL CANNOT BE INCREASE0 BEYOND THE PREWIRED VALUE (±11.75 ± .25V), ONLY DECREASED TOWARDS ZERO VOLTS.

TO UNBALANCE THE CLAMPING ACTION, TIE A RESISTOR, RU, BETWEEN AVCU1 AND AVCOTT FOR THE NEGATIVE CLAMP LEVEL ADJUSTMENT, OR BETWEEN AVCU2 AND AVCOTT FOR THE POSITIVE CLAMP LEVEL ADJUSTMENT.

THE NEW CLAMP LEVEL FOR THE AFFECTED POLARITY WILL BE

$$\left[\frac{(R\ (K\Omega) + 0.562 + 0.7)}{3.92}\right] X \left[\frac{3.92 \times RU\ (K\Omega)}{3.92 + RU\ (K\Omega)}\right] + 0.7$$

VOLTS, ± 0.25V, WHERE R(KΩ) IS THE RESISTANCE BETWEEN AVCA1 AND AVCAP IN KILOHMS, AND RU (KΩ) IS THE UNBALANCE RESISTANCE BETWEEN AVCU1 OR AVCU2 AND AVCOTT, IN KILOHMS. NOTE THAT THE QUANTITY IN BRACKETS AT THE RIGHT OF THE PAGE IS JUST 3.92 IN PARALLEL WITH RU (KΩ). THE OTHER POLARITY WILL NOT BE AFFECTED, STILL CLAMPING AT [R(KΩ) + 0.562 + 1.4] VOLTS ±0.25V. BOTH POLARITIES CAN BE VARIED SIMULTANEOUSLY BY THE UNBALANCE METHOD. THE ADJUSTABLE CLAMP IS RESTRICTED TO USEAGE WITH ONE GENERAL PURPOSE OP-AMP CIRCUIT AT A TIME ON THE NOAA CARD. VARIABLE RESISTANCE ON THE DAUGHTER-BOARD MAY BE USED FOR R(KΩ) OR RU (KΩ). LEADS OF THE ADJUSTABLE CLAMP CIRCUIT SHOULD NOT BE EXTERNALLY DRIVEN.

REV. 1	REV. 4	REV. 7	PRINTS TO PRI DL102	ENGINEER 6-21-79 TR Saldrici	GENERAL ELECTRIC DRIVES SYSTEMS DEPT. SALEM, VA. U.S.A.
REV. 2	REV. 5	ISSUED 8-24-79			APPLICATION NOTES DS3800NOAA CONT. ON SH. 2KA SH. NO. 2JA
REV.3	REV. 6	MADE BY R.A.SALIBRICI			

DSD 0151 (09-76)

CARD

2.1.4 Dp-AMP SATURATION OETECTOR

A SATURATION DETECTOR CIRCUIT- THAT MONITORS ON-BOARD OP-AMPS FOR FAULTED OUTPUT CONDITIONS IS PROVIDED ON THE CARD. THE DETECTOR INDICATOR LED, IMOK (YELLOW), IS NORMALLY ON AND EXTINGUISHES WHEN A MONITORED OP-AMP OUTPUT REACHES A FAULT SATURATION LEVEL, AN, Op-AMP CAN BE OVERDRIVEN BY A FACTOR OF TWO (TWICE THE INPUT VOLTAGE THAT JUST CAUSES THE OUTPUT TO CLAMP APPLIED TO THE INPUT FOR ANY INVERTING GAIN. CONFIGURATION) WITHOUT TRIPPING THE SATURATION DETECTOR. THE IMOK LED DOES NOT LATCH IN THE TRIPPED (OFF) STATE.

2.2 INTERFACE REQUIREMENTS

OP-AMP OUTPUT VOLTAGES RANGE OVER ±12V MAXIMUM. INPUT VOLTAGES SHOULD NOT EXCEED ±13.5V TO ACOM. OUTPUT DRIVE CAPABILITY IS 10MA OR A 1.5KΩ LOAD OVER THE FULL SWING FOR GENERAL PURPOSE OP-AMPS, AND 4MA OR A 2.74KΩ LOAD OVER A ±11.0V SWING FOR THE PRECISION OP-AMPS. OUTPUTS SHOULD NOT BE EXTERNALLY DRIVEN.

2.3 FUNCTIONAL DESCRIPTION

REFER TO THE BLOCK DIAGRAM FOR REFERENCE.

2.3.1 INPUTS

MNEMONIC	PIN NO..	DESCRIPTION
AFBA	PA4	Dp-AMP FEEDBACK LEAD FOR AMPLIFIER A
AFBB	PA16	OP-AMP FEEDBACK LEAD FOR AMPLIFIER B
AFBC	PA26	OP-AMP FEEDBACK LEAD FOR AMPLIFIER C
AFBD	PA40	OP-AMP FEEDBACK LEAD FOR AMPLIFIER D
AFBE	PA52	OP-AMP FEEDBACK LEAD FOR AMPLIFIER E
AFBF	PA60	Op-AMP FEEDBACK LEAD FOR AMPLIFIER F
AIIA1	PA6	OP-AMP INVERTING INPUT #1 FOR AMP. A
AI IA2	PA8	Dp-AMP INVERTING INPUT #2 FOR AMP. A
AIIB1	PA18	OP-AMP INVERTING INPUT #1 FOR AMP. B
AI IB2	PA20	OP-AMP INVERTING INPUT #2 FOR AMP. B
AIIC1	PA30	OP-AMP INVERTING INPUT #1 FOR AMP. C
AI IC2	PA32	Op-AMP INVERTING INPUT #2 FOR AMP. C
AIID1	PA42	OP-AMP INVERTING INPUT #1 FOR AMPL. D
AII02	PA44	OP-AMP INVERTING INPUT #2 FOR AMP: D
AI IE	PA54	OP-AMP INVERTING INPUT FOR AMP: E
AI IF	PA62	OP-AMP INVERTING INPUT FOR AMP: F
ASJA	PA2	OP-AMP SUMMING JUNCTION FOR AMP: A

REV. 1	REV. 4	REV. 7	PRINTS TO PPI DL102	ENGINEER 6-21-79 R Salibrici	GENERAL ELECTRIC DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.
REV. 2	REV. 5	ISSUED 8-24-79			APPLICATION NOTES
REV. 3	REV. 6	MADE BY R.A.SALIBRICI			DS3800NOAA CONT. ON SH. 2LA SH. NO. 2KA

DSD 0151 (09-76)

2.3.1 INPUTS (CONT'D)

MNEMONIC	PIN NO.	DESCRIPTION
ASJB	PA14	OP-AMP SUMMING JUNCTION FOR AMP. B
ASJC	PA26	OP-AMP SUMMING JUNCTION FOR AMP. C
ASJD	PA38	OP-AMP SUMMING JUNCTION FOR AMP. D
ASJE	PA50	OP-AMP SUMMING JUNCTION FOR AMP. E
ASJF	PA58	DP-AMP SUMMING JUNCTION FOR AMP. F
ANIA	PA10	OP-AMP NON-INVERTING INPUT FOR AMP.A
ANIB	PA22	OP-AMP NON-INVERTING INPUT FOR AMP.B
ANIC	PA34	OP-AMP NON-INVERTING INPUT FOR AMP.C
ANID	PA46	OP-AMP NON-INVERTING INPUT FOR AMP.D
ANIE	PA66	OP-AMP NON-INVERTING INPUT #1 FOR AMP.F
ANIF1	PA64	OP-AMP NON-INVERTING INPUT #2 FOR AMP.F
AVCOTT	PA76	VARIABLE CL-AMP OUTPUT TIE LEAD
AVCA1	PA70	VARIABLE CLAMP ADJUSTMENT LEAD #1
AVCA2	PA72	VARIABLE CLAMP ADJUSTMENT LEAD #2
AVCU1	PA74	VARIABLE CLAMP UNBALANCE LEAD #1
AVCUZ	PA78	VARIABLE CLAMP UNBALANCE LEAD #2

2.3.2 OUTPUTS

MNEMONIC	PIN NO.	DESCRIPTION
AOTA	PA12	OP-AMP OUTPUT FDR AMPLIFIER A
AOTB	PA24	OP-AMP OUTPUT FOR AMPLIFIER B
AOTC	PA36	OP-AMP OUTPUT FOR AMPLIFIER C
AOTD	PA48	OP-AMP OUTPUT FOR AMPLIFIER D
AOTE	PA56	OP-AMP OUTPUT FOR AMPLIFIER E
AOTF	PA68	OP-AMP OUTPUT FOR AMPLIFIER F
AVCIIT	PA80	VARIABLE CLAMP INVERTING INPUT TIE LEAD

2.3.3 LEO INDICATORS

THERE IS ONLY ONE LED INDICATOR ON THE CARD.

MNEMONIC	COLOR	DIODE NO.	DESCRIPTION
IMOK	YELLOW	CR30	NORMALLY ON;OP-AMP SATURATION DETECTION

2.3.4 MAINTENANCE TEST POINTS

THERE ARE THIRTEEN MAINTENANCE TEST POINTS ON THE FRONT EDGE OF THE CARD.

TEST POINT	DESCRIPTION
TP1	TEST INVERTING INPUT FOR AMPLIFIER A
TP2	ROTA TEST OUTPUT
TP3	TEST INVERTING INPUT FOR AMPLIFIER B
TP4	AOTB TEST OUTPUT
TP5	TEST INVERTING INPUT FOR AMPLIFIER C
TP6	AOTC TEST OUTPUT
TP7	TEST INVERTING INPUT FOR AMPLIFIER D
TP8	AOTD TEST OUTPUT
TP9	TEST INVERTING INPUT FOR AMPLIFIER E.
TP10	AOTE TEST OUTPUT
TP11	TEST INVERTING INPUT FOR AMPLIFIER F
TP12	AOTF TEST OUTPUT
TP13	ACOM (ANALOG GROUND)

2.3.5 MAINTAINABILITY

THE NOAA REQUIRES NO MAINTENANCE OTHER THAN INITIAL CALIBRATION. THE CARD MAY BE PERIODICALLY RECALIBRATED.

CARS

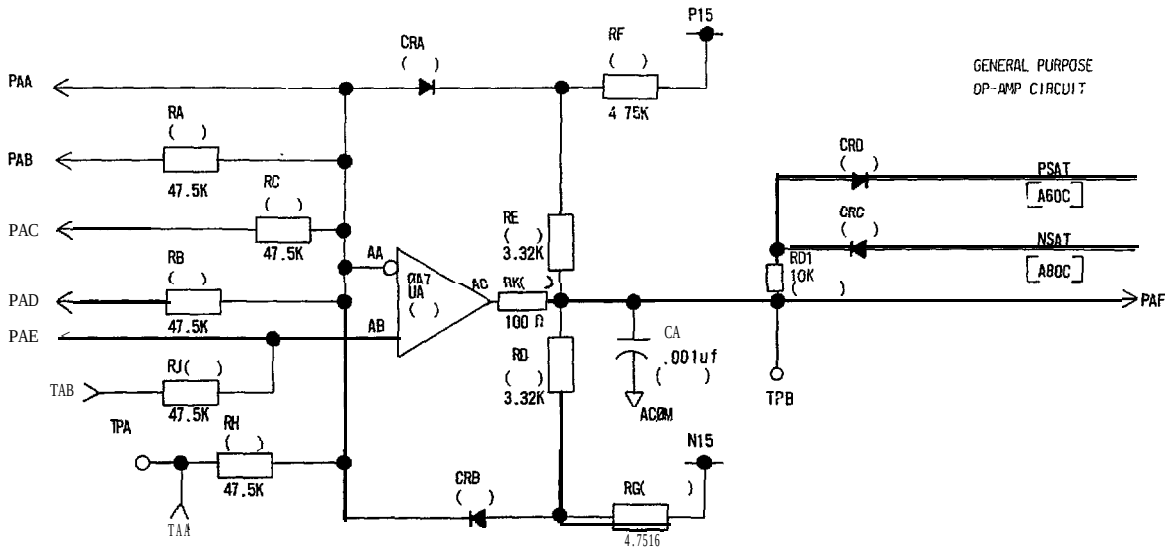
2.4 APPLICATION DATA

2.4.1 POWER REQUIREMENTS

VOLTS	AMPS	WATTS
+15V ± 1%	.045	0.675
-15v ± 1%	050	0.750

2.4.2 INSTALLATIONS PROCEOURE AND ADJUSTMENTS

SEE SECTION 2.1, THEORY OF OPERATION.



I/O PINS							
CKT	PAA	PAB	PAC	PAD	PAE	PAF	
A	2 ASJA	4 AFBA	6 A11A1	8 A11A2	10 ANIA	12 AOTA	
B	14 ASJB	16 AFBB	18 A11B1	20 A11B2	22 ANIB	24 AOTB	
C	26 ASJC	28 AFBC	30 A11C1	32 A11C2	34 ANIC	36 AOTC	
D	38 ASJD	40 AFBD	42 A11D1	44 A11D2	46 ANID	48 AOTD	

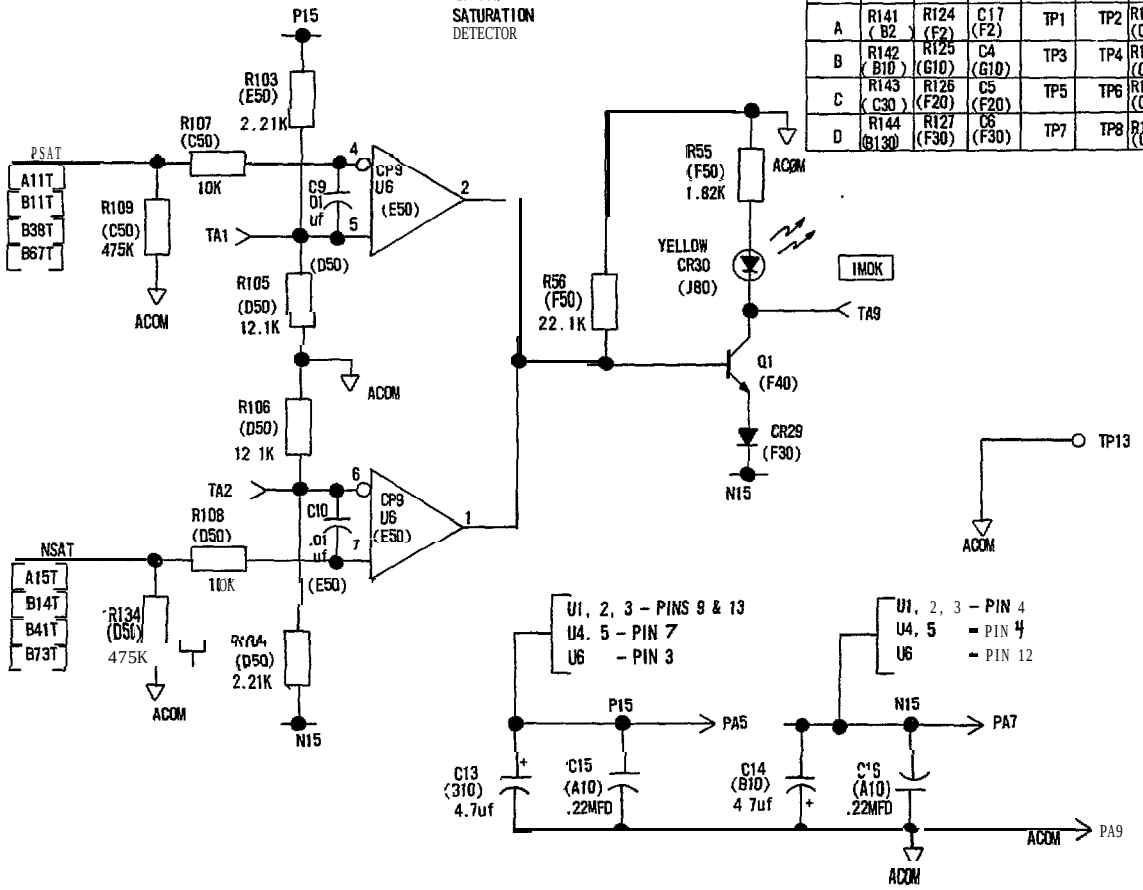
DIODES			
CRA	CRB	CRC	CRD
CR1 (D2)	CR5 (C2)	CR9 (F2)	CR13 (G2)
CR2 (D10)	CR6 (C10)	CR10 (G10)	CR14 (H10)
CR3 (D20)	CR7 (C20)	CR11 (F20)	CR15 (G20)
CR4 (D30)	CR8 (C30)	CR12 (G30)	CR16 (G30)

RESISTORS										
RA	RB	RC	RD	RE	RF	RG	RD1	UA	AA	AB
R25 (B2)	R29 (C2)	R33 (B2)	R37 (E2)	R41 (E2)	R45 (D2)	R49 (C2)	R110 (E2)	U1 (E10)	1	2
R26 (A10)	R30 (B10)	R34 (B10)	R38 (F10)	R42 (E10)	R46 (C10)	R50 (C10)	R111 (F10)	U1 (E10)	7	6
R27 (B20)	R31 (C20)	R35 (B20)	R39 (E20)	R43 (E20)	R47 (D20)	R51 (C20)	R112 (E20)	U2 (E30)	1	2
R28 (B30)	R32 (C30)	R36 (B30)	R40 (F30)	R44 (E30)	R48 (D30)	R52 (C30)	R113 (F30)	U2 (E30)	7	6

AC
12
10
12
10

CKT	RJ	RK	CA	TPA	TPB	RH
A	R141 (B2)	R124 (F2)	C17 (F2)	TP1	TP2	R118 (C2)
B	R142 (B10)	R125 (G10)	C4 (G10)	TP3	TP4	R119 (C10)
C	R143 (C30)	R126 (F20)	C5 (F20)	TP5	TP6	R120 (C20)
D	R144 (B130)	R127 (F30)	C6 (F30)	TP7	TP8	R121 (C30)

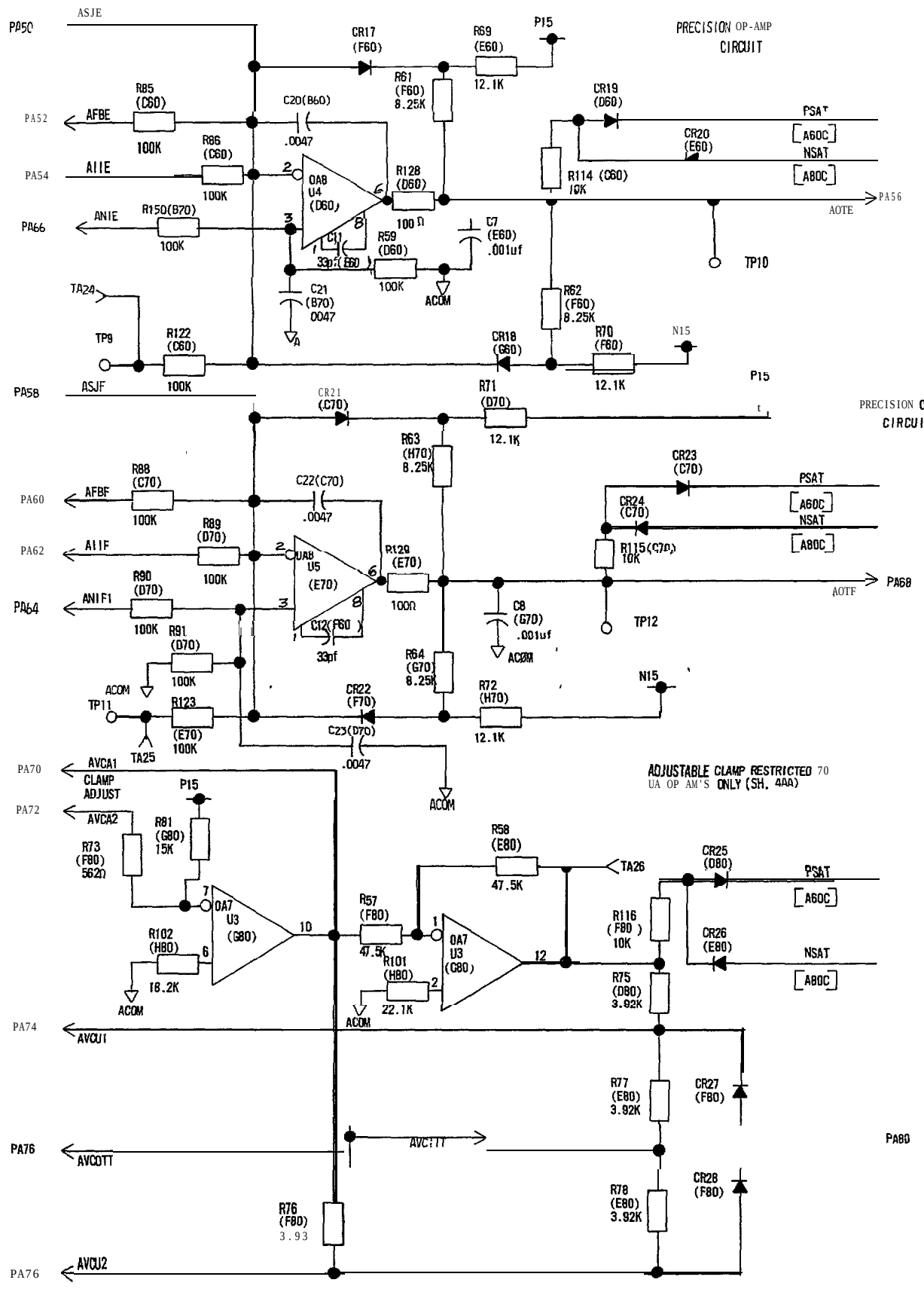
TAA	TAB
4	21
5	14
3	12
6	7



EV.1 OEE WRuch 10/25/77	REV. 4	REV. 5	PRINTS TO PRI 53-36, TRO, DL104 BHEL	ENGINEER 8-22-77 R9 Salbries	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.
EV.2 OEL JVG 780630	ISSUED August 22, 1977				DS 3800N0AA
EV.3 OFJ WB 820913	MADE BY O. RHODES		DOM PLUS		CONT. ON SH. 4BA SH. NO. AA

ELEMENTARY DIAGRAM
GENERAL PURPOSE OP AMPS

D 8152 (09-76)



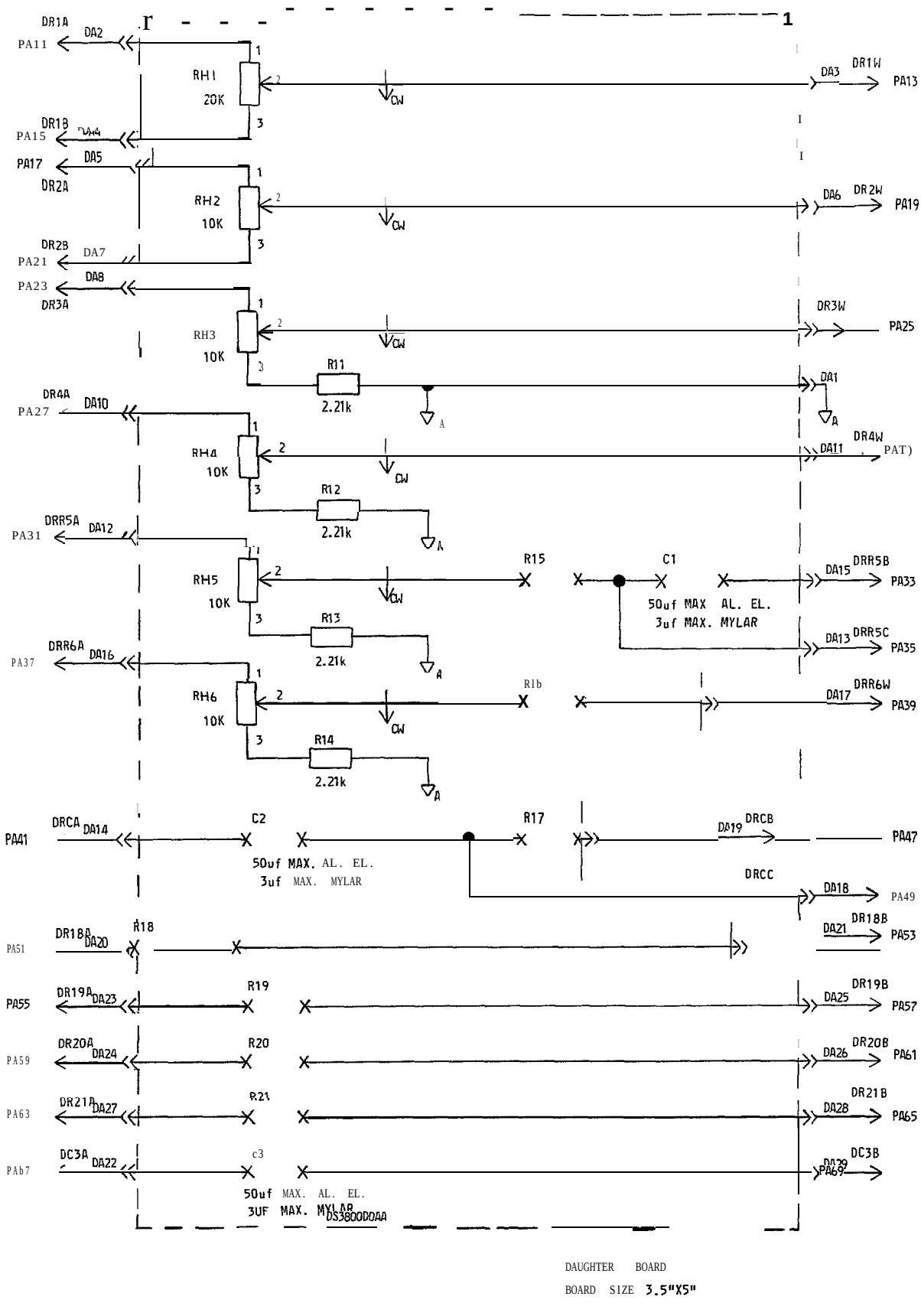
REV 1 CES mRich 10/25/77	REV 4 OEJS [illegible] 12/16/77	REV.5 OFJ QAB 820913	PRINTS TO PRI 53-36, TRQ, DL104 BHEL	ENGINEER 8-22-77 R.A. Sahlmen	GENERAL ELECTRIC
REV 2 OEL JVG 6/30/78	ISSUED August 22, 1977			DOM PLUS	DRIVE SYSTEMS DEPARTMENT
REV.3 OEQ [illegible] 781103	MADE BY O. RHODES				SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM
GENERAL PURPOSE OP AMPS

DS380ONDAA

(CONT. ON SH. 4CA SH. NO. 4BA

DSD 0152 (09 76)

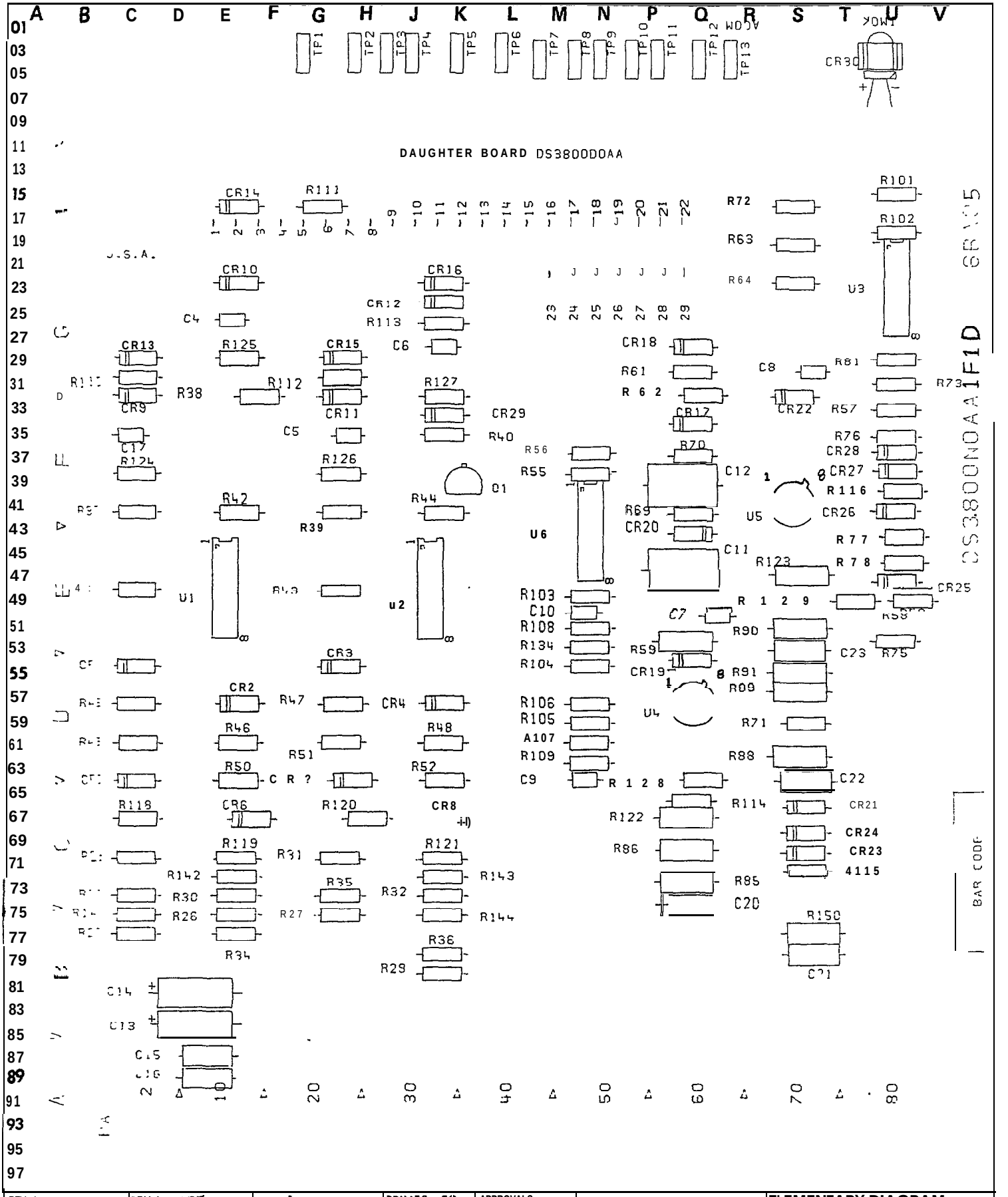


DAUGHTER BOARD

BOARD SIZE 3.5"X5"

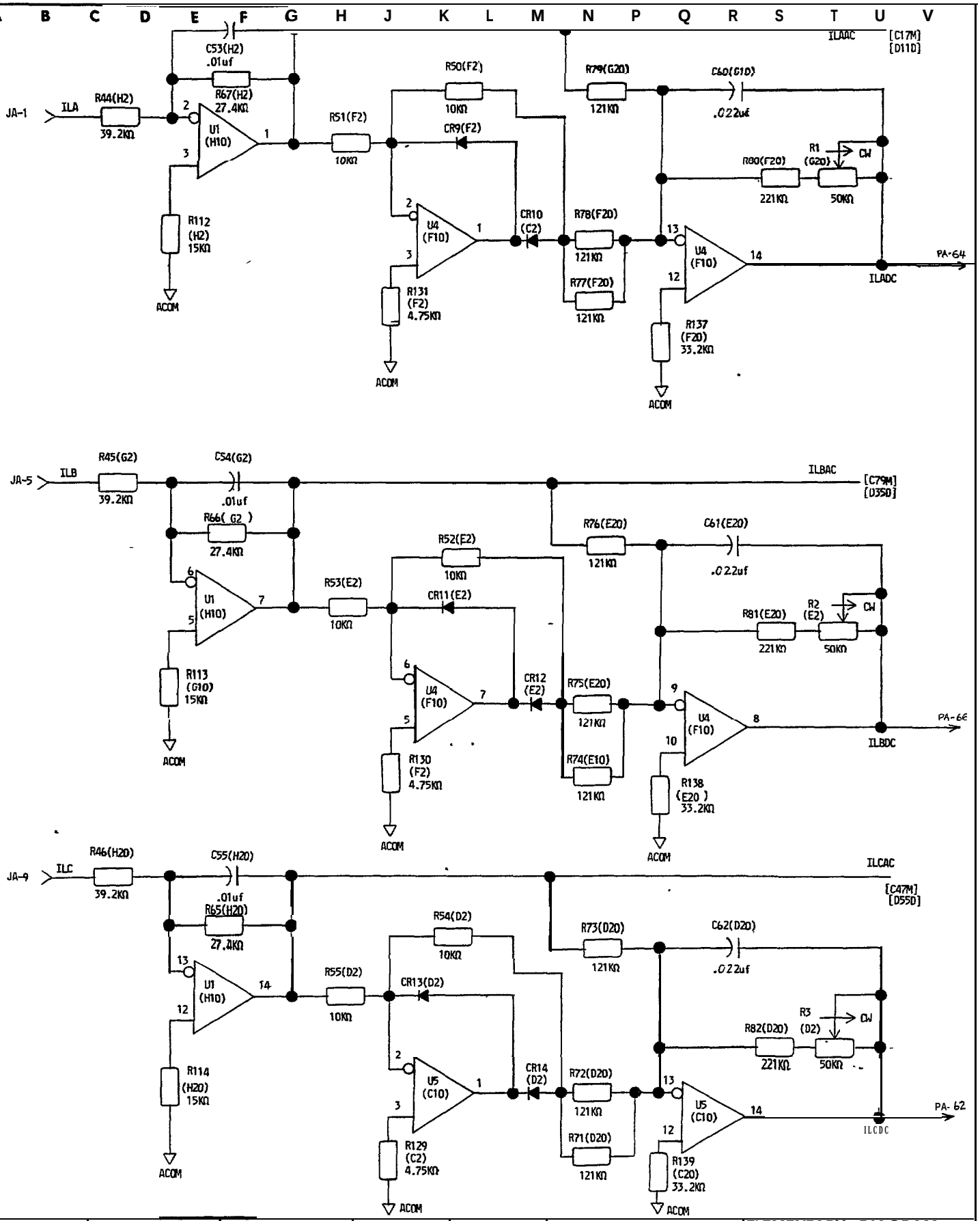
REV 1 MLO 780824	REV.4	REV 5	PRINTS TO PRI BHEL DL104	ENGINEER 8-22-77 R.A. Salibrici	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM DAUGHTER BOARD
REV.2 OEW SHD 6/27/79	ISSUED August 25 1977			DOM PLUS		DS3800N0AA
REV 3	MADE BY ROBERT A. SALIBRICI					CONT. ON SH. 4DA SH. NO. 4CA

DSD 0152 (09-76)

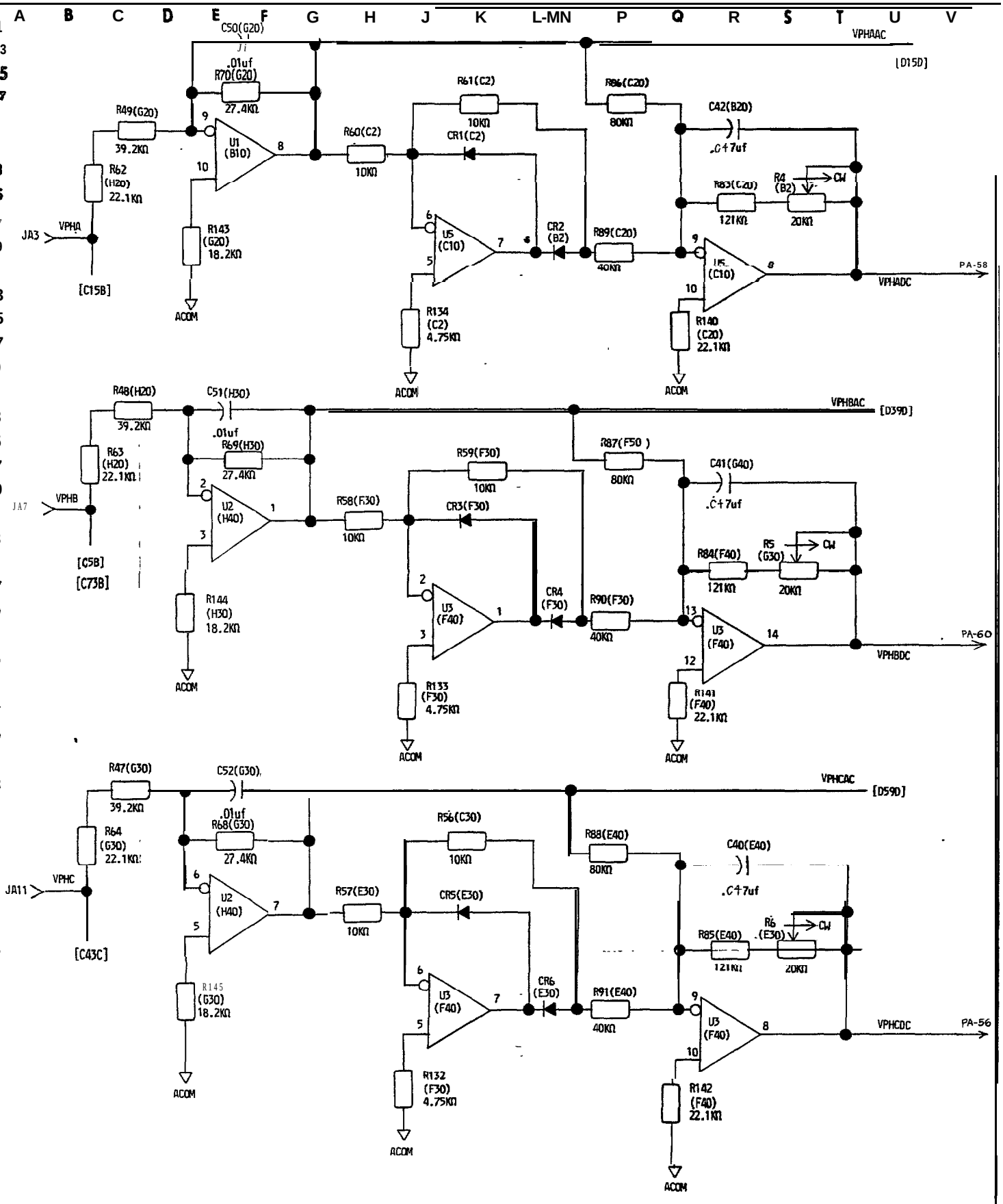


REV 1	REV 4 OFJ alb 4/20 4, 13	REV 5	PRINTS TO CAEL D2104 PRI	APPROVALS ab	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV2 RE ISSUED	April 16, 1982			FIRST MADE FOR	DRIVE SYSTEMS DEPT. SALEM, VA., U.S.A.	DS3800NOAA
REV.3 MOTEC TEIC 244	MADE BY A. Boehler			I.C. NO.		CONT. ON SH. FNL SHEET

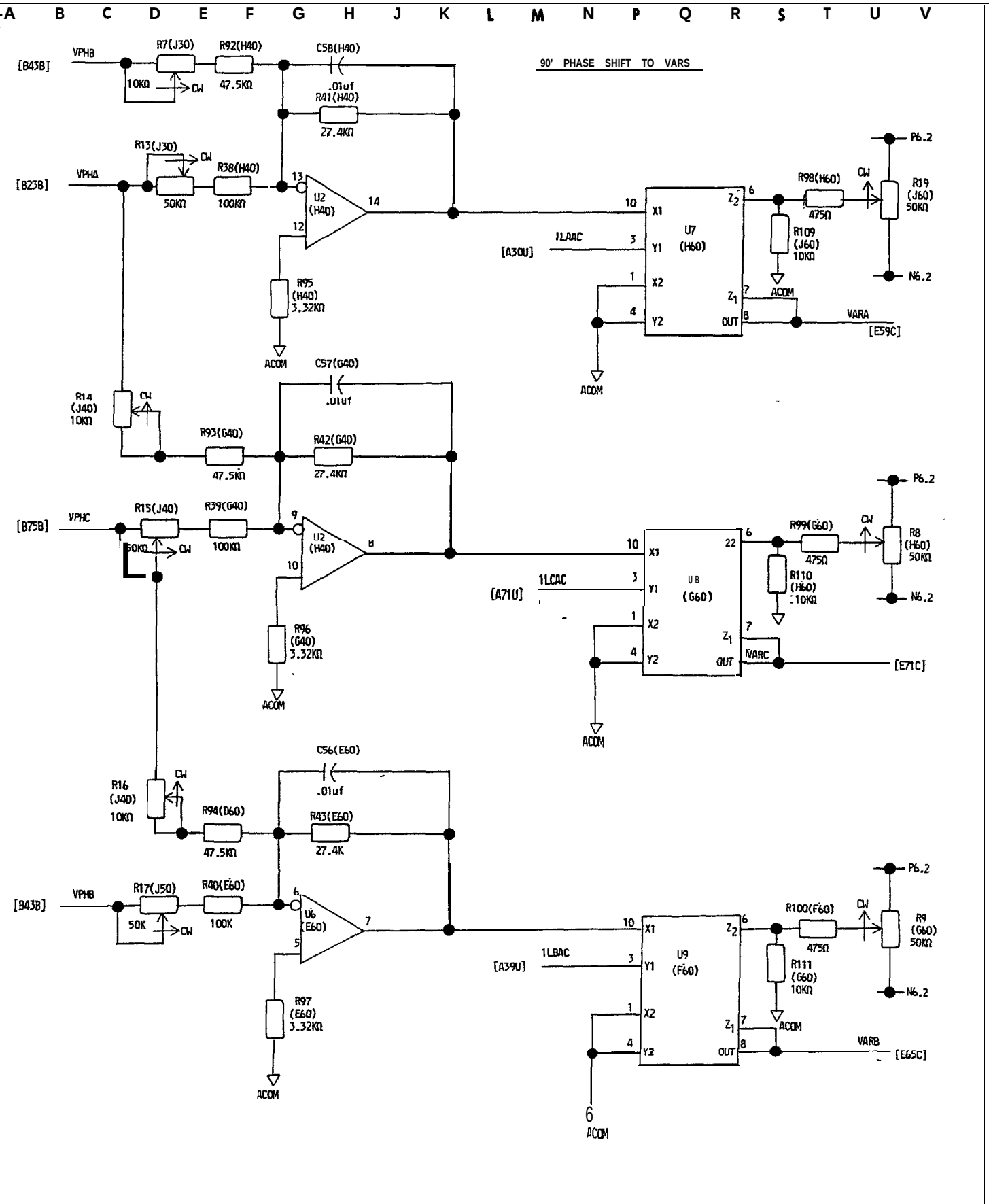
FORM IC 2346 (04-76)



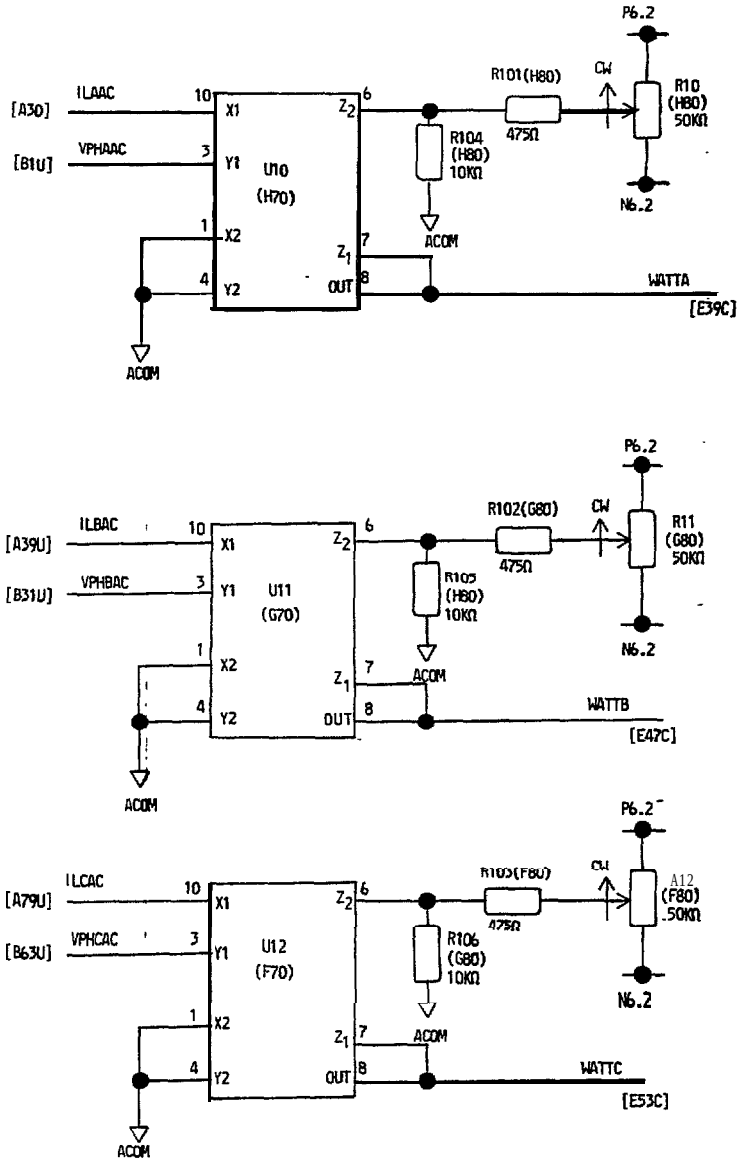
	REV.4	REV.5	PRINTS TO DL104	ENGINEER LSB	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM COMPLEX POWER TRANSDUCER
REV.2	ISSUED 11-12-86					DS3800NPTB
REV.3	MADE BY L. S. BOEBAUM					CONT. ON SH. 48A SH. NO. 4AA



	REV. 4	REV. 5	PRINTS TO DL104	ENGINEER LSB	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED 11-12-86				DRIVE SYSTEMS DEPARTMENT	COMPLEX POWER TRANSDUCER
REV. 3	MADE BY L. S. BLOEBAUM				SALEM, VA. U.S.A.	DS3800NPTB
						CONT. ON SH. 4CA SH. NO. 4B



	REV.4	REV. 5	PRINTS TO DL104	ENGINEER LSB	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED 11-12-86				DRIVE SYSTEMS DEPARTMENT	COMPLEX POWER TRANSDUCER
REV. 3	MADE BY L S. PLOEBAUM				SALEM, VA. U.S.A.	DS3800NPTB
						CONT. ON SH. 4 DASH NO. ACA



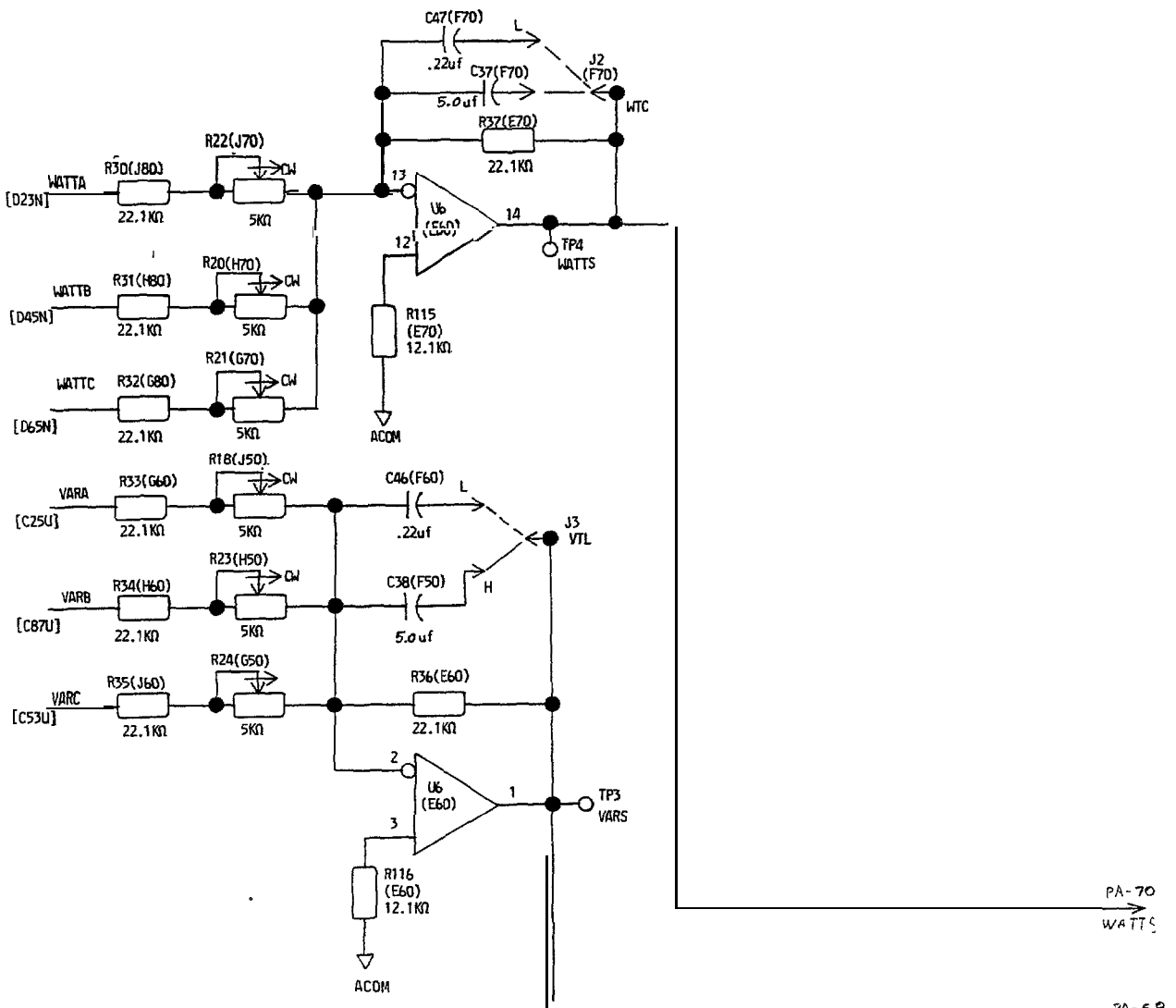
	REV.4	REV.5	PRINTS TO	ENGINEER
			DL104	
REV 2	ISSUED 11-12-86			
REV. 3	MADE BY L.S. BLOEBAUM			

GENERAL ELECTRIC
DRIVE SYSTEMS DEPARTMENT
SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM
COMPLEX POWER TRANSDUCER

DS3800NPTB
CONT. ON SH. 4EA SH. NO. 4D1

AF 10527

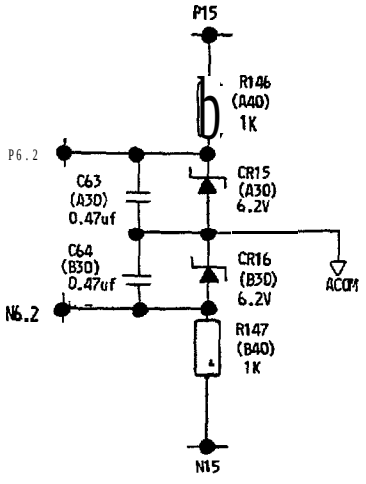


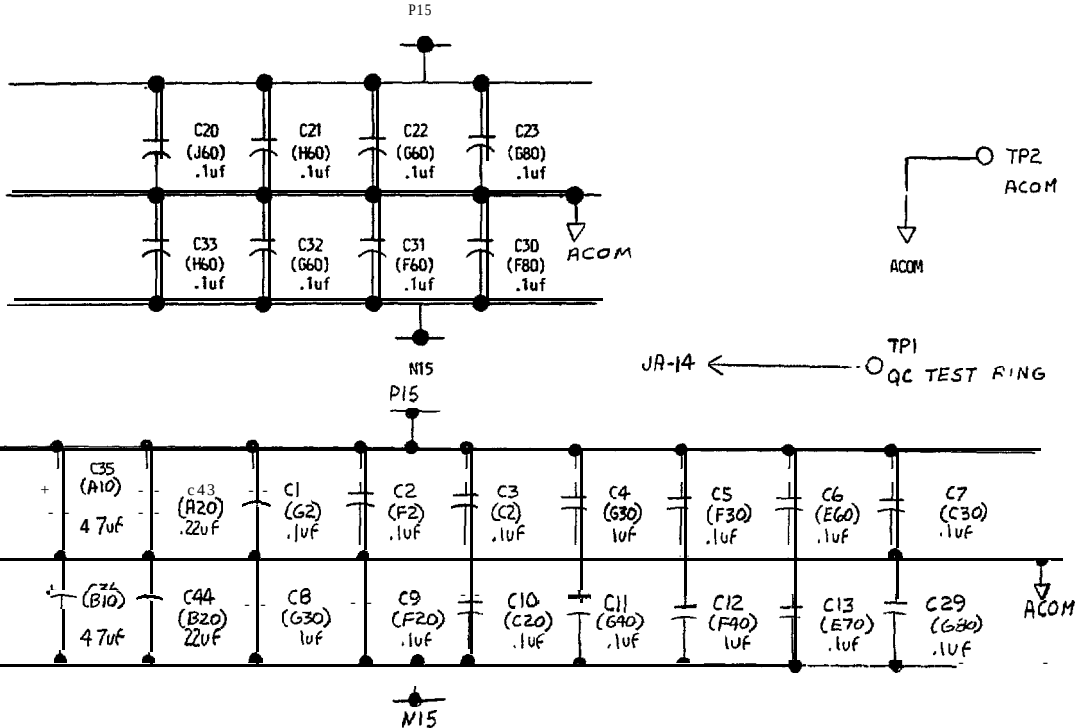
	REV.4	REV.5	PRINTS10 D1104	ENGINEER	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM COMPLEX POWER TRANSDUCER
REV. 2	ISSUED 11-12-86					DS3800NPTB
REV. 3	MADE BY L.S.					CONT. ON SH. 4GA SH. NO. 4EA

DED 0182 (09-76)

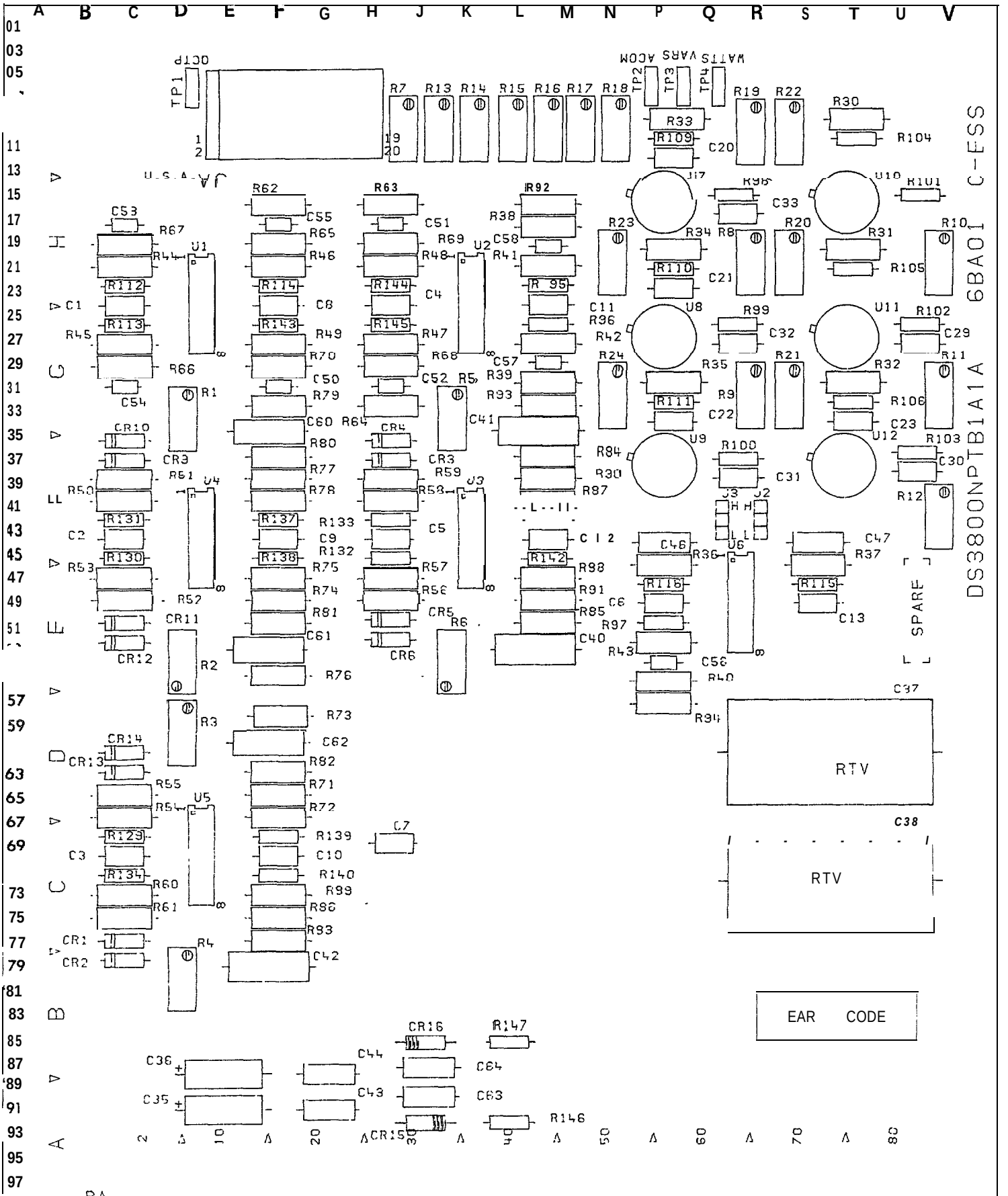
AF 10527

P15	N15	ACOM
U1-6 P4	U1-6 P11	
U7-12 P9	U7-12 P5	JA-2,4,6,8,10,12



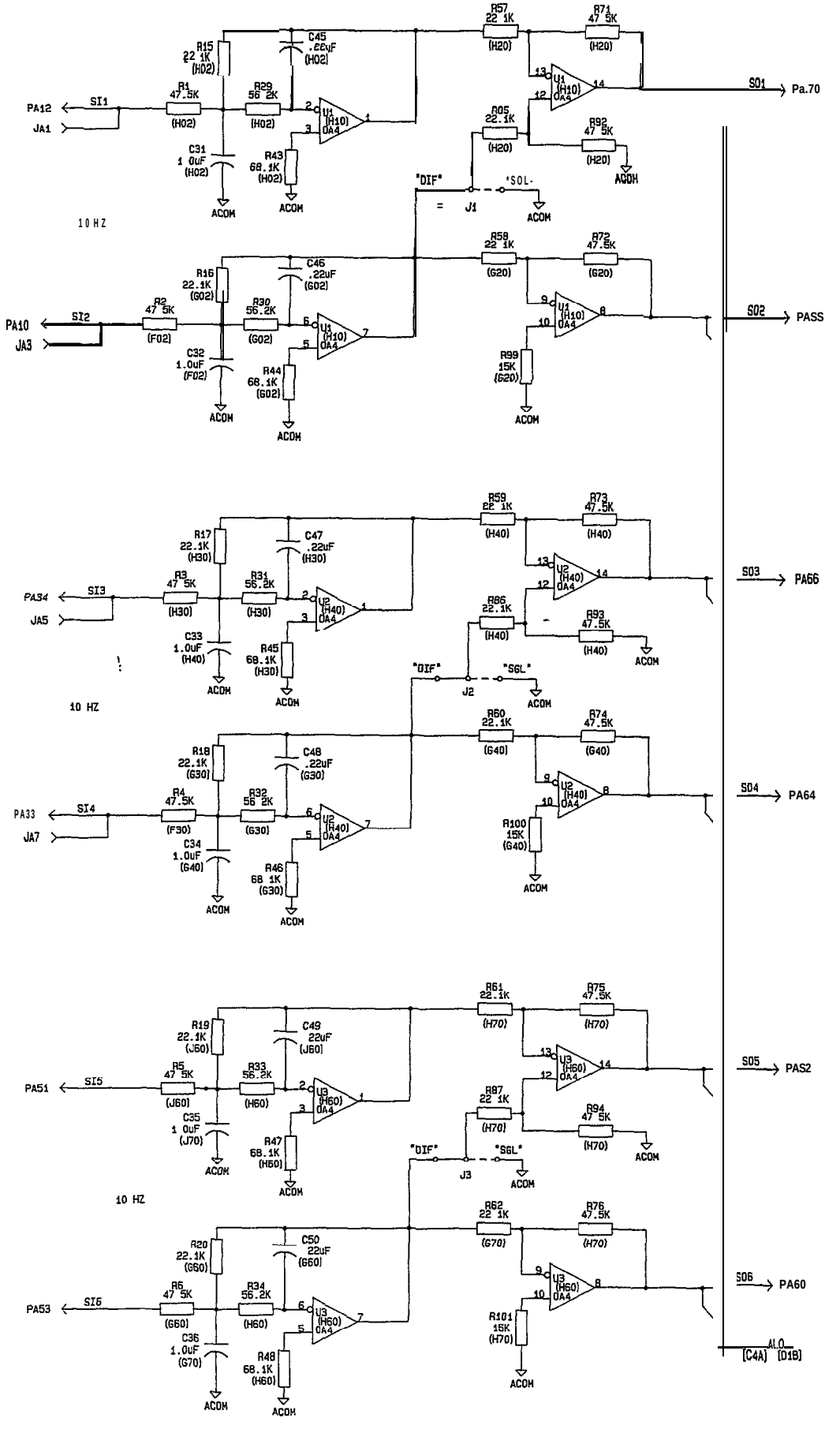


REV. 4	REV.5	PRINTS TO D1 104	ENGINEER LSB	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM COMPLEX POWER TRANSDUCER
REV. 2	ISSUED 11-12-86				DS3800NPTB
REV.3	MADE BY L. [illegible]				CONT. ON SH. 4HA SH. NO. 4GA



1 PA	REV 4	REV 5	PRINTS TO DL104	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM COMPLEX POWER TRANSDUCER
REV 2	ISSUED 11-12-86			FIRST MADE FOR	DRIVE SYSTEMS DEPT. SALEM, VA., U.S.A.	DS3800NPTB
REV 3	MADE BY E.F. ONTIVEROS			I.C NO		CONT. ON SH. FL SH. NO. 4HA

FORM IC 2348 (04-76)



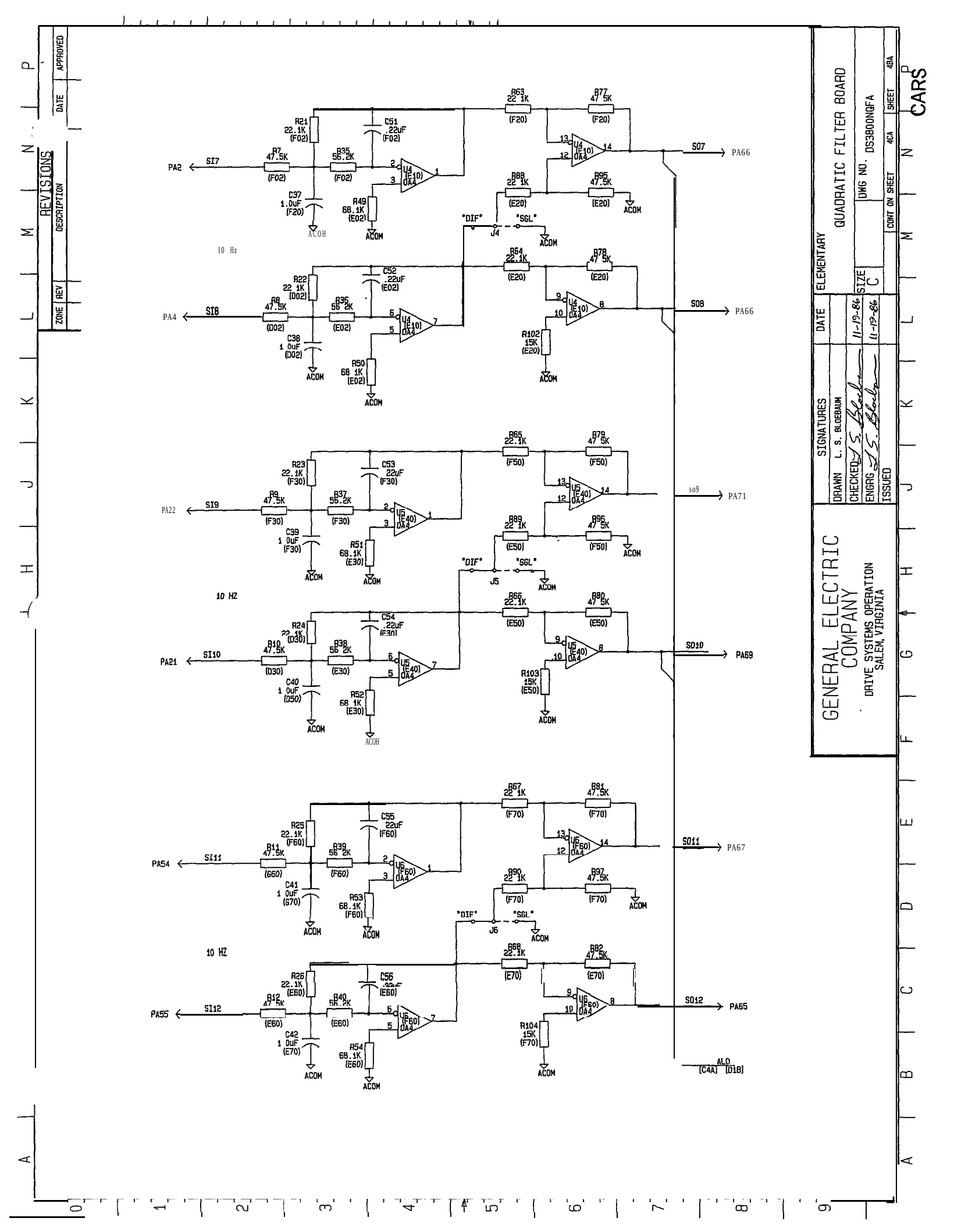
ZONE	REV	DESCRIPTION	DATE	APPROVED

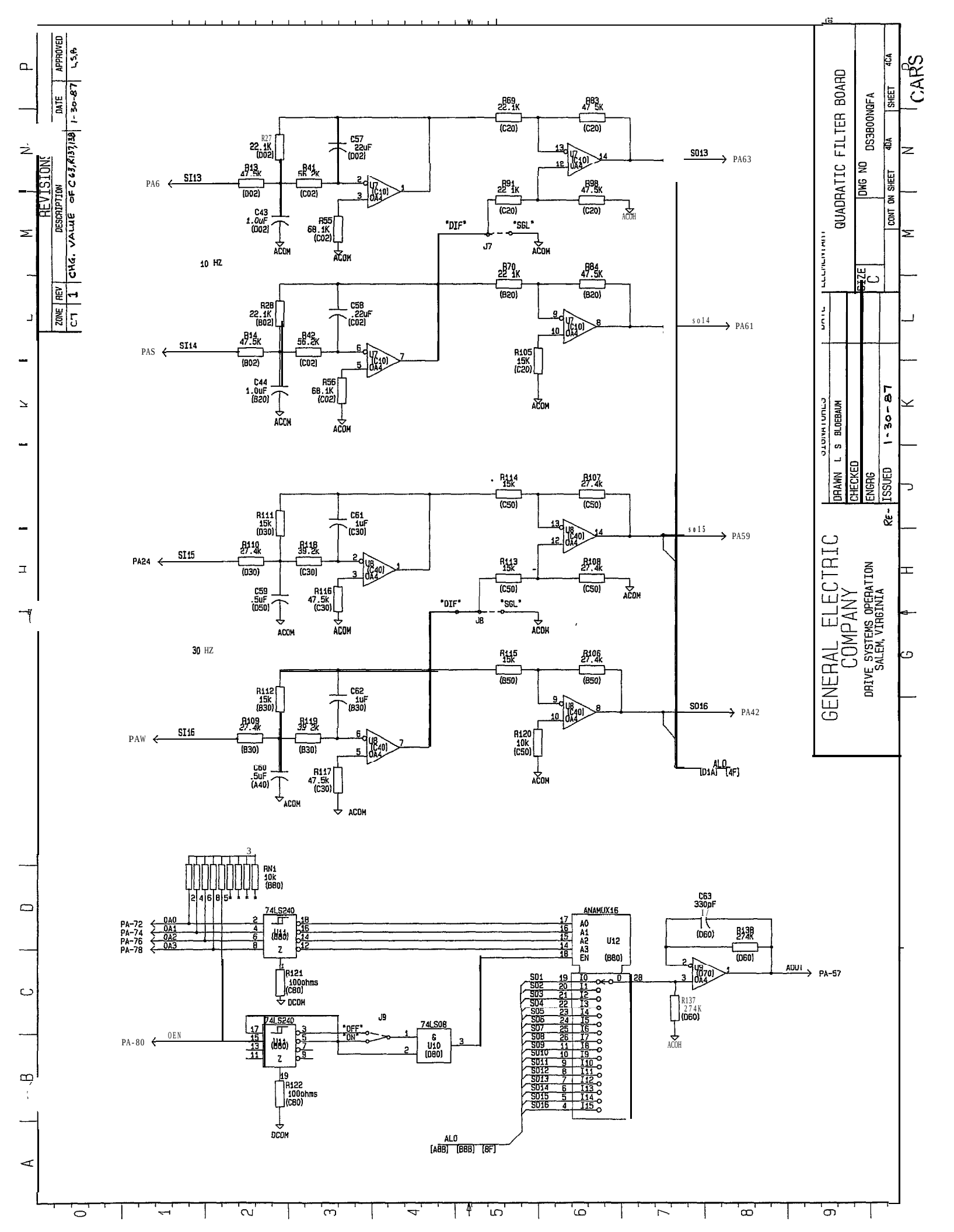
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS OPERATION
SALEM, VIRGINIA

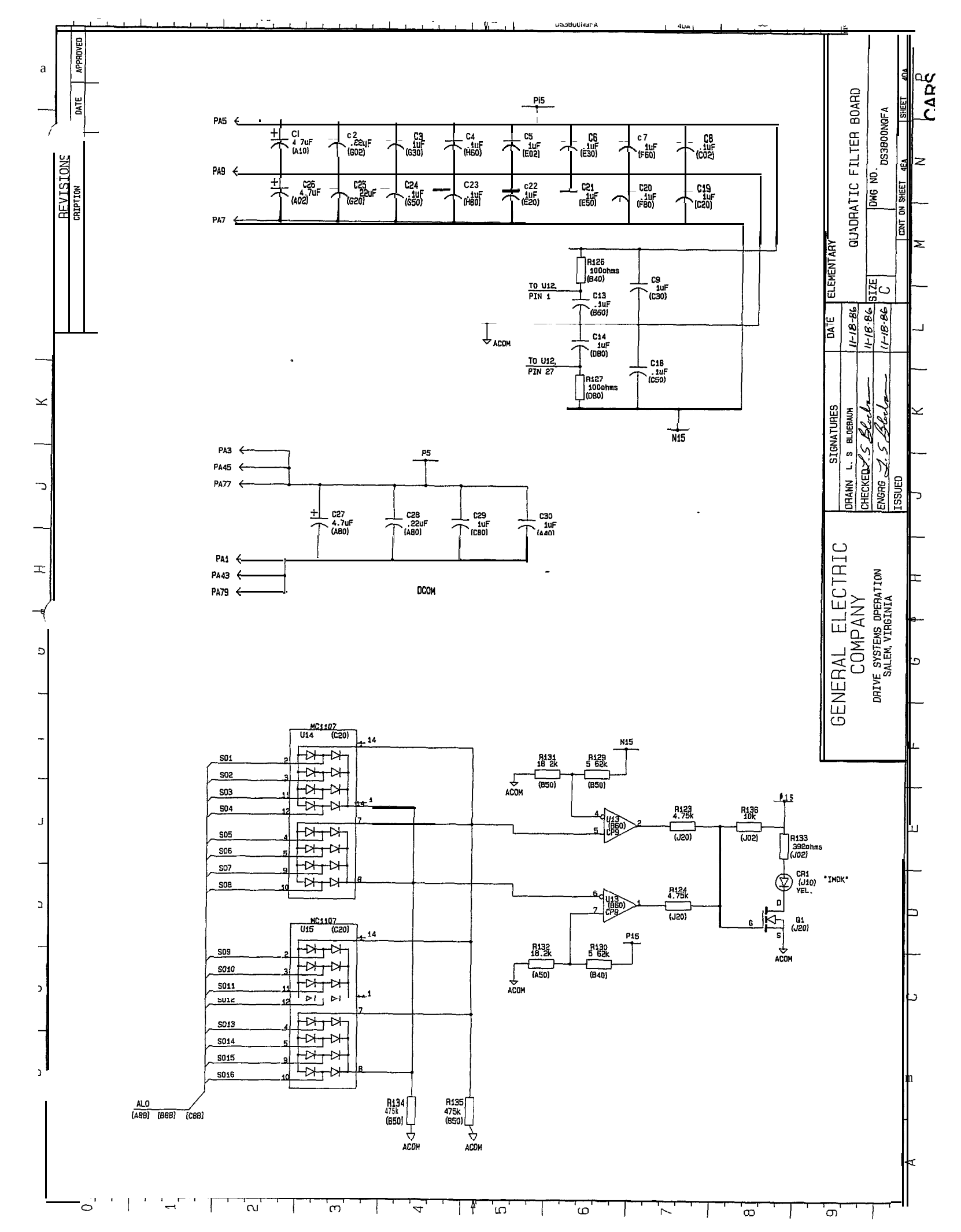
QUADRATIC FILTER BOARD

SIGNATURES	DATE
DRAWN L. S. BLOEBAUM	11-18-86
CHECKED L. S. Bloebaum	11-18-86
ENGRG L. S. Bloebaum	11-18-86
ISSUED	

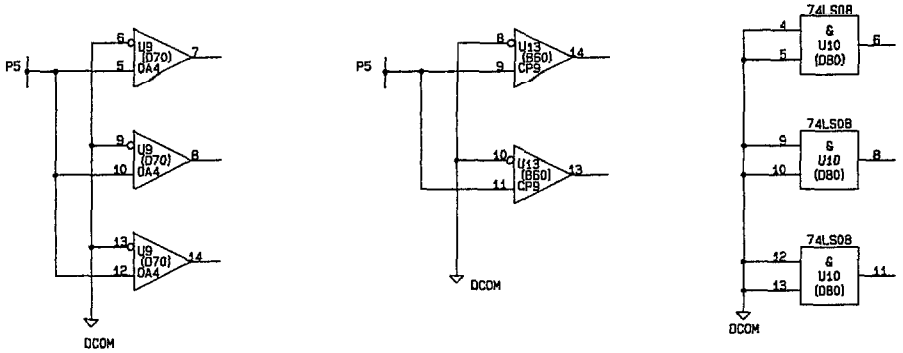
ELEMENTARY
SIZE C
DWG NO. DS3800NQFA
CONT ON SHEET 4BA
SHEET 1AA


REVISIONS
ZONE
REV
DESCRIPTION
DATE
APPROVED
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS OPERATION
SALEM, VIRGINIA
QUADRATIC FILTER BOARD
ELEMENTARY
DWG NO. DS3800NGFA
SIZE C
SHEET 4BA
CONT ON SHEET 4CA
SIGNATURES
DRAWN L. S. BLOEBAUM
CHECKED
ENGRG
ISSUED
DATE 11-19-86
PA2
SI7
PA4
SI8
PA22
SI9
PA21
SI10
PA54
SI11
PA55
SI12
10 Hz
10 HZ
10 HZ
"DIF"
"SGL"
J4
J5
J6
ACOM
ACOH
S07
PA66
S08
PA66
so9
PA71
SD10
PA69
SD11
PA67
SD12
PA65
ALO
(C4A) (D1B)


REVISIONS
ZONE | REV | DESCRIPTION | DATE | APPROVED
C7 | 1 | CHG. VALUE OF C43,R137,138 | 1-30-87 | L.S.B
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS OPERATION
SALEM, VIRGINIA
QUADRATIC FILTER BOARD
DWG NO DS3800NGFA
SIZE C
SHEET 4DA
CONT ON SHEET 4DA
DRAWN L S BLOEBAUM
CHECKED
ENGRG
ISSUED 1-30-87
10 HZ
30 HZ
SI13
SI14
SI15
SI16
SO13
SO14
SO15
SO16
PA6
PAS
PA24
PAW
PA63
PA61
PA59
PA42
"DIF"
"SGL"
J7
J8
J9
ACOM
DCOM
ALO
ADJ1
PA-57
PA-72
PA-74
PA-76
PA-78
PA-80
OEN
74LS240
ANAMUX16
74LS08
RN1
10k
(B80)
C63
330pF
R121
100ohms
R122
100ohms
R137
274K


P15
PA5
PA9
PA7
C1 4.7uF (A10)
C2 .22uF (G02)
C3 1uF (G30)
C4 1uF (H60)
C5 1uF (E02)
C6 .1uF (E30)
C7 1uF (F60)
C8 .1uF (C02)
C26 4.7uF (A02)
C25 .22uF (G20)
C24 .1uF (G50)
C23 1uF (H80)
C22 1uF (E20)
C21 1uF (E50)
C20 .1uF (F80)
C19 1uF (C20)
R126 100ohms (B40)
TO U12, PIN 1
C13 .1uF (B60)
C9 1uF (C30)
ACOM
C14 1uF (D80)
TO U12, PIN 27
R127 100ohms (D80)
C18 .1uF (C50)
N15
PA3
PA45
PA77
P5
C27 4.7uF (A80)
C28 .22uF (A80)
C29 1uF (C80)
C30 1uF (A40)
PA1
PA43
PA79
DCOM
MC1107
U14 (C20)
SO1
SO2
SO3
SO4
SO5
SO6
SO7
SO8
MC1107
U15 (C20)
SO9
SO10
SO11
SO12
SO13
SO14
SO15
SO16
ALO
(A88) (B88) (C88)
R134 475k (B50)
R135 475k (B50)
ACOM
ACOM
R131 18.2k (B50)
R129 5.62k (B50)
ACOM
N15
U13 (B50) CP9
R123 4.75k (J20)
R136 10k (J02)
R133 392ohms (J02)
CR1 (J10) YEL.
"IMOK"
U13 (B50) CP9
R124 4.75k (J20)
Q1 (J20)
ACOM
R132 18.2k (A50)
R130 5.62k (B40)
P15
ACOM
REVISIONS
DESCRIPTION
DATE
APPROVED
ELEMENTARY
QUADRATIC FILTER BOARD
DWG NO. DS3800N0FA
SIZE C
SHEET 4DA
CONT ON SHEET 4EA
SIGNATURES
DRAWN L. S BLOEBAUM
CHECKED
ENGRG
ISSUED
DATE 11-18-86
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS OPERATION
SALEM, VIRGINIA

ZONE	REV	DESCRIPTION	DATE	APPROVED
		REVISIONS		



POWER & GROUND CONNECTIONS

	P15	N15	P5	ACOM	DCOM
JA-2, 4, 6, 8	—	—	—	✓	—
U1 - U9	4	11	—	—	—
U10	—	—	14	—	7
U11	—	—	20	—	10
U12			—	—	12
U13	3	12	—	—	—
U14, U15	NONE				

NUMBERS REPRESENT PINS ON THE ICs

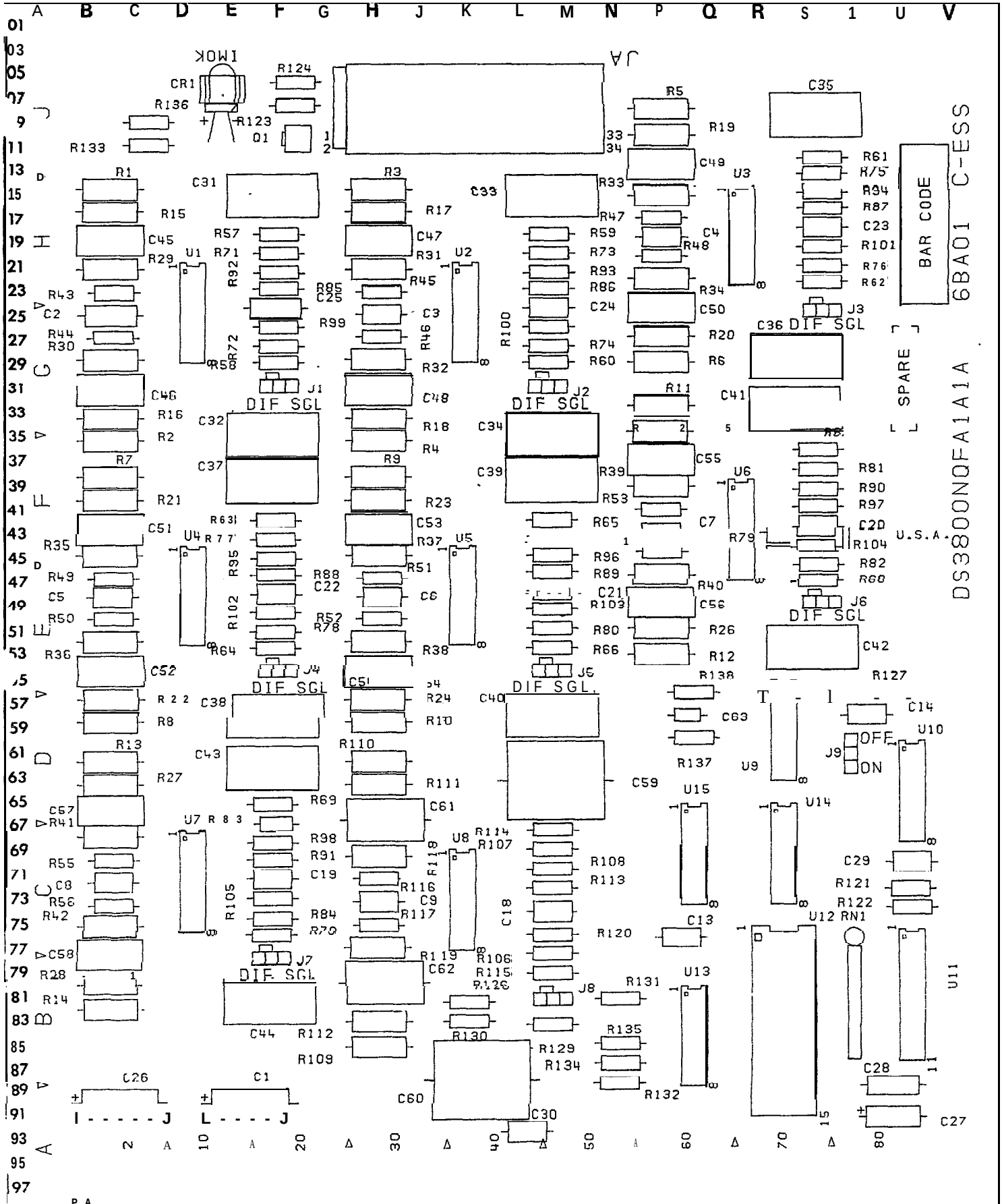
DRAWN	L. S. BLOEBAUM	11-18-86	
CHECKED	L S Bloebaum	11-18-86	SIZE C
ENGRG	L S Bloebaum	11-18-86	
ISSUED			

QUADRATIC FILTER BOARD

DWG NO DS3800NQFA

CONT ON SHEET LJFA PC SHEET 4EA

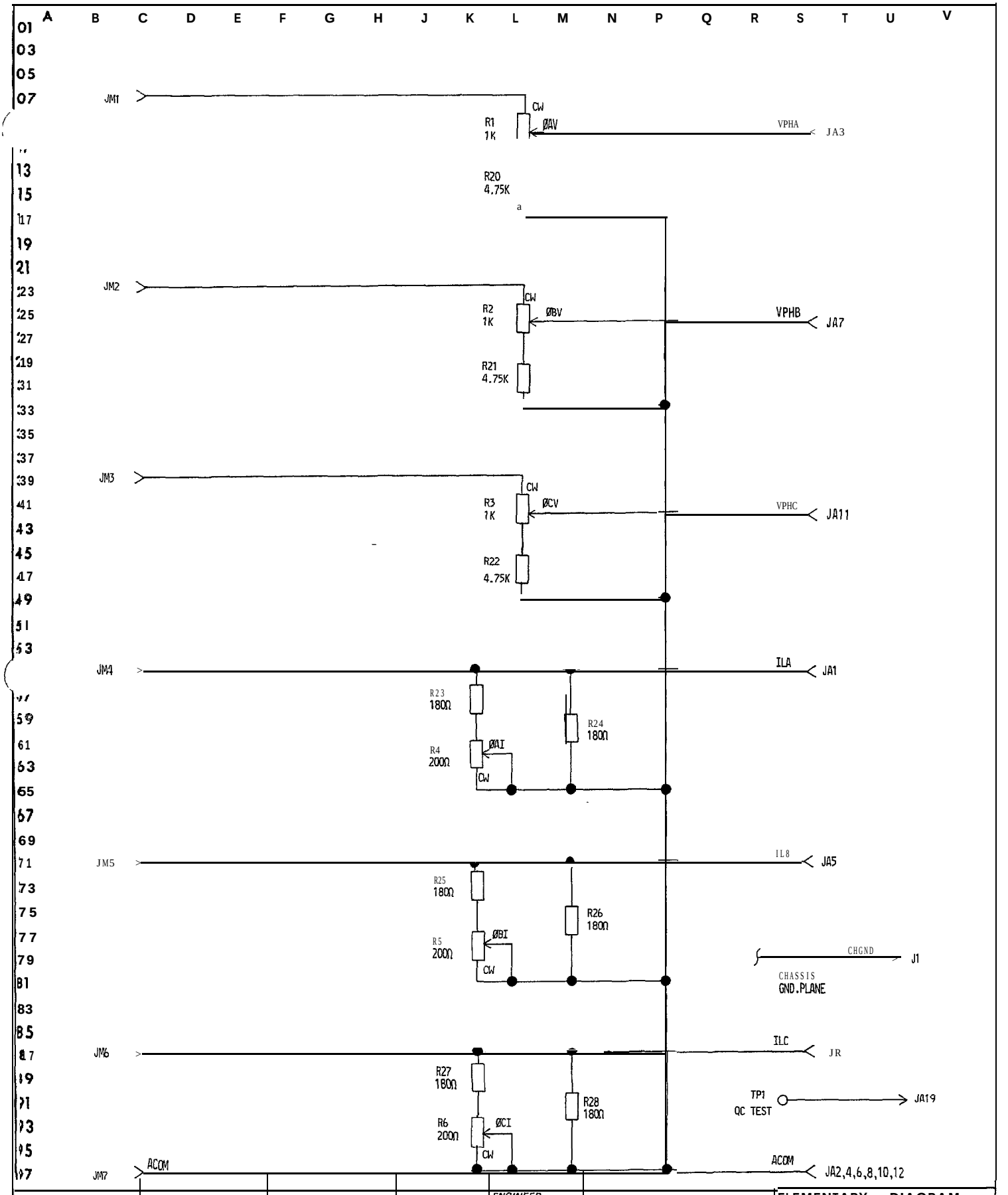
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS OPERATION
SALEM, VIRGINIA



PEV 1	REV4	REV 5	PRINTS TO DL104 (mE)	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV 2	ISSUED DEC 9, 1986			FIRST MADE FOR	DRIVE SYSTEMS DEPT. SALEM, VA., U.S.A.	DS 38 00 NQFA
REV 3	MADE BY M. REICH			I.C. NO		CONT. ON SH. FL SH. NQ4FA

FORM IC 2349 (04-76)

CARS



1	REV 4	REV 5	PRINTS TO DL104	ENGINEER
REV 2	ISSUED 82-06-25			TGSI
REV 3	MADE BY T.A. KOCHER			

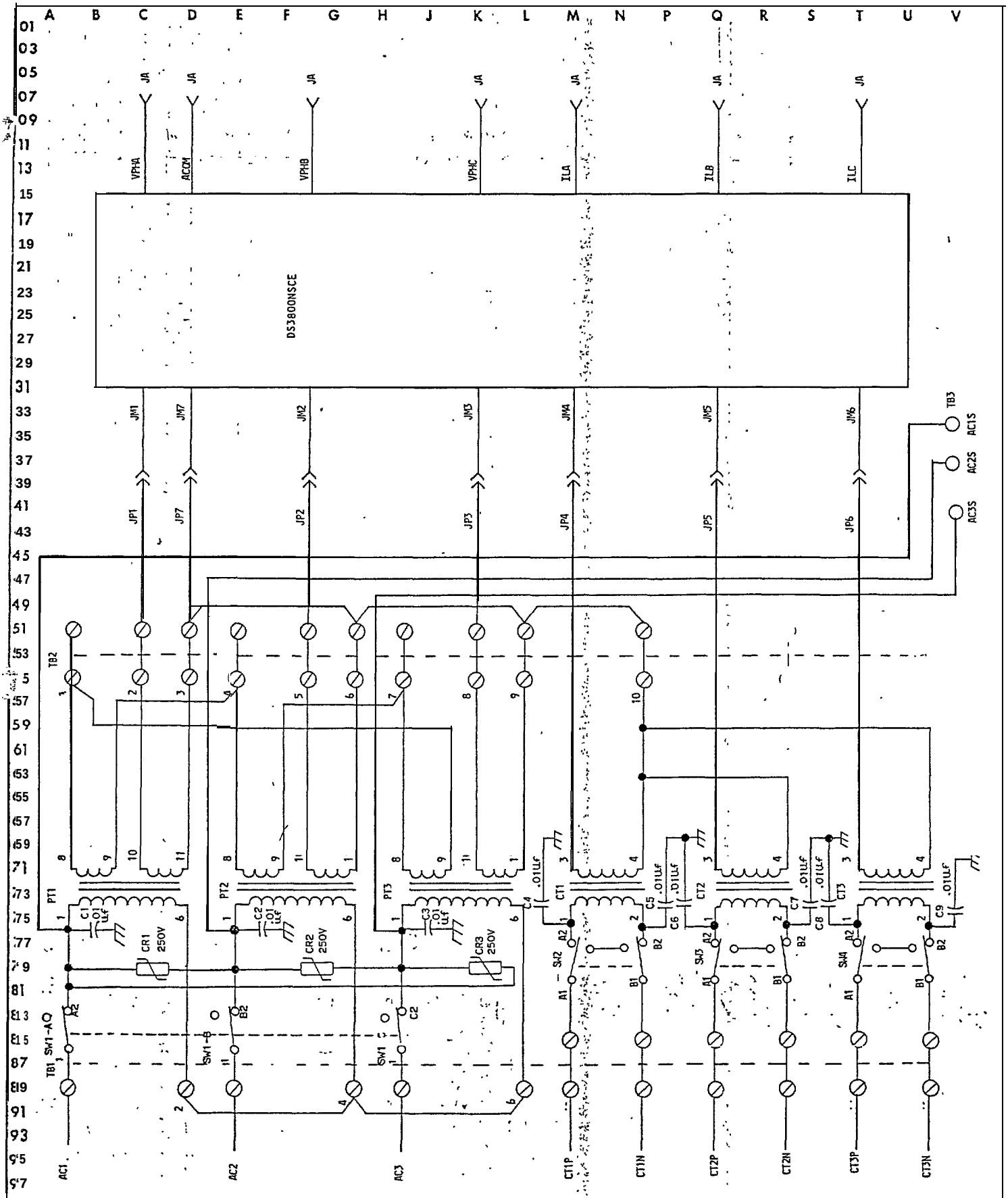
GENERAL@ ELECTRIC
DRIVE SYSTEMS DEPARTMENT
SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM
PASSIVE SIGNAL CONDITIONING BOARD

DS380ONSCE

CONT. ON SH. 4BA SH. NO. 4AA

USD 0142 (09 76)



REV. 1 OEA 9-21-82	REV. 4	REV. 5	PRINTS TO DL114	ENGINEER D.W.S	GENERAL@ ELECTRIC -DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM POWER CONDITIONING MODULE
REV. 2	ISSUED 8/11/82			TGSI		DS3820PCN
REV. 3	MADE BY D.SEXTON					CONT. ON SH. SH. NO. 4A1

9.1.0 Scope

This document establishes the performance requirement and recommended tests for the Power Conversion Module identified as: DS3820PCNA.

This specification will check analog transfer functions and component tolerances.

9.2.0 Test Equipment and Documentation

9.2.1 Standard Equipment Required:

Test equipment shall be provided which meets the requirements and accuracies prescribed in this specification.

9.3.0 Setup and Initial Loading

9.3.1 Before starting the test, make sure that the module's printed wire board, DS3800NSCE, is properly connected to the module.

9.3.2 Each of the following outputs should be loaded with a 25.12 k ohm ± .5%, 1/4W resistive load to ACOM (JA2, 4, 6, 8, 10, 12), VPHA (JA3), VPHB (JA7), and VPHC (JA1).

9.3.3 The outputs ILA (JA1), ILB (JA5), and ILC (JA9) should each be loaded with a 39.2 k ohm resistor *to* ACOM.

9.3.4 Daughter Board - None

9.4.0 Signal Levels

9.4.1 TTL Input Levels - None.

9.4.2 Process Input Levels - as defined in test.

9.4.3 TTL Output Levels - None.

9.4.4 Process Output Levels - As defined in test.

9.5.0 Test Procedure

9.5.1 Preliminary Inspection - The module and card should be inspected prior to application of power to verify that it is assembled according to the assembly drawing.

9.5.2 Digital Tests - Not applicable.

9.5.3 Hybrid Tests - Not applicable.

9.5.4 Analog Tests

4.1 With the outputs of the printed circuit board loaded as prescribed in Section 9.3 and with SW1 in the closed position, apply 3-phase, 115V RMS line to line voltage between terminals AC1, AC2, and AC3.

4.2 Turn potentiometer R1 clockwise until the voltage from ACOM (JA7) to VPHA (JA1) is 12.00V ± 0.012V.

4.3 Repeat step 4.2 for both VPHB (JA2) and VPHC (JA3) by adjusting potentiometers R2 and R3, respectively.

4.4 With SW2 in the closed position, apply a 5A RMS current between terminals CT1N and CT1P. Adjust potentiometer R4 in the clockwise direction until the voltage from ACOM (JA2) to ILA (JA1) is 5.00V RMS ± 0.005V RMS.

4.5 With SW2 in the closed position, apply a 5A RMS current between terminals CT2N and CT2P. Adjust potentiometer R5 in the clockwise direction until the voltage from ACOM (JA2) to ILB (JA) is 5.00V RMS ± 0.005V RMS.

4.6 With SW4 in the closed position, apply a 5A RMS current between terminals CT3N and CT3P. Adjust potentiometer R6 in the clockwise direction until the voltage from ACOM (JA2) to ILC (JA9) is 5.00V RMS ± 0.005V RMS.

4.7 Check *to* see that the resistance between *the* chassis ground plane on the card (J1) and the module chassis is 0.00 ohms.

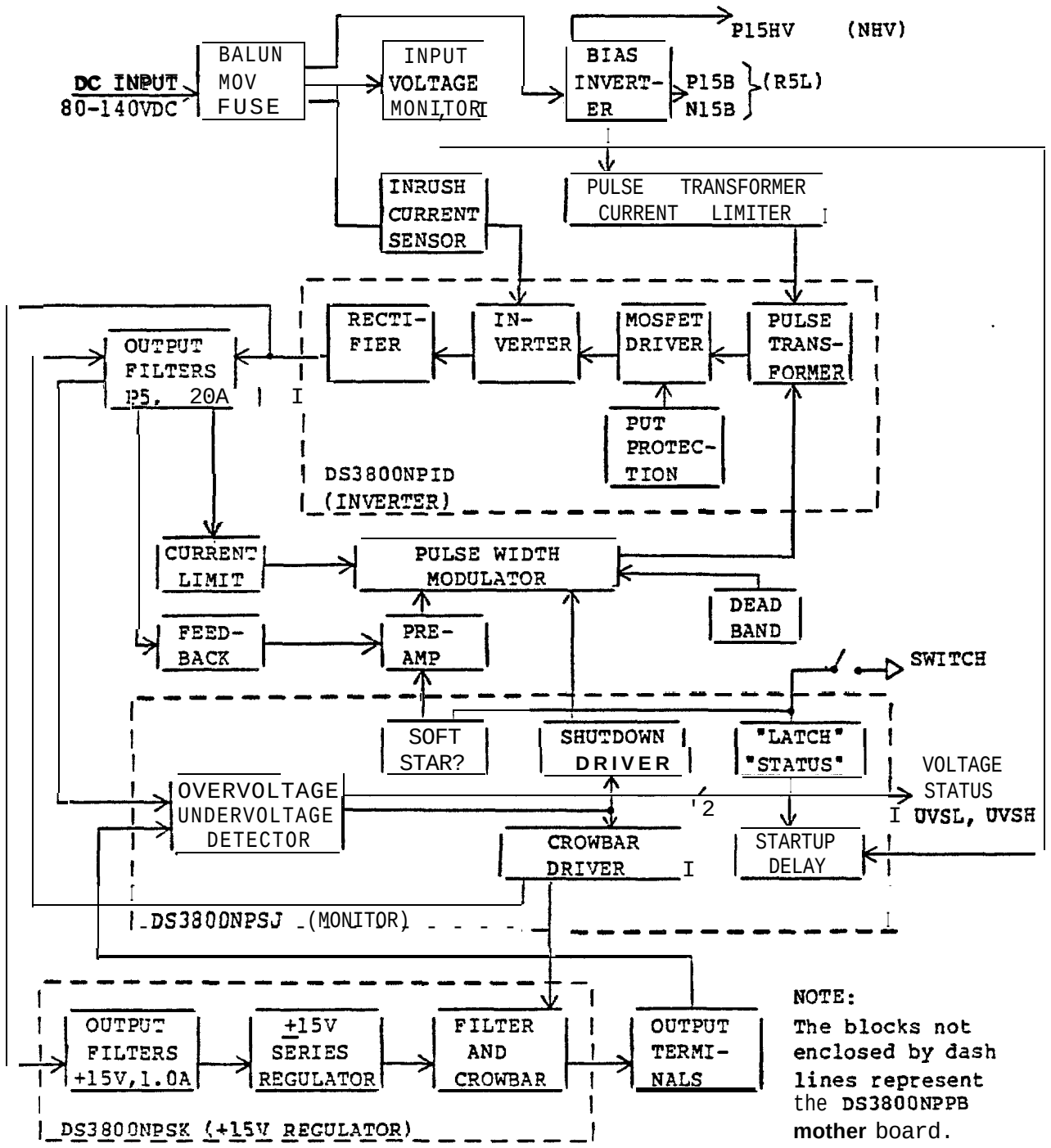
9.5.5 Special Tests - None.

9.5.6 Temperature Tests - This module and card shall be tested at ambient room temperature only.

9.5.7 Supplemental Tests - None

END OF TEST

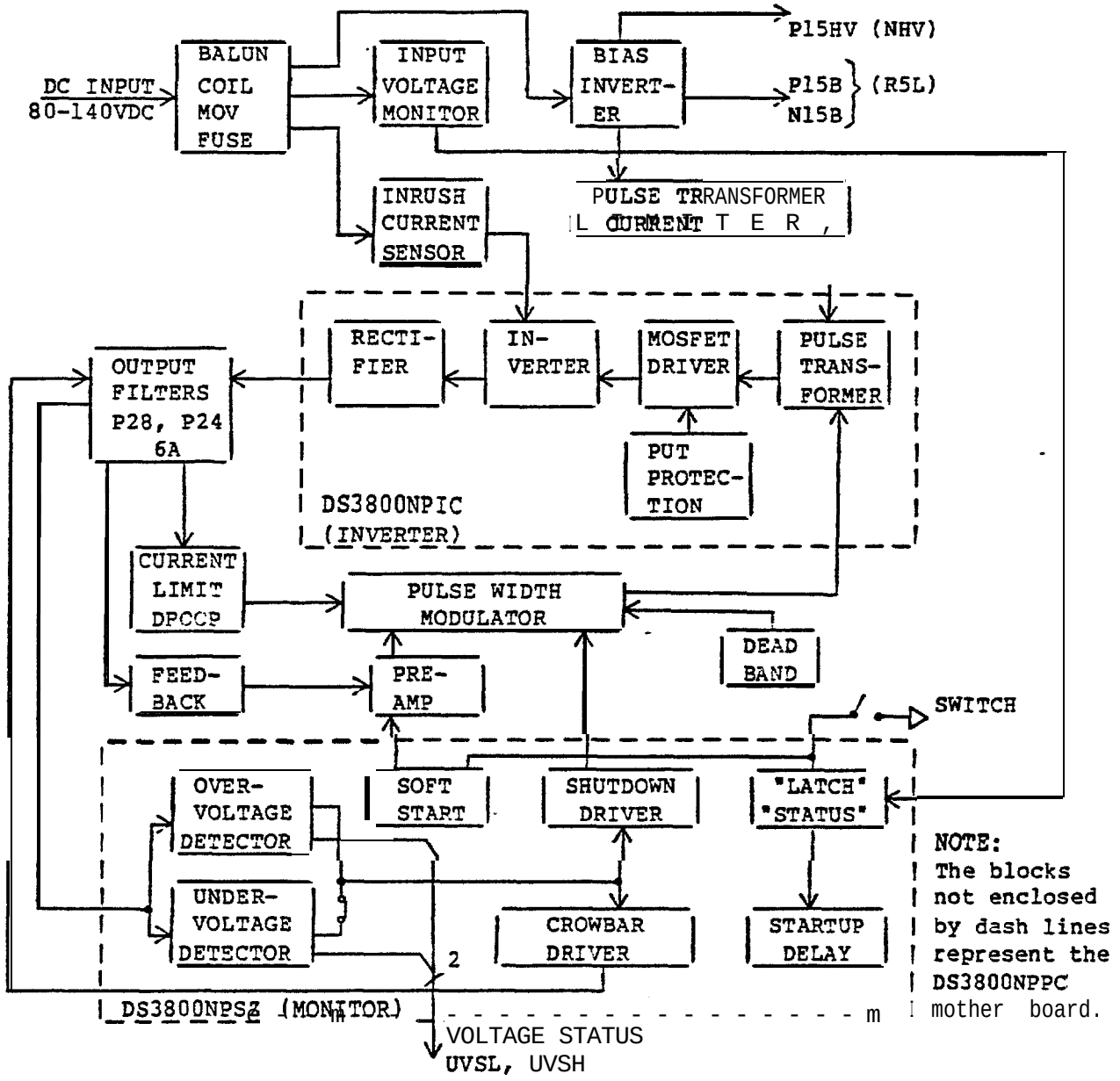
Figure 1
DS3820PSCB BLOCK DIAGRAM



NOTE:
The blocks not enclosed by dash lines represent the DS3800NPPB mother board.

REV.1	REV.4	(REV.7	PRINTS TO ENGINEER DLL12	DESCRIPTION OF OPERATION AND APPLICATION DATA	
REV.2	REV.5	RE-ISSUED January 25 1984	GENERAL ELECTRIC	Powe r Supply	
REV.3	REV.6	MADE BY J. R. Halo	COMPANY. SALEM, VA	DS3820PSC CON". ON SH. 2BA SH. NO. 2AA	

Figure 2
DS3820PSCC/D BLOCK DIAGRAM



REV.1	REV.4	REV.7	PRINTS TO DL112-[illegible]	ENGINEER J.R. Hale	DESCRIPTION OF OPERATION AND APPLICATION DATA
REV.2	REV.5	RE-ISSUED January 25 1984	GENERAL ELECTRIC I		Power Supply
REV.3	jREV.6	MADE BY J. R. Hale	COMPANY SALEM, VA		DS3820PSC
					CONT.ON SH. 2CA SH.NO. 2BA

DEFINITION AND FUNCTION

The DS3820PSC Power Supply module exists in 3 forms to provide different power outputs:

- DS3820PSCB - P5, P15, N15
- DS3820PSCC - P28
- DS3820PSCD - P24

The power supply accepts an 80VDC to 140VDC input, which may be floated or grounded. All busses are regulated to a stabilized operating point so that the output loads and input lines vary within their respective limits.

THEORY OF OPERATION

(Refer to the PSC block diagrams, Figure 1 for PSCB and Figure 2 for PSCC and PSCD.)

The PSC module forms *consist* of the following printed *circuit* boards:

	MOTHER	INVERTER	MONITOR	+15VDC REGULATOR
PSCB -	DS3800NPPB	DS3800NPID	DS3800NPSJ	DS3800NPSK
PSCC & PSCD -	DS3800NPPC	DS3800NPIC	DS3800NPSZ	--------

NOTE: The PSCC and PSCD module forms contain identical components, but differ in their voltage output (P28 for PSCC, P24 for PSCD). Refer to the Application Data section for adjustment procedures that change a PSCC to a PSCD.

The PSC module receives a voltage input via the mother board's (NPPB or NPPC) JD connector. Each input line contains two 5A fuses. Switch Sl, located at *the* front end of the module, controls the inverter regulator so that the output busses can be either enabled or disabled. Note that the DISABLE position does not disconnect the input lines from the power supply.

When the PSC receives a voltage input, an internal auxiliary power supply provides coarsely regulated +15VDC to the switching regulator control circuits via the mother board's dc/dc converter. The ±15VDC for the control circuits is designated P15B and N15B, and can be monitored on the NPSJ board via TP3 to TP4 and TP8 to TP4.

A push-pull switching regulator serves as the 'power supply's heart. It functions with a pair of MOSFETs (metal oxide semiconductor field effect transistors1 and a pulse width modulator (PWM). When switch SW1 is enabled, the inverter regulator starts up slowly from zero volts with a *very* narrow duty cycle to provide a soft start during turn-on. The PWM's outputs drive the MOSFET gates through pulse transformers. In turn, the MOSFETs switch, producing an alternating current input that excites the power transformer.

While the power transformer is ac excited, output voltages appear on its secondaries. The power supply then rectifies and filters them, producing P5 or P28/24 output busses (depending of the supply form). In addition, form PSCB contains series regulators that control the P15 and N15 output busses.

REV.1	REV.4	REV.7	PRINTS TO DL112	ENGINEER J.R. Hale	DESCRIPTION OF OPERATION AND APPLICATION DATA
REV.2	REV.5	RE-ISSUED January 25 1984		GENERAL ELECTRIC	Power Supply
REV.3	REV.6	MADE BY J. R. Hale		COMPANY SALEM,VA	DS3820PSC CONT.ON SH. 2DA SH.NO. 2CA

Fault Detection and Current Limit

The PSC mother board (NPPB or NPPC) and monitor board (NPSJ or NPSZ) provide internal and external fault -detection to protect the power supply during normal operation. The Power supply shuts down if any of its output busses exceed the undervoltage or overvoltage threshhold. The yellow IMOK LED turns off when the supply shuts down.

Current limit lines CL+ and CL- connect a low resistance current sensor to the PWM regulator. If the bus current exceeds the limit (see Table 1), the sensor causes the regulator to limit current. Once the PWM regulator limit occurs, all output voltages decrease with increasing load. If they near the undervoltage thresholds, the undervoltage crowbar fires and shuts down the power supply.

Berg jumpers (see Table 2) can be used to isolate the faults.

APPLICATION DATA

POWER SPECIFICATIONS AND FEATURES

Table 1
POWER OUTPUT

MODULE FORM	OUTPUT VOLTAGE	CURRENT			CURRENT LIMIT
		either	or-	or*	
PSCB	P5	20A	15A	12A	24A
	P15	1A	1.5A	2A	-----
	N15	1A	1.5A	2A	-----
PSCC	P28	6A			7.2A
PSCD	P24	6A			-----

- These currents can only be achieved if 100 CFM is passed through the module. This can be done by mounting a muffin fan (218A4491P1) directly above or below the supply.

NOTE: The application engineer determines the PSCB's current according to the total load required.

NOTE: To change the PSCC to a PSCD, adjust the PSCC's voltage output from 28VDC to 24VDC using the following steps:

1. Remove the NPPC's B2, B3 Berg jumper
2. Adjust the NPSZ's R5 potentiometer for 3.4VDC on TP4 and TP6
3. Adjust the NPPC's R1 pot for a 24VDC ± .05VDC output measured across it's J1 and J2 pin 8 or across the NPSJ/Z's TP4 and TP11
4. Replace the NPPC's B2, B3 jumper.

REV.1	JREV.4	REV.7	PRINTS TO DL112	ENGINEER J.R. Hale	DESCRIPTION OF OPERATION AND APPLICATION DATA
REV.2	REV.5	RE-ISSUED January 25 1984	I-GENERAL ELECTRIC		Power Supply
REV.3	REV.6	MADE BY J. R. Hale	COMPANY SALEM,VA		DS3820PSC CONT.ON SH. 2EA SH.NO. 2DA

- VOLTAGE INPUT: 80VDC to 140VDC input via the mother board's (NPPB or NPPC) JD connector
- Rides through a 2 millisecond (ms) dip to zero volts from a 98VDC input or greater, while staying within its rated performance specifications.
- Reverse input polarity protection. No damaging arc is drawn when the input is disconnected.
- Does not malfunction upon 100% load removal. The power supply does not shut down with 25% of rated load step removal or application.
- PSCB's P5 bus requires remote sense wires.
- PSCC/D contain a voltage droop circuit that allows the power supply to operate in parallel with another PSCC/D and share load within their ratings.
- The Low Voltage Status High (LVSH) and Low Voltage Status Low (LVSL) signals are used to indicate P5 bus status at the user's remote control center.
- When the NPSZ is used (PSCC/D), there is a 3ms delay on the undervoltage (UV) trip. If UV is enabled, F1-F2 is jumpered.

COMPONENT SPECIFICATIONS

Switching regulator switching frequency: 20KHz for MOSFETs
40KHz for ripple

Power transistor (MOSFET) type: HEXFET, IRF 350

Pulse width modulator type: SG3524

SWITCHES

Switch S1 - Logic control switch for operation.
ENABLE position enables the output bus (supply on)
DISABLE position disables the output bus (supply off)

INDICATING LIGHTS

Yellow IMOK LED, on the NPSJ/Z, turns on when the PSC's outputs are within limits.

OPERATING AND STORAGE ENVIRONMENT

See GEK-61680.

REV.1	REV.4	REV.7	PRINTS TO DL112	ENGINEER J.R.Hale	DESCRIPTION OF OPERATION AND APPLICATION DATA
REV.2	(REV.5	IRE-ISSUED January 25 1984		GENERAL ELECTRIC	Power Supply
REV.3	REV.6	[MADE BY J. R. Hale		COMPANY SALEM,VA	DS3820PSC CONT.ON SH. 2FA SH.NO. 2EA

BERG JUMPERS

Table 2

MODULE FORM	JUMPER POSITION/ NORMAL	TEST	LOCATION	BERG JUMPER'S FUNCTION IN THE NORMAL POSITION
PSCB	A2-A3	A1-A2	NPPB	To disable monitor shutdown
	B1-B2	-----	NPPB	Use this position or shutdown will occur, for other application
	D1-D2	D2-D3	NPSJ	To disable shutdown
PSCC & PSCD	B2-B3	B1-B2	NPPC	To disable monitor shutdown
	C2-C3	-----	NPPC	Non-droop output
	C1-C2	-----	NPPC	Droop for parallel operation
	D1-D2	D2-D3	NPSJ*	To disable shutdown
	F1-F2	-----	NPSZ	Normal overvoltage and undervoltage shutdown
	F2-F3	-----	NPSZ	Overvoltage shutdown only with UVSL and UVSH signal outputs indicating undervoltage

* NOTE: PSC power supplies used prior to August 1983 contain an NPSJ as the monitor board. Later power supplies use the NPSZ (which does not contain the D jumpers) as the monitor board.

POTENTIOMETERS

Table 2

MODULE FORM	SUPPLY	POT	LOCATION
PSCB	P5	R1	NPPB
	P15	R6	NPSK
	N15	R7	NPSK
PSCC/ PSCD	P28	R1	NPPC
	P24	R1	NPPC

NOTE: All other potentiometers located on the PSC are for factory adjustment only. All voltages are referenced to R5L.

REV.1	REV.4	REV.7	PRINTS TO	ENGINEER	DESCRIPTION OF OPERATION
			DL112	J.R. Hale	AND APPLXCATION DATA
(REV.2	REV.5	(RF-ISSUED	Hale	--GENERAL	Power Supply
		January 25 1984		ELECTRIC	
REV.3	REV.6	MADE BY		COMPANY	DS3820PSC
		J. R. Hale		SALEM,VA	CONT.ON SH. 3AA SH.NO. 2FA

3.00 TROUBLESHOOTING

DUE TO COMPLEX FACTORY SETTINGS AND MODULE TEST, THE PSCB AND PSCC POWER SUPPLIES ARE NOT FIELD REPAIRABLE. DEFECTIVE MODULES MUST BE RETURNED TO THE FACTORY FOR REPAIR. SOME SYMPTOMS AND CAUSE ANALYSIS IS PROVIDED TO DIAGNOSE A FAILED MODULE. NOTE THAT ALL TEST POINTS ARE REFERED TO THE NPSJ MONITOR AND CONTROL CARD UNLESS OTHERWISE INDICATED.

3.10 THIS TROUBLE SHOOTING PROCEDURE IS BASED ON THE ASSUMPTION OF A SINGLE POINT FAILURE. IN CERTAIN SITUATIONS THE RECOMMENDED ACTION WILL NOT CORRECT THE PROBLEM. IF THIS OCCURS, REPLACE THE POWER SUPPLY MODULE.

3.11 SYMPTON: INTERMITTANT SWITCHING OF LVSL SIGNAL

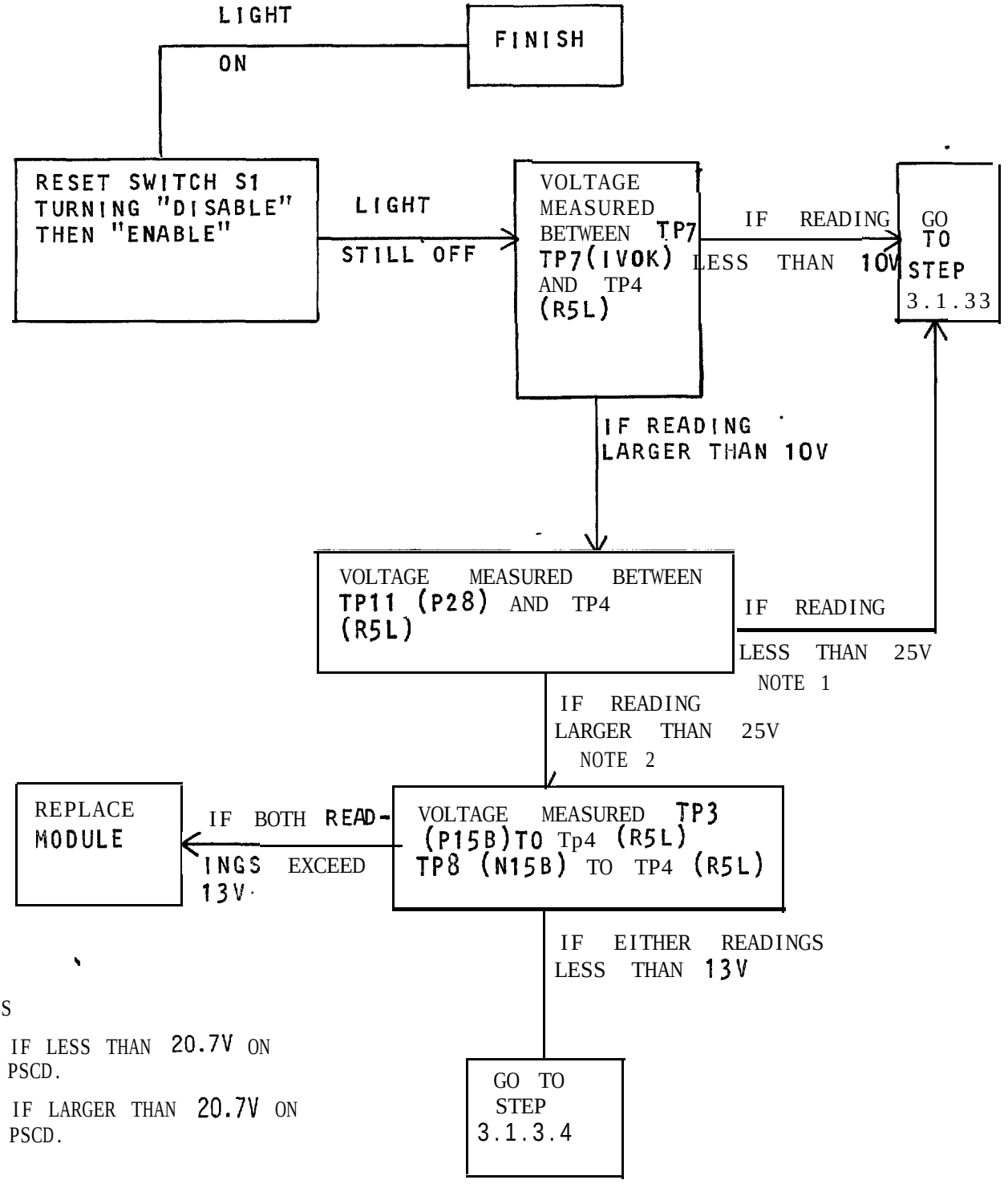
DIAGNOSIS: INPUT VOLTAGE MAY BE MARGINALLY LOW. VERIFY INPUT TO BE IN RANGE OF 80 TO 140 VDC. CHECK R5L TIED TO DCOM TO ACOM, AND R5H TIED TO P5 AT LOAD TERMINAL WHEN PSCB IS USED.

3.12 SYMPTON : NO STATUS, NO OUTPUT

DIAGNOSIS: REMOVE POWER FROM INPUT -CONNECTOR JD. CHECK TO SEE IF THE FUSES FU1 AND FU2 HAVE BLOWN-OUT. THEY ARE LOCATED NEAR THE JD CONNECTOR. REPLACE IF DEFECTIVE.

3.1.3 SYMPTON : SWITCH "S1" IS "ENABLE", BUT STATUS LIGHT IS OFF.

3.1.3.1. DIAGNOSIS AND RECOMMONDATION (PSCC, PSCD ONLY)

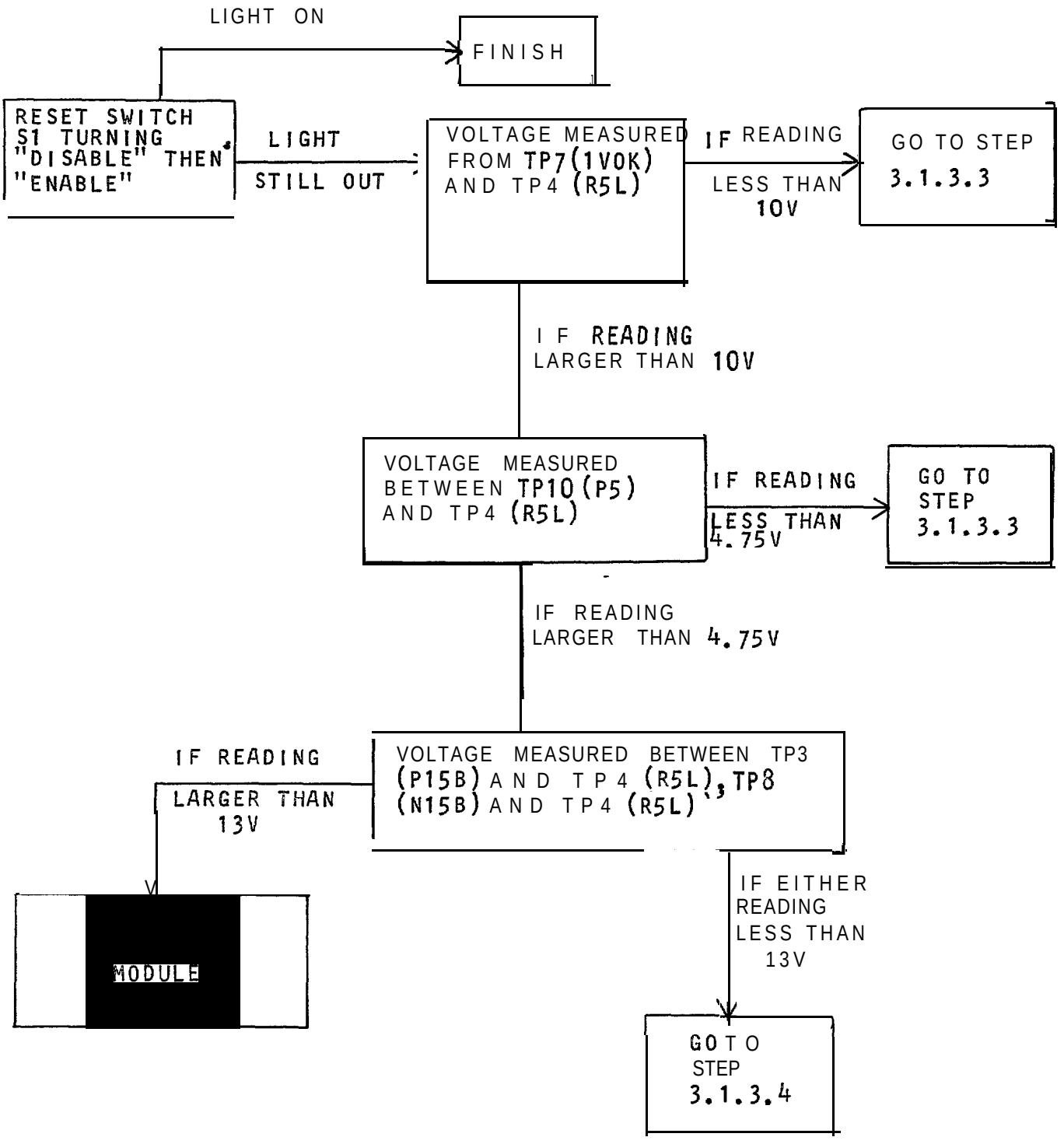


NOTES

1. IF LESS THAN 20.7V ON PSCD.
2. IF LARGER THAN 20.7V ON PSCD.

J.1 OGA 4-21-83 PSI	REV.4	REV. 7	PRINTS TO DL113 BHEL 2520	ENGINEER JPI 7/2/81	GENERAL ELECTRIC SALEM, VA. U.S.A.	POWER SUPPLY
REV 2	REV. 5	ISSUED 7-3-81				DS3820PSC
REV 3	REV. 6	MADE BY R.H. HUANG				CONT. ON SH. 3AC SH. NO. 3AE

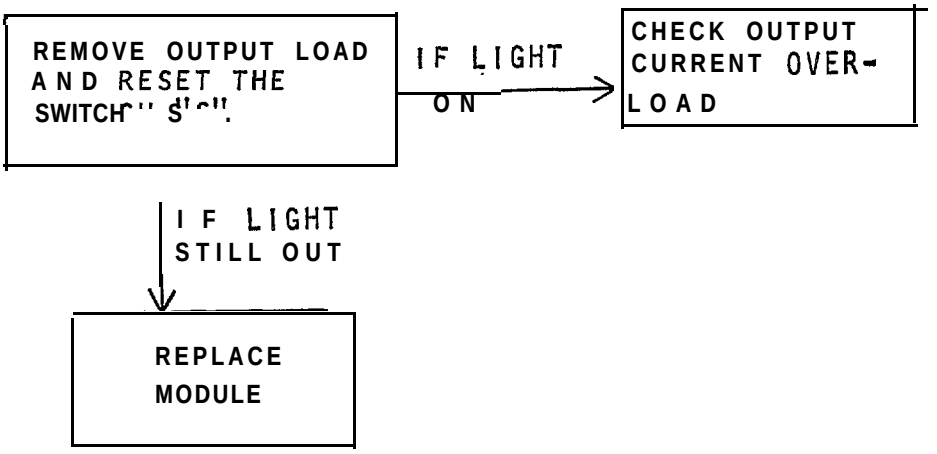
3.1.3.2 DIAGNOSIS AND REMEDIAL ACTION (PSCB ONLY)



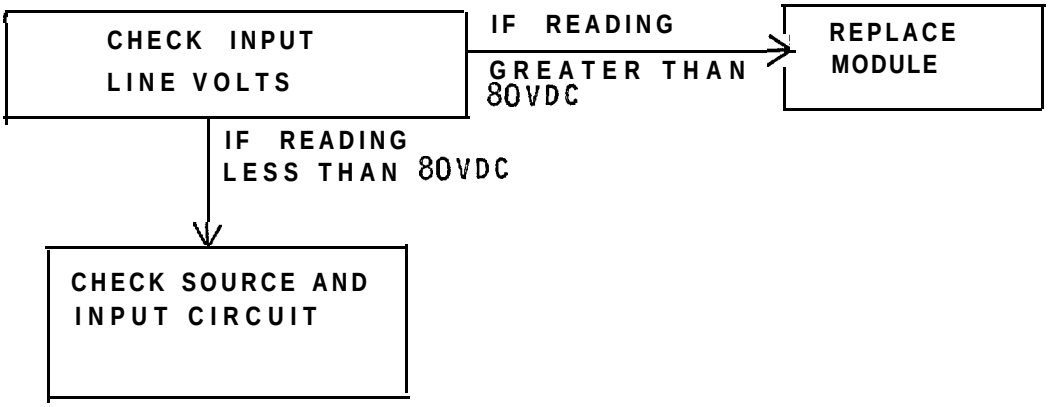
J 1	REV. 4	REV. 7	PRINTS TO DL113 2520 BHEL	ENGINEER RH 7/9/81	GENERAL@ ELECTRIC	POWER SUPPLY
REV 2	REV. 5	ISSUED 7-3-81				
REV 3	REV. 6	MADE BY K.H. HUANG			SALEM. VA. U.S.A.	DS3820PSC 3AD SH NO. 3A

CONT. ON SH 2

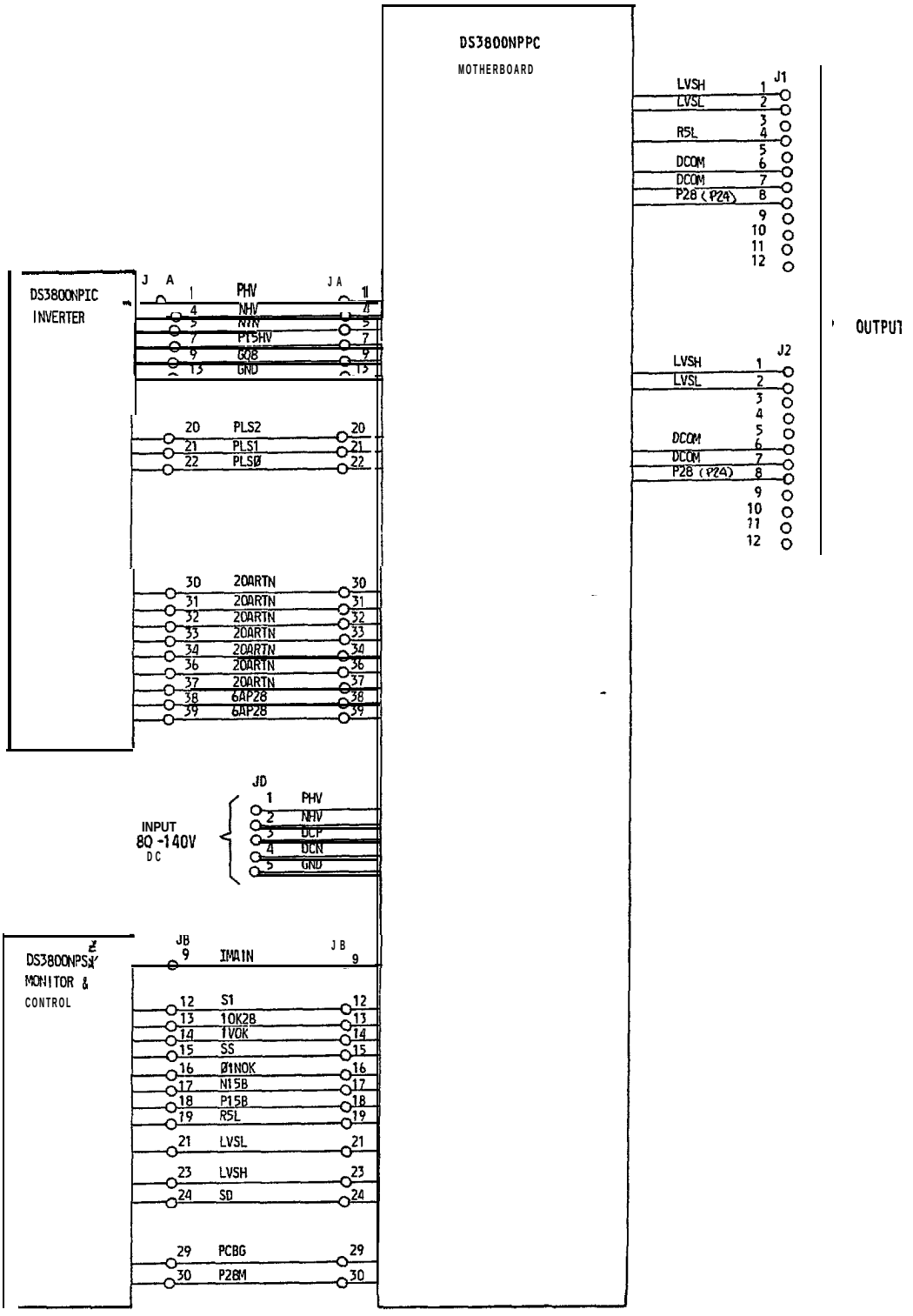
3.1.3.3 DIAGNOSIS AND REMEDIAL ACTION:



3.1.3.4 DIAGNOSIS AND REMEDIAL ACTION:



AF 10527



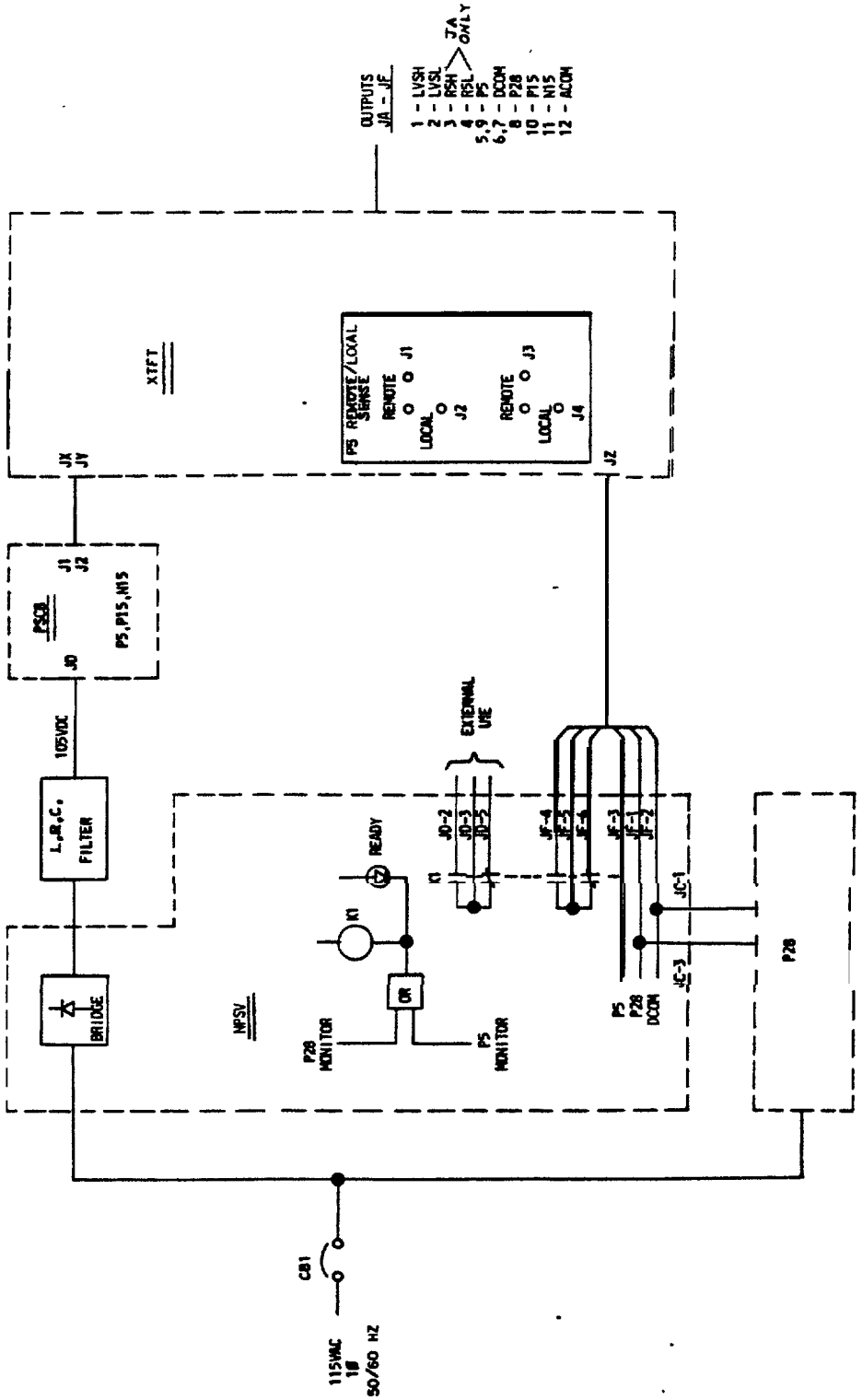
REV. 1 OGC 4-21-83	REV. 4	REV. 5	PRINTS TO BKET DL114	ENGINEER R. H. 7/2/81
REV. 2	ISSUED July 2, 1981			
REV. 3	MADE BY REX HUANG (JAF)			

GENERAL ELECTRIC
DRIVE SYSTEMS DEPARTMENT
SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM
Power Supply
FORM C, D

DS3820PSC

CONT. ON SH. FL, SH. NO. 4BA



23/1983	REV. 4	REV. 5	PRINTS TO DL114	ENGINEER RA Shiflett	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM Power Supply, Form B BLOCK DIAGRAM
REV. 2	ISSUED 2-4-83					D33620P3L
REV. 3	MADE BY R.G. SHIFLETT					CONT. ON SH. 2BA SH. NO. 2AA

CARS

DEFINITION AND FUNCTION

The DS3820PSL Power Supply module, Form B, provides P5, P28, P15, N15, ACOM, OCOM, and Low Voltage Sense outputs from a single-phase, 115VAC input. The module contains a P28 power supply, a DS3800NPSV monitor board, an L-R-C fllter, a DS3820PSCB power supply module, and a DS3800XTFT distribution board. Power is received via termlnal board connections, and is output v-la the terminal board and the XTFT board's connector plugs. (The DS3820PSL module, Form A, excludes the DS3820PSCB module.)

THEORY OF OPERATION

P28 POWER SUPPLY

The PSL module (Form B) contains a P28 power supply that transmits +28VDC to the XTFT distribution board through the NPSV board. This power supply receives a single-phase, 115VAC Input vla the PSL module's CB1 connection and through an MOV (metal oxide varlator).

DS3800NPSV MONITOR BOARD

The NPSV board monitors both +5VDC from the DS3820PSCB power supply and +28VDC from the P28 power supply. If both supplies are okay, an on-board status LEO (labelled READY) turns on and an on-board relay (labelled Kl) picks up; if either supply fails, the READY light turns off and Kl drops out. The relay outputs are provlded on terminal board TB2 (see Table 2), and one set of normally open contacts are used for LVSL Inputs in the Low Voltage Sense circuit.

The NPSV transmits +28VDC to the DS3800XTFT distribution board. Additionally, the board contains a dlode brldge that converts the module's 115VAC Input to DC, then transmits the voltage to an L-R-C filter.

L-R-C FILTER

The PSL module's L-R-C (Inductor-resistor-capacitor) assembly filters the dlode bridge output to produce a 105VDC output, then transmits the voltage to terminal board TB1 (see Table 2) and to the DS3820PSCB power supply.

DS3820PSCB POWER SUPPLY MODULE

The PSCB module supplles P5, P15, and N15 to the DS3800XTFT distribution board. The PSCB contains a yellow status LEO that Is "on" when Its 3 voltage outputs are wlthln limits.

Berg jumpers located on boards wlthln the PSCB must be connected as follows to enable the PSCB fault circuitry: On the DS3800NPSJ board, place a jumper In the D1-D2 position; on the DS3800NPPB board, place a jumper In the A2-A3 position.

For more detalled information about the PSCB module, refer to the DS3820PSCB Descrlptlon of Operation and Application Data (section 2).

REV.1 Sept. 1983	REV.4	REV.7	PRINTS TO I DL112	ENGINEER RHS	DESCRIPTION OF OPERATION AND APPLICATION DATA
REV.2	REV.5	ISSUED May 25. 1983		GENERAL ELECTRIC	Power Supply Form B
(REV.3	(REV.6	MADE BY R. G. Shiflett		COMPANY SALEM,VA	DS3820PSL CONT.ON SH. 2CA SH.NO. 2BA

DS3800XTFT DISTRIBUTION BOARD

The XTFT board receives power inputs from within the PSL module to distribute that power externally to multiple loads via connector plugs JA through JF (see Table 3). Additionally, the XTFT contains Berg jumpers that control P5 sensing as follows: To enable "remote" P5 sensing, connect jumpers J1 (R) and J3 (R), with J2 and J4 disconnected; to enable "local" P5 sensing, connect jumpers J2 (L) and J4 (L), with J1 and J3 disconnected.

Law Voltase Sense

The XTFT provldes Low Voltage Sense outputs (LVSL and LVSH) via connector plugs JA through JF, pins 1 and 2 respectively on each plug. For the LVSH output signal, the XTFT transfers the PSCB module's output to plugs JA through JF, pin 1. For the LVSL output, however, the NPSV board's signal and the PSCB's signal connect in sertes (on the XTFT) so that the XTFT outputs (pin 2) on all plugs will go high if either LVSL signal goes high.

The NPSV's Low Voltage Sense signals are a "normally open relay contact" that indicate the PS and P28 pawer; the PSCB module's Low Voltage Sense signals are open collector that indicate the PS, P15, and N15 power.

APPLICATION DATA

POWER SPECIFICATIONS

Input: 115VAC ±10%, single phase, 50/60 Hz. via TB1 (see Table 2)

Output: +5VDC, ±15VDC, +28VDC, vta XTFT plugs JA through JF (see Table 3);
+28VDC, via TB2 (see Table 2)

WARNING: THIS EQUIPMENT CONTAINS A POTENTIAL HAZARD OF ELECTRICAL SHOCK OR BURN. ONLY PERSONNEL WHO ARE ADEQUATELY TRAINED AND THOROUGHLY FAMILIAR WITH THE EQUIPMENT AND THE INSTRUCTIONS SHOULD INSTALL, OPERATE, OR MAINTAIN THIS EQUIPMENT.

TO PREVENT HAZARD OF ELECTRICAL SHOCK OR BURN, APPROVED GROUNDING PRACTICES AND PROCEDURES MUST BE ADHERED TO.

MECHANICAL DATA

Size

Helght: 13.75 inches
Width: 13.26 inches
Depth: 13.3 Inches
Weight: 38 lbs

Mount1 nq

Panel mounted

REV.1 mREV. 4 Sept. 19831	JREV.7	PRINTS TO DL112	ENGINEER RAJ	DESCRIPTION OF OPERATION AND APPLICATION DATA
REV.2 / REV.5	ISSUED May 25, 1983		GENERAL ELECTRIC	Power Supply Form B
REV.3 / REV.6	MADE BY R. G. Shiflett		COMPANY SALEM,VA	DS3820PSL CONT.ON SH. 2CB SH.NO. 2CA

P28 Adjustments

The P28 power supply requires no adjustments. The P28 output should measure 28VDC ± 1VDC from TB2-1 (+28VDC) to TB2-5 (DCOM).

P5, P15, and N15 Adjustments

If voltages appear to be out of tolerance, it Is strongly recommended that any measurements and corrections be made in reference to R5L (TP4) on the NPSJ board (located in the PSCB module). Table 1 lists the adjustment data for the P5, P15, and N15.

All busses are monitored in reference to R5L (TP4) on the NPSJ board. P5 is remotely monitored In reference to the NPSJ board's R5H (TP10), but that point Is connected to TP4. The P15 and N15 busses are locally sensed at the PSCB power supply.

Table 1
P5, P15, and N15 ADJUSTMENT DATA*

PSCB OUTPUT	POT NUMBER ON BOARD	LIMITS	TEST POINT I ON NPSJ
P5	R1 on NPPB	±50mV	TP10
P15	R6 on NPSK	±100mV	TP2
N15	R7 on NPSK	±100mV	TP3

* Note: PSCB measurements are with respect to TP4 on the NPSJ board.

ACOM and DCOM do not connect together directly at the power supply, but usually connect at the load. The NPSK board (In the PSCB module) contains back to back diodes between R5L and ACOM.

I/O CONNECTIONS

Table 2
TERMINAL BOARD CONNECTIONS

SIGNAL NAME	PIN NO.	DIRECTION	DEFINITION
DCPOS	TB1-1	Output	Positive DC output from diode bridge
DCN	TB1-2	Output	Negative DC output from diode bridge
GND	TB1-3	Input	Power supply chassis ground
AC1	TB1-4	Input	AC Power input to PSL module
AC2	TB1-5	Input	AC Power input to PSL module
P28	TB2-1	Output	+28VDC output from P28 Power supply
C	TB2-2	Output	K1 relay output. common connection
NC	TB2-3	Output	K1 relay output. normally closed connection
NO	TB2-4	Output	K1 relay output, normally open connection
DCOM	TB2-5	Output	Digital common from P28 power supply

Table 3
XTFT POWER OUTPUT
,-CONNECTORS JA, JB, JC, JD, JE, and JF

SIGNAL NAME	PIN NO.	DEFINITION
LVSH	1	Low voltage sense high
LVSL	2	Low voltage sense low
R5H*	3	Remote P5 high, enabled by jumper J1
R5L*	4	Remote P5 low, enable by lumper J3
P5	5	Power +5VDC
DCOM	6	Power digital common
DCOM	7	Power digital common
P28	8	Power +28VDC
P5	9	Power +5VDC
P15	10	Power +15VDC
N15	71	Power -15VDC
ACOM	12	Power analog common

* NOTE: R5H and R5L are provided only on the JA connector; pins 3 and 4 on connectors JB through JF are not used. (The R5 sensing function must be enabled by connecting the XTFT board's J1 and J3 Berg jumpers.)

OPERATING AND STORAGE ENVIRONMENT

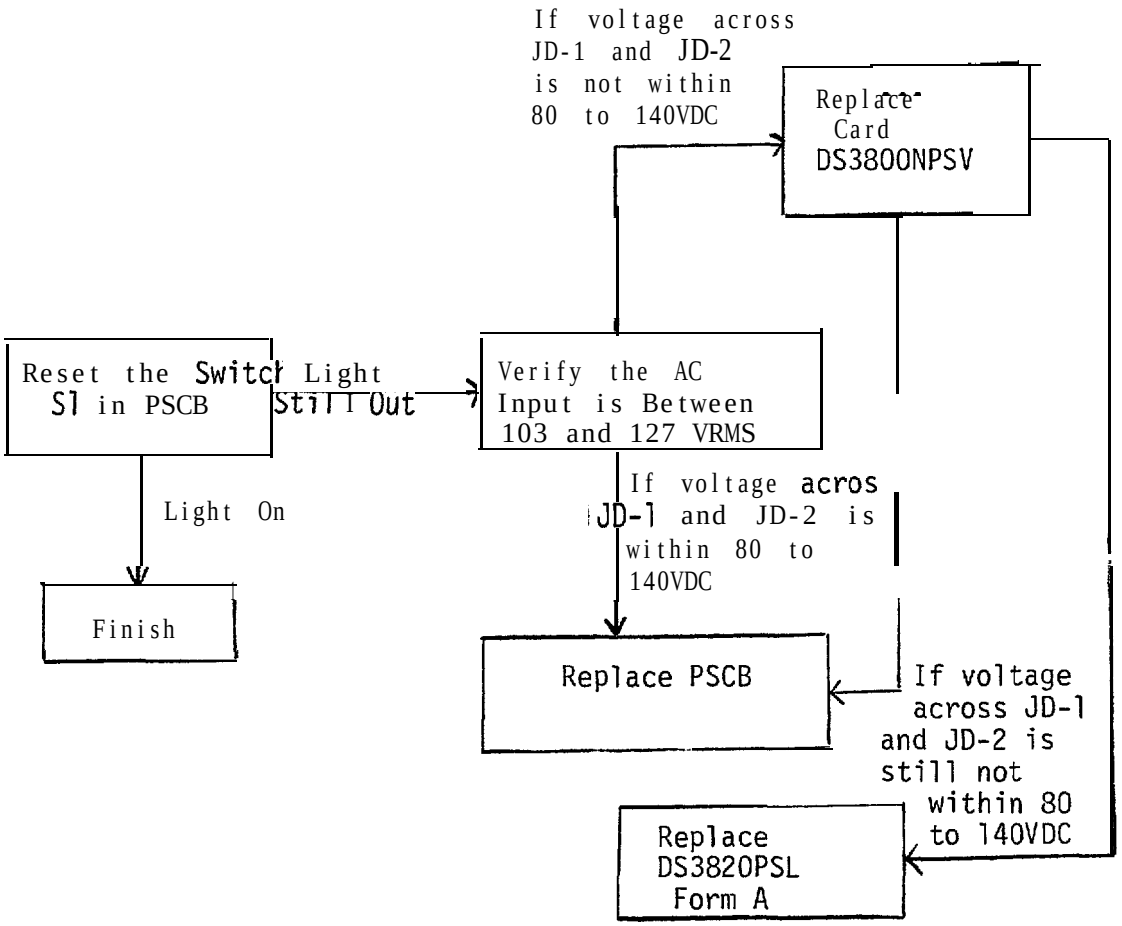
See instruction book GEK-61680.

REV.1 JB Sept. 19831	REV.4	REV.7	PRINTS TO DL112	ENGINEER RJ	DESCRIPTION OF OPERATION AND APPLICATION DATA
REV.2	REV.5	ISSUED May 25. 1983		GENERAL ELECTRIC	Power Supply Form B
REV.3	REV.6	MADE BY R. G. Shlflett		COMPANY SALEM, VA	DS3820PSL CONT.ON SH. FL SH.NO. 2DA

3.0 Troubleshooting

This power supply makes accessible to the user multiple test points and visual aids which determine the power supply status. The supply is not repairable in the field. When it is determined that the supply is defective, it should be replaced as a unit and returned to a repair depot. A list of symptoms and possible causes as well as remedial action or depot repair required is given below:

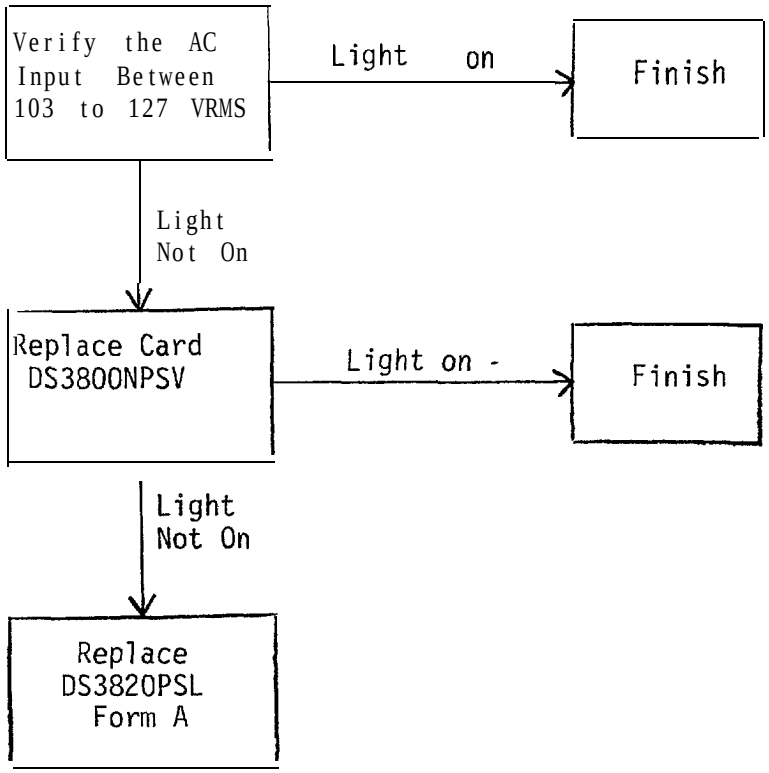
3.1 Symptom: Status on DS3820PSCB



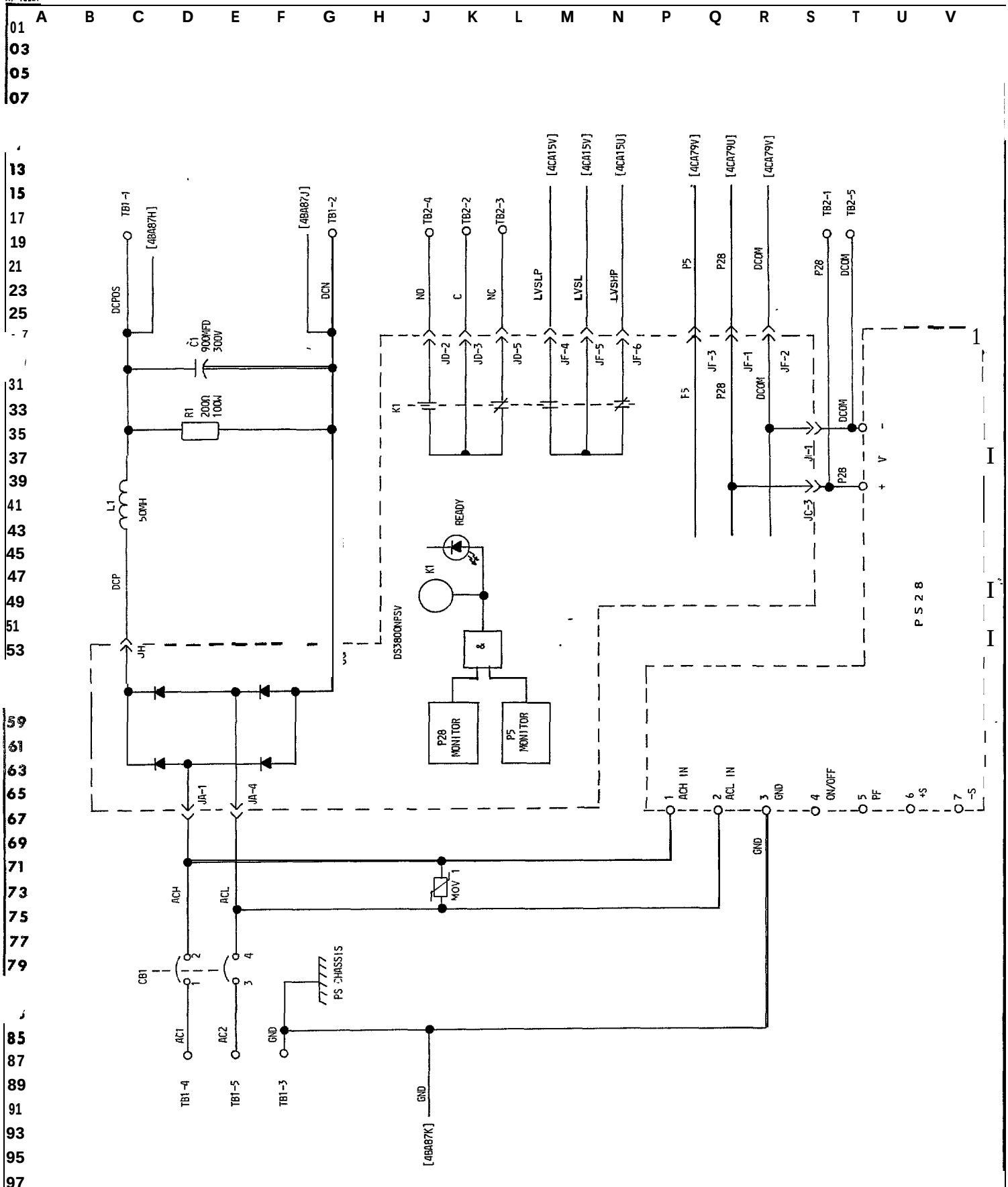
REV. 1	REV. 4	REV. 7	PRINTS TO DL113	ENGINEER P.S.P.	GENERAL ELECTRIC	TROUBLESHOOTING INSTRUCTIONS
REV. 2	REV. 5	ISSUED 2-3-83				
REV. 3	REV. 6	MADE BY RG SHIFLETT			SALEM. VA. U.S.A.	DS3820PSL CONT ON SH. 3AB SH. NO. 3AA

DSD 0151 (09-76)

3.2 Symptom: Status light on NPSV "off" but status light on PSCB is "on."



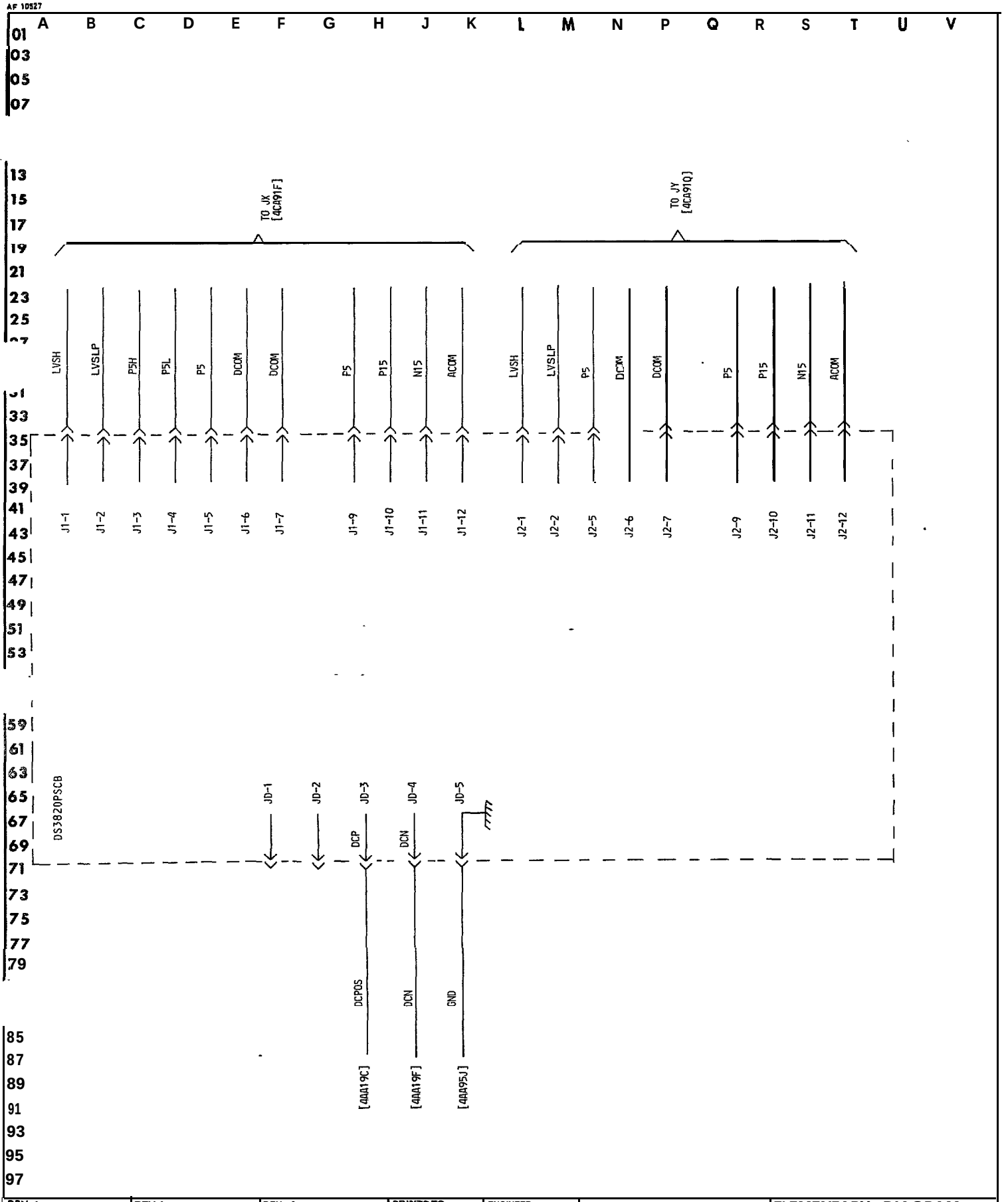
AF 10527



V.1	REV. 4	REV. 5	PRINTS TO DL114	ENGINEER RA 2-4-83	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
V.2	ISSUED 2-4-83				DRIVE SYSTEMS DEPARTMENT	FOR A & B
REV. 3	MADE BY R.G.SHIFLETT				SALEM, VA. U.S.A.	DS3820PSL
						CONT. ON SH. 4BA SH. NO 4AA

DSD 0152 (09 76)

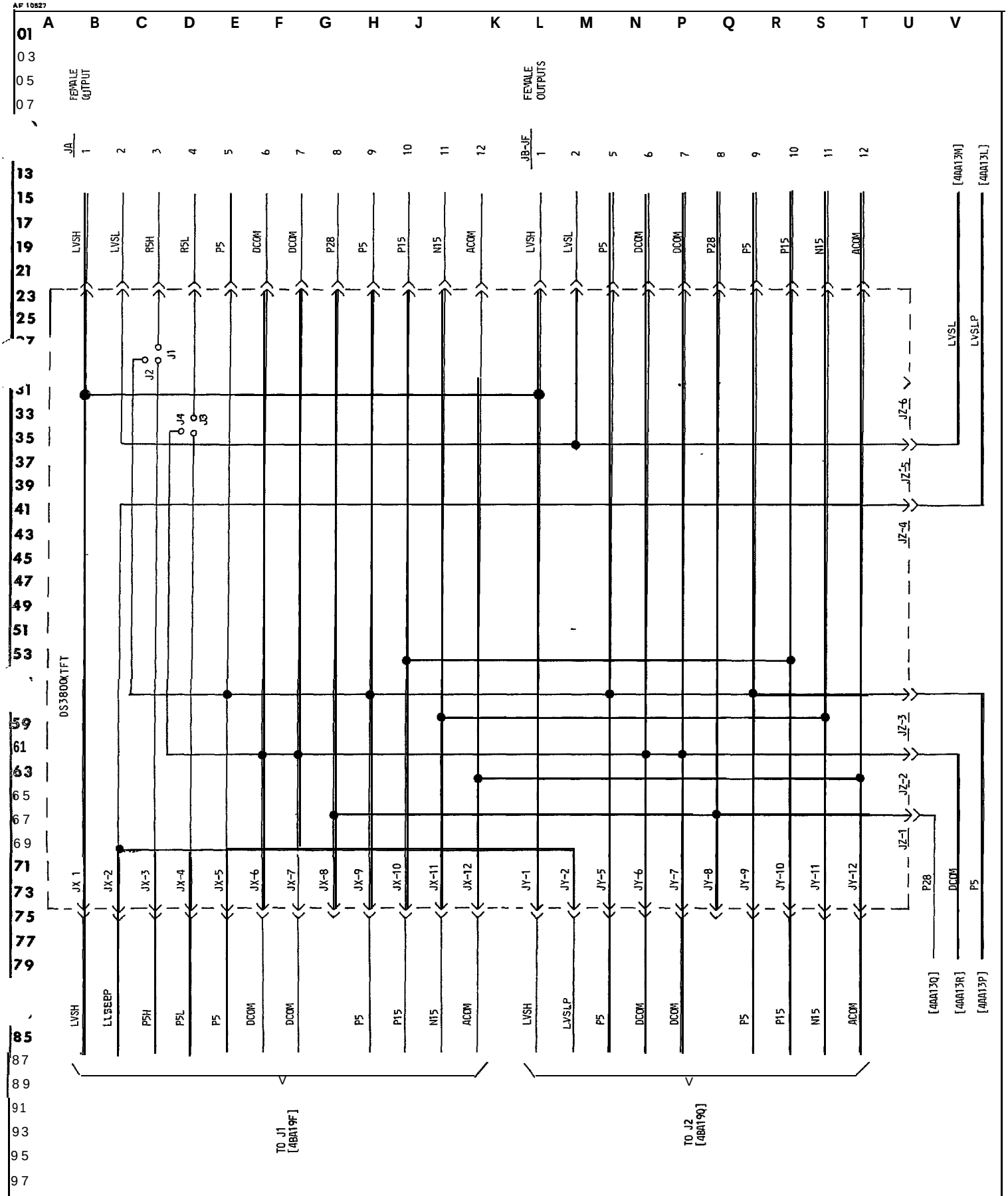
AF 10527



REV.1	REV.4	REV. 6	PRINTS TO DL114	ENGINEER RB DD Fleet	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM FORM B ONLY
.2	ISSUED 2-4-83					
REV.3	MADE BY R.G.SHIFLETT					DS3820PSL
						CONT. ON SH.4CA SH. NO.4BA

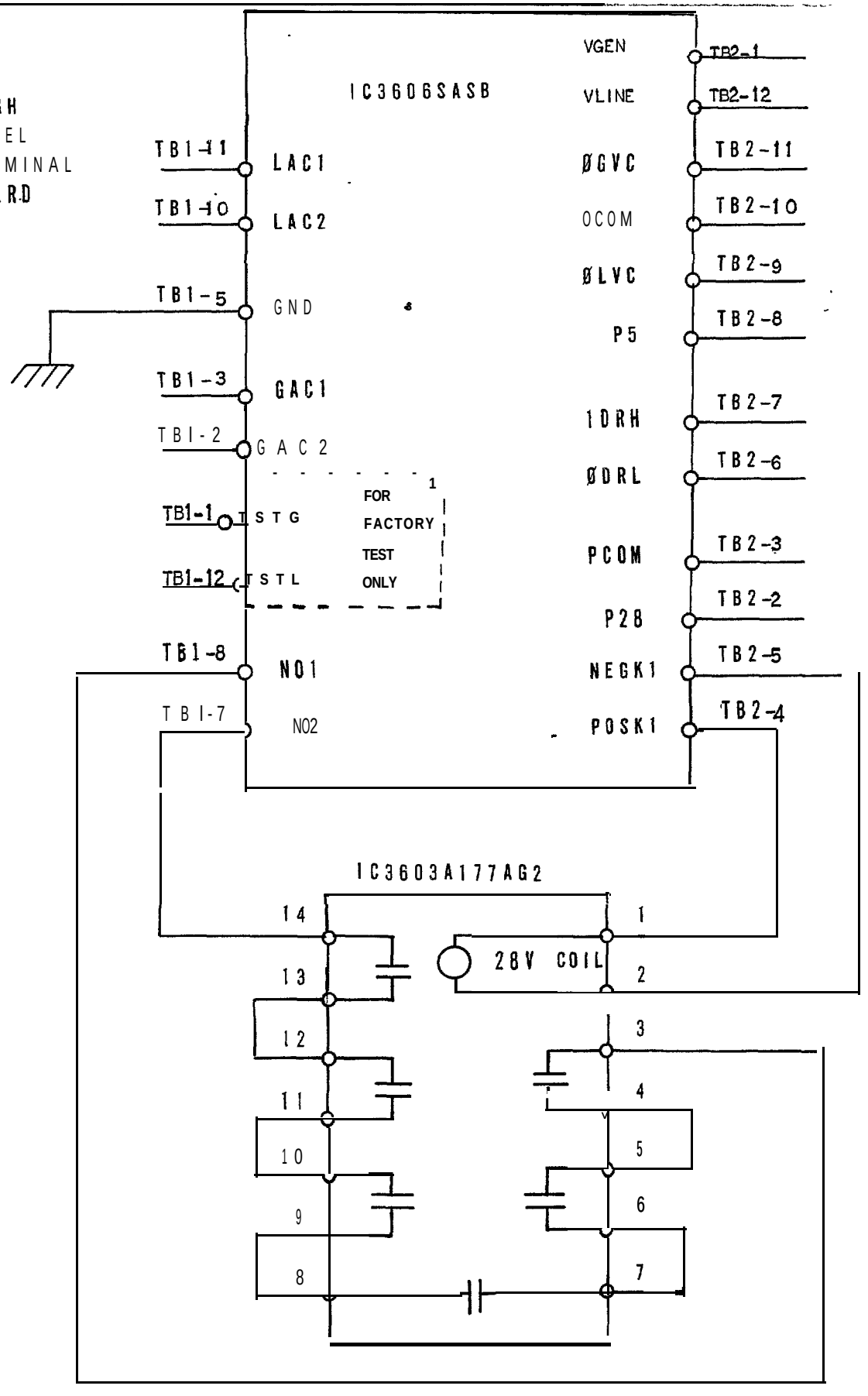
DSD 0193 [09 76]

AF 10527



REV. 1 OGH-2 11-2-83 PSP	REV. 4	REV. 5	PRINTS TO DL114	ENGINEER RA	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM FORM A & B
√ 2	ISSUED 4-4-83					
AEV.3	MADE BY R.G.SHIFLETT					DS3820PSL CONT. ON SH. FL. SH. NO. 4CA

DBD 0152 (09 76)



EV.1	REV. 4	REV. 7	PRINTS TO 3EF1 4QA5 4EB2 4RA1	ENGINEER RE Grubbs	GENERAL ELECTRIC	MICROSYNCHRONIZER ISOLATION MODULE
REV. 2	REV. 5	ISSUED 2 / 11		CUSTOM CONTROL	DRIVE SYSTEMS DEPT.	IC3602A178
REV. 3	REV. 6	MADE BY ROBERT E. GRUBBS			SALEM, VA. U.S.A.	CONT. ON SH. 3.1 SH. NO. 3.0

DSD 0151 (09-76)

APPLICATION NOTES FOR IC3602A178

1.0 DEFINITION AND PURPOSE

THE IC3602A178 MICROSYNCHRON'I ZER ISOLATION MODULE IS THE INPUT AND OUTPUT ISOLATION MODULE FOR THE GAS TURBINE MICROSYNCHRONIZER SYSTEM. THE MODULE CONSISTS OF A PWB, IC3606SASB, A KMR RELAY, AND A MOUNTING PLATFORM AND COVER.

THIS MODULE ISOLATES TWO 120 VOLT AC SIGNALS (LINE VOLTAGE AND GENERATOR VOLTAGE) FROM THE PAGE SIGNAL ELECTRONICS. THESE ISOLATED SIGNAL VOLTAGES ARE CONVERTED TO DIGITAL LEVELS BY ZERO VOLTAGE CROSS DETECTOR CIRCUITS AND OUTPUT TO THE MICROSYNCHRONIZER, IC3600SASC. VOLTAGE LEVEL PERMISSIVE CIRCUITS PREVENT SYNCHRONIZATION ATTEMPTS AT LOW AC VOLTAGES BY SHUTTING OFF THE DIGITAL OUTPUT IF THE AC VOLTS DROPS TOO LOW. SASB ALSO PROVIDES A RELAY DRIVER CIRCUIT TO DRIVE A "BREAKER CLOSE" RELAY MOUNTED AS PART OF THE ISOLATION MODULE.

MAJOR CIRCUIT FUNCTIONS ARE.

- 2 TRANSIENT SUPPRESSJON FILTERS
- 2 ISOLATION SIGNAL TRANSFORMERS
- 2 ZERO VOLTS COMPARATORS
- 2 VOLTAGE LEVEL COMPARATORS
- 2 VOLTAGE PERMISSIVE LED'S
- RELAY DRIVER WITH REDUNDANT INPUTS
- "BREAKER CLOSE" RELAY

2.0 THEORY OF OPERATION

REFER TO THE FUNCT IONAL BLOCK DIAGRAM ON SHEET 3.7. THE SASB CONTAINS TWO IDENTICAL ISOLATION CIRCUITS. ONE TO ISOLATE THE LINE VOLTAGE AND ONE TO ISOLATE THE GENERATOR VOLTAGE. EACH CIRCUIT HAS A HIGH VOLTAGE SUPPRESS ION NETWORK CONSISTING OF A MOV AND A RESISTOR CAPACITOR DIVIDER. THE 120 VAC SIGNAL IS TRANSFORMER ISOLATED AND REDUCED DOWN TO ABOUT 8 VAC . THE 8 VAC S IGNAL IS I NPUT TO' A COMPARATOR WITH A REFERENCE OF ZERO VOLTS - THUS A ZERO VOLTS CROSS DETECTOR. EACH POSITIVE GOING ZERO VOLTS CROSSING GENERATES A HIGH TO: LOW LOGIC TRANSITION - THUS THE OUTPUT OF THE COMPARATOR IS A TTL SQUARE WAVE, INVERTED, AND IN PHASE WITH THE INCOMING AC SIGNAL. THE 8 VAC SIGNAL IS ALSO RECTIFIED AND INPUT TO A LEVEL COMPARATOR WHICH ACTS AS A PERM ISS IVE FOR THE ZERO VOLTS CROSS DETECTOR OUTPUT. THIS PERM ISS I VE LEVEL PREVENTS SYNCHRONIZATION IF THE LINE VOLTAGE IS BELOW 60 VAC OR IF THE GENERATOR VOLTAGE IS BELOW 85 VAC . TWO LED' S I NDICATE WHEN THE PERMISSIVE LEVELS ARE PRESENT. LVP 1 ND ICATES LINE VOLTAGE PERMISS IVE IS ENABLED AND GVPINDICATES GENERATOR VOLTAGE PERMISSIVE IS ENABLED.

2.0 THEORY OF OPERATION (CONT D)

IN ADDITION TO THE AC ISOLATION CIRCUITRY, SASB ALSO CONTAINS A RELAY DRIVER CIRCUIT FOR DRIVING THE "BREAKER CLOSE" RELAY. TO ENERGIZE THE RELAY, TWO PERM ISSIVES ARE REQUIRED (2 FOR REDUNDANCY). 1DRH MUST BE DRIVEN HIGH BY AN ACTIVE HIGH DRIVER AND ØDRL MUST BE PULLED LOW. IF EITHER SIGNAL IS NOT TRUE OR OPEN, THE RELAY WILL NOT ENERGIZE. A SUPPRESS ION NETWORK CONS IST I NG OF A SERIES. RESI STOR AND I CAPACITOR ARE PROVIDED TO CONNECT ACROSS THE RELAY CONTACTS TO REDUCE CONTACT WEAR DUE TO BOUNC ING. THE NORMALLY OPEN CONTACTS OF THE "BREAKER CLOSE" RELAY, K1, ARE USED TO ENERGIZE AN EXTERNAL LINE CONTACTOR TO PUT THE TURBINE GENERATOR ON LINE.

REFER TO t C3600SASC, MICROSYNCHRONI ZER CARD, FOR MDRE DETA I LED SYNCHRONIZER SYSTEM INFORMATION.

3.0 FUNCTIONAL DESCRIPTION

3.1 INPUTS

MNEMONIC	PIN NO.	DESCRIPTION
*LAC1, LAC2	TB1-11,10	AC LINE VOLTAGE INPUT 150 VAC MAX. 20MA MAX. MUST MATCH PHASI NG WITH GAC1, GAC2.
*GAC1, GAC2	TB1-3,2	AC GENERATOR VOLTAGE INPUT 150 VAC MAX., 20MA MAX. MUST MATCH PHAS I NG WITH LAC1, LAC2
1DRH	TB2-7	RELAY DRIVER INPUT, LOGIC "1" ENABLES HALF OF DR IVER. $V_{1H(MIN)}$ = 2.4V AT 1.5MA $V_{1L(MAX)}$ = .5V AT OMA
ØDRL	TB2-6	RELAY DRIVER INPUT, LOGIC "0" ENABLES HALF OF DRIVER $V_{1L(MAX)}$ = 0.4V AT -1.5MA $V_{1H(MIN)}$ = 3.5V AT OMA

* WARNING . IF PHASES ARE REVERSED, SERIOUS DAMAGE MAY BE DONE TO TURBINE GENERATOR IF SYNCHRONIZATION IS ATTEMPTED. ALWAYS VER IN PROPER PHASE RELATIONSHIP BEFORE OPERAT t NG MICROSYNCHRONIZER.

3.0 FUNCTIONAL DESCRIPTION (CONT D)

TSTL,TSTG	TB1-12,1	ISOLAT I ON TRANSFORMER PRIMARY CENTER TAPS, TEST I NPUTS ONLY.
GND	TB1-5	SUPPRESS ION BYPASS TO GROUND. 'MUST BE TIED TO GROUND OR CHASSIS TO PROVIDE PATH FOR TRANSIENT CURRENTS. NORMALLY TIED TO CHASSIS IN ISOLATION MODULE, THUS USE STAR WASHERS TO MOUNT MODULE TO PANEL WALL.

3.2 OUTPUTS

MNEMONIC	PIN NO	DESCR I PT I ON
ØLVC	TB2-9	LINE VOLTAGE ZERO CROSS LOGIC OUTPUT. TTL SQUARE WAVE IN PHASE WITH AC LINE VOLTAGE, THE FALLING EDGE IS USED TO GENERATE A LINE INTERRUPT ON MICRO-SYNCHRON I ZER CARD. IC3600SASC. USE SHIELDED, TWISTED PAIR TO CONNECT OLVC TO SASC. OUTPUT STAYS LOGIC "1" WHEN LAC1,2 I S BELOW 60 VAC. $V_{OH(MIN)}$ = 2.4V AT -3MA. $V_{OL(MAX)}$ = 0.4V A T 12MA.
ØGVC	TB2-11	GENERATOR VOLTAGE ZERO CROSS LOGIC OUTPUT. TTL SQUARE WAVE IN PHASE WITH AC GENERATOR VOLTAGE. THE FALLING EDGE IS USED TO CAUSE A GENERATOR INTERRUPT ON THE MICROSYNCHRONIZER CARD. IC3600SASC. USE SHIELDED, TWISTED PAIR TO CONNECT ØGVC TO SASC. OUTPUT STAYS LOGIC "1" WHEN GAC1,2 IS BELOW 85 VAC. $V_{OH(MIN)}$ = 2.4V A T -3MA $V_{OL(MAX)}$ = 0.4 VOLT 12MA.

3.0 FUNCTIONAL DESCRIPTION (CONT'D)

3.2 OUTPUTS (CONT'D)

MNEMONIC	PIN NO	DESCRIPTION
POSK1, NEGK1	TB2-4,5	FOR INTERNAL MODULE WIRING ONLY. RELAY DRIVER OUTPUT TO "BREAKER- CLOSE" RELAY, K1.
VLINE	TB2-12	SECONDARY OF LINE VOLTAGE ISOLATION TRANSFORMER, RESISTOR BUFFERED THRU 2.21KΩ. OUTPUT IS 9 VAC ±0.5 VAC FOR 120 VAC INPUT ON LACY ,2.
VGEN	TB2-1	SAME AS VLINE, EXCEPT FOR GENERATOR VOLTAGE SECONDARY.
NO 1, NO 2	TB1-8,7	SIX NORMALLY OPEN CONTACTS IN SERIES ON THE "BREAKER CLOSE" RELAY K1. USE THIS INTERLOCK TO ENERGIZE AN EXTERNAL LINE CONTACTOR TO PUT THE TURBINE GENERATOR ON LINE. CONTACTS RATED TO CLOSE A 6 AMP, 140MH CIRCUIT ONLY. DO NOT ATTEMPT TO OPEN CIRCUIT UNDER LOAD.

3.3 INDICATING LEDS

LED	DESCRIPTION
ØLVP	THIS LED IS ON IF THE LINE VOLTAGE IS ABOVE 60 VAC AND ØLVC IS ENABLED.
ØGVP	THIS LED IS ON IF THE GENERATOR VOLTAGE IS ABOVE 85 VAC AND ØGVC IS ENABLED.

3.4 TEST POINTS

POINT	DESCRIPTION
ØLVP	LOGIC "0" IF ØLVC OUTPUT IS ENABLED. OUTPUT ONLY.
ØGVP	LOGIC "0" IF ØGVC OUTPUT IS ENABLED. OUTPUT ONLY.
DCOM	SASB DCOM

V.1	REV. 4	REV.7	PRINTS TO 3EF1 4QA5 4EB2 4RA1	ENGINEER R E Grubbs	GENERAL ELECTRIC	MICROSYNCHRONIZER ISOLATION MODULE
REV. 2	REV. 5	ISSUED 2-13-79 2520			SALEM, VA. U.S.A.	IC3602A178
REV. 3	REV. 6	MADE BY R.E. GRUBBS				CONT. ON SH. 3 SH. NO. 3

DSD 0151 (09-76)

3.0 FUNCTIONAL DESCRIPTION (CONT'D)

3.5 INSTALLATION AND MAINTAINABILITY

THE IC3606SASB IS MOUNTED INSIDE THE MICROSYNCHRONIZER ISOLATION MODULE. THERE ARE NO ADJUSTMENTS NECESSARY FOR INSTALLATION.

TWO TERMINAL BMRDS ARE PROVIDED FOR CONNECTION OF INPUTS AND OUTPUTS. TB1 IS FOR HIGH LEVEL INPUTS AND OUTPUTS, TB2 IS FOR LOW LEVEL INPUTS AND OUTPUTS. GND (TB1-5) IS JUMPERED TO CHASSIS AS PART OF THE MODULE WIRING. THUS USE STAR WASHERS TO MOUNT THE ISOLATOR MODULE TO THE PANEL WALL. DO NOT WIRE TO TSTL OR TSTG. THESE ARE FOR FACTORY TEST ONLY. SIGNALS ØGVC AND ØLVC SHOULD CONNECT TO THE MICROSYNCHRONIZER CARD. SASC, THRU TWO SHIELDED, TWISTED PAIRS (ØGVC, DCOM AND ØLVC, DCOM).

DCOM AND PERMISSIVE SIGNALS ØLVP AND ØGVP ARE AVAILABLE AS MAINTENANCE TEST POINTS.

3.6 POWER REQUIREMENTS

P5 : 5 VDC ± .25 VDC AT 85MA. MAX. MUST USE P5 FROM MICROSYNCHRONIZER CARD, IC3600SASC.
P28 : 28 VDC ± .7 VDC AT 100MA. MAX.

4.0 APPLICATION DATA

4.1 INPUT VOLTAGES

LINE VOLTAGE (LAC1-2)	150 VAC MAX. 50/60HZ
GENERATOR VOLTAGE (GAC1-2)	150 VAC MAX. 50/60HZ
LINE VOLTAGE PERMISSIVE LEVEL	60 VAC ± 6 VAC
GENERATOR VOLTAGE PERMISSIVE LEVEL	85 VAC ± 6 VAC

4.2 PHASE ERROR

TYPICALLY THE GENERATOR AND LINE VOLTAGE ARE MATCHED BEFORE SYNCHRONIZING FOR THIS CONDITION THE PHASE ERROR IS SMALL. WHEN 120 VAC IS APPLIED ACROSS, LAC1 AND LAC2 WITH LAC1 TIED TO GAC1 AND LAC2 TIED TO GAC2, THE PHASE DIFFERENCE BETWEEN THE FALLING EDGES OF ØLVC AND ØGVC WILL BE LESS THAN 10 µSEC (0.22°) TYPICAL, 46 µSEC (1.04°) MAXIMUM. FOR MISMATCHED LINE AND GENERATOR VOLTAGES OF ABOUT 35 VAC, THE TYPICAL ERROR IS LESS THAN 30 µSEC (0.65°), 75 µSEC (1.62°) MAXIMUM. FOR MISMATCHED LINE AND GENERATOR VOLTAGES OF ABOUT 65 VAC, THE TYPICAL ERROR: IS LESS THAN 48 µSEC (1.04°), 100 µSEC (2.16°) MAXIMUM.

4.3 TRANSIENT VOLTAGE SUPPRESSION

THE AC HIGH VOLTAGE INPUTS ARE DESIGNED TO WITHSTAND THE FOLLOWING SIMULATED LIGHTING/NOISE TESTS.

- 6000V VOLTAGE SURGE TEST PER UL 943
- THE ANSI C37.90A - 1974 SURGE WITHSTAND CAPACITY TEST (SWC).

V.1 BC938A 4 '30 80 PJR	REV. 4	REV. 7	PRINTS TO 3EF1 4QA5 4EB2 4RA1	ENGINEER R E Grubbs	GENERAL ELECTRIC	MICROSYNCHRONIZER ISOLATION MODULE
REV. 2 PJR DU938A485 2/18/80	REV. 5	ISSUED 2-13-79 2526				IC3602A178
REV. 3	REV. 6	MADE BY R.E. GRUBBS			SALEM, VA. U.S.A.	CONT ON SH. 3.6 SH. NO. 3.5

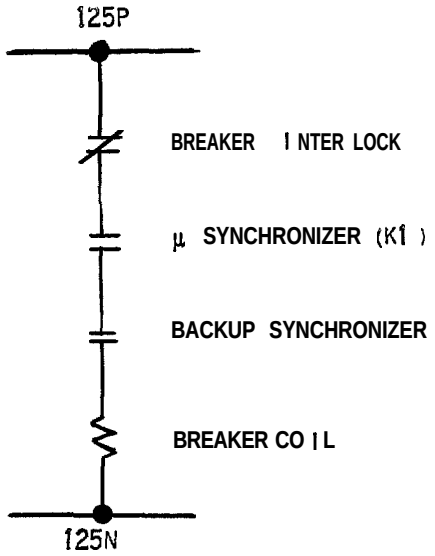
DSD 0151 (09-76)

4.4 TEMPERATURE RAWE

SASB WILL OPERATE PROPERLY OVER THE 0°C TO 70°C TEMPERATURE RANGE.

4.5 "BREAKER CLOSE" RELAY CONTACT RATING

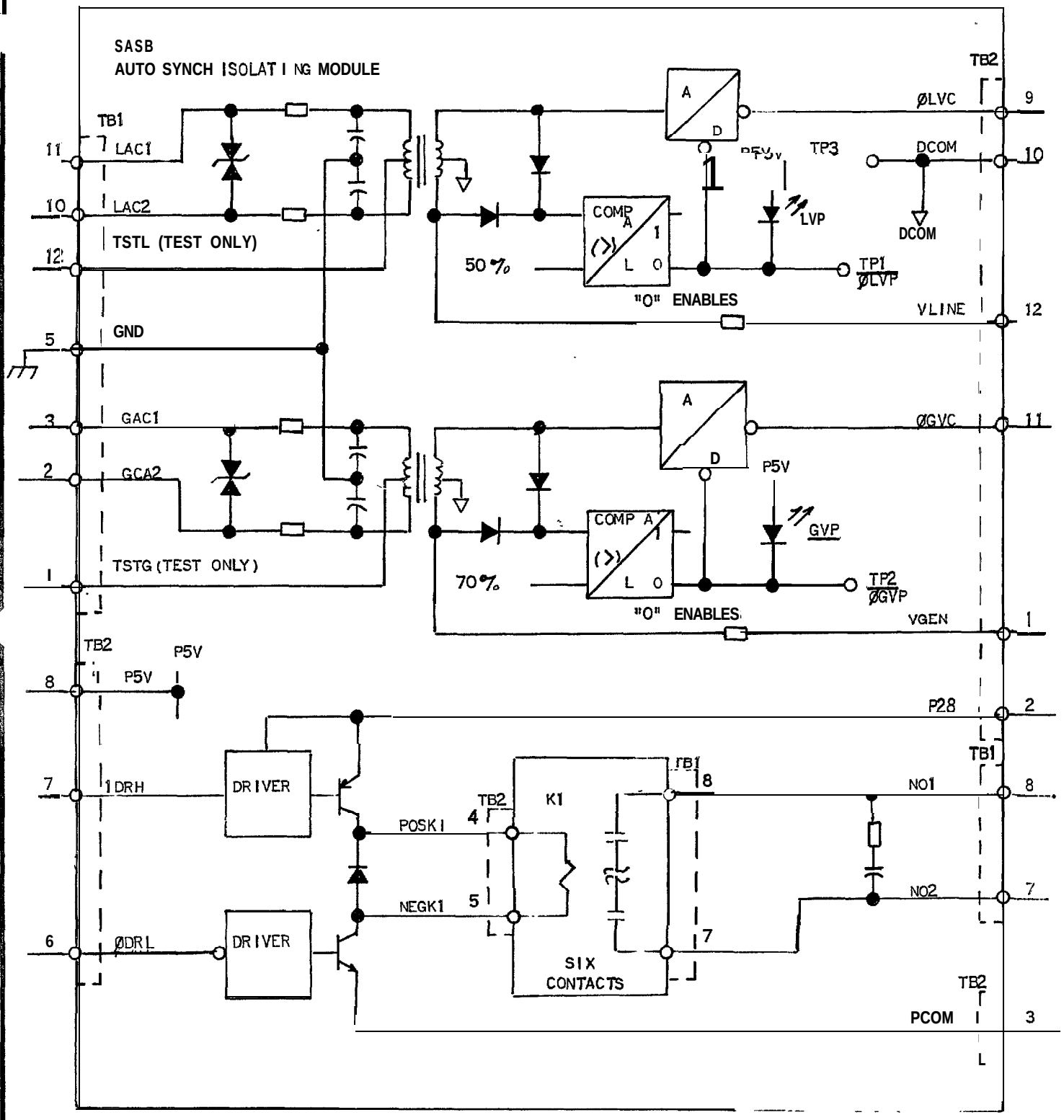
THE "BREAKER CLOSE" t NTERLOCK CONSISTS OF SIX NORMALLY OPEN CONTACTS CONNECTED IN SERIES. THE RELAY HAS BEEN TESTED TO MAKE A 125 VDC, 6 AMP, 140MH C IRCUIT FOR '10,000 OPERATIONS. THE SUPPRESS ION PROVIDED ACROSS NO 1, NO 2 IS NOT SUFFICIENT TO OPEN THIS CIRCUIT UNDER LOAD, AS ARCING WILL OCCUR. -THUS AN EXTERNAL BREAKER INTERLOCK IS REQUIRED TO BREAK THE CIRCUIT. SEE BELOW., (DUR ING FAULT TESTING, K1 SUCCESSFULLY OPENED THE ABOVE LOAD MORE THAN 12 TIMES, ALTHOUGH NOT RATED TO DO SO.)



V. 1 BU938AQJ 4/30/80 PJR	REV. 4	REV. 7	PRINTS TO 3EF1 4QA5 4EB2 4RA1	ENGINEER RE Grubbs	GENERAL ELECTRIC	MICROSYNCHRONIZER ISOLATION MODULE
REV. 2	REV. 5	ISSUED 2-13-79 2520				IC3602A178
REV. 3	REV. 6	MADE BY R.E. GRUBBS			SALEM, VA. U.S.A.	CONT. ON SH. 3.7 SH. NO. 3.6

DSD 0151 (09-76)

5.0 MICROSYNCHRONIZER ISOLATION MODULE BLOCK DIAGRAM



REV. 1 BU938AQJ 4/30/80 PJR	REV. 4	REV. 7	PRINTS TO 3EF1 4QA5 4EB2 4RA1	ENGINEER R E Grubbs	MICROSYNCHRONIZER ISOLATION MODULE
REV. 2	REV. 5	ISSUED 2-13-79		GENERAL ELECTRIC CARS SALEM, VA. U.S.A.	1C3602A178
REV. 3	REV. 6	MADE BY R. E. GRUBBS			CONT. ON SH. FL. SH. NO. 3.7

APPENDIX F

SYSTEMELEMENTARYDIAGRAM

F-l. INTRODUCTION

This appendix contains the elementary (electrical) diagram for the μPSS, groups G5, G6, G7, and G8. Elementary sheet 01B contains the Table of Contents for the elementary.

APPENDIX E

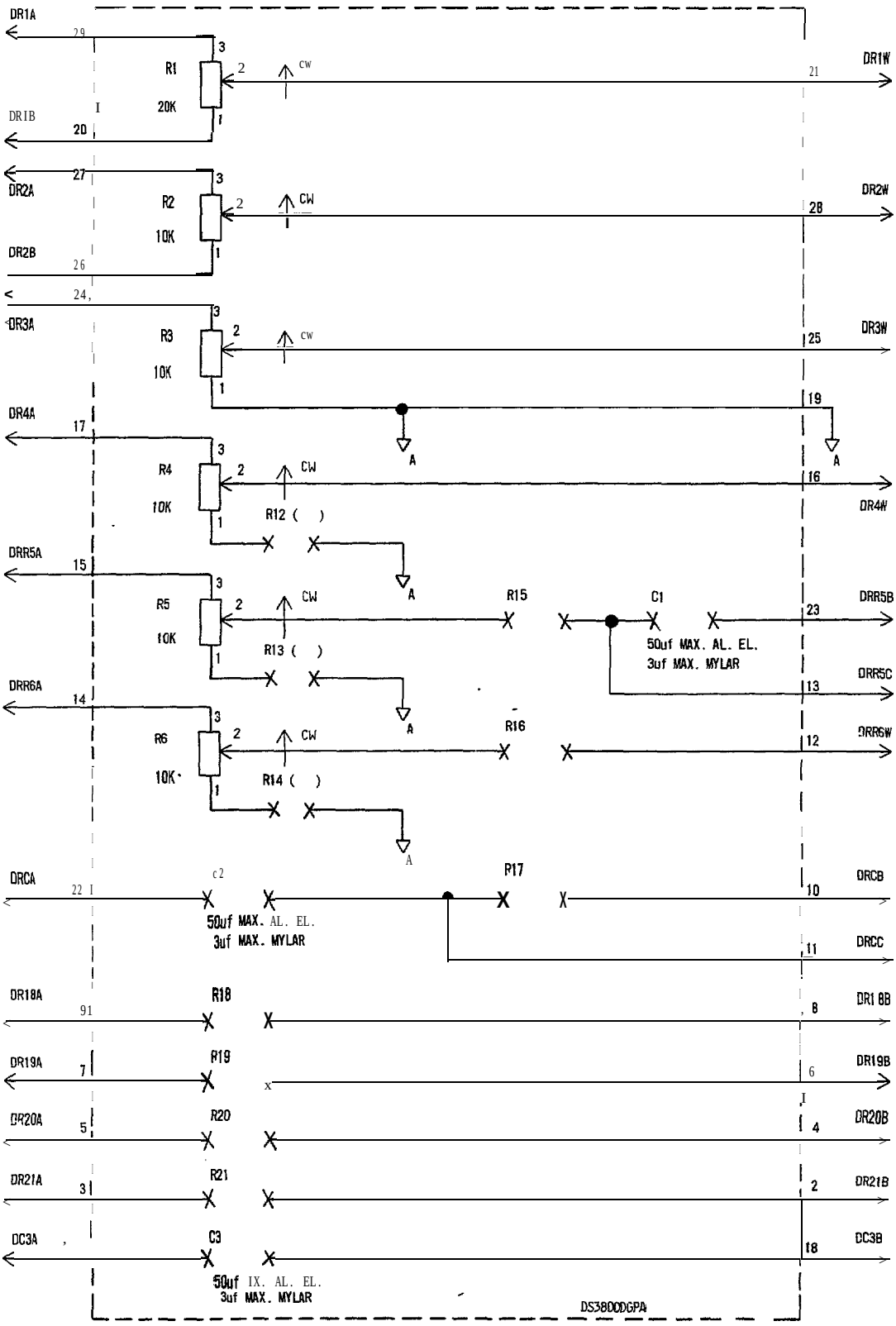
PRINTED WIRING BOARD/MODULE DRAWINGS

E-l. INTRODUCTION

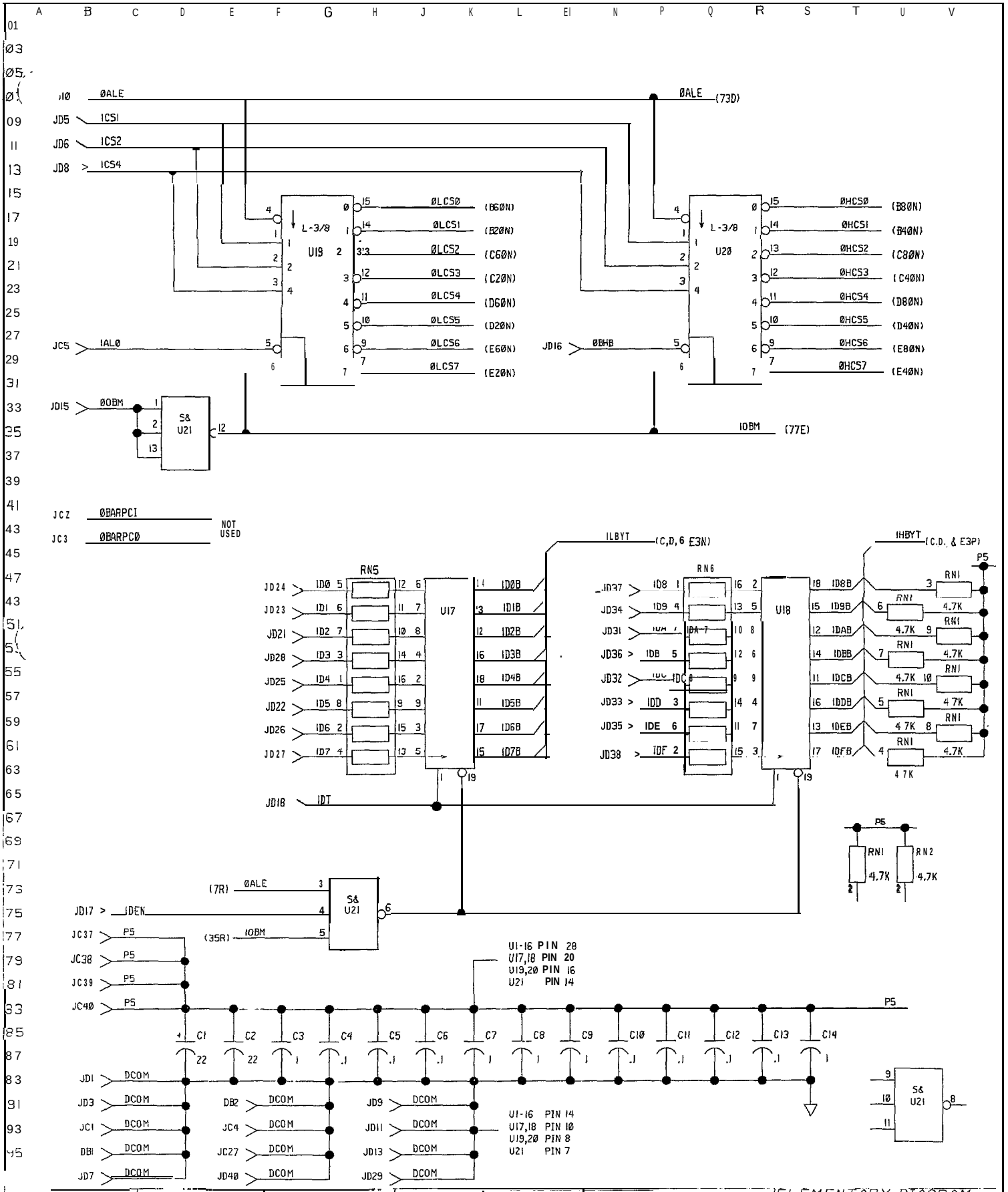
This appendix contains the descriptions of operation and elementary diagrams for printed wiring boards and modules used in the μPSS. These are engineering drawings for use in troubleshooting.

These drawings are arranged alphabetically as follows:

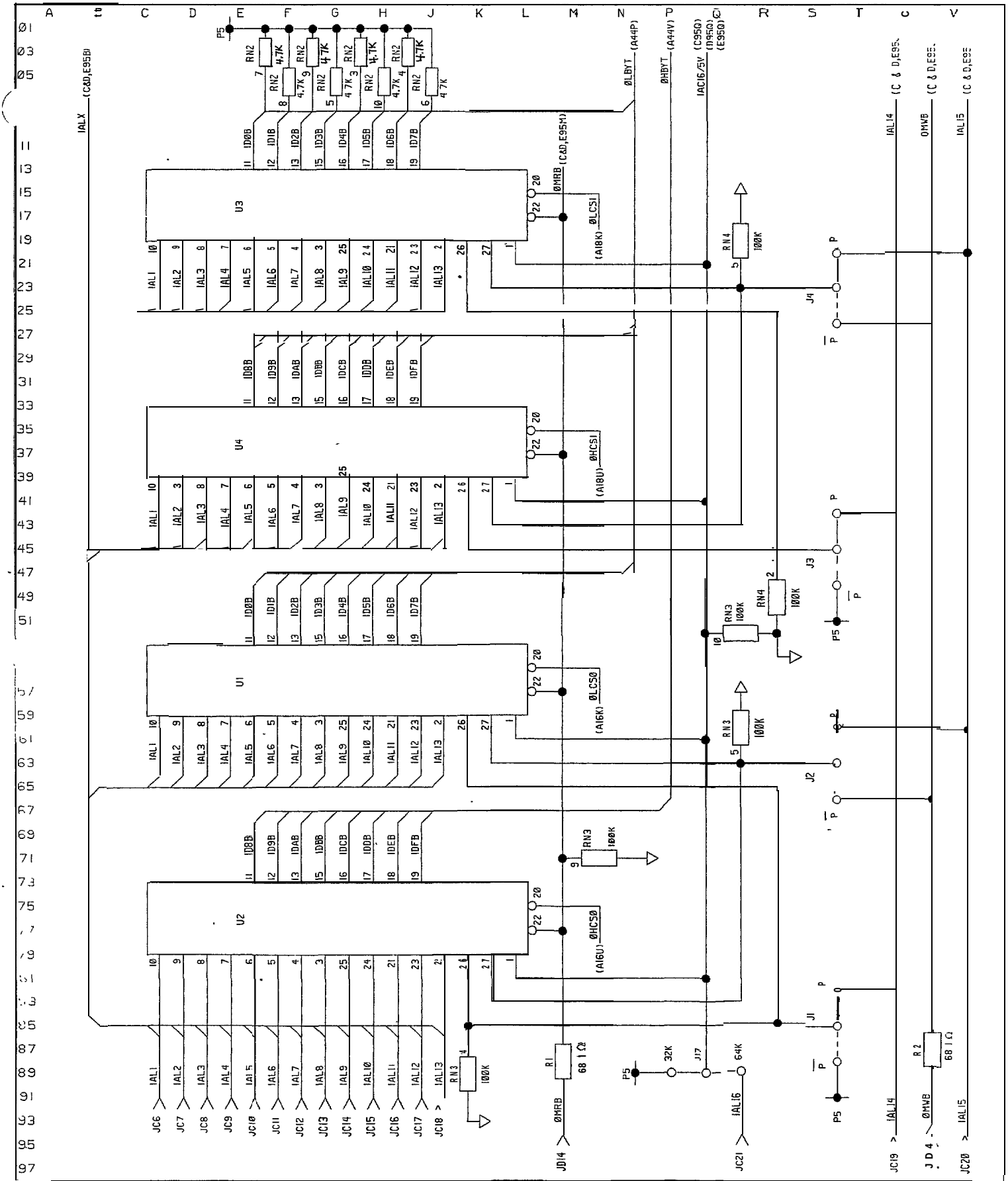
1. DS3800DGPA Gain Programming Board (DS3800NGMA Daughter Board)
2. DS3800DMPK CPU Memory Board (DS3800HMPK Daughter Board)
3. DS3800HADA Address Decoder Board
4. DS3800HAIC Analog Input Board
5. DS3800HMPK CPU Board
6. DS3800HRRB Relay Output Board
7. DS3800HSCC Signal Conditioning Board
8. DS3800HUMB Universal Memory B Board
9. DS3800HXPD CPU Expander Board
10. DS3800NDAC Digital-to-Analog Converter Board
11. DS3800NFTA Frequency Transmitter Board
12. DS3800NGMA General Purpose Multiplier Board
13. DS3800NOAA Operational Amplifier Board
14. DS3800NPTB Complex Power Transducer Board
15. DS3800NQFA Quadratic Filter Board
16. DS3800NSCE Signal Conditioning Board
17. DS3820PCN/PCNA Power Conditioning Module
18. DS3820PSC Power Supply
19. DS3820PSL Power Supply
20. IC3602A178 Microsynchronizer Isolation Module



1 OEC VG 78.20	REV 4	REV 5	PRINTS TO DL104 SHEC PRI	ENGINEER 8-11-77 R.A. Salibrici	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA U.S.A.	ELEMENTARY DIAGRAM GAIN PROGRAMMING DAUGHTER BOARD
REV 2 OEF VG 780728	ISSUED August 12, 1977			DOM PLUS		DS380DDGPA
REV 3	MADE BY ROBERT A.SALIBRICI					CONT. ON SH. SH. NO. 4AA

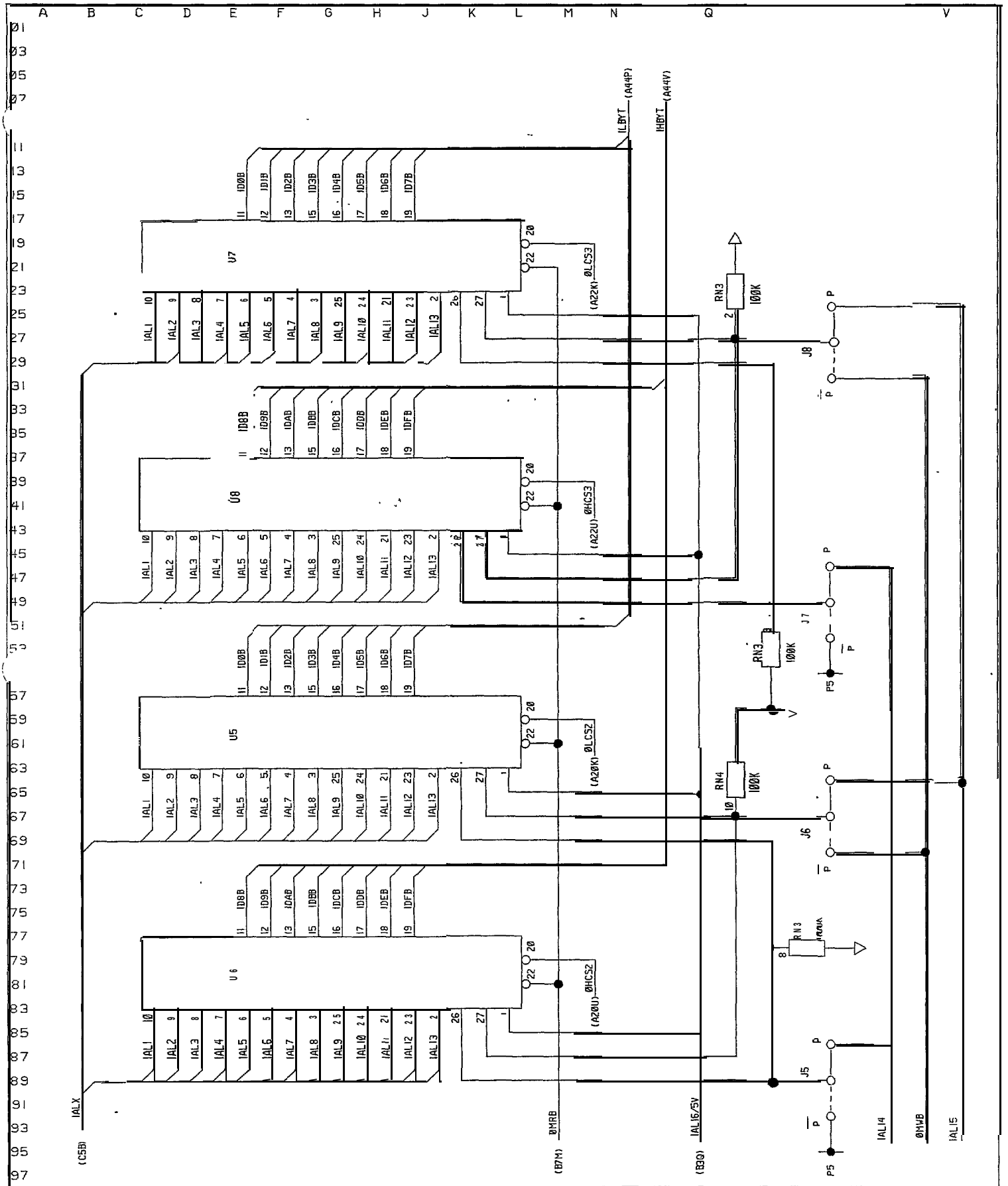


OEC	REV.4	REV.5	PRINTS TO DL104 PRI	ENGINEER E.D./AB	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM CPU MEMORY
MAC 860513 REV 2 OEH	ISSUED 86-07-07			FIRST MADE FOR		
CWF 880519 REV.3 OEN	REISSUED 1-11-89				DRIVE SYSTEMS DEPT.	DS3800DMPK
	MADE BY E.DILLON			C.NO	SALEM, VA. U.5.A	CONT.ON SH. 4AB SH.NO. 4AD



	REV.4	REV 5 PRINTS TO DL104 PRI	ENGINEER EJD	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM CPU MEMORY
REV.2	ISSUED 86-03-07		FIRST MADE FOR	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	DS3800DMPK
REV 3	MADE BY E.DILLON		I.C.NO.		CONT.ON SH. 4AC SH.NO. 4AB

CARS



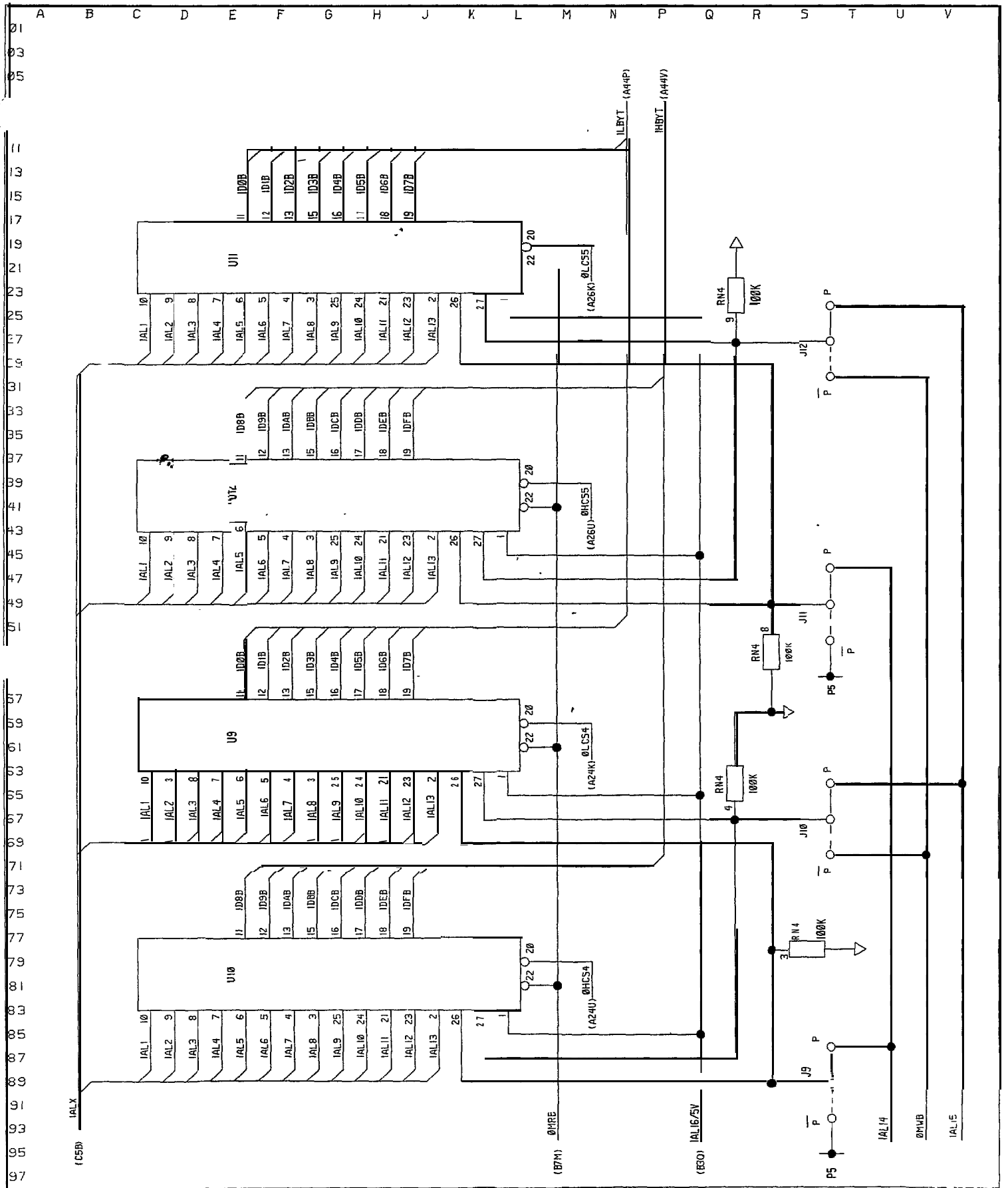
	REV 4	REV.5	PRINTS TO DL104 PR1	ENGINEER EID	GENERAL ELECTRIC COMPANY DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.
REV.2	ISSUED 86-03-07			FIRST MADE FOR	
REV.3	MADE BY E.DILLON			I.C NO.	

ELEMENTARY DIAGRAM
CPU MEMORY

DS3800DMPK

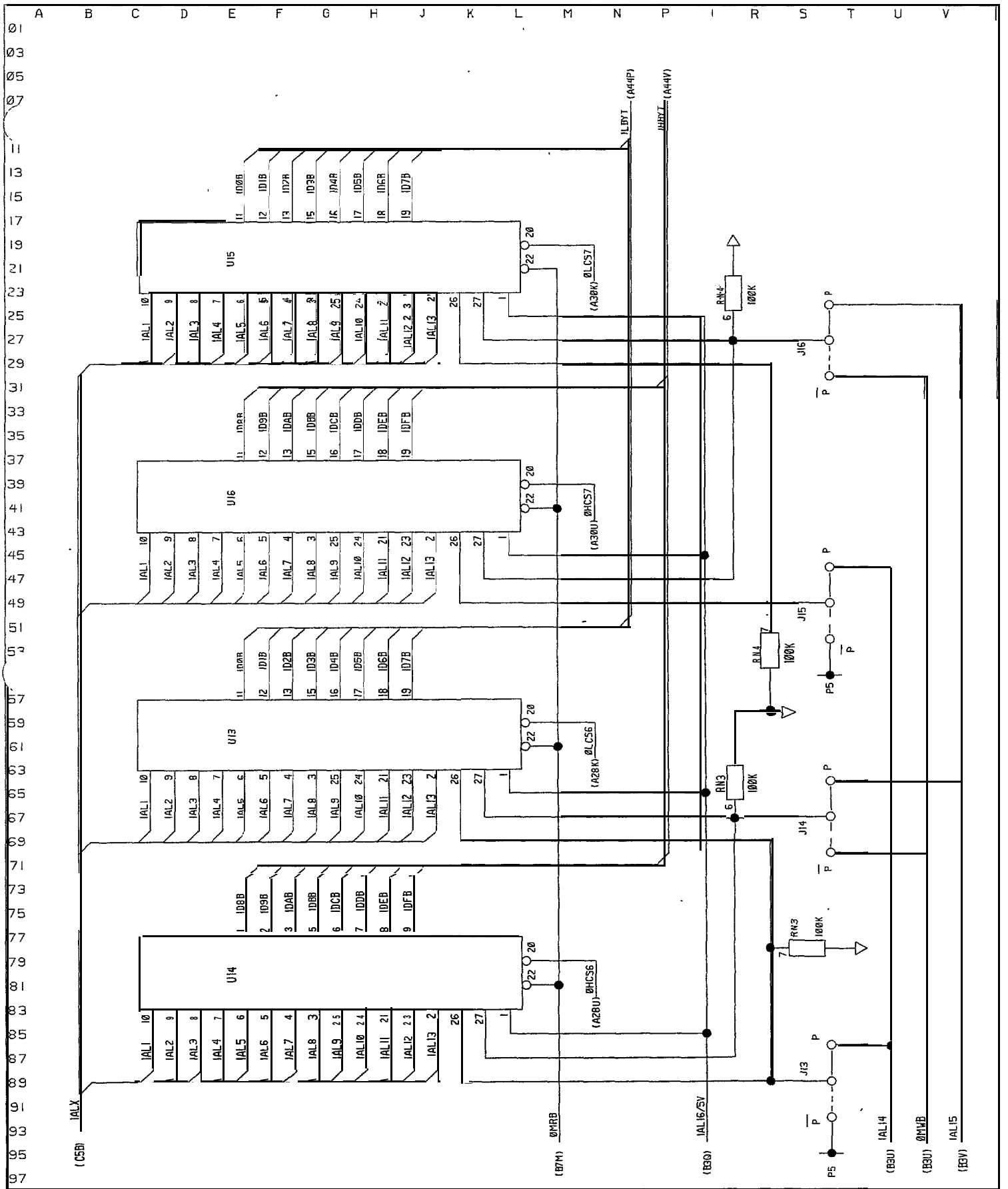
CONT.ON SH. 4AD SH.NO. 4AC

CARS



V.1	REV 4	REV.5	PRINTS TO DL104 PRI	ENGINEER EJQ	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM CPU MEMORY
REV 2	ISSUED 86-03-07			FIRST MADE FOR	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	DS3800DMPK
REV 3	MADE BY E.DILLON			I.C.NO.		CONT.ON SH. 4AE SH.NO. 4AD

CARS



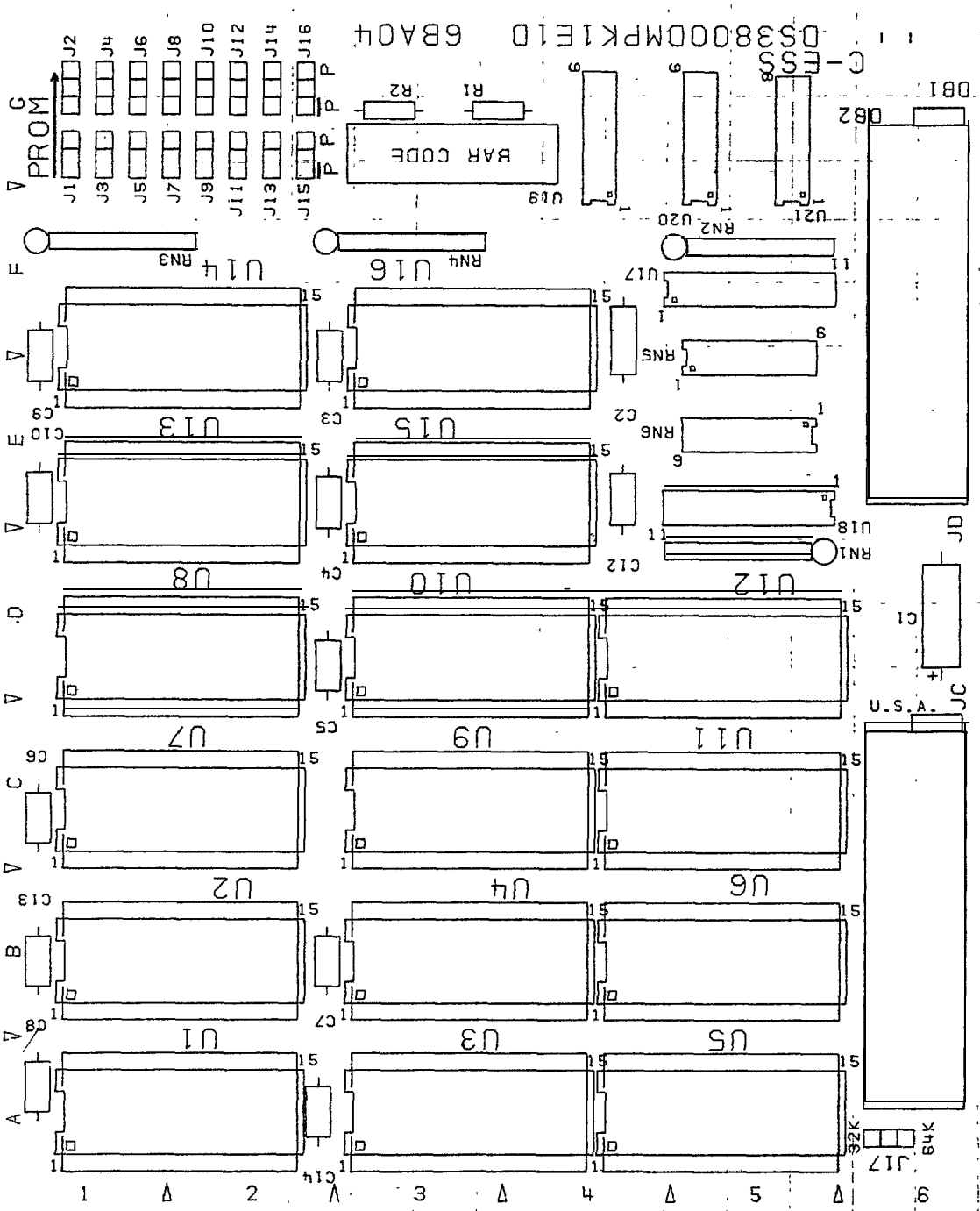
.1	REV.4	REV.5	PRINTS TO DL104 PRI	ENGINEER	GENERAL ELECTRIC COMPANY
REV 2	ISSUED 86-03-07			FIRST MADE FOR	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.
REV 3	MADE BY F DILLON			I.C.NO.	

ELEMENTARY DIAGRAM
CPU MEMORY

DS3800DMPK

CONT ON S.L. 445 SH NO 40F

CARTS

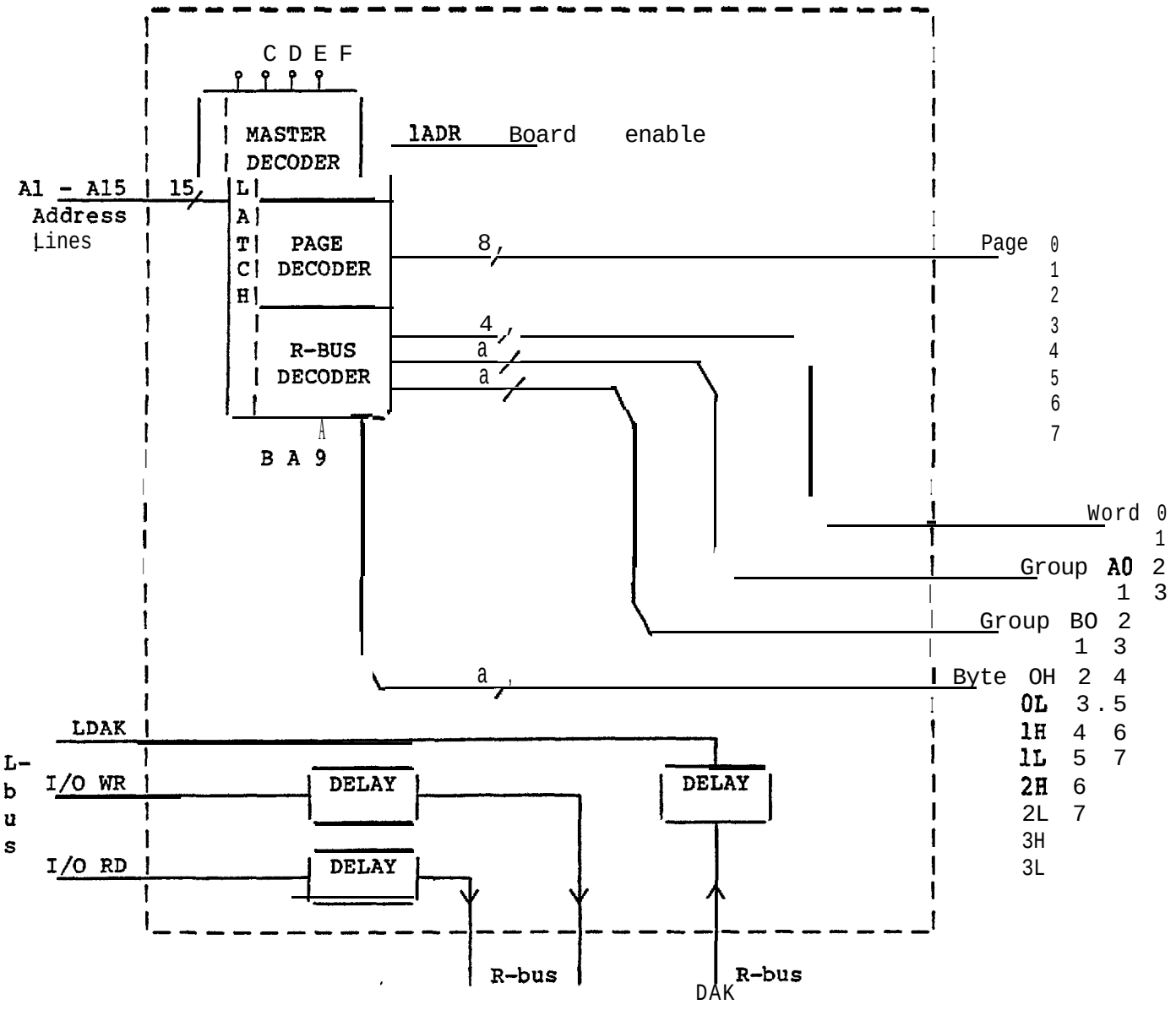


OEC ...HC 860513	REV 4	REV 5	PRINTS TO PRI D2104	APPROVALS alB	GENERAL ELECTRIC	ELEMENTARY DIAGRAM CPU MEMORY
REV 2 OEN alB 881130	ISSUED 86-03-07 REISSUED 88-11-30 alB			FIRST MADE FOR	DRIVE SYSTEMS DEPT, SALEM, VA,, U.S.A.	D53800DMPK
REV 3	MADE BY W. LUNSFORD			TE NO		CONT. ON SH. FL SH. NO. 4 AF

FORM IC 2349 (09-76)

CADS

Figure 1
DS3800HADA BLOCK DIAGRAM



REV.1	REV.4	REV.7	PRINTS TO DL102 PR	ENGINEER GLR	DESCRIPTION OF OPERATION AND APPLICATION DATA
REV.2	REV.5	ISSUED December 13, 1983		GENERAL ELECTRIC	Address Decoder
REV.3	REV.6	MADE BY P. L. Roark		COMPANY SALEM,VA	DS3800HADA CONT.ON SH. 2BA SH.NO. 2AA

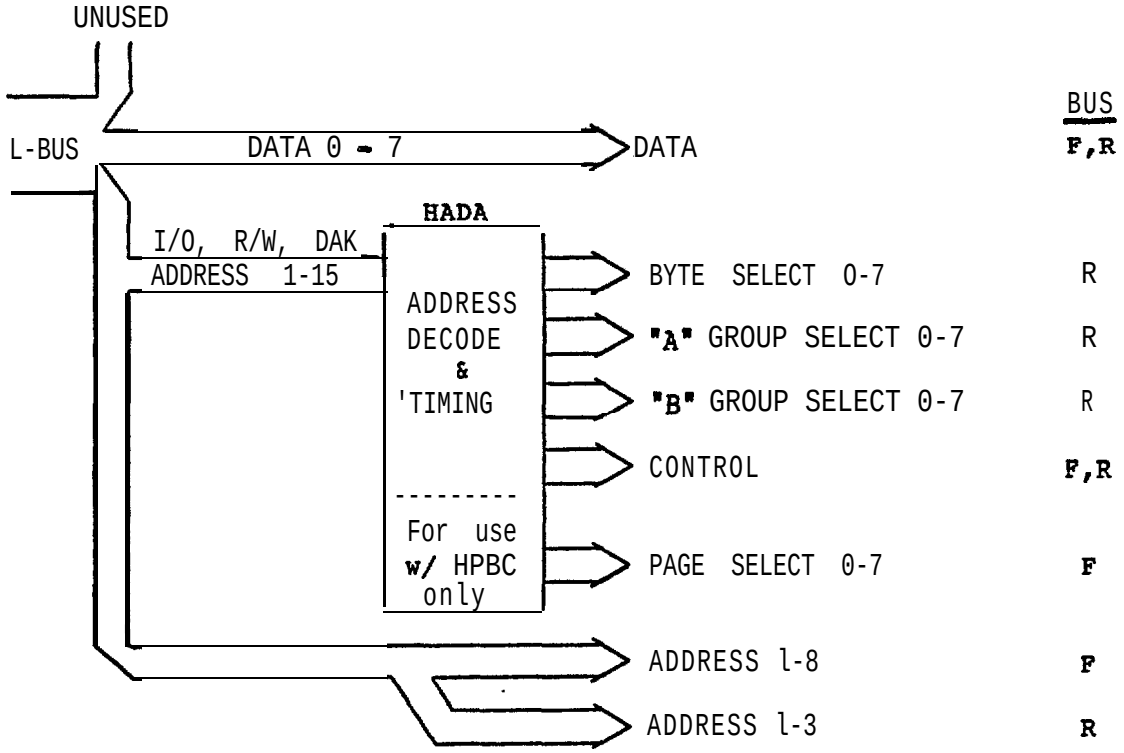
DEFINITION AND FUNCTION

The DS3800HADA Address Decoder board allows use of R-bus I/O boards within the I/O address space of the L-bus. When used alone, the HADA generates 16 group selects (128 bytes) with appropriate word/byte selects for connection to R-bus boards within a single module. Up to 8 DS3800HPBC Parallel Buffer Decoder boards can be added to permit the resultant F-bus or R-bus to be run 25 feet to a remote I/O module or I/O stick. The HADA cannot be plugged into a standard L-bus slot.

THEORY OF OPERATION

The HADA can be used 2 ways: alone to directly drive 200-level I/O boards using' the R-bus, or by adding DS3800HPBC boards that drive the F-bus or R-bus up to 25 feet (see Figure 2). When driving I/O directly (see Figure 3), the HADA decodes the L-bus address and control inputs into R-bus group and byte selects, and I/O read and write strobes. When used with HPBC boards (see Figure 4), the HADA simultaneously transmits the I/O read and write strobes to all the HPBCs. The page select, however, is transmitted to each board individually when that board is addressed so that only 1 HPBC is active at a time. In turn, each HPBC creates an R-bus or F-bus as required.

Figure 2
HADA L-BUS DECODE



REV.1	REV.4	REV.7	PRINTS TO DL102 PRI	ENGINEER P&R	DESCRIPTION OF OPERATION AND APPLICATION DATA
(REV.,2	JREV.5	ISSUED December 13, 1983		GENERAL ELECTRIC	Address Decoder
REV.3	REV.6	MADE BY P. L. Roark		COMPANY SALEM,VA	DS3800HADA CONT.ON SH. 2CA SH.NO. 2BA

Figure 3
HADA USED FOR DIRECT R-BUS I/O

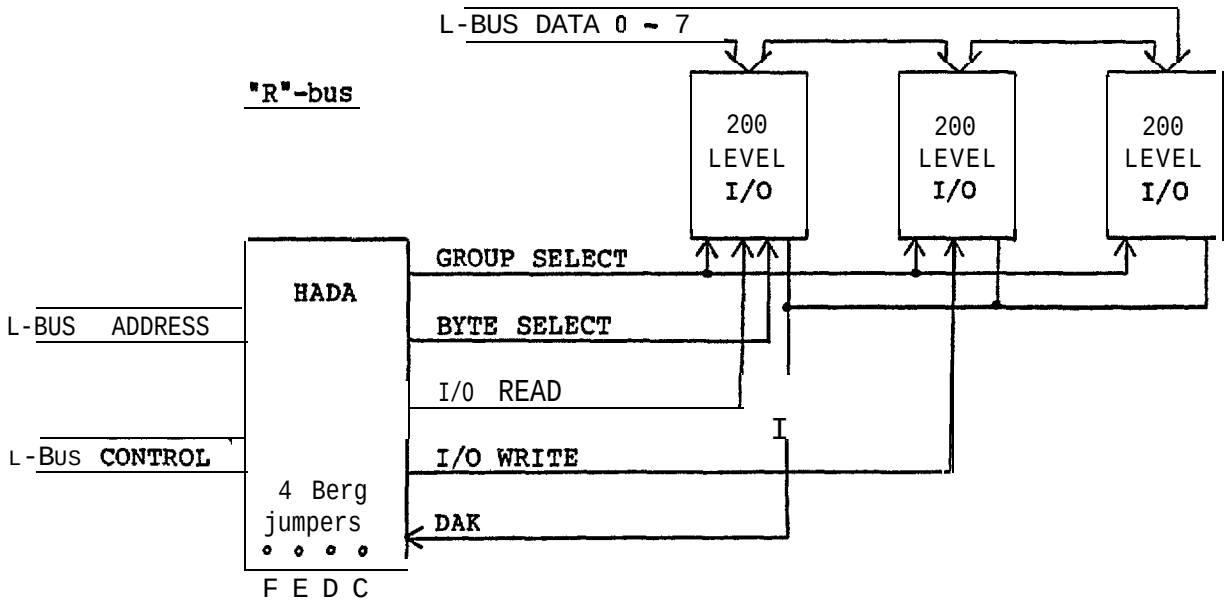
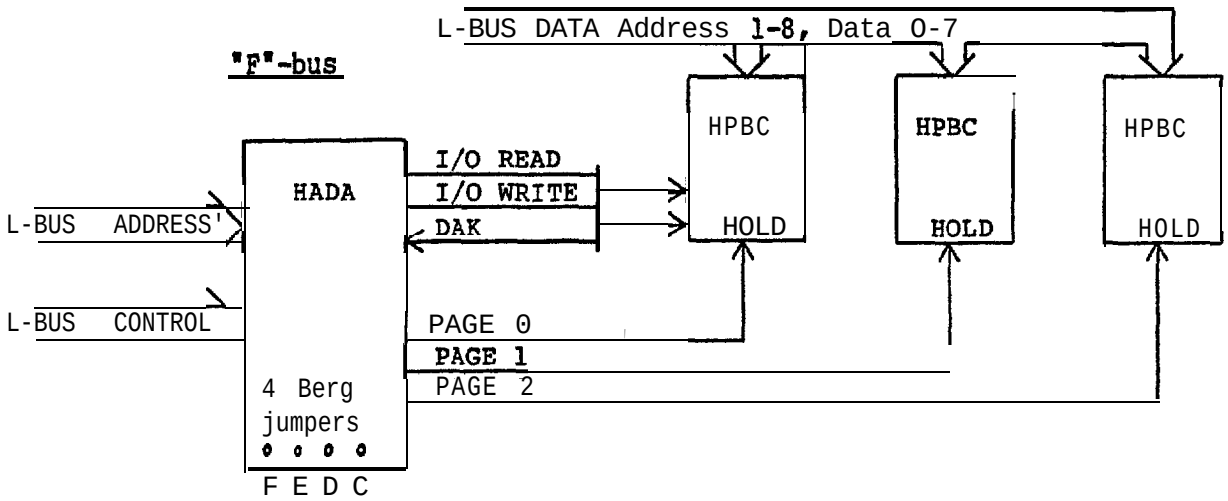


Figure 4
HADA USED FOR HPBC DRIVEN I/O

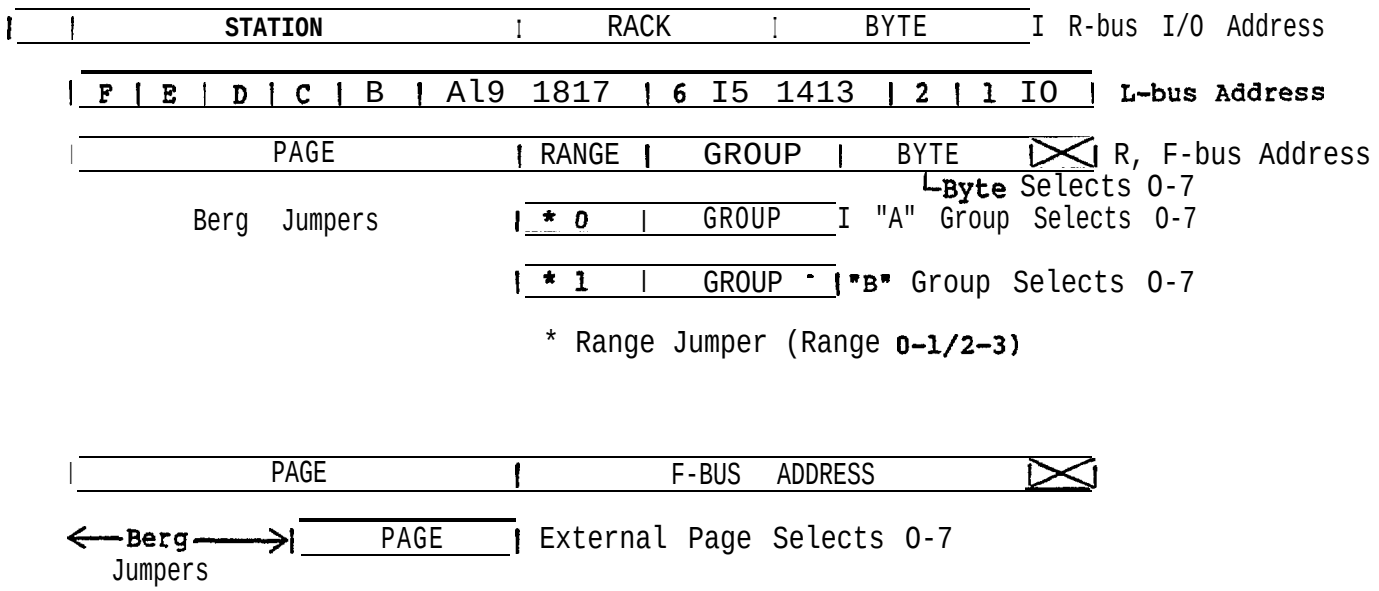


REV.1	REV.4	JREV.7	PRINTS To DL102 PR	ENGINEER PZR	DESCRIPTION OF OPERATION AND APPLICATION DATA
REV.2	REV.5	ISSUED December 13 , 1983		GENERAL ELECTRIC	Addresa Decoder
(REV.3	REV.6	MADE BY P. L. Roark		COMPANY SALEM,VA.	DS3800HADA CONT.ON SH. 2DA SH.NO. 2CA

ADDRESSING

(Refer to Figure 5.) The R-bus and F-bus contain an 8-bit address bus, and the L-bus contains a 16-bit address bus. The HADA decodes the upper 7 L-bus address bits (F through 9) into the F-bus or R-bus page and the I/O station address selected for the board. It decodes the next 8 lower address bits (8 through 1) into range, byte, group A, and group B selects for the R-bus. Address bit 0 is not used. Address lines 8 through F provide the "B" inputs to a comparator, which contains Berg jumpers on the "A" inputs. The jumpers, J1 through J8, allow the user to either pull high or ground each A input for selecting page, range, and station addresses.

Figure 5
ADDRESSING



REV.1	REV.4	REV.7	PRINTS To DL102 PR	ENGINEER P2R	DESCRIPTION OF OPERATION (AND APPLICATION DATA
REV.2	REV.5	ISSUED December 13, 1983	GENERAL ELECTRIC		Address Decoder
REV.3	REV.6	MADE BY P. L. Roark	COMPANY SALEM,VA.		DS3800HADA CONT.ON SH. 2EA SH.NO. 2DA

Address Jumperinq

(Refer to Figure 6.) The R-bus contains 8 group select lines, labelled 0 through 7, for selecting a 64 byte (8 x 8) block of memory space. The HADA produces 2 sets of these 8 group select lines, set A and B, from address lines 4 through 6. Both groups (128 bytes total) reside in either of 2 ranges: range l-2 or range 3-4. Address bits 7 and 8 select the range, and Berg jumper Jl corresponds to bit 8.

The R-bus address space is divided into eight 256 byte pages, with the 2 ranges (1-2 and 3-4) residing in 1 of the pages. The HADA provides 8 page select lines, and address lines 9 through B to select the page. Jumpers J2 through J4 correspond with these 3 address lines for the multiplexer input.

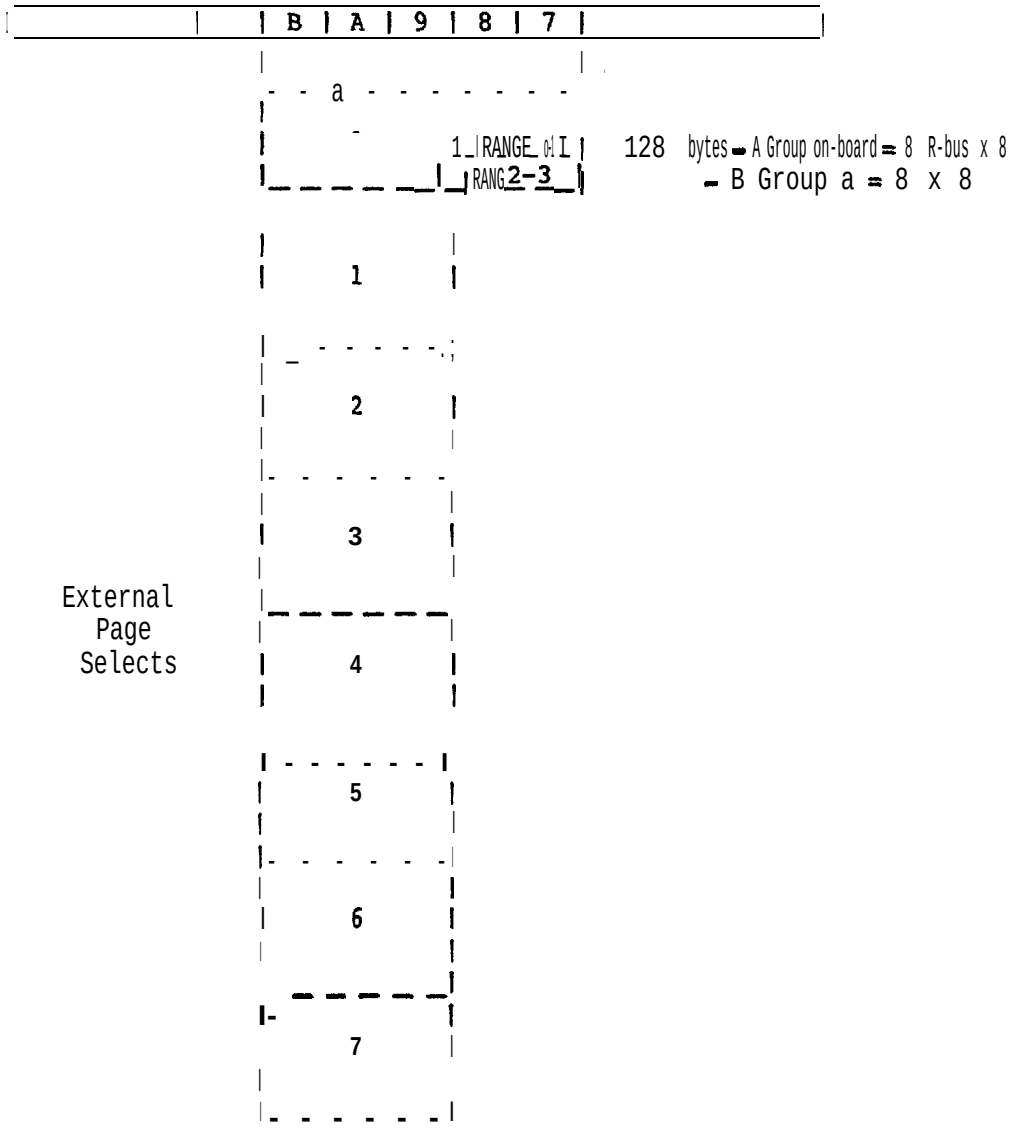
NOTE: The setting of jumpers J2, J3, and J4 determine the page address assigned to the HADA's R-bus decode. When using HPBC boards, the user cannot assign that particular page to any HPBC.

The HADA decodes the remaining upper address lines (C through F) into the station address for the board. Jumpers J5 through J8 correspond with these 3 address lines for the multiplexer input.

Address Latching

The HADA latches address lines 4 through 7 and address lines 9 through B using the L-bus read/write strobes for latch clocking. During "word" operations, this function ensures that the group select lines remain valid when the L-bus address changes to accept the second byte of the word.

Figure 6
PAGE/RANGE SELECT JUMPERS



READ AND WRITE STROBE DELAY

Because HPBC and 200-level I/O boards operate at slower rates than the L-bus speed, the **HADA** delays the R-bus read, write, and data outputs to these boards. The **HADA** buffers the L-bus read or write inputs, then sends them to a counter. As it counts down, the counter is slowing the output onto the R-bus. When the **HADA** receives a DAK from the R-bus or F-bus, it sends the signal through a counter before transmitting it to the processor*

REV.1	REV.4	(REV.7	PRINTS TO I DL102 PR	ENGINEER P2R	DESCRIPTION OF OPERATION AND APPLICATION DATA
REV.2	REV.5	ISSUED December 13, 1983		GENERAL ELECTRIC	Address Decoder
REV.3	REV.6	MADE BY P. L. Roark		COMPANY SALEM;VA	DS3800HADA CONT.ON SH. 2GA SH.NO. 2FA

POWER AND COOLING REQUIREMENTS

+5VDC: .62 amps maximum
.40 amps typical

Heat dissipation: 3.1 watts maximum
2.0 watts typical

OPERATING AND STORAGE ENVIRONMENT

See GEK-61680.

I/O SIGNAL DEFINITIONS

Table 1*

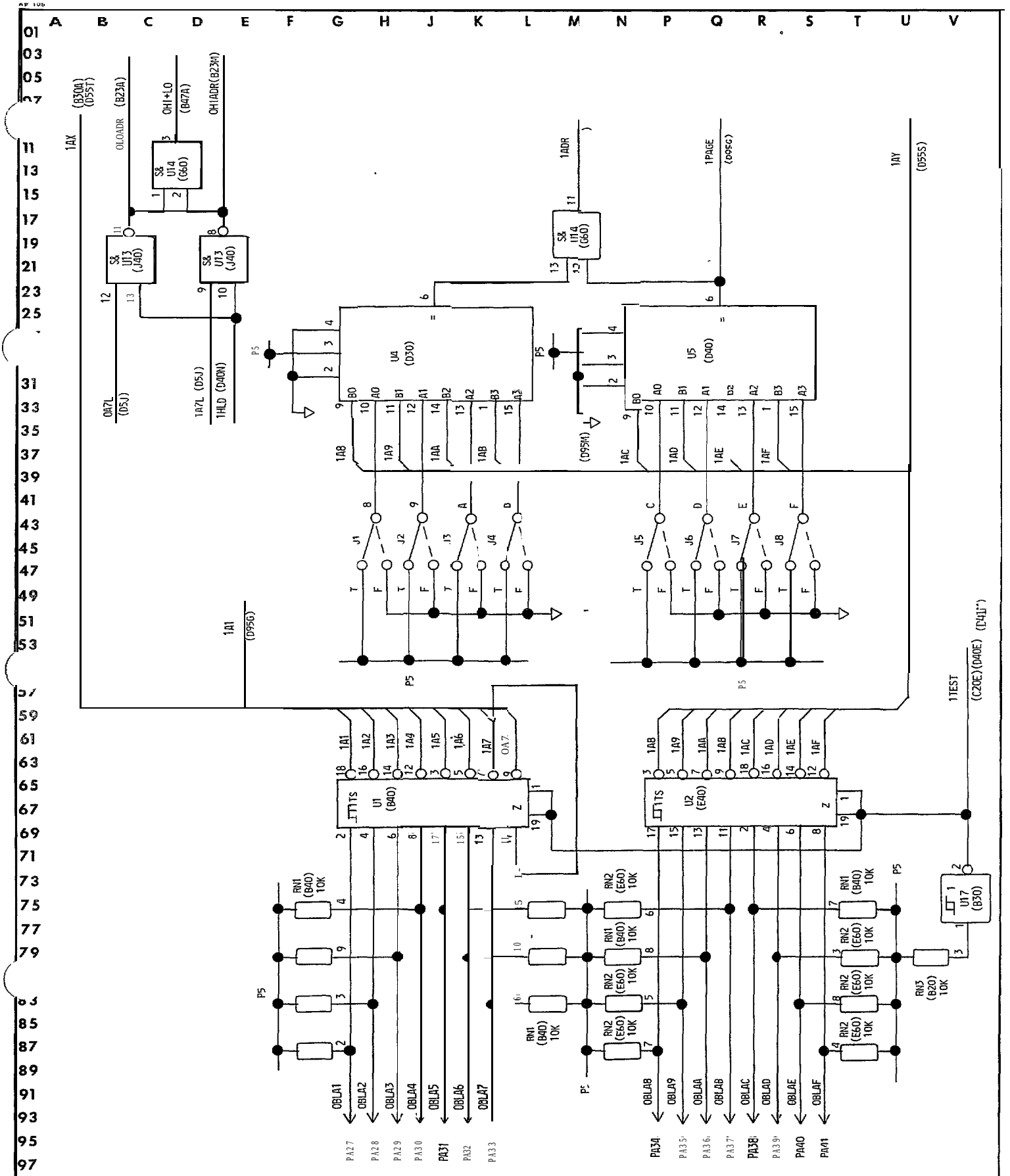
SIGNAL NAME	PIN NO.	DIRECTION	DEFINITION
DCOM	PA1	Input	Power digital common
P5	PA3	Input	Power +5VDC
OBLCIOR	PA6	Input	I/O read
OBLCIOW	PA8	Input	I/O write
OBLCDAK	PA10	Output	Device acknowledge
OBCRDA	PA12	Input	Device acknowledge
OBCRRS	PA13	Output	Read strobe
OBCRWS	PA14	Output	Write strobe
OBLCRST	PA15	Input	Reset
OGBSO	PA18	Output	Group Select B line 0
OGBS1	PA19	Output	Group Select B line 1
OGBS2	PA20	Output	Group Select B line 2
OGASO	PA21	Output	Group Select A line 0
OGAS1	PA22	Output	Group Select A line 1
OGAS2	PA23	output	Group Select A line 2
OGAS3	PA24	Output	Group Select A line 3
OGAS4	PA25	Output	Group Select A line 4
OBLA1	PA27	Input	Address line 1
OBLA2	PA28	Input	Address line 2
OBLA3	PA29	Input	Address line 3
OBLA4	PA30	Input	Address line 4
OBLA5	PA31	Input	Address line 5
OBLA6	PA32	Input	Address line 6
OBLA7	PA33	Input	Address line 7
OBLA8	PA34	Input	Address line 8

* Continued on next page.

Table 1 (Continued)

SIGNAL NAME	PIN NO.	DIRECTION	DEFINITION
OBLA9	PA35	Input	Address line 9
OBLAA	PA36	Input	Address line A
OBLAB	PA37	Input	Address line B
OBLAC	PA38	Input	Address line c
OBLAD	PA39	Input	Address line D
OBLAE	PA40	Input	Address line E
OBLAF	PA41	Input	Address line F
DCOM	PA43	Input	Power digital common
P5	PA45	Input	Power +5VDC
OGBS3	PA48	Output	Group Select B line 3
OGBS4	PA49	Output	Group Select B line 4
OGBS5	PA50	Output	Group Select B line 5
OGBS6	PA51	Output	Group Select B line 6
OGBS7	PA52	Output	Group Select B line 7
OBYTOAL	PA53	Output	Low byte address line 0
OBYTOAM	PA54	Output	High byte address line 0
OBYT1AL	PA55	Output	Low byte address line 1
OBYT1AM	PA56	Output	High byte address line 1
OBYT2AL	PA57	Output	Low byte address line 2
OBYT2AM	PA58	Output	High byte address line 2
OBYT3AL	PA59	Output	Low byte address line 3
OBYT3AM	PA60	Output	High byte address line 3
OPAGE0	PA61	Output	Page address line 0
OPAGE1	PA62	Output	Page address line 1
OPAGE2	PA63	Output	Page address line 2
OPAGE3	PA64	Output	Page address line 3
OPAGE4	PA65	Output	Page address line 4
OPAGE5	PA66	Output	Page address line 5
OPAGE6	PA67	Output	Page address line 6
OPAGE7	PA68	Output	Page address line 7
OGAS5	PA70	Output	Group Select A line 5
OGAS6	PA71	Output	Group Select A line 6
OGAS7	PA72	Output	Group Select A line 7
OWORDA0	PA74	Output	Word address line 0
OWORDA1	PA76	Output	Word address line 1
P5	PA77	Input	Power +5VDC
OWORDA2	PA78	Output	Word address line 2
DCOM	PA79	Input	Power digital common
OWORDA3	PA80	Output	Word address line 3

REV.1	REV.4	REV.7	PRINTS TO DL102 PRI	ENGINEER P2K	DESCRIPTION OF OPERATION (AND APPLICATION DATA
REV.2	REV.5	ISSUED December 13, 1983		GENERAL ELECTRIC	Address Decoder
REV.3	REV.6	MADE BY P. L. Roark		COMPANY SALEM, VA.	DS3800HADA CONT.ON SH. FL SH.NO. 2HA

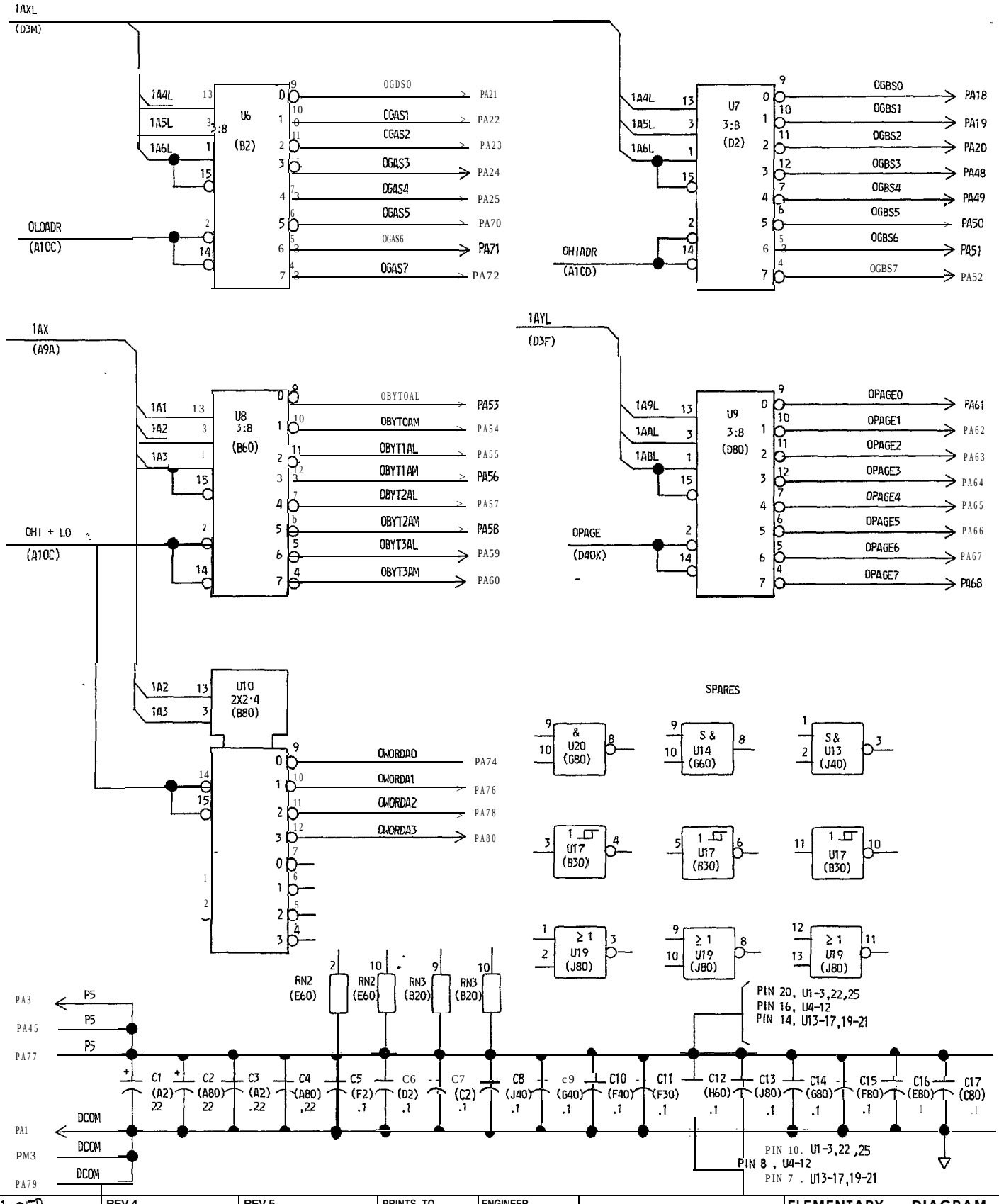


2 9861008 AC 10/6/86	REV. 4	REV 5	PRINTS TO DL104 PRI	ENGINEER EG
REV 2	ISSUED 4-8-83			
REV 3	MADE BY E. DILLON			

GENERAL ELECTRIC
DRIVE SYSTEMS DEPARTMENT
SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM
L - ADR DECODE
DS3800HADA
CONT. ON SH. 4BA SH. NO 4AA

AF 10527



REV 1 OED 9/01/08 BC 10/14/86	REV 4	REV 5	PRINTS TO DL104 PRI	ENGINEER ED	GENERAL@ ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.
REV2	ISSUED 4-8-83				
REV 3	MADE BY E. DILLON				

ELEMENTARY DIAGRAM

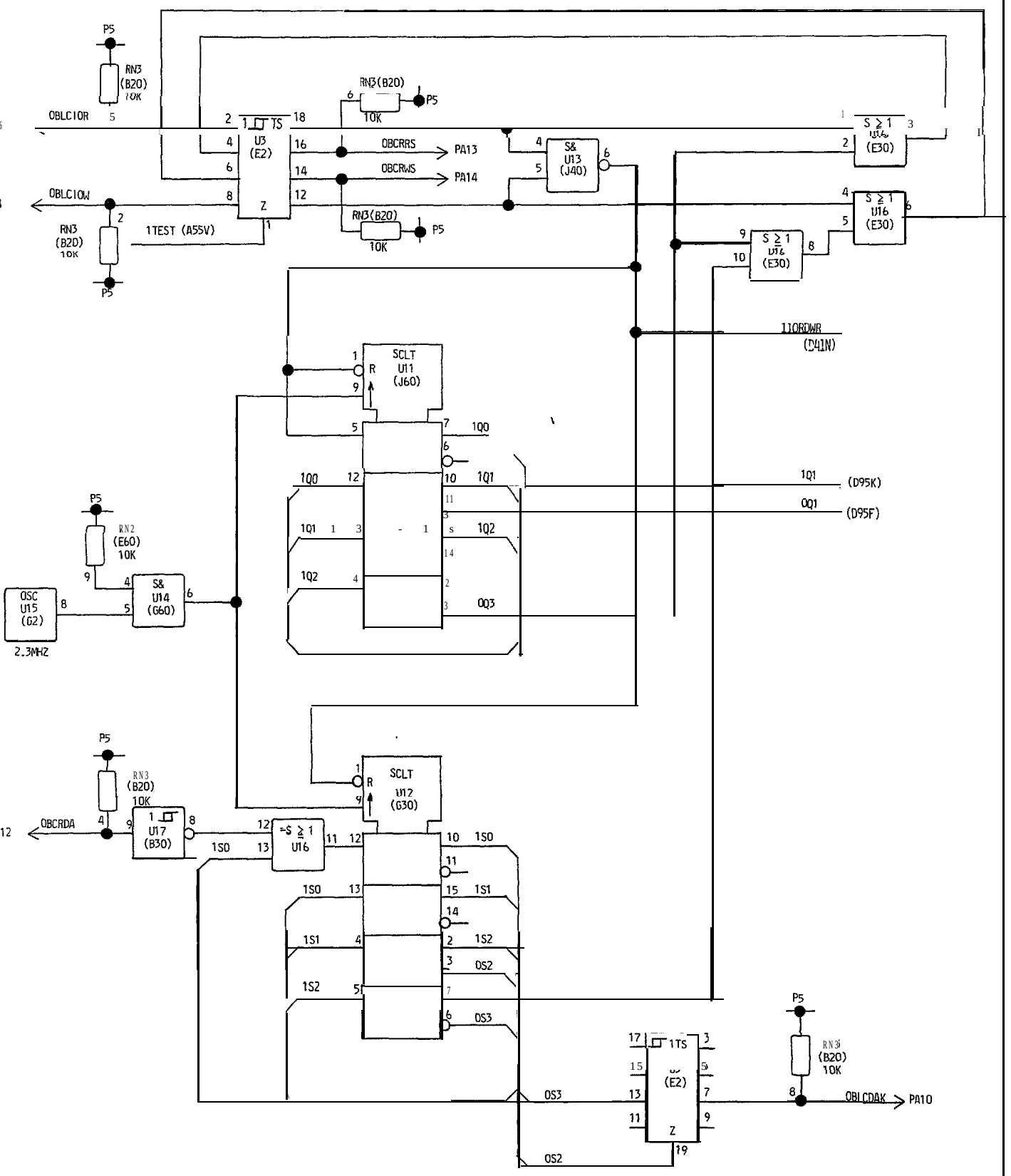
L * ADR DECODE

DS3800HADA

CONT. ON SH. 4CA SH. NO. ABA

OBD 0152 (09 76)

AF 10527



OED 861008 10/6/86	REV 4	REV.5	PRINTS TO DL104 PRI	ENGINEER EL	GENERAL@ ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.
REV.2	ISSUED 4-8-83				
REV 3	MADE BY E. DILLON				

ELEMENTARY DIAGRAM

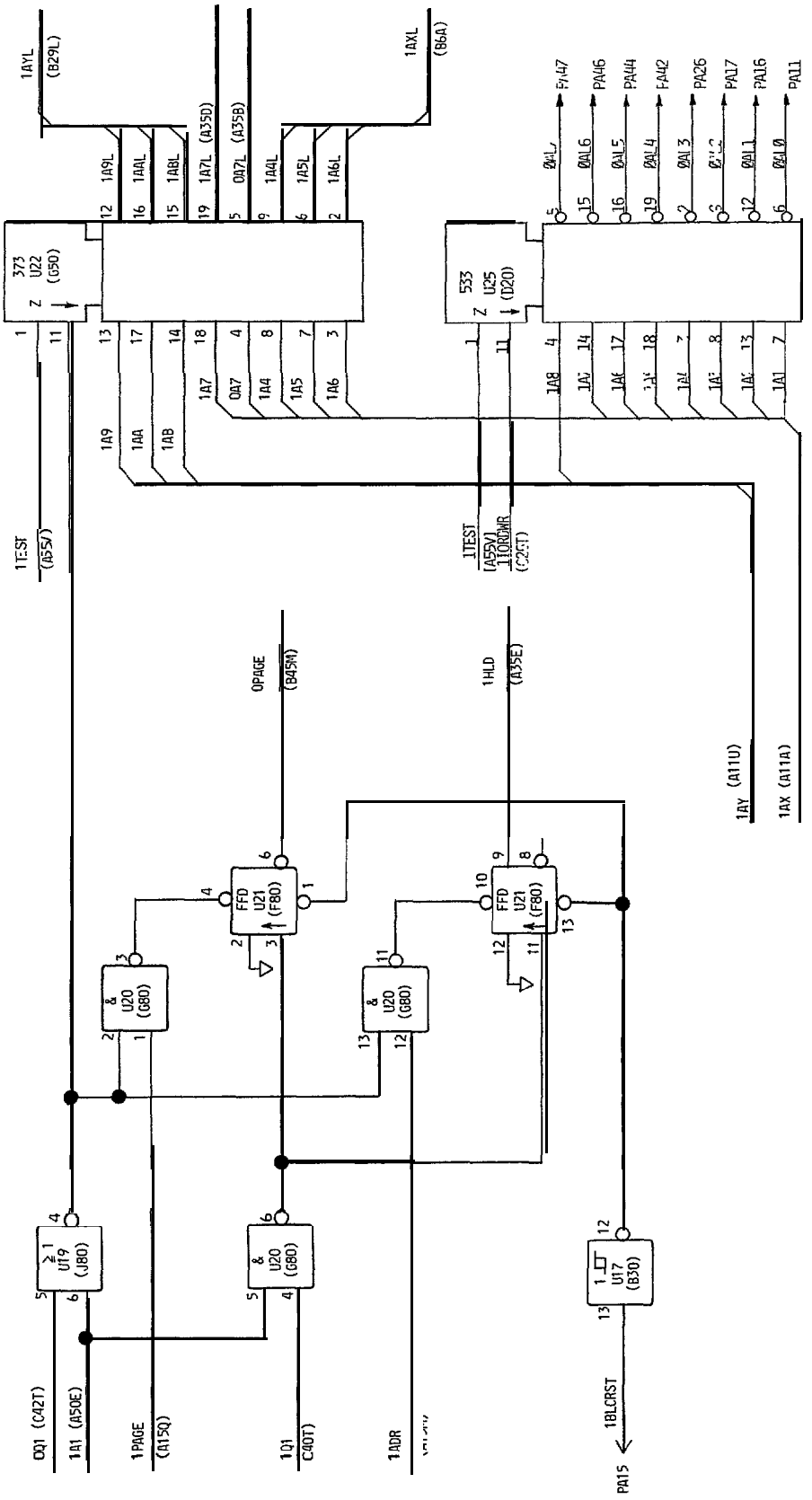
L-ADR DECODE

DS3800HADA

CONT. ON SH. 4DA SH. NO. 4CA

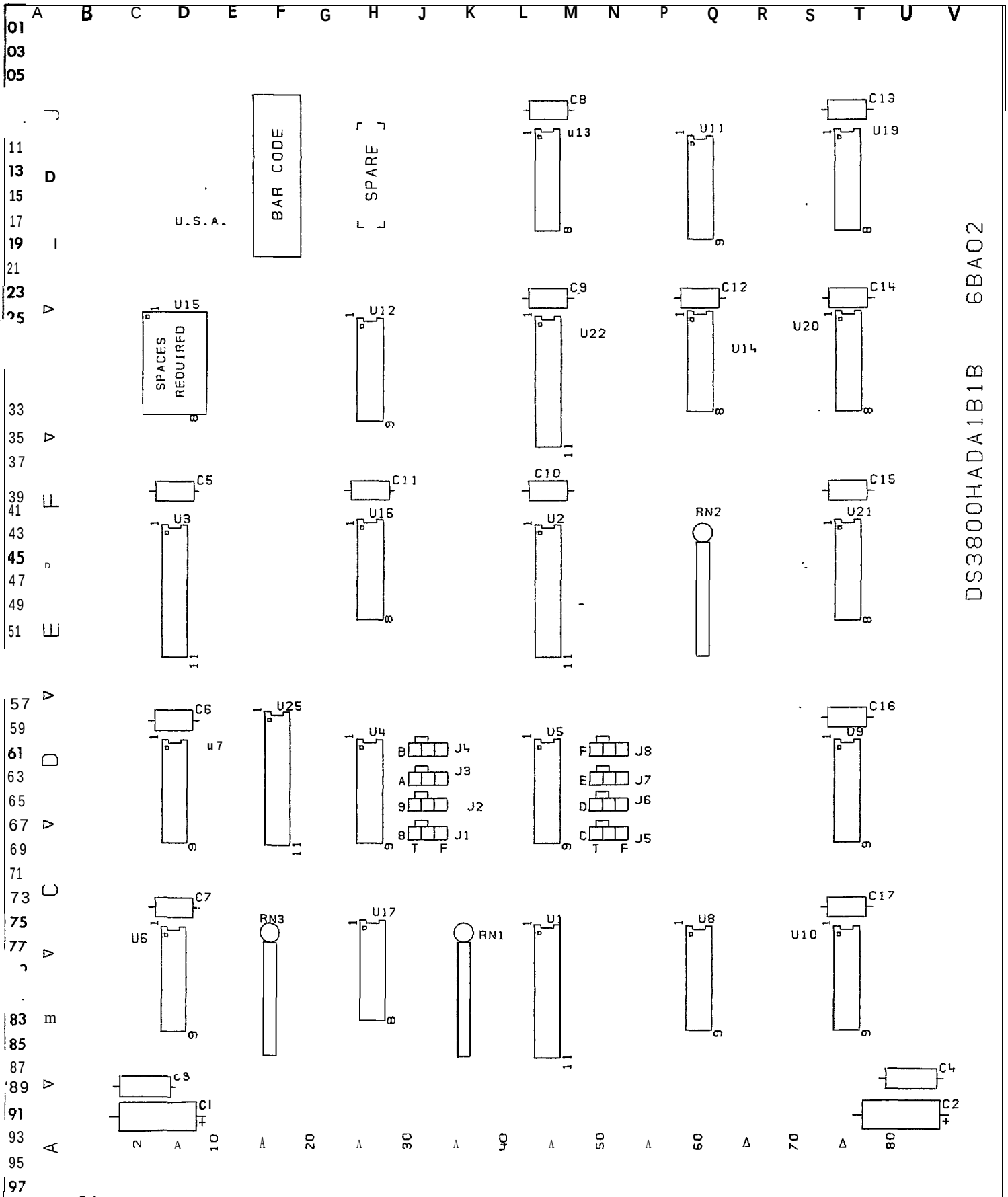
OED 0192 (0976)

AF 10527



OED 12 961008	REV 4	REV 5	PRINTS TO DL104 PRI	ENGINEER ES	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.
REV 2	ISSUED 4-8-83				ELEMENTARY DIAGRAM L = ADR DECODE
REV 3	MADE BY E. DILLON				DS3800HADA CONT. ON SH. 4 SH. NO. 4DA

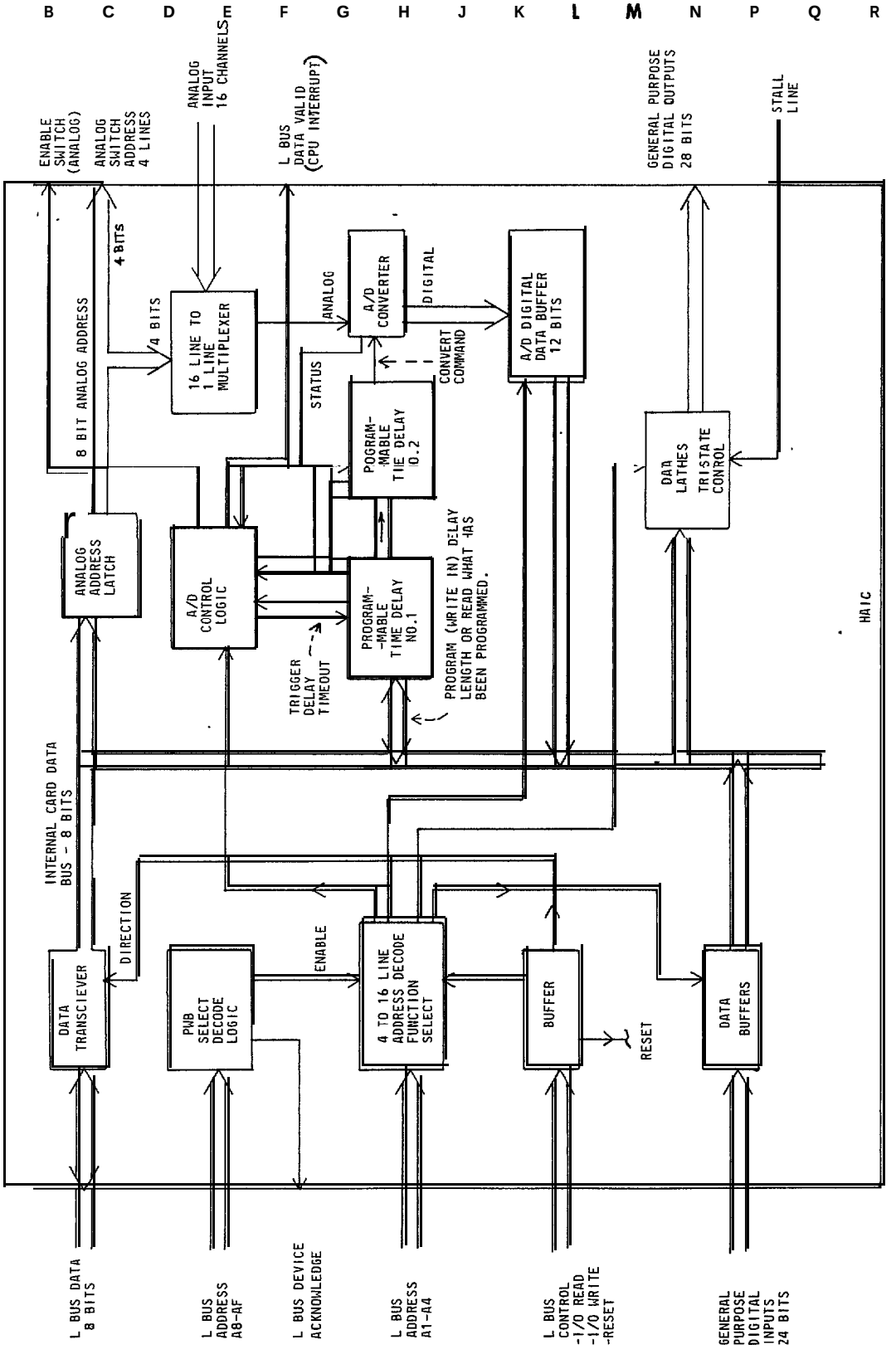
GSD 0152 (0976)



	REV4	REV 5	PRINTS TO DL104	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV 2	ISSUED Oct, 8, 1986			FIRST MADE FOR	DRIVE SYSTEMS DEPT. SALEM, VA., U.S.A.	DS 3800 HADA
REV3	MADE BY M.L. REICH			IC NO.		CONT. ON SH. FL SH. NO. 4EA

FORM IC 2349 (04-76)

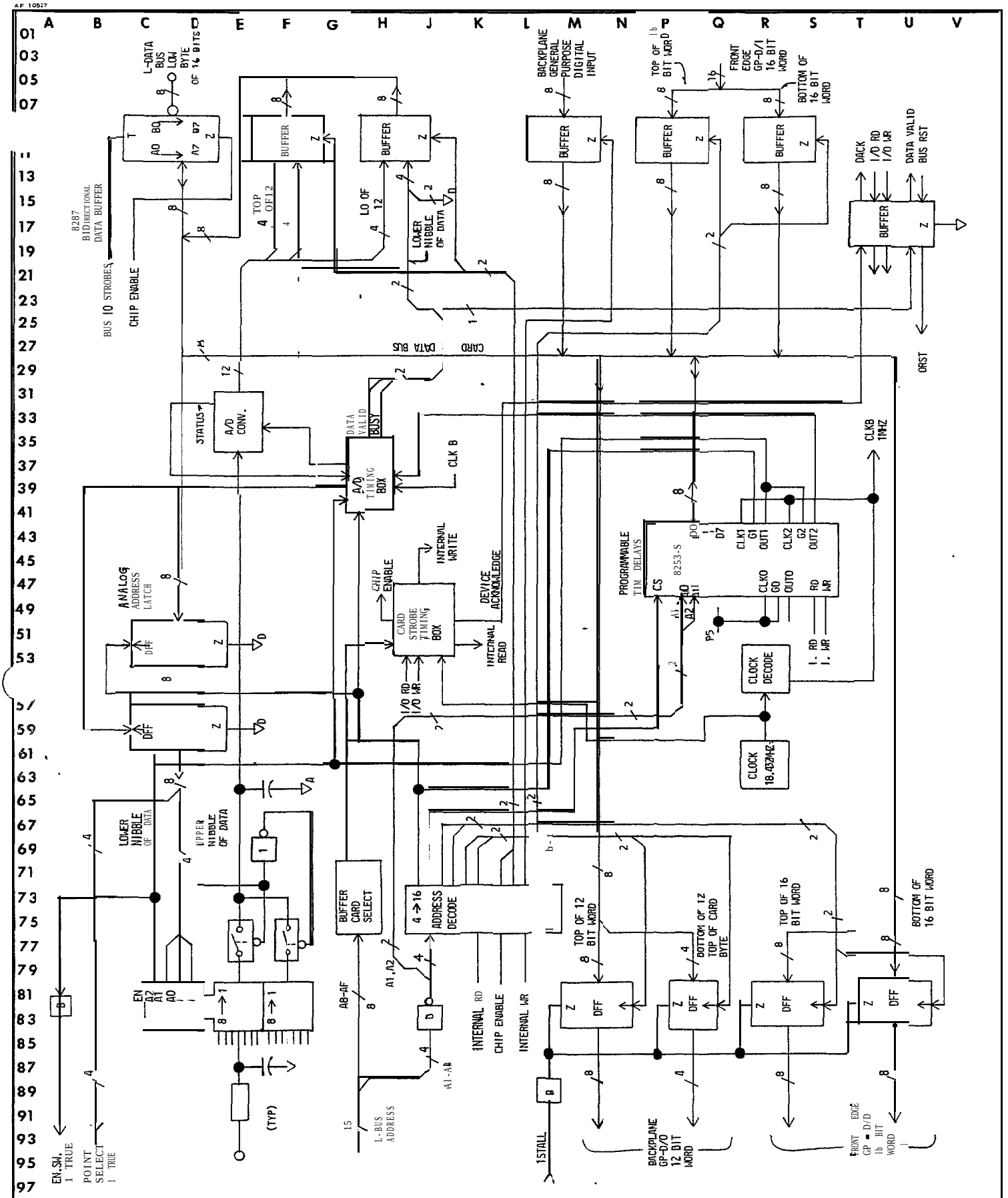
AF 10527



1	REV 4	REV 5	PRINTS TO 2520 DL102 PRI	ENGINEER 5.5.83 C.H Sweeley	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM. VA. U.S.A.	ELEMENTARY DIAGRAM SIMPLIFIED BLOCK DIAGRAM
REV.2	ISSUED 83-05-12					DS3800HAIC
REV 3	MADE BY R.A. SALIBRICI					CONT. ON SH. 2BA SH. NO. 2AA

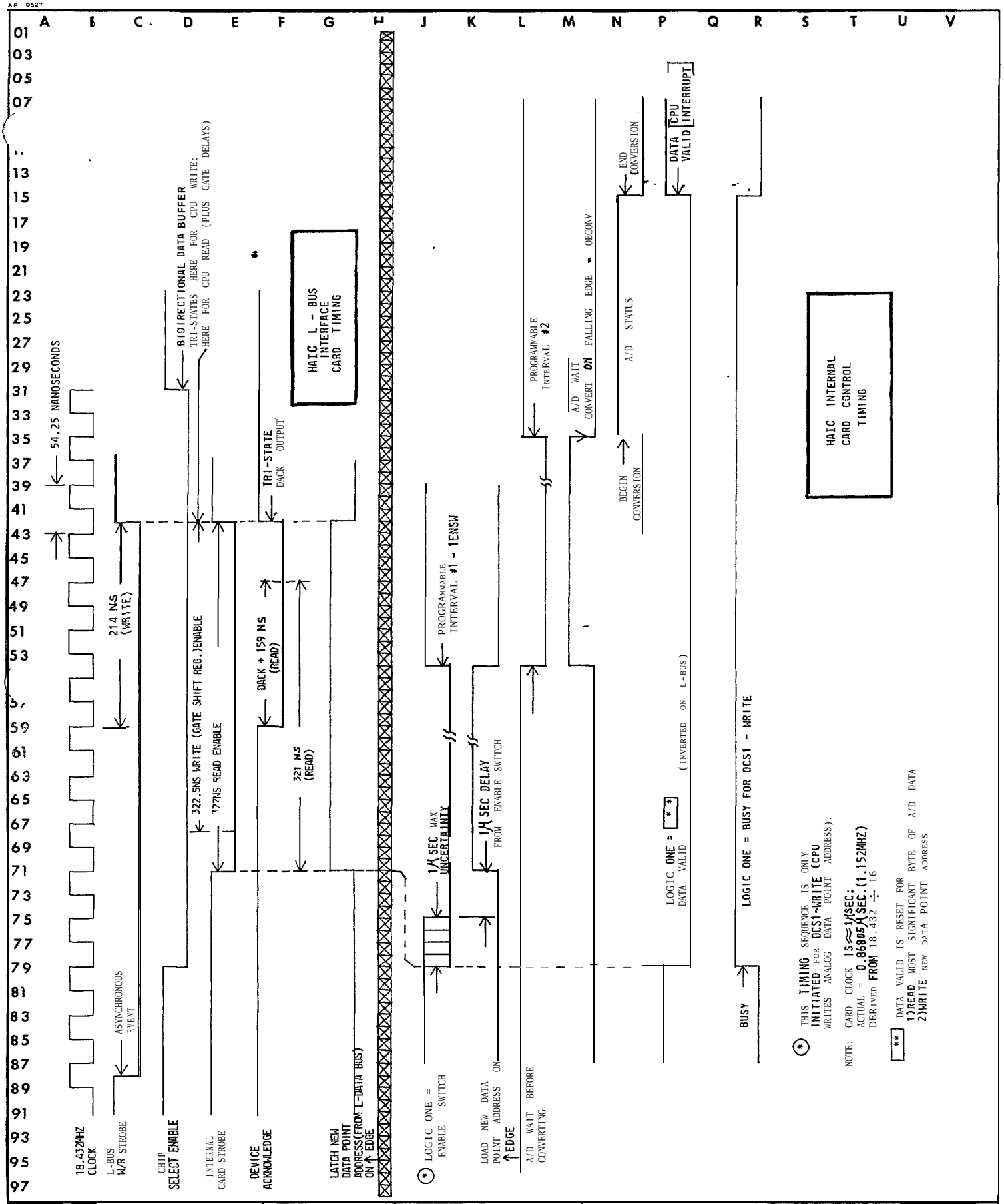
DSD 0132 (80 76)

C4RS



1	REV 4	REV 5	PRINTS TO DL102 2520 PR1	ENGINEER 5-5 83 C.H Swabe	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.
REV 2	ISSUED 83-05-12				ELEMENTARY DIAGRAM DETAILED BLOCK DIAGRAM
REV 3	MADE BY R.A. SALIBRICI				DS38D0HAIC CONT ON SH 2 OF SH NO 2

AF 0527



1	REV 4	REV 5	PRINTS TO 2520 DL102 PRI	ENGINEER 5-5-83 C.H. Sweeden	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA U.S.A.
REV 2	ISSUED 83-05-12				ELEMENTARY DIAGRAM TIMING DIAGRAM
REV 3	MADE BY R.A. SALIBRICI				DS3800HAIC CONT. ON SH. 2DA SH. NO. 2CA

CARS

2.0 BASIC FUNCTION

THE HAIC CARD ACCEPTS 16 ANALOG INPUT SIGNALS, MULTIPLEXES THEM TO ONE LINE, AND CONVERTS TO A TWO BYTE DIGITAL WORD. IT ALSO ROUTES GENERAL PURPOSE ONE BYTE, DIGITAL INPUTS AND OUTPUTS WITH RESPECT TO THE MICROPROCESSOR DATA BUS. BOTH FUNCTIONS OCCUR UNDER THE DIRECT CONTROL OF A MICROPROCESSOR (DS3800HMPC).

2.1 THEORY OF OPERATION

2.1.1 ANALOG INPUTS

HAIC DOES NOT USE A SAMPLE/HOLD CIRCUIT PRECEEDING THE A/D CONVERTER, THE SELECTED ANALOG INPUTS GOES DIRECTLY TO THE A/D MODULE. THE A/D IS OF THE SUCCESSIVE APPROXIMATION VARIETY AND EXECUTES A COMPLETE CONVERSION IN 20uSEC. MAXIMUM. TO PREVENT ALIASING ERRORS; INPUT FREQUENCY SHOULD BE LESS THAN 10HZ; THIS LIMITATION IS A RESULT OF THE DELAY BEFORE CONVERTING REQUIRED TO ALLOW CARD INPUT FILTERS TO SETTLE IN ADDITION TO THE 20uSEC..
FILTERING CONSISTS OF A SERIES 10K RESISTOR AND .00133MF TO ACOM ON THE SELECTED CHANNEL (330PF CERAMIC BEFORE THE MUX FOR EACH ANALOG LINE AND 1000PF MYLAR AFTER IT). FILTERING ENABLES USE OUTSIDE THE MODULE FOR THE ANALOG INPUTS.

THE ANALOG INPUT LINES ARE SINGLE-ENDED AND REFERENCED TO ACOM. FULL SCALE INPUT IS ±10V WITH NO OVER-RANGE; ie THE A/D 'OUTPUT REACHES FULL COUNT AT +9.9927V, -9.9976V AND STAYS FULL BEYOND THAT RANGE, SO THAT THERE IS ND DIGITAL INDICATIONS OF ANALOG OVER-RANGE.
MAXIMUM SAFE INPUT RANGE TO THE CARD WITH-OUT DAMAGE IS ±15V (THE CARD POWER RAILS), ALTHOUGH AS DISCUSSED, THE DIGITAL

CARS

2.1.1. ANALOG INPUTS (CONT'D)

OUTPUT WILL REFLECT NO MORE THAN FULL COUNT. SMALL SIGNAL INPUT IMPEDANCE ON ALL ANALOG CHANNELS IS 100 MEGΩ OVER TEMPERATURE, WHEN SELECTED (LOW FREQUENCY}; DC LEAKAGE IS 500NA MAX. OVER-TEMPERATURE SELECTED, 50NA MAX. NOT SELECTED.

2.1.2 DIGITAL ARCHITECTURE OVERVIEW

HAIC INTERNAL CARD DATA BUS IS 8 BITS (1 BYTE) WIDE DESIGNATED IDX: 100-107. IT PHYSICALLY INTERFACES THE LOWER OROER BYTE OF THE 16 BIT MICROPROCESSOR DATA BUS (PART OF L-BUS), AND IS INVERTED BY THE DRIVER/RECEIVER BUFFER TO MEET BUS PROTOCOL. THE MICROPROCESSOR DOES 2 I/O CYCLES TO GET MORE THAN B BITS OF DATA IN ANY INFORMATION EXCHANGE OR ROUTING OPERATION.CARD BIT 100, THE LEAST SIGNIFICANT BIT (LSB), IS WIRED TO THE MICROPROCESSOR DATA BUS LSB.

2.1.3 ADDRESSING OVERVIEW

THE 16 BIT MICROPROCESSOR ADDRESS BUS (PART OF L-BUS) SELECTS ALL HAIC CARD FUNCTIONS. THE TOP BUS BYTE, A8-AF, SELECTS HAIA CARD FOR MICROPROCESSOR INTERFACING, COMPARED AGAINST A BERG JUMPER SELECTABLE BIT PATTERN (SETTABLE ON HAIC), AS AN I/O PORT. OF THE LOWER BUS B BITS, ONLY AI-A4 ARE WIRED TO HAIA, WHICH ARE THEN EXPANDED INTO 16 CHIP SELECT LINES BY A 4 TO 16 LINE DECODER. ELEVEN OF THE CHIP SELECT LINES (OCS0-OCS10) ARE USED, ENABLING VARIOUS CARD FUNCTIONS ON DEMAND FROM THE CPU. THE REMAINING CHIP SELECT LINES ARE NOT USED.

2.1.4 A/O CONVERSION CYCLE

TO INITIATE AN A/O ANALOG POINT CONVERSION, THE MICROPROCESSOR WRITES AN B BIT ANALOG POINT ADDRESS TO HAIC ON THE MICRO-PROCESSOR DATA BUS (LOWER BYTE, D0-D7) SPECIFYING WHICH ANALOG INPUT TO CONVERT. THE ADDRESS BUS LINES SELECT HAIC CARD, OCS1 FUNCTION, FOR CONSIDERATION BY THE CPU.

2.1.4 A/D CONVERSION CYCLE (CONT'D)

THIS SELECTION THEN TRIGGERS THE CONVERSION CYCLE ON HAIC. THE 8 BIT POINT ADDRESS ON THE MICROPROCESSOR DATA BUS (WRITTEN BY THE MICROPROCESSOR) IS CAPTURED IN AN ANALOG ADDRESS LATCH (0-FF), AND 1ENSW (THE ENABLE SWITCH LINE) GOES LOW AT THE MICROPROCESSOR WRITE STROBE EDGE (INTERNALLY GATED). A LOW ON 1ENSW OPENS ALL ANALOG MULTIPLEXING SWITCHES (ie, EACH OF THE 16 HAIC ANALOG INPUT LINES CAN GO TO A "POINT" CARD WHERE IT MAY BE MULTIPLEXED BY 16 AGAIN). A PROGRAMMABLE TIMER (TRIGGERED BY OCS1 EVENT) WAITS A TIME DELAY (DELAY #1) AND THEN RESETS 1ENSW HIGH, ENABLING ALL ANALOG MULTIPLEXING SWITCHES.

APPROXIMATELY ONE MICROSECOND (0.868uSEC. OR 1 CARD CLOCK PULSE) AFTER 1ENSW WENT LOW, THE B BIT ANRLDG POINT ADDRESS GETS CLOCKED FROM THE BUS CAPTURE LATCH INTO THE LOAD ADDRESS LATCH. THE TOP FOUR BITS (MOST SIGNIFICANT) DF THE ANALOG ADDRESS ARE USED BY THE HAIC 16 TO 1 MUX, WHILE THE BOTTOM FOUR (LEAST SIGNIFICANT) ARE SENT OFF CARD TO ALL POINT CARDS AS A SUBSYSTEM BUS, ALONG WITH THE ENABLE SWITCH LINE. WHEN 1ENSW GOES HIGH AFTER THE FIRST DELAY, A UNIQUE PAIR OF MULTIPLEX SWITCHES CLOSES, SPECIFIED BY THE ANALOG POINT ADDRESS (STORED IN THE LOAD LATCH). AN ANALOG SIGNAL IS NOW CONNECTED INTO THE A/D MODULE, WHERE IT IS FIRST BUFFERED BY A FOLLOWER OP-AMP, THEN APPLIED TO THE SUCCESSIVE APPROXIMATION A/D CIRCUIT. NOTE THE DUAL REGISTER STRUCTURE OF THE ANALOG POINT ADDRESS LATCH. AS OUTLINED, THE FIRST LATCH CAPTURES DATA FROM THE MICROPROCESSOR BUS (AT THE WRITE STROBE) AND HOLDS IT IN TEMPORARY STORAGE. THE SECOND LATCH LOADS THE DATA INTO THE ANALOG SUBSYSTEM, ONE CARD CLOCK PULSE AFTER THE ANALOG SWITCHES WERE DISABLED. THIS SEQUENCE OF EVENTS INSURES THAT THE DISABLE SWITCH COMMAND ARRIVED AT ITS DESTINATION, AND WAS ACTED ON, BEFORE SENDING OUT A NEW POINT ADDRESS. THIS SEQUENCE ALONG WITH DELAY #1 ENSURES THAT TWO ANALOG CHANNELS WILL NDT BE SHORTED TOGETHER WHILE ONE SET OF

2.1.4 A/D CONVERSION CYCLE (CONT'D)

SWITCHES IS OPENING AND THE OTHER IS CLOSING (INCLUDES THE POSSIBILITY OF RELAY SWITCHES).

WHEN THE FIRST DELAY COMPLETES AND 1ENSW GOES HIGH, A SECOND DELAY IS TRIGGERED, THE WAIT BEFORE CONVERT DELAY. THIS DELAY ALLOWS THE SELECTED ANALOG CHANNEL TO SETTLE AND STABALIZE BEFORE CONVERTING (CARD FILTER CAPACITORS ON HAIC, LINE DISTRIBUTED CAPACITANCE, DISABLED TO ENABLED AMPLIFIER SETTLING TIME ON POINT CARDS AND STRAY CAPACITANCE). THIS DELAY SHOULD BE AT LEAST 200 MICROSECONDS TO PERMIT AMPLIFIER AND FILTER SETTLING, BASED ON LABORATORY AND CALCULATED DATA.

THE END OF THE SECOND DELAY COMMANDS THE A/D TO CONVERT, WHICH TYPICALLY TAKES 10 MICROSECONDS , 20 MICROSECONDS MAXIMUM. AT THE COMPLETION OF CONVERSION, 12 DIGITAL BITS OF INFORMATION (11 MAGNITUDE AND ONE SIGN BIT) ARE LATCHED AT THE A/D MODULE OUTPUT AND ARE THE-N AVAILABLE TO BE READ, IN 2'S COMPLIMENT FORM, FROM THE CARD DATA BUS (INVERTED ONTO THE MICROPROCESSOR BUS).

THE A/D MODULE ALSO OUTPUTS A STATUS FLAG, SIGNALLING THE END OF A CONVERSION, WHICH TRIGGERS A MICROPROCESSOR INTERRUPT ON THE DATA VALID LINE, 1DAVAL. THE MICROPROCESSOR CAN THEN RETURN TO THE HAIC, AT ITS LEISURE, AND READ THE DATA IT PREVIOUSLY REQUESTED WHEN IT INITIATED THE CONVERSION CYCLE.

ONCE STARTED, THE CONVERT CYCLE SHOULD NOT BE DISTURBED BY ANOTHER OCSI EVENT; IT WILL NOT DAMAGE ANYTHING BUT IS MAY CAUSE ERRONEOUS PERFORMANCE FOR THAT CYCLE. (IT WILL PROBABLY BE IGNOREO, BUT MAY HAPPEN TO CATCH THE RIGHT SPOT IN A FRAME AND BE DISRUPTIVE). ANY OTHER CARD OPERATION DURING A CONVERSION CYCLE IS PERMITTED (OTHER THAN OCSI), INCLUDING GENERAL PURPOSE I/O OR 'DATA READING.

2.1.5 BUSY SIGNAL AND DATA VALID SIGNAL

WHEN THE MICROPROCESSOR TRANSMITS A CONVERSION COMMAND TO HAIC (OCS1) THE BUSY SIGNAL LINE (1BUSY) GOES HIGH AND STAYS HIGH UNTIL THE A/D IS DONE CONVERTING, THEN GOES LOW. THIS SIGNAL IS AVAILABLE ON THE LOWER BYTE OF A/D DATA (EVENT OCS2) AS BIT 1D1; I.E. BECAUSE THERE ARE 12 BITS OF DATA AND 16 BITS IN THE 2 BYTE A/O READ FRAME. 1BUSY CAN BE SOFTWARE READ AS A HANDSHAKE BY THE MICROPROCESSOR.

THE DATA VALID LINE (1DAVAL.) IS THE MICROPROCESSOR INTERRUPT SIGNAL, WIRED TO THE MICROPROCESSOR INTERRUPT CIRCUITRY VIA THE BACKPLANE. IT GOES HIGH WHEN THE A/D FINISHES CONVERTING, ESTABLISHING THE INTERRUPT REQUEST (INVERTED ON THE MICRO-PROCESSOR BUS). IT IS RESET LOW WHEN THE MICROPROCESSOR READS THE MOST SIGNIFICANT BYTE OF A/D DATA, OR IF ANOTHER A/D CONVERT CYCLE IS INITIATED. 1DAVAL CAN BE SOFTWARE READ AS A HANDSHAKE BY THE MICROPROCESSOR ON THE LOWER BYTE OF A/B DATA, BIT 1D0.

2.1.6 CARD CLOCK SCHEME

THE CARD CLOCK SOURCE WHICH CONTROLS BUS INTERFACING IS AN 18.432 MHZ QUARTZ CRYSTAL. THIS FREQUENCY IS THEN DIVIDED DOWN BY 16 TO 1.152 MHZ AND BECOMES THE BASIC CLOCK FOR A/D CONTROL LOGIC (0.868 MICROSECOND).

2.1.7 TIME DELAYS

CONVERSION CYCLE TIME DELAYS ON HAIC ARE GENERATED BY AN INTEL 6253-5 PROGRAMMABLE TIMER CHIP. IT COMES WITH THREE SEPARATE, INDEPENDENT TIMERS, TWO OF WHICH ARE USED. EACH TIMER IS A 16 BIT PRELOADABLE COUNTDOWN REGISTER, WITH TWO DIFFERENT AVENUES OF INTERFACE; AN EIGHT BIT DIGITAL PORT WHICH CONNECTS TO THE HAIC- INTERNAL DATA BUS (ONE PORT PER CHIP - COMMON TO ALL TIMER REGISTERS), AND A SET OF THREE TIMING CONTROL PINS PER REGISTER. THE DATA BUS PORT ALLDWS MICROPROCESSOR SOFT-WARE TO SET OPERATING MODE (2) AND PRELOAD A TIME DELAY COUNT

2.1.7 TIME DELAYS (CDNT'D)

TO EACH REGISTER, CAPTURED BY A COROLLARY LATCH. THE THREE CONTROL PINS; GATE, CLOCK IN, AND OUTPUT, MANIPULATE THE TIMING EVENT ITSELF.

COUNT PRELOADING NEED ONLY BE DONE ONCE, AS AN INITIALIZATION, SINCE THE REGISTER COROLLARY LATCH HOLDS THE SOFTWARE COUNT VALUE FOR REPEATED USE. COUNTS ARE PROGRAMMABLE FROM N = 4 TO N = $(2^{16} - 1)$ = 65,535 (THE CHIP DOES NOT LIKE N = 3) EACH DELAY REGISTER TAKES A CARD CLOCK INPUT (0.868 MICROSECOND), A GATE INPUT, AND CONTROLS A SIGNAL OUTPUT PIN. ON HAIC, THE CHIP OPERATES IN MODE 2, FREE-RUNNING COUNTDOWN; THE PRELDADED REGISTER DECREMENTS ON EACH CLOCK PULSE UNTIL COUNT ZERO IS REACHED AT WHICH POINT, THE OUTPUT PIN GOES LOW. IF NOTHING ELSE HAPPENED, IT WOULD STAY LOW FOR ONE CLOCK PULSE, RETURN HIGH AND BEGIN ANOTHER FULL COUNTDOWN. HOWEVER, ON HAIC, WHEN THE OUTPUT PIN GOES LOW, IT DRIVES THE GATE LOW, WHICH FREEZES COUNTING AND FO-RCES THE OUTPUT NIGH (ANYTIME THE GATE GOES LOW, COUNTING IS FROZEN AND THE OUTPUT IS FORCED HIGH IF IT IS NOT ALREADY HIGH).

WHEN THE NEXT A/D CONVERSION CYCLE IS INITIATED (OCSI) THE GATE WILL BE BROUGHT HIGH, WHICH RELDADS FULL COUNT INTO THE REGISTER AND BEGINS A NEW COUNTDOWN. .

DN THE FIRST DELAY (IENSW), THE DELAY LENGTH IS N + 2 PLUS AN UNCERTAINTY OF UP TO ONE ADDITIONAL COUNT, DUE TO THE ASYNCH-RONOUS NATURE OF THE MICROPROCESSOR WRITE STROBE WITH RESPECT TO HAIA INTERVAL CLOCKS (MICROPROCESSOR WRITE STROBE INITIATES OCSI). THE SECOND DELAY, WAIT BEFORE CONVERT, IS N + 2 COUNTS IN LENGTH. RECALL THAT EACH COUNT IS 0.868 MICROSECONDS LDNG. TIME DELAYS CAN THUS BE VARIED FROM (4+2)X0.868 = 5.21 MICRO-SECOND TO (65,535+2)X.868 = 56.886 MILLISECONDS.

2.1.8 MICROPROCESSOR BUS INTERFACE TIMING

THE HAIC TO MICROPROCESSOR BUS INTERFACE MECHANISM PIVOTS ON A HIGH SPEEO SHIFT REGISTER (SR). THE SR IS CLOCKED BY 18.432 MHZ AND ENABLED BY CARD SELECT, WITH A MICROPROCESSOR READ/WRITE BUS STROBE. SEVERAL CARD ENABLING SIGNALS ARE TAPPED OFF THE SR AND IMPLEMENT ACTUAL DEVICE OPERATION ON CARD. THE LAST SIGNALGENERATED IS OBLCDAK, DEVICE ACKNOWLEDGE, WHICH TELLS THE MICROPROCESSOR THAT HAIA RECOGNIZES HAVING BEEN ADDRESSED, AND IS ACTING ON IT.

2.1.9 GENERAL PURPOSE INPUTS

THERE ARE THREE 8 BIT, GENERAL PURPOSE DIGITAL INPUT PORTS ON HAIC. 8 BITS ARE BROUGHT IN FROM THE BACKPLANE UNDER OCS4 CONTROL, AND 16 BITS ARRIVE THROUGH FRONT EDGE CONNECTOR JA; THE LOWER BYTE UNDER OCS5 CONTROL AND THE UPPER BYTE ON OCS6. EACH INPUT BIT LINE IS FILTERED BY A SERIES 100Ω RESISTOR AND 0.1uf TO DCOM, PRECEEDED BY A 10KPULLUP. THE FILTER TIME-CONSTANT RANGES FROM 5 MICROSECOND TO 20 MICROSECOND {DUE' TO THE LARGE VARIANCE IN CERAMIC CAPACITANCE). THESE INPUTS CAN BE USED OUTSIDE THE MODULE. ALL GENERAL PURPOSE INPUTS DUMP ONTO THE CARD INTERNAL DATA BUS, TO BE REAO BY THE MICROPROCESSOR (DATA BUS). LOGIC ONE INPUT IS TRUE AND IS NOT INVERTED ONTO THE INTERNAL CARD DATA BUS. FIVE VOLTS IS SENT OUT ON IA THROUGH A DIODE AND 100Ω BUFFER RESISTOR, USED BY THE INPUT SOURCES.

2.1.10 GENERAL PURPOSE OUTPUTS

THERE ARE THREE AND ONE HALF 8 BIT GENERAL PURPOSE DIGITAL OUTPUT PORTS ON HAIC. TWELVE BITS ARE ROUTED OUT THE BACKPLANE, THE TOP 8 BITS ON OSC7 AND THE REMAINING 4 BITS ON OCS8. SIXTEEN BITS GO OUT FRONT EOGE CONNECTOR JB; THE LOW BYTE ON OCS9, THE HIGH BYTE ON OCS10.

OUTPUT DATA ARE CAPTURED BY OCTAL OFF (LATCHING) REGISTERS AND HELD UNTIL REWRITTEN TO BY THE MICROPROCESSOR. DATA ARE AVAILABLE WHILE ISTALL, THE STALL LINE IS LOW; A HIGH ON ISTALL

2.1.10 GENERAL PURPOSE OUTPUTS (CONT'O)

TRISTATES ALL GENERAL PURPOSE OUTPUT BITS. THE BOTTOM BYTE OF THE THE FRONT EDGE (JB) OUTPUTS CAN BE FORCEO TO CLOCK OUT THE INTERNAL CARD DATA BUS AT ANYTIME BY DRIVING TP7 LOW MANUALLY: DATA IS CAPTURED ON THE FALLING EDGE.
LOGIC ONE OUTPUTS ARE TRUE, AND ARE NOT INVERTED FROM THE INTERNAL CARD DATA BUS TO THE OUTPUT PORT PINS.

2.1.11 STALL LINE

ALL GENERAL PURPOSE OUTPUT TRISTATE CONTROL IS WIRED TO THE STALL LINE, 1STALL. IT IS INTENDED FOR USE DURING POWER UP TO PREVENT RANDOM SPURIOUS DUTPUT PATTERNS FROM BEING TRANSMITTED, AND AFFECTS ALL OUTPUT BITS. IN NORMAL OPERATION, 1STALL IS CONTRDLLEO BY THE MICROPROCESSOR, AND IS HELD LOW TO ENABLE.THE OUTPUT PORTS (A HIGH TRISTATES OR "STALLS"' THE PDRTS). A BERG JUMPER CAN FORCE THE STALL LINE LOW IF THIS FUNCTION IS NOT DESIRED (J10) FUR THE BACKPLANE OUTPUTS ONLY; 1BOGPK.

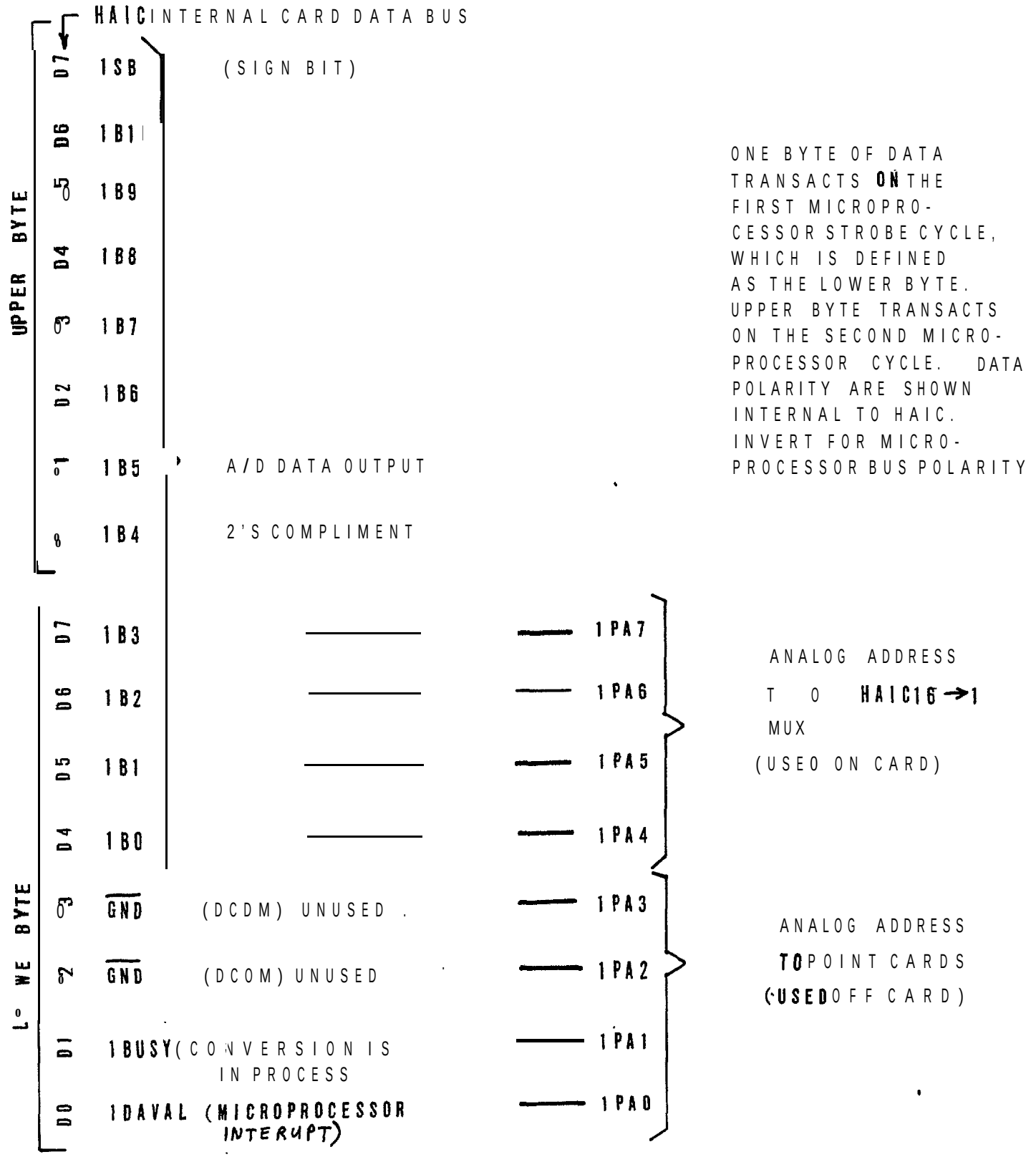
2.1.12 ERROR BUDGET FOR A./D CONVERSION.

TOTAL HAIC CARD ERROR FOR THE ANALOG TO DIGITAL CONVERSION PROCESS FROM ANALOG INPUT TO DIGITAL OUTPUT APPEARS IN THE CHART BELOW. THE ERROR FIGURE TAKES TWO FORMS; NO SOFTWARE OFFSET COMPENSATION, AND INCLUDING SOFTWARE OFFSET COMPENSATION. THE COMPENSATION EMPLOYS ONE ANALOG CHANNEL AS A ZERO REFERENCE BY SHORTING IT TO ACOM, AND THEN SOFTWARE SUBTRACTS OFFSET MEASURED ON THE SHORTED REFERENCE CHANNEL FROM THE REMAINING ACTIVE CHANNELS.

TYPE	WORST CASE	RSS
OFFSET CANCELLING	±9.598LSB(±0.478%)	±5.400LSB(±0.264%)
NON CANCELLING	±12.253LSB(±0.599%)	±6.072LSB(±0.297%)

2.2 INTERFACE REQUIREMENTS

2.2.1 HAIC DATA FIELD



V. 1	REV. 4	REV. 7	PRINTS TO PRI DL102,2520	ENGINEER RAS	GENERAL ELECTRIC DSD SALEM, VA. U.S.A.	DESCRIPTION OF OPERATION
REV. 2	REV. 5	ISSUED 83-05-12				DS3800HAIC
REV. 3	REV. 6	MADE BY R.A.SALIBRICI				PRINT NO SH. 2NA SH. NO. 2MA

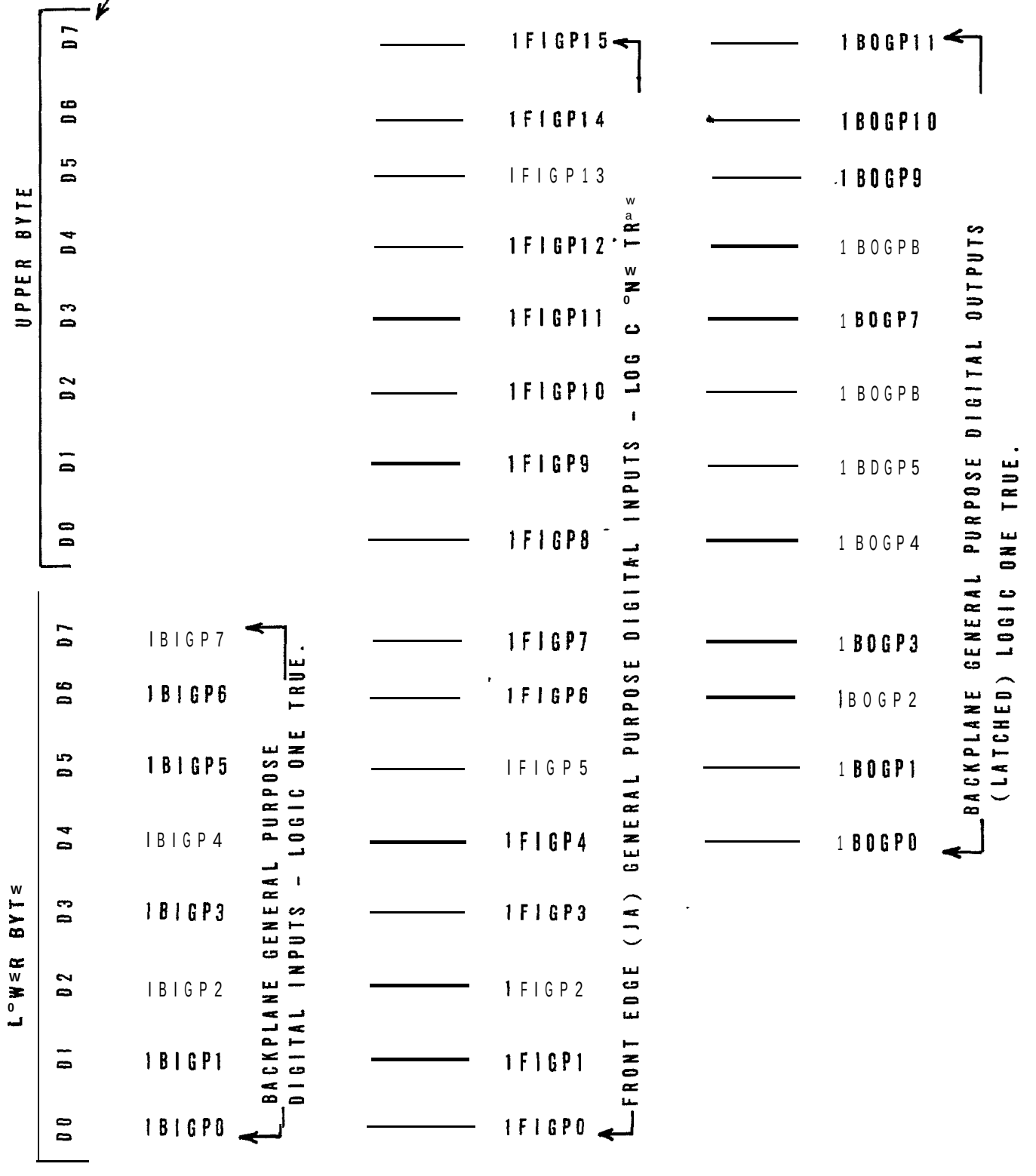
DSD 0151 (09-76)

CARS

2.2 INTERFACE REQUIREMENTS(CONT'D)

2.2.1 HAIC DATA FIELD

HAIC CARD DATA BUS



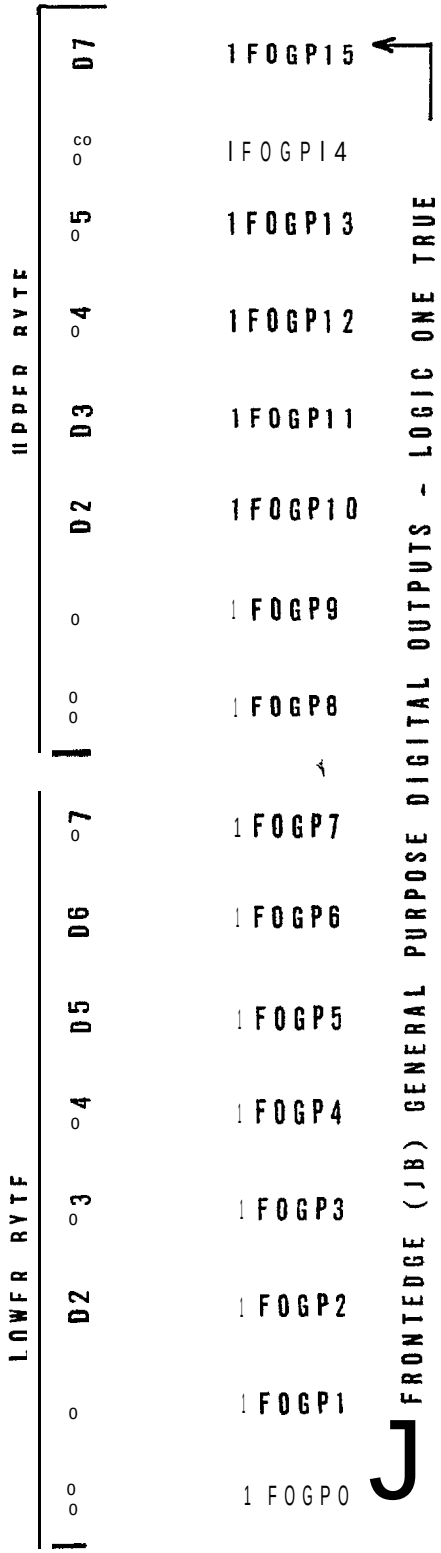
AEV. 1	REV. 4	REV. 7	PRINTS TO PR DL1 02,2520	ENGINEER RAS	GENERAL ELECTRIC DSD SALEM, VA. U.S.A.	DESCRIPTION OF OPERATION
REV. 2	REV. 5	ISSUED 83-05-12				DS3800HAIC
AEV. 3	REV. 6	MADE BY R.A.SALIDRICI				CONT ON SH. 2PA SH. NO. 2NA

CARS

2.2 INTERFACE REQUIREMENTS (CONT'D)

2.2.1 HAIC DATA FIELD

HAIC CARD DATA BUS



[SEE INTEL DATA CATALOGUE FOR TIMER CHIP 8253-5 DATA FIELD.]

REV. 1	REV. 4	REV. 7	PRINTS TO PRI DL102520	ENGINEER RAS	GENERAL ELECTRIC DSD SALEM, VA. U.S.A.	DESCRIPTION OF OPERATION DS3800HAIC
REV. 2	REV. 5	ISSUED 83-05-12				
REV. 3	REV. 6	MADE BY R.A.SALIBRICI				CONF. FINISH. 2QA SH. NO. 2PA

DSD 0151 (09-76)

CARS

2.2 INTERFACE REQUIREMENTS (CONT'D)

2.2.2 HAIC FUNCTION ADDRESS FIELD

AB THRU AD ARE BERG JUMPER SELECTABLE CARD ADDRESS PATTERN.

AE (PA34) MUST BE A "ONE" TO SELECT HAIC CARD.

AF (PA38) MUST BE A "ZERO" TO SELECT HAIC CARD.

FUNCTION	A4*	A3*	A2*	A1*	RD/WR	NAME
1FOGP0-P7	0	a	0	0	WR	OCS10
1FOGP8-P15	0	0	0	1	WR	OCS9
1BOGP0-P3	0	0	1	0	WA	ocsa
1BOGP4-P11	0	0	1	1	WR	ocs7
1FIGP8-P15	0	1	0	0	RD	OCS6
1FIGP0-P7	0	1	0	1	RD	OCS5
1BIGP0-P7	0	1	1	0	RD	OCS4
AN. PT.ADD. & START A/O CONV.CYCLE	0	1	1	1	WR	OCS1
READ A/D LSB	1	0	0	0	RD	OCS2
READ A/D MSB	1	0	0	1	RD	ocs3
8253-5 ENABLE	1	1	X	X	X	OCS0
(WR DELAY ① LENGTH)			↓ 0	↓ 1	X	ACESS COUNTER 1
(WR DELAY ② LENGTH)			1	0	X	ACESS COUNTER 2
			1	1	WR	CONTROL WORD

X IS DON'T CARE STATE, CAN BE EITHER 0 OR 1.

DELAY ① ENABLE SWITCH DELAY (1ENSW)

DELAY ② WAIT BEFORE CONVERTING DELAY (OECONV)

* A1-A4 SHOWN ABOVE REPRESENT THE LOGIC STATES INTERNAL TO HAIC. INVERT FOR PROPER POLARITY AT THE CARD PINS (BUS POLARITY):

A4-PA59, A3-PA58, A2-PA60, A1-PA56

NOTE THAT SINCE A5-A7 ARE NOT CONNECTED, MULTIPLE BUS ADDRESSES WILL SELECT EACH FUNCTION.

2.2 INTERFACE REQUIREMENTS (CONT'D)

2.2.3 HAIC ANALOG ADDRESS FIELD

CHANNEL	PIN	IPA7*	IPA6*	IPA5*	IPA4*
D	PA16	D	D	D	D
1	PA19	0	0	D	1
2	PA16	0	0	1	D
3	PA20	D	0	1	1
4	PA21	0	1	0	0
5	PA22	0	1	D	1
6	PA23	0	1	1	D
7	PA24	0	1	1	1
6	PA2	1	0	0	D
9	PA4	1	0	0	1
10	PA13	1	0	1	D
11	PA15	1	0	1	1
12	PA6	1	1	0	0
13	PA17	1	1	0	1
14	PA10	1	1	1	0
15	PA12	1	1	1	1

* IPA4 - IPR7 SHOWN ABOVE REPRESENT THE LOGIC STATES INTERNAL TO HAIC. INVERT FOR PROPER POLARITY ON THE MICROPROCESSOR DATA BUS.

IPAD - IPA3 CONTROL ANALOG POINT CARD MULTIPLEXING, THE ADDRESSING PROTOCOL HERE DEPENDS ON THE POINT CARD SELECTED, ALTHOUGH THE NORMAL MODE WILL BE INCREASING CHANNEL NUMBER FOR INCREASING BINARY COUNT (AS ABOVE).

IPAD - IPA3 AND IENSW ARE CAPABLE OF DRIVING 24 POINT CARDS UNDER THE FOLLOWING GUIDELINES: THE INPUT BUFFER ON THE POINT CARD MUST BE AN LS244, LS2400 OR EQUIVALENT (0.2MA INPUT LOAD FOR LOGIC LOW) WITH A 12.1K PULLUP RESISTOR FOR IPA0 - IPA3 AND 15K FOR IENSW.

CARS

2.3 FUNCTIONAL DESCRIPTION

* INDICATES MICROPROCESSOR BUS PIN

2.3.1 BACKPLANE

MNEMONIC	PIN PA-	DESCRIPTION
DCOM	1	LOGIC GROUND
CH8	2	ANALOG INPUT CHANNEL 8
P5	3	LOGIC POWER
CH 9	4	ANALOG INPUT CHANNEL 9
P15	5	AN'ALOG POWER
"1 BLCRST	6	MICROPROCESSOR BUS RESET LINE
N15	7	ANALOG POWER
CH 12	8	ANALOG INPUT CHANNEL 12
ACOM	9	ANALOG GROUND
CH 14	10	ANALOG INPUT CHANNEL 14
*OBLCIOW	11	MICROPROCESSOR I/O WRITE LINE
CH 15	12	ANALOG INPUT CHANNEL 15
CH 10	13	ANALOG INPUT CHANNEL 10
*OBLCIOR	14	MICROPROCESSOR I./O READ LINE
CH 11	15	ANALOG INPUT CHANNEL 11
CH 0	16	ANALOG INPUT CHANNEL 0
CH 13	17	ANALOG INPUT CHANNEL 13
CH 2	18	ANALOG INPUT CHANNEL 2
CH 1	19	ANALOG INPUT CHANNEL 1
CH 3	2D	ANALOG INPUT CHANNEL 3
CH 4	21	ANALOG INPUT CHANNEL 4
CH 5	22	ANALOG INPUT CHANNEL 5
CH 6	23	ANALOG INPUT CHANNEL 6
CH 7	24	ANALOG INPUT CHANNEL 7
1 BOGP7	25	BACKPLANE GENERAL PURPOSE OUTPUT, BIT 7, LOGIC 1 TRUE
1 BOGP6	26	OUTPUT, BIT 6, LOGIC 1 TRUE
1 BOGP9	27	OUTPUT, BIT 9, LOGIC 1 TRUE
1 BOGP10	28	OUTPUT, BIT 10, LOGIC 1 TRUE
1 BOGP8	29	OUTPUT, BIT 8, LOGIC 1 TRUE
1 BOGP11	30	OUTPUT, BIT 11, LOGIC 1 TRUE
*OBLAA	31	MICROPROCESSOR ADDRESS BUS BIT 1)

V. 1	REV. 4	REV. 7	PRINTS TO PRI DL102 2521	ENGINEER RAS	GENERAL ELECTRIC DSD SALEM, VA. U.S.A.	DESCRIPTION OF OPERATION
REV. 2	REV. 6	ISSUED 83-05-12				DS3800HAIC
REV. 3	REV. 6	MADE BY R.A.SALIBRICI				CONTROL NO. 2UA SH. NO. 2TA

CARS

2.3 FUNCTIONAL DESCRIPTION (CONT'D)

2.3.1 BACKPLANE (CONT'D)

MNEMONIC	PIN PA-	DESCRIPTION
1 BOGP5	32	BACKPLANE GENERAL PURPOSE OUTPUT, BIT 5, LOGIC 1 TRUE.
*OBLAD	33	MICROPROCESSOR BUS BIT 13-ADDRESS
*DBLAE	34	MICROPROCESSOR BUS BIT 14-ADDRESS
"OBLAB	35	MICROPROCESSOR BUS BIT 11-ADDRESS
*OBLA8	36	MICROPROCESSOR BUS BIT 8-ADDRESS
1 BOGP4	39	BACKPLANE GENERAL PURPOSE OUTPUT, BIT 4, LOGIC 1 TRUE.
*OBLAF	38	MICROPROCESSOR BUS BIT 15-ADDRESS
1PAO	39	ANALOG POINT CARD ADDRESS BIT 0
*OBLA9	40	MICROPROCESSOR ADDRESS BUS BIT 9
1 PA1	41	ANALOG POINT CARD ADDRESS BIT 1
*OBLAC	42	MICROPROCESSOR ADORESS BUS BIT 12
DCOM	43	LOGIC GROUND
1BIGP7	44	BACKPLANE GENERAL PURPOSE INPUT, BIT 9, LOGIC 1 TRUE.
P5	45	LOGIC POWER
1BIGP5	46	BACKPLANE GENERAL PURPOSE INPUT, BIT 5, LOGIC 1 TRUE.
1PA'3	47	ANALOG POINT CARD ADDRESS BIT 3
1BIGP4	48	BACKPLANE GENERAL PURPOSE INPUT, BIT 4, 'LOGIC 1 TRUE.
1PA2	49	ANALOG POINT CARD ADDRESS BIT 2
1BIGP3	50	BACKPLANE GENERAL PURPOSE INPUT, BIT 3, LOGIC 1 TRUE.
1BIGP6	51	INPUT BIT 6, LOGIC 1 TRUE.
1BIGP2	52	INPUT BIT 2, LOGIC 1 TRUE.
	53	
1BIGP1	54	INPUT BIT 1, LOGIC 1 TRUE.
1BIGP0	55	INPUT BIT 0, LOGIC 1 TRUE.
*OBLA1	56	MICROPROCESSOR ADDRESS BUS BIT 1
1STALL	57	GENERAL PURPOSE OUTPUT TRISTATE CONTROL LINE
*OBLA3	58	MICROPROCESSOR ADDRESS BUS BIT 3

2.3 FUNCTIONAL DESCRIPTION (CONT'D)

2.3.1 BACKPLANE (CONT'D)

MNEMONIC	PIN PA-	DESCRIPTION
*OBLA4	59	MICROPROCESSOR BUS BIT 4 -ADDRES
*OBLA2	60	MICROPROCESSOR BUS BIT 2-ADDRES
*OBLD5	61	MICROPROCESSOR DATA BUS BIT 5
ODAVAL	62	MICROPROCESSOR INTERRUPT REQUEST LINE
———	63	———
*OBLD3	64	MICROPROCESSOR DATA BUS BIT 3
IENSW	65	ENABLE SWITCH LINE (MULTIPLEXERS) LOGIC 1 TRUE.
*OBLD1	66	MICROPROCESSOR DATA BUS BIT 1
*OBLCDAK	67	DEVICE ACKNOWLEDGE LINE (TO MICROPROCESSOR CONTROL BUS).
*OBLD0	68	MICROPROCESSOR DATA BUS BIT 0
1BOGP1	69	BACKPLANE GENERAL PURPOSE OUTPUT BIT 1, LOGIC 1 TRUE.
*OBLD7	70	MICROPROCESSOR DATA BUS BIT 7
1BOGP2	71	BACKPLANE GENERAL PURPOSE OUTPUT BIT 2, LOGIC 1 TRUE.
*OBLD6	72	MICROPROCESSOR DATA BUS BIT 6
———	73	(SAFETY GROUND-UNUSED).
*OBLD2	74	MICROPROCESSOR DATA BUS BIT 2
	75	(P28 -UNUSED)
*OBLD4	76	MICROPROCESSOR DATA BUS BIT 4
P5	77	LOGIC POWER
1BDGP0	78	BACKPLANE GENERAL PURPOSE OUTPUT BIT 0, LOGIC 1 TRUE.
DCOM	73	LOGIC GROUND
1BOGP3	80	BACKPLANE GENERAL PURPOSE OUTPUT BIT 3, LOGIC 1 TRUE.

CARS

2.3 FUNCTIONAL DESCRIPTION (CONT'D)

2.3.2 FRONTEDGE

MNEMONIC	PIN #	DESCRIPTION
P5R	A1	P5 REMOTE
DCOM	A2	LOGIC RETURN (REMOTE)
1FIGP4	A3	FRONTEDGE GEN. PURPOSE INPUT, BIT 4; LOGIC 1 TRUE
1FIGP5	A4	FRDNTEDGE GEN. PURPOSE INPUT,, BIT 5, LOGIC 1 TRUE
1FIGP3	A5	FRONTEDGE GEN. PURPOSE INPUT, BIT 3, LOGIC 1 TRUE
1FIGP6	A6	FRONTEDGE GEN. PURPOSE INPUT, BIT 6, LOGIC 1 TRUE
1FIGP2	A7	FRONTEDGE GEN. PURPOSE INPUT, BIT 2, LOGIC 1 TRUE
1FIGP7	A8	FRONTEDGE GEN. PURPOSE INPUT, BIT 7, LOGIC 1 TRUE
1FIGP0	A9	FRONTEDGE GEN. PURPOSE INPUT, BIT 0, LOGIC 1 TRUE
1FIGP1	A10	FRONTEDGE GEN. PURPOSE INPUT, BIT 1, LOGIC I TRUE
P5R	A11	P5 REMOTE
DCOM	A12	LOGIC 'RETURN (REMOTE)
1FIGP9	A13	FRONTEDGE GENERAL PUR'POSE INPUT, BIT 9, LOGIC 1 TRUE.'
1FIGP8	A14	FRONTEDGE GENERAL PURPOSE INPUT, BIT 8, LOGIC 1 TRUE
1FIGP15	A15	FRONTEOGE GENERAL PURPOSE INPUT, BIT 15, LOGIC 1 TRUE
1FIGP10	A16	FRONTEDGE GENERAL PURPOSE INPUT, BIT 10, LOGIC 1 TRUE
1FIGP14	A17	FRONTEDGE GENERAL PURPOSE INPUT, BIT 14, LOGIC 1 TRUE
1FIGP11	A18	FRONTEDGE GENERAL PURPOSE INPUT; BIT 11, LOGIC 1 TRUE
1FIGP13	A19	FRONTEDGE GENERAL PURPOSE INPUT, BIT 13, LOGIC 1 TRUE
1FIGP12	A20	FRONTEOGE GENERAL PURPOSE INPUT, BIT 12, LOGIC 1 TRUE
1FOGP11	B1	FRONTEDGE GENERAL PURPOSE OUTPUT, BIT 11, LOGIC 1 TRUE
DCOM	62	LOGIC RETURN (REMOTE)
1FOGP15	63	FRONTEDGE GENERAL PURPOSE OUTPUT, BIT 15, LOGIC 1 TRUE
1FOGP10	B4	FRONTEDGE GENERAL PURPOSE OUTPUT, BIT 10, LOGIC 1 TRUE
1FOGP13	B5	FRONTEOGE GENERAL PURPOSE OUTPUT, BIT 13, LOGIC 1 TRUE

MNEMONIC	PIN #	DESCRIPTION
1FOGP14	B6	FRONTEDGE GENERAL PURPOSE OUTPUT, BIT 14, LOGIC1 TRUE
1FOGP9	B7	FRONTEDGE GENERAL PURPOSE OUTPUT, BIT 9, LOGIC1 TRUE
1FOGP12	B8	FRONTEDGE GENERAL PURPOSE OUTPUT., BIT12, LOGIC 1 TRUE
1FOGP0	B9	FRONTEDGE GENERAL PURPOSE OUTPUT, BIT 0, LOGIC 1 TRUE
1FOGP8	B10	FRONTEDGE GENERAL PURPOSE OUTPUT, BIT 8, LOGIC 1 TRUE
IFOGP4	B11	FRONTEDGE GENERAL PURPOSE OUTPUT, BIT 4, LOGIC 1 TRUE
1FOGP1	B12	FRDNTEDGE GENERAL PURPOSE OUTPUT, BIT 1, LOGIC 1 TRUE
1FOGP6	B13	FRONTEDGE GENERAL PURPDSE DDTPUT, BIT 6, LOGIC 1 TRUE
1FOGP5	B14	FRONTEDGE GENERAL PURPOSE OUTPUT, BIT-5, LOGIC 1 TRUE
1FOGP3	B15	FRONTEDGE GENERAL PURPOSE OUTPUT, BIT 3, LOGIC 1TRUE
1FOGP7	B16	FRONTEDGE GENERAL PURPOSE OUTPUT, BIT 7, LOGIC 1 TRUE
———	B17	———
1FOGP2	B18	FRONTEDGE GENERAL PURPOSE DUTPUT, BIT 2, LOGIC 1 TRUE
———	B19	———
———	B20	———

2.3 FUNCTIONAL DESCRIPTION (CDNT'D)

2.3.3 TEST LOOPS

TESTPOINT	NAME	DESCRIPTION
TP1	CLK	CARD CLOCK: 1..152MHZ
TP2	ES	1ENABLESWITCH; PA65
TP3	BSY	1BUSYSIGNAL
TP4	DV	DDATA VALID SIGNAL; PA62
TP5	OB	DATA BUFFER DIRECTION; HIGH TRANSMITS TO MICROPROCESSOR, LOW RECEIVES.
TP6	AN	ANALOG SIGNAL JUST BEFORE A:/D CONVERSION (AFTER OP-AMP)
TP7	FOG	FRONTEDGE GENERA'L PURSPOSE OUTPUT CONTROL LINE; DRIVE LOW TO LATCH INTERNAL CARD DATA BUS.
TPB	ACOM	

2.3.4 POTENTIOMETERS

POT	NAME	DESCRIPTION
R1	OFFSET	A:/D MODULE OFFSET ADJUSTMENT
R2	GAIN	A:/D MODULE GAIN ADJUSTMENT

THESE POTS ARE FACTORY SET AND SHOULD NOT BE TAMPERED WITH.

2.4 APPLICATION DATA

2.4.1 POWER REQUIREMENTS

VOLTS	TOLERANCE	AMPS(MAX)	WATTS (MAX)
+5V	±5%	875MA	4.594
+15V	±1%	70MA	1.06
-15V	±1%	70MA	1.06
		TOTAL WATTS	6.715

2.4.2 TTL INPUTS

ALL LOGIC INPUTS TAKE STANDARD LOW POWER SCHOTTKY TTL VOLTAGE LEVELS. ALL EXCEPT THOSE 'NOTED BELOW TAKE 0.2MA LOGIC LOW INPUT CURRENT, 20ua LOGIC HIGH:

OBLAE-OBLAF (PA36,40,31,35,42,33,34,38) TAKE 0.4MA LOW AND 20ua HIGH, AS DOES 1STALL(PA57);

V. 1	REV. 4	REV. 7	PRINTS TO PRI DL102,2520	ENGINEER RAS	GENERAL ELECTRIC	DESCRIPTION OF OPERATION
REV. 2	REV. 5	ISSUED 83-05-12			DSD	DS3800HAIC
REV. 3	REV. 6	MADE BY R.A.SALIBRICI			SALEM, VA. U.S.A.	SCH. SH. 2ZA SH. NO. 2YA

DSD 0151 (09-76)

2.4.2 TTL INPUTS (CONT'D)

OBLD0 - OBLD7 (PA68, 66, 74, 64, 76, 61, 72, 70) TAKE 0.2MA LOW, 50ua HIGH;

ALL GENERAL PURPOSE INPUTS HAVE A 4.7K RESISTOR TO P5 (+5V) IN ADDITION TO DRAWING 0.2MA LOGIC LOW. THUS THE LOW INPUT CURRENT IN THESE LEADS IS 1.3MA EACH.

2.4.3 TTL OUTPUTS

ALL LOGIC OUTPUTS SUPPLY STANDARD LOW POWER SCHOTTKY VOLTAGE LEVELS:

LOGIC LOW = 0.5V MAX.
LOGIC HIGH = 2.7V MIN.

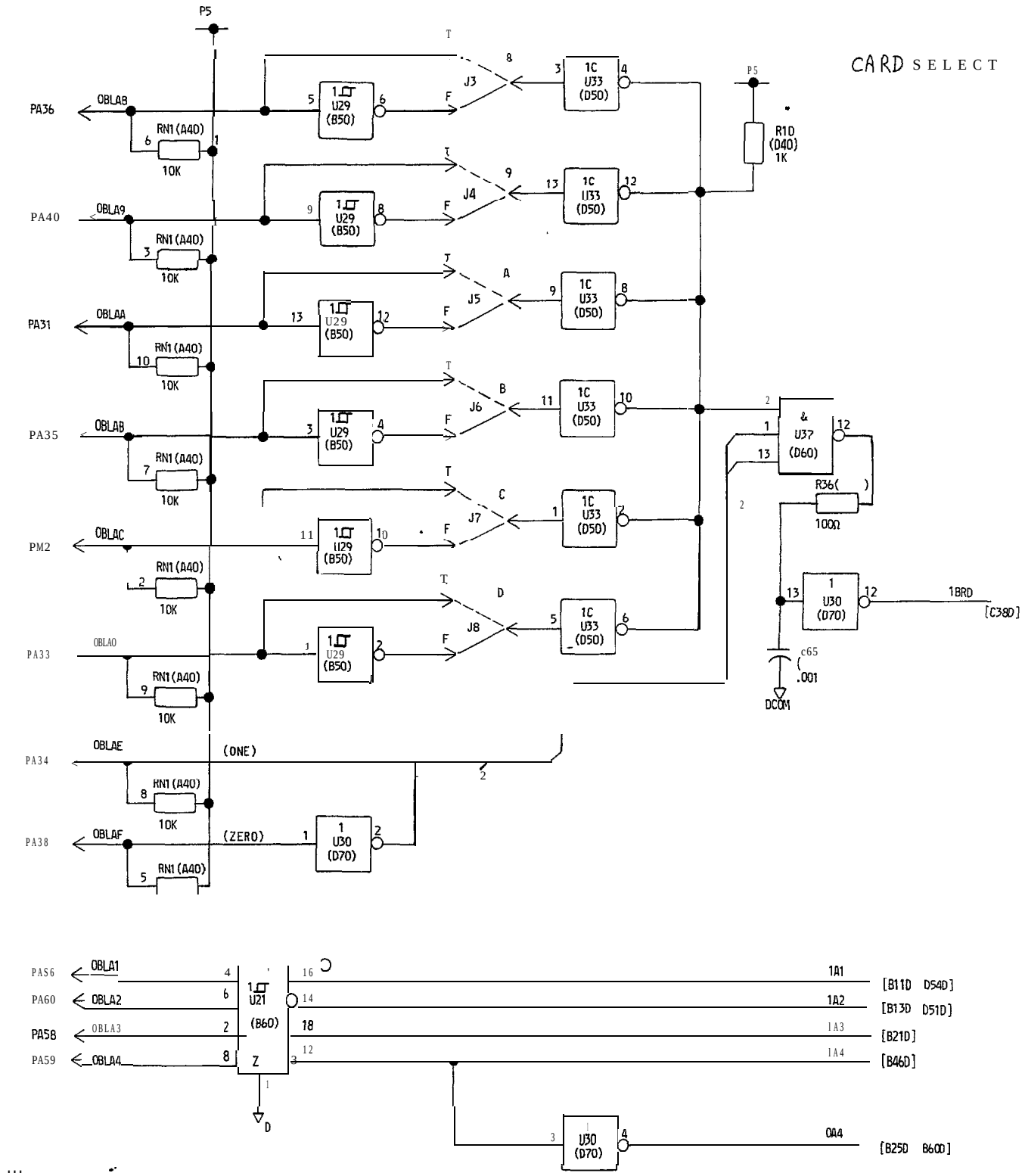
OUTPUT DRIVE CAPABILITY IS 24MA LOW EXCEPT FOR THE FOLLOWING:

OBLCDAK (PA67), 1ENSW (PA65), ODAVAL (PA62) WHICH DRIVE 16MA LOW;

OBLD0 - OBLD7 (PA68, 66, 74, 64, 76, 61, 72, 70) WHICH DRIVE 32MA LOW.

1PA0 - 1PA3 (PA39, 41, 49, 47) AND 1ENSW (PA65) ARE INTENDED TO DRIVE 24 POINT CARDS UNOER THE FOLLOWING CONDITIONS: THE INPUT BUFFER ON THE POINT CARD MUST BE AN LS240, 13244 OR EQUIVALENT, (0.2MA INPUT LOAD FOR LOGIC LOW) WITH A 12.1K PULLUP RESISTOR FOR 1PA0 - 1PA3 AND 15K FOR 1ENSW.

AF 10527

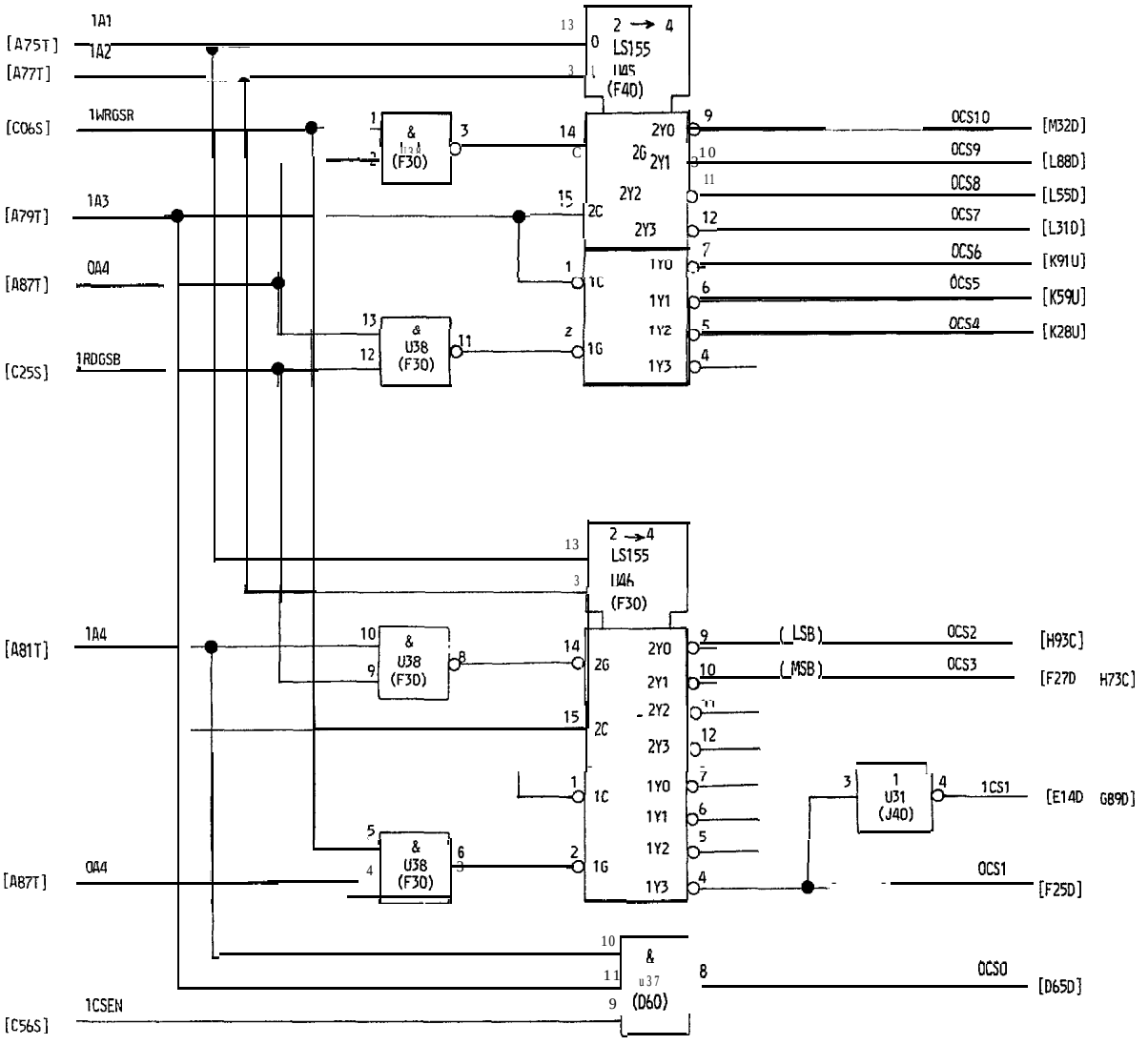


REV 1	REV 4	REV 5	PRINTS TO 2520 DL104 PRI	ENGINEER 5-5-83 C.N. Smoke	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM ANALOG CONVERSION
REV 2	ISSUED 83-05-12			FIRST MADE FOR MK IV SPEEDTRONIC		DS3800HAIC
REV 3	MADE BY R.A. SALIBRICI					CONT. ON SH. 4BA SH. NO. 4AA

CARS

AF 1052"

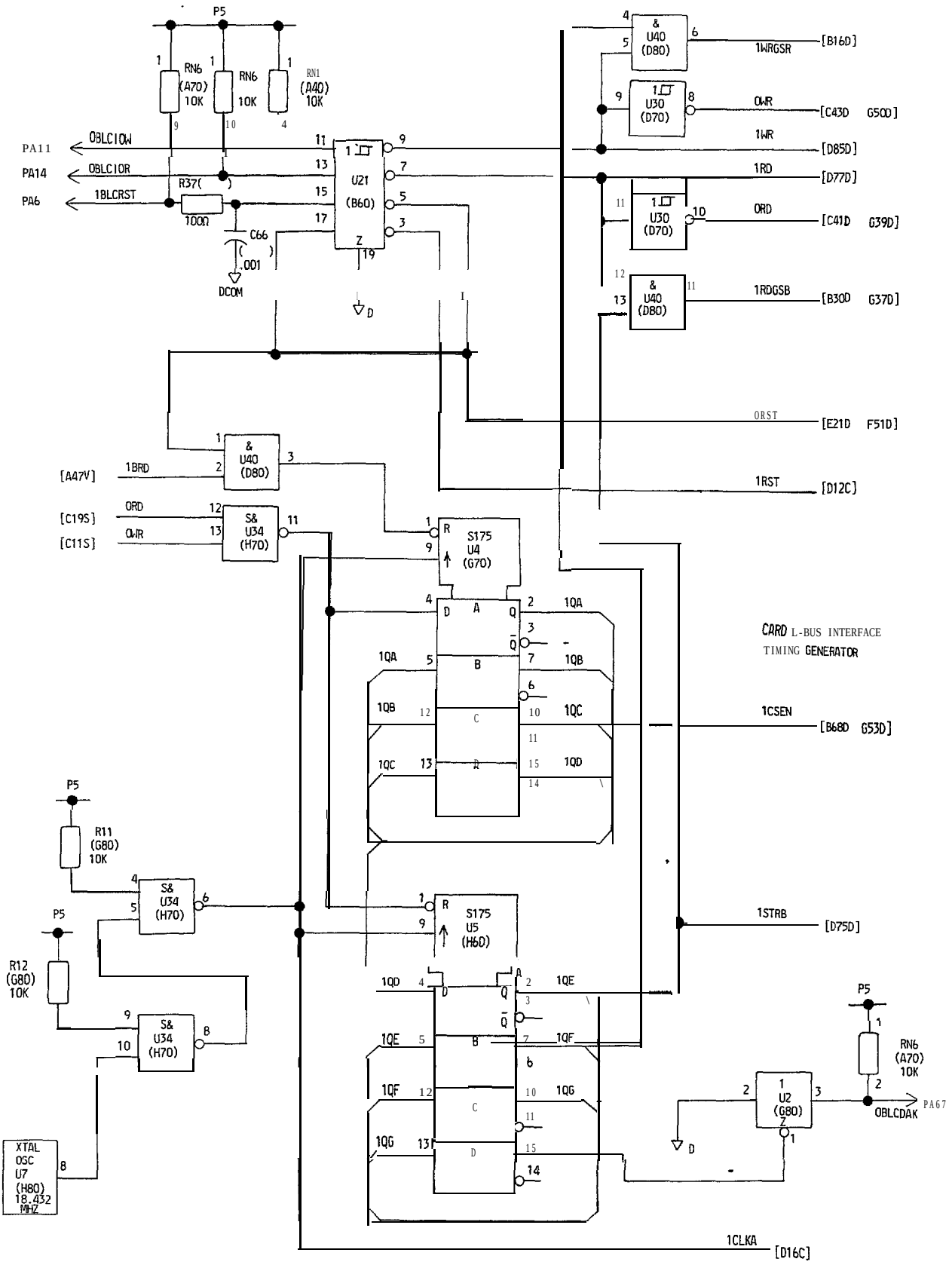
ADDRESS DECODE



1	REV 4	REV 5	PRINTS TO 2520 DL104 PRI	ENGINEER S 5-83 C.H Semeche	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.5.A	ELEMENTARY DIAGRAM ANALOG CONVERSION
REV 2	ISSUED 83-05-12					DS3800HAIC
REV 3	MADE BY R.A SALIBRICI 67					CONT. ON SH 4CA SH. NO 4BA

CARS

AF 10527



1	REV 4	REV 5	PRINTS TO 2520 DL104 PRI	ENGINEER 5-5-83 C.H Sweeke	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA U.S.A.	ELEMENTARY DIAGRAM ANALOG CONVERSION
REV 2	ISSUED 83-05-12					DS3800HAIC
REV 3	MADE BY R.A. SALIBRICI GT					CONT. ON SH. 4DASH. NO. 4C

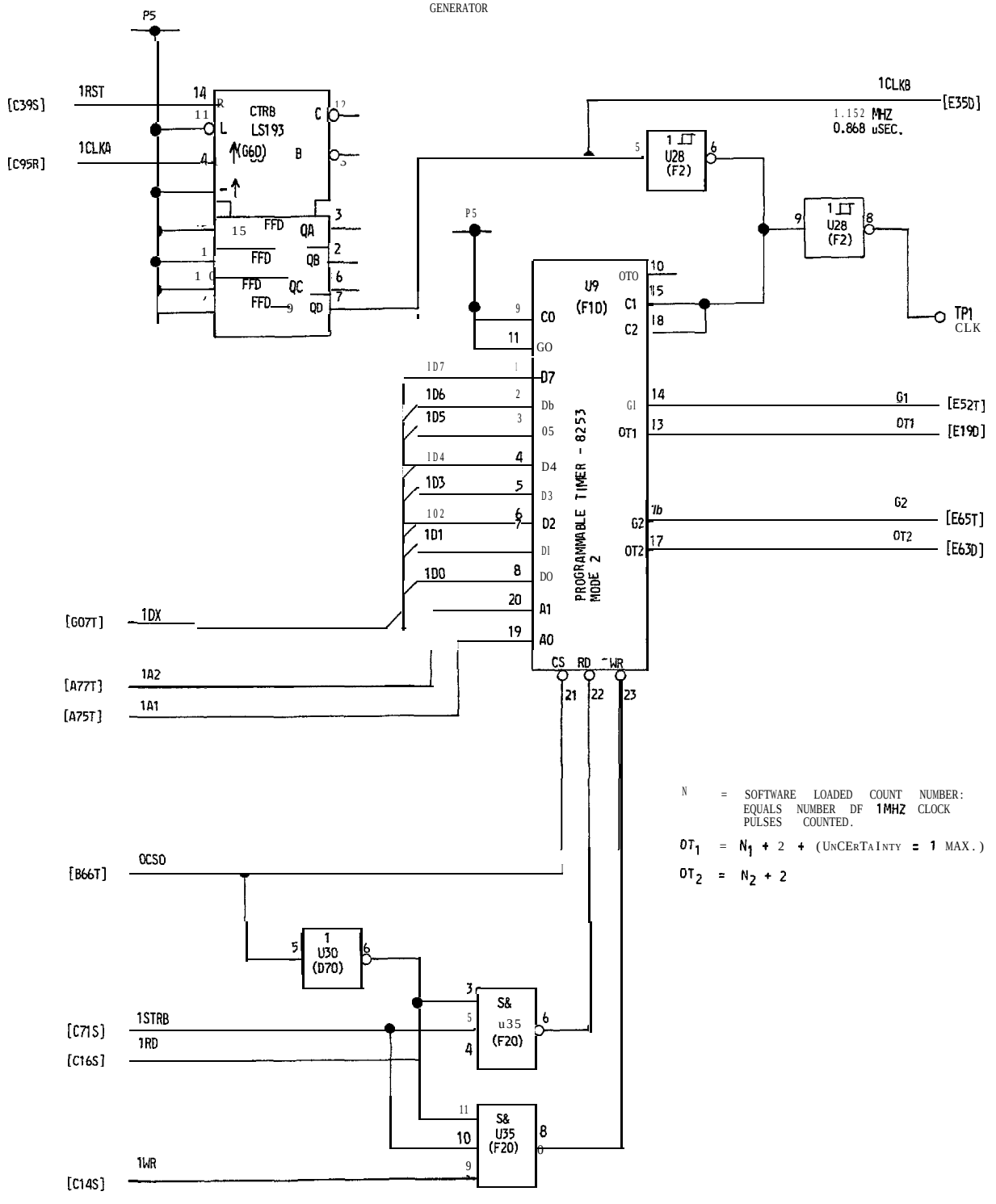
GSD 8153 (09 76)

CARS

AF 10527

A B C D E F G H J K L M N P Q R S T U V

ENABLE SWITCH AND
A/D CONVERTER DELAY
GENERATOR



N = SOFTWARE LOADED COUNT NUMBER: EQUALS NUMBER DF 1MHZ CLOCK PULSES COUNTED.

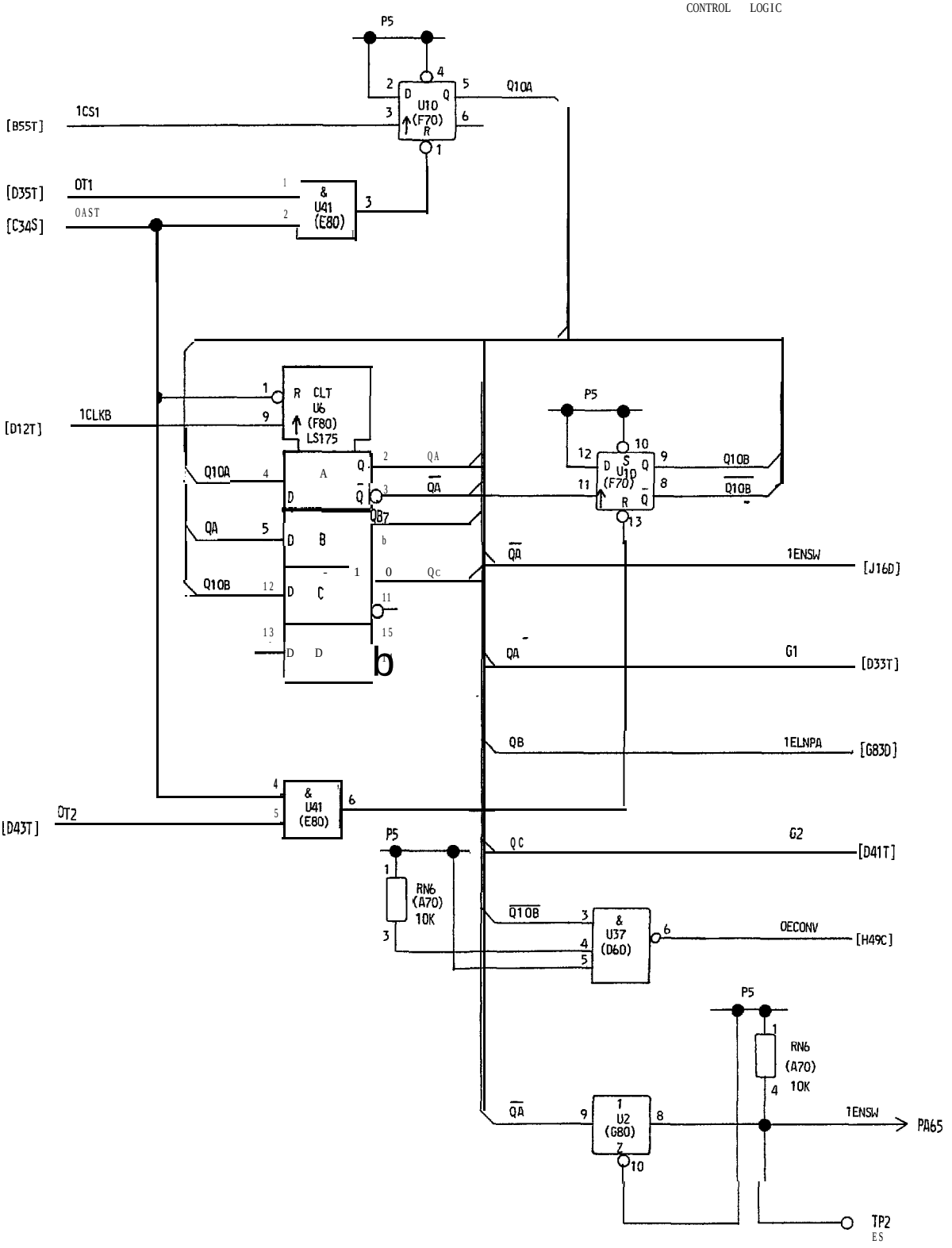
$OT_1 = N_1 + 2 + (\text{UNCERTAINTY} = 1 \text{ MAX.})$

$OT_2 = N_2 + 2$

1	REV 4	REV 5	PRINTS TO 2520 DL104 PRI	ENGINEER 5-5-83 C.A. Lemke	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U S.A.	ELEMENTARY DIAGRAM ANALOG CONVERSION
REV 2	ISSUED 83-05-12					DS3800HAIC
REV 3	MADE BY R.A. SALIBRICI					CONT. ON SH. 4EA SH. NO. 4DA

CARS

A/D CONVERSION
CONTROL LOGIC

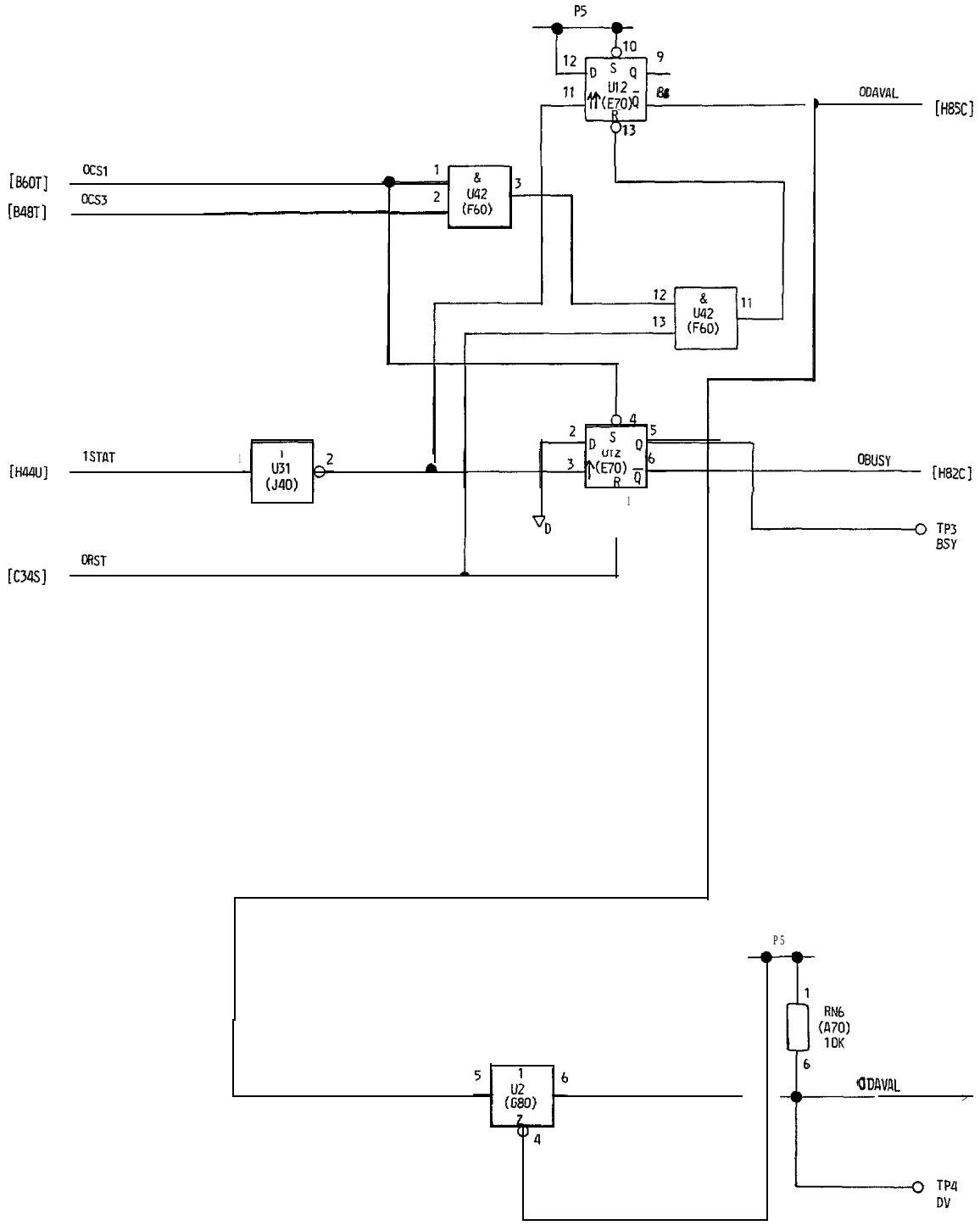


	REV 4	REV 5	PRINTS TO 2520 DL104 PRI	ENGINEER 5-5-83 C.H. Sweeke	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM ANALOG CONVERSION
REV 2	ISSUED 83-05-12					DS3800HAIC
REV 3	MADE BY R.A. SALIBRICI					CONT. ON SH. 4FA SH. NO. 4E

DSD 0152 (09 78)

CARS

A/D CONVERSION CONTROL LOGIC (CONTD.)



1	REV 4	REV 5	PRINTS TO 2500 DL104 PR1	ENGINEER 55 83 C.A. Sweede	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA U S A	ELEMENTARY DIAGRAM ANALOG CONVERSION
REV 2	ISSUED 83-05-12					DS3800HAIC
REV 3	MADE BY R.A. SALIBRICI GT					CONT. ON SH. 4GA SH. NO. 4F1

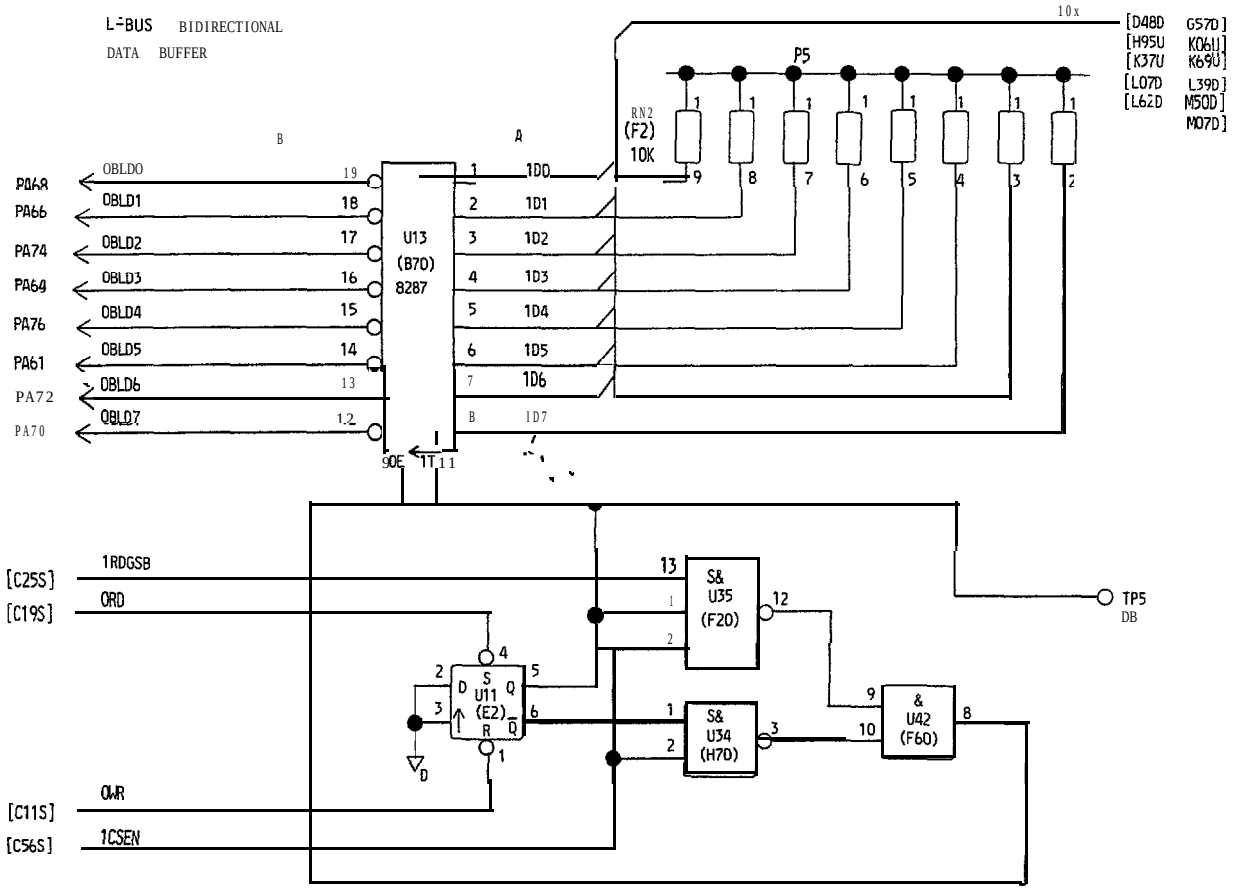
DSD 0153 (09 78)

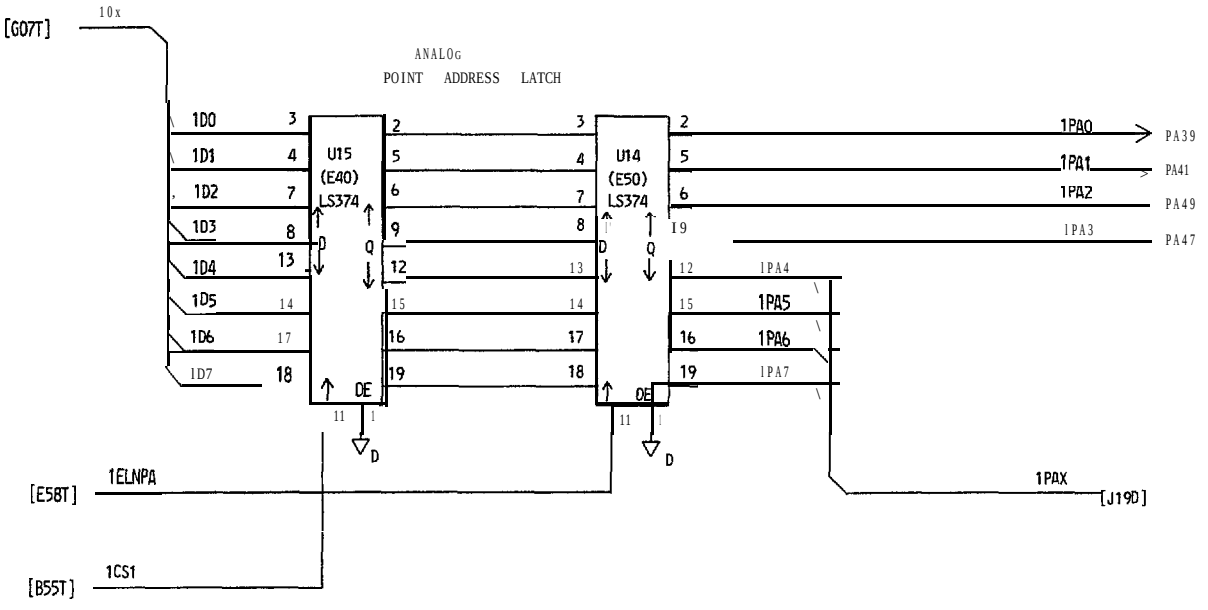
AF 10527

A 8 C D E F G H J K L M N P C? R S T U V

01 03 05 07 11 13 15 17 19 21 23 25 27 29 31 33 35 37 39 41 43 45 47 49 51 53 57 59 61 63 65 67 69 71 73 75 77 79 81 83 85 87 89 91 93 95 97

L-BUS BIDIRECTIONAL DATA BUFFER



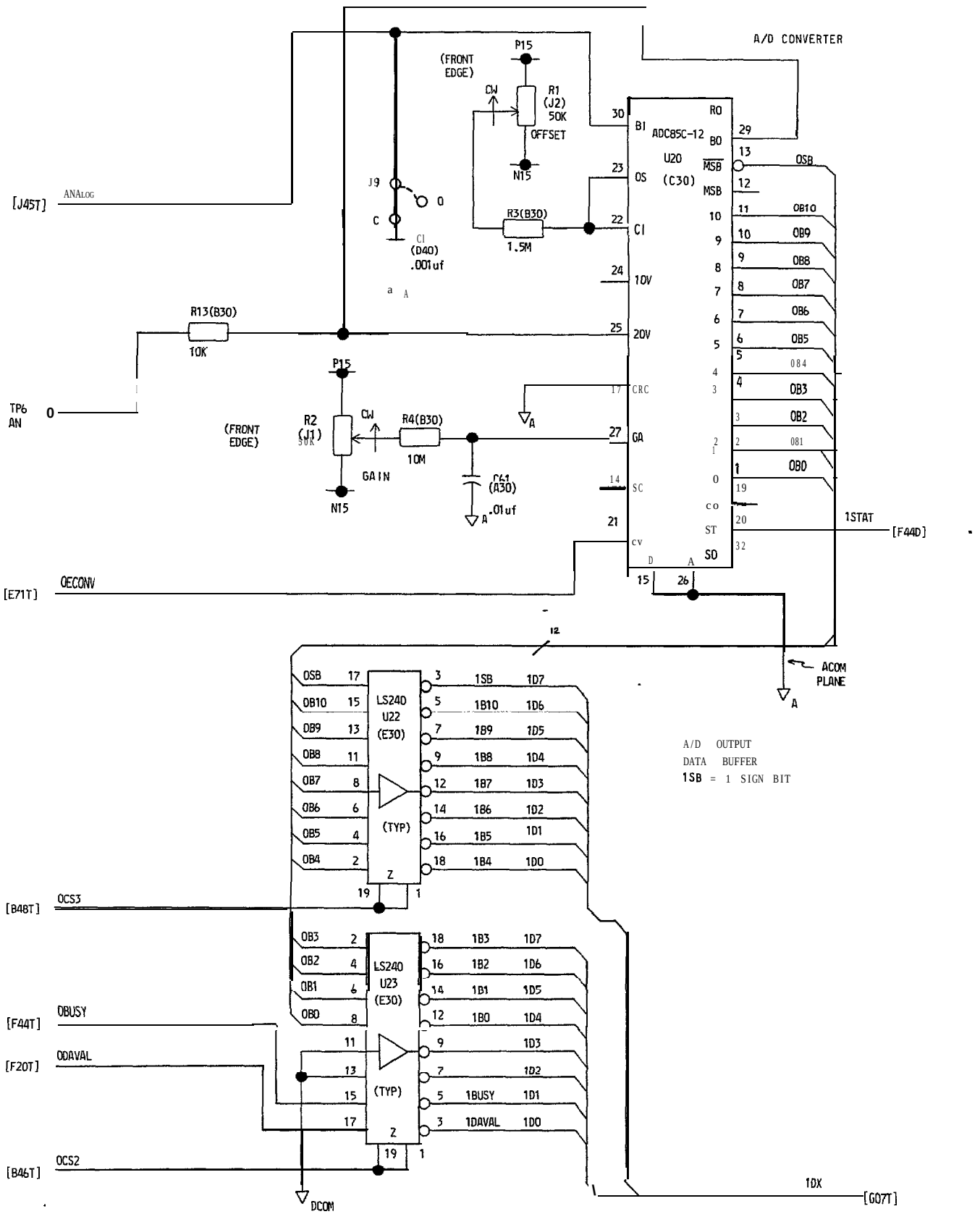


1	REV 4	REV 5	PRINTS TO 3520 DL104 PR1	ENGINEER 5-5-83 e.m. Swedro	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM ANALOG CONVERSION
REV 2	ISSUED 83-05-12					DS3800HAIC
REV 3	MADE BY R.A. SALIBRICI					CONT. ON SH. 4HASH. NO. 4G

OSD 0152 ID. 741

CAPS

AF 10527



1	REV 4	REV 5	PRINTS TO 2520 DL104 PRI	ENGINEER 5-5-83 R.A. Salibrici	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.
REV 2	ISSUED 83-05-12				
REV 3	MADE BY R.A. SALIBRICI				

ELEMENTARY DIAGRAM

ANALOG CONVERSION

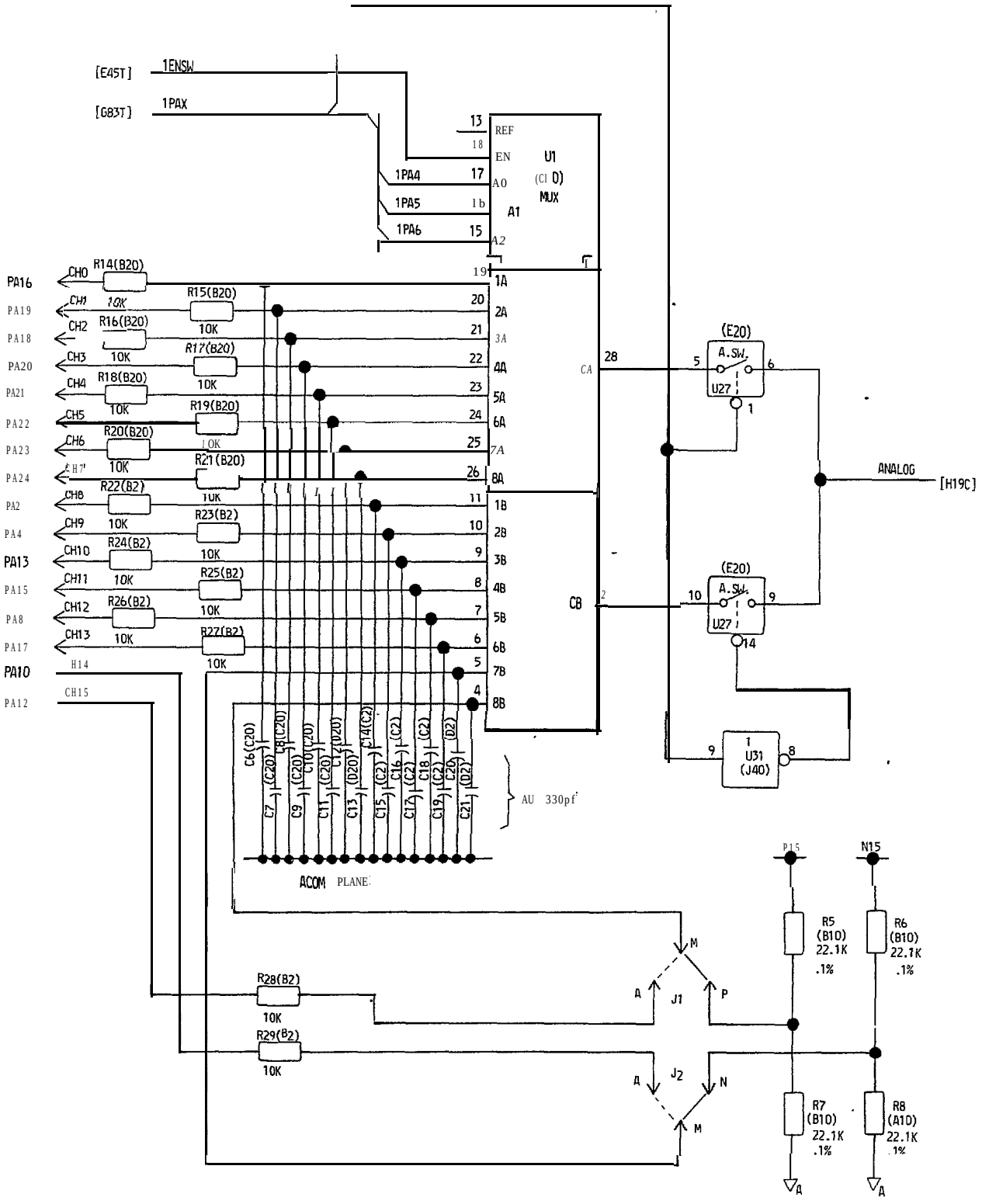
DS3800HAIC

CONT. ON SH. 4JA SH. NO. 4H

DSD 0153 (09 74)

AR 10527

ANALOG MULTIPLEXER



	REV 4	REV 5	PRINTS TO 2520 DL104 PR1	ENGINEER 5-5-83 C.H. Sweeley	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.
REV 2	ISSUED 83-05-12				
REV 3	MADE BY R.A. SALIBRICI				

ELEMENTARY DIAGRAM

ANALOG CONVERSION

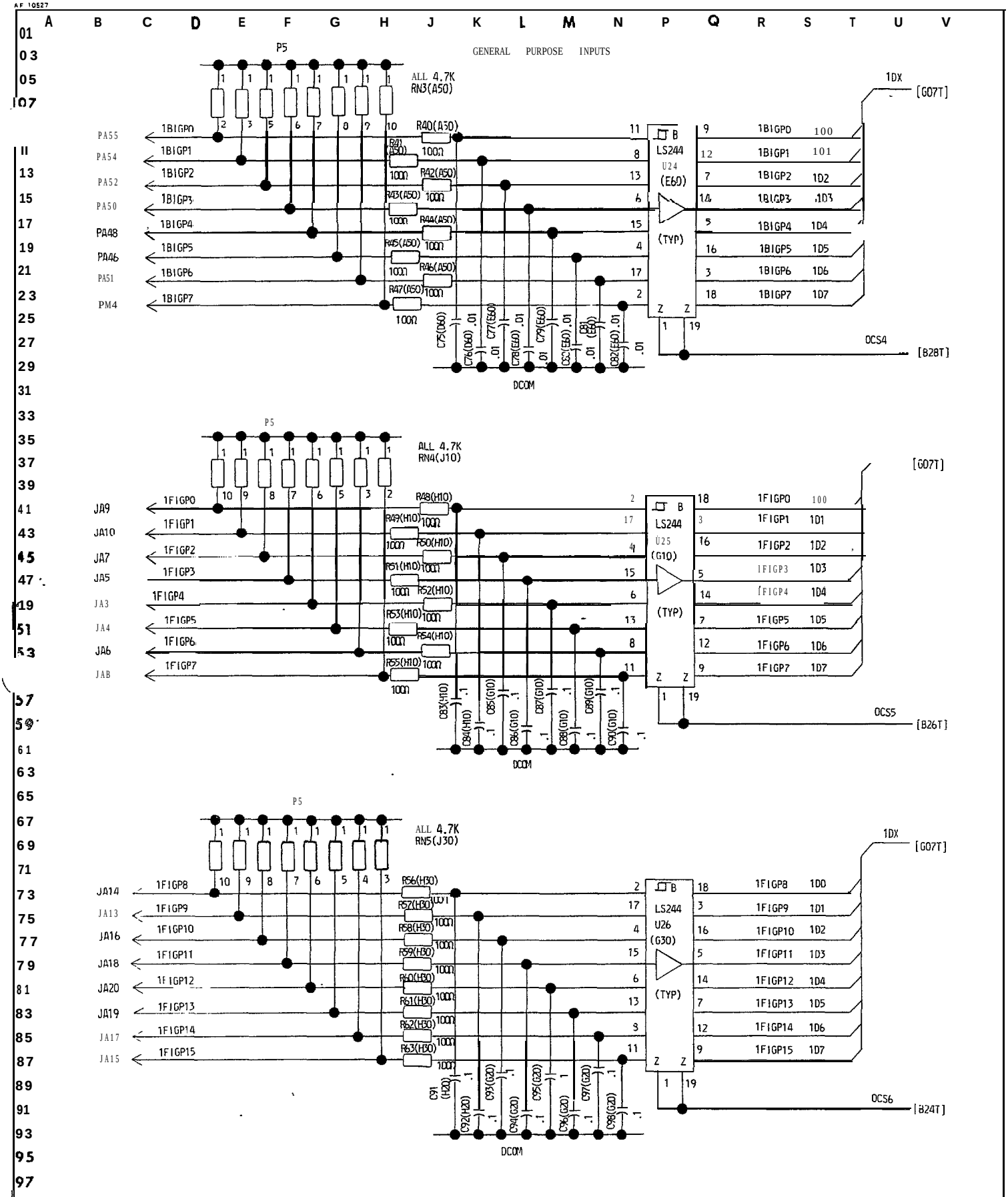
DS3800HAIC

CONT. ON SH. 4KA SH. NO. 4JA

CARS

DSD 0157 109 761

AF 10527



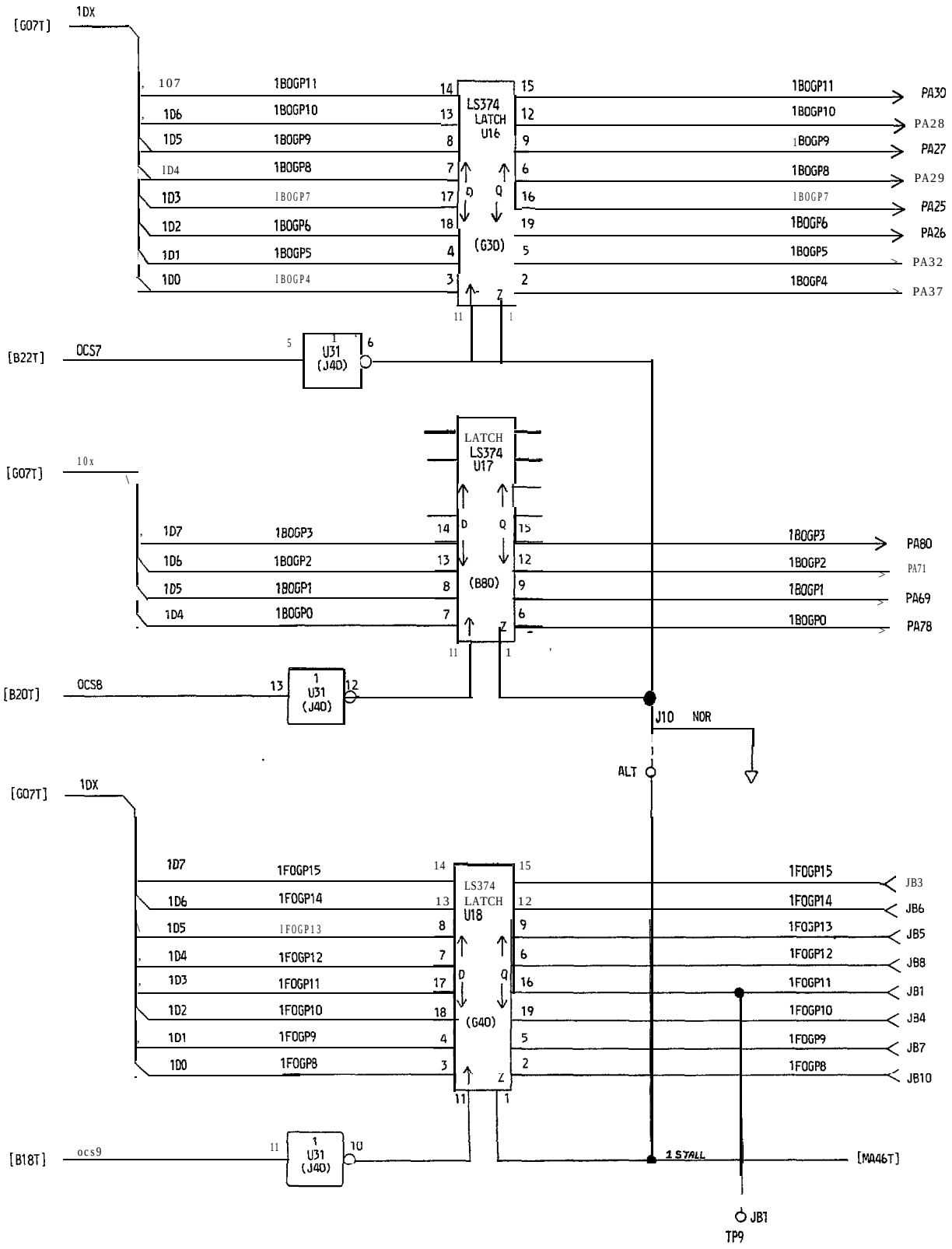
1	REV 4	REV 5	PRINTS TO DL104 PRI	ENGINEER 5-5-83	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA U.S.A.	ELEMENTARY DIAGRAM ANALOG CONVERSION DS3800HAJC CONT. ON SH. 4LA SH. NO 4KA
REV 2	ISSUED 83-05-12					
REV 3	MADE BY R.A. SALIBRICI					

AF 10527

A B C D E F G H J K L M N P Q R S T U V

01 03 05 07 11 13 15 17 19 21 23 25 27 29 31 33 35 37 39 41 43 15 47 19 51 53 57 59 61 63 65 67 69 71 73 75 77 79 81 83 85 87 89 91 93 95 97

GENERAL PURPOSE OUTPUTS



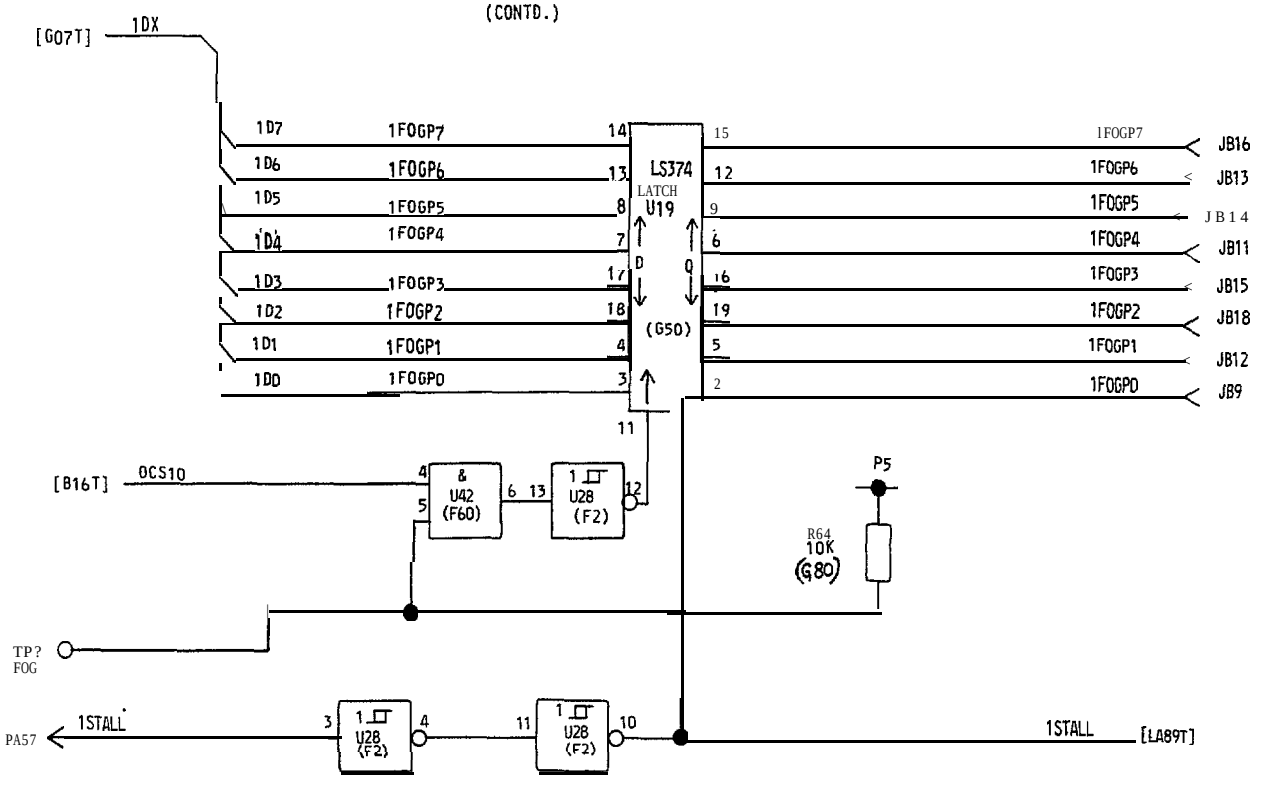
1	REV 4	REV 5	PRINTS TO 2520 DL104 PRI	ENGINEER 5-5-83 C.N. Sumalee	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U S.A	ELEMENTARY DIAGRAM ANALOG CONVERSION
REV 2	ISSUED 83-05-12					DS3800HAIC
REV 3	MADE BY R.A SALIBRICI					CONT. ON SH. 4MA SH. NO. 4L

DSD 0152 (09 76)

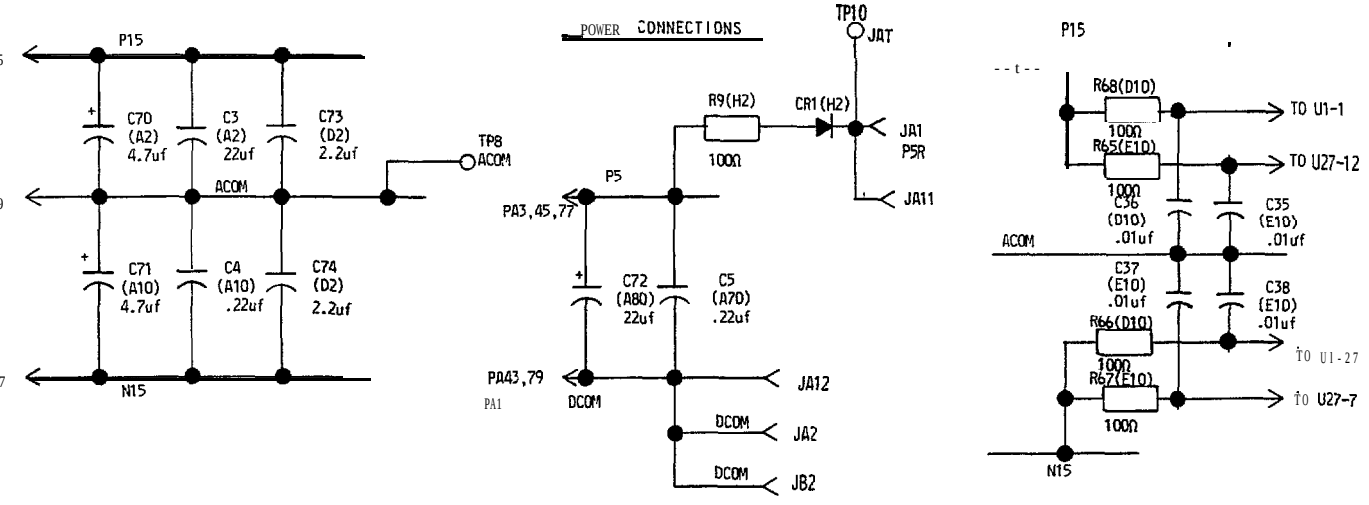
CARS.

AF 10527

GENERAL PURPOSE OUTPUTS
(CONTD.)



POWER CONNECTIONS



V 1	REV 4	REV 5	PRINTS TO 2520 DL104 PR1	ENGINEER 5-5-83 C.H. Sweeley	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV 2	ISSUED 83-05-12				DRIVE SYSTEMS DEPARTMENT	ANALOG CONVERSION
REV 3	MADE BY R.A. SALIBRICI				SALEM, VA. U.S.A.	DS3800HAIC
						CONT. ON SH. 4 SH. NO. 4MA

DSD 0152 (09 76)

CARS

A B C D E F G H J K L M N P Q R S T U V

01 03 05 07 09 11 13 15 17 19 21 23 25 27 29 31 33 35 37 39 41 43 45 47 49 51 53 55 57 59 61 63 65 67 69 71 73 75 77 79 81 83 85 87 89 91 93 95 97

POWER CONNECTIONS TO IC'S

DCOM TO PIN 7
P5 TO PIN 14 : U2,U7,U10-U12,U28-U31, U33-U35,U37,U38,U40-U42

DCOM TO PIN 8
P5 TO PIN 16 . U4-U6,U45,U46,U8

DCOM TO PIN 10
P5 TO PIN 20 . U13-U19,U21-U26

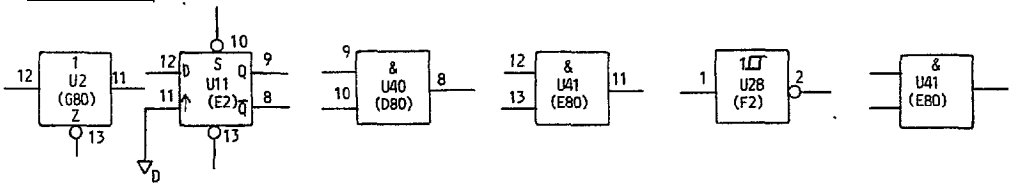
IN ADDITION:
P5 TO PIN 16 OF U20, & PIN 24 OF U9
DCOM TO PIN 12 OF U9

P15
U20-PIN 28

N15 ACOM
U20-PIN 31 U1-PIN 12
U20-PINS 26.15
U27-PIN 3

SPECIAL
U1 - SEE SH. 4MA ZONE 85T
U27- SEE SH. 4MA ZONE 85T

SPARE IC'S



FILTER CAPACITANCE ON POWER BUSES

THE FOLLOWING CHIPS MUST HAVE THE LISTED CAPACITANCE PLACED AS CLOSE TO THEM AS IS PHYSICALLY POSSIBLE. P5 TO DCOM.

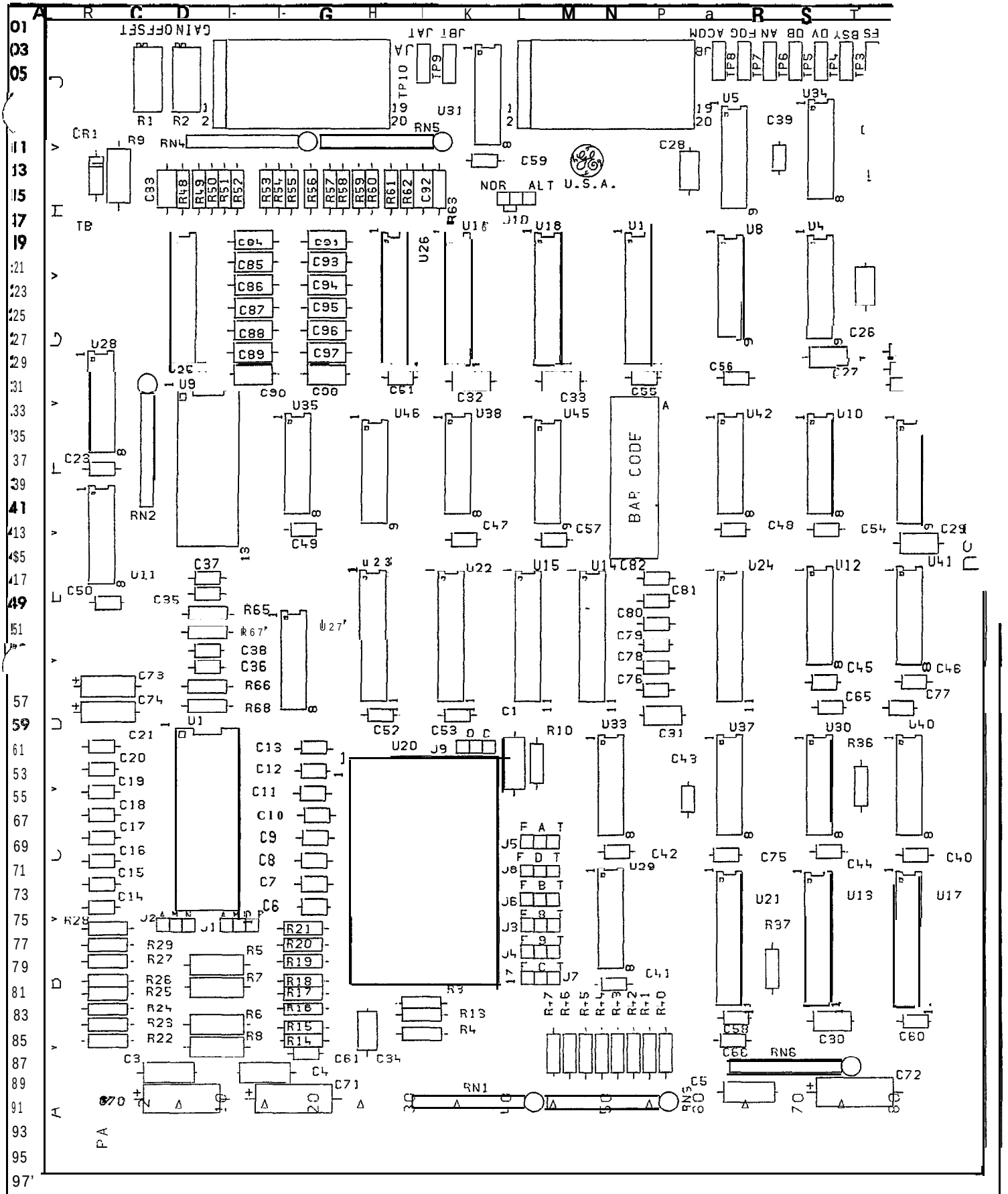
U2,U4,U5,U6,U13,U14,U16,U18,U20 (C26-C34) (.1uf)
U7,U28 (C22,C23) (.001uf)
U34,U40 (C39-C40) (.01uf)

IN ADDITION, DISTRIBUTE 20 (.01uf) 68A091PG10Z CAPACITORS ACROSS THE BOARD THROUGHOUT THE LOGIC, P5 TO DCOM. (C41-C60)

NOTE; TA * - R : SUFFIX "R" MEANS "READ (THIS POINT) ONLY"
TA * - W : SUFFIX "W" MEANS "WRITE (LOGIC LEVELS) TO THIS POINT"

1	REV 4	REV 5	PRINTS TO 2520 DL104 PRI	ENGINEER 5-5-83 C.H. Swabe	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM ANALDG CONVERSION
REV 2	ISSUED 83-05-12					DS3800HAIC
REV 3	MADE BY GT R.A. SALIBRICI					CONT. ON SH.4PA SH. NO.4NA

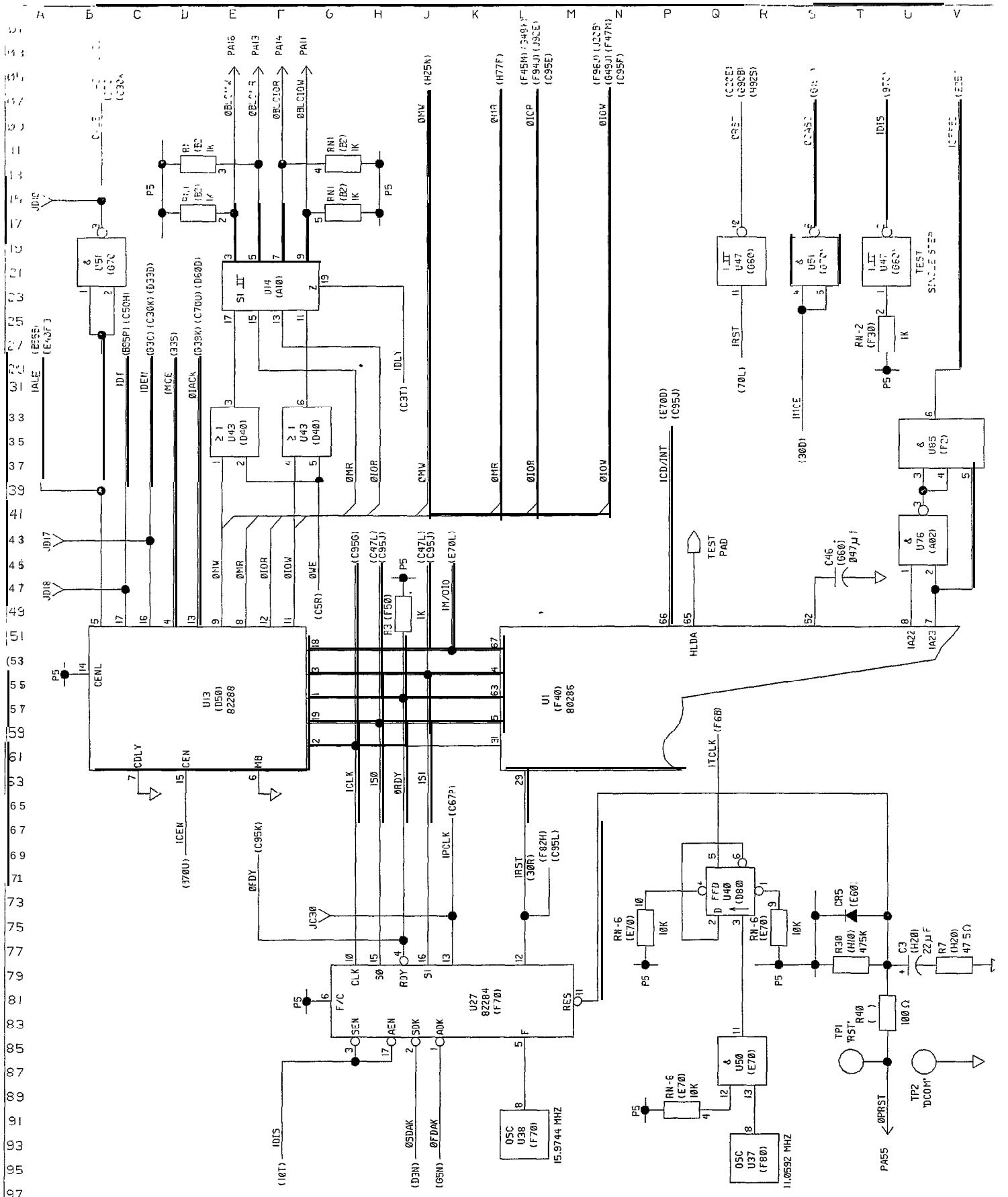
OSD 0152 (06 76)



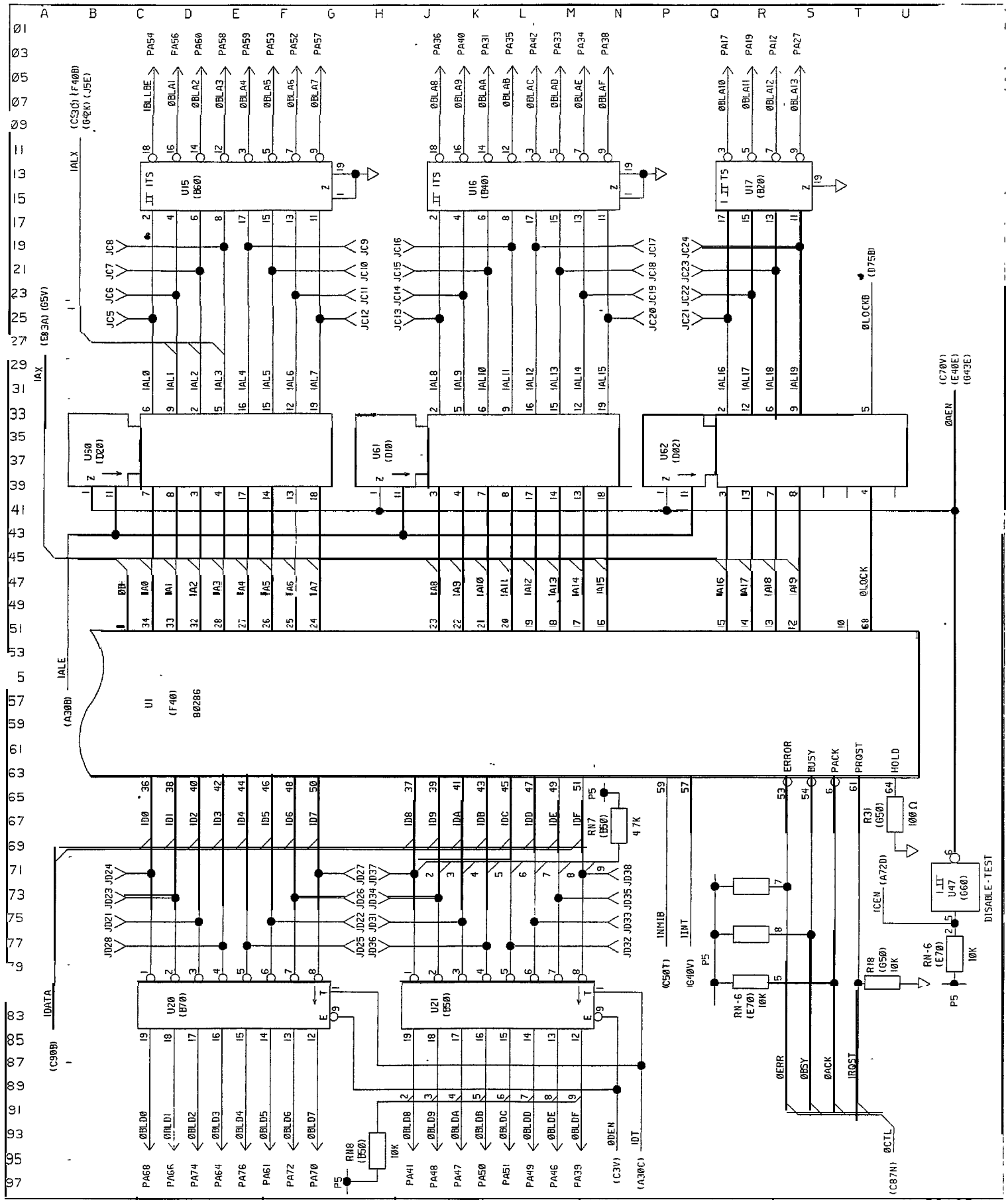
	REV.4	REV 5	PRINTS TO DL104 PRI	APPROVALS WLL	GENERAL ELECTRIC	ELEMENTARY DIAGRAM ANALOG CONVERSION
REV 2	ISSUED 83-05-12			FIRST MADE FOR	DRIVE SYSTEMS DEPT. SALEM, VA., U.S.A.	DS3800HAIC
REV.3	MADE BY LUNSFORD			TC. NO		CONT. ON SH. FL SH. NO. 4 PA

FORM IC 2349 (04-76)

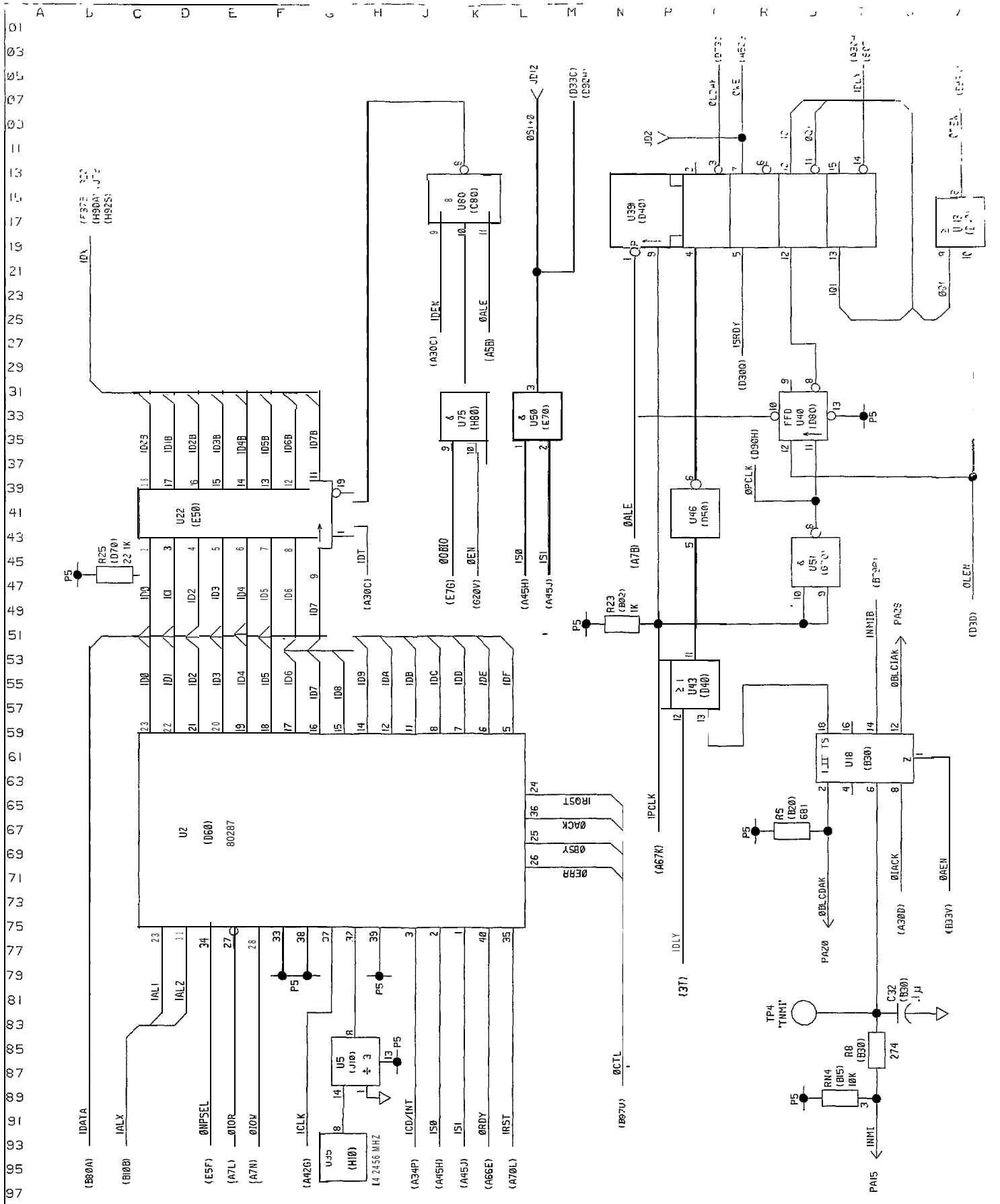
CARS



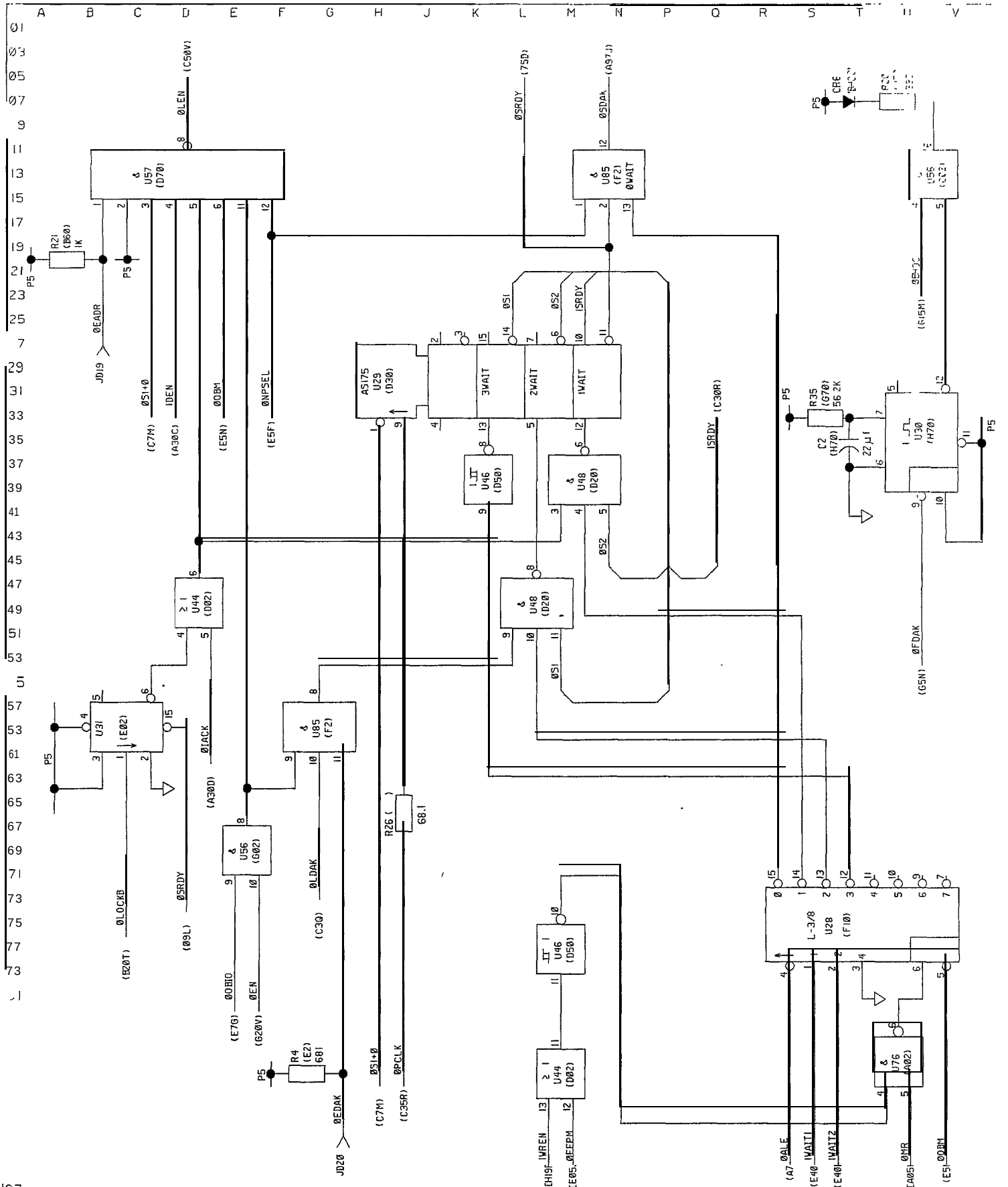
REV.1 ØEK MLO 860530	REV 4	REV 5	PRINTS TO DL104	ENGINEER	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM CPU
REV 2 OET CWF	ISSUED			FIRST MADE FOR 880518.	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	DS3800HMPK
REV.3	MADE BY E.DILLON			I.C NO		CONT.ON SH. 4 BA SH.NO. 4A



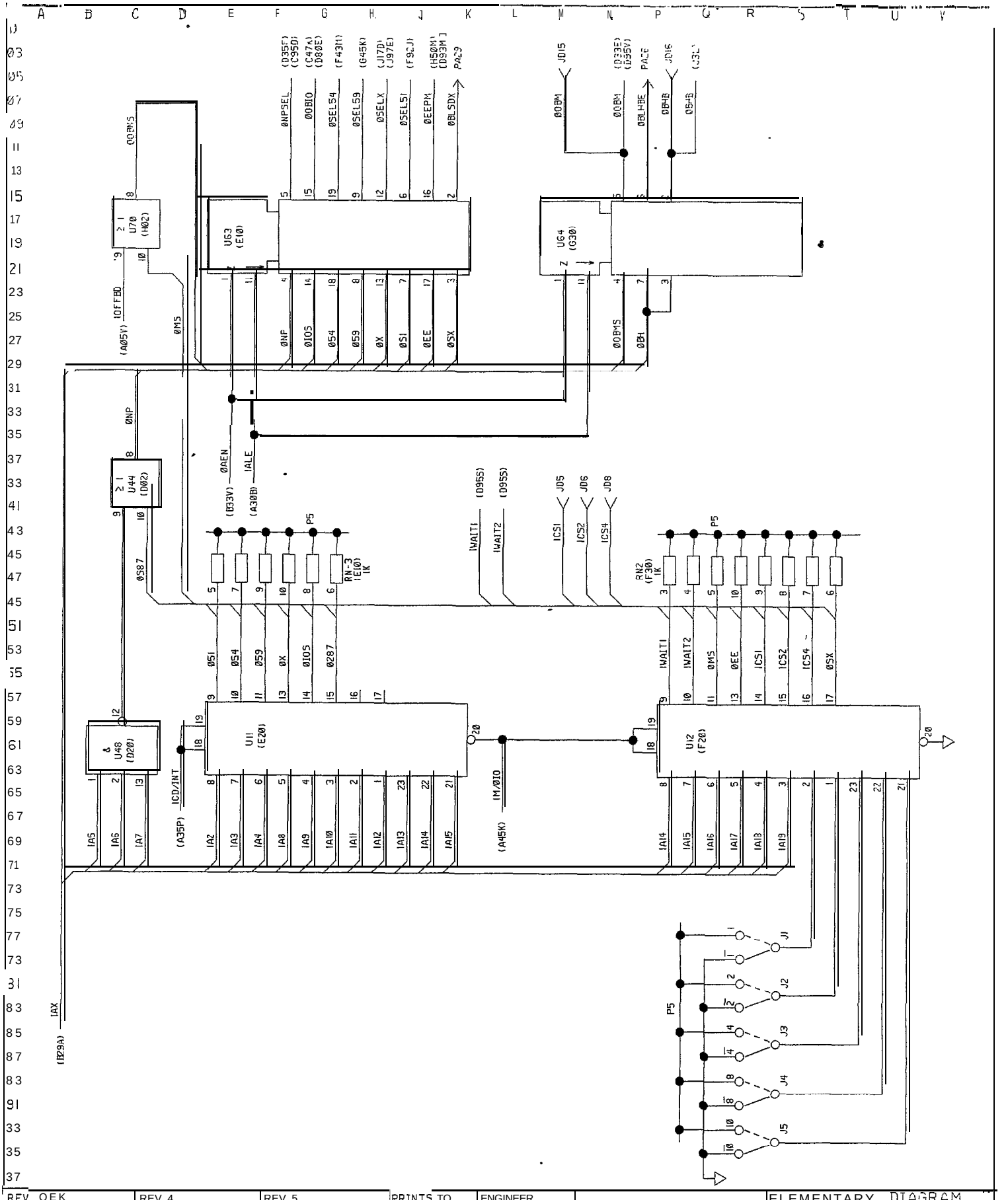
REV.10EF	REV 4	REV 5	PRINTS TO DL104 R1	ENGINEER EJD	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM CPU
REV 2	ISSUED May 23, 1986			FIRST MADE FOR	DRIVE SYSTEMS DEPT SALEM, VA. U.S.A.	DS3800HMPK
REV.3	MADE BY E.DILLON			I.C NO.		CONT.ON SH. 4CA SH NO 45



REV.1 OET CwF 880518	REV 4	REV 5	PRINTS TO DL104	ENGINEER	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM CPU
REV 2	ISSUED			FIRST MADE FOR		
REV.3	MADE BY E.DILLON			I C.NO	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	DS3800HMPK
						CONT.ON SH. 4DA SH.NO. 4C



'EV I OEK MLO 860930	REV 4	REV 5	PRINTS TO DL104 PRI	ENGINEER ED.	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM CPU
REV 2	REISSUED OCT 24,1986			FIRST MADE FOR		DS3800HMPK
REV.3.	MADE BY E.DILLON			I.C.NO.	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	CONT.ON SH. 4EA SH.NO. 4D5

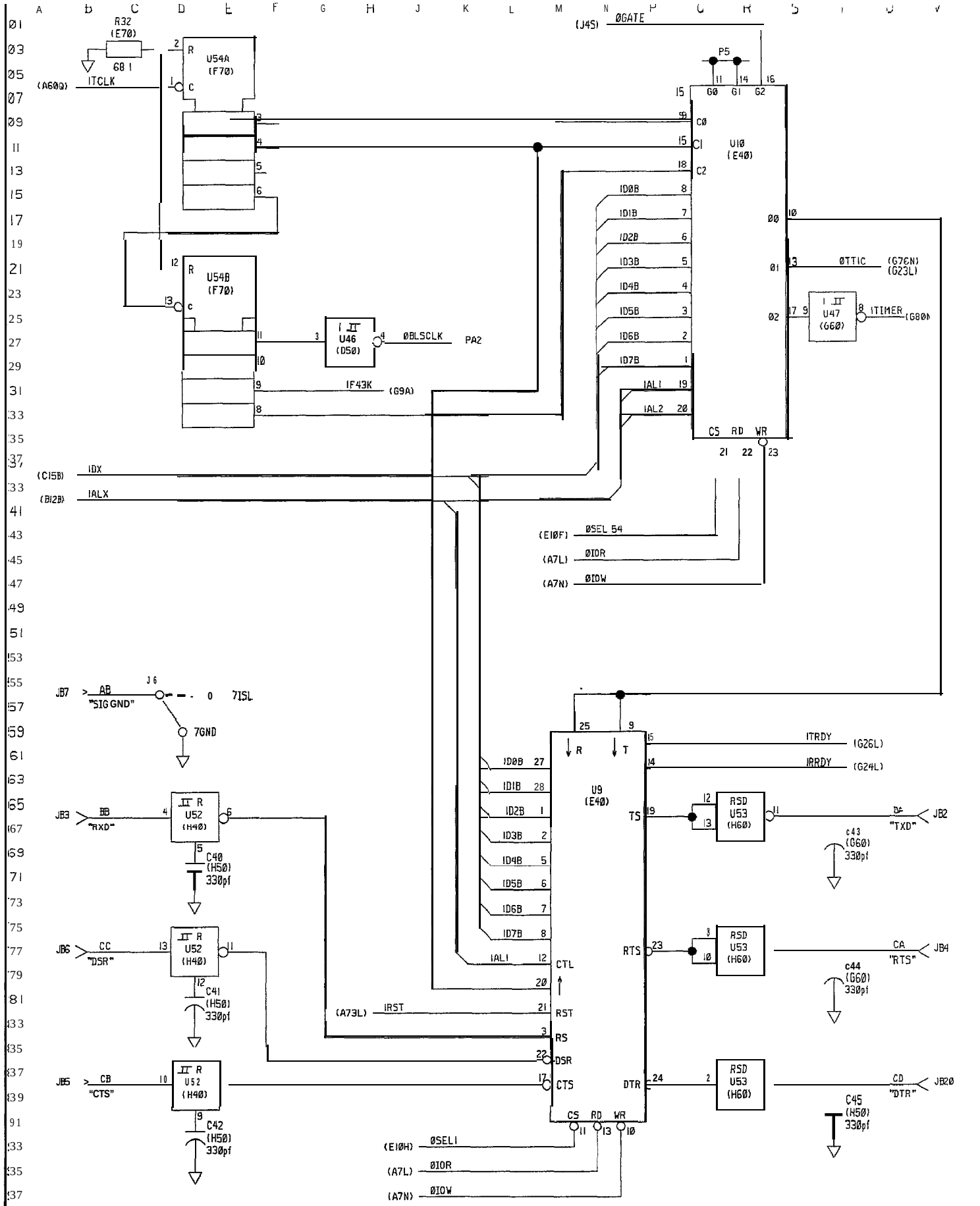


REV OEK MLO 860930	REV 4	REV 5	PRINTS TO DL104 FRI	ENGINEER E.D	GENERAL ELECTRIC COMPANY
REV.2	RE-ISSUED OCT 24, 1986			FIRST WAFER	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.
REV.3	MADE BY E.DILLON			I.C.NO	

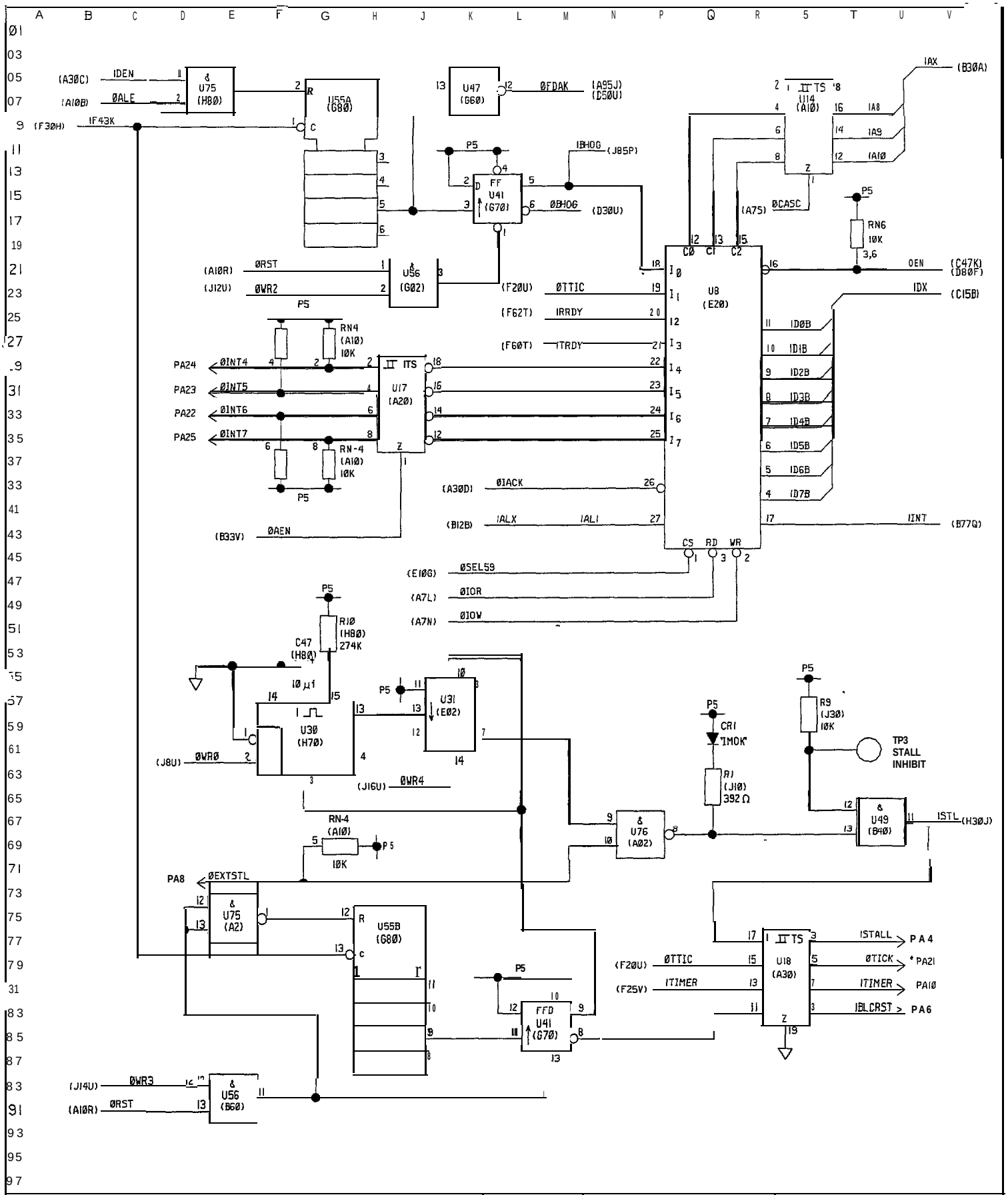
ELEMENTARY DIAGRAM
CPU

DS3800HMPK

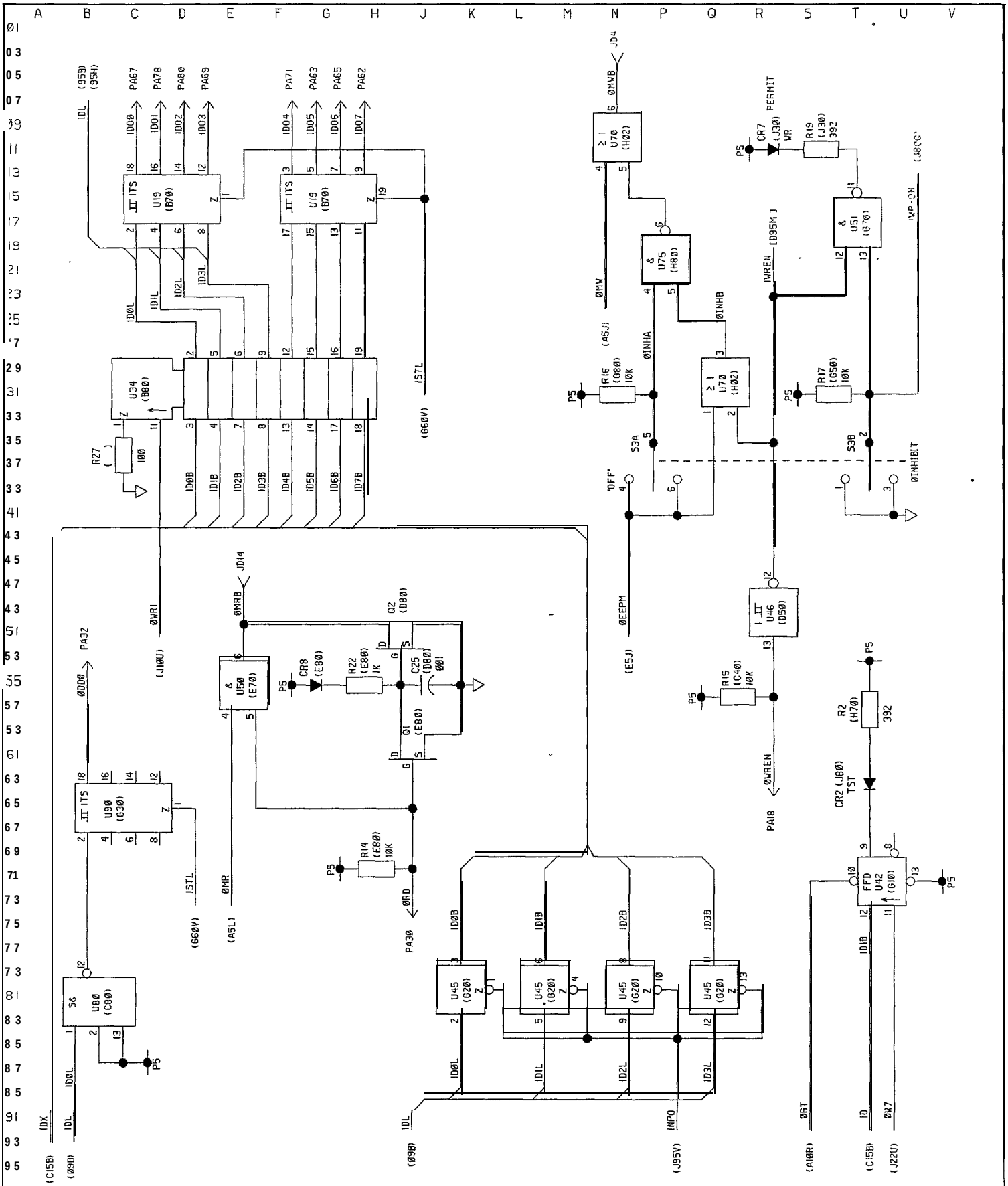
CONT.ON SH. 4FA SH.NO. 4EA



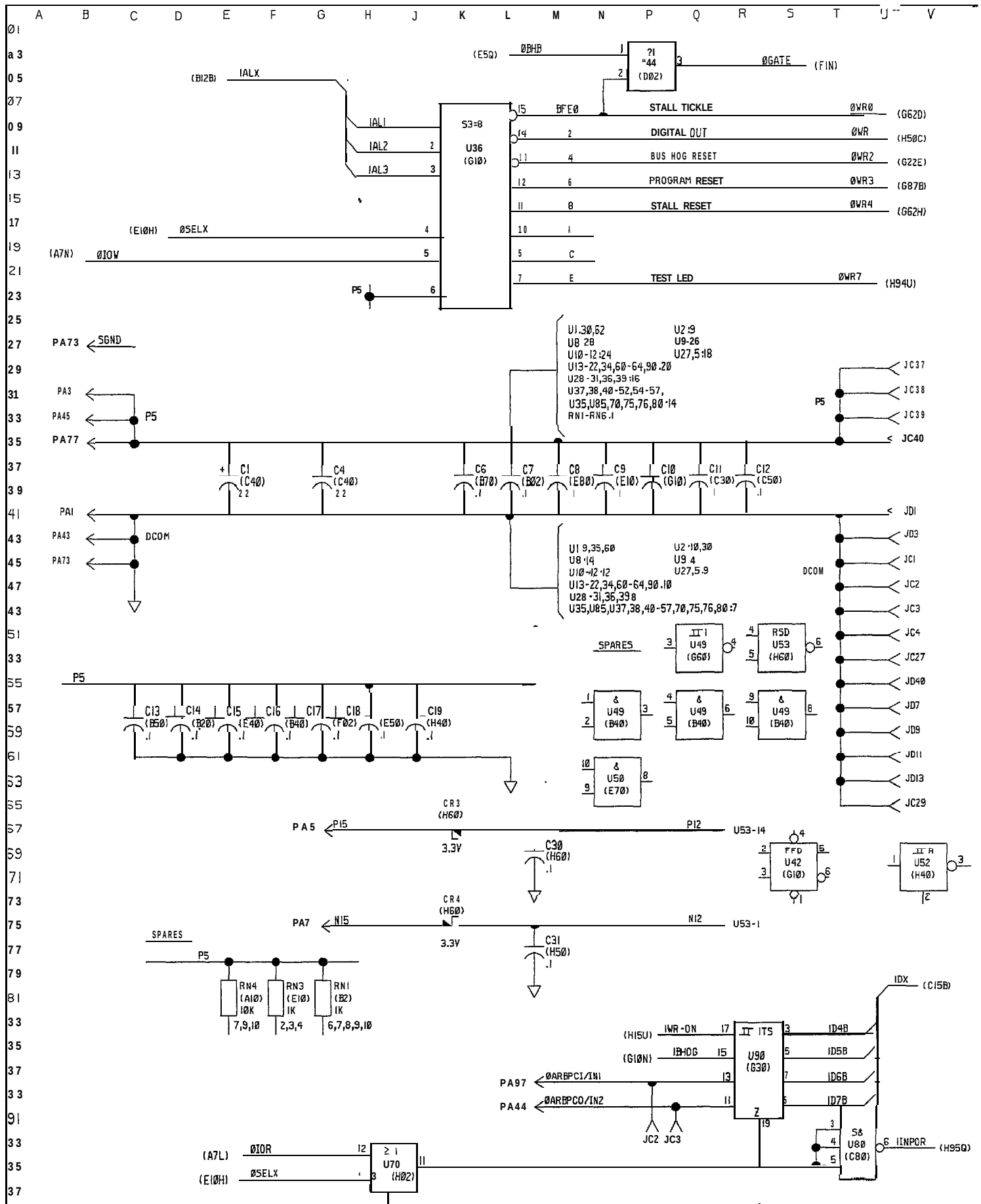
REV.1	REV4	REV.5	PRINTS TO DL104 PRI	ENGINEER EJD	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM CPU
REV.2	ISSUED April 22, 1986			FIRST MADE FOR		DS3800HMPK
REV.3	MADE BY E.DILLON			I.C.NO.	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	CONT.ON SH.4GA SH.NO.4F



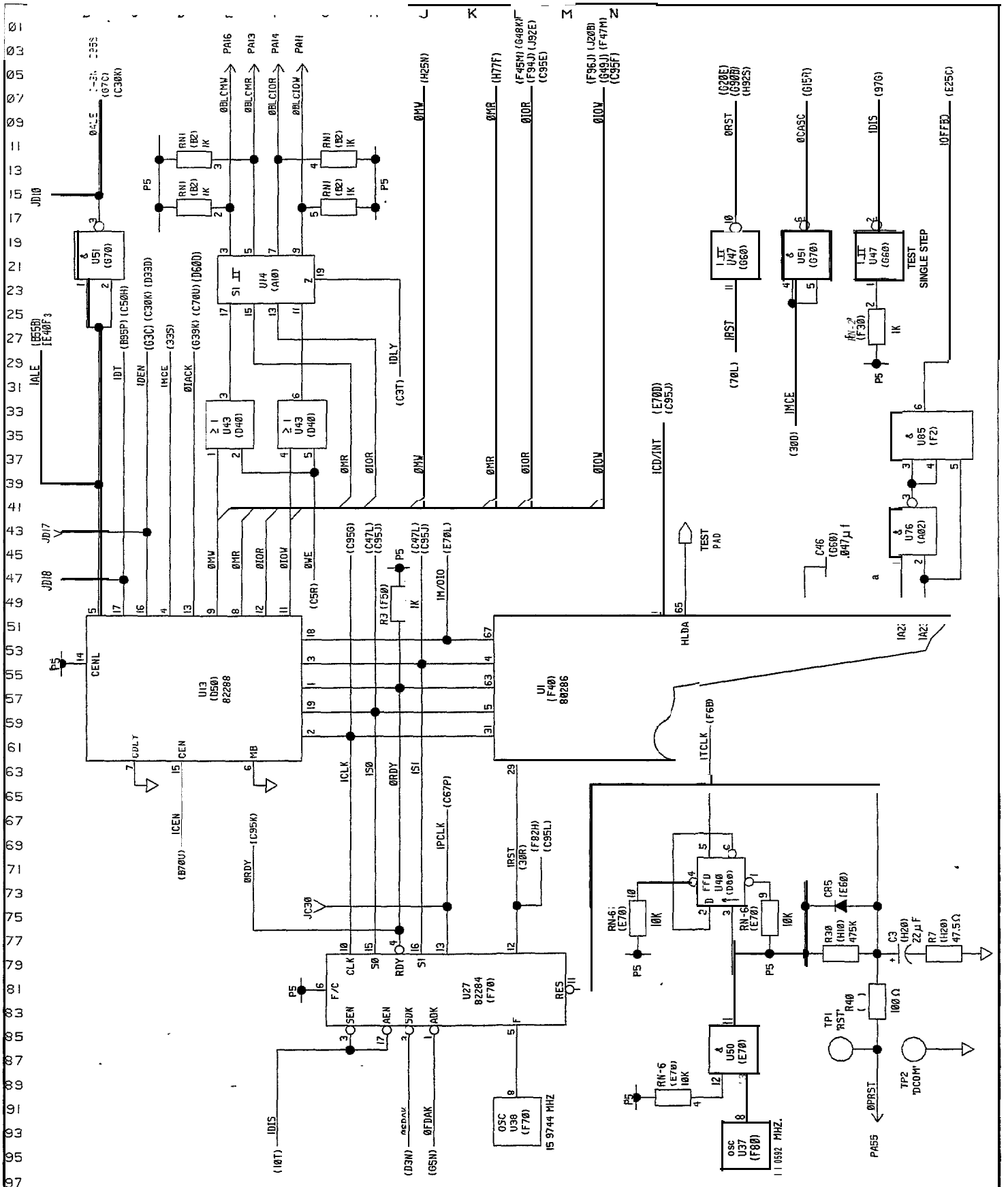
REV.1 OEW MAC 890116	REV.4	REV.5	PRINT5 TO DL104 prl	ENGINEER	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM CPU
REV 2	ISSUED			FIRST MADE FOR		
REV.3	MADE BY E.DILLON			I C.NO	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	DS3800HMPK CONT.ON SH. 4HA SH NO.4GA



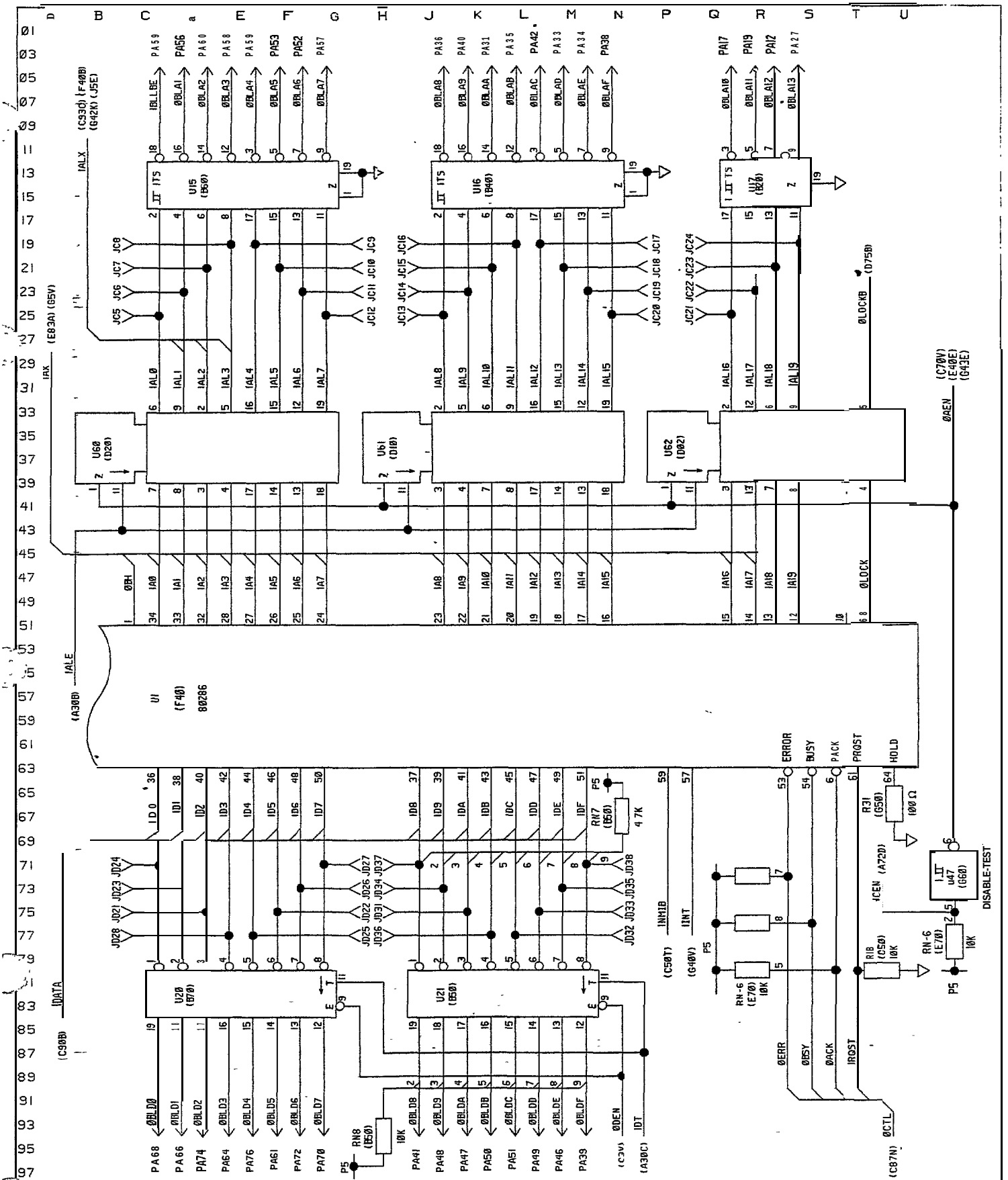
REV.1 OEK MLO 860930	REV 4	REV 5	PRINTS TO DL104 PRI	ENGINEER E O.	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM CPU
REV.2	REISSUED OCT 24, 1986			FIRST MADE FOR		
REV.3	MADE BY E.DILLON			I.C.NO.	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	DS3800HMPK CONT.ON SH. 4JA SH.NO. 4HA



REV I OEK MLO 860938	REV.4	REV.5	PRINT5 TO DL104 pr1	ENGINEER	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM CPU
REV.2 OEW MAC 890116	ISSUED			FIRST MADE FOR	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	DS3800HMPK
REV.3	MADE BY E.DILLON			I C NO.		CONT.ON SH. 4KA SH.NO.4JA

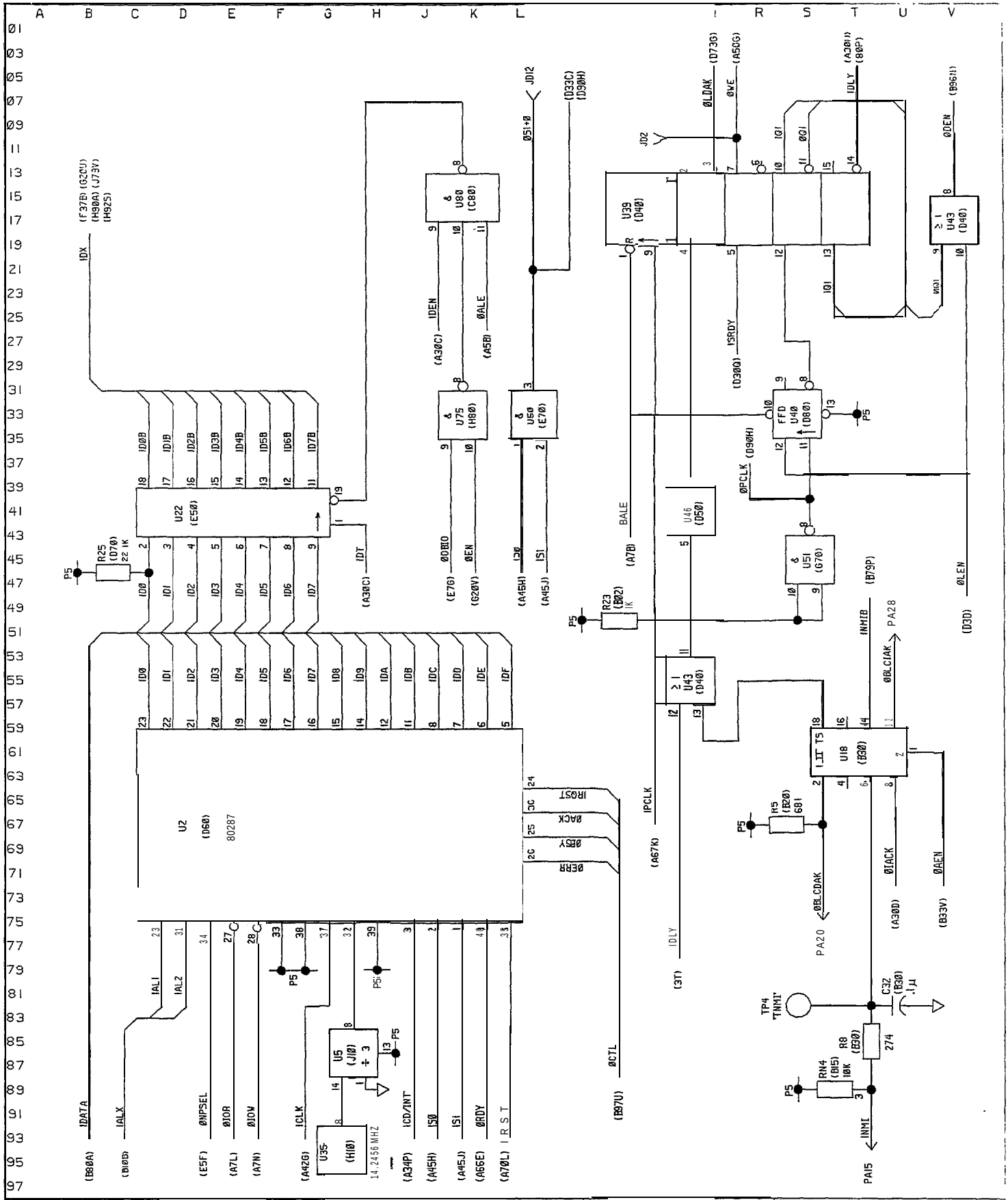


REV.1 ØEK MLJ 860930	REV.4	REV.5	PRINTS TO DL104	ENGINEER	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM CPU
REV.2 OET GWF 880518	ISSUED			FIRST MADE FOR	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	DS3800HMPK
REV.3	MADE BY E.DILLON			I.C.NO.		CONT.ON SH. 4BA SH.NO. 4AA



EV.1 DEF 1L0 860429	REV.4	REV.5	PRINTS TO DL104 PRI	ENGINEER EJP	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM CPU
REV.2	ISSUED May 23 1986			FIRST MADE FOR		
REV.3	MADE BY E.DILLON			I.C.NO.	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	DS3800HMPK
						CONT.ON SH.4CA SH.NO.4BA

PARS



REV.1 OET CwF 880518	REV 4	REV.5	PRINTS TO DL104	ENGINEER
REV.2	ISSUED			FIRST MADE FOR
REV.3	MADE BY E.DILLON			I.C.NO.

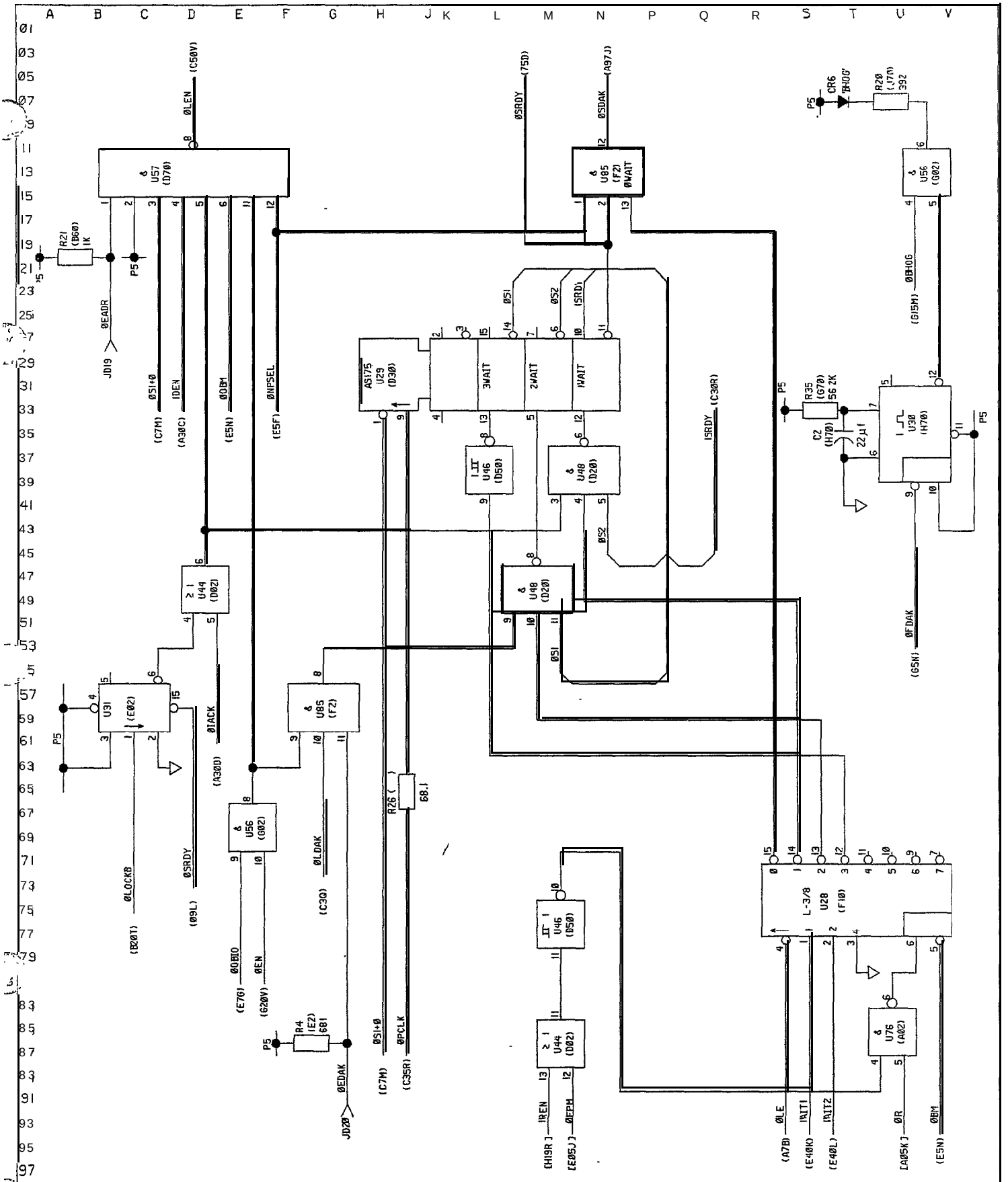
GENERAL ELECTRIC COMPANY

DRIVE SYSTEMS DEPT.

SALEM, VA. U.S.A.

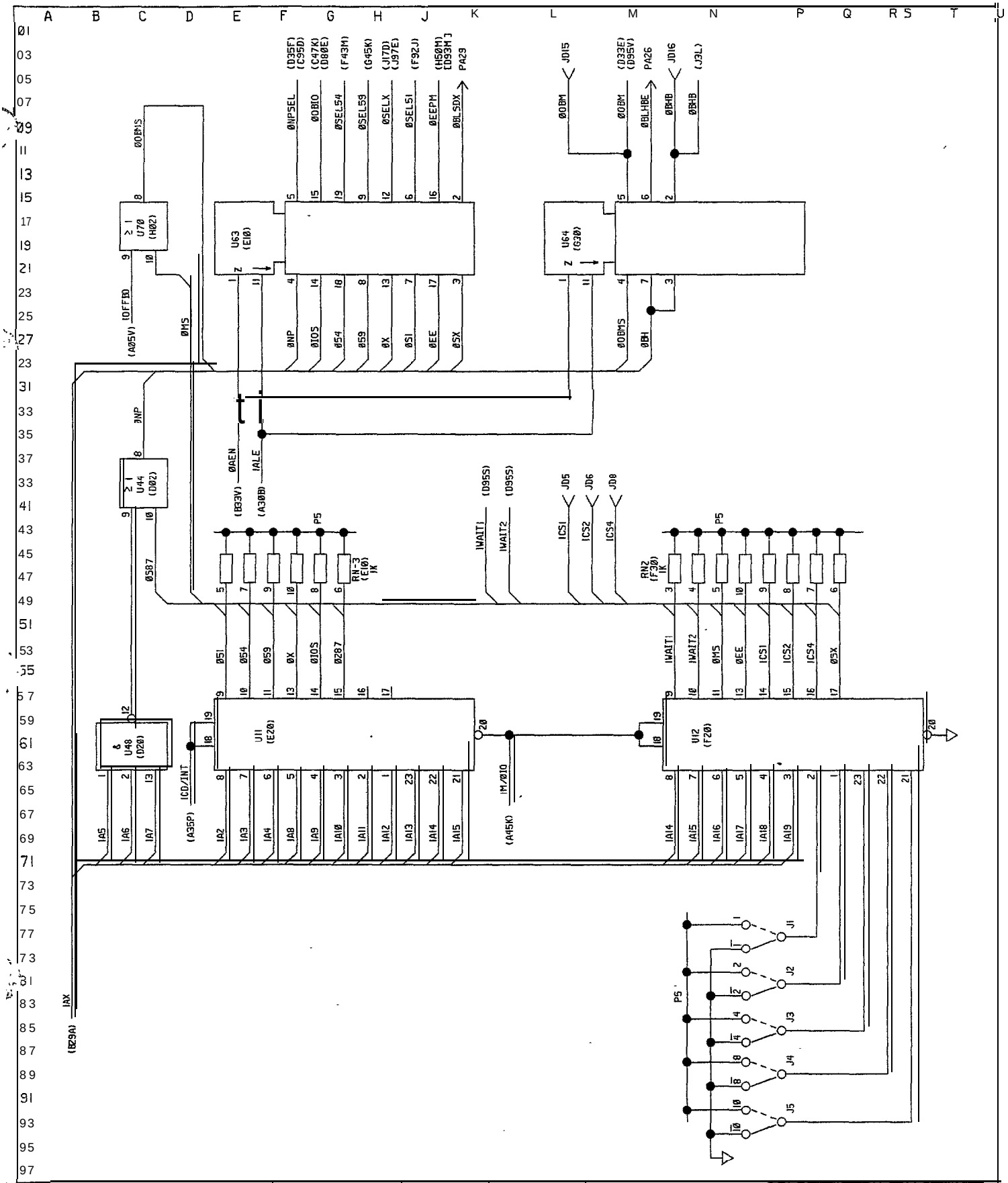
ELEMENTARY DIAGRAM
CPU

DS3800HMPK

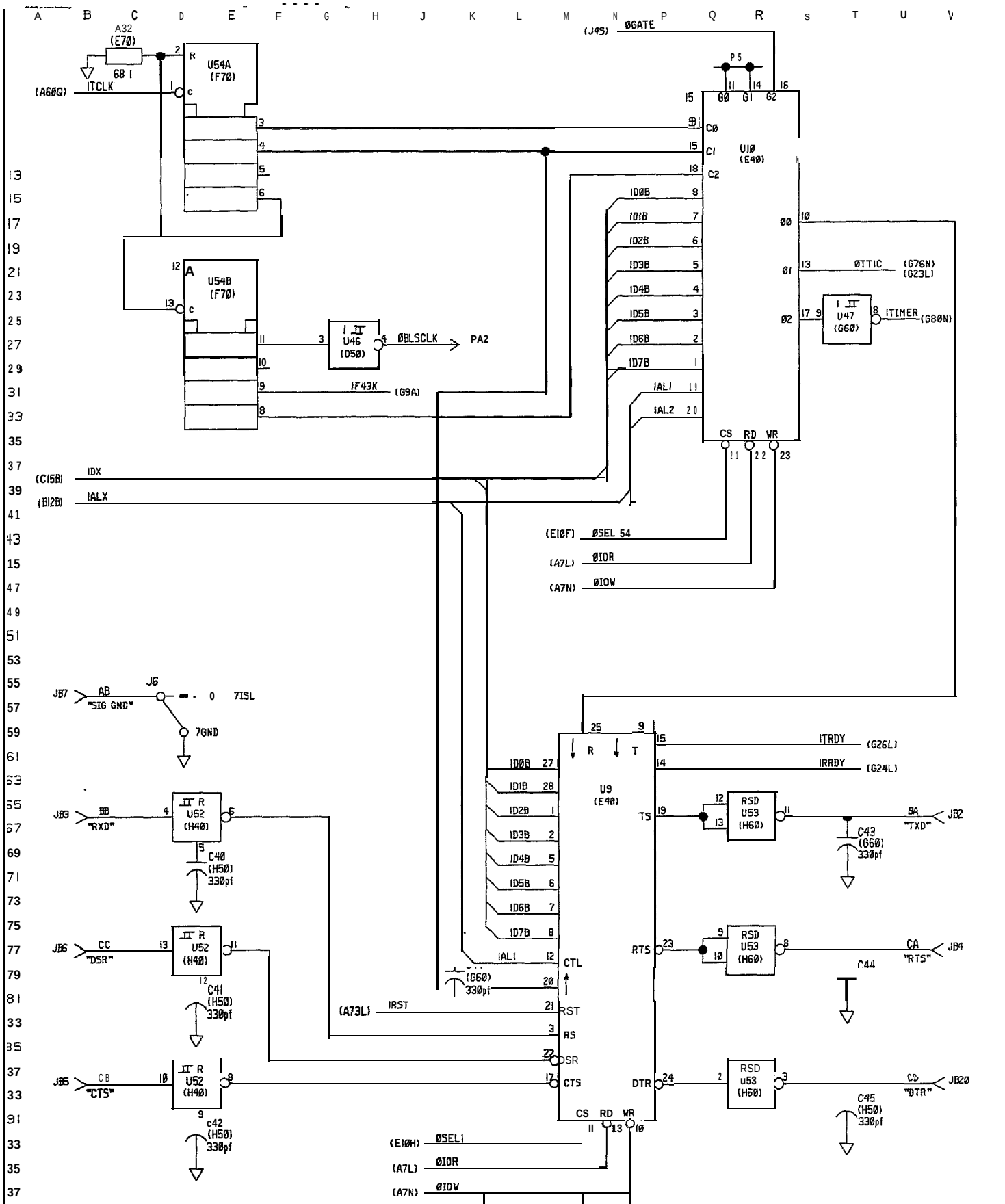


REV.1 O E K MLD 860930	REV 4	REV.5	PRINTS TO DL104 PRI	ENGINEER ED.	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM CPU
REV.2	REISSUED OCT 24,1986			FIRST MADE FOR	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	DS3800HMPK
REV.3	MADE BY E.DILLON			I.C.NO.		CONT.ON SH. 4EA SH.NO. 4DA

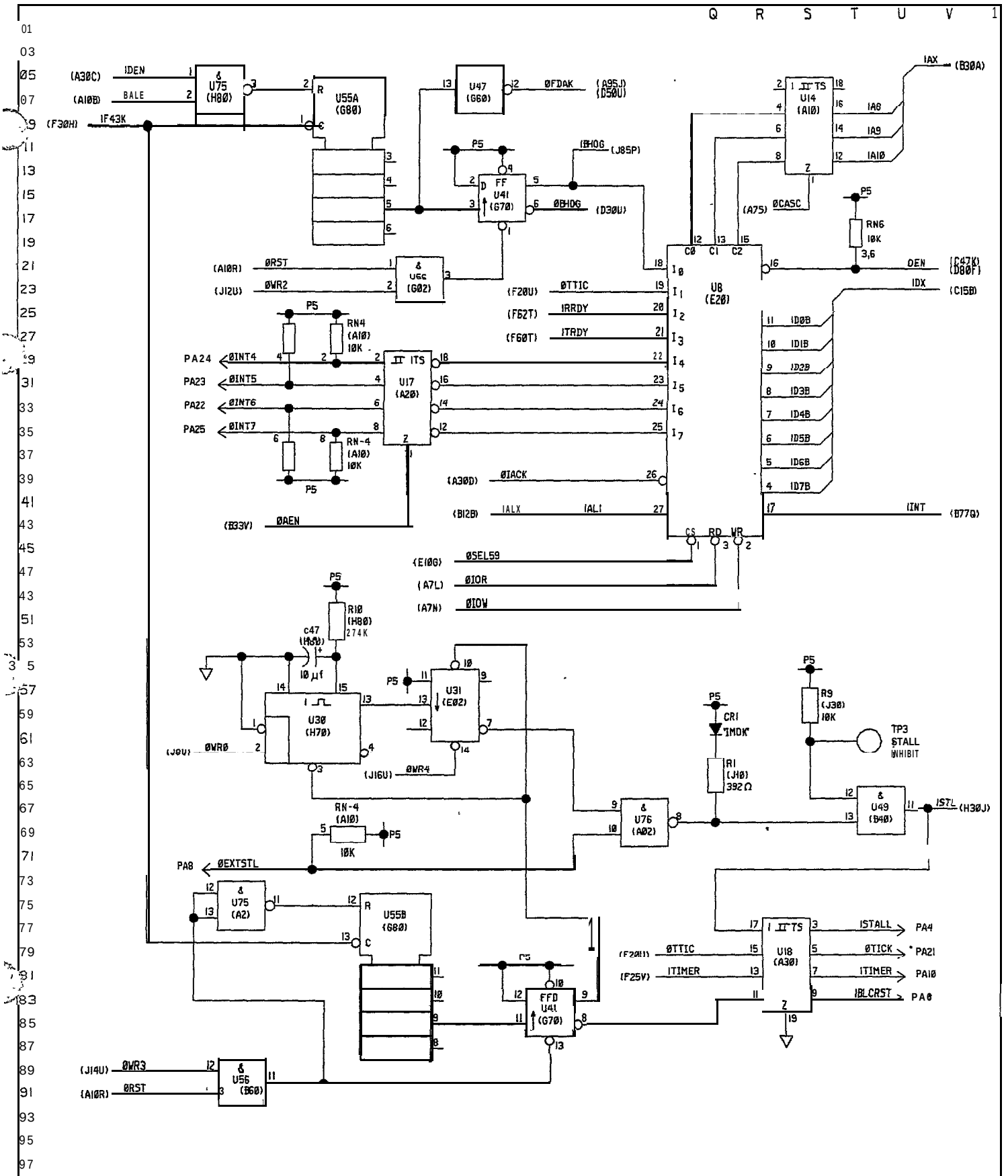
CARS



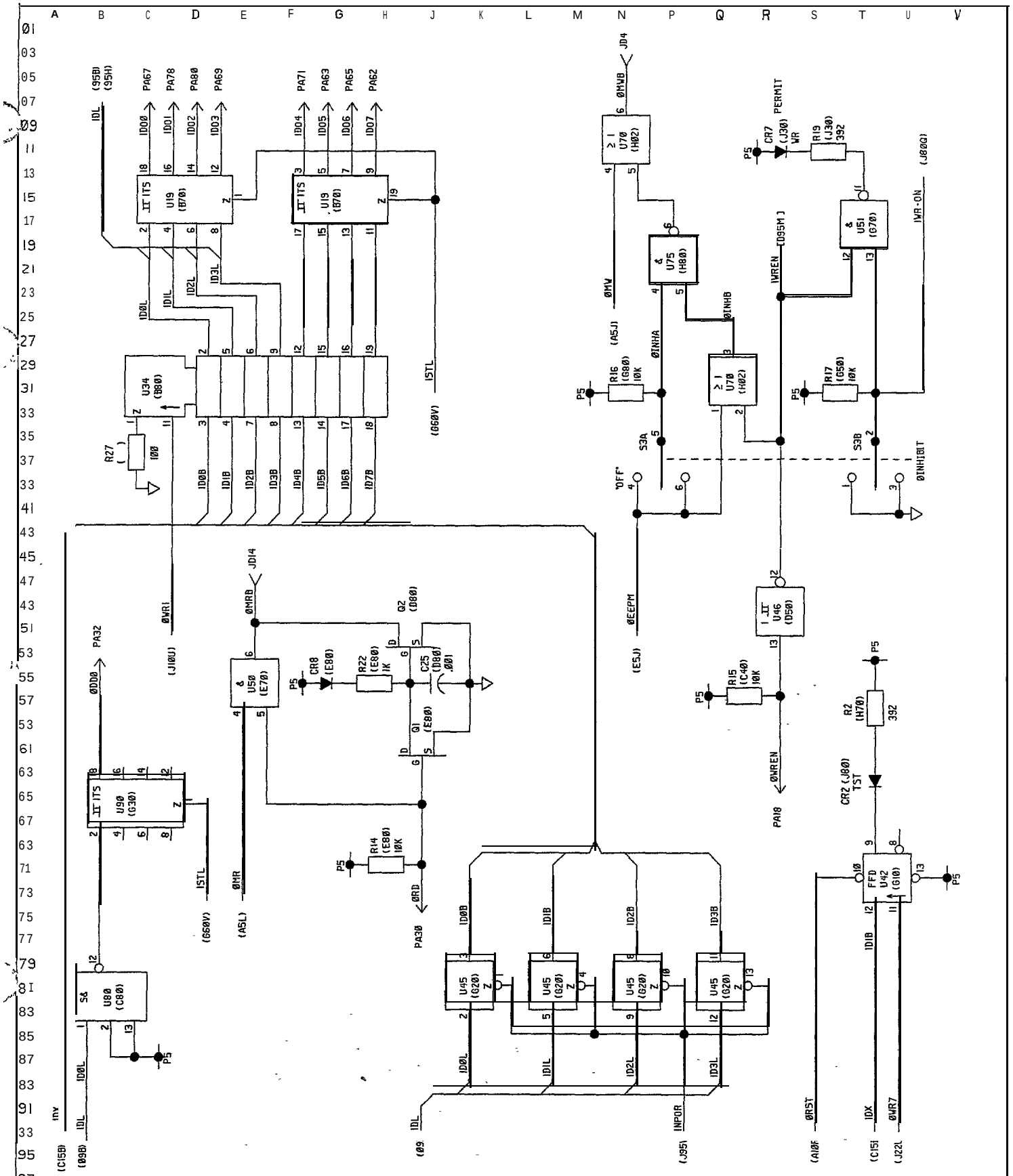
REV.1 OEK MLO 860930	REV.4	REV.5	PRINTS TO DL104 PRI	ENGINEER E.D.	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM CPU
REV.2	Re-ISSUED OCT 24, 1986			FIRST MADE FOR	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	DS3800HMPK
REV.3	MADE BY E.DILLON			I.C.NO.		CONT.ON SH.4FA SH.NO.4E



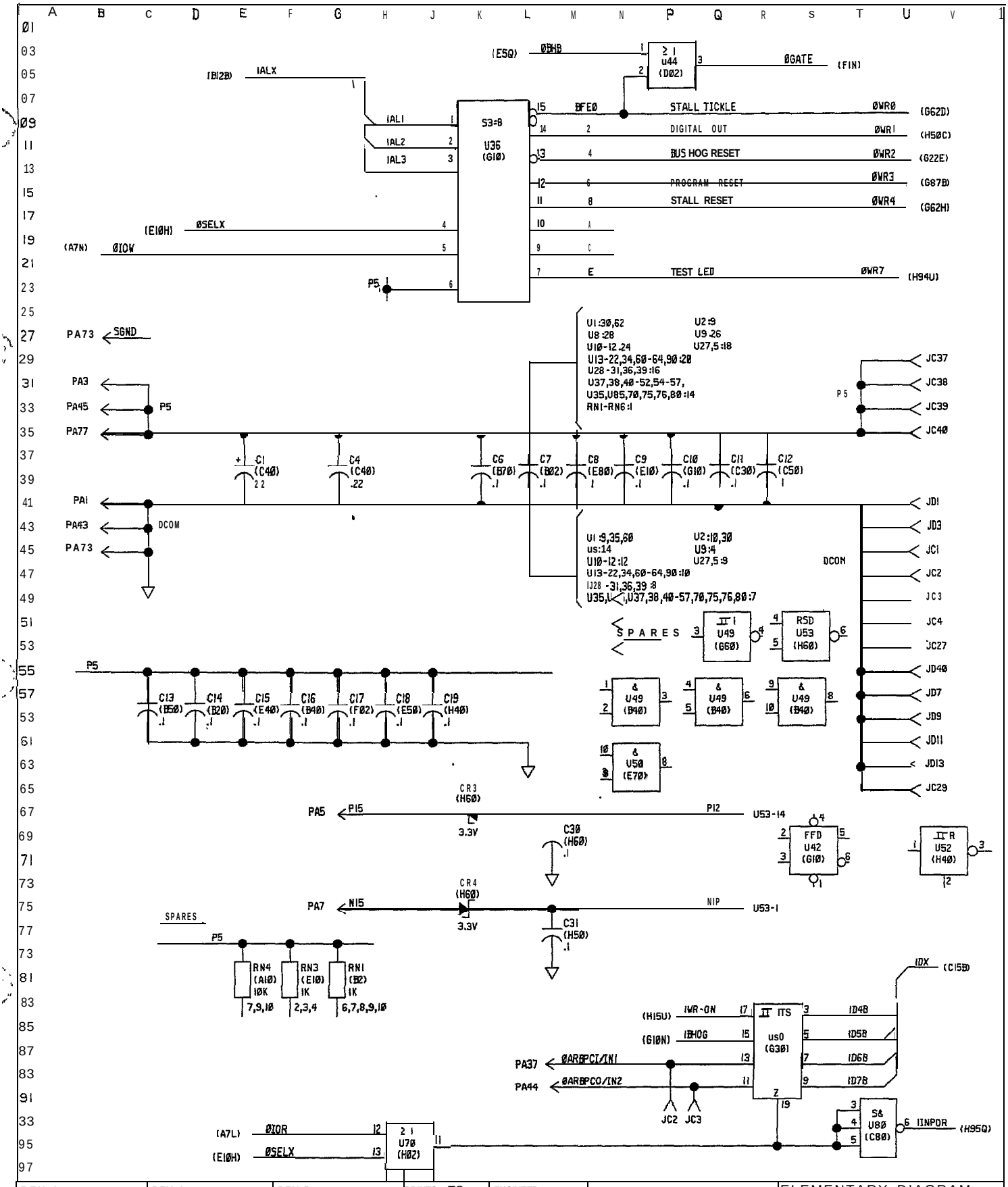
REV.1	REV.4	REV.5	PRINTS TO DL104 PRI	ENGINEER EJQ	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM CPU
REV.2	ISSUED April 22, 1986			FIRST MADE FOR	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	DS3800HMPK
REV.3	MADE BY E.DILLON			I.C.NO.		CONT.ON SH.4GA SH.NO.4FA



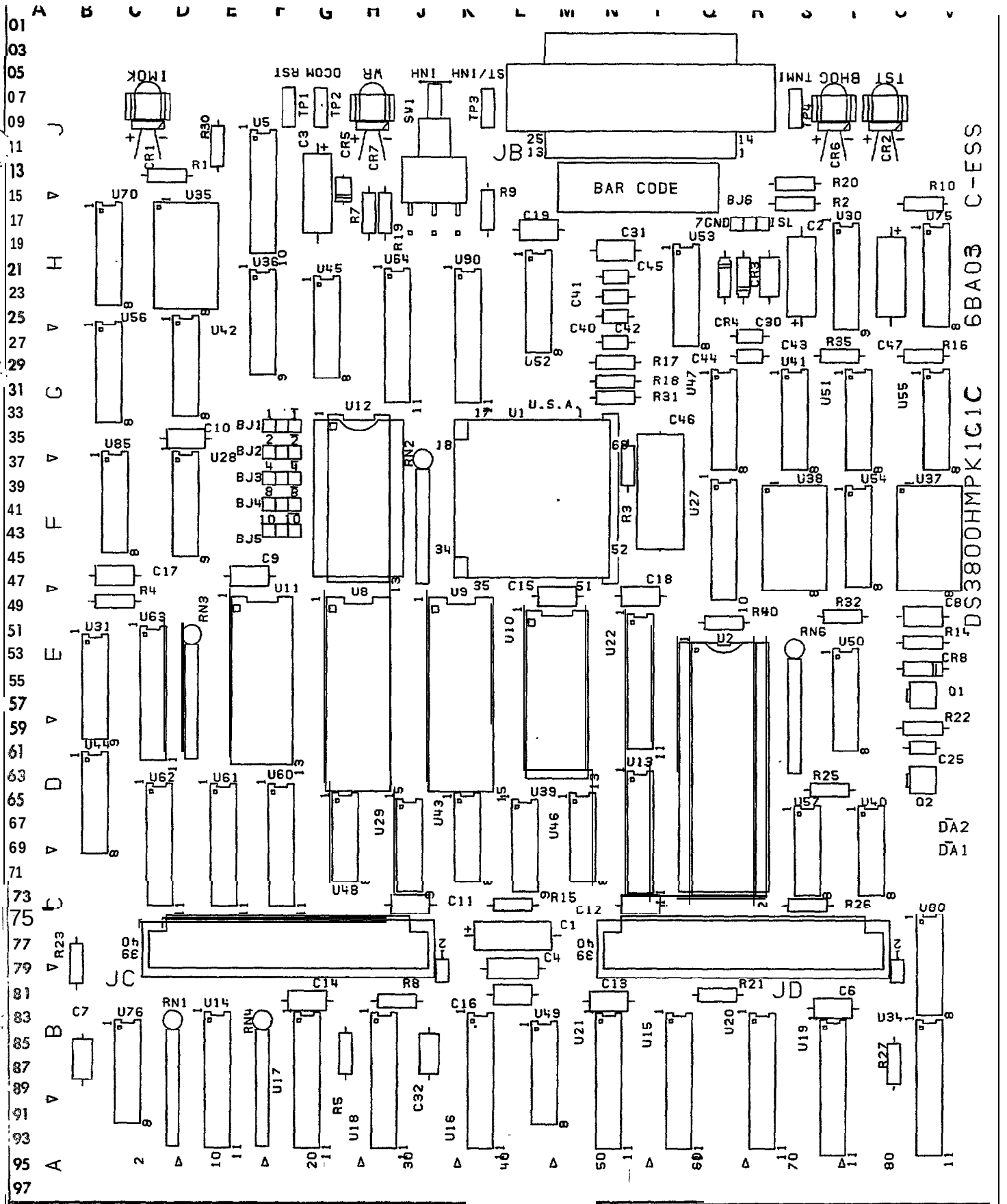
REV.1 DEW MAC 890116	REV.4	REV.5	PRINTS TO DL104 pR1	ENGINEER	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM CPU
REV.2	ISSUED			FIRST MADE FOR		
REV.3	MADE BY E.DILLON			I.C.NO.	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	DS3800HMPK
						CONT.ON SH.4HA SH.NO.4GA



"REV., OEK MLO 860930	REV.4	REV.5	PRINTS TO DL104 PRI	ENGINEER E.D.	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM CPU
REV.2	RE-ISSUED OCT. 24, 1986			FIRST MADE FOR	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	DS3800HMPK
REV.3	MADE BY: E.DILLON			I.C.NO.		CONT.ON SH. 4JA SH.NO. 4HA



REV.1 OEK MLO 860930	REV.4	REV.5	PRINTS TO DL104 pr1	ENGINEER	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM CPU
REV.2 OEW MAC B90116	ISSUED			FIRST MADE FOR	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	DS3800HMPK
REV.3	MADE BY E.DILLON			I.C.NO.		CONT.ON SH.4KA SH.NO.4JA



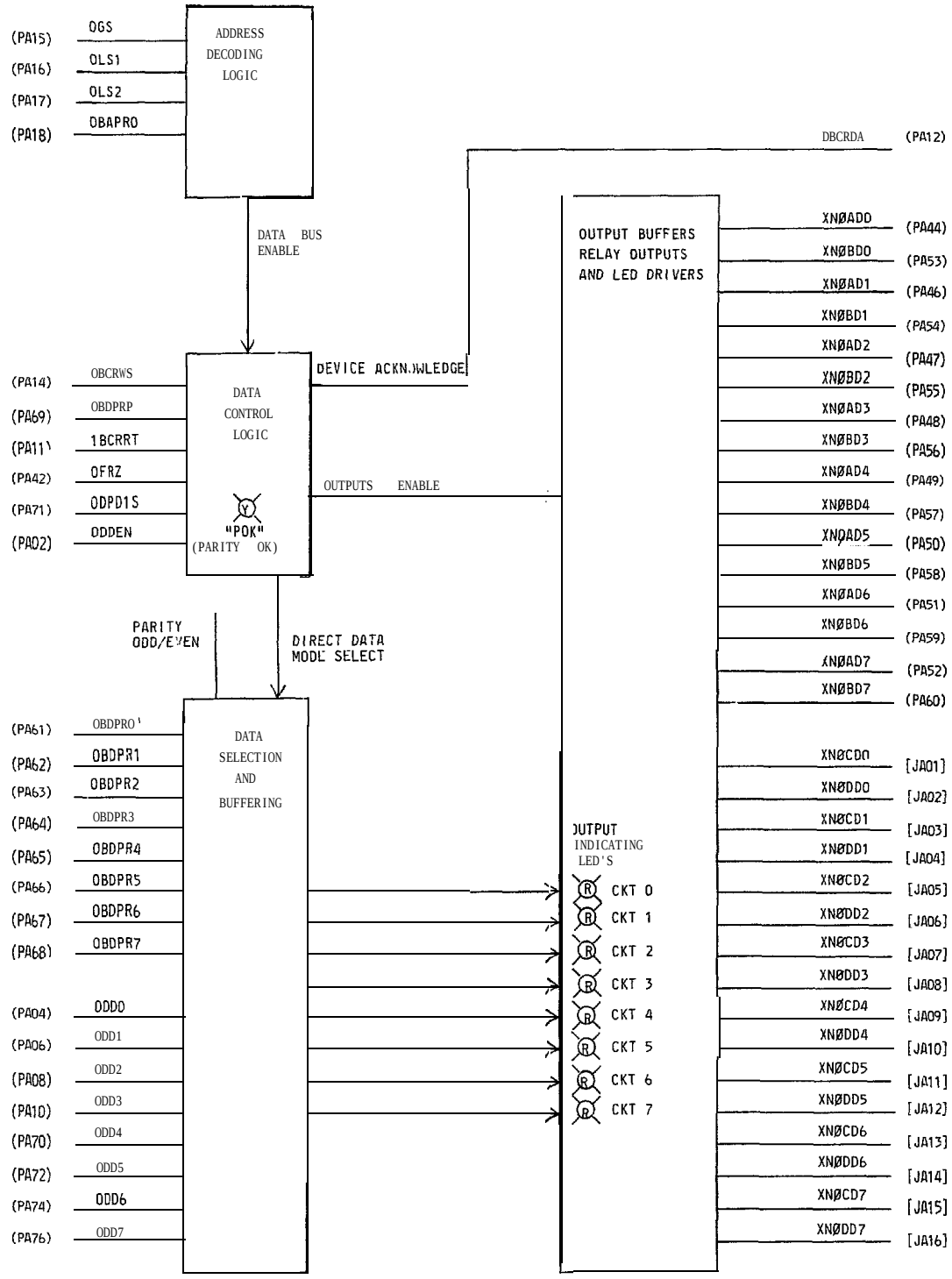
REV 1 OEK DA	REV 4	REV 5	PRINTS TO	APPROVALS
MLO 860930			DL104 PRI	
REV. 2!	ISSUED			FIRST MADE FOR
	April 22, 1986			
REV 3	MADE BY		I.C. NO.	
	E.DILLON			

GENERAL ELECTRIC
DRIVE SYSTEMS DEPT.
SALEM, VA., U.S.A.

ELEMENTARY DIAGRAM
CPU
DS3800HMPK
CONT. ON SH. FL SH. NO. 4KA

FORM IC 2349 (04-78)

CARS CARS



11 OEJ KEH 830726	REV 4	REV 5	PRINTS TO CHEL DL102 PKT	ENGINEER GL STIRK	GENERAL@ ELECTRIC	ELEMENTARY DIAGRAM
HE" 2	ISSUED 5-25-78				DRIVE SYSTEMS DEPARTMENT	HRRB BLOCK DIAGRAM
REV 3	MADE BY G L. STIRK				SALEM, VA U.S.A	DS3800HRRB
						CONT. ON SH. 2BA SH. NO. 2AA

2.0 DEFINITION OF PURPOSE

THE HRRB (RELAY OUTPUT CARO) IS A SINGLE SIZED PRINTED WIRE BOARD DESIGNED TO PROVIDE INTERFACE BETWEEN LOGIC INPUT AND RELAY CONTACT OUTPUTS. WHILE THE PRIMARY APPLICATION OF THE RELAYS IS INTENDED TO BE USED IN LOW LEVEL OR ANALOG CIRCUITS WHEN A TRUE OPEN OR SHORT CONDITIONS ARE. NECESSARY, THEY CAN BE USED IN 115VAC CIRCUITS. FOR MORE DETAILED INFORMATION ON THE CONTACT RATING REFER TO THE APPLICATION DATA (SEC.2.4). THE OUTPUTS CONSIST OF A SET OF B N.O. CONTACTS BROUGHT TO THE FRONT EDGE CONNECTOR (JA), AND A DIFFERENT SET OF 8 N.O. CONTACTS BROUGHT TO THE BACKPLANE CONNECTOR (PA).

2.1 THEORY OF OPERATION

AS MENTIONED IN THE PREVIOUS SECTION, THE HRRB PROVIDES THE INTERFACE BETWEEN LOGIC INPUTS AND RELAY CONTACT OUTPUTS. THE LOGIC INPUTS ARE RECEIVED FROM EITHER DIRECT DATA INPUTS (ODDO-7) OR FROM THE R-BUS (OBDPRO-7) (REFER TO BLOCK DIAGRAM SH. 2AA). FOR EXPLANATION OF ADDRESSING AND SYSTEM INTERFACE. SCHEMES REFER TO GEK-61680.

IF IT IS DESIRED TO OUTPUT DATA WITHOUT ADDRESSING THE CARD AND FEED DATA UNLATCHED TO THE OUTPUTS, THE DIRECT DATA MODE OF OPERATION MAY BE USED. TO USE THE DIRECT DATA MODE THE BIRECT DATA MODE SELECT LINE (ODDEN, PA02) SHOULD BE TIED TO DCOM. ZERO TRUE DATA TD BE OUTPUTTED SHDULD NOW BE BROUGHT ONTO DIRECT DATA INPUTS (ODDO-7). ZERO TRUE DATA APPLIED TD THE DIRECT DATA INPUT LINES WILL APPEAR AT THEIR RESPECTIVE OUTPUTS AFTER A MAXIMUM DELAY OF 7 MILLISECONDS.

2.2 INTERFACE REQUIREMENTS

THE INPUTS AND OUTPUTS OF THE HRRB ARE LOW LEVEL TTL TYPE WITH THE EXCEPTION OF THE RELAY CONTACT DUTPUTS. THE LOW LEVEL SIGNALS ARE EXPECTED TO BE USED WITHIN AN ELECTRONICS MODULE OR VERTICAL TERMINATION ASSEMBLY. THE RELAY CONTACT OUTPUTS MAY BE RUN AS FAR AS ALLOWED BY THE APPLICATION.

2.3 FUNCTIONAL DESCRIPTION

2.3.1 INPUTS (ALL INPUTS FLOAT TO A LOGICAL ONE)

ALL TTL SIGNALS ARE ZERO ("0") TRUE EXCEPT FOR 1BCRRT WHICH IS ONE ("1") TRUE.

MNEMONIC	PIN NO.	DESCRIPTION (SEE BLOCK DIAGRAM 2AA)
0GS	PA15	G SELECT ADDRESS
0LS1	PA16	L SELECT 1 ADDRESS
0LS2	PA17	L SELECT 2 ADDRESS
0BAPR0	PA18	UPPER/LOWER BYTE ADDRESS
0BCRWS	PA14	WRITE STROBE
1BCRRT	PA11	BUS RESET
0FRZ	PA42	FREEZE ON ERROR INPUT (RESET CONTROL FOR PARITY ERROR).
0DPD1s	PA71	
0BDPR0	PA61	BUS DATA BIT 0
0BDPR1	PA62	BUS DATA BIT 1
0BDPR2	PA63	BUS DATA BIT 2
0BDPR3	PA64	BUS DATA BIT 3
0BDPR4	PA65	BUS DATA BIT 4
0BDPR5	PA66	BUS DATA BIT 5
0BDPR6	PA67	BUS DATA BIT 6
0BDPR7	PA68	BUS DATA BIT 7
0BDPRP	PA69	BUS DATA PARITY
0000	PA04	DIRECT DATA BIT 0
0DD1	PA06	DIRECT DATA BIT 1
0DB2	PA08	DIRECT DATA BIT 2
0DD3	PA10	DIRECT DATA BIT 3
0DB4	PA70	DIRECT DATA BIT 4
0DD5	PA72	DIRECT DATA BIT 5
0DD6	PA74	DIRECT DATA BIT 6
0DD7	PA76	DIRECT DATA BIT 7
0DDEN	PA02	DIRECT DATA MODE SELECT

REV. 1	REV. 4	REV. 7	PRINTS TO PRI DL102 REL1	ENGINEER GLS	GENERAL @ ELECTRIC	RELAY OUTPUT PWB
REV. 2	REV. 5	ISSUED 5-25-78		DIRECTO-MATIC PLUS	DRIVE SYSTEMS DEPT.	DS3800HRRB
REV. 3	REV. 6	MADE BY G.L. STI RK			SALEM, VA. U.S.A.	CONT. ON SH. 2DA SH. NO. 2CA

2.3 FUNCTIONAL DESCRIPTION

2.3.2 OUTPUTS ALL OUTPUT ON BACKPLACE CONNECTOR

MNEMONIC	PIN NO.	DESCRIPTION (SEE BLOCK DIAGRAM 2AA)
OBCRDA	PA12	DEVICE ACKNOWLEDGE
XNØAD0	PA44	N.O. RELAY CONTACT BIT 0
XNØBD0	PA53	N.O. RELAY CONTACT BIT 0
XNØAD1	PA46	N.O. RELAY CONTACT BIT 1
XNØBD1	PA54	N.D. RELAY CONTACT BIT 1
XNØAD2	PA47	N.O. RELAY CONTACT BIT 2
XNØBD2	PA55	N.O. RELAY CONTACT BIT 2
XNØAD3	PA46	N.O. RELAY CONTACT BIT 3
XNØBD3	PA56	N.D. RELAY CONTACT BIT 3
XNØAD4	PA49	N.O. RELAY CONTACT BIT 4
XNØBD4	PA57	N.O. RELAY CONTACT BIT 4
XNØAD5	PA50	N.O. RELAY CONTACT BIT 5
XNØBD5	PA56	N.D. RELAY CONTACT BIT 5
XNØAD6	PA51	N.O. RELAY CONTACT BIT 6
XNØBD6	PA59	N. O. RELAY CONTACT BIT 6
XNØAD7	PA52	N. O. RELAY CONTACT BIT 7
XNØBD7	PA60	N.O. RELAY CONTACT BIT 7
XNØCD0	JA01	N.O. RELAY CONTACT BIT 0
XNØDD0	JA02	N.O. RELAY CONTACT BIT 0
XNØCD1	JA03	N.O. RELAY CONTACT BIT 1
XNØDD1	JA04	N.O. RELAY CONTACT BIT 1
XNØCD2	JA05	N.O. RELAY CONTACT BIT 2
XNØDD2	JA06	N.O. RELAY CONTACT BIT 2
XNØCD3	JA07	N.O. RELAY CONTACT BIT 3
XNØDD3	JA08	N.O. RELAY CONTACT BIT 3
XNØCD4	JA09	N.O. RELAY CONTACT BIT 4
XNØDD4	JA10	N.O. RELAY CONTACT BIT 4
XNØCD5	JA11	N.O. RELAY CONTACT BIT 5
XNØDD5	JA12	N.D. RELAY CONTACT BIT 5
XNØCD6	JA13	N.O. RELAY CONTACT BIT 6
XNØDD6	JA14	N.O. RELAY CONTACT BIT 6
XNØCD7	JA15	N.O. RELAY CONTACT BIT 7
XNØDD7	JA16	N.O. RELAY CONTACT BIT 7

PA CONNECTOR = BACKPLANE CDNNECTOR
IA CONNECTOR = FRONT EDGE CONNECTOR

EV. 1	REV. 4	REV. 7	PRINTS TO DL102 PRI BHEL	ENGINEER GLS	GENERAL ELECTRIC	RELAY OUTPUT PWB
REV. 2	REV. 5	ISSUED 5-25-78		DIRECTO-MATIC	DRIVE SYSTEMS DEPT.	DS3800HRRB
REV. 3	REV. 6	MADE BY G.L. STIRK		PLUS	SALEM, VA. U.S.A.	CONT. ON SH. 2EA SH. NO. 2DA

DSD 0151 (09-76)

2.3.3 INDICATING LED'S

THERE ARE NINE LIGHT-EMITTING DIODES (LED) ON THE FRONT EDGE OF THE HRRA, TO INDICATE THE STATE, OF THE RELAY OUTPUTS AND PARITY CONDITION. THE LED'S ARE IDENTIFIED ACCORDING TO THE ASSOCIATED BIT OUTPUT OR FUNCTION.

LED	DESCRIPTION	COLOR
WPOK	LUMINOUS WHEN WRITE PARITY OK	YELLOW
0	LUMINOUS WHEN BIT 0 RELAY PICKED-UP	RED
1	LUMINOUS WHEN BIT 1 RELAY PICKED-UP	RED
2	LUMINOUS WHEN BIT 2 RELAY PICKED-UP	RED
3	LUMINOUS WHEN BIT 3 RELAY PICKED-UP	RED
4	LUMINOUS WHEN BIT 4 RELAY PICKED-UP	RED
5	LUMINOUS WHEN BIT 5 RELAY PICKED-UP	RED
6	LUMINOUS WHEN BIT 6 RELAY PICKED-UP	RED
7	LUMINOUS WHEN BIT 7 RELAY PICKED-UP	RED

2.3.4 MAINTAINABILITY

THE HRRB HAS BEEN DESIGNED TO REQUIRE A MINIMUM OF MAINTANANCE TOOLS. IF A PARTICULAR CARD IS SUSPECTED OF BEING FAULTY, ITS INPUTS (EITHER BUS OR DIRECT DATA). SHOULD BE FORCED TO A KNOWN STATE(S) AND THE OUTPUT CHECKED FOR THE CORRECT RESPONSE. THE OUTPUTS CAN BE CHECKED BY OBSERVING THE LED'S AND VOLTAGE CHECK OF THE OUTPUT..

When A PWB Has Been Keset The Parity Led Will Remain Off Until The Card Has Been Addressed And Strobed With Correct Data And Parity.

EV. 1 OEJ REH 830726	REV. 4	REV. 7	PRINTS TO PRI BL100 BHEL	ENGINEER GLS	GENERAL ELECTRIC	RELAY OUTPUT PWB
REV. 2	REV. 5	ISSUED 5-25-78		DIRECTO-MATIC PLUS	DRIVE SYSTEMS DEPT.	DS3800HRRB
REV. 3	REV. 6	MADE BY G.L. STIRK			SALEM, VA. U.S.A.	CONT. ON SH. 2GA SH. NO. 2FA

2.4 APPLICATION DATA

2.4.1 POWER REQUIREMENTS

P5	347 MA
P15	3 MA
P28	250 MA

2.4.2 HEAD DISSIPATED

6.1 WATTS

2.4.3 INSTALLATION PROCEDURE AND ADJUSTMENTS

NONE

2.4.4 TTL INPUTS

LOGIC ZERO = +0.4±0.4 VDC

LOGIC ONE = +3.75±1.75 VDC

2.4.4.1 SIGNALS OBDPR0-OBDPR7

THESE SIGNALS HAVE THE CAPABILITY OF SOURCING 0.58MA (.83SLL) AND SINKING 0.020MA.

2.4.4.2 SIGNALS OLS1, OLS2, OGS, OBCRWS, IBCRRT, AND OBAPR0

THESE SIGNALS HAVE THE CAPABILITY OF SOURCING 0.2 MA. (.28SLL) AND SINKING 0.0MA.

2.4.4.3 SIGNALS DDD0-ODD7 AND OBDPRP

THESE SIGNALS HAVE THE CAPABILITY OF SOURCING 0.7MA. (1.0 SLL) AND SINKING (0.0MA).

2.4.4.4 SIGNALS OFRZ, ODDEN, IEVEN, AND ODPDIS

THESE SIGNALS HAVE THE CAPABILITY OF SOURCING 1.5MA. 2.2 (SLL) AND SINKING 0.0MA.

2.4.5 TTL OUTPUTS

LOGIC ZERO = +0.25±0.25 VDC

LOGIC ONE = +4.0±1.5 VDC

HI-Z = HIGH IMPEDANCE STATE WITH +20 AND -200ua LEAKAGE.

2.4.5.1 SIGNAL OBCRDA

THIS SIGNAL CAN SINK 19MA (27 SLL) AND SOURCE 12MA. (>27 SLL).

2.4.6 PROCESS OUTPUTS

ALL PROCESS OUTPUTS ARE RELAY CONTACT OUTPUTS WITH THE FOLLOWING CHARACTERISTICS.

2BV AT 2 AMPS	DC RESISTIVE
115V AC AT .5 AMPS	L/R LESS THAN 25 MSEC.
*125V DC AT 150MA.	L/L LESS THAN 5 MSEC.-

*125V DC RATING CAN GO UP TO 145 VDC AT 150MA BUT SHOULD NOT EXCEED 1/4 OF THE RATED # OF CYCLES (100,000). CONTACTS SHOULD NOT BE USED ON LOW LEVEL SIGNALS AFTER THEY HAVE BEEN USED ON 115 VAC OR 125V DC CKTS.

CONTACTS CAPABLE OF ", DRY CIRCUIT ", OPERATION.

JA CONNECTOR

VOLTAGE FROM JA CDNNECTOR CONTACTS TO GROUND SHOULD BE LESS THAN 200V PEAK.

JA CONNECTOR CONTACTS ARE FUSED FOR CURRENT IN EXCESS OF 3 AMPS.

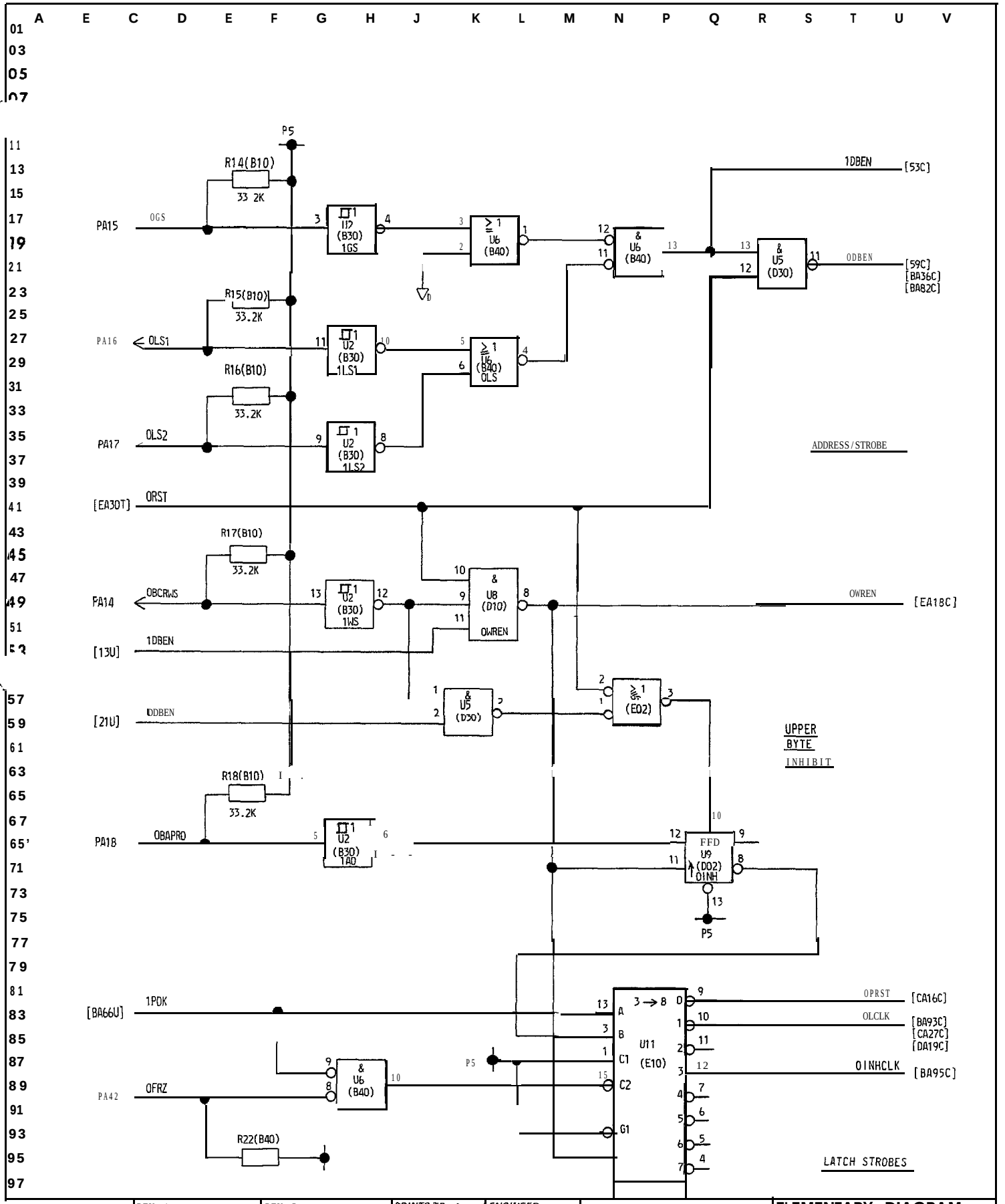
SUPPRESSORS ARE RECOMMENDED FOR ALL RELAY LOAD APPLICATIONS (EXCEPT TO IC3600KMR, 115VAC COIL, RELAYS)

PA CONNECTORS

THE PA CONNECTOR CONTACT OUTPUTS SHOULD BE USED WITH VOLTAGE LESS THAN 15VDC.

REV. 1 OEJ REH 830726	REV. 4	REV. 7	PRINTS TO DL102 BHEL PR 1	ENGINEER GLS	GENERAL ELECTRIC	RELAY OUTPUT PWB
REV. 2	REV. 5	ISSUED 5-25-78			DRIVE SYSTEMS DEPT.	DS3800HRRB
REV. 3	REV. 6	MADE BY G.L. STIRK			SALEM, VA. U.S.A.	CONT. ON SH. FL. SH. NO. 2HA

DSD 0151 (09-76)



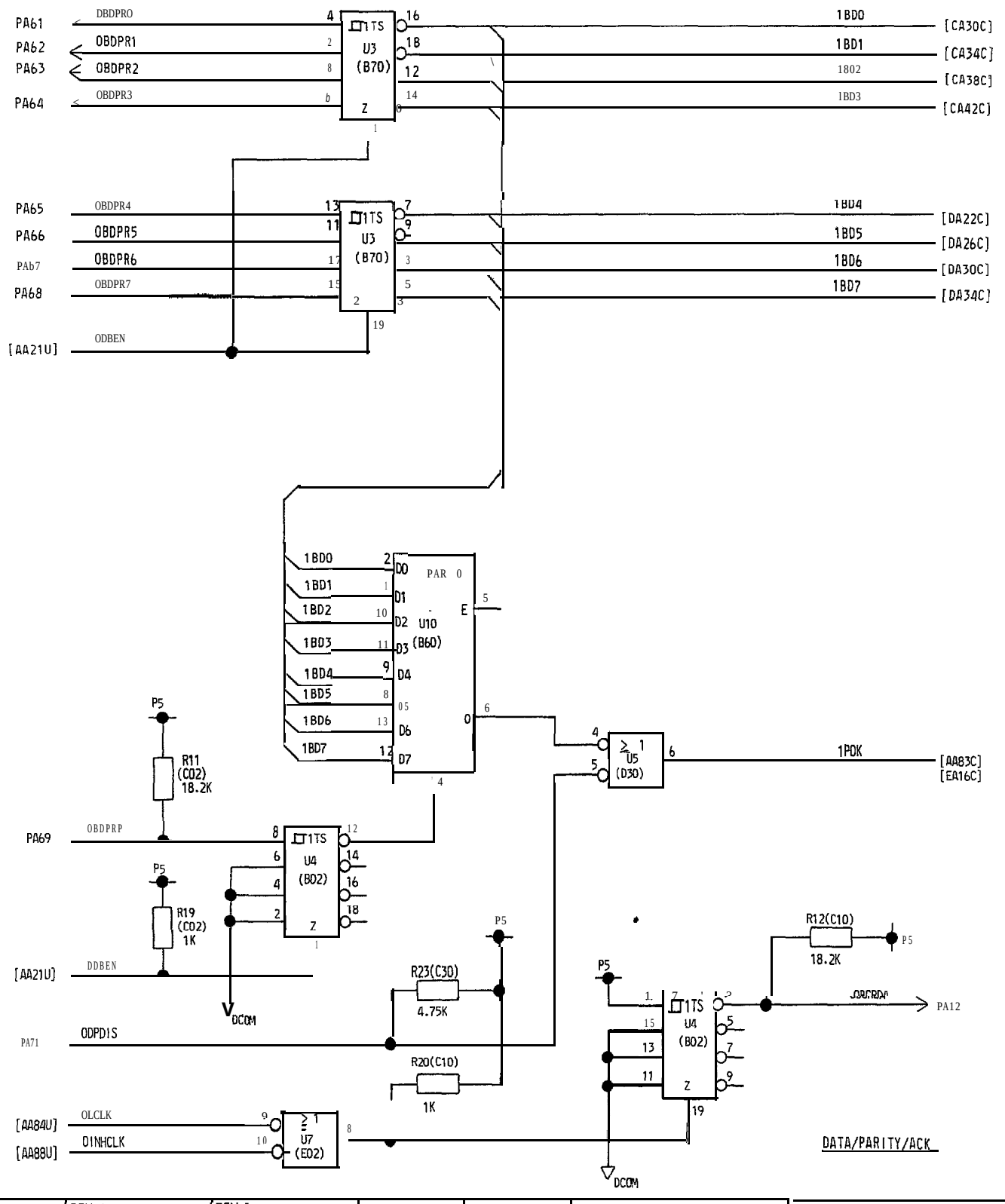
1 of C	REV 4 RE	REV 5	PRINTS TO DL104	ENGINEER 780202	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.
REV 2 OEK REH 830726	ISSUED February 6, 1978				
REV 3	MADE BY R.W. MILLER				

ELEMENTARY DIAGRAM

RELAY OUTPUT

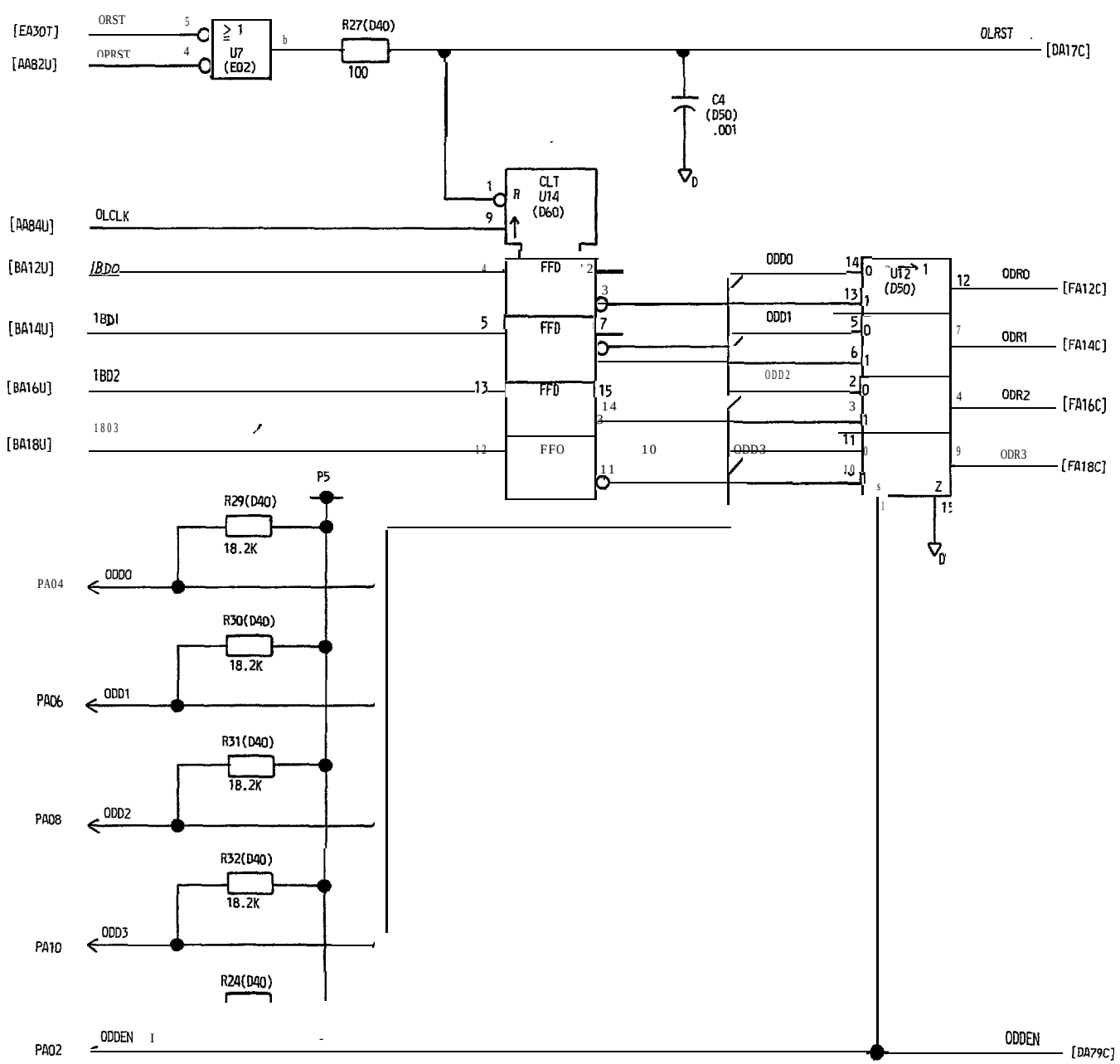
DS3800HRRB

CONT. ON SH. 4BA SH. NO. 4AA



1 DEC MH 2/1/78	REV 4 RE-	REV 5	PRINTS TO BHEL DL104 PAI	ENGINEER Rumiller 780202	GENERAL@ ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A	ELEMENTARY DIAGRAM RELAY OUTPUT
REV 2 OEK REH 830726	ISSUED Feb 6, 1978					D S 3 8 0 0 H R R B
REV 3	MADE BY R W. MILLER ER					CONT. ON SH. 4CA SH. NO. 4BA

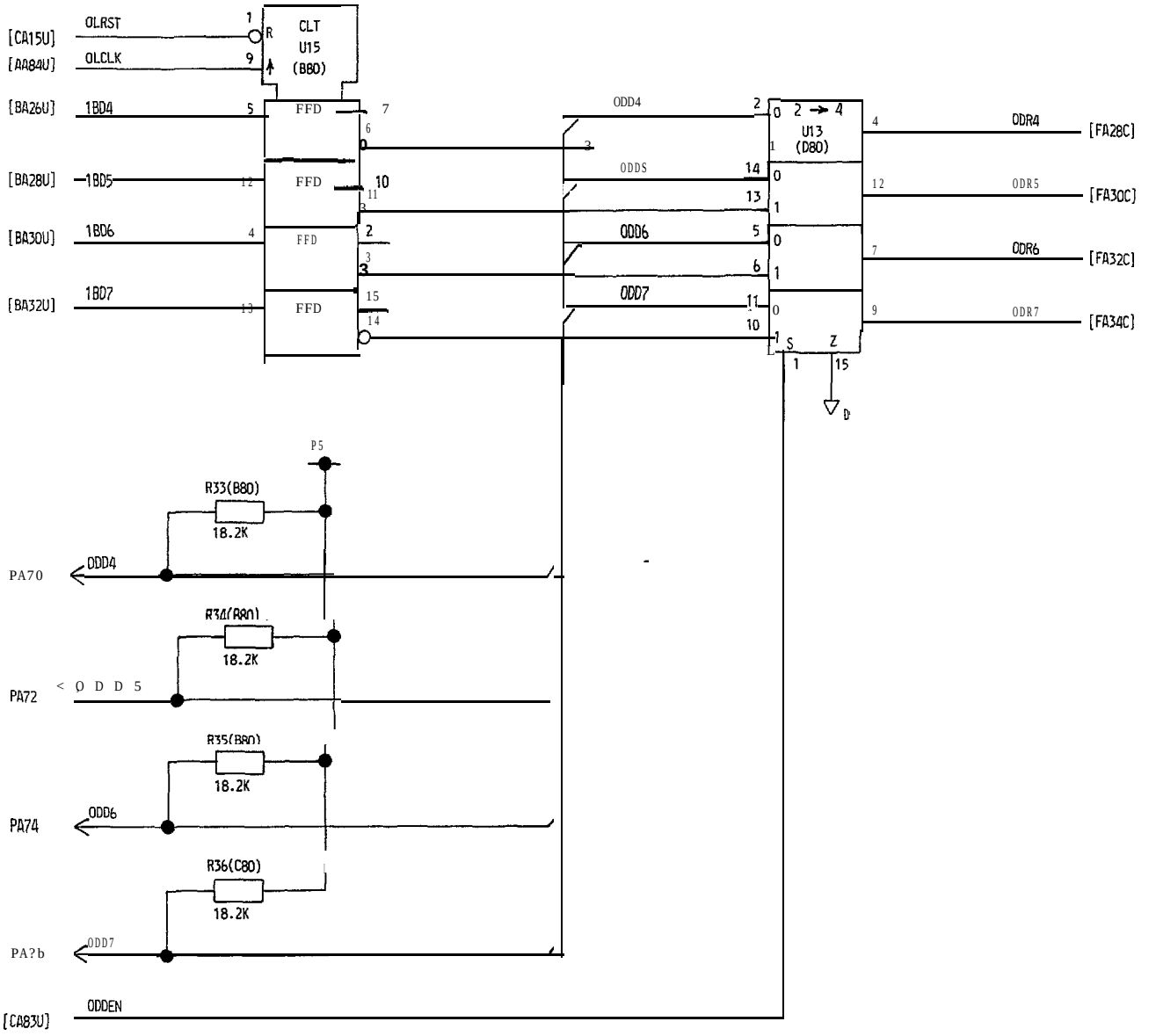
AF 10527



DIRECT DATA-1

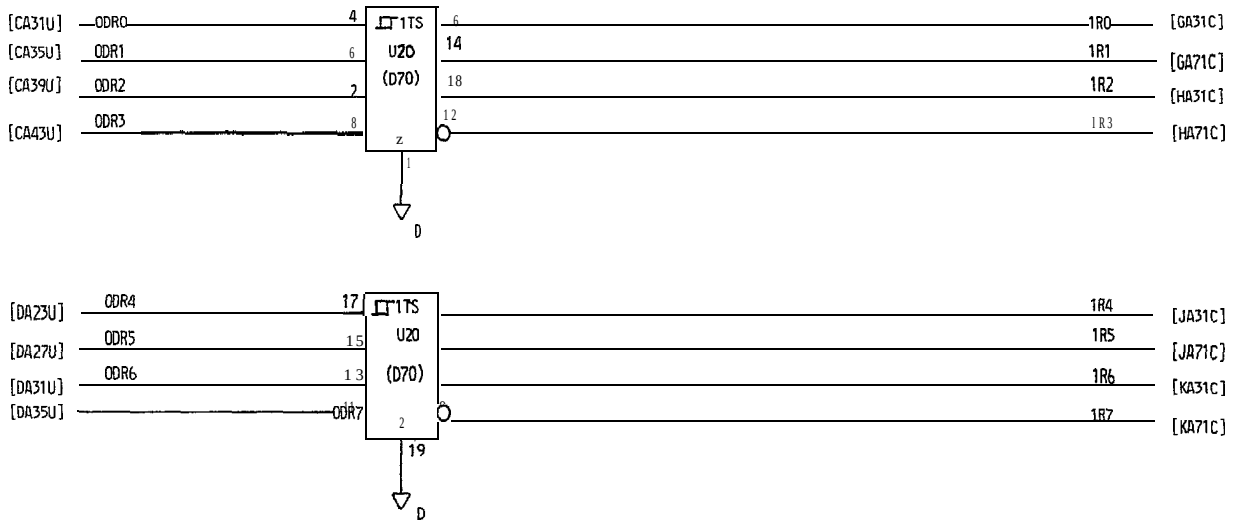
1 OEC NH 2/2/78	REV 4	REV 5	PRINTS TO B#EL DL104	ENGINEER REH/RWM	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM RELAY OUTPUT
REV 2 OEK REH 830726	ISSUED RE - 10-8-81					
REV 3	MADE BY R W MILLER					D S 3 8 0 0 H R R B
						CONT. ON SH. 4DA SH. NO. 4CA

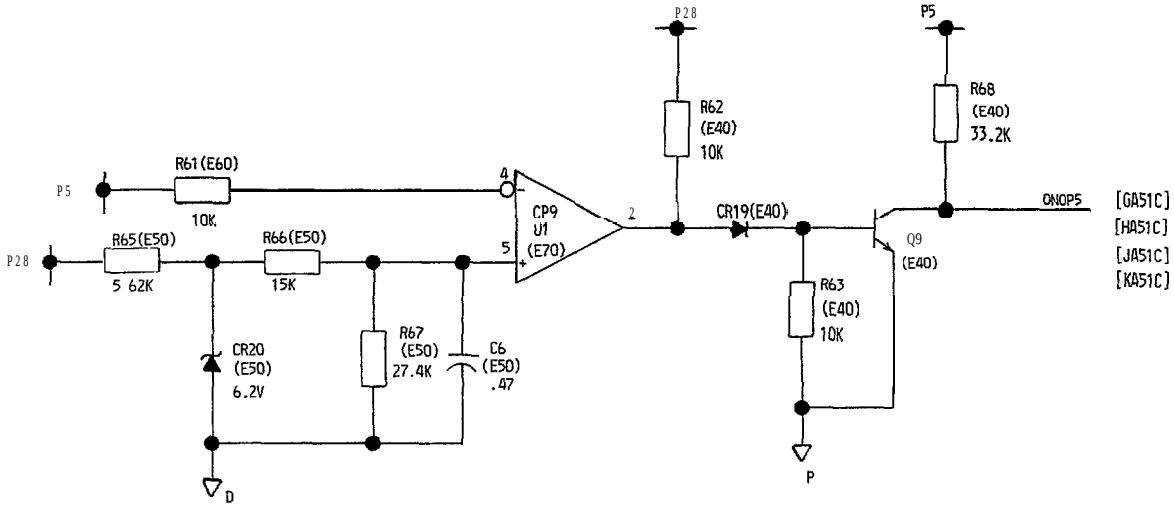
DSD 0152 (09 76)



DIRECT DATA • 2

V 1 OEC 11:15 2/2/78 REV 2 OEK REV 830726 REV 3	REV 4 ISSUED 2-6-78 MADE BY R.W. MILLER	REV 5	PRINTS TO BHEL DL104 FKL	ENGINEER 2/02/02	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM RELAY OUTPUT DS3800HRRB CONT. ON SHEA SH. NO. 40





PWR HOLD OFF

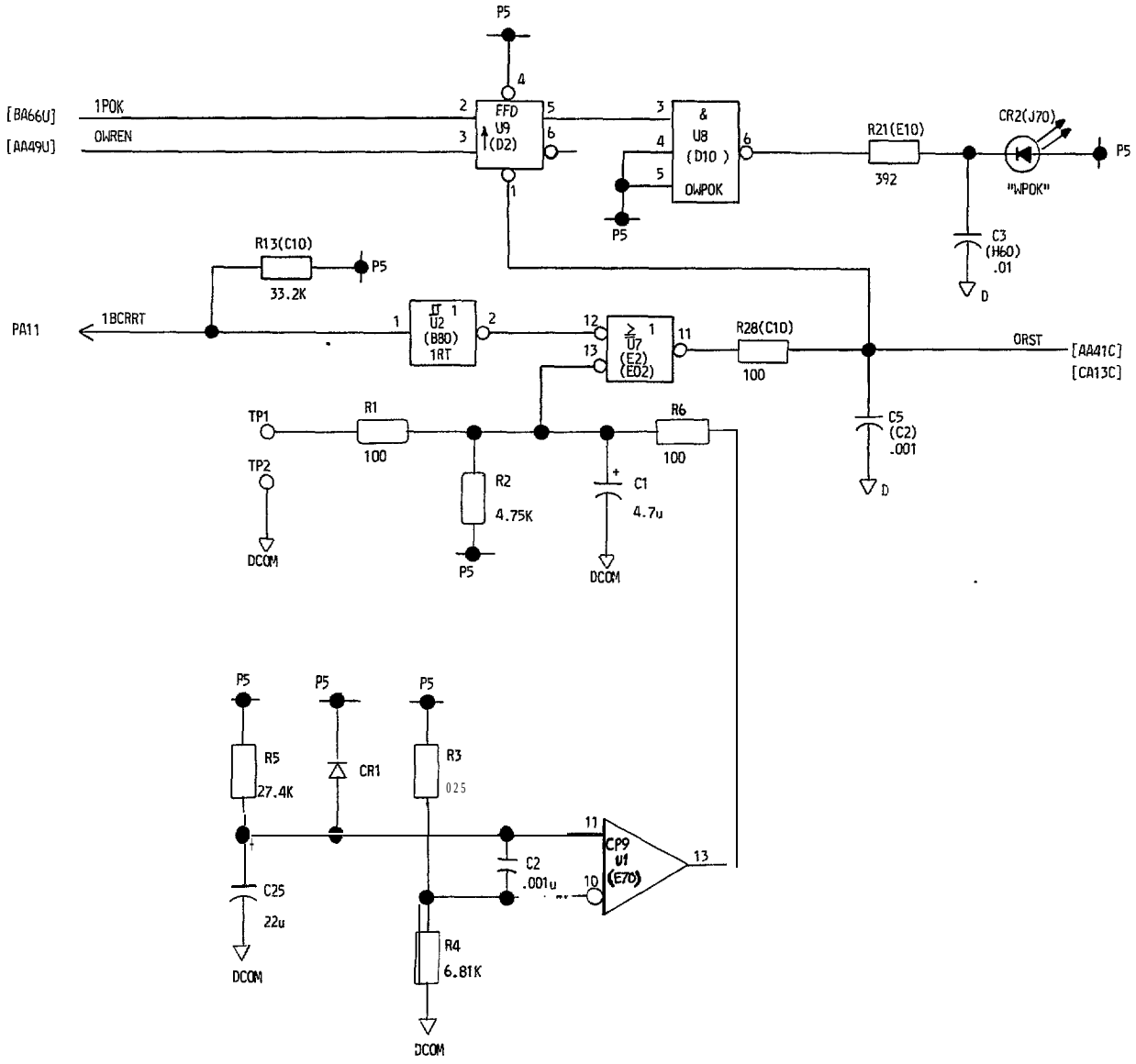
1 OEK -LH 830726	REV 4	REV 5	PRINTS TO BHEL DL104 PKI	ENGINEER Rwmiller 780202	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U S.A
REV 2	ISSUED 2-6-78				
REV 3	MADE BY R.W. MILLER				

ELEMENTARY DIAGRAM

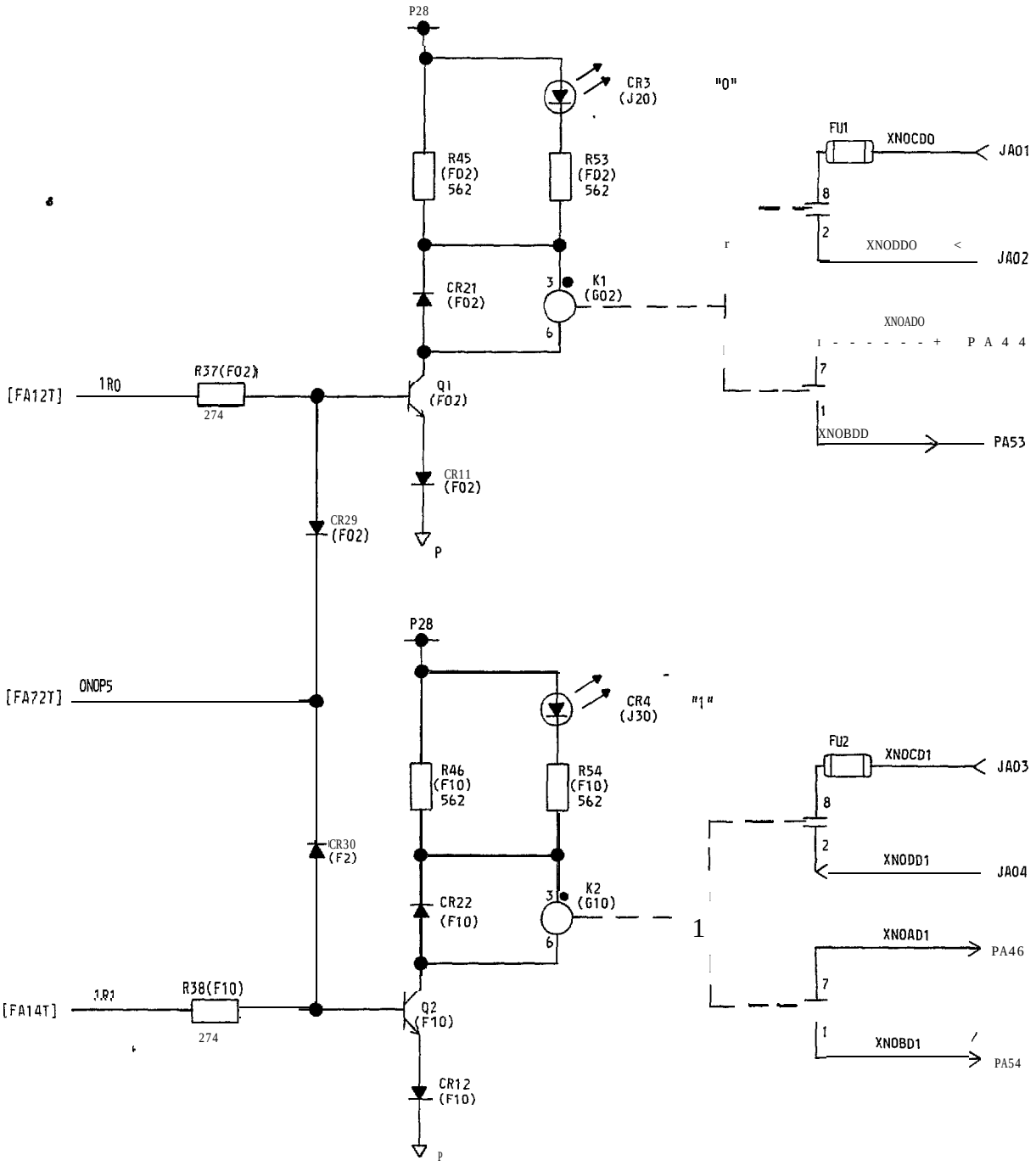
RELAY OUTPUT

DS3800HRRB

CONT. ON SH. 4GA SH. NO. 4F

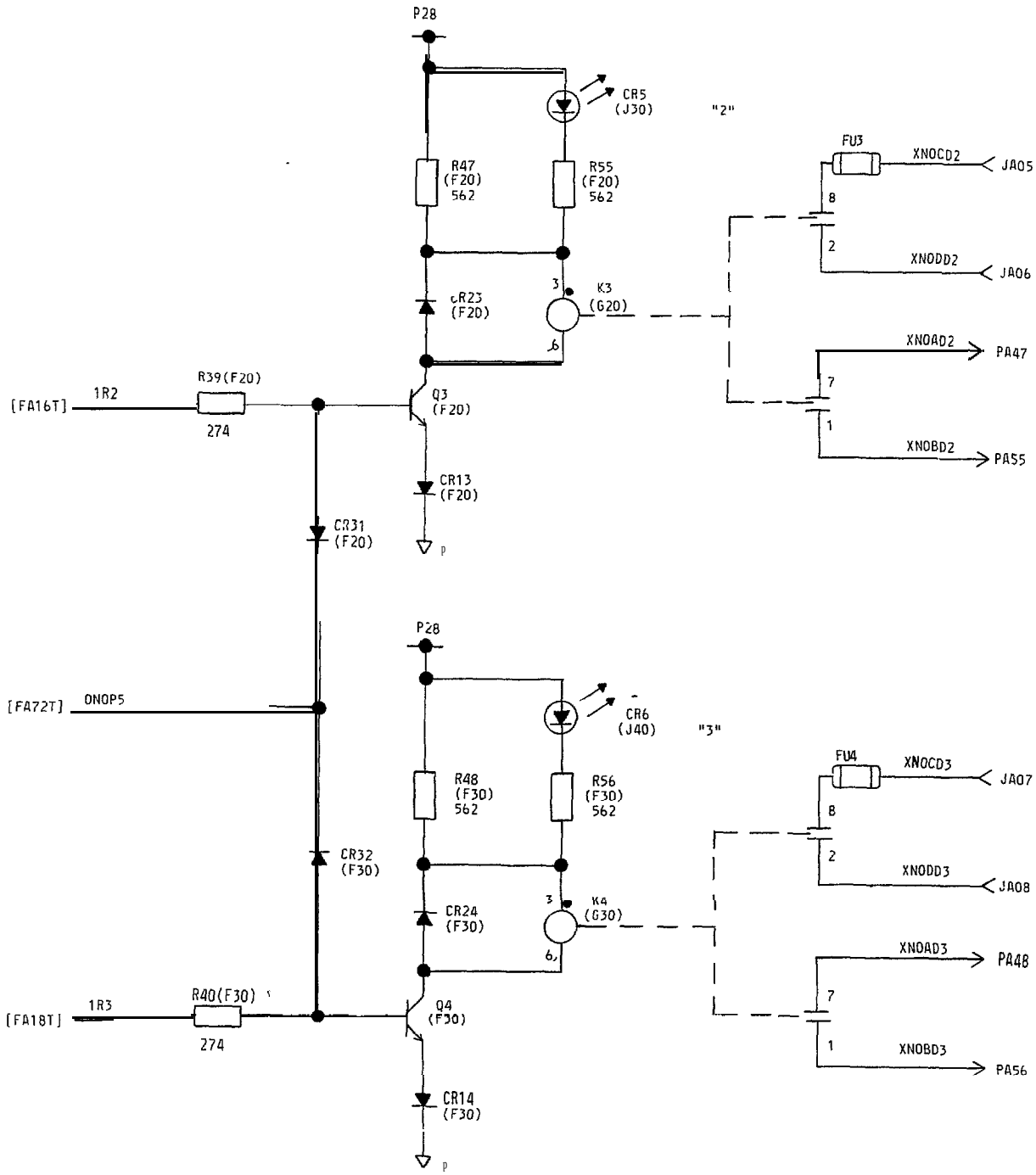


MH 2/2/78	REV 4 RE-	REV 5	PAINTS TO BHEL DL104	ENGINEER Rumilla 780203	GENERAL@ ELECTRIC	ELEMENTARY DIAGRAM
REV 2 OEK KSH 830726	ISSUED 2-6-78		PKI		DRIVE SYSTEMS DEPARTMENT	RELAY OUTPUT
REV 3 OER acb 840806	MADE BY R W. MILLER				SALEM. VA. U.S.A	DS 3800HRRB
						CONT. ON SH. 4FA SH. NO. 4EA



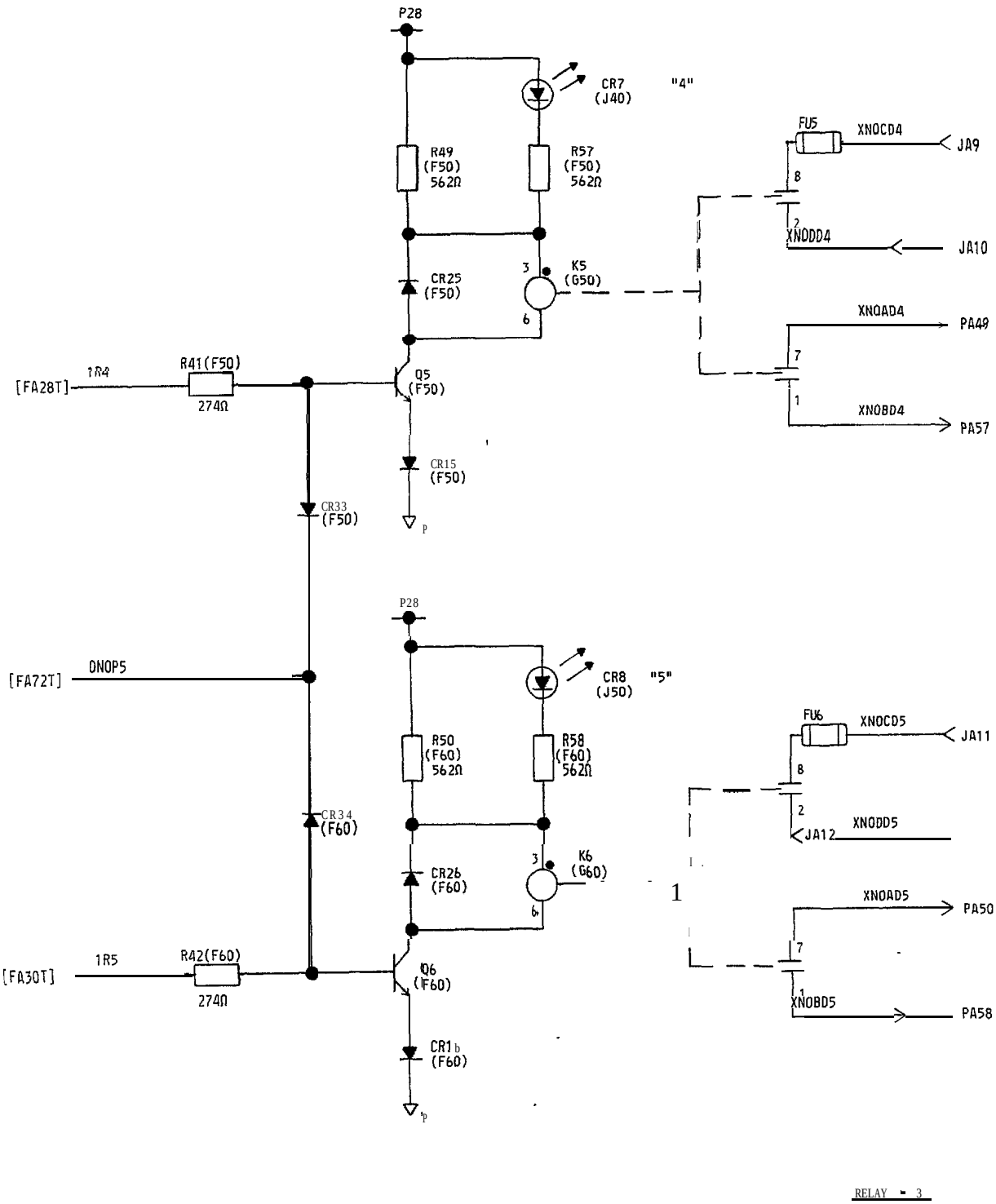
DEK REH 830726	REV 4	REV 5	PRINTS TO BHEL DL104 D 1	ENGINEER 780202	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA U.S.A.	ELEMENTARY DIAGRAM RELAY OUTPUT
REV 2	ISSUED 2-6-78					DS3800HRRB
REV 3	MADE BY R.W. MILLER					CONT. ON SH. 4HA SH. NO. 4GA

DSD 0152 (09 76)

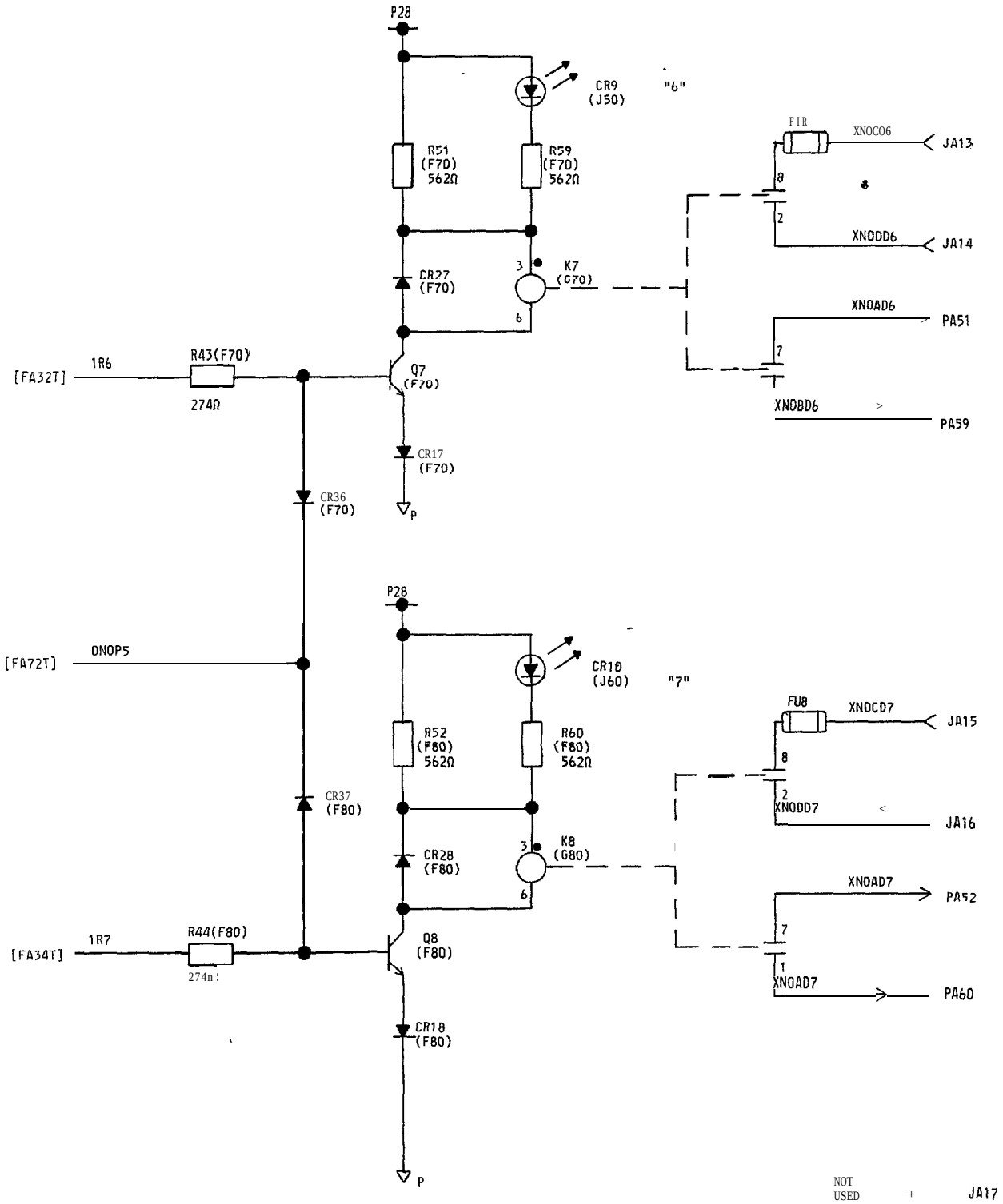


RELAY - 2

PER H 830726	REV 4	REV 5	PRINTS TO 8HEL DL104 JKI	ENGINEER R.W. Miller 783232	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A	ELEMENTARY DIAGRAM RELAY OUTPUT
REV 2	ISSUED 2-6-78					DS3800HRRB
REV 3	MADE BY RW MILLER					CONT. ON SH. 4JA SH. NO. 4

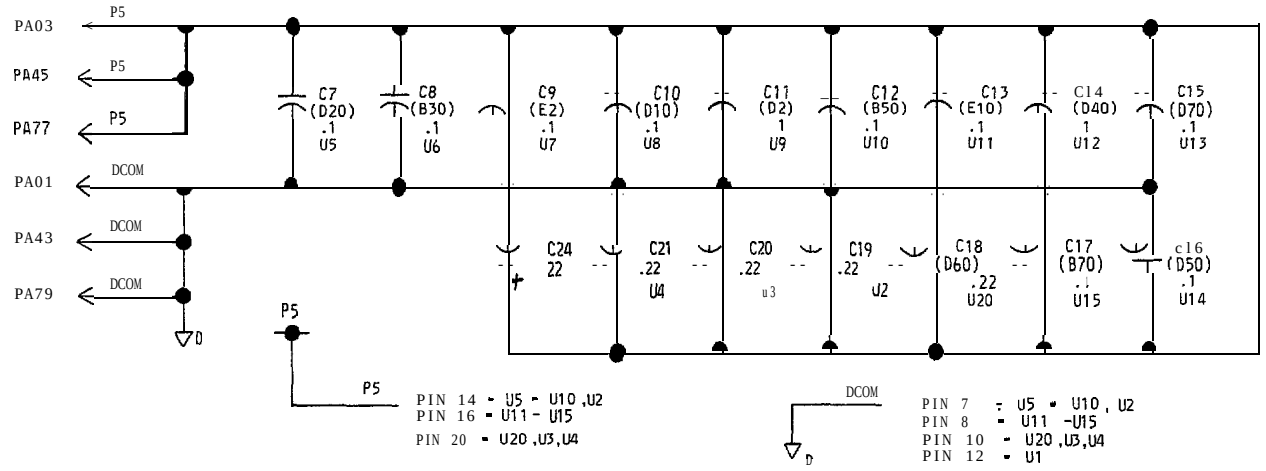


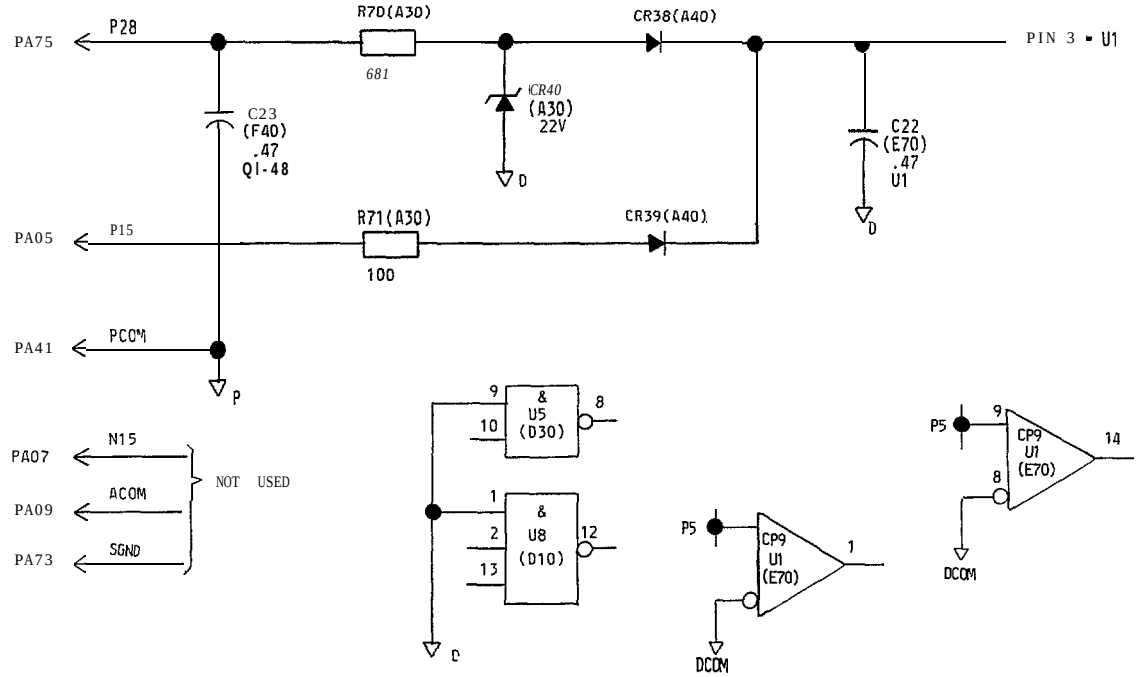
REV EL ECN 830726 REV 2	REV 4	REV 5	PRINTS TO BHEL DL104 PRI	ENGINEER 780202	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA U.S.A
REV 3	ISSUED J - d - 78 MADE BY R.W.MILLER				ELEMENTARY DIAGRAM RELAY OUTPUT DS3800HRRB CONT ON SH 4KA SH NO 4JA



RELAY - 4

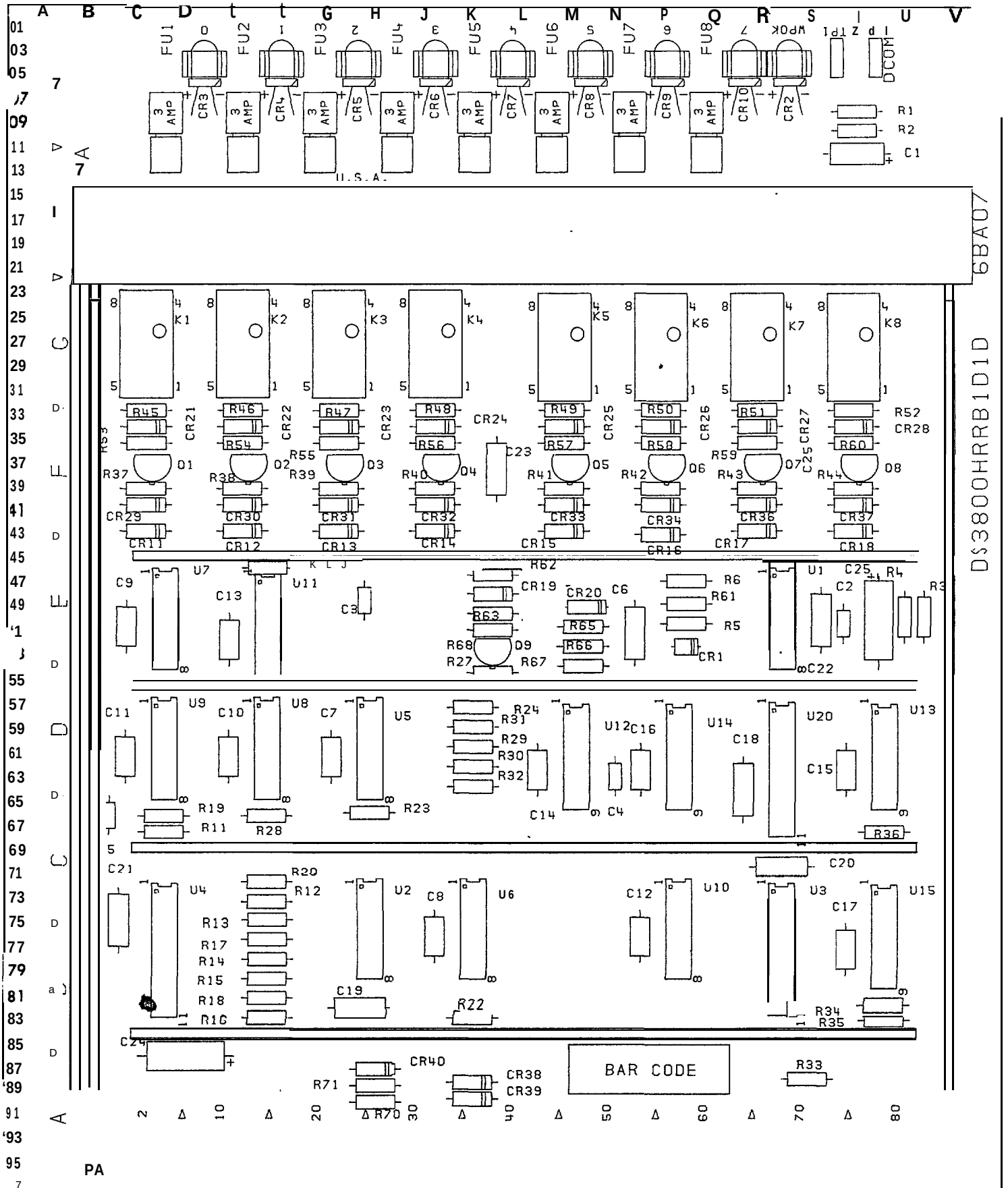
JEL REH 830726	REV 4	REV 5	PRINTS TO BHEL DL104 P.R1	ENGINEER R.W.Miller 780202	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV 2	ISSUED 2-6-78				DRIVE SYSTEMS DEPARTMENT	RELAY OUTPUT
REV 3	MADE BY R.W.MILLER				SALEM, VA. U.S.A	DS3800HRRB CONT. ON SH. 4LA SH. NO. 4KA



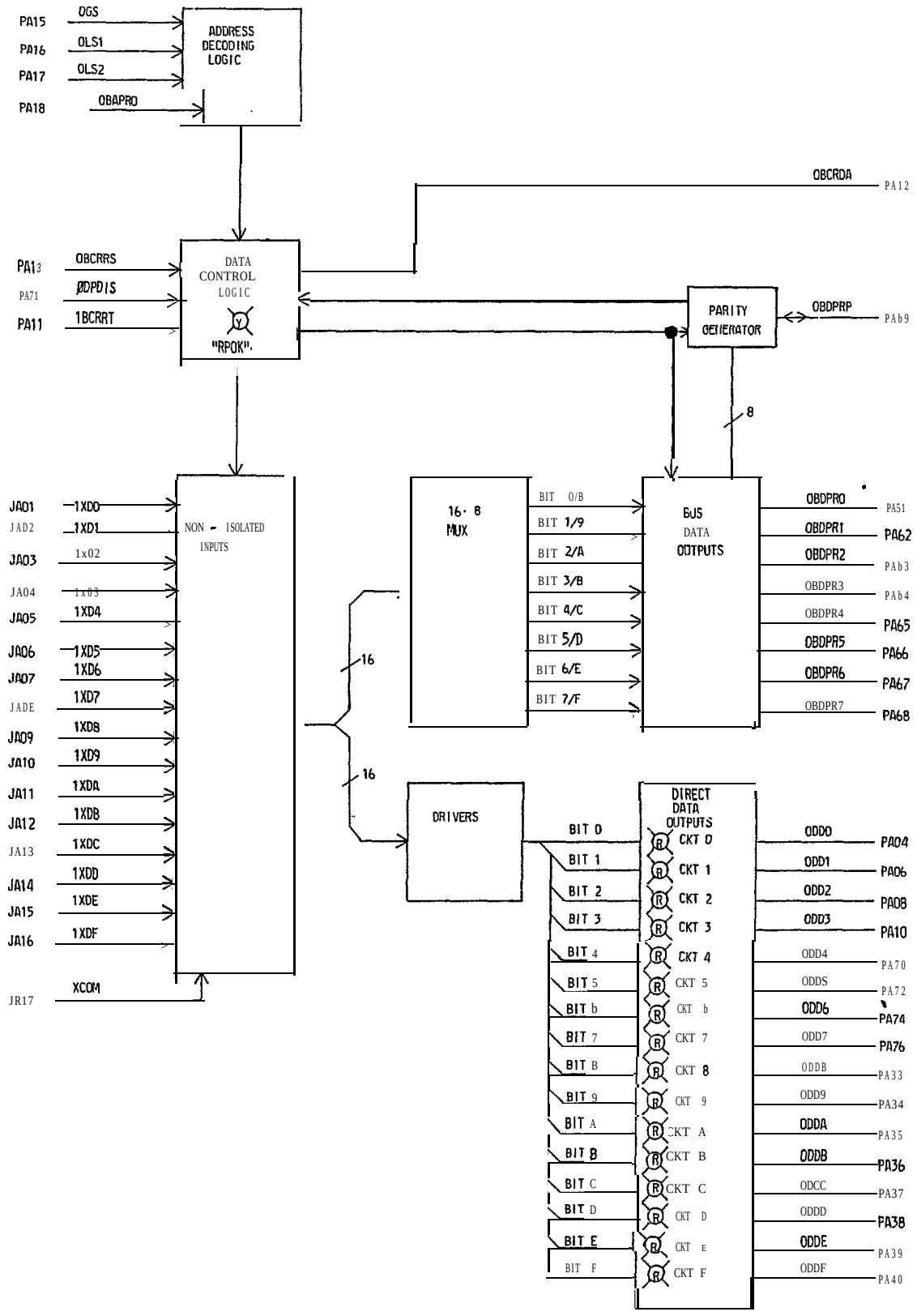


PWR/BYPASS

V 1 6-28-78	REV 4	REV 5	PRINTS TO BHEL DL10A P/1	ENGINEER Miller 78202	GENERAL@ ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA U.S A	ELEMENTARY DIAGRAM RELAY OUTPUT
REV 2 OEL eff 830726	ISSUED 2-6-79					DS3800HRRN
HE" 3 OER eff 840806	MADE BY R.W. MILLER					CGNT. ON SH 4MA SH NO 4L



REV 1 OEL REH 7-26-83	REV 4	REV 5	PRINTS TO BHEL DL104 PRT	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV 2 OER ALB 8-06-84	ISSUED RE 7-26-83/8-6-84			FIRST MADE FOR	DRIVE SYSTEMS DEPT. SALEM, VA., U.S.A.	RELAY OUTPUT
REV 3	MADE BY 7-26-83 R. HANNAH			T.C. NO		DS3800 HRRB
						CONT. ON SH. FL SH. NO. 4MA



REV. 1	REV 4	REV.5	PRINTS TO	ENGINEER	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
T DAVIDSON 8-9-82			DL102		DRIVE SYSTEMS DEPARTMENT	28VDC NON-ISOLATED INPUT
REV 2 OEP	ISSUED				SALEM, VA. U.S.A	PWB BLOCK DIAGRAM
REH 831130	12\12\78					DS3800HSCC
REV 3	MADE BY					CONT. ON SH2BA SH. NO. 2AP
	R.E. VANDERPOOL					

2.0 DEFINITION AND PURPOSE

THE HSCC IS A 28VDC NON-ISOLATED SIGNAL CONDITIONING CARD ON A SINGLE SIZE DIRECTOMATIC PLUS PRINTED WIRE BOARD THE SIXTEEN FRONTEDGE INPUTS, AFTER CONDITIONING, ARE AVAILABLE AS LOGIC OUTPUT SIGNALS FOR DIRECT USE OR CAN BE READ VIA THE DRIVE LEVEL INTERFACE.

2.1 THEORY OF OPERATION

THE HSCC FUNCTIONALLY CONSISTS OF SIXTEEN FILTERED 28 VOLT INPUTS, LOGIC OUTPUT DRIVERS, DRIVE LEVEL INTERFACE AND A BUS ISOLATION CIRCUIT.

THE ACTIVE LOW INPUT SIGNALS ARE FILTERED WITH A RC NETWORK AND BUFFERED WITH A SCHMITT INPUT GATE. THIS SIGNALS ARE BUFFERED AGAIN AND PRESENTED TO THE BACKPLANE PINS FOR DIRECT USE. A LED IS CONNECTED TO EACH OUTPUT DRIVER.

EITHER GROUP OF EIGHT PROCESS INPUTS CAN BE READ VIA THE DRIVE LEVEL INTERFACE. IF THE CARD IS NOT BEING RESET AND 'OGS', 'OBCRRS' AND EITHER 'OLS1' OR 'OLS2' ARE ACTIVE LOW, THE LOWER EIGHT BITS WILL APPEAR ON THE 'OBDPRD'-'OBDPRP' DATA LINES. THE ABOVE PLUS 'OBAPRO' ACTIVE LOW WILL CAUSE THE UPPER EIGHT BITS TO BE PRESENTED (SEE GEK-61680).

2.2 INTERFACE REQUIREMENTS

THE INPUTS AND OUTPUTS OF THE HSCC ARE LOW LEVEL TTL TYPE WITH THE EXCEPTION OF THE 28VDC NON-ISOLATED INPUTS. THE LOW LEVEL SIGNALS ARE EXCEPTED TO BE USED WITHIN AN ELECTRONICS MODULE OR VERTICAL TERMINATION ASSEMBLY. THE 28VDC NON-ISOLATED INPUT WIRING SHOULD BE NO LONGER THAN 100 FEET.

2.3 FUNCTIONAL DESCRIPTION

ALL TTL SIGNALS ARE ZERO ("0") TRUE EXCEPT FOR 1 BCRRT WHICH IS ONE ("1") TURE. THE NON-ISOLATED PROCESS DATA INPUTS ARE "1" TRUE.

2.3.1. INPUTS (ALL TTL INPUTS FLOAT TO A 'LOGIC ONE>

MNEMONIC	PIN NO. (SEE NOTE 1)	DESCRIPTION (SEE BLOCK DIAGRAM SH. 2AA)
0GS	PA15	GROUP SELECT
0LS1	PA16	LINEAR SELECT 1
0LS2	PA17	LINEAR SELECT 2
0BCRRS	PA13	READ STROBE
1 BERRT	PA1 1	BUS RESET
0B0PRP *	PA69	.0ATA BUS PARITY BIT
0BAPR0	PA18	UPPER/LOWER BYTE ADDRESS
1XD0	JA01	NON-ISOLATED PROCESS DATA BIT 0
1XD1	JA02	NON-ISOLATED PROCESS DATA BIT 1
1XD2	JA03	NON-ISOLATED PROCESS DATA BIT 2
1XD3	JA04	NON-ISOLATED PROCESS DATA BIT 3
1XD4	JA05	NON-ISOLATED PROCESS DATA BIT 4
1XD5	JA06	NON-ISOLATED PROCESS DATA BIT 5
1XD6	JA07	NON-ISOLATED PROCESS DATA BIT 6
1XD7	JA08	NON-ISOLATED PROCESS DATA BIT 7
1XD8	JA09	NON-ISOLATED PROCESS DATA BIT 8
1 x09	JA10	NON-ISOLATED PROCESS DATA BIT 9
1XDA	JA11	NON-ISOLATED PROCESS DATA BIT A
1XDB	JA12	NON-ISOLATED PROCESS DATA BIT B
I XOC	JA13	NON-ISOLATED PROCESS DATA BIT C
1XDD	JA14	NON-ISOLATED PROCESS DATA BIT 0
1X0E	JA15	NON-ISOLATED PROCESS DATA BIT E
1 X0F	JA16	NON-ISOLATED PROCESS DATA BIT F
XCOM	JA17	NON-ISOLATED PROCESS DATA FILTER COMMON
ØD P0 IS	PA71	PARITY DISABLE

NOTE 1: PA CONNECTOR = BACKPLANE CONNECTOR
JA CONNECTOR = FRONT EDGE CONNECTOR
* BI-DIRECTIONAL SIGNAL

REV. 1 OEP EH 831130	REV. 4	REV. 7	PRINTS TO DL102	ENGINEER DSM 12/12/78	GENERAL ELECTRIC DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	28VDC NON- I SOLATED INPUT PWB
REV. 2	REV. 5	ISSUED 12/12/78				DS3800HSCC
REV. 3	REV. 6	MADE BY R.E. VANDERPOOL				CONT. ON SH. 2DA SH. NO. 2CA

2.3.2 OUTPUTS

MNEMONIC	PIN ND.	DESCRIPTION (SEE BLOCK DIAGRAM SH. 2AA)
DBCRDA	PA12	DEVICE ACKNOWLEDGE
OBDPR0	PA61	DATA BUS BIT 0
OBDPR1	PA62	DATA BUS BIT 1
OBDPR2	PA63	DATA BUS BIT 2
DBDPR3	PA64	DATA BUS BIT 3
OBDPR4	PA65	DATA BUS BIT 4
OBDPR5	PA66	DATA BUS BIT 5
OBDPR6	PA67	DATA BUS BIT 6
OBDPR7	PA66	DATA BUS BIT 7
OBDPRP *	PA69	DATA BUS PARITY BIT
ODD0	PA04	DIRECT DATA BIT 0
ODD1	PAD6	DIRECT DATA BIT 1
ODD2	PA08	DIRECT DATA BIT 2
ODD3	PA10	DIRECT DATA BIT 3
ODD4	PA70	DIRECT DATA BIT 4
ODD5	PA72	DIRECT DATA BIT 5
ODD6	PA74	DIRECT DATA BIT 6
ODD7	PA76	DIRECT DATA BIT 7
ODD8	PA33	DIRECT DATA BIT 8
ODD9	PA34	DIRECT DATA BIT 9
ODDA	PA35	DIRECT DATA BIT A
ODDB	PA36	DIRECT DATA BIT B
ODDC	PA37	DIRECT DATA BIT C
ODDD	PA38	DIRECT DATA BIT D
ODDE	PA39	DIRECT DATA BIT E
ODDF	PA40	DIRECT DATA BIT F

* BI-DIRECTIONAL SIGNAL

2.3.3 INDICATING LED'S

THERE ARE SEVENTEEN LIGHT-EMITTING DIODES (LEO) LOCATED ON THE FRONT EDGE OF THE HSCC TO INDICATE THE STATE OF PROCESS INPUT LINES AND BUS PARITY CONDITION. THE LED'S ARE IDENTIFIED ACCORDING TO THEIR ASSOCIATED, FUNCTION.

LEO	DESCRIPTION	COLOR
RPOK	LUMINOUS WHEN READ PARITY OK	YELLOW
0	LUMINOUS WHEN PROCESS INPUT 0 IS "1"	RED
1	LUMINOUS WHEN PROCESS INPUT 1 IS "1"	RED
2	LUMINOUS WHEN PROCESS INPUT 2 IS "1"	RED
3	LUMINOUS WHEN PROCESS INPUT 3 IS "1"	RED
4	LUMINOUS WHEN PROCESS INPUT 4 IS "1"	RED
5	LUMINOUS WHEN PROCESS INPUT 5 IS "1"	RED
6	LUMINOUS WHEN PROCESS INPUT B IS "1"	RED
7	LUMINOUS WHEN PROCESS INPUT 7 IS "1"	RED
8	LUMINOUS WHEN PROCESS INPUT 8 IS "1"	RED
9	LUMINOUS WHEN PROCESS INPUT 9 IS "1"	RED
A	LUMINOUS WHEN PROCESS INPUT A IS "1"	RED
B	LUMINOUS WHEN PROCESS INPUT B IS "1"	RED
C	LUMINOUS WHEN PROCESS INPUT C IS "1"	RED
D	LUMINOUS WHEN PROCESS INPUT D IS "1"	RED
E	LUMINOUS WHEN PROCESS INPUT E IS "1"	RED
F	LUMINOUS WHEN PROCESS INPUT F IS "1"	RED

2.3.4 MAINTAINABILITY

THE HSCC HAS BEEN DESIGNED TO REQUIRE A MINIMUM OF MAINTENANCE TOOLS. IF A PARTICULAR CARD IS SUSPECTED OF BEING FAULTY, IT MAY BE TESTED USING THE FOLLOWING DIAGNOSTIC PROCEDURE. ESTABLISH KNOWN INPUTS TO THE HSCC AND VERIFY PROPER DIRECT DATA OUTPUTS VIA THE LED'S ALSO, AGAIN WITH KNOWN INPUTS ESTABLISHED AND WITH THE USE OF A SUITABLE MAINTENANCE AID, READ THE PWB AND VERIFY PROPER BUS OPERATION.

2.4 APPLICATION DATA

2.4.1 POWER REQUIREMENTS

P5 .633 AMPS MAXIMUM
P15 .003 AMPS MAXIMUM
P28 .012 AMPS MAXIMUM

2.4.2 HEAT DISSIPATION

THE MAXIMUM POWER DISSIPATION IS 4.36 WATTS WITH ALL PROCESS INPUTS AT 28V.

2.4.3 INSTALLATION PROCEDURE AND ADJUSTMENTS

NONE

2.4.4. TTL INPUTS

LOGIC ZERO = 0.4 ±0.4VDC AT .58MA MAX
LOGIC ONE = 3.75 ±1.75VDC AT 0.0MA MAX
HYSTERESIS= .400 VOLTS MINIMUM
EACH INPUT IS EQUAL TO 83% OF ONE STANDARD LOGIC LOAD (0.83 SLL)

2.4.5 TTL OUTPUTS

LOGIC ZERO = 0.25 ± 0.25VDC
LOGIC ONE = 4.0 ± 1.5VDC
HI-Z = HIGH IMPEDANCE STATE WITH +20 OR -20ua LEAKAGE.

2.4.5.1 SIGNALS OBDPRO-OBDPRP

SINK 19MA (27 SLL)
SOURCE 12MA (>27 SLL)

2.4.5.2 OBCRDA

SINK 19.0MA (27 SLL)
SOURCE 0.12MA (>27 SLL)

REV. 1 T.D. 8-9-82	REV. 4	REV. 7	PRINTS TO DL1 02	ENGINEER	GENERAL@ ELECTRIC	28VDC NON-ISOLATED INPUT PWB
REV. 2	REV. 6	ISSUED 12/12/78			DRIVE SYSTEMS DEPT.	DS3800HSCC
REV. 3	REV. 6	MADE BY R. E. VANDERPOOL			SALEM, VA. U.S.A.	CONT. ONSH. 2HA SH. NO. 2GA

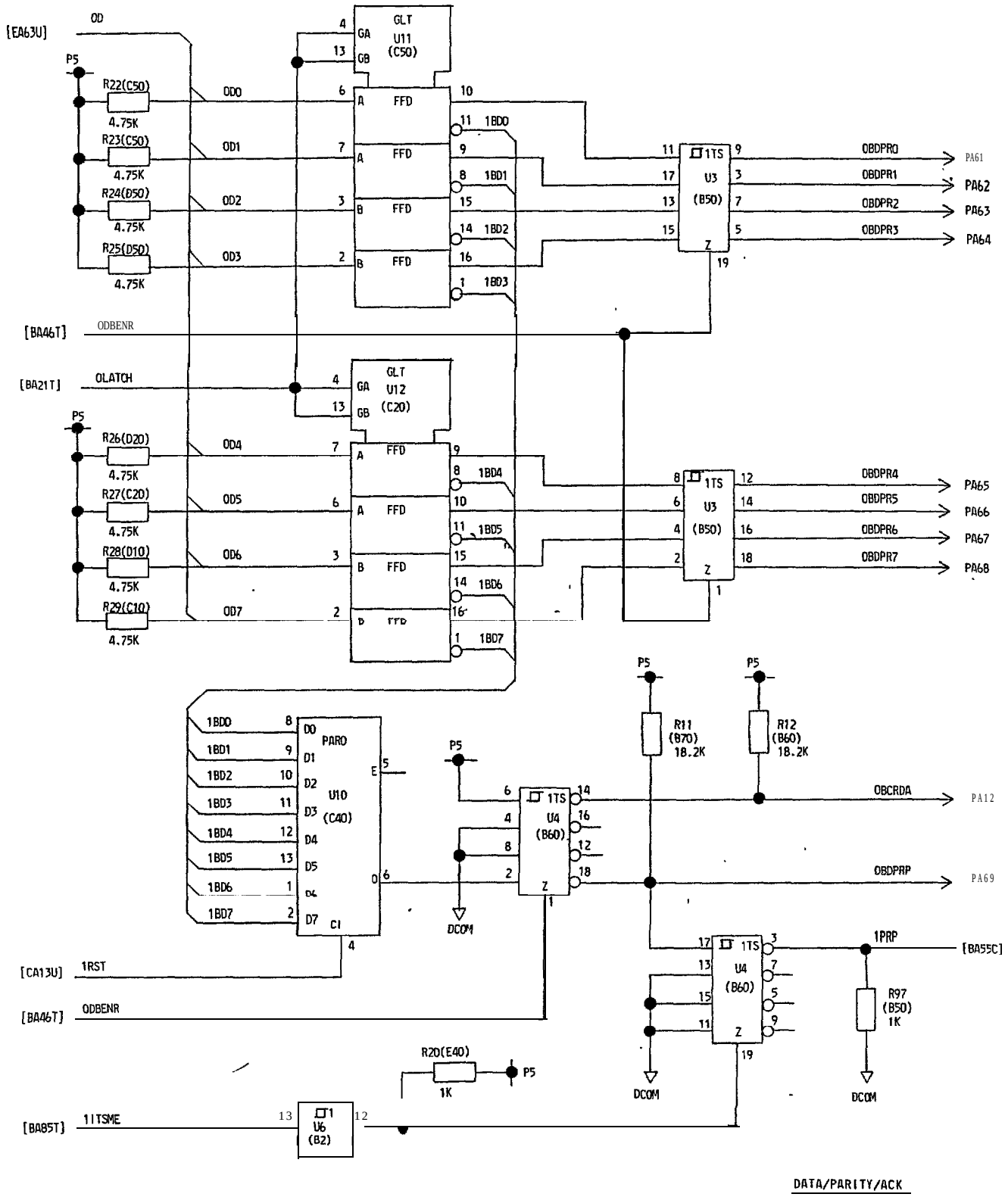
2.4.5.2 SIGNALS ODD0-ODDF

SINK . 10MA(1 4 SLL)
SOURCE 10MA(>1 4 SLL)

2.4.6 PROCESS INPUTS

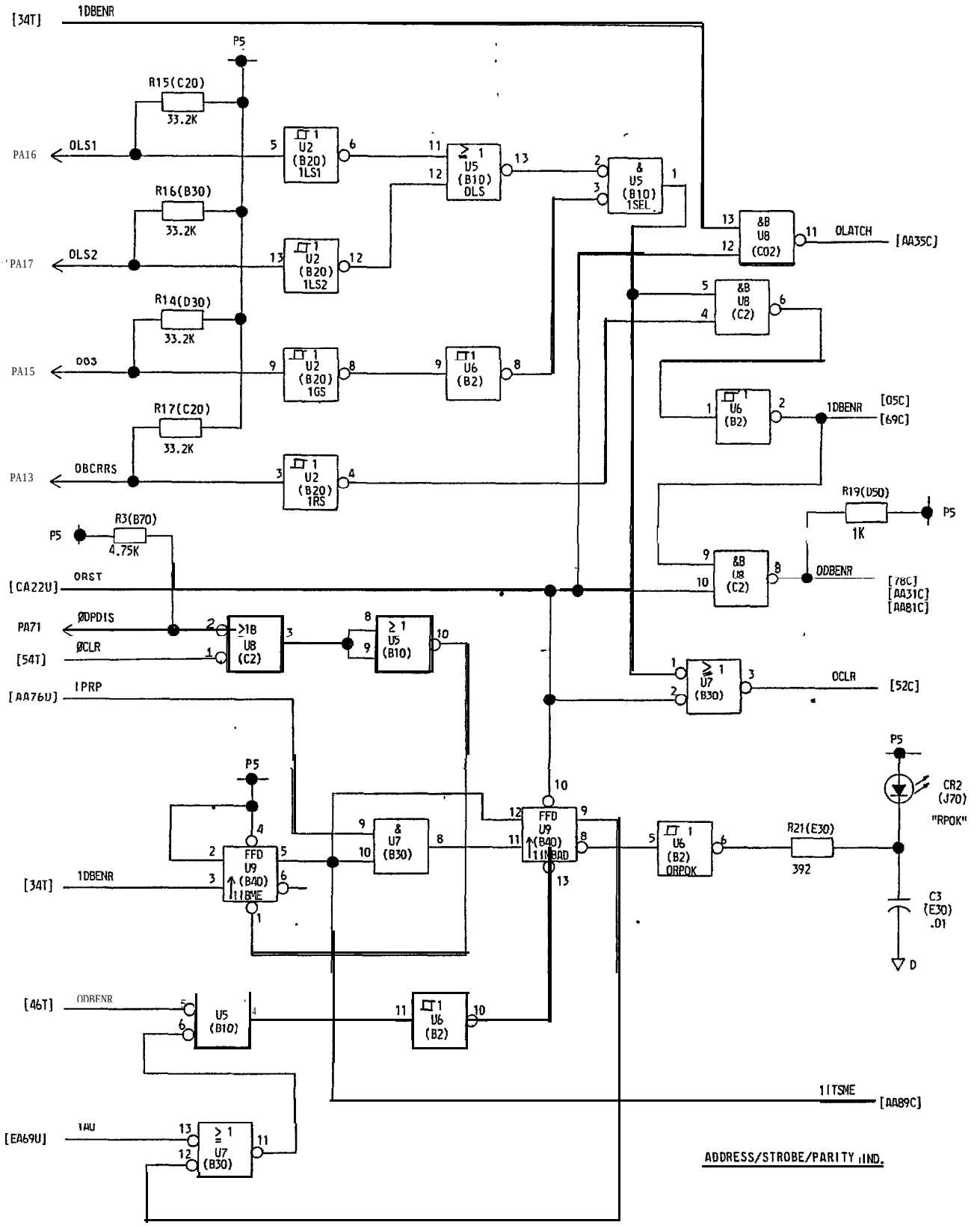
THE PROCESS INPUTS HAVE A TYPICAL "0" TO "1" THRESHOLD VOLTAGE OF 11.45V, AND A "1" TO "0" THRESHOLD VOLTAGE OF 2.841. THE INPUT AT 28V DRAWS APPROXIMATELY 1.5 MILLIAMP (1 9Ω LOAD).

HEV. 1	REV. 4	REV. 7	PA INTS TO DL102	ENGINEER	GENERAL ELECTRIC	28VDC NON-ISOLATED INPUT PWB
REV. 2	REV. 5	ISSUED 12/12/78				
REV. 3	REV. 6	MADE BY R.E. VANDERPOOL			SALEM, VA. U.S.A.	DS3800HSCC CONT. ON SH. FL, SH. NO. 2HA



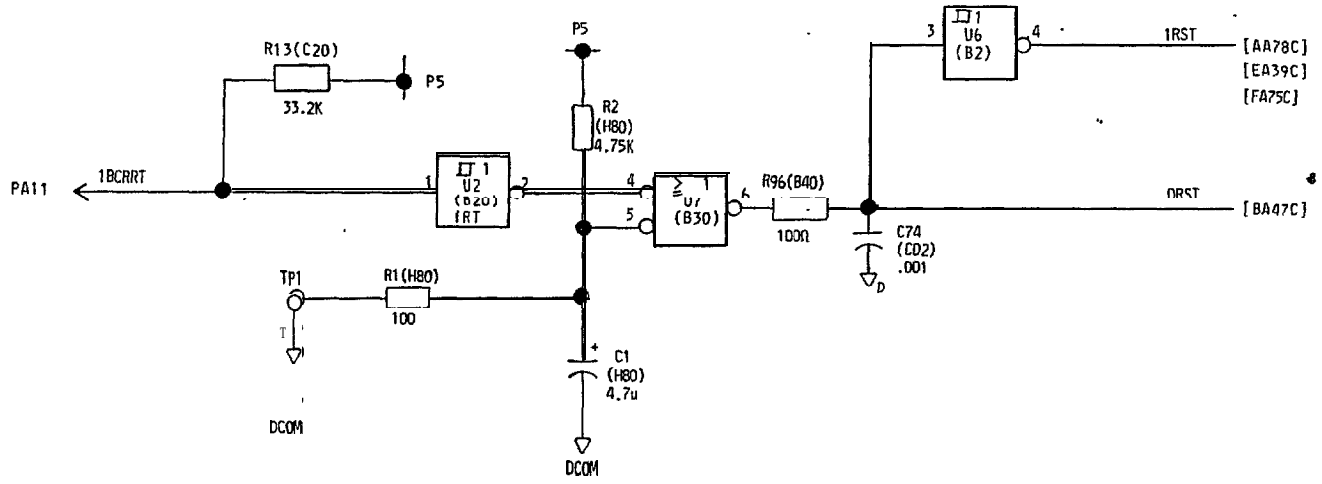
REV.1 OEQ REN 831130	REV. 4	REV 5 ,	PRINTS TO DL104	ENGINEER YLC	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM 28 VDC NONISOLATED INPUT
REV.2	ISSUED 4-14-78					
REV 3	MADE BY R. E. VANDERPOOL 4/5/78					DS3800HSCC CONT. ON SH. 4BA SH. NO. 4DA

CARS



REV 1 OEQ EEH 831130	REV 4	REV 5	PRINTS TO DL104	ENGINEER YLC	GENERAL@ ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM 28 VDC NONISOLATED INPUT
EV 2	ISSUED 4-14-78					
REV.3	MADE BY R. E. VANDERPOOL 4/5/78					DS3800HSCC CONT. ON SH. 4CA SH. NO. 4BA

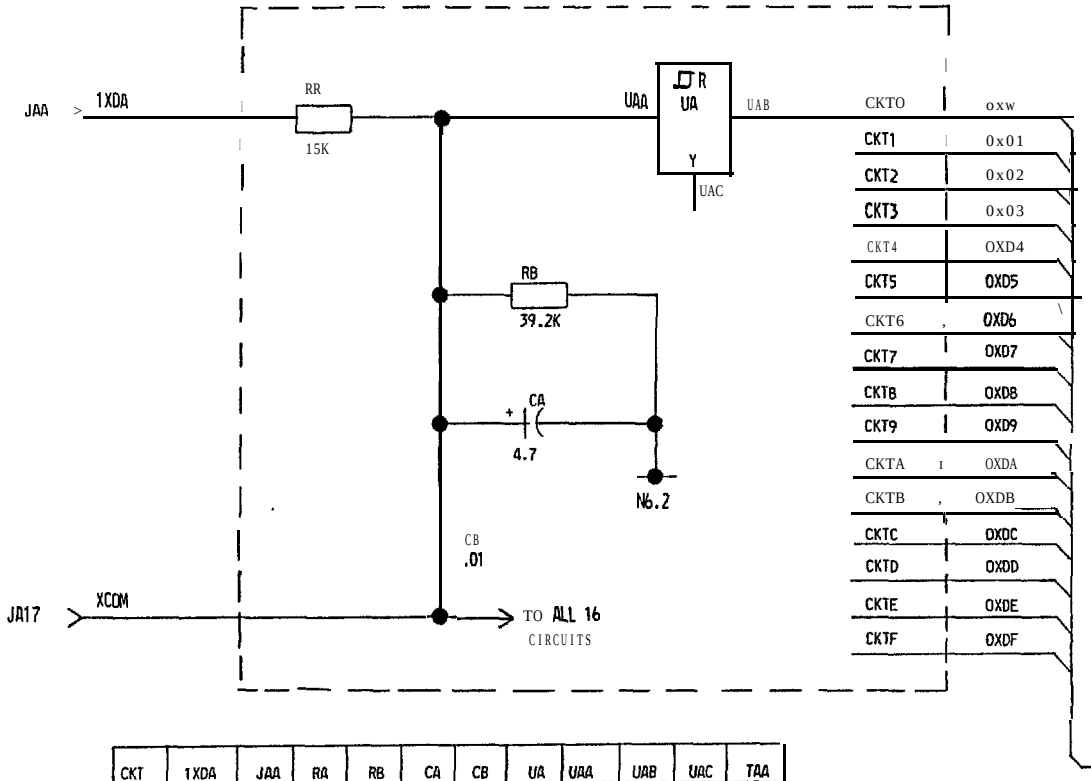
DSD 0192 (09 76)



REV. 1 OEQ REH 831130	REV. 4	REV 5	PRINTS TO DL104	ENGINEER YLC
REV 2	ISSUED 4-14-78			
REV 3	MADE BY RL R.E. VANDERPOOL 4/5/78			

GENERAL ELECTRIC
DRIVE SYSTEMS DEPARTMENT
SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM
28 VDC NONISOLATED INPUT
DS3800HSCC
CONT. ON SH. 4DA SH. NO. 4CA



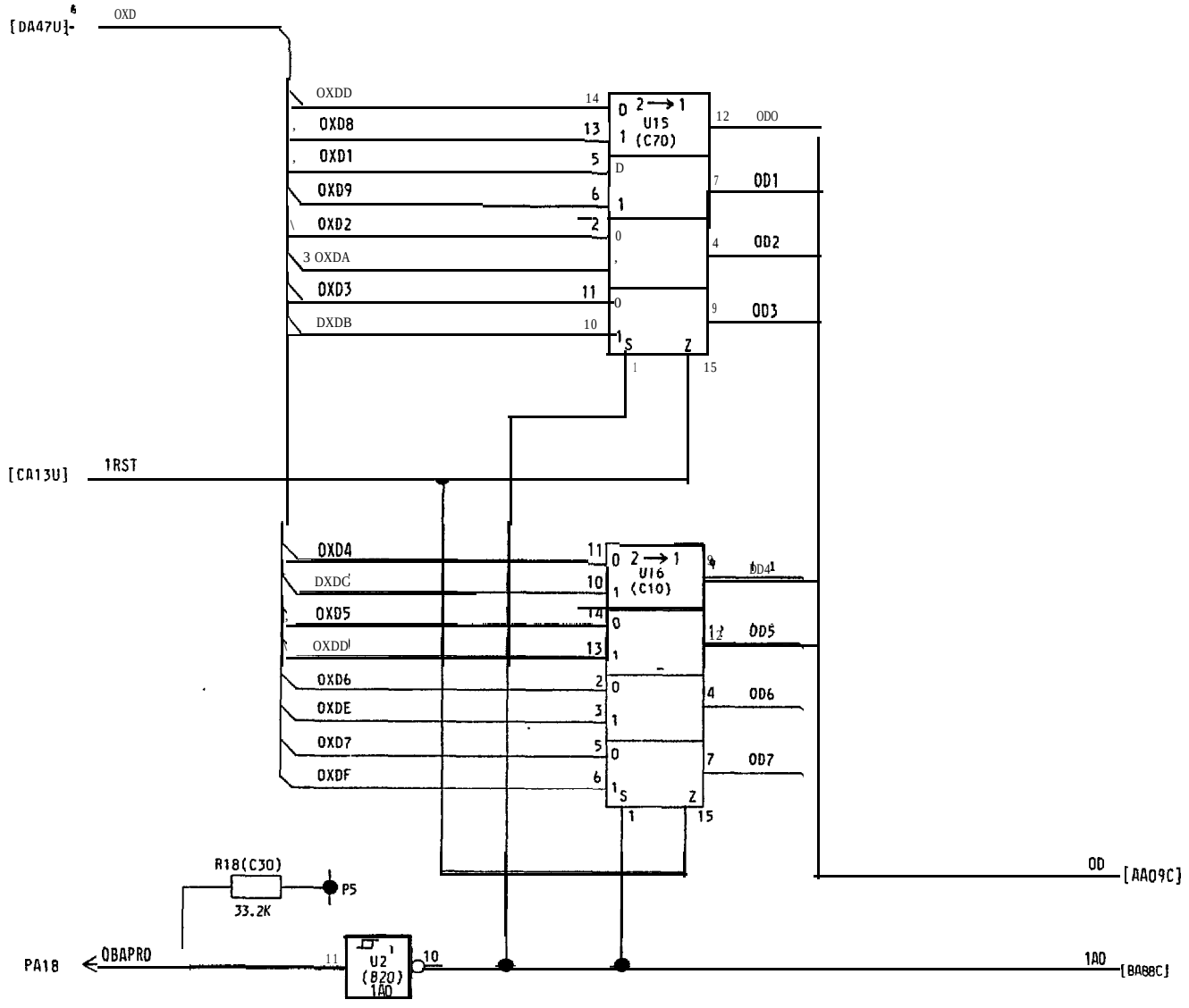
CKT	1XDA	JAA	RA	RB	CA	CB	UA	UAA	UAB	UAC	TAA
0	1XD0	JA01	R46 (G02)	R62 (F02)	C4 (F02)	C20 (G02)	U17 (E02)	1	3	2	1
1	1XD1	JA02	R47 (G02)	R63 (G02)	C5 (F02)	C21 (G10)	U17 (E02)	4	6	5	3
2	1XD2	JA03	R48 (G10)	R64 (G02)	C6 (F10)	C22 (G10)	U17 (E02)	10	8	9	11
3	1XD3	JA04	R49 (G10)	R65 (G10)	C7 (F10)	C23 (G20)	U17 (E02)	13	11	12	2
4	1XD4	JA05	R50 (G20)	R66 (G10)	C8 (F20)	C24 (G20)	U18 (E20)	1	3	2	8
5	1XD5	JA06	R51 (G20)	R67 (G20)	C9 (F20)	C25 (G30)	U18 (E20)	4	6	5	10
6	1XD6	JA07	R52 (G30)	R68 (G20)	C10 (F30)	C26 (G30)	U18 (E20)	10	8	9	4
7	1XD7	JA08	R53 (G30)	R69 (G30)	C11 (F30)	C27 (G40)	U18 (E20)	13	11	12	9
8	1XD8	JA09	R54 (G40)	R70 (G30)	C12 (F40)	C28 (G40)	U19 (E50)	1	3	2	15
9	1XD9	JA10	R55 (G40)	R71 (G40)	C13 (F40)	C29 (G50)	U19 (E50)	4	6	5	17
A	1XDA	JA11	R56 (G50)	R72 (G40)	C14 (F50)	C30 (G50)	U19 (E50)	10	8	9	25
B	1XDB	JA12	R57 (G50)	R73 (G50)	C15 (F50)	C31 (G60)	U19 (E50)	13	11	12	16
C	1XDC	JA13	R58 (G60)	R74 (G50)	C16 (F60)	C32 (G60)	U20 (E80)	4	6	5	24
D	1XDD	JA14	R59 (G60)	R75 (G60)	C17 (F70)	C33 (G70)	U20 (E80)	1	3	2	22
E	1XDE	JA15	R60 (G70)	R76 (G60)	C18 (F70)	C34	U20 (G80)	(E80)	10	8	9	26
F	1XDF	JA16	R61 (G70)	R77 (G70)	C19 (F80)	C35 (G80)	U20 (E80)	13	11	12	23

PROCESS INPUT
SIGNAL CONDITIONING

REV 1	REV 4	REV.5	PRINTS TO DL104	ENGINEER YLC	GENERAL@ ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A
REV 2	ISSUED 4-14-78				
REV 3	MADE BY . R. E. VANOERPOOL 4/5/78				

ELEMENTARY DIAGRAM
28 VOC NONISOLATED INPUT
DS3800HSCC
CONT. ON SH. 4EA SH. NO. 4DA

DSD 6193 (09 76)

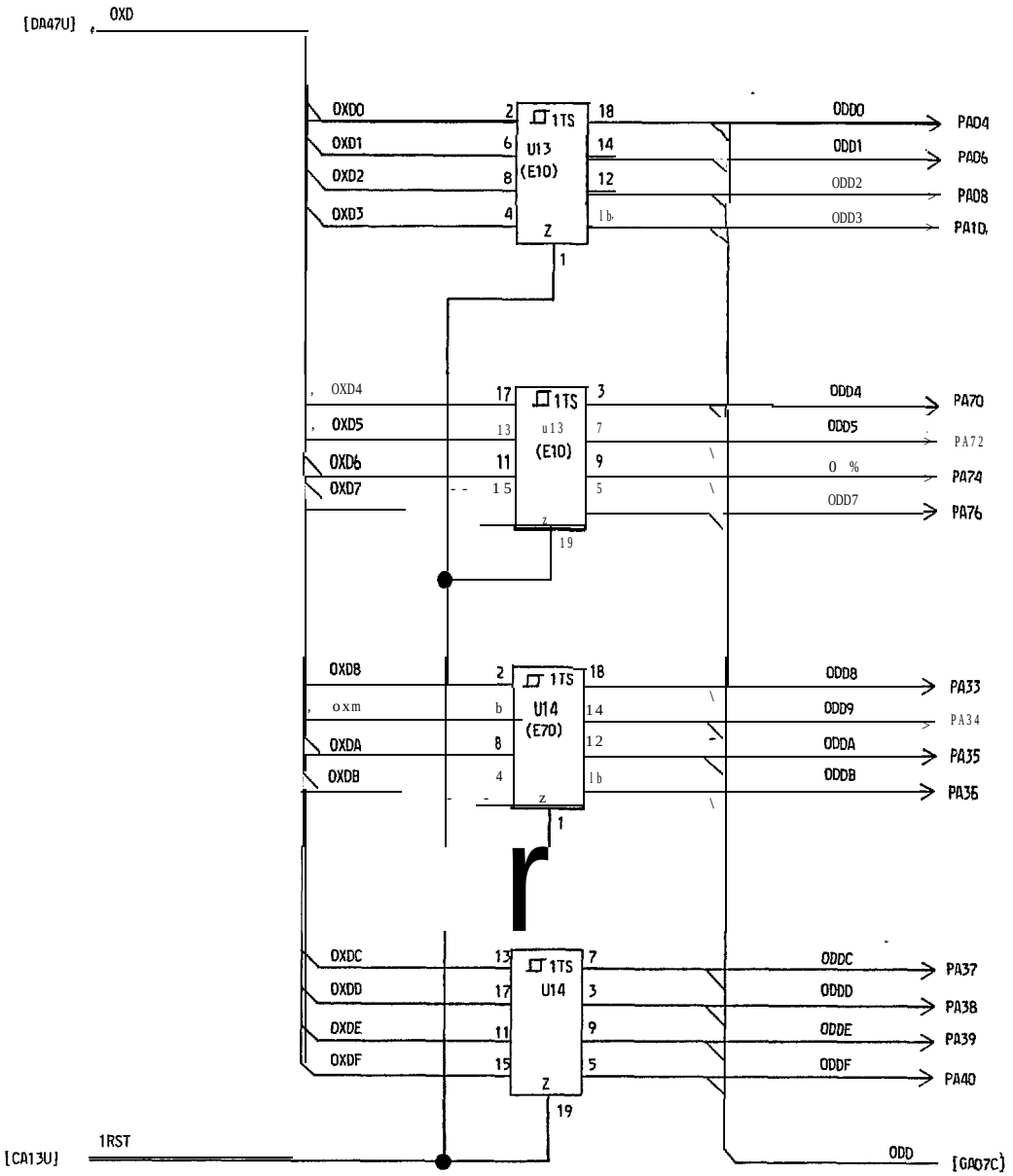


BUS DATA MUX

16:8

REV 1	REV. 4	REV.5	PRINTS TO DL1D4	ENGINEER YfC	GENERAL@ ELECTRIC	ELEMENTARY DIAGRAM
REV 2	ISSUED 4-14-78				DRIVE SYSTEMS DEPARTMENT	28 VDC NONISOLATED INPUT
REV 3	MADE BY R. E. VANDERPODL 4/5/78				SALEM, VA. U.S.A.	DS3800HSCC CONT. ON SH. 4FA SH. NO. 4EA

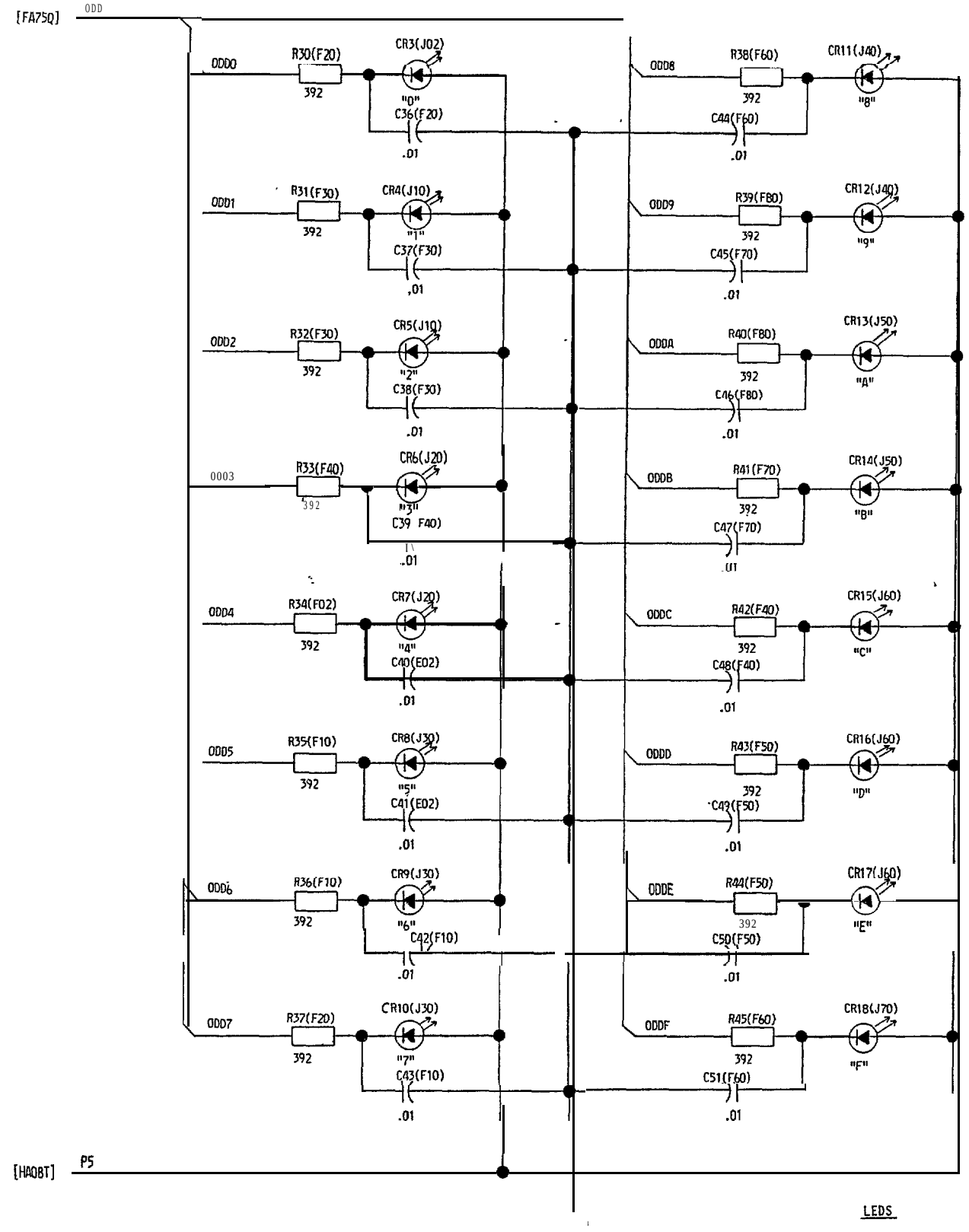
DSD 0192 (09 76)



DIRECT DATA

REV. 1	REV 4 REV.5	PRINTS TO DL184	ENGINEER YLC	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A	ELEMENTARY DIAGRAM 28 VDC NONISOLATED INPUT
REV 2	ISSUED 4-14-78				
REV 3	MADE BY RL R. E. VANDERPOOL 4/5/78				DS 3800HSCC CONT. ON SH. 4GA SH. NO. 4FA

OSD 0195 (09 78)

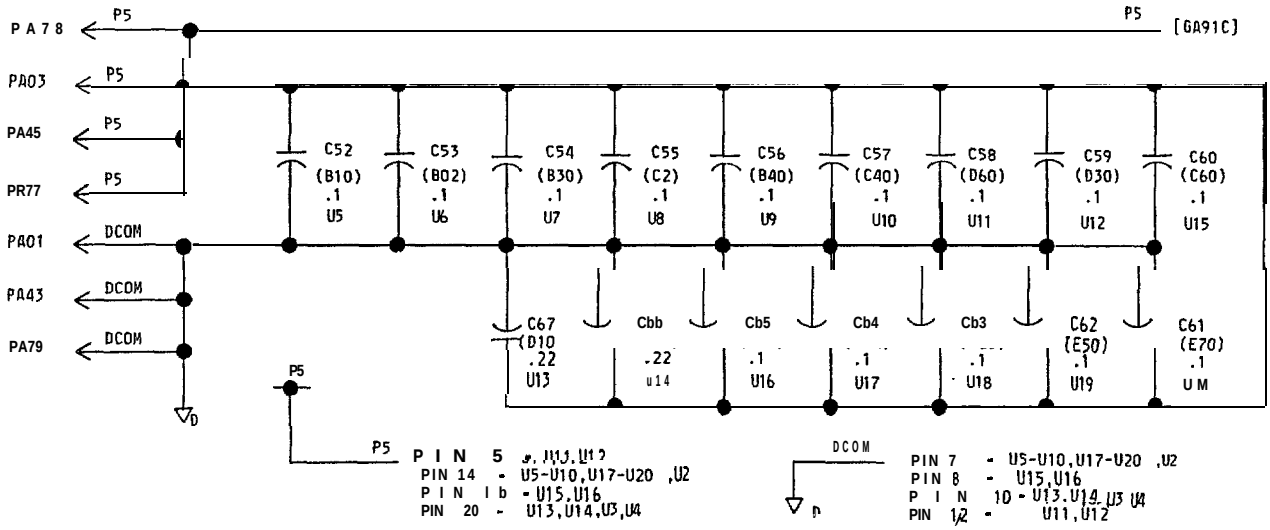


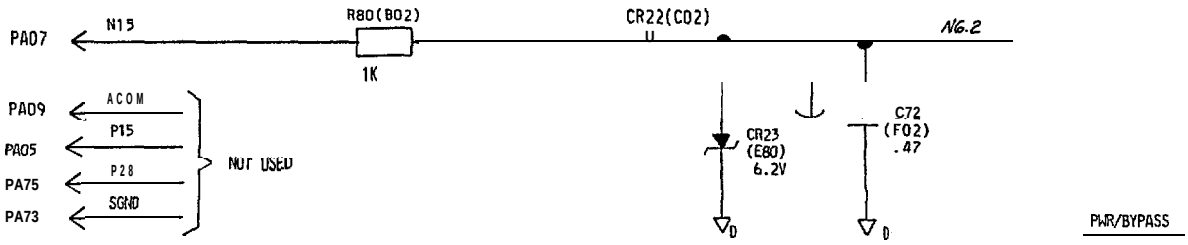
REV 1	REV 4	REV.5	PRINTS TO DL104	ENGINEER YLC	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM 28 VDC NONISOLATED INPUT
REV 2	ISSUED 4-14-78					
REV 3	MADE BY R. E. VRNDERPOOL 4/5/78					D S 3 8 0 0 H S C C CONT. ON SH. 4HA SH. NO 4GA

DSD 0152 (09 76)

A B C D E F G H J K L M N P Q R S T U V

NOTE; THIS P5 TIES THE LEDS TO PA78 AND THEN TO PA77 AT PA CONN.



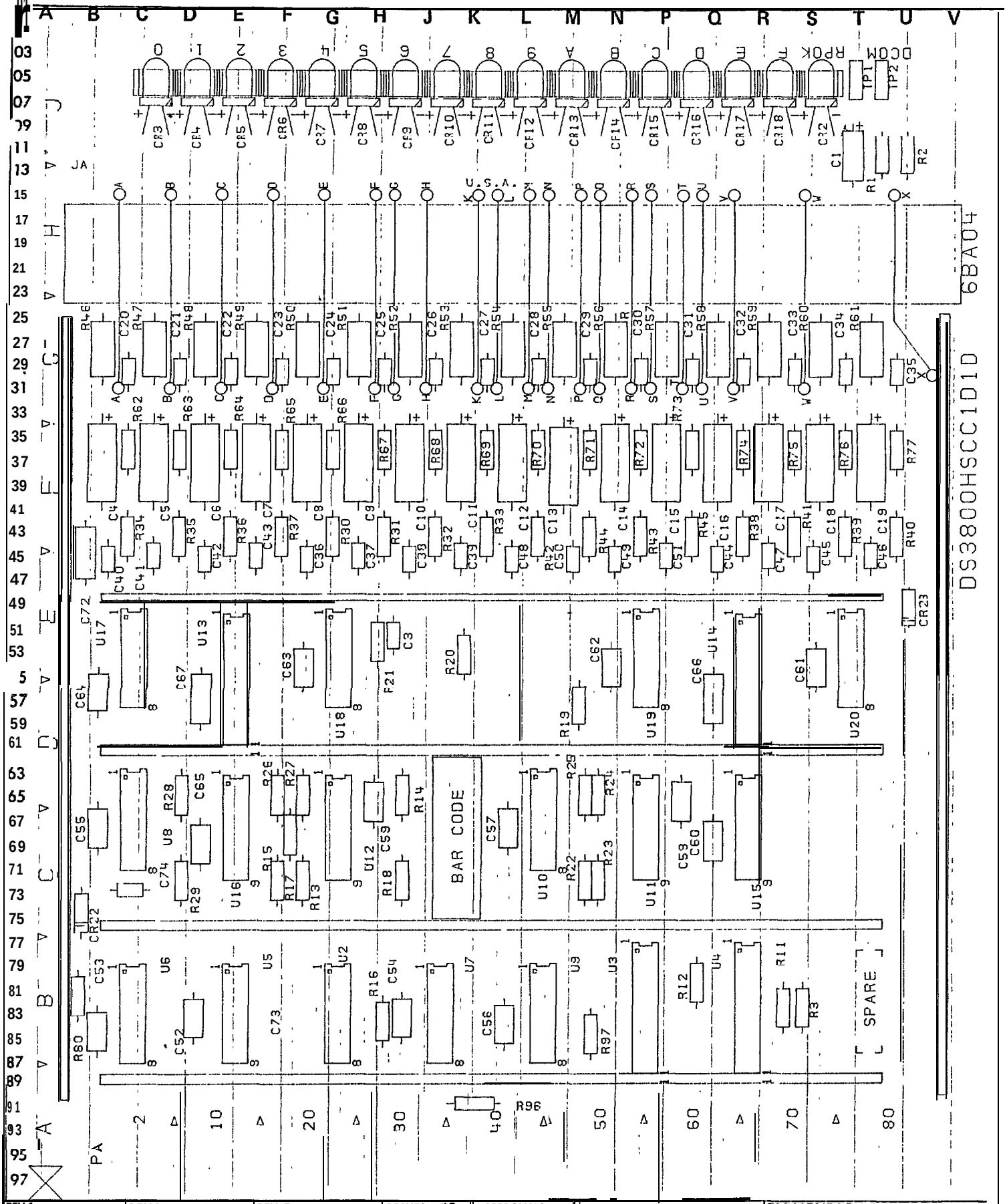


REV.1 OEQ REV 9311 30	REV 4	REV 5	PRINTS TO DI.104	ENGINEER YEC	GENERAL@ ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.
REV 2	ISSUED 4-14-78				
REV 3	MADE BY R. E. VANDERPOOL 4/5/78				

ELEMENTARY DIAGRAM
28 VDC NONISOLATED INPUT

DS 3800HSCC

CONT. ON SH. 4 JA SH. NO. 4HA



REV 1	REV4	REV 5	PRINTS TO DL100	APPROVALS RSH	GENERAL ELECTRIC
REV.2	ISSUED 11-30-83			FIRST MADE FOR	DRIVE SYSTEMS DEPT. SALEM, VA., U.S.A.
REV3 CEP REH 831130	MADE BY R.E. HANNAH			TC.NO	

ELEMENTARY DIAGRAM
ASSEMBLY PATTERN
28VDC NONISOLATED INPUT
DS3800HSCC
CONT. ON SH. FL. SH. NO. 4 OF 4

FORM IC 2349 (04-76)

POWER SYSTEM STABILIZER

WITH GENERIC INTERFACE

FOR 3Ø SENSING

3S7932LA4ØØ

G5 WITH MODEM AND TRANSIENT EVENT RECORDER

G6 WITH MODEM, NO TRANSIENT EVENT RECORDER

G7 NO MODEM, NO TRANSIENT EVENT RECORDER
(MODEM ON SH. 10E NOT INCLUDED IN. SYSTEM)

G8 NO MODEM, INCLUDE5 TRANSIENT EVENT RECORDER
(MODEM ON SH. 10E NOT INCLUDED IN SYSTEM)

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT
SALEM, VA USA

TITLE SH ET
238B1249 Ø1A

I. GIBBS
1 STANDARD
4932

ORIG

CONT ON SH 018

238B249

TABLE OF CONTENTS

SHEET	DESCRIPTION
000	TITLE SHEET
01A	TABLE OF CONTENTS
01B	SYSTEM ADDRESSES
OZA.E	BERG JUMPERS
10A	POWER SUPPLY
10B	GROUNDING
10C	<PCM > POWER CONVERSION MODULE
10D	<SYN > SYNCHRONIZING MOD.
10E	< MDM > MODEM
10F	<LCP > LOCAL CONTROL DS3820DPAD
12A	CONTACT INPUTS
12B	RELAY CONTACTOUTPUTS
	<MEM1> CIRCUIT BOARDS
20A	<MEM> CARD LOCATIONS
20B,C	<HEM> BACKPLANE
	<MEM> ROW I
31A	OPTIONAL PLNF
31B	DS3800HADA BUS CONV.
31C	DS3800NGMA WITH DAUGHTER BOARD DS3800DGPA
31D	OPTIONAL HUMB
31E	OPTIONAL HUMB
31F	DS3800HAIC ANALOG INPUT ED.
31G	OPTINAL HUMB
31H	DS4815PSSD MEMORY BD
31J	DS4815PSSA MEMORY BD.
31K	DS3800HXPD INTERRUPT EXPANDER
31L	DS3815DMCC MICROPROCESSOR
31LA	DS3815DMCC DAUGHTER BOARD
	<MEM> ROW 2
32A	DS3800HSCC CONTACT INPUTS
32B	DS3800HRRB RELAY BD. EXT. USE
32C	DS3800NGMA WITH DAUGHTER BOARD DS3800DGPA
32D	DS3800IIRRB RELAY ED. INT USE
32E	DS3800NOAA ANALOG OUTPUT
32F	DS3800NDAC D/A CONVERTER
32J	DS3800NFTA FREQUENCY TRANSDUCER
	DS3800NOFA QUADRATIC FILTER
32L	DS3800NPTB COMPLEX POWER BD.

SIZE	CD
DRAWING NO	238B 249
CONT ON SH	
TITLE	TABLE OF CONTENTS
	GENERAL ELECTRIC COMPANY
	DRIVE SYSTEMS DEPT SALEM, VA U.S.A.
ENGINEER	I. GIBBS
DRAWN	LP 3-23-89
CHECKED	
ISSUED	03-30-88
APPROVED	I. GIBBS
MFG APPROVAL	
QUALITY APPROVAL	
PRODUCTION RELEASE	STANDARD
TECH PUBS RELEASE	
PRINTS TO	
TECH PUBS NO	4932
MFG PART NO	
NEXT ASSY	
USED ON	

SYSTEM ADDRESSES

MEMORY

		ADDR
HMPK (IL) PROM	DMC PROD CODE	F0000
HMPK (IL) RAM	FIXED MEM	
	DYNAMIC MEM	E8000
PSSD (IH) EEPROM	CBL LIBRARY BLOCKS	E0000
	VARDEF SYMBOL TABLE	D0000
	////////////////// /'\/\A/\/\/\/\/\ /'\/\/\/\/\/\/\A //////////////////	40000
PSSA (IJ) RAM	RAM	
	//////////////////	30000
DP RAM	RES. FOR HLNA	2C000
PSSD (IH) EEPROM	EEPROM BASE SYSTEM	28000
PSSA (IJ) EPROM	DAS CODE FOR USER MONITOR	20000
	DAS CONSTANTS	
HMPK (IL)	DAS RAM	10000
	STACK DMC BASE SYSTEM (SEGMENT CODE)	4000
	DMC VARIABLES	0000

HAIC SLOT IF I/O ADDR AS00 I/O

DIGITAL INPUTS: I/O ADDR A90A

JA	CKT	IFIGP	INPUT
9	1	0	LOCAL OFFLINE PB
10	2	1	LOCAL ONLINE PB
4	6	5	LOCAL TEST KEYSWITC

DIGITAL OUTPUTS: I/O ADDR A902

JB	CKT	IFOGP *	OUTPUT
1	12	11	LOCAL ALRM LED
4	11	10	LOCAL TEST EN. CED
7	10	9	LOCAL OFFLINE LED
10	9	8	LOCAL ONLINE LED

ANALOG IN : I/O ADDR A90E

HEM	LOC	CH	APPLICATION
0000		0	V PH A
0001		1	V P H B
0002		2	v PH c
0003		3	1 LINE A
0004		4	I LINE B
0005		5	I LINE C
0006		6	WATTS
0007		7	VARS
0008		8	FIELD VOLTS
0009		9	DELTA F
000A		10	FREQ
0008		11	PSS OUT
000C		12	HALF HERTZ TST SIG.
OOOD		13	SPARE
000E		14	NO INPUT
OOOF		15	NO INPUT

RELAY OUTPUTS FRONT CONN. EXTERNAL USE

HRRB SLOT 2B REF SHEET 12B

ODD NUMBERED TERMINALS ARE FUSED FOR 3A.

I/O ADDR : 8040

JA	PSSTB	RELAY	BIT	APPLICATION
1-2	IS-20	K1	0	ALARM
3-Y	21-22	K2	1	ALARM
5-6	23-24	K3	2	5PAR E
7-B	25 -26	K4	3	EXT ONLINE LAMP
9-10	27-28	K5	4	EXT OFFLINE LAMP
11-12	—	K6	5	PHA_PA
13-14	—	K7	6	PHB_PA
IS-16	—	K8	7	PHC_PA

RELAY OUTPUTS: BACK CONN. INTERNAL USE

HRRB SLOT 2D REF. SHEET 32E & J

i/O ADDRESS : 8042

RELAY	BIT	APPLICATION
K1	0	PSS ONLINE
K2	1	DIGITAL TEST
K3	2	MODULATION TEST
K4	3	MOD. TEST WITH OSC
KS	4	GP-PSS PA
K6	5	GP_PSS
K7	6	PA-TEST
K8	7	VTM_AC

ANALOG OUTPUTS: NDAC EF, Stl. 32F

OUTPUT	ADDR	APPLICATION
2	8000/2	PSS OUTPUT
3	8004/6	0.5 HZ TEST SIGNAL
	8008/A	JOG TEST SIGNAL
4	800C/E	SPARE

HSCC/2A I/O ADDR 8050/8052 SH. 12A

CONTACT INPUTS

JA	HI PSSTB	LO	I/O ADDR	BIT	CONTACT
1	15	10	8050	0	OPER. PB ONLINE
2	15	11	8050	1	OPER PBOFFLINE
3	16	12	8050	2	52G
4	16	13	8050	3	FAULT-LIST-RST
5	17	14	8050	4	SPARE
6	CLA-	CLA+	8050	5	CLA_TRIP
7	CLB-	CLB+	8050	6	CLB_TRIP
8	CLC-	CLC+	8050	7	CLC,TRIP
1-16	NC	NC	8052	*	NOT AVAILABLE

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA U.S.A.

SYSTEM ADDRESSES

238B1249 02A

SH NO. 01B

LP 3-23-89 I. GIBBS
STANDARD
4932

03-30-88 I. GIBBS

238B1249 02A

HADA			MEM	
BERG	BOARD LOC.	POSITION CHOICES	SLOT 1B	APPLICATION
J1	D30	T,F	F	8
J2	D30	T,F	F	9
J3	D30	T,F	F	A
J'I	D30	T,F	F	B — I/O ADDRESS 8000
J5	D50	T,F	F	C
J6	D50	T,F	F	D
J7	D50	T,F	F	E
J8	D50	T,F	T	F

DS3800HAIC			MEM		
BERG	BOARD LOC.	POSITION CHOICES	SLOT IF		APPLICATION
J1	B10	P,A	A		P15 MONITOR OR REMOTE INPUT
J2	B10	N,A	A		N15 MONITOR OR REMOTE INPUT
J3	B45	T,F	T		8
J4	C45	T,F	F		9
J5	C45	T,F	F		A — I/O ADDRESS A300
J6	C45	T,F	T		B
J7	B45	T,F	F		C
J8	c45	T,F	T		D
J9		O,C	0		
J1B		NOR,ALT	ALT		NOR - DISABLE INHIBIT ALT - ENABLE INHIBIT

BERG JUMPER5
DS3800HAIC
DS3800HADA

MEM

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT
SALEM, VA U.S.A.

BERG JUMPERS
238B1249 02B
SW NO 02A
CONT ON SH

DRAWN J. GIBBS
PRINTS TO STANDARD
I.C. NO 4932
ISSUED 03-30-88
MADE BY J. GIBBS

REV.1
REV.2
REV.3
REV.4
REV.5

ORIG

HUMB		MEM								
		DS4815PSSD				DS4815PSSA				
BERG	CHOICES	SLOT IH (B02) IN . = QUAD				SLOT IJ (B02) IN • = QUAD				
		A	B	C	D	A	B	C	D	
J1	IN,OUT	(D10)	(E10)	(G10)	(J10)	(D10)	(E10)	(G10)	(J10)	
J*	EN,DIS	EN	EN	EN	EN	EN	EN	EN	EN	
J5*	2,3,4	(A02) 4	(B10) 4	(B02) 4	(B10) 4	(A02) 2	(B10) 2	(B02) 2	(B10) 2	
JD*	T,F	(D10) F	(E10) F	(G10) F	(H10) F	(D10) F	(E10) F	(G10) F	(H10) F	
JE*	T,F	(D10) F	(E10) F	(G10) F	(H10) F	(D10) F	(E10) F	(G10) F	(H10) F	
JF*	T,F	(C10) F	(E10) T	(F10) F	(H10) F	(C10) F	(E10) T	(F10) F	(H10) T	
J10*	T,F	(C10) T	(E10) T	(F10) F	(H10) F	(C10) T	(E10) T	(F10) T	(H10) T	
J11*	T,F	(C10) F	(E10) F	(F10) T	(H10) T	(C10) F	(E10) F	(F10) T	(H10) T	
J12*	T,F	(C10) T	(E10) T	(F10) T	(H10) F	(C10) F	(E10) F	(F10) F	(H10) F	
J13*	T,F	(C10) T	(D10) T	(F10) T	(H10) F	(C10) F	(D10) F	(F10) F	(H10) F	
DAUGHTERBD. 304A84836 #		G9	G9	G9	G9	G4	G4	G5	G5	
MEMORY TYPE SIZE		EEPROM 8K X 8				PROM 8K X 8		RAM 8K X 8		

NOTE: CHARACTERS IN PARENTHESES ARE PHYSICAL BOARD LOCATION
X JUMPER IS A DON'T CARE ; NOT REQUIRED.

APPLICATION	
OSCILLATOR IN/OUT	
QUAD ENABLE/DISABLE	
SPEED SELECT 250, 350, 450 nSECONDS	

3 MSB OF 2ND HEX DIGIT

1ST HEX DIG

SLOT	QUAD	ADDR
-1H-	A	D0000
-1H-	B	D8000
-1H-	C	E0000
-1H-	D	20000
-1J-	A	10000
-1J-	B	18000
-1J-	C	30000
-1J-	D	38000

BERG JUMPERS (HUMB)
MEM

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM VA. U.S.A.

BERG JUMPERS

238B1249 02B 02C

ENGINEER I. GIBBS
DRAWN
CHECKED
FIRST ASSY FOR REL
STANDARD
LTR NO 4932
REQ
ISSUED 03-30-88
MADE BY I. GIBBS

HXPD				MEM	
BERG	BOARD LOC.	POSITION CHOICES		SLOT 1K	APPLICATION
RAM	B15	$\overline{R}$,0,1,2 D,E,F		$\overline{R}$	RAM MEMORY SEGMENT
PROM	B15	$\overline{P}$,0,1,2 D,E,F		$\overline{P}$	PROM MEMORY SEGMENT
PD	D5	T,F		X	PROM STARTING ADDRESS
PE	D5	T,F		X	
PF	D5	T,F		X	
	D10	64,128		X	8K OR 16K PROMS
RD	D10	T,F		X	RAM STARTING ADDRESS
RE	D15	T,F		X	
RF	D15	T,F		X	
		2K,8K		8K	RAM CHIP SIZE
J15	F40	2K,8K		8K	RAM CHIP SIZE
J17		RUN,TEST		RUN	

X = DON'T CARE, NOT REQUIRED

HXPD (CONTINUED)				MEM	
BERG	BOARD LOC.	POSITION CHOICES		SLOT 1K	APPLICATION
8	G10	T,F		F	2ND DIGIT / LSB 1ST DIGIT: I/O ADDRESS 9000H
9	G10	T,F		F	
A	G10	T,F		F	
B	G10	T,F		F	
C	G10	T,F		T	
D	G10	T,F		F	
1	H10	1,BLANK		1	DIGITAL INPUTS FROM
2	H10	2,BLANK		2	
4	H10	4,BLANK		4	
8	H10	8,BLANK		8	
16	H10	16,BLANK		16	
32	H10	16,BLANK		32	
64	H10	16,BLANK		64	
128	H10	16,BLANK		128	

BERG JUMPERS
DS3800HXPD

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT
SALEM, VA U.S.A.

ELEMENTARY DIAGRAM
BERG JUMPERS

238B1249 02D

SH NO 02C

ENGINEER J GIBBS
PRINTS TO STANDARD
I.C. NO 4932
REV.1
REV.2
ISSUED 03-30-88
MADE BY J GIBBS
REV.3
REV.4
REV.5

DS3815DMCC
(HMPK: U12= PSP3800HMPK

HMPK		MEM
BERG	CHOICE	SLOT IL
J1	1, $\overline{1}$	$\overline{1}$
J2	2 $\overline{2}$	2
J3	4 $\overline{4}$	4
J4	8 $\overline{8}$	$\overline{8}$
J5	10 $\overline{10}$	$\overline{10}$
J6	GND ISL	ISL

DMPK		MEM
BERG	CHOICE	SLOT IL
J1	PROM, PROM	PROM
J2	PROM, PROM	PROM
J3	PROM, PROM	PROM
J4	PROM, PROM	PROM
J5	PROM, PROM	PROM
J6	PROM, PROM	PROM
J7	PROM, PROM	PROM
J8	PROM, PROM	PROM
J9	PROM, PROM	PROM
J10	PROM, PROM	PROM
J11	PROM, PROM	PROM
J12	PROM, PROM	PROM
J13	PROM, PROM	PROM
J14	PROM, PROM	PROM
J15	PROM, PROM	PROM
J16	PROM, PROM	PROM
J17	32K, 64K	32K

BERG JUMPERS
(HMPK)

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM
BERG JUMPERS
238B1249 02E
SH NO. 02D

ENGINEER J. GIBBS
PRINTS TO
FIRST MADE FOR APPLICATION STANDARD
I.C. NO 4932

REV 1
REV 2 ISSUED 03-30-88 MADE BY J. GIBBS
REV 3
REV 4
REV 5

ORIG

DS3800NFTA			FREQ. TRANS	SIG. GEN.	
			5LOT		
BJ	POS	LOC.	2J	2K	COMMENTS
1	FT/SG	H10	FT	SG	SIGNAL GENERATOR
2	FT/SG	H10	FT	SG	OR
3	FT/SG	H20	FT	S G	FREQUENCY
4	FT/SG	H20	FT	SG	TRANSDUCER
5	FT/SG	H40	FT	SG	
6	FT/SG	H40	FT	SG	
7	BJ7	G60	BJ7		WASHOUT TIME 1 JUMPER ONLY 10
8	BJ8	G60		ANY	10= 51D. 3
5	BJ9	G60			DON'T CARE FOR .5
10	BJ10	G60			SIG. GEN. 0
24	LO/HI	C20	HI	X	X: DON'T CARE
31	WR/N	C30	N	N	
32	50/60	B70			SET
33	50/60	870			FOR
34	50/60	C70			SYSTEM
35	50/60	D70			FREQUENCY
27	FT/SG	C30	FT	SG	

DS3800NQFA			SLOT	
BJ	POS.	LOC.	2_H	COMMENT
J1	D/S		D	SINGLE OR
J2	D/S		S	DOUBLE ENDED INPUTS
J3	D/S		s	
J4	D/S		s	
J5	D/S		s	
J6	D/S		s	
J7	D/S		S	
J8	D/S		s	
J9	ON/OFF		X	NOT USED, DON'T CARE

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.

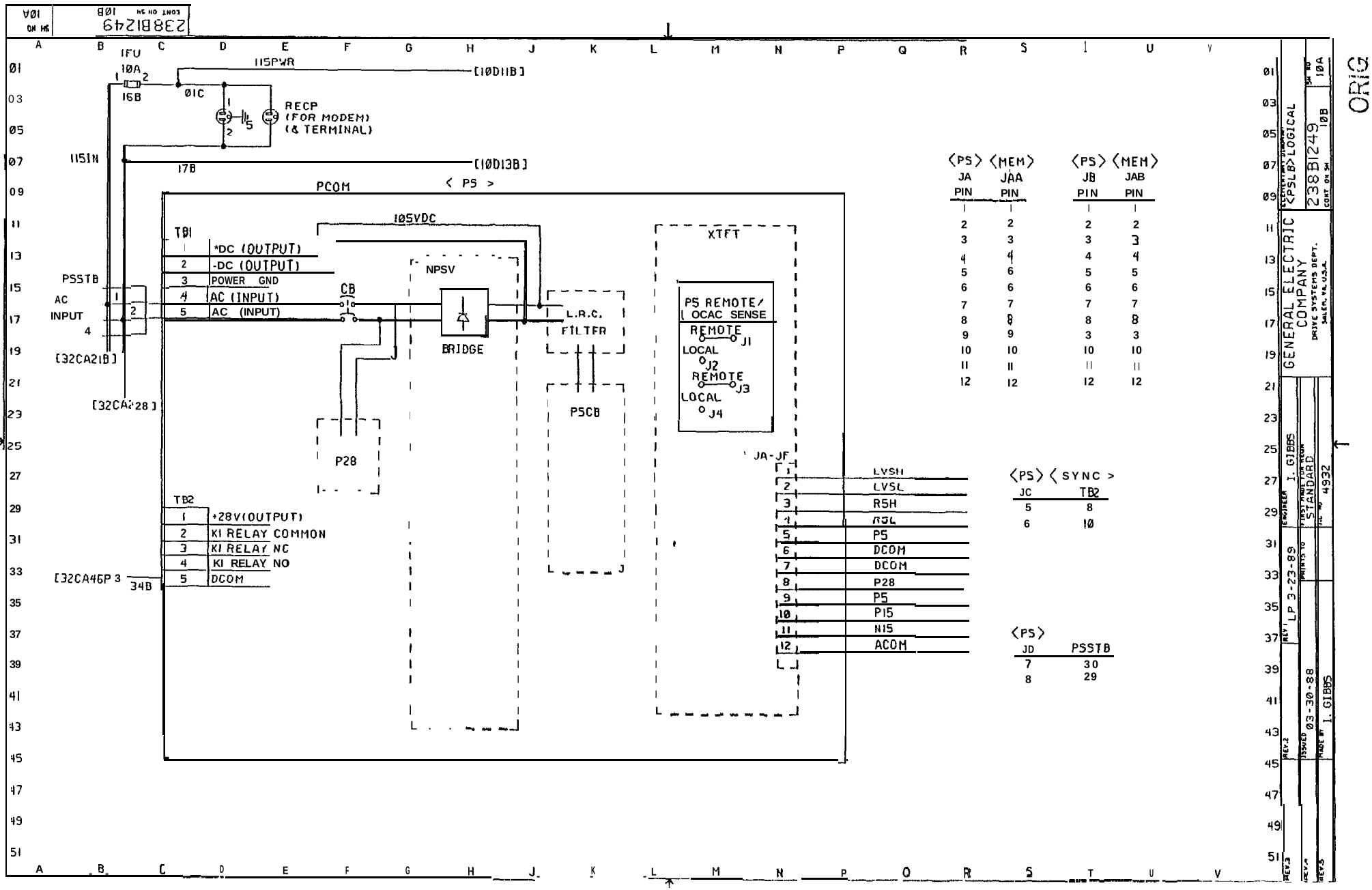
238B1249 10A
CONT ON SH 02E

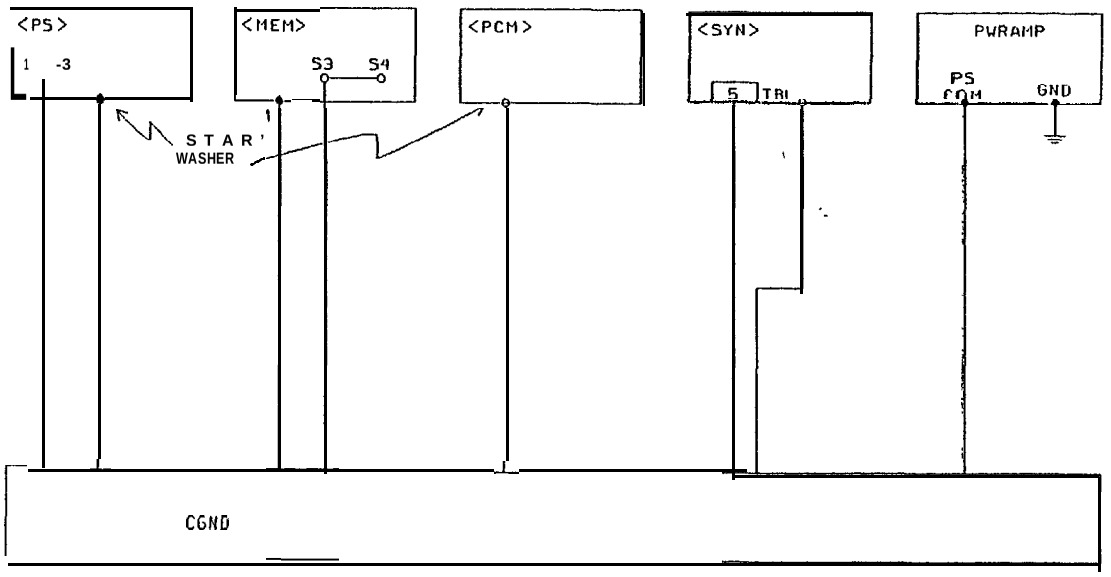
BERG JUMPERS

ENGINEER I. GIBBS
FIRST MADE FOR MSN STANDARD
PRINTS TO
DWG NO 4932

REV'D LP 3-23-89
REV'D
ISSUED 03-30-88
MADE BY I. GIBBS
REV'S
REV'S

238B1249 10A
CONT ON SH 02E


238B1249
10B
CONT ON SH
10A
ON SH
IFU
10A
16B
01C
115PWR
[10D11B]
RECP
(FOR MODEM)
(& TERMINAL)
115IN
17B
[10D13B]
PCOM
< PS >
105VDC
TB1
*DC (OUTPUT)
-DC (OUTPUT)
POWER GND
AC (INPUT)
AC (INPUT)
PSSTB
AC
INPUT
[32CA21B]
[32CA28]
CB
NPSV
BRIDGE
L.R.C.
FILTER
PSCB
P28
XTFT
PS REMOTE/
OCAC SENSE
REMOTE
J1
LOCAL
J2
REMOTE
J3
LOCAL
J4
JA-JF
LVSH
LVSL
R5H
R5L
P5
DCOM
DCOM
P28
P5
P15
N15
ACOM
TB2
+28V(OUTPUT)
KI RELAY COMMON
KI RELAY NC
KI RELAY NO
DCOM
[32CA46P3 34B]
<PS> <MEM>
JA JAA
PIN PIN
2 2
3 3
4 4
5 6
6 6
7 7
8 8
9 9
10 10
11 11
12 12
<PS> <MEM>
JB JAB
PIN PIN
2 2
3 3
4 4
5 5
6 6
7 7
8 8
3 3
10 10
11 11
12 12
<PS> < SYNC >
JC TB2
5 8
6 10
<PS>
JD PSSTB
7 30
8 29
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
<PSLB> LOGICAL
238B1249
10B
10A
ENGINEER I. GIBBS
STANDARD
LP 3-23-89
ISSUED 03-30-88
MADE BY I. GIBBS
DWG NO 4932
REV.2
REV.3
REV.4
REV.5



GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT
SALEM, VA U.S.A.

ELEMENTARY DIAGRAM
GROUNDING

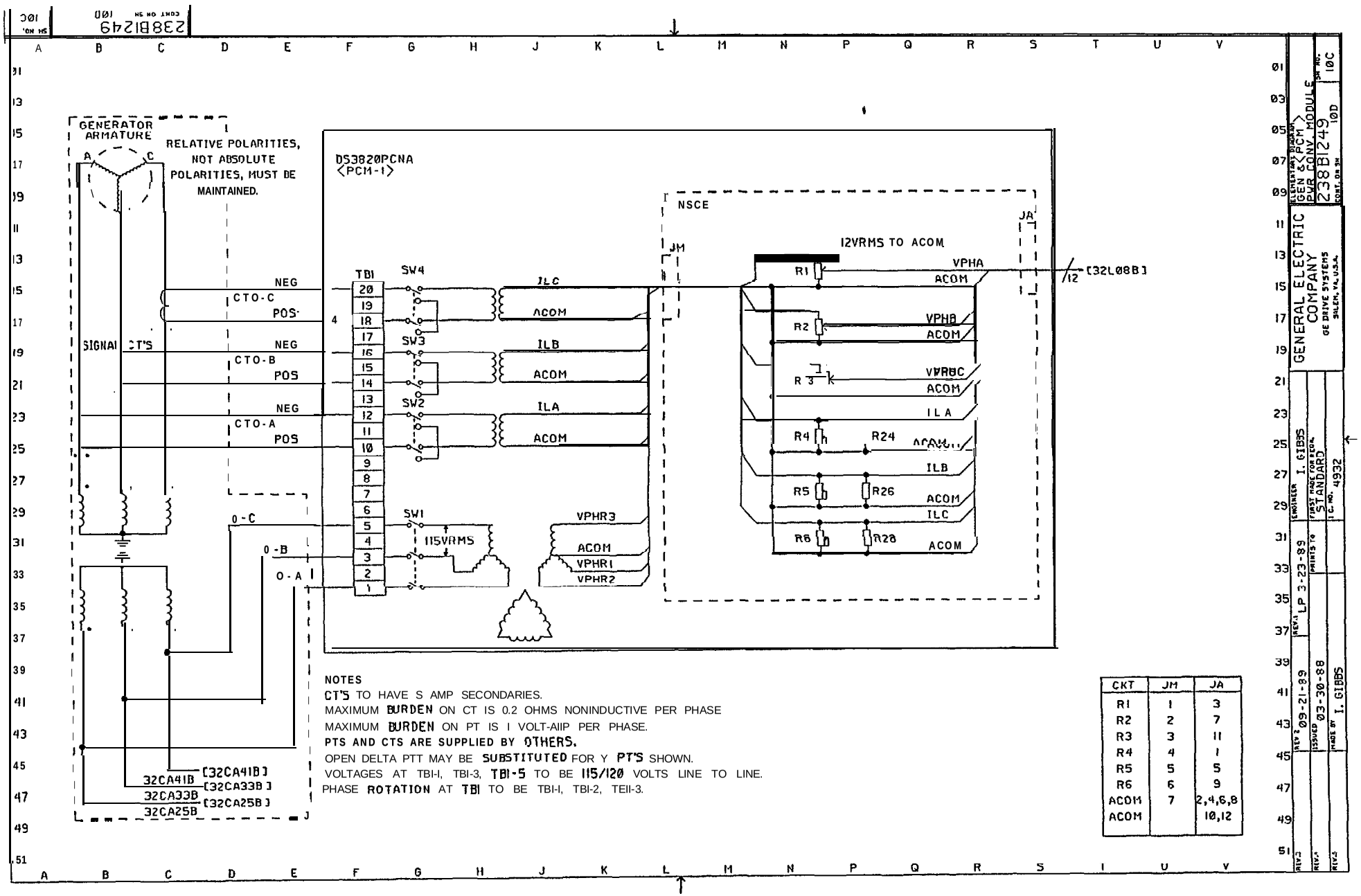
238B1249 10C

SH NO. 10B

ENGINEER T. GIBBS
FIRST MADE FOR ROM STANDARD
I.C. NO 4932
PRINTS TO
ISSUED 03-30-88
MADE BY T. GIBBS

REV 1
REV 2
REV 3
REV 4
REV 5

238B1249 10C
CONT ON SH
10B
ON SH

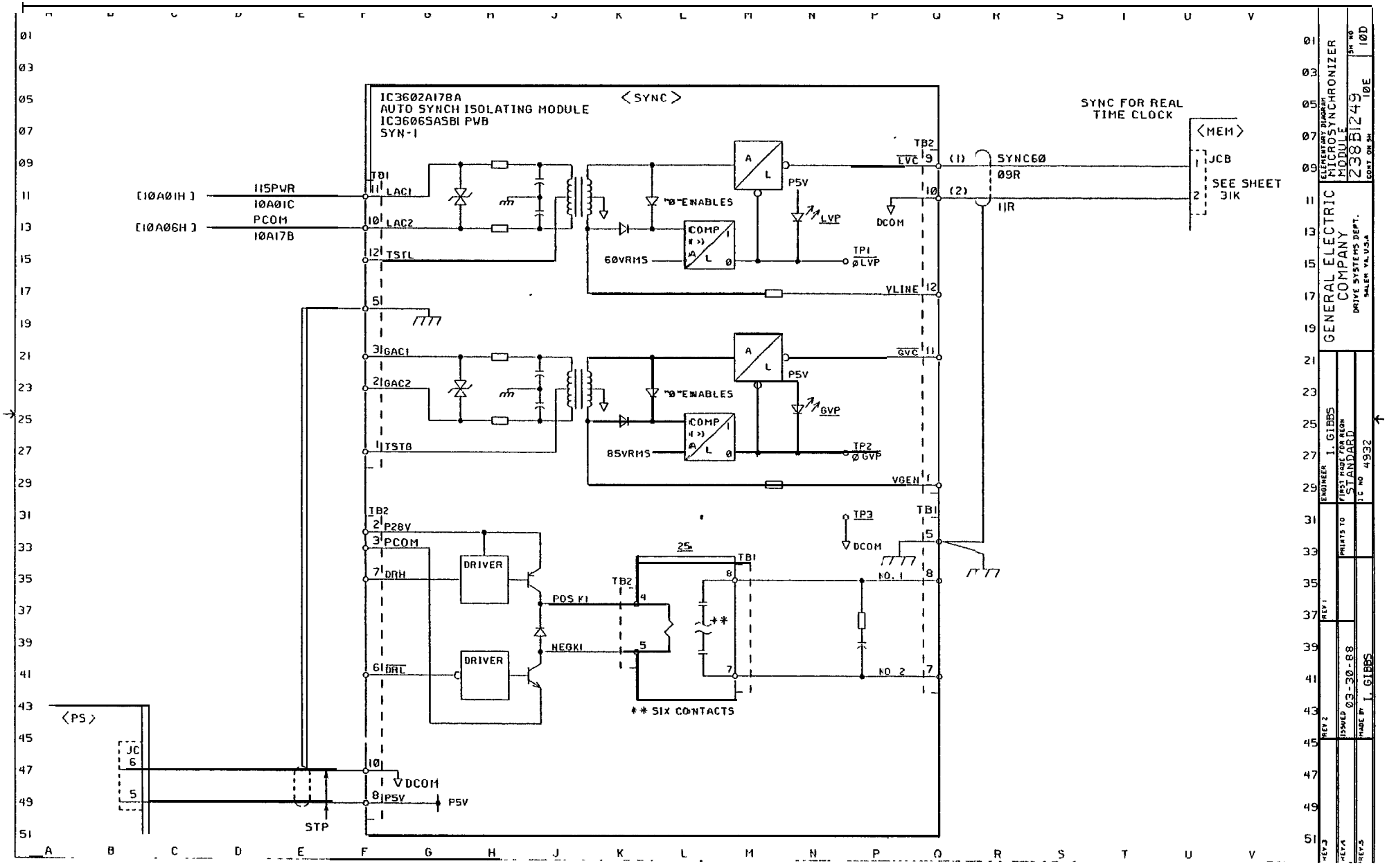


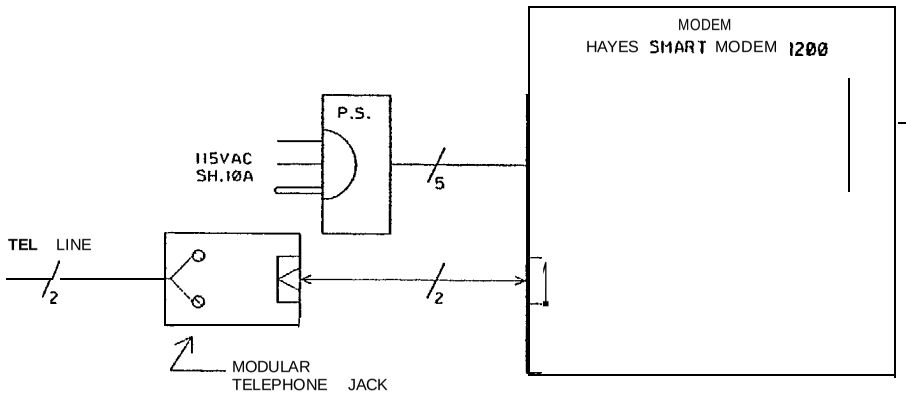
NOTES

CT'S TO HAVE 5 AMP SECONDARIES.
MAXIMUM BURDEN ON CT IS 0.2 OHMS NONINDUCTIVE PER PHASE
MAXIMUM BURDEN ON PT IS 1 VOLT-AIIP PER PHASE.
PTS AND CTS ARE SUPPLIED BY OTHERS.
OPEN DELTA PTT MAY BE SUBSTITUTED FOR Y PT'S SHOWN.
VOLTAGES AT TBI-I, TBI-3, TBI-5 TO BE 115/120 VOLTS LINE TO LINE.
PHASE ROTATION AT TBI TO BE TBI-I, TBI-2, TEII-3.

CKT	JM	JA
R1	1	3
R2	2	7
R3	3	11
R4	4	1
R5	5	5
R6	6	9
ACOM	7	2,4,6,8
ACOM		10,12

GENERAL ELECTRIC COMPANY
GE DRIVE SYSTEMS
SALEM, VA, U.S.A.
ELEMENTARY DIAGRAM
GEN & <PCM>
PWR CONV MODULE
238B1249
CONT. ON SH 10D
SH NO. 10C
ENGINEER I. GIBBS
FIRST PAGE FOR REQM'T STANDARD
I.C. NO. 4932
REV-1 LP 3-23-89
PRINTS TO
REV-2 09-21-89
ISSUED 03-30-88
MADE BY I. GIBBS
REV-3
REV-4
238B1249
CONT ON SH 10D
SH NO. 10C


IC3602A178A
AUTO SYNCH ISOLATING MODULE
IC3606SASB1 PWB
SYN-1
<SYNC>
SYNC FOR REAL TIME CLOCK
<MEM>
JCB
SEE SHEET 31K
SYNC60
09R
11R
[10A01H]
115PWR
10A01C
[10A06H]
PCOM
10A17B
LAC1
LAC2
TSTL
60VRMS
COMP
ENABLES
P5V
LVP
DCOM
TP1
ØLVP
LVC
YLINE
GAC1
GAC2
TSTB
85VRMS
TP2
ØGVP
GVC
VGEN
P28V
PCOM
DRH
DRL
DRIVER
POS K1
NEG K1
25
NO. 1
NO. 2
TP3
** SIX CONTACTS
<PS>
JC
6
5
STP
DCOM
P5V
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM VA. U.S.A.
MICROSYNCHRONIZER MODULE
238B1249
10D
10E
I. GIBBS
03-30-88
4932
STANDARD



POWFR SUPPLY & PLUG TERMINATED TELEPHONE CABLE SUPPLIED WITH MODEM.

NOTE: MODEM NOT INCLUDED WITH G7. OR G8.

9 277A5771G27L045 C31L425 1

PIN	MODEM DESIGNATION	DMCC DESIGNATION
1	FG	FG
2	TXD	TXD
3	RXD	RXD
4	RTS	RTS
5	CT5	CTS
6	DSR	DSR
7	SG	SG
8	CXR	CXR
9	+V	
10	-V	
11		
12	CI	
13		
14		
15	SCT	
16		
17	SCR	
18	CN	
19		
20	DTR	DTR
21	RDL	
22	RI	
23	SS	
24	SCTE	
25	TM/CN	

ONLY THESE 3 PIN5 ARE WIRED.

MODEM END 25 PIN MALE D CONNECTOR
DMCC END 25 PIN FEMALE D CONNECTOR

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT
SALEM, VA U.S.A.

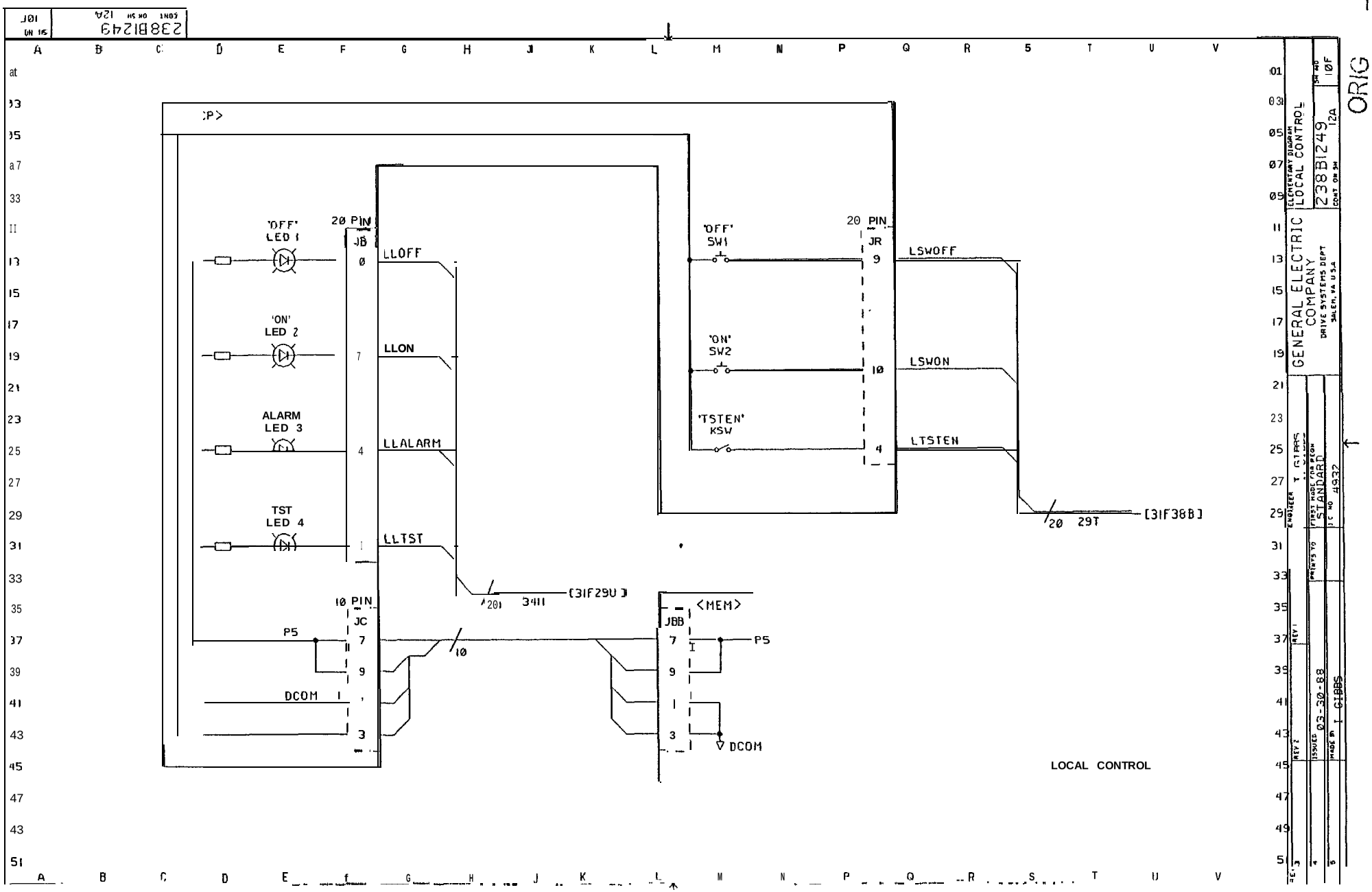
ELEMENTARY DIAGRAM
MODEM

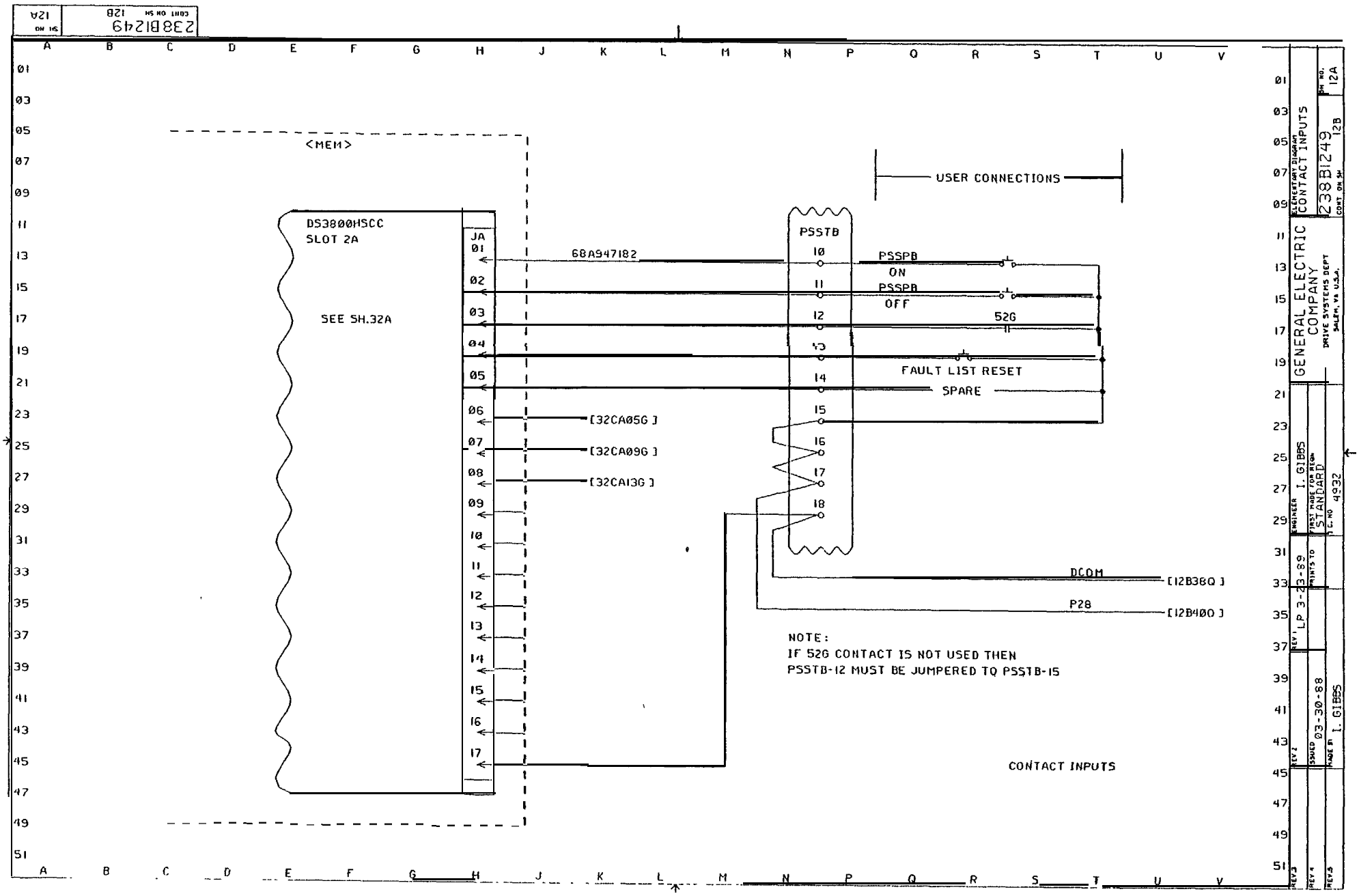
238B1249 10F

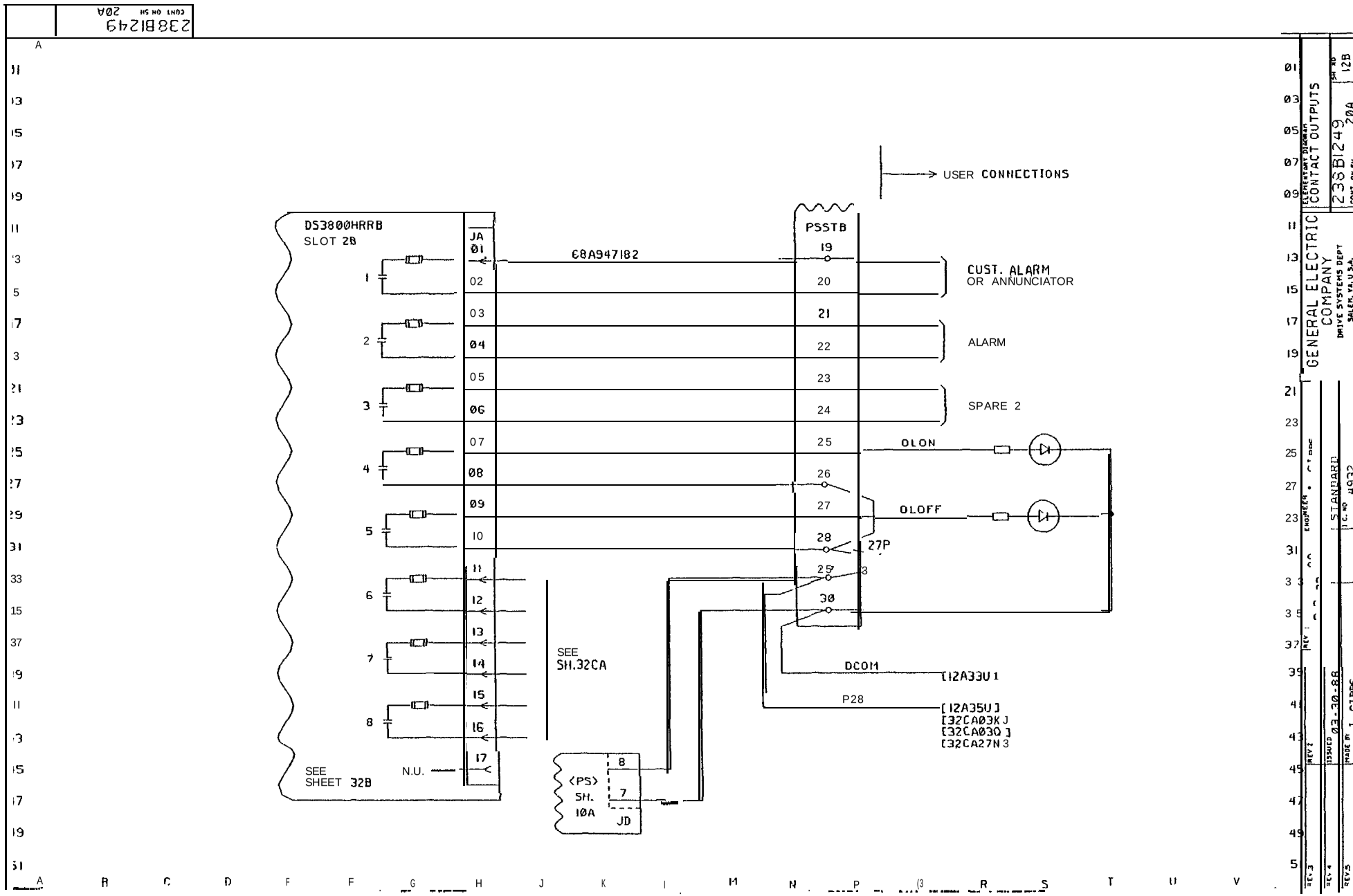
SH NO 10E

ENGINEER T. GTARR

REV.
REV.
REV.3


:P>
'OFF' LED 1
20 PIN JB
LLOFF
'ON' LED 2
LLON
ALARM LED 3
LLALARM
TST LED 4
LLTST
(3IF29U)
10 PIN JC
P5
DCOM
'OFF' SW1
20 PIN JR
LSWOFF
'ON' SW2
LSWON
'TSTEN' KSW
LTSTEN
(3IF38B)
<MEM>
JBB
P5
DCOM
LOCAL CONTROL
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT
SALEM, VA U.S.A.
238B1249
12A
10F
03-30-88
STANDARD
4932


<MEM>
DS3800HSCC
SLOT 2A
SEE SH.32A
JA
01
02
03
04
05
06
07
08
09
10
11
12
13
14
15
16
17
68A947182
[32CA05G]
[32CA09G]
[32CA13G]
USER CONNECTIONS
PSSTB
10
11
12
13
14
15
16
17
18
PSSPB
ON
PSSPB
OFF
52G
FAULT LIST RESET
SPARE
DCOM
[12B38Q]
P28
[12B40Q]
NOTE:
IF 52G CONTACT IS NOT USED THEN
PSSTB-12 MUST BE JUMPERED TO PSSTB-15
CONTACT INPUTS
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT
SALEM, VA U.S.A.
ELEMENTARY DIAGRAM
CONTACT INPUTS
238B1249
SH NO. 12A
CONT ON SH 12B
ENGINEER I. GIBBS
FIRST MADE FOR REQ STANDARD
PRINTS TO
T.C. NO 4932
REV'D LP 3-23-89
ISSUED 03-30-88
MADE BY I. GIBBS
REV.2
REV.3
REV.4
REV.5


USER CONNECTIONS
DS3800HRRB
SLOT 2B
JA
PSSTB
68A947182
CUST. ALARM
OR ANNUNCIATOR
ALARM
SPARE 2
OLON
OLOFF
27P
DCOM
[12A33U1
P28
[12A35U]
[32CA03KJ
[32CA03Q]
[32CA27N3
SEE
SH.32CA
SEE
SHEET 32B
N.U.
<PS>
SH.
10A
JD
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT
SALEM, VA, U.S.A.
CONTACT OUTPUTS
238B1249
20A
12B
STANDARD
03-30-88
1 GTR65
4932

ROW 1 (UPPER RIGHT)	SLOT	ROW 2
WIRED & RESERVED FOR (DS3815PLNF)	A	DS3800HSCC
DS3800HADA	B	DS3800HRRB
	C	DS3800NGMA DS3800DGPA
WIRED & RESERVED FOR (HUMB)	D	DS3800HRRB
WIRED & RESERVED FOR (HUMB)	E	DS3800DOAA DS3800NOAA
DS3800HAIC	F	DS3800NDAC
WIRED & RESERVED FOR (HUMB)	G	
DS4815PSSD	H	DS3800NOFA
DS4815PSSA	J	DS3800NFTA
DS3800HXPD	K	(OPTIONAL SIGNAL GEN.) DS3800NFTA
DS3815DMCC	L	DS3800NPTB
RESERVED	M	DS3800NGMA DS3800DGPA

LOWER LEFT

FRONT VIEW

<MEM> CARD LAYOUT

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT
SALEM VA. U.S.A.

ELEMENTARY DIAGRAM <MEM> LAYOUT

238B1249 20B

SH NO 20A

ENGINEER J. GIBBS
FIRST MADE FOR RCAH STANDARD
DC NO 4932

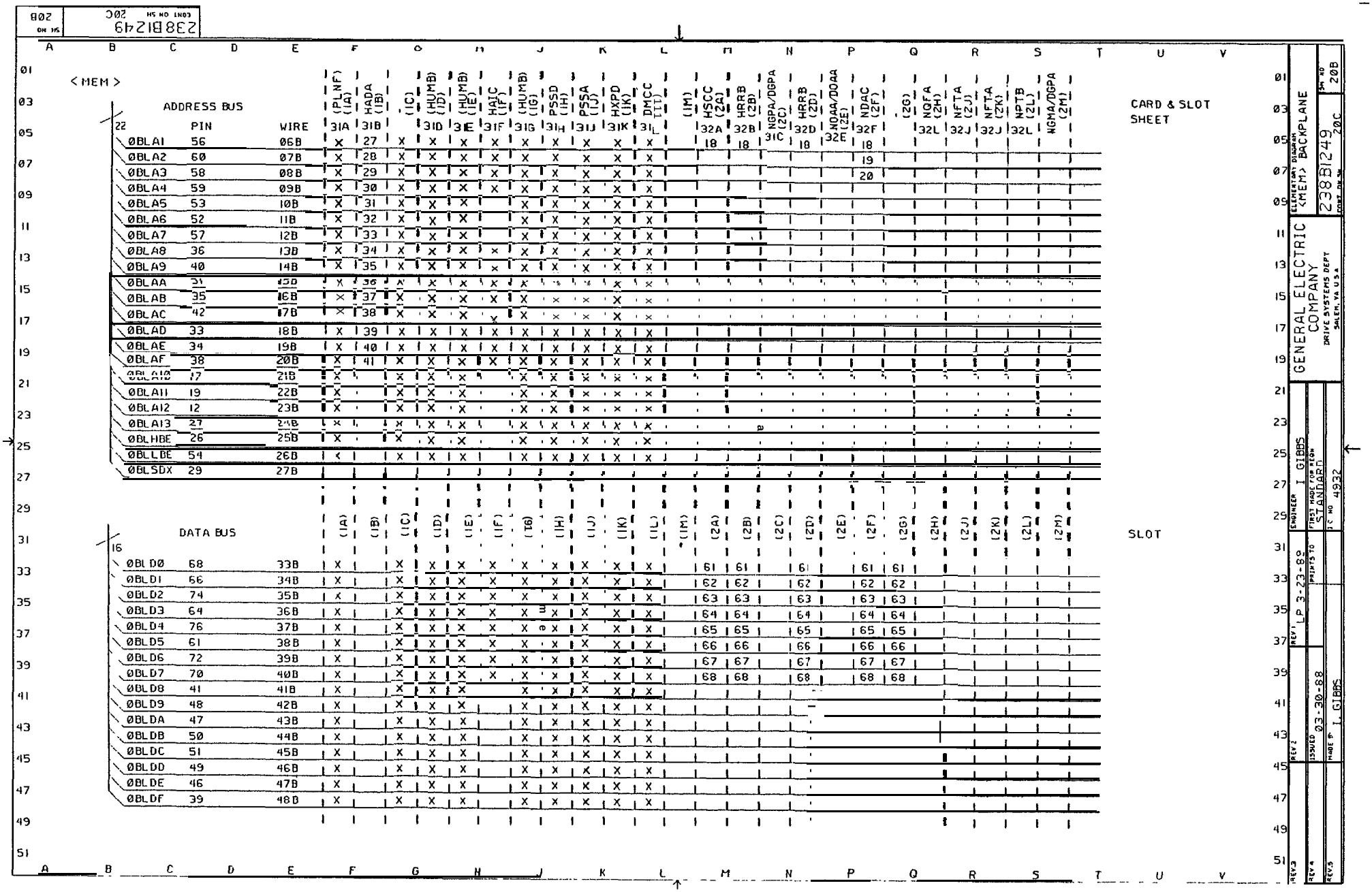
REV 1 LP 3-23-89 PRINTS TO

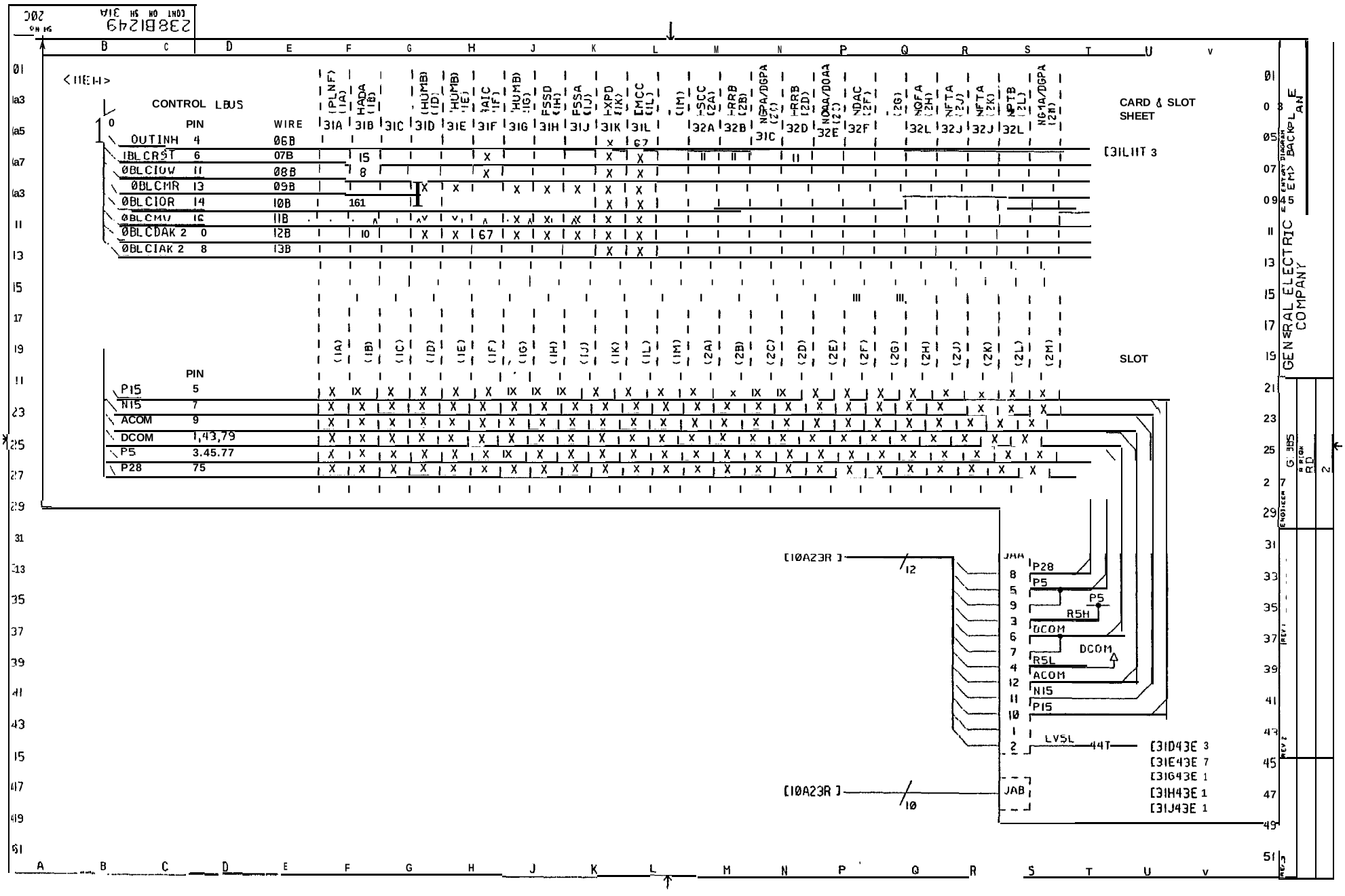
REV 2 ISSUED 03-30-88 MADE BY J. GIBBS

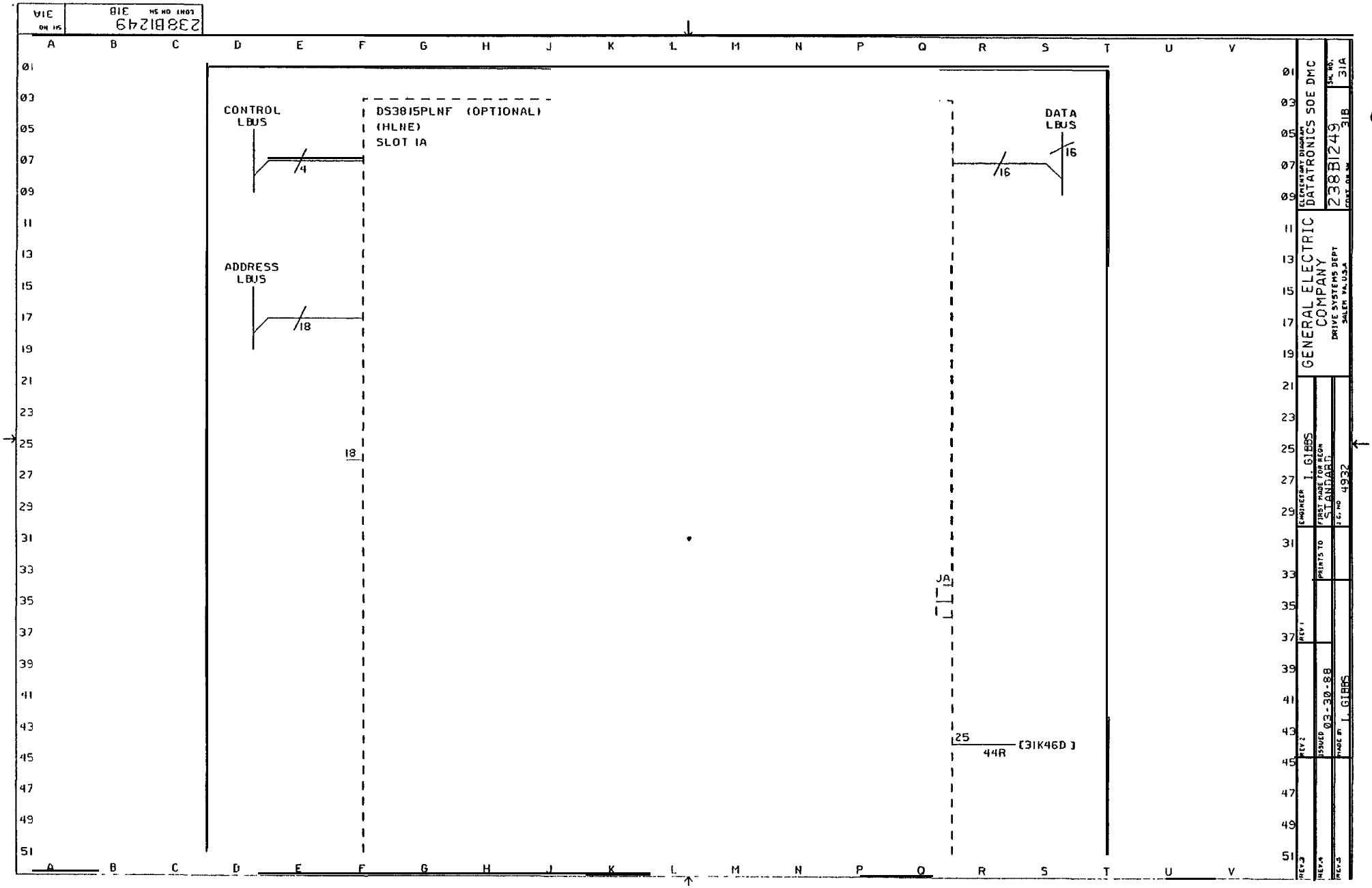
REV 3

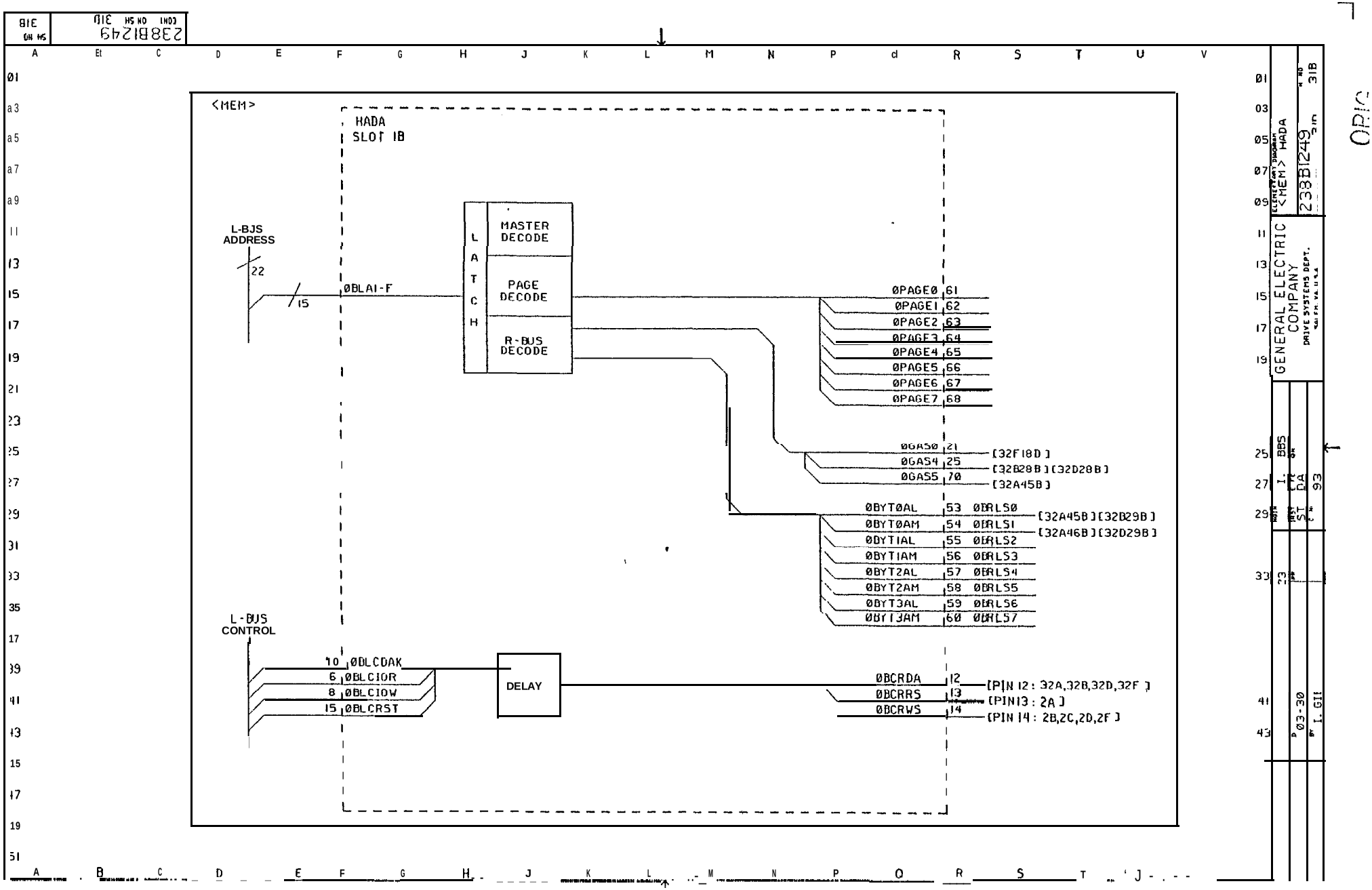
REV 4

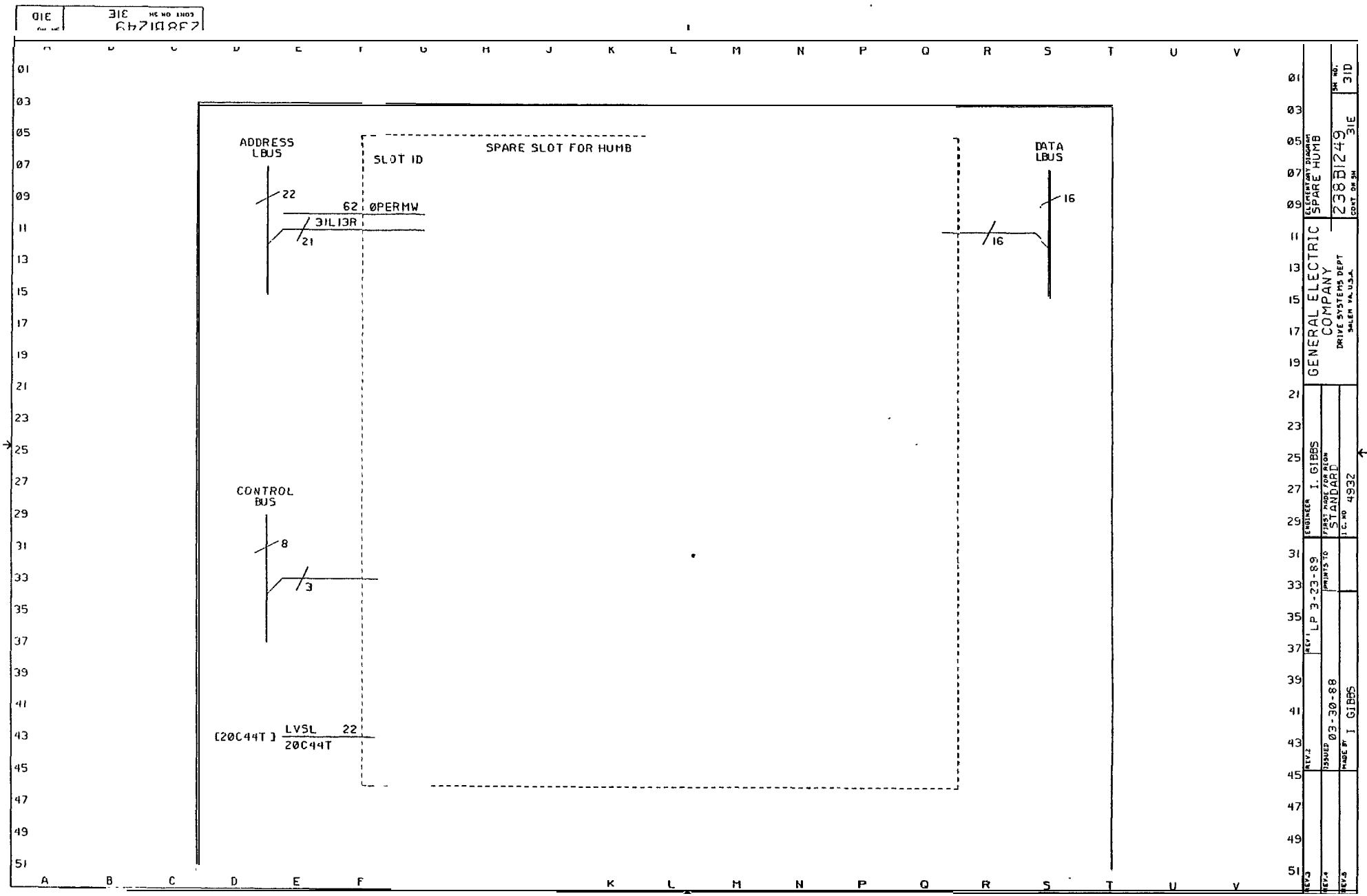
REV 5


<MEM>
ADDRESS BUS
DATA BUS
CARD & SLOT
SHEET
SLOT
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT
SALEM, VA U.S.A.
ELEMENTARY DIAGRAM <MEM> BACKPLANE
238B1249
20B
20C
ENGINEER J. GIBBS
DRAWN BY J. GIBBS
4932

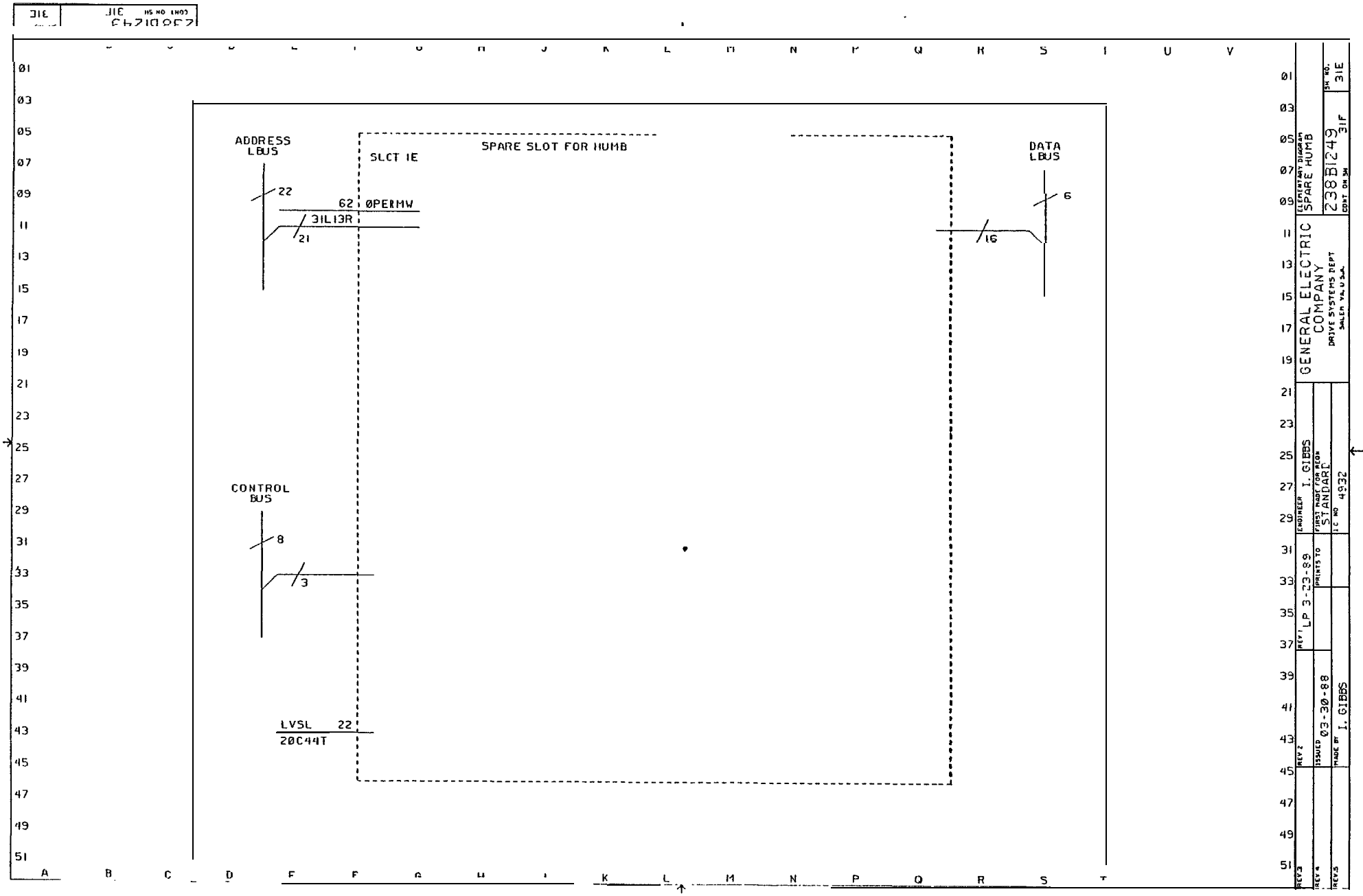

CONTROL LBUS
PIN
WIRE
OUTINH 4 06B
IBLCRST 6 07B
ØBLCIOV 11 08B
ØBLCMR 13 09B
ØBLCIOR 14 10B
ØBLCMIV 16 11B
ØBLCDAK 2 0 12B
ØBLCIAK 2 8 13B
CARD & SLOT
SHEET
SLOT
P15 5
N15 7
ACOM 9
DCOM 1,43,79
P5 3.45.77
P28 75
P28
P5
R5H
DCOM
R5L
ACOM
N15
P15
LVSL
44T
JAB
GENERAL ELECTRIC COMPANY

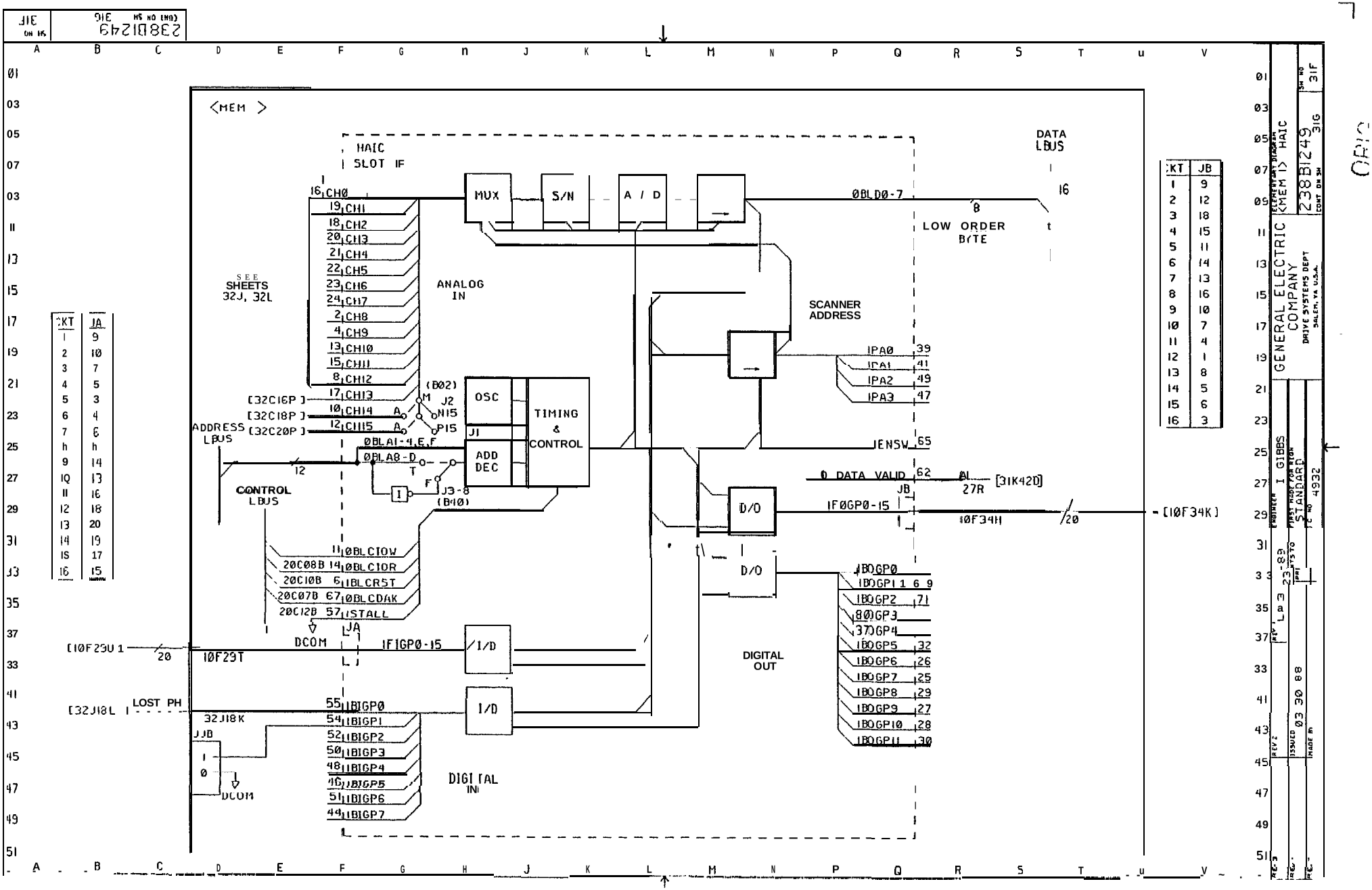

CONTROL LBUS
4
DS3815PLNF (OPTIONAL)
(HLNE)
SLOT 1A
ADDRESS LBUS
18
18
DATA LBUS
16
16
JA
25
44R
(31K46D)
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT
SALEM VA.U.S.A.
DATATRONICS SOE DMC
238B1249
3 1B
3 1A
J. GIBBS
4932
03-30-88
J. GIBBS

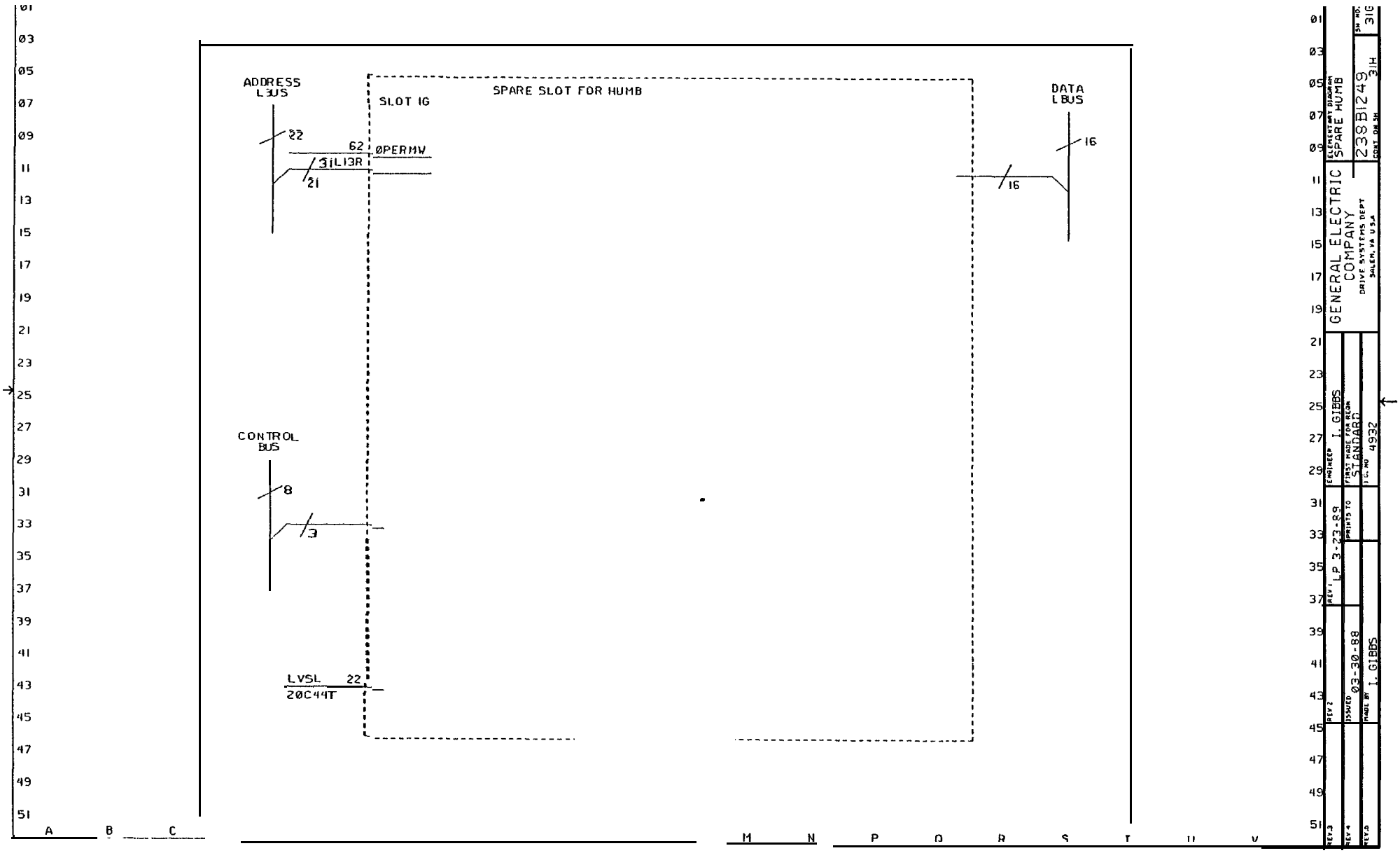

<MEM>
HADA
SLOT 1B
L-BJS ADDRESS
22
15
ØBLAI-F
LATCH
MASTER DECODE
PAGE DECODE
R-BUS DECODE
ØPAGE0 61
ØPAGE1 62
ØPAGE2 63
ØPAGE3 64
ØPAGE4 65
ØPAGE5 66
ØPAGE6 67
ØPAGE7 68
ØGAS0 21 [32F18D]
ØGAS4 25 [32B28B][32D28B]
ØGAS5 70 [32A45B]
ØBYT0AL 53 ØBRLS0 [32A45B][32B29B]
ØBYT0AM 54 ØBRLS1 [32A46B][32D29B]
ØBYT1AL 55 ØBRLS2
ØBYT1AM 56 ØBRLS3
ØBYT2AL 57 ØBRLS4
ØBYT2AM 58 ØBRLS5
ØBYT3AL 59 ØBRLS6
ØBYT3AM 60 ØBRLS7
L-BUS CONTROL
10 ØBLCDAK
6 ØBLCIOR
8 ØBLCIOW
15 ØBLCRST
DELAY
ØBCRDA 12 [PIN 12: 32A,32B,32D,32F]
ØBCRRS 13 [PIN 13: 2A]
ØBCRWS 14 [PIN 14: 2B,2C,2D,2F]

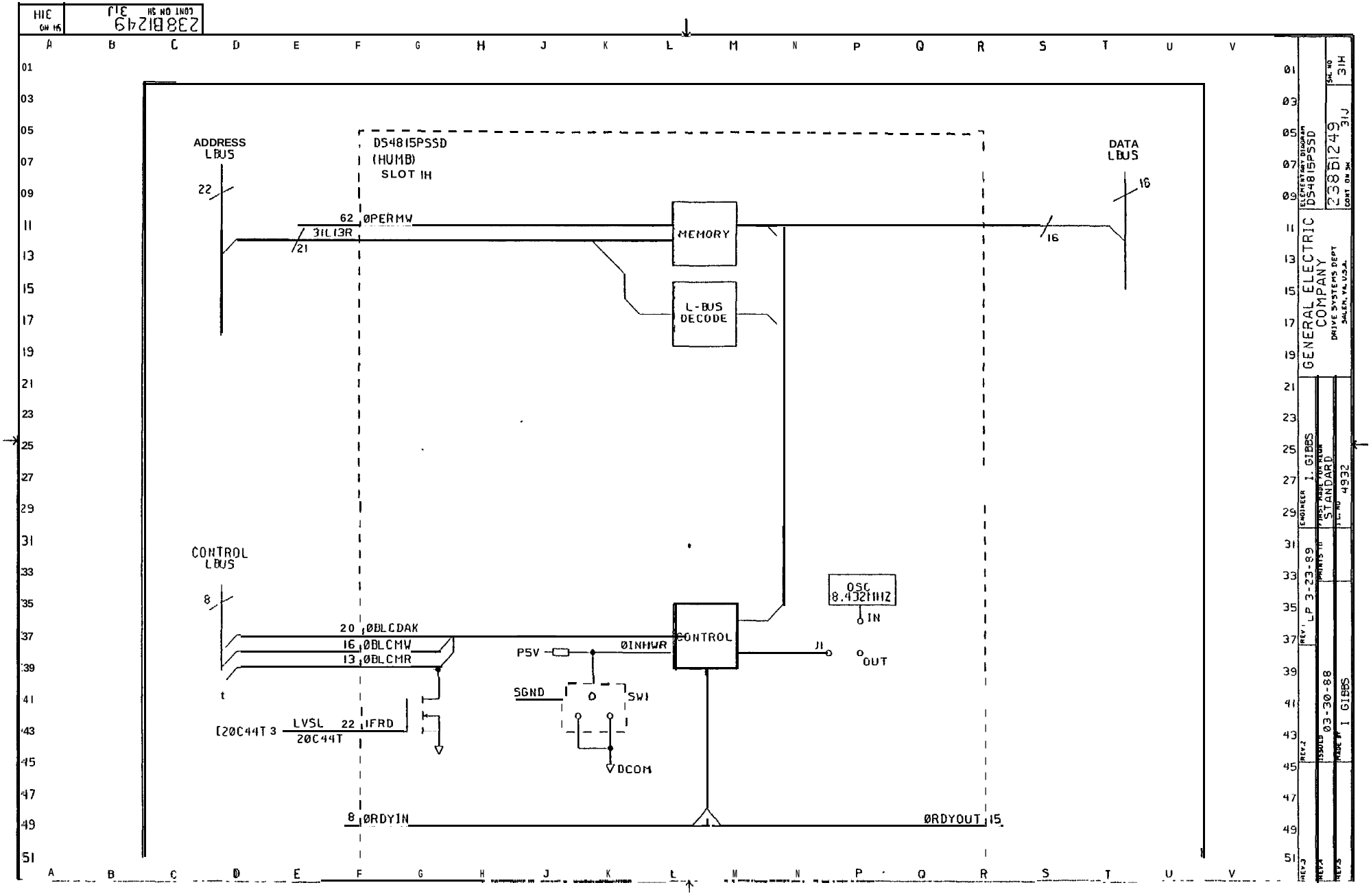

SPARE SLOT FOR HUMB
SLOT ID
ADDRESS LBUS
22
62 ØPERMW
31L13R
21
DATA LBUS
16
16
CONTROL BUS
8
3
[20C44T] LVSL 22
20C44T
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT
SALEM VA. U.S.A.
ELEMENTARY DIAGRAM
SPARE HUMB
238B1249
SH NO. 31D
31E
ENGINEER I. GIBBS
REV'D LP 3-23-89
PRINTS TO
FIRST MADE FOR REON
STANDARD
I.C. NO 4932
ISSUED 03-30-88
MADE BY I GIBBS
REV.A
REV.B
REV.C
REV.D

ORIG

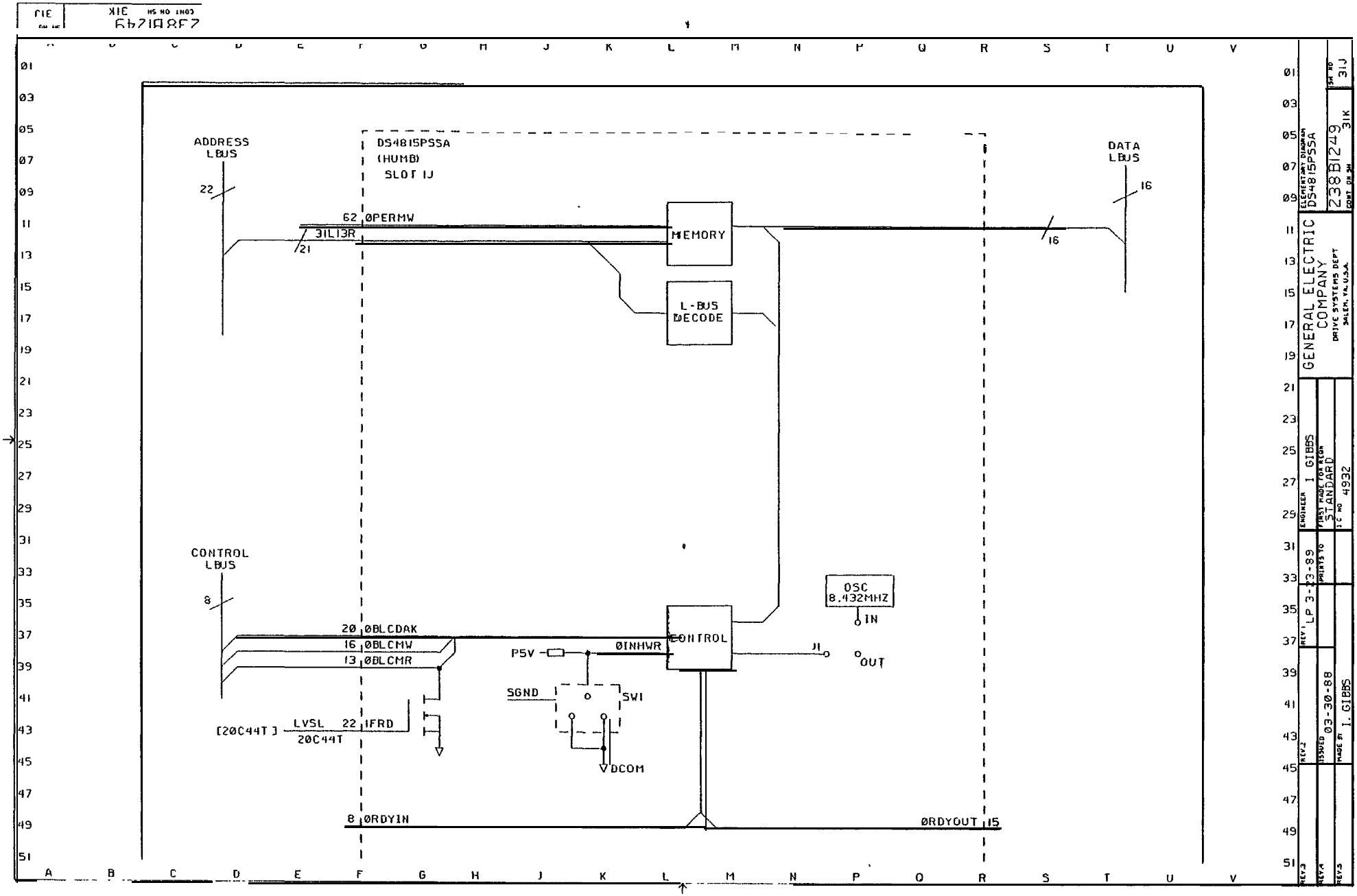

SPARE SLOT FOR NUMB
SLCT IE
ADDRESS LBUS
22
62 ØPERMW
31L13R
21
DATA LBUS
6
16
CONTROL BUS
8
3
LVSL 22
20C44T
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT
SALEM VA, U.S.A.
ELEMENTARY DIAGRAM
SPARE NUMB
238B1249
31F
31E
ENGINEER I. GIBBS
FIRST MADE FOR REQN STANDARD
I.C. NO 4932
REV 1 LP 3-23-89
PRINTS TO
REV 2 ISSUED 03-30-88
MADE BY I. GIBBS

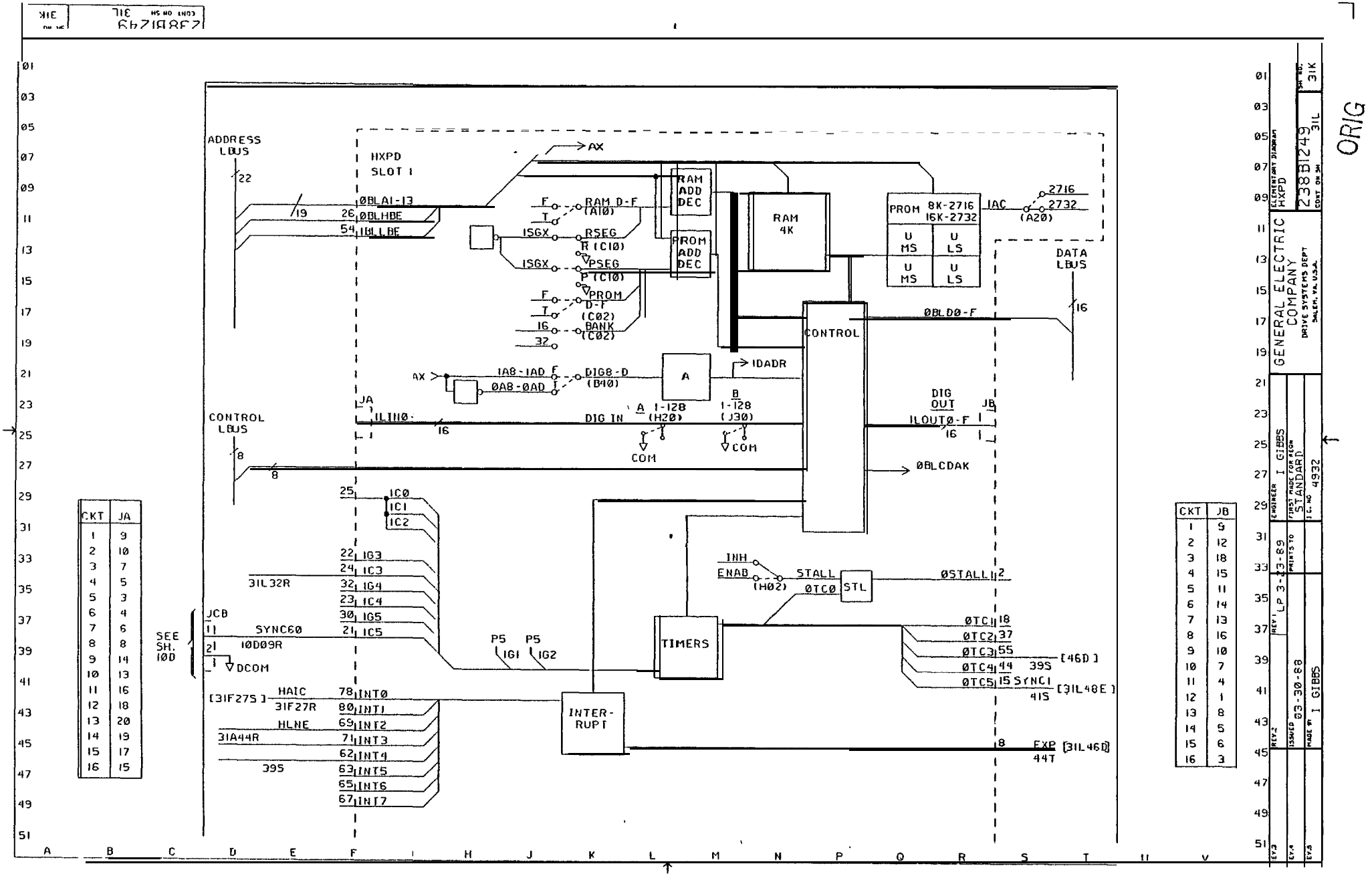

<MEM>
HAIC
SLOT IF
MUX
S/N
A / D
DATA LBUS
ØBLDØ-7
LOW ORDER BYTE
SEE SHEETS 32J, 32L
ANALOG IN
SCANNER ADDRESS
IPAØ
IPA1
IPA2
IPA3
OSC
TIMING & CONTROL
ADDRESS LBUS
CONTROL LBUS
ADD DEC
ØBLA1-4,E,F
ØBLA8-D
IENSW
Ø DATA VALID
27R
[31K42D]
D/O
IFØGPØ-15
10F34H
[10F34K]
ØBLC1OW
ØBLC1OR
IBLCR5T
ØBLCDAK
ISTALL
DCOM
[10F23U]
10F23T
I/D
DIGITAL OUT
IBOGPØ
IBOGP1
IBOGP2
IBOGP3
IBOGP4
IBOGP5
IBOGP6
IBOGP7
IBOGP8
IBOGP9
IBOGP10
IBOGPU
[32J18L]
LOST PH
32J18K
IBIGPØ
IBIGP1
IBIGP2
IBIGP3
IBIGP4
IBIGP5
IBIGP6
IBIGP7
DIGITAL IN
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT
SALEM, VA U.S.A.
238B1249
31G
31F
ENGINEER I GIBBS
STANDARD
ISSUED 03 30 88


SPARE SLOT FOR HUMB
SLOT 16
ADDRESS BUS
22
62
ØPERMW
31LI3R
21
DATA BUS
16
16
CONTROL BUS
8
3
LVSL 22
20C44T
ELEMENTARY DIAGRAM
SPARE HUMB
238B1249
SH NO. 316
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT
SALEM, VA U.S.A.
ENGINEER J. GIBBS
LP 3-23-89
FIRST MADE FOR REGN STANDARD
J.C. NO 4932
ISSUED 03-30-88
MADE BY J. GIBBS


DS4815PSSD
(HUMB)
SLOT 1H
ADDRESS
LBUS
22
62 ØPERMW
31L13R
21
MEMORY
L-BUS
DECODE
16
DATA
LBUS
16
CONTROL
LBUS
8
20 ØBLCDAK
16 ØBLCMW
13 ØBLCMR
CONTROL
OSC
8.432MHZ
IN
OUT
J1
P5V
ØINHWR
SGND
SW1
DCOM
[20C44T 3
LVSL 22 IFRD
20C44T
8 ØRDYIN
ØRDYOUT 15
ELEMENTARY DIAGRAM
DS4815PSSD
238B1249
31J
CONT ON SH
SH. NO
31H
GENERAL ELECTRIC
COMPANY
DRIVE SYSTEMS DEPT
SALEM, VA. U.S.A.
ENGINEER J. GIBBS
PRINTS TO STANDARD
DWG NO 4932
REV 1 LP 3-23-89
ISSUED 03-30-88
PAGE # 1 GIBBS
REV 2
REV 3
REV 4
REV 5

ORIG

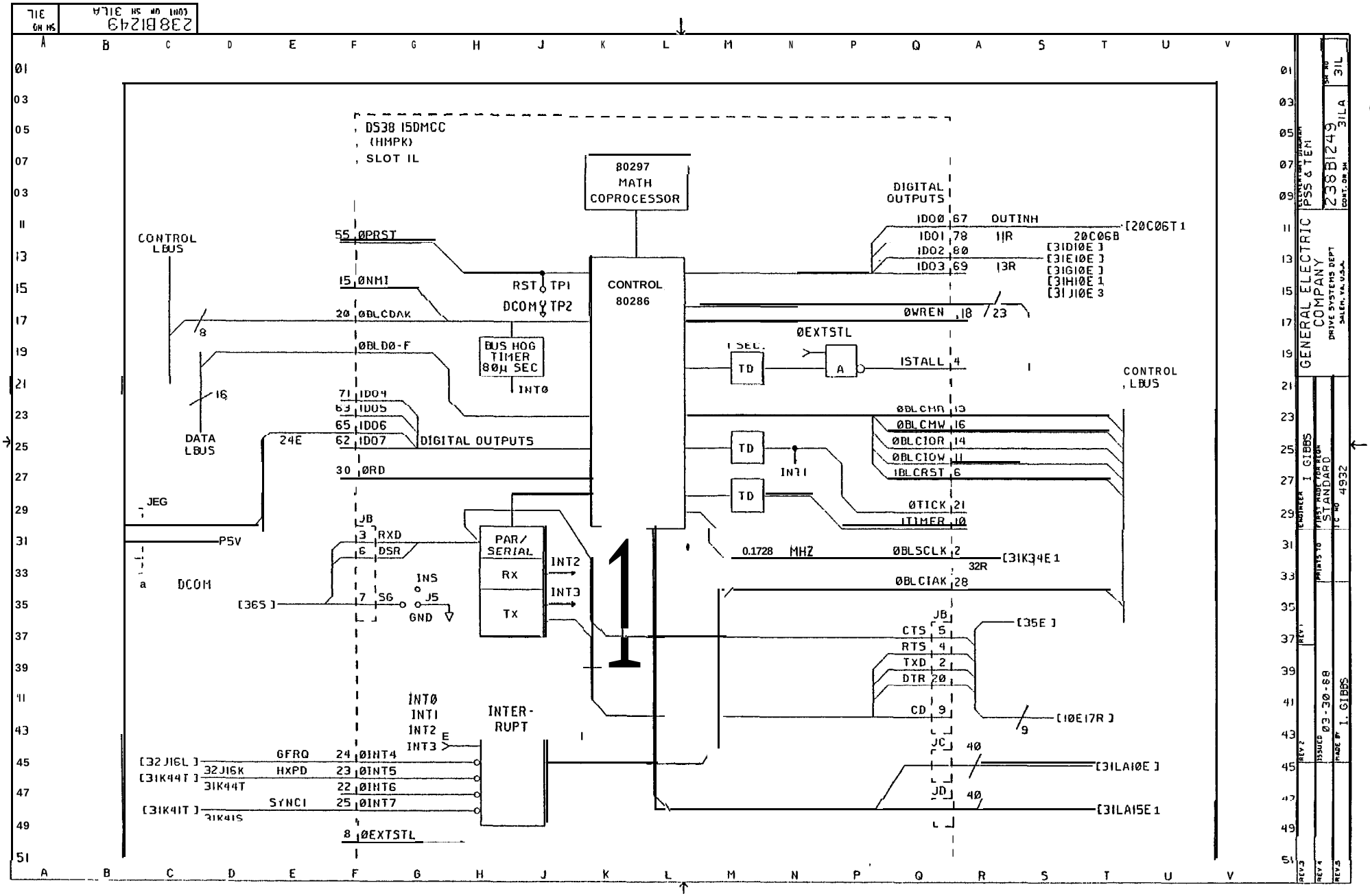

ADDRESS LBUS
22
DS4815PSSA
(HUMB)
SLOT 1J
62 ØPERMW
31L13R
21
MEMORY
L-BUS DECODE
DATA LBUS
16
16
CONTROL LBUS
8
20 ØBLCDAK
16 ØBLCMW
13 ØBLCMR
ØINHWR
CONTROL
P5V
OSC 8.432MHZ
IN
J1
OUT
SGND
SW1
[20C44T]
LVSL 22 IFRD
20C44T
DCOM
8 ØRDYIN
ØRDYOUT 15
ELEMENTARY DIAGRAM
DS4815PSSA
238B1249
31K
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT
SALEM, VA. U.S.A.
ENGINEER J GIBBS
FIRST PAGE FOR RIGN STANDARD
ECC NO 4932
REV1 LP 3-23-89
PRINTED
REV2
ISSUED 03-30-88
MADE BY J. GIBBS
REV3
REV4
REV5



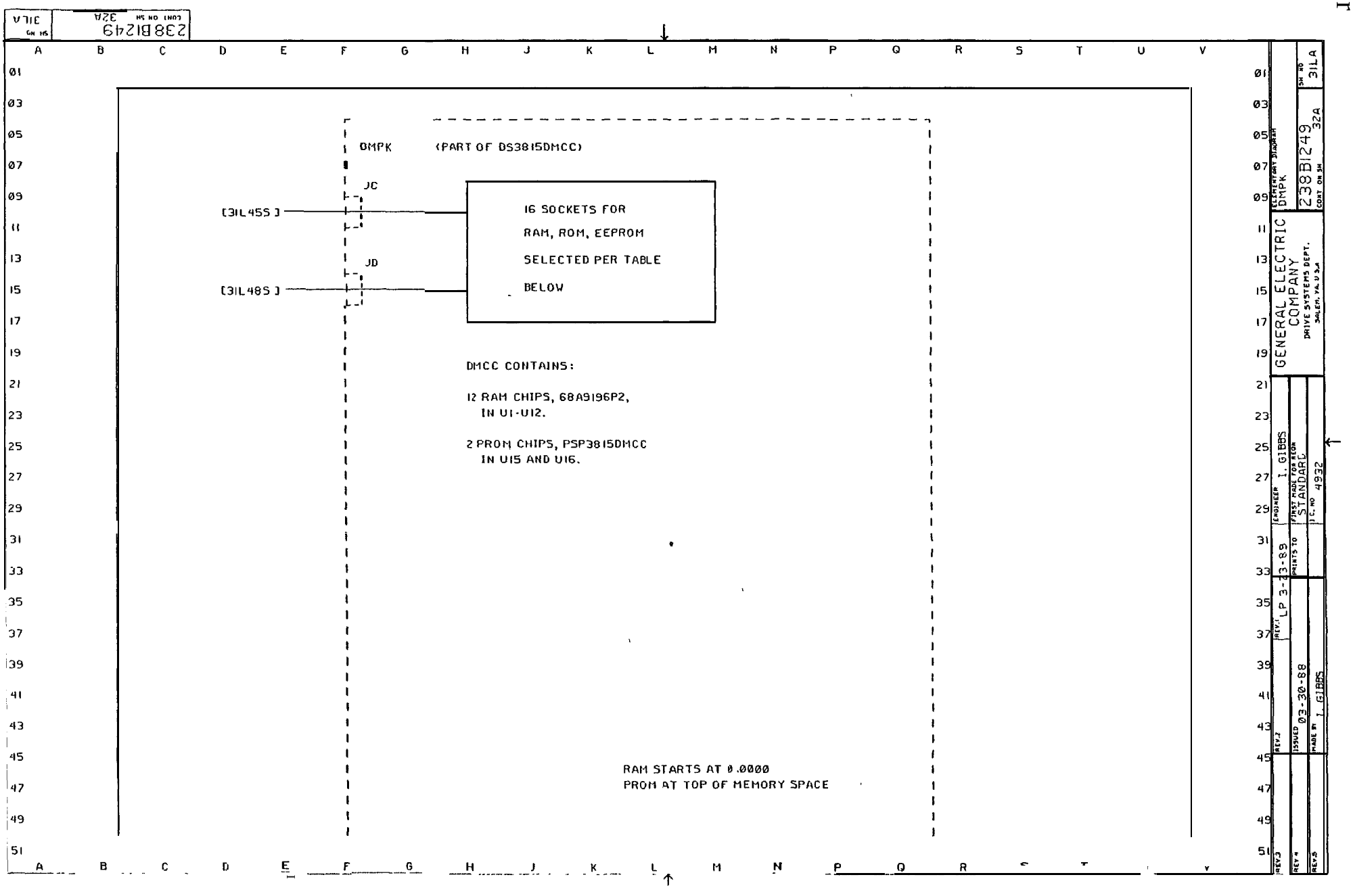
CKT	JA
1	9
2	10
3	7
4	5
5	3
6	4
7	6
8	8
9	14
10	13
11	16
12	18
13	20
14	19
15	17
16	15

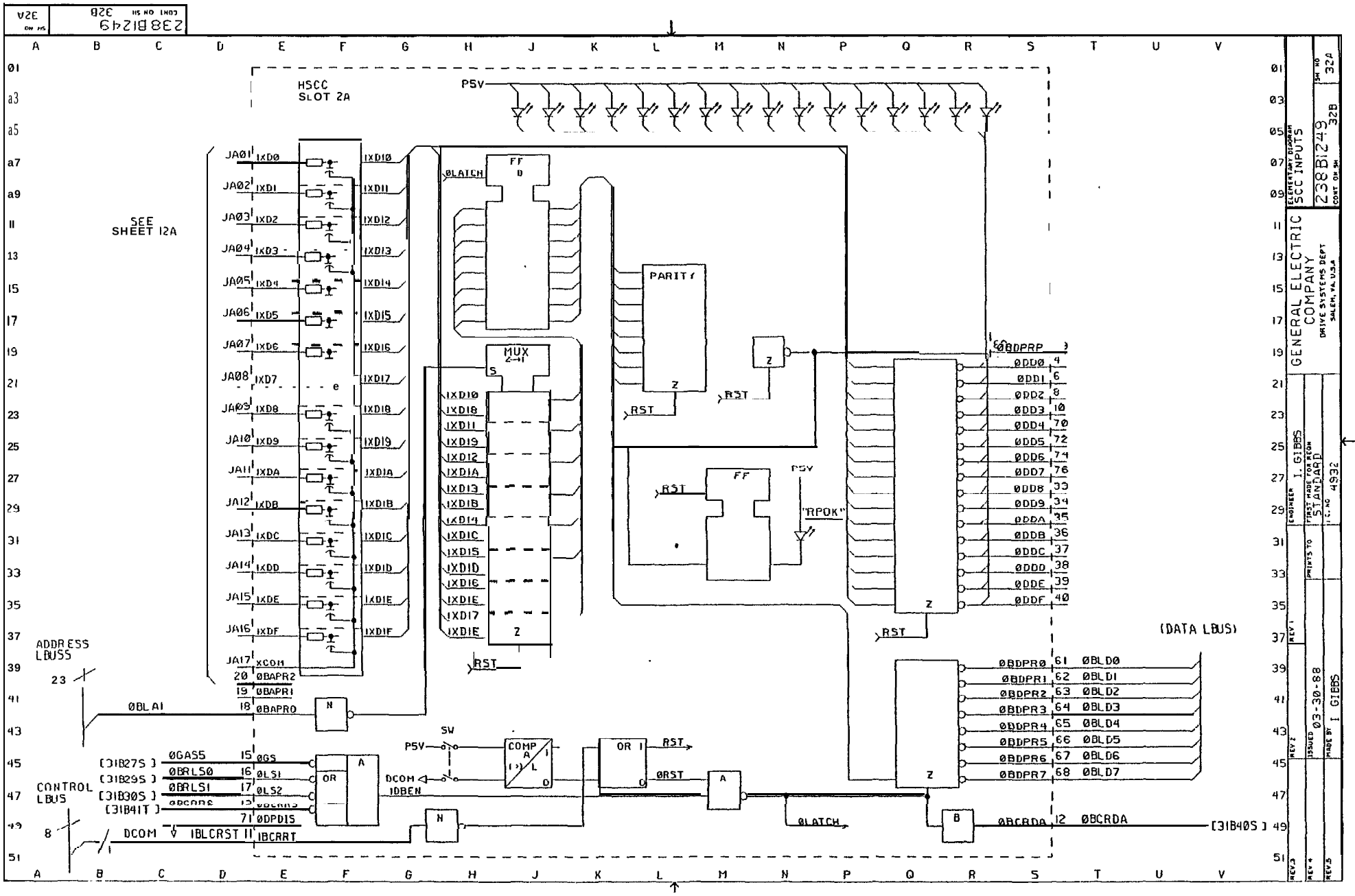
CKT	JB
1	9
2	12
3	18
4	15
5	11
6	14
7	13
8	16
9	10
10	7
11	4
12	1
13	8
14	5
15	6
16	3

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT
SALEM, VA. U.S.A.
ELEMENT/UNIT DIAGRAM HXPD
238B1249
SH. NO. 31K
CONT ON SH 31L
ORIG
ENGINEER I GIBBS
FIRST MADE FOR REGN STANDARD
DRG. NO. 4932
REV.1 LP 3-13-89
PRINTS TO
REV.2
ISSUED 03-30-88
MADE BY I GIBBS

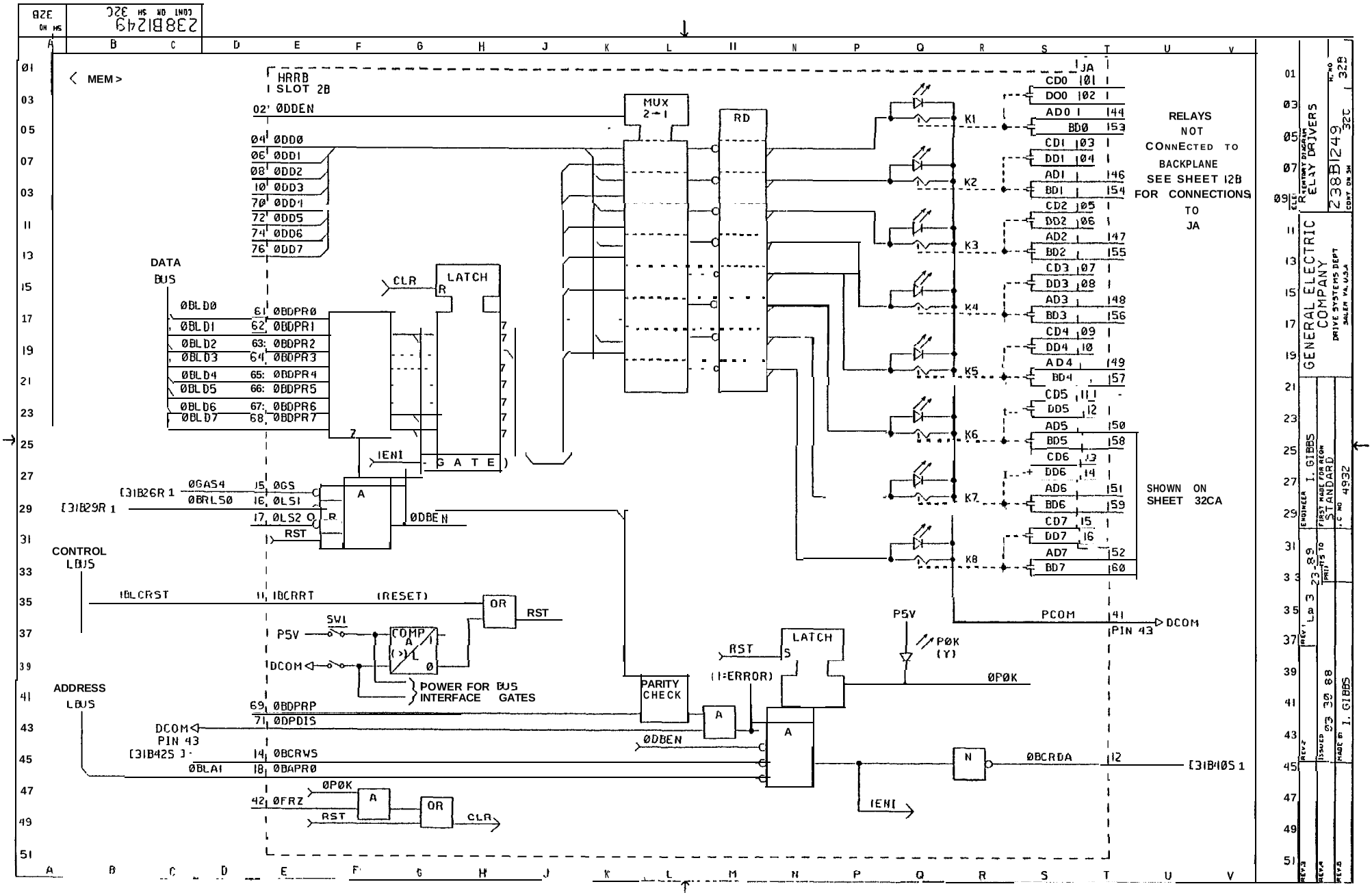

, DS38 I5DMCC
, (HMPK)
, SLOT IL
80297
MATH
COPROCESSOR
CONTROL
80286
CONTROL
LBUS
DATA
LBUS
55 ØPRST
15 ØNMI
20 ØBLCDAK
ØBLD0-F
RST TP1
DCOM TP2
BUS HOG
TIMER
80µ SEC
INT0
71 ID04
63 ID05
65 ID06
62 ID07
DIGITAL OUTPUTS
30 ØRD
24E
JEG
PSV
DCOM
JB
3 RXD
6 DSR
7 SG
INS
J5
GND
[36S]
PAR/
SERIAL
RX
TX
INT2
INT3
INT0
INT1
INT2
INT3
INTER-
RUPT
E
GFRQ
HXPD
SYNCI
32J16K
31K44T
31K41S
[32J16L]
[31K44T]
[31K41T]
24 ØINT4
23 ØINT5
22 ØINT6
25 ØINT7
8 ØEXTSTL
DIGITAL
OUTPUTS
ID00 67 OUTINH
ID01 78 IIR
ID02 80
ID03 69 I3R
[20C06T1
20C06B
[3ID10E]
[3IE10E]
[3IG10E]
[3IH10E 1
[3IJ10E 3
ØWREN 18 / 23
ØEXTSTL
TSEL
TD
A
ISTALL 4
CONTROL
, LBUS
ØBLCMR 13
ØBLCMW 16
ØBLCIOR 14
ØBLCIOW 11
IBLCRST 6
TD
INT1
TD
ØTICK 21
ITIMER 10
0.1728 MHZ
ØBLSCLK 2
[31K34E1
32R
ØBLCIAK 28
JB
CTS 5
RTS 4
TXD 2
DTR 20
CD 9
[35E]
[10E17R]
9
JC
40
[31LAI0E]
JD
40
[31LAI5E1
1

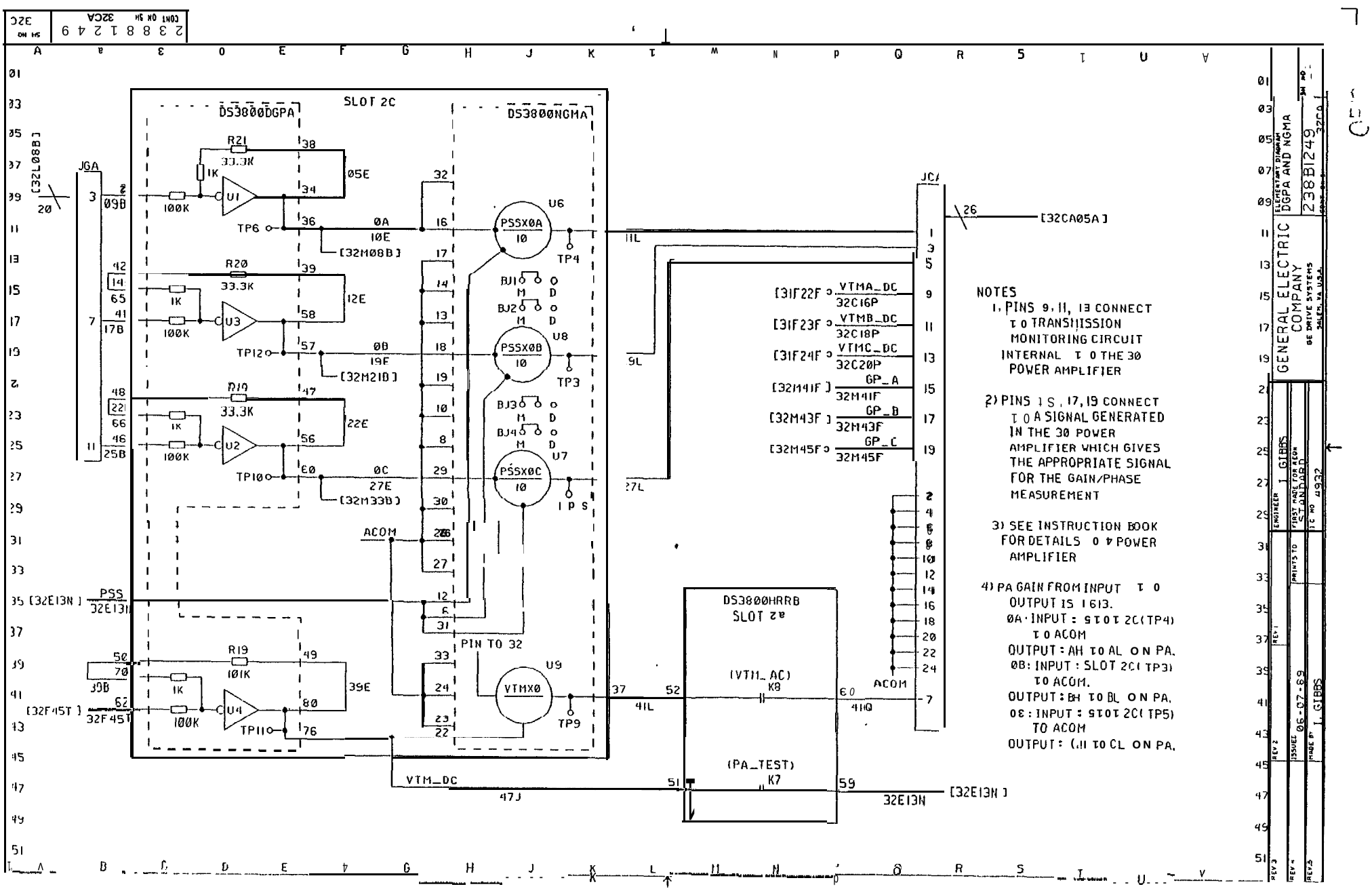
ORIG

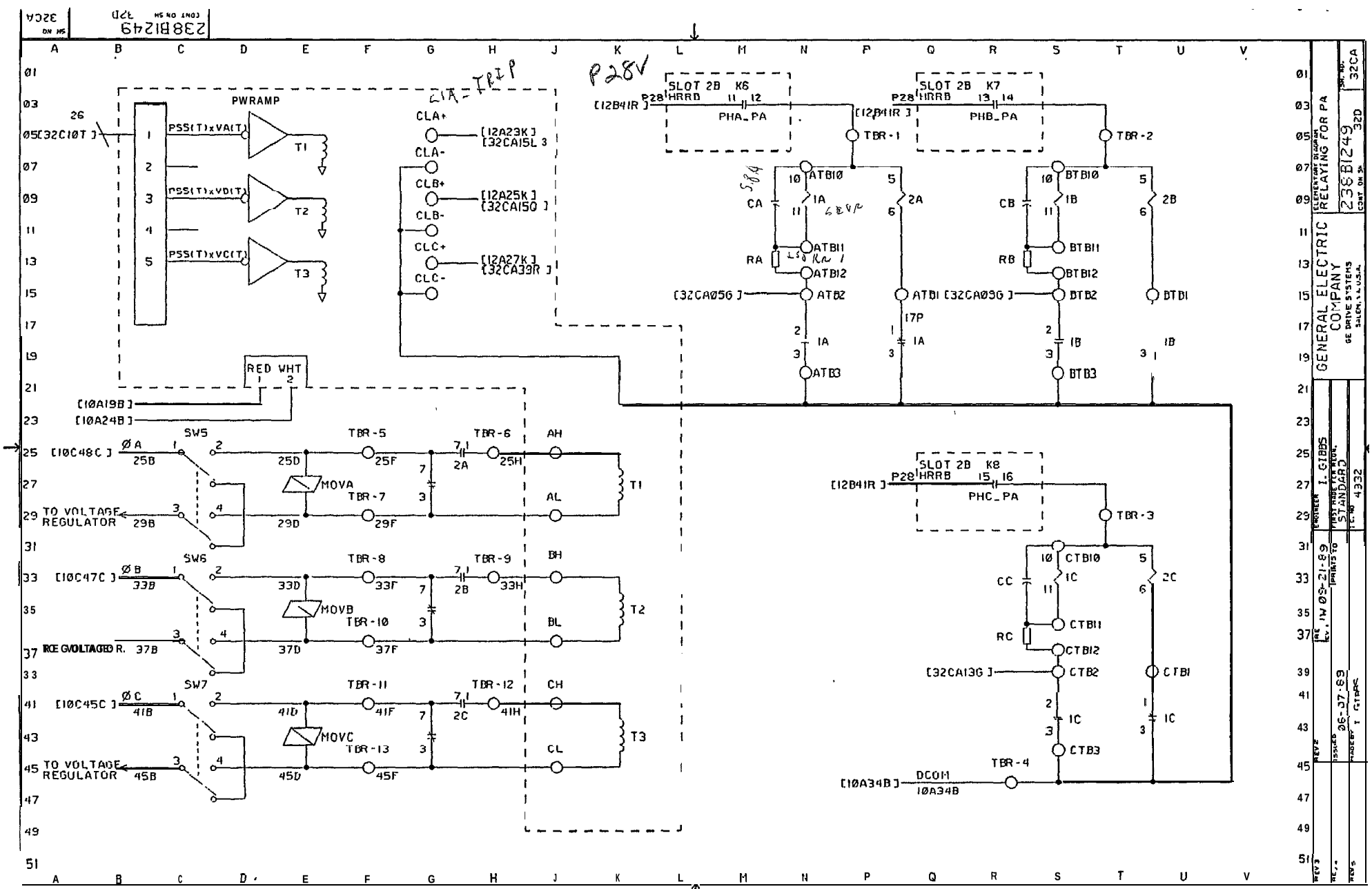

238B1249 32A
31LA
DMPK (PART OF DS3815DMCC)
JC
[31L45S]
JD
[31L48S]
16 SOCKETS FOR
RAM, ROM, EEPROM
SELECTED PER TABLE
BELOW
DMCC CONTAINS:
12 RAM CHIPS, 68A9196P2,
IN U1-U12.
2 PROM CHIPS, PSP3815DMCC
IN U15 AND U16.
RAM STARTS AT 0.0000
PROM AT TOP OF MEMORY SPACE
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
DMPK
238B1249 32A
31LA
LP 3-13-89
03-30-88
J. GIBBS
4932

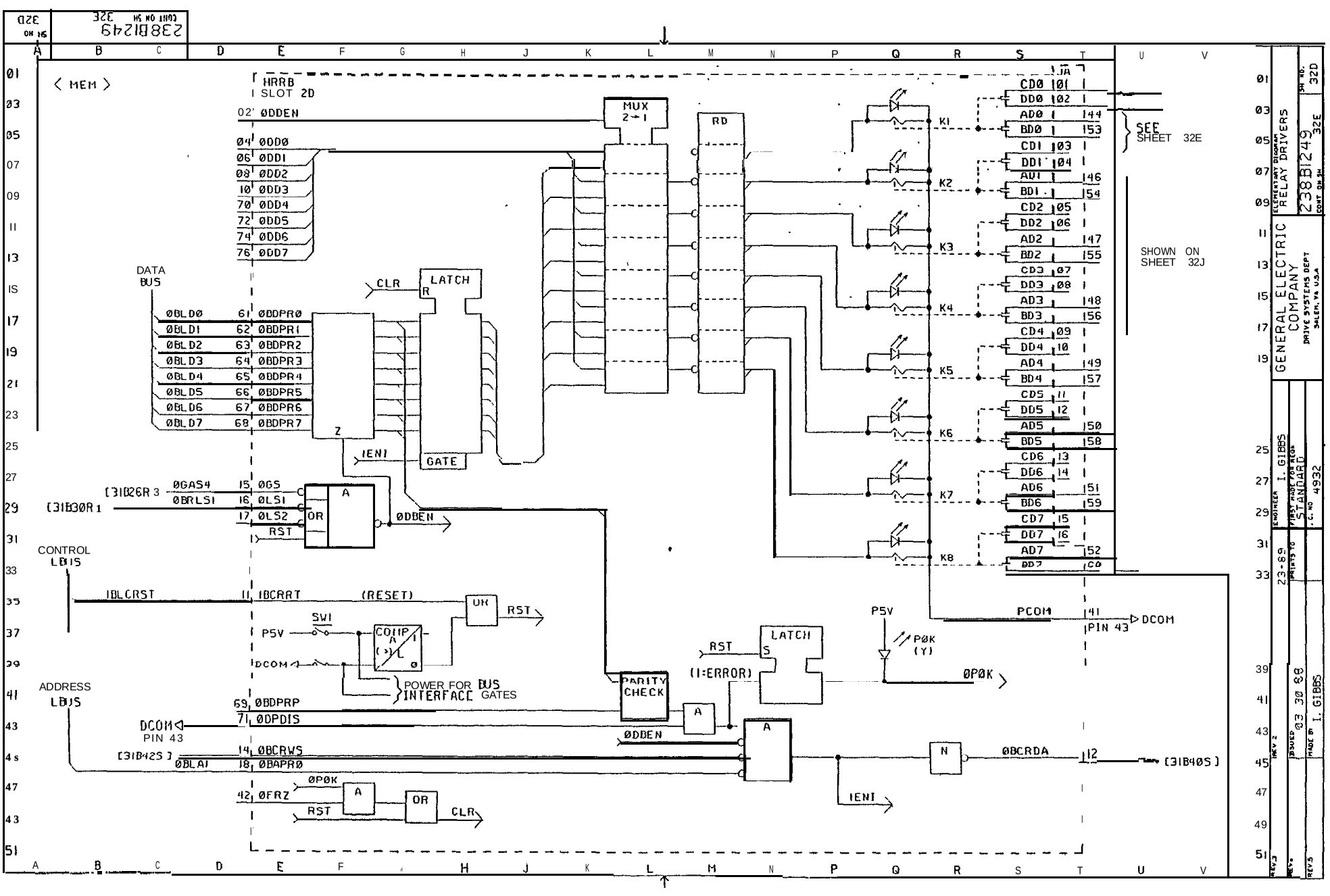

HSCC
SLOT 2A
PSV
SEE
SHEET 12A
JA01 IXD0
JA02 IXD1
JA03 IXD2
JA04 IXD3
JA05 IXD4
JA06 IXD5
JA07 IXD6
JA08 IXD7
JA09 IXD8
JA10 IXD9
JA11 IXDA
JA12 IXDB
JA13 IXDC
JA14 IXDD
JA15 IXDE
JA16 IXDF
JA17 XCOM
IXD10
IXD11
IXD12
IXD13
IXD14
IXD15
IXD16
IXD17
IXD18
IXD19
IXD1A
IXD1B
IXD1C
IXD1D
IXD1E
IXD1F
20 ØBAPR2
19 ØBAPR1
18 ØBAPRO
ØLATCH
FF
D
MUX
S
PARITY
Z
RST
FF
"RPOK"
ØBDPRP
ØDD0 4
ØDD1 6
ØDD2 8
ØDD3 10
ØDD4 70
ØDD5 72
ØDD6 74
ØDD7 76
ØDD8 33
ØDD9 34
ØDDA 35
ØDDB 36
ØDDC 37
ØDDD 38
ØDDE 39
ØDDF 40
(DATA LBUS)
ØBDPR0 61 ØBLD0
ØBDPR1 62 ØBLD1
ØBDPR2 63 ØBLD2
ØBDPR3 64 ØBLD3
ØBDPR4 65 ØBLD4
ØBDPR5 66 ØBLD5
ØBDPR6 67 ØBLD6
ØBDPR7 68 ØBLD7
ADDRESS
LBUSS
23
ØBLAI
CONTROL
LBUS
8
[31B27S] ØGAS5 15 ØGS
[31B29S] ØBRLS0 16 ØLS1
[31B30S] ØBRLS1 17 ØLS2
[31B41T] 71 ØDPD15
DCOM
IBLCRST
IBCRRT
SW
COMP
OR
ØRST
IDBEN
ØBCRDA 12 ØBCRDA
[31B40S]
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT
SALEM, VA. U.S.A.
SCC INPUTS
238B1249
32A
32B
I. GIBBS
ENGINEER
PRINTS TO
FIRST MADE FOR MEDIUM
STANDARD
4932
ISSUED 03-30-88
MADE BY I GIBBS
REV 1
REV 2
REV 3
REV 4
REV 5

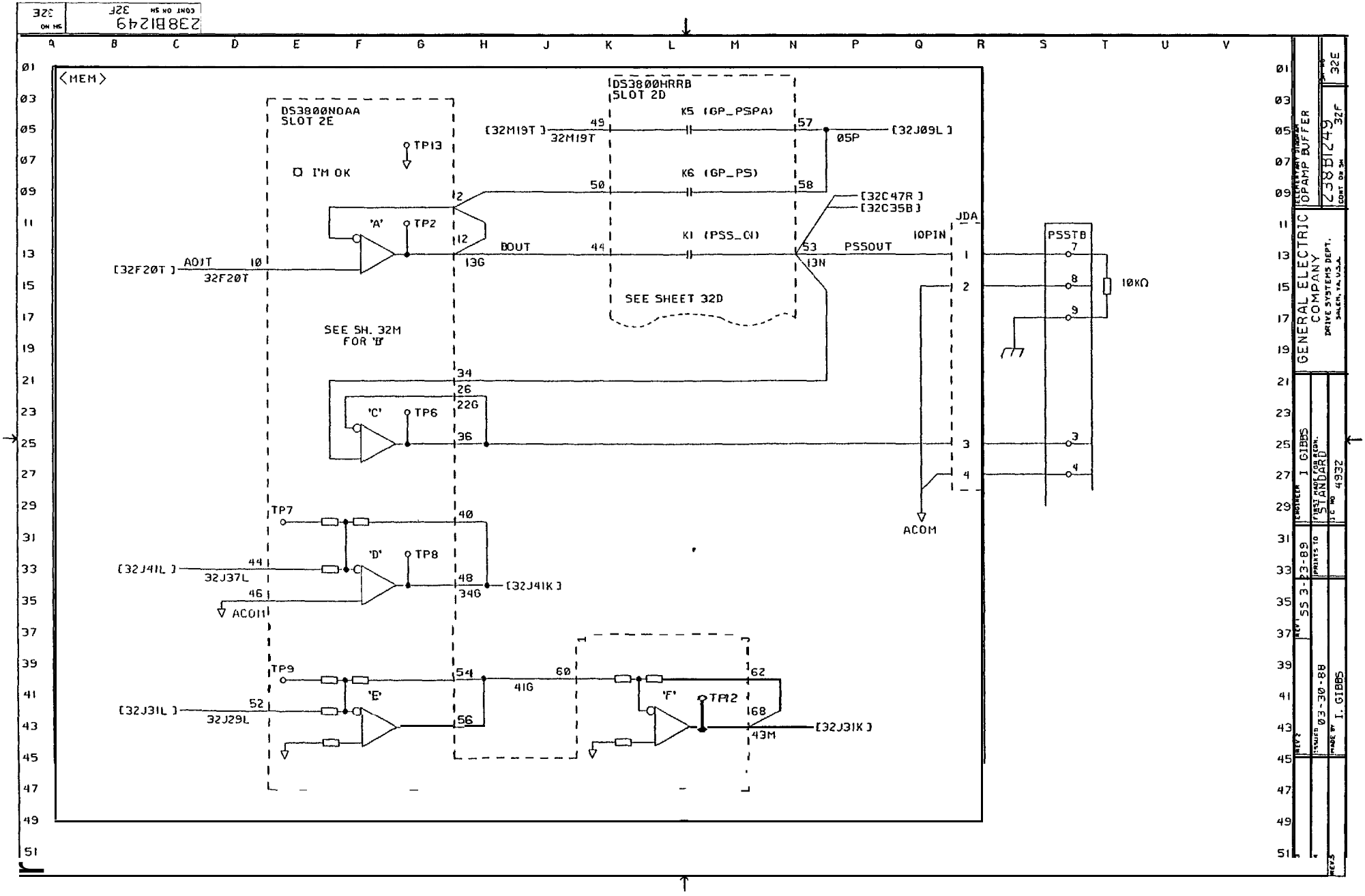
ORIG

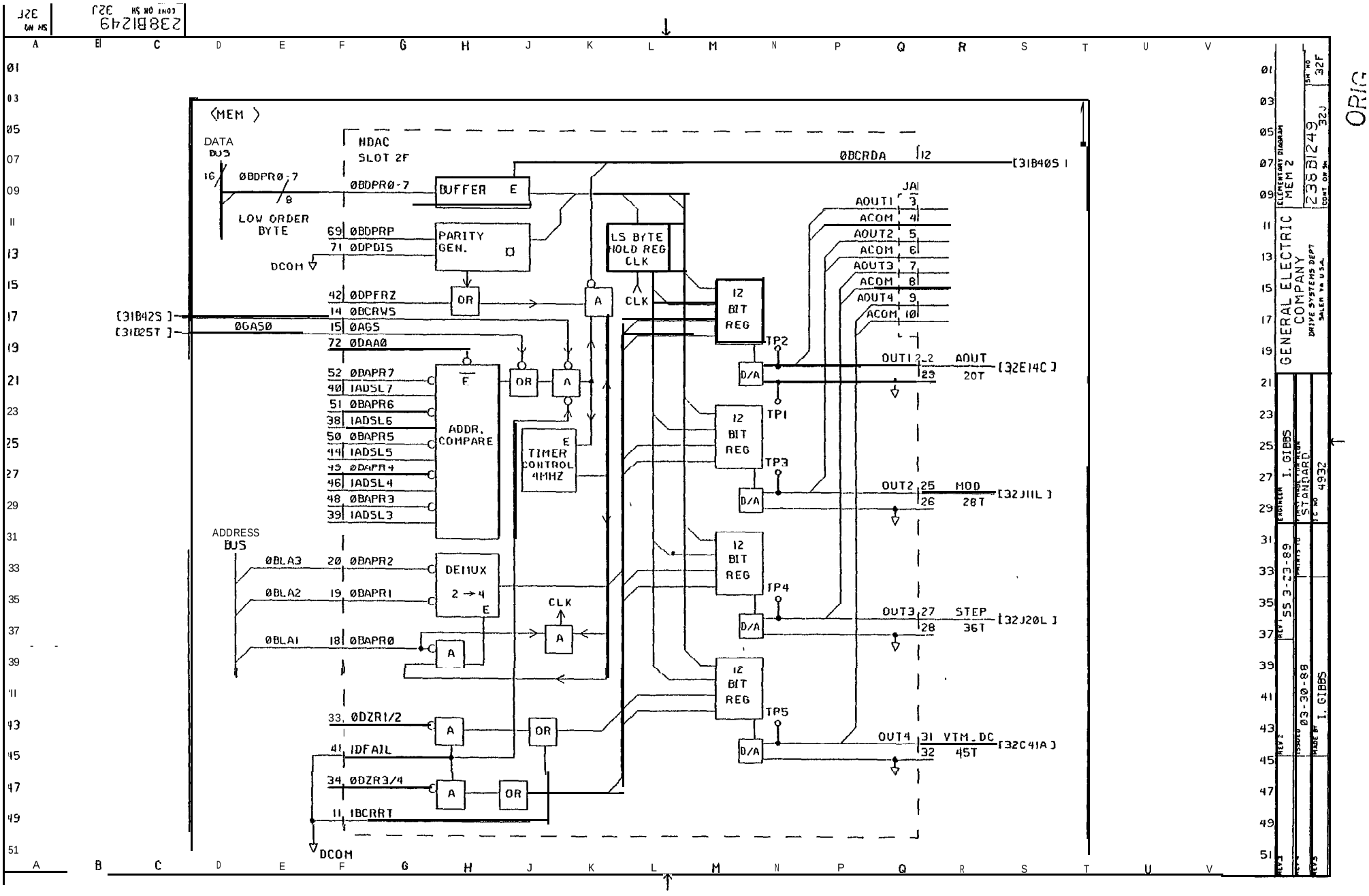

MEM>
HRRB
SLOT 2B
02' 0DDEN
04' 0DD0
06' 0DD1
08' 0DD2
10' 0DD3
70' 0DD4
72' 0DD5
74' 0DD6
76' 0DD7
MUX 2→1
RD
DATA BUS
0BLD0
0BLD1
0BLD2
0BLD3
0BLD4
0BLD5
0BLD6
0BLD7
61' 0BDPR0
62' 0BDPR1
63' 0BDPR2
64' 0BDPR3
65' 0BDPR4
66' 0BDPR5
67' 0BDPR6
68' 0BDPR7
CLR
LATCH
IENI
GATE
[31B26R1
0GAS4
0BRLS0
15' 0GS
16' 0LS1
17' 0LS2
RST
0DBEN
[31B29R1
CONTROL BUS
IBLCRST
11' IBCRRT
(RESET)
OR
RST
SW1
P5V
COMP
DCOM
POWER FOR BUS INTERFACE GATES
ADDRESS BUS
69' 0BDPRP
71' 0DPDIS
PARITY CHECK
DCOM PIN 43
[31B42S1]
14' 0BCRWS
0BLA1
18' 0BAPR0
0P0K
42' 0FRZ
RST
OR
CLR
RST
LATCH
S
(1=ERROR)
0P0K
P5V
P0K (Y)
0DBEN
N
0BCRDA
12
[31B40S1]
IENI
K1
K2
K3
K4
K5
K6
K7
K8
JA
CD0 101
DD0 102
AD0 144
BD0 153
CD1 103
DD1 104
AD1 146
BD1 154
CD2 105
DD2 106
AD2 147
BD2 155
CD3 107
DD3 108
AD3 148
BD3 156
CD4 109
DD4 110
AD4 149
BD4 157
CD5 111
DD5 112
AD5 150
BD5 158
CD6 113
DD6 114
AD6 151
BD6 159
CD7 115
DD7 116
AD7 152
BD7 160
PCOM 41
PIN 43
DCOM
RELAYS NOT CONNECTED TO BACKPLANE SEE SHEET 12B FOR CONNECTIONS TO JA
SHOWN ON SHEET 32CA
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT
SALEM VA. U.S.A.
ENGINEER I. GIBBS
FIRST MADE FOR REGN STANDARD
DRAWN BY I. GIBBS
ISSUED 03 30 88
2388249
32C
32B


SLOT 2C
DS3800DGPA
DS3800NGMA
JGA
JC1
R21
R20
R19
U1
U3
U2
U4
PSSX0A
PSSX0B
PSSX0C
VTMX0
U6
U8
U7
U9
TP6
TP12
TP10
TP11
TP4
TP3
TP9
ACOM
PIN TO 32
PSS
32E13I
32F45T
VTM_DC
47J
DS3800HRRB
SLOT 2E
(VTM_AC)
K8
(PA_TEST)
K7
[32E13N]
32E13N
[32CA05A]
[31F22F] VTMA_DC 32C16P
[31F23F] VTMB_DC 32C18P
[31F24F] VTMC_DC 32C20P
[32M41F] GP_A 32M41F
[32M43F] GP_B 32M43F
[32M45F] GP_C 32M45F
[32L08B]
[32M08B]
[32M21B]
[32M33B]
NOTES
1. PINS 9, 11, 13 CONNECT TO TRANSMISSION MONITORING CIRCUIT INTERNAL TO THE 30 POWER AMPLIFIER
2) PINS 15, 17, 19 CONNECT TO A SIGNAL GENERATED IN THE 30 POWER AMPLIFIER WHICH GIVES THE APPROPRIATE SIGNAL FOR THE GAIN/PHASE MEASUREMENT
3) SEE INSTRUCTION BOOK FOR DETAILS OF POWER AMPLIFIER
4) PA GAIN FROM INPUT TO OUTPUT IS 1613.
0A: INPUT: SLOT 2C (TP4) TO ACOM
OUTPUT: AH TO AL ON PA.
0B: INPUT: SLOT 2C (TP3) TO ACOM.
OUTPUT: BH TO BL ON PA.
0E: INPUT: SLOT 2C (TP5) TO ACOM
OUTPUT: CH TO CL ON PA.
GENERAL ELECTRIC COMPANY
DGPA AND NGMA
238B1249
J. GIBBS
06-07-89

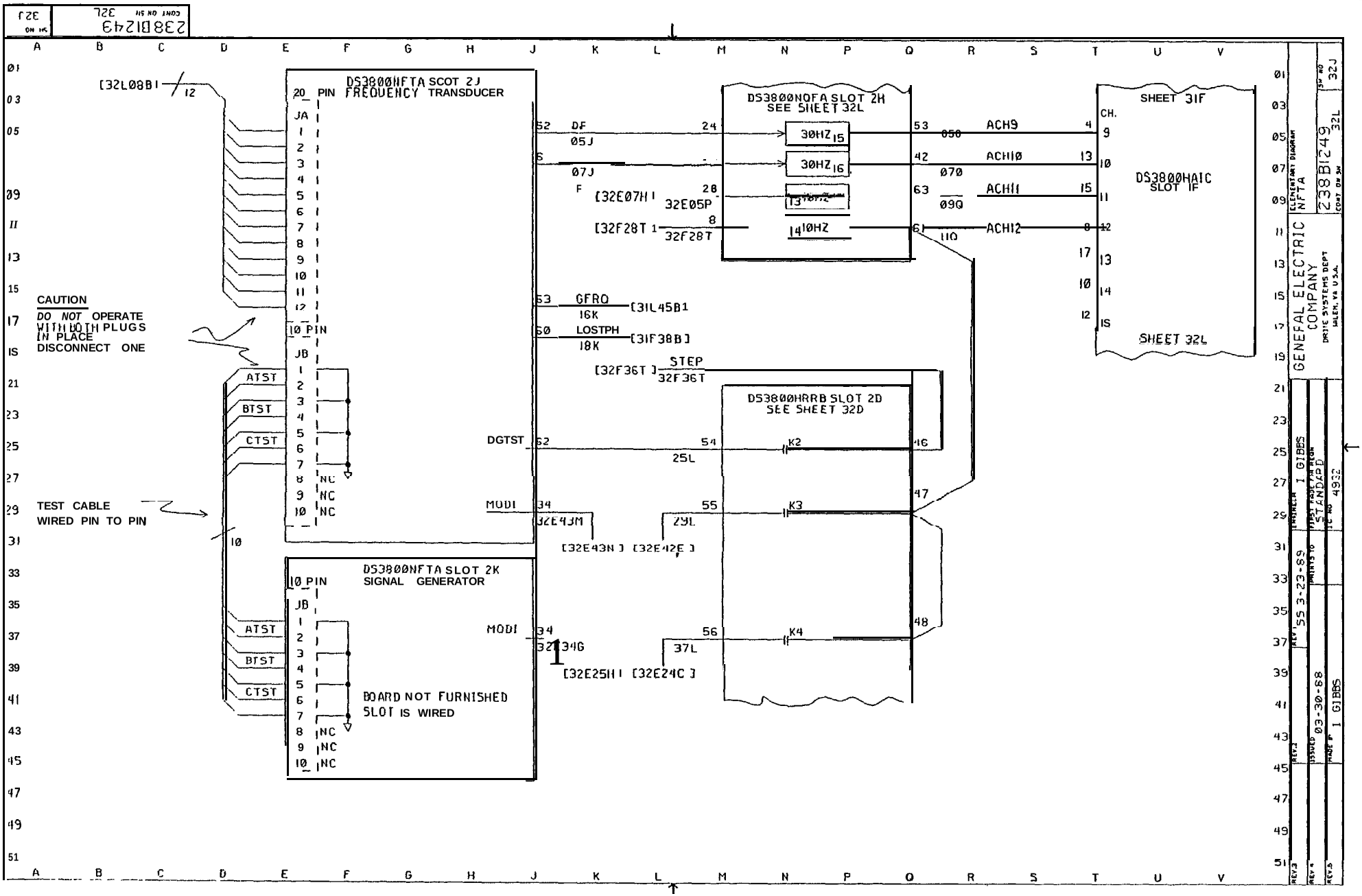

PWRAMP
PSS(T)xVA(T)
PSS(T)xVB(T)
PSS(T)xVC(T)
T1
T2
T3
CLA+
CLA-
CLB+
CLB-
CLC+
CLC-
[12A23K] [32CA15L 3]
[12A25K] [32CA15O]
[12A27K] [32CA33R]
SLOT 2B K6
SLOT 2B K7
SLOT 2B K8
P28 HRRB
PHA_PA
PHB_PA
PHC_PA
[12B41R]
TBR-1
TBR-2
TBR-3
TBR-4
ATBI0
ATBI1
ATBI2
ATB2
ATB3
ATBI
BTBI0
BTBI1
BTBI2
BTB2
BTB3
BTBI
CTBI0
CTBI1
CTBI2
CTB2
CTB3
CTBI
1A
2A
17P
1B
2B
1C
2C
CA
RA
CB
RB
CC
RC
[32CA05G]
[32CA09G]
[32CA13G]
RED WHT
[10A19B]
[10A24B]
SW5
SW6
SW7
ØA
ØB
ØC
[10C48C]
[10C47C]
[10C45C]
TBR-5
TBR-6
TBR-7
TBR-8
TBR-9
TBR-10
TBR-11
TBR-12
TBR-13
MOVA
MOVB
MOVC
AH
AL
BH
BL
CH
CL
TO VOLTAGE REGULATOR
DCOM
[10A34B]
10A34B
P28V
GENERAL ELECTRIC COMPANY
GE DRIVE SYSTEMS
SALEM, VA U.S.A.
RELAYING FOR PA
238B1249
32CA
32D
I. GIBBS
STANDARD
4332
06-07-89
09-21-89

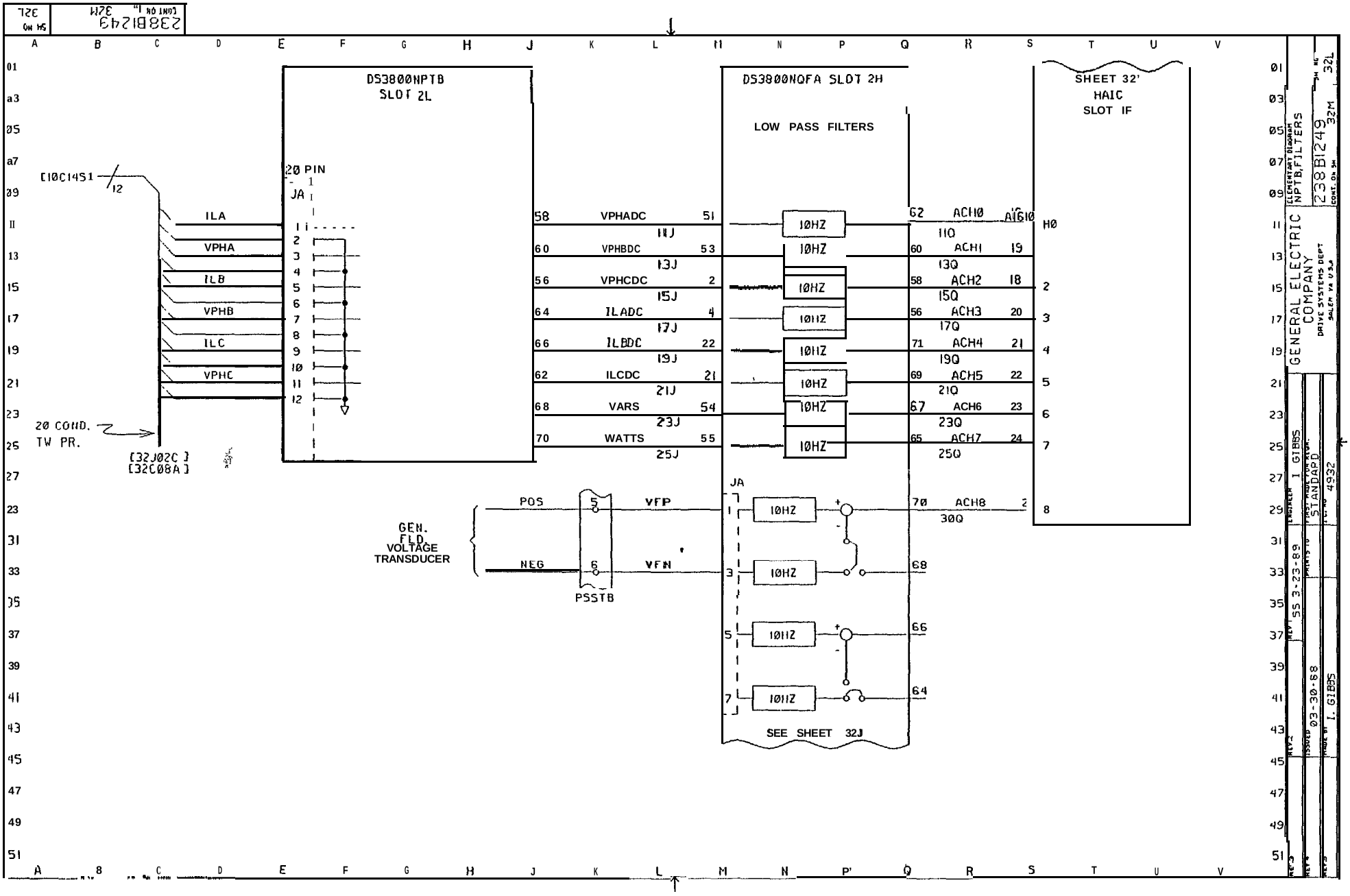

< MEM >
HRRB
SLOT 2D
02' ØDDEN
04' ØDD0
06' ØDD1
08' ØDD2
10' ØDD3
70' ØDD4
72' ØDD5
74' ØDD6
76' ØDD7
MUX 2→1
RD
DATA BUS
CLR
LATCH
R
ØBLD0 61 ØBDPR0
ØBLD1 62 ØBDPR1
ØBLD2 63 ØBDPR2
ØBLD3 64 ØBDPR3
ØBLD4 65 ØBDPR4
ØBLD5 66 ØBDPR5
ØBLD6 67 ØBDPR6
ØBLD7 68 ØBDPR7
Z
IENI
GATE
[31B26R3 ØGAS4 15 ØGS
ØBRLSI 16 ØLSI
[31B30R1 17 ØLS2
OR
A
RST
ØDDEN
CONTROL LBUS
IBLCRST 11 IBCRRT (RESET)
OR
RST
SWI
P5V
COMP
DCOM
POWER FOR BUS INTERFACE GATES
ADDRESS LBUS
69 ØBDPRP
71 ØDPDIS
DCOM PIN 43
[31B42S] 14 ØBCRWS
ØBLAI 18 ØBAPRØ
PARITY CHECK
A
ØDDEN
LATCH
S
RST
(1:ERROR)
P5V
PØK (Y)
ØPØK
A
N
ØBCRDA 12
[31B40S]
ØPØK
42 ØFRZ
A
OR
RST
CLR
IENI
CD0 101
DD0 102
AD0 144
BD0 153
CD1 103
DD1 104
AD1 146
BD1 154
CD2 105
DD2 106
AD2 147
BD2 155
CD3 107
DD3 108
AD3 148
BD3 156
CD4 109
DD4 110
AD4 149
BD4 157
CD5 111
DD5 112
AD5 150
BD5 158
CD6 113
DD6 114
AD6 151
BD6 159
CD7 115
DD7 116
AD7 152
K1
K2
K3
K4
K5
K6
K7
K8
PCOM 141
PIN 43 DCOM
SEE SHEET 32E
SHOWN ON SHEET 32J
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT
SALEM, VA U.S.A.
RELAY DRIVERS
238B1249
32D
32E
I. GIBBS
STANDARD
4932
23-89
03 30 88
I. GIBBS

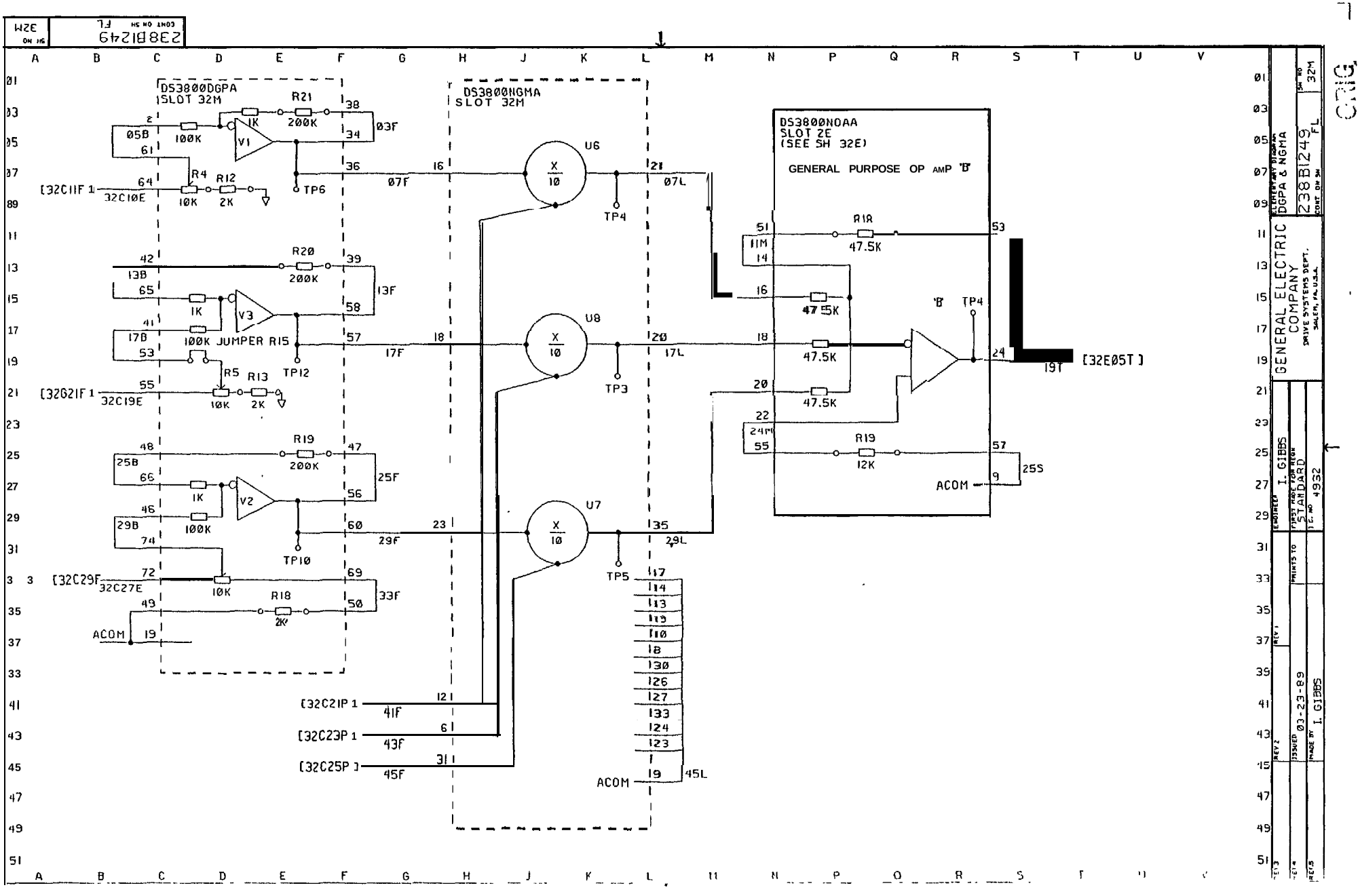

<MEM>
DS3800NOAA
SLOT 2E
DS3800HRRB
SLOT 2D
TP13
I'M OK
'A'
TP2
'B'
'C'
TP6
'D'
TP8
'E'
'F'
TP12
TP7
TP9
K5 (GP_PSPA)
K6 (GP_PS)
K1 (PSS_ON)
[32M19T]
32M19T
05P
[32J09L]
[32C47R]
[32C35B]
JDA
10PIN
PSSTB
PSSOUT
BOUT
13G
13N
ADJT
[32F20T]
32F20T
10kΩ
SEE SHEET 32D
SEE SH. 32M
FOR 'B'
22G
34G
41G
43M
ACOM
ACOM
[32J41L]
32J37L
[32J41K]
[32J31L]
32J29L
[32J31K]
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
OPAMP BUFFER
238B1249
J. GIBBS
STANDARD
03-30-88
SS 3-23-89


(MEM)
DATA BUS
16
0BDPR0-7
8
LOW ORDER BYTE
NDAC SLOT 2F
0BDPR0-7
BUFFER E
69 0BDPRP
71 0DPDIS
PARITY GEN.
DCOM
42 0DPFRZ
14 0BCRWS
15 0AGS
72 0DAA0
[31B42S]
[31B25T]
0GAS0
OR
A
CLK
LS BYTE HOLD REG CLK
0BCRDA 12
[31B40S]
12 BIT REG
D/A
TP2
TP1
TP3
TP4
TP5
AOUT1 3
ACOM 4
AOUT2 5
ACOM 6
AOUT3 7
ACOM 8
AOUT4 9
ACOM 10
OUT1 2-2 AOUT 20T [32E14C]
OUT2 25 MOD 28T [32J11L]
OUT3 27 STEP 36T [32J20L]
OUT4 31 VTM_DC 45T [32C41A]
52 0BAPR7
40 IADSL7
51 0BAPR6
38 IADSL6
50 0BAPR5
44 IADSL5
45 0BAPR4
46 IADSL4
48 0BAPR3
39 IADSL3
F
ADDR. COMPARE
E
TIMER CONTROL 4MHZ
ADDRESS BUS
0BLA3
0BLA2
0BLA1
20 0BAPR2
19 0BAPR1
18 0BAPR0
DEMUX 2→4
33 0DZR1/2
41 IDFAIL
34 0DZR3/4
11 IBCRRT
DCOM
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT
SALEM VA U.S.A.
ELEMENTARY DIAGRAM MEM 2
2388249 32J
32F
ENGINEER I. GIBBS
STANDARD
DWG NO 4932
REV'S 3-23-89
ISSUED 03-30-88
DRAWN BY I. GIBBS

ORIG


DS3800NFTA SCOT 2J
20 PIN FREQUENCY TRANSDUCER
[32L08B1] 12
JA
JB
10 PIN
CAUTION
DO NOT OPERATE WITH BOTH PLUGS IN PLACE
DISCONNECT ONE
TEST CABLE
WIRED PIN TO PIN
ATST
BTST
CTST
NC
DS3800NFTA SLOT 2K
SIGNAL GENERATOR
BOARD NOT FURNISHED
SLOT IS WIRED
DS3800NOFA SLOT 2H
SEE SHEET 32L
30HZ 15
30HZ 16
10HZ 14
[32E07H]
32E05P
[32F28T]
32F28T
DF
05J
07J
GFRO
16K
[31L45B1]
LOSTPH
18K
[31F38B]
STEP
[32F36T]
32F36T
DS3800HRRB SLOT 2D
SEE SHEET 32D
DGTST
MODI
25L
29L
37L
K2
K3
K4
[32E43N] [32E42E]
[32E25H] [32E24C]
SHEET 31F
SHEET 32L
DS3800HAIC
SLOT 1F
CH.
ACH9
ACH10
ACH11
ACH12
050
070
090
110
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT
SALEM, VA U.S.A.
ELEMENTARY DIAGRAM
NFTA
238B1249
32J
32L
DRAWN 1 GIBBS
REV. SS 3-23-89
ISSUED 03-30-88
SHEET 1 GIBBS


DS3800NPTB
SLOT 2L
DS3800NQFA SLOT 2H
LOW PASS FILTERS
SHEET 32'
HAIC
SLOT IF
20 PIN
JA
[10C14S1
12
ILA
VPHA
ILB
VPHB
ILC
VPHC
20 COND.
TW PR.
[32J02C]
[32C08A]
58 VPHADC 51
11J
60 VPHBDC 53
13J
56 VPHCDC 2
15J
64 ILADC 4
17J
66 ILBDC 22
19J
62 ILCDC 21
21J
68 VARS 54
23J
70 WATTS 55
25J
10HZ
10HZ
10HZ
10HZ
10HZ
10HZ
10HZ
62 ACH0 AI610
11Q
60 ACH1 19
13Q
58 ACH2 18
15Q
56 ACH3 20
17Q
71 ACH4 21
19Q
69 ACH5 22
21Q
67 ACH6 23
23Q
65 ACH7 24
25Q
H0
2
3
4
5
6
7
8
JA
POS 5 VFP
NEG 6 VFN
PSSTB
GEN.
FLD.
VOLTAGE
TRANSDUCER
1
3
5
7
10HZ
10HZ
10HZ
10HZ
70 ACH8 2
30Q
68
66
64
SEE SHEET 32J
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT
SALEM VA U.S.A.
238B1249
32M
32L
I. GIBBS
4932
STANDARD
SS 3-23-89
03-30-88
I. GIBBS



GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
DGPA & NGMA
238B1249 FL
SH NO 32M
CONT ON SH
ENGINEER I. GIBBS
STANDARD
DRAWN BY I. GIBBS
ISSUED 03-23-89

```
;----------------------------------------------------------------------------
;..NAME: PGE_VARDEF_0002
;..REVISED_FROM: PGE_VARDEF_0001
;
;..AUTHOR:GIBBS
;
;..COPYRIGHT:    COPYRIGHT (C) 1986
;                BY GENERAL ELECTRIC COMPANY, U.S.A
;
;..KEYWORDS:DHC
;
;..ABSTRACT:
;
;..DESCRIPTION:       This module defines to a DMC VARDEF
;                    used in a DATATRONIC DAS.
;-----------------------------------------------------------------------
;
;  *_MEANS_DOWNLOAD(Y/N)  :  N
;
;   Physical      Symbolic      Constant    Fixed-point
;     name          name          value       format        Comments
;   ========      ========      ========    ===========    ========
;
;   NOTE: If a line start's with semicolon sign then the whole line
;         is treated as comment.

;-----------------------------------------------------------------------

;SEGMENT NAMES
     . SEGMENT              RCRDER
     . SEGMENT              T_RCRDR

;-----------------------------------------------------------------------

;                VARIABLES
;
;-----------------------------------------------------------------------
;   LOGIC FOR POWER AMPLIFIER
    .L              PA-FAULT          ;INDICATES POWER AMPLIFIER PHASE FAULT.
    .CC             K_LPA_ENABLE      0                ;SET TO 1 TO ENABLE.
    .L              LPA_ENABLE        ;ENABLES PA PROTECTIDN.
    .L              PA-ENABLE         ;POWER AMPLIFIER ENABLE.
    .L              ALL-TRIP          ;CAUSES ALL PA CHANNELS TO TRIP.
    .L              TMA_TRIP          ;CONDITIONS FOR TN TRIP PHA NECESSARY.
    .L              TMB_TRIP          ;CONDITIONS FOR TM TRIP PHB NECESSARY.
    .L              TMC_TRIP          ;CONDITIONS FOR TM TRIP PHC NECESSARY.
    .L              A-OUT             ;INDICATES PH A TRIPPED.
    .L              B-OUT             ;INDICATES PH B TRIPPED.
    .L              C_OUT             ;INDICATES PH C TRIPPED.
    .L              ONE_OUT           ;INDICATES DNE PHASE TRIPPED.
    .L              TWO-DUT           ;INDICATES TWO PHASES TRIPPED.
    .cc             K-TRIP-ONE        0                ;SET TO 1 TO ENABLE.
    .L              TRIP-ONE-DUT      ;IF TRUE PH TRIPS. ALL PH WILL TRIP.
    .CC             K_TRIP_TWO        0                ;SET TO 1 TO ENABLE.
    .L              TRIP-TWO-DUT      ;IF TRUE PH'S TRIP, ALL PH WILL TRIP.
;
    .L              TMA_TRIP_EN       ;TMA PH A WILL TRIP AFTER VTM-DELAY.
    .L              TMB_TRIP_EN       ;TMA PH B WILL TRIP AFTER VTM-DELAY.
    .L              TMC_TRIP_EN       ;TMA PH C WILL TRIP AFTER VTM-DELAY.

    .L              LPA_TEST          ;SET - TO ADJUST PA OUTPUT USING VTM_PRESET.
    .L              LPA_TEST_ONL      ;SET - CONNECTS PA TO SENSING FOR PA-TEST.
;
;                   FAULTS(6-11)      INDICATES A PA FAULT.
```

```
.CC     K_PH_ROT        0               ;SET TO 1 TO ENABLE.
.L      PH_ROT_REV      ;IF SET. INDICATES SYSTEM PH ROT IS ACB.
        FAULTS(12)      FAULT MESSAGE FOR PHASE ROTATION.

.CC     K_VTM_1         0               ;SET TO 1 TO ENABLE.
.L      VTM_1           ;IF SET TM WORKS WITH UP TO 1 FAULT.
.cc     K_VTM_2         0               ;SET TO 1 TO ENABLE.
.L      VTM_2           ;IF SET TM WORKS WITH UP TO 2 FAULTS.
.L      VTM_ENABLE      ;ENABLES TM.
.L      VTMO_ENABLE     ;ENABLES TM TRIPS.
.cc     VTMO-DELAY      10000           ;DELAY MSEC FOR ENABLE TM TRIP.
.vc     CVTMO_DELAY                     ;CURRENT VTMO-DELAY.
.cc     K_LVTM          0               ;SET TO 1 TO ENABLE.
.L      LVTM_DISABLE    ;IF TRUE TM PROTECTION DISABLED.

.cc     K_LCL           0               ;SET TO 1 TO ENABLE.
.L      LCL_DISABLE     ;IF TRUE DISABLES CURRENT LIMIT (PULL RELAYS).
.L      CL-DISABLE      ;FUNCTIONS A CL DISABLE.
.cc     CL-DELAY        3000            ;DELAY MSEC OR CL ON.
.VC     CCL-DELAY                       ;CURRENT CL-DELAY.
.L      CL-ON           ;CURRENT LIMIT ON.

.L      CLA_TRIP_L      ;CURRENT LIMIT TRIP PH A LOGICAL.
.L      CLB_TRIP_L      ;CURRENT LIMIT TRIP PH B LOGICAL.
.L      CLC_TRIP_L      ;CURRENT LIMIT TRIP PH C LOGICAL.

.VR     VTMA_DC         ;COUNT VALVE OF TM PHA IN VOLTS DC.
.VR     VTMB_DC         ;COUNT VALVE OF TM PHB IN VOLTS DC.
.VR     VTMC_DC         ;COUNT VALVE OF TM PHC IN VOLTS DC.

.VR     VTMA-ERROR      ;VTMA_DC - VTM-AVE.
.VR     VTMB_ERROR      ;VTMB_DC - VTM_AVE.
.VR     VTMC-ERROR      ;VTMC_DC - VTM_AVE.
.CR     VTM-AVE         3.0     ;SET = AVE TM VOLTAGE.
.CR     VTM_ERROR       1.5     ;ALLOWABLE ERROR VDC.

.VR     ABS_VTMA_ERR    ;ABSOLUTE VALUE OF VTMA_ERROR.
.VR     ABS_VTMB_ERR    ;ABSOLUTE VALUE OF VTMB_ERROR.
.VR     ABS_VTMC_ERR    ;ABSOLUTE VALUE OF VTMC-ERROR.

.cc     VTM_DELAY       1       ;MILLSECOND DELAY FOR TM.

.VC     VTMA_CDELAY     ;CURRENT VALUE OF DELAY TMA.
.vc     VTMB_CDELAY     ;CURRENT VALUE OF DELAY TMB.
.vc     VTMC_CDELAY     ;CURRENT VALUE OF DELAY TMC.

.CR     VTM_DC          1.0     ; SET = TM OUTPUT VOLTAGE.
.CR     VTM_PRESET      0.0     ;SET = TH PRESET OUTPUT VOLTAGE.
.VR     VTM_COUNT_IN            ;INPUT TO A-C CONVERTER.
.VC     VTM_ COUNT              ;DEFINE COUNT VALVE.
.CR     VTH-SCALE       .0048888        ;A/D SCALING.
.CR     VTMD_SCALE      204.8   ;D/A VTH SCALING.

.CR     GPSS_NORM       1.0     ;GPSS MULT. = 0 CHANNELS OUT.
.CR     GPSS_ONE_OUT    1.5     ;GPSS MULT. = 1 CHANNEL OUT.
.CR     GPSS_TWO_OUT    3.0     ;GPSS MULT. = 2 CHANNELS DUT.
.VR     PA_GPSS                 ;SELECTED GPSS

.L      INTERF_ON
.VC     CDELAY
.cc     TDELAY          1500    ; MILLISECONDS FOR PA ON.

.L      SS_FLT_RST              ;SINGLE SHOT HARDWARE FAULT RST.
.L      TMP_FLT_RST             ;TEMP STORAGE FOR SINGLE SHOT.
```

```
LOGIC FOR HECDRDER TRIGGER
.L              RCRD_T
.L              RCRD_0
COUNT FOR RECORDER TRIGGER
.vc             RCRD_TRIG
;  DUMMY FOR RECORDER TRIG STATUS
.L              T-STAT
LOGICAL VALUE FOR KEY SNITCH PERMISSIVE
.L              LTSTC
LOGICALS FOR PSS ON/OFF COMMANDS
.L              ON-PERM         ;PERMISSIVE TD TURN ON.
.L              LOFFC           ;LOGICAL OFF COMMAND.
.L              LONC            ;LOGICAL ON COMMAND.
.L              LOLC            ;LOGICAL ONLINE COMMAND.
.L              LPSSONC         ;LOGICAL PSS O N  COMMAND.
.L              LPSSOFC         ;LOGICAL PSS OFF COMMAND.
LOGICAL FOR POWER LEVEL SENSING AUTOMATIC ON/OFF
.L              LAUTOC
.L              LPWRONC         ;LOGICAL POWER ON COMMAND.
.L              PONCMD          ;POWER ON COMMAND = PONX MUST BE TRUE.
.L              POFCMD          ;POWER O F F COMMAND - POFX MUST BE TRUE.
.L              PONX            ;POWER LEVEL TO TURN ON REACHED.
.L              POFX            ;POWER LEVEL TO TURN OFF REACHED.
.L              L52GX           ;52 BREAKER LOGICAL.

.vc             PDNCNTR         ;CURRENT DELAY FOR PONCMD TO BE TRUE.
.vc             POFCNTR         ;CURRENT DELAY FOR PDFCHD TO BE TRUE.
INPUT DEFINITIONS FOR HSCC2A
.I10000         R-ON            ;REMOTE PSS O N .
.I10000         CIN000
.I10000         R_OFF           ;REMOTE PSS O F F .
.I10001         CIN001
 110002         G52             ;SENSES 52 BREAKER INPUT = SETS LOGICAL.
: 110002        CIN002
.I10003         CIN003
.I 10003        FLT_LST_RST     ;REMOTE FAULT LIST RESET.
.110004         CIN004
.I10005         CIN005
.I10005         CLA_TRIP        ;SENSES CURRENT LIMIT PH. A TRIP.
.I10006         CIN006
.I10006         CLB_TRIP        ;SENSES CURRENT LIMIT PH. B TRIP.
 110007         CIN007
: 110007        CLC_TRIP        ;SENSES CURRENT LIMIT PH. C TRIP.
.I10010         CIN010
.I10011         CIN011
.I10012         CIN012
.I10013         CIN013
.110014         CIN014
.I10015         CIN015
.I10016         CIN016
.I10017         CIN017
.I10017         EXT_TST         ;IF CONNECTED USED TO TRIGGER TRANSIENT
                                 EVENT RECORDER WITH A CONTACT.

INPUT DEFINITIONS FOR HAIC
.I10100         L-OFF           ;LOGICAL OFF COMMAND FROM PSS LOCAL CONTROL.
                                 (PSSLC)
.I10100         CIN100
.I10101         L_ON            ;LOGICAL ON COMMAND FROM PSSLC.
 110101         CIN101
: 110102        CIN102
.I10103         CIN103
.110104         CIN104
```

```
 I10105      TSTKSW          ;KEY IS TEST POSITION FROM PSSLC.
: 110105     CIN105
.I10106      CIN106
.I10107      CINf07
.I10110      LOSS-PH         ;LOSS OF A SENSING PHASE VOLTAGE.
.I10110      CIN110
.I10111      CIN111
.I10111      RCDR_ENABLE
.I10112      CIN112
.I10113      CIN113
.I10114      CIN114
.I10115      CIN115
.I10116      CIN116
.I10117      CIN117

OUTPUT DEFINITIONS FOR HRRBs

.D10000      RLYOOO
.D10000      ALRM1
.D10001      RLY001
.D10001      ALRM2
.D10002      RLY002
.D10003      RLY003
.D10003      RL_ON           ;EXTERNAL PSS ON.
.a10004      RLYO04
.D10004      RL_OFF          ;EXTERNAL PSS OFF.
.D10005      RLYOOS
.D10005      PHA_PA          ;PH.A POWER AMPLIFIER - CLOSED IS ON.
.D10006      RLY006
.D10006      PHB_PA          ;PH. B POWER AMPLIFIER - CLOSED IS ON.
.D10007      RLY007
.D10007      PHC_PA          ;PH.C POWER AMPLIFIER = COLSED IS ON.
.D10010      RLY010
.D10010      PSS_ON          ;PSS IS ON.
.D10011      RLY011
.D10011      DIGTST          ;DIGITAL TEST INDICATOR.
.D10012      RLY012
.D10012      MTSTSYS         ;MODULATION TEST INDICATOR.
.D10013      RLY013
.D10013      MTSTSP          ;MODULATION TEST WITH OSCILLATION.
.D10014      RLY014
.D10014      GP_PSSPA        ;CLOSES FOR PSS + PA GAIN/PHASE.
.D10015      RLY015
.D10015      GP_PSS          ;CLOSES FOR PSS ONLY GAIN/PHASE.
.D10016      RLY016
.D10016      PA-TEST         ;CLOSES FOR PA TESTS - INITIAL SET UP.
.D10017      RLY017
.D10017      VTM_AC          ;CLOSES FOR TRANSMISSION MONITOR - PUTS
                              SIGNAL INTO POWER AMPLIFIER.

OUTPUT DEFINITIONS FOR HAIC JB

.D10100      LO100
.D10100      LL_OFF
.D10101      LO101
.D10101      LL_ON
.D10102      LO102
.D10102      LL_ALRM
.D10103      LO103
.D10103      LL_TST
.D10104      LO104
.D10105      LO105
.D10106      LO106
.D10107      LO107
```

```
;  TRANSIENT  RECORDER BLOCKS STATES
   .V00E0C              XSTER_STATES          *12
   . V00E0C             RCRD_1_STATE
   .V00E2C              RCRD_2_STATE
   .V00E4C              RCRD_3_STATE
   .V00E6C              RCRD_4_STATE
   . V00EBC             RCRD_5_STATE
   . V00EAC             RCRD_6_STATE

   CURRENT SCAN VALUES TABLE
   .V0010C      SCAN-VL          *16
   .V0010C      VOLT-4
   .V0012C      VOLT_B
   .V0014C      VOLT-C
   .V0016C      CURR_A
   .V0018C      CURR_B
   .V001AC      CURR_C
   .V001CC      VARS
   .V001EC      POWER
   .V0020C      VOLT-FIELD
   .V0022C      DELTA-F
   .V0024C      DC-F
   .V0026C      PSS_OUT
   .V0028C      TST_SINE
   .V002AC      TMA_COUNT
   .V002CC      TMB_COUNT
   .V002EC      TMC_COUNT

;  CURRENT  OPERATING  CONDITIONS

   .VR          CURR_VOLT_A
   .VR          CURR_VOLT_B
   .VR          CURR_VOLT_C
   .VR          CURR_CURR_A
   .VR          CURR_CURR_B
   .VR          CURR_CURR_C
   .VR          CURR_VARS
   .VR          CURR_POWER
   .VR          CURR_V_FIELD
   .VR          CURR_DELTA_F
   .VR          CURR_DC_F
   .VR          CURR_PSS_OUT

;  EVENT LIST FULL LOGICALS IN WORD FORMAT
   .VC          LIST-I-FULL
   .vc          LIST-P-FULL
   .vc          LIST_3_FULL
   .vc          LIST_4_FULL
    vc          LIST_5_FULL
   :vc          LIST_6_FULL

; LIST LAST ENABLED BY THE LIST CONTROL BLOCK
   .vc          L-L-EN

; LOGICALS ( IN WDRD FORMAT ) USED TO CONTROL
; RECORDER LISTS 4 AND 5
   .vc          L_4_EN
   .vc          L_5_EN

;  LOGICAL (IN WORD FORMAT I TO CONTROL
; TEST LIST
   .vc          T-L-EN
```

```
;  DUMMY LOGICAL ENABLE OUTPUT
   .L
;  VARIABLES USED TO INDICATE RECORDER EVEN1
   .L            RCRD_EVENT
   .L            TRIGGER
   .L            OLD-EVENT
;  VARIABLE USED TO INDICATE PSS EVENT
   .L            TST_EVENT
   .L            INT_TST
   .L            NX_EDGE
   .L            PX_EDGE
   .L            NO-EDGE
   .L            TMP_LOG1
   .L            OX_TST
   .VC           JMP_ARR                 *2
;  VARIABLE ARRAY FOR PSS CALCULATION
   .vc           FVAR            *256
;  PSS PREFAULT ARRAY
   .L            PSS_ERR          *2
;                 PSS_ERR(1). .  PSS OUTPUT CEILING FAULT
;                 PSS_ERR(2). .  CWFT OUTPUT CEILING FAULT
;  SUBSYSTEM PREFAULT ARRAY
   .L            SUB-ERR          *2
;                 SUB_ERR(1) .. DMC OFFLINE
;                 SUB ERR(2) .. PSS SEGHENT NOT RUNNING
; SIZE OF THE FAULT-LIST
   .cc           ND-OF-FAULTS              13
   .L            FLT_RST                   ;FAULT LIST RESET
;  FAULT LIST ARRAY FOR ALARM LIST DISPLAY
   .L            FAULTS           *20
;                       FAULT(1). . PSS OUTPUT CEILING FAULT
;                       FAULT (2). . LOSS OF PHASE FAULT
;                       FAULT(3). . DMC OFFLINE
;                       FAULT (4). . CWFT CEILING FAULT
;                       FAULT(5). . PSS SEGMENT NC? RUNNING
;-----------------------------------------------------------------------
;
;                 CONSTANTS
;
;-----------------------------------------------------------------------
;  CONSTANTS  ARRAY FOR PSS CALCULATION
   .C0010R       FCON                    0.4             ;TOTAL GAIN = 0.4 • CWFT GAIN
   .C0014R       T1                      10.0            ;WASHOUT T C
   .C0018R       T2                      0.25            ;LEAD
   .C001CR       T3                      0.030249        ; LAG
   .C0020R       T4                      0.25            ; LEAD
   .C0024R       T5                      0.030249        ; LAG
   .C0028R       TX0                     .0740931        ;TORS. COEFFICIENT
   .C002CR       TX!                     -.0038163       ;TORS. CDEFFICIENT
   .C0030R       TX2                     .056341         ;TORS. CDEFFICIENT
   .C0034R       TX3                     -.061294        ;TORS. COEFFICIENT
   .C0038R       TX4                     .0045176        ;TORS. COEFFICIENT
   .C003CR       TX5                     -.069842        ;TORS. COEFFICIENT
   .C0040R       TY1                     -2.75887        ;TORS. CDEFFICIENT
   .C0044R       TY2                     3.67106         ;TORS. COEFFICIENT
   .C0048R       TY3                     -2.970299       ;TORS. COEFFICIENT
   .C004CR       TY4                     1.39707         ;TORS. COEFFICIENT
   .C0050R       TY5                     -.33026843      ;TORS. COEFFICIENT
```

```
.C0054R     A                       17.93221        ;
.C0058R     B                       -33.5506        ;
.C005CR     C                       15.69310
.C0060R     D                       -1.13608
.C0064R     E                       .322675         ;
.C0068C     TFN_SEL                 0               ;0=BILINEAR,1=STEP INVARIANT
.C006AC     ETST_CNT                300             ;EXCIT. TEST: 60 CNTS = 1 SEC
.C006CR     W1                      0.999167        ;
.C0070R     w2                      -0.99833        ;
.C0074C     UOL                     32767           ;UPPER OUT LIMIT 10%=10V
.C0076C     LOL                     -32767          ;LOWER OUT LIMIT -10%=-10V
.C0078C     OVL                     5024            ;OVER VOLTAGE LIMIT 20% OF 115V
.C007AC     UVL                     -5024           ;UNDER VOLTAGE LIMIT -20% DF 115V
.C007CC     VLTC                    60              ;VOLTAGE LIMIT TC=1SEC*60
.C007EC     UCL                     16383           ;UPPER CEILING LIMIT +5V
.C0080C     LCL                     -16383          ;LOWER CEILING LIMIT -5V
.C0082C     CLTC                    120             ;CEILING LIMIT TC=2SEC*60
.C0084C     O_U_OPT                 1               ;OVER/UNDER VDLTAGE OPT. ENABLE
.C0086C     TOR_DIS                 0               ;TORSIONAL FILTER DISABLE (1=DISABLED
.C0088C     ESTEP                   3275            ;EXC. TST STEP 1X = 1V
.C008AC     DSTEP                   -327            ;DIG. JOG TST -1V = .02% DELTA F
.C008CC     ASINE                   819             ;SINE WAVE AMPLITUDE .05% (1% = 5V)
.C008EC     ROVL                    2512            ;RCRD OVERVOLT. LIMIT 10% OF 115 V
.C0090C     RUVL                    -2512           ;RCRD UNDERVOLT. LIMIT -10% DF 115 V
.C0092C     RPSSL                   16384           ;RCRD PSSOUT LIMIT .5 PU
.C0094C     RCRD_OPT                1               ;RECORDER OPTION ENABLED
.C0096C     PDN_LVL                 2409            ;POWER DN LEVEL 16%
.C0098C     POF_LVL                 2168            ;POWER OFF LEVEL 901 OF ON
.C009AC     PLVL_OPT                0               ;POWER LEVEL OPTIDN
.C009CC     RCRDRATE                3               ;RECORDER SCHEDULING RATE
.C009EC     TST_DUR                 7200            ;TEST DURATION (60=1SEC)

    SPARE DA OUTPUT CHANNEL
.CP         DA4OUT                  VTM_COUNT

    EVENT RECORDER SCAN TIME IN MILLISECONDS
.CC         SCAN-MSECS              50

    LOGICAL USED TO CLEAR OR ENABLE TRANSIENT EVENT RECDRDERS
.CC         LIST-EN                 0  ; FILLED OUT BY U MON INIT
                                       AT INITIALIZATION AND OVER WRITTEN
                                       TO OFFH EVERY TIME RCRDER SEG RUNS

    CONSTANT COUNT VALUES
.CC         CNT_0                   0
.CC         CNT_1
.CC         CNT_2                   2
.CC         CNT_3                   3
.CC         CNT_4                   4
.CC         CNT_5                   5
.CC         CNT_6                   6
.CC         CNT_7                   7
.CC         CNT_8                   8
.CC         CNT_9                   9
.CC         CNT_10                  10
.CC         CNT_11                  11
.CC         CNT_OFF                 255
.CC         CNT_10TH                10000

;   NUMBER OF TRANSIENT EVENT RECORDER BLOCKS IN THE SYSTEM
.CC         NO-XSTER                0 ; FILLED OUT BY U MON INIT

;   DOUBLE WORD POINTERS TO TRANSIENT EVENT LISTS
```

```
. COOAOP              EVENT_LIST_1              .C0000C   FILLED  OUT  BY  U  MON  INIT
.C00A2P               DUMMY_PTR_1               .C0000C   FILLED  OUT  BY  U  MON  INIT
.C00A4P               EVENT_LIST_2              .C0000C   FILLED  OUT  BY  U  MON  INIT
 C00A6P               DUMMY_PTR_2               .C0000C   FILLED  OUT  BY  U  MON  INIT
:C00A8P               EVENT_LIST_3              .C0000C ; FILLED  OUT  BY  U  MON  INIT
. COOAAP              DUMMY_PTR_3               .C0000C   FILLED  OUT  BY  U  MON  INIT
.COOACP               EVENT_LIST_4              .C0000C   FILLED  OUT  BY  U  MON  INIT
. COOAEP              DUMMY_PTR_4               .C0000C   FILLED  OUT  BY  U  MON  INIT
.C00B0P               EVENT_LIST_5              .C0000C   FILLED  OUT  BY  U  MON  INIT
.C00B2P               DUMMY_PTR_5               .C0000C   FILLED  OUT  BY  U  MON  INIT
.C00B4P               EVENT_LIST_6              .C0000C   FILLED  OUT  BY  U  MON  INIT
. C00B6P              DUMMY_PTR_6               .C0000C   FILLED  OUT  BY  U  MON  INIT

STATE CDDES FOR THE XSTER BLOCKS
.cc           INITIALIZE              1
.cc           QUIESCENT               2
.CC           PRE_TRIGGER             3
.CC           POST-TRIGGER            4
.CC           COMPLETE                5

 OFFSET TABLE  - IN THE ORDER THE VARIABLES APPEAR IN THE DISPLAY LIST
                 NOT THE SCANNED ORDER
.CR           OFFSETS                 *16
              &                       0.0             ; VOLT  A
              &                       0.0             ; VOLT  B
              &                       0.0             ; VOLT  C
              &                       0.0             ;CURRENT A
              &                       0.0             ;CURRENT B
              &                       0.0           ;CURRENT C
              &                       0.0             ; POWER
              &                       0.0             ; VARS
              &                       0.0             ; PSS  OUTPUT
              &                       0.0             ;PSS FREQ.
              &                       0.0             ;VOLT FIELD
              &                       0.0             ;TEST SINE
              &                       0.0
              &                       0.0
              &                       0.0
              &                       0.0

GAIN  TABLE  - IN THE ORDER THE VARIABLES APPEAR IN THE DISPLAY LIST
               NOT THE SCANNED ORDER
.CR           GAINS                   *16
              &                       .0006369        ; VOLT  A
              &                       .0006369        ; VOLT  B
              &                       .0006369        ; VOLT  C
              &                       .0008138        ;CURRENT A
              &                       .0008138        ;CURRENT B
              &                       .0008138        ;CURRENT C
              &                       .00084912       ; POWER
              &                       .00084912       ; VARS
              &                       .004885197851   ; PSS  OUTPUT
              &                       .00097703957    ; P S S FREQ.
              &                       .00097703957    ;VOLT FIELD
              &                       .00097703957    ; TEST SINE
              &                       1.0
              &                       1.0
              &                       1.0
              &                       1.0

NUMBER OF EVENT RECORDS BEFORE THE TRIGGER
.CC           NO-BEFORE-RD            10

NUMBER DF EVENT RECORDS AFTER THE TRIGGER
```

```
.CC          NO_AFTER_RD            200

POINTS TO SCAN TABLE
.cc          POINTS                 *17
             &                      12      ;NO. OF PTS. TO SCAN
             &                      0       ;FIRST POINT TO SCAN
             &                      1
             &                      2
             &                      3
             &                      4
             &                      5
             &                      7
             &                      b
             &                      11
             &                      9
             &                      8
             &                      12
             a                      0
             &                      0
             &                      0
             &                      0

STATION ID
.CC          STAT_ID                *40
             &                      20053
             &                      20043
             &                      22351
             &                      8270
             &                      21587
             &                      21569
             &                      20297
             &                      8270
             &                      6224
             &                      8224
             &                      8224
             &                      a224
             &                      8224
             &                      8224
             &                      8224
             &                      a224
             &                      8224
             &                      8224
             &                      8224
             &                      8224
             &                      8224
             &                      8224
             &                      8224
             &                      8224
             &                      8224
             &                      0224
             &                      8224
             &                      8224
             &                      8224
             &                      8224
             &                      8224
             &                      8224
             &                      8224
             &                      8224
             &                      8224
             &                      8224
             &                      8224
```

UNIT I D

.CC	UNIT-ID	*40
	&	20053
	&	10043
	&	22351
	&	8270
	&	20053
	&	21577
	&	8224
	&	a224
	&	a224
	&	8224
	&	a224
	&	a224
	&	a224
	&	a224
	&	a224
	&	a224
	&	a224
	&	a224
	&	8224
	&	a224
	&	a224
	&	8224
	&	8224
	&	a224
	&	a224
	&	8224
	&	a224
	&	a224
	&	8224
	&	a224
	&	a224
	&	8224
	&	8224
	&	8224
	&	8224
	&	8224
	&	8224
	&	a224
	&	8224
	&	8224

PASSWORD CHARACTER COUNT

.cc	PWRD_CHR	6

PASSWORD

.CA	PASSWRD	*7
	&	'PLEASE'

MACHINE RATINGS ARRAY

.CR	RATING	*9	
	&		0.0
	&		0.0
	&		0.0
	&		0.0
	&		0.0
	&		0.0
	&		0.0
	&		0.0
	&		0.0

```
;-----------------------------------------------------------------------

                        **** PART 3:  SCREEN DATA ****

; Use the following header format in this Variables Definition file
;to define the Screen data to be displayed. Max 10 Screens are allowed.

;-----------------------------------------------------------------------
;  Screen-Display-Data.
;
;  Screen-# : 1
;
;  Screen-Header (Max 4D ch) : CURRENT DPERATIDN
;
;             Position-On-Screen          Variable-Displayed
;                (1 t h r u 24)
;
;                    1                          CURR_VOLT_A
;                    2                          CURR_VOLT_B
;                    3                          CURR_VOLT_C
;                    4                          CURR_CURR_A
;                    5                          CURR_CURR_B
;                    6                          CURR_CURR_C
;                    7                          CURR_VARS
;                    8                          CURR_POWER
;                    9                          CURR_V_FIELD
;                   10                          CURR_DELTA_F
;                   11                          CURR_PSS_OUT
;  End_of_Screen_Display_Data.
;
;
```

```
;-----------------------------------------------------------------------
;  Screen-Display-Data.

;  Screen_#  :  2

;  Screen-Header (Max 40 ch) :POWER  AMPLIFIER  PROTECTION
;
;             Position_On_Screen          Variable-Displayed
;                (1 thru 24)

                      1                          LPA_ENABLE
                      2                          PSS_ON
;                     3                          ALL-TRIP
;                     4                          A-OUT
;                     5                          PHA_PA
;                     6                          B_OUT
;                     7                          PHB_PA
;                     8                          C-OUT
;                     9                          PHC_PA
;                    10                          CL-DISABLE
;                    11                          VTM_ENABLE
;                    12                          ONE_OUT
;                    13                          TWO_OUT
;                    14                          VTM_1
;                    15                          VTM_2
;                    16                          TRIP-ONE-OUT
;                    17                          TRIP_TWO_OUT
;                    18                          PA-TEST
;                    19                          CLA_TRIP
;                    20                          CLA_TRIP_L
;                    21                          CLB_TRIP
;                    22                          CLB_TRIP_L
;                    23                          CLC_TRIP
;                    24                          CLC_TRIP_L
;  End_of_Screen_Display_Data.
;
```

```
;-------------------------------------------------------------------------

;  Screen_Display_Data.
;
;  Screen-# : 3
;
;  Screen-Header (Max 40 ch) :TRANSMISSION MONITOR I/O
;
;            Position_On_Screen          Variable-Displayed
                (1 thru 24)

                    1                     VTM_DC
                    2                     VTM_PRESET
                    3                     VTM_ENABLE
                    4                     VTM_COUNT_IN
                    5                     VTM_SCALE
                    6                     VTM_COUNT
                    7                     VTM_AVE
                    8                     VTM_ERROR
                    9                     TMA_COUNT
                   10                     VTMA_DC
                   11                     VTMA_ERROR
                   12                     ABS_VTMA_ERR
                   13                     TMA_TRIP_EN
                   14                     TMA_TRIP
                   15                     TMB_COUNT
                   16                     VTMB_DC
                   17                     VTMB_ERROR
                   18                     ABS_VTMB_ERR
                   19                     TMB_TRIP
                   20                     TMC_COUNT
                   21                     VTMC_DC
                   22                     VTMC_ERROR
                   23                     ABS_VTMC_ERR
                   24                     TMC_TRIP
  End_of_Screen_Display_Data.
```

```
;-----------------------------------------------------------------------

;    Screen-Display-Data.
;
;  Screen_# :   4
;
;  Screen-Header (Max 40 ch) :SCREEN 4

;              Position_On_Screen          Variable-Displayed
;                 (1 thru 24)

;                      1                              PA-TEST
;  End-of-Screen-Display-Data.
;
```

```
;-----------------------------------------------------------------------
;
;  Screen-Display-Data.
;
;  Screen-#   :  5
;
; Screen-Header (Max 40 ch) :SCREEN 5
;
;             Position_On_Screen          Variable-Displayed
;                (1 thru 24)
;
;                    1                          PA-TEST
;  End-of-Screen-Display-Data.
;
```

```
;-------------------------------------------------------------------------
; Screen-Display-Data.
;  Screen_#  :  6
;
; Screen-Header (Max 40 ch) :SCREEN 6
;              Position-On-Screen              Variable-Displayed
                 (1 thru 24)
;                       1                              PA-TEST
;     End-of-Screen-Display-Data.
;Sample
;-------------------------------------------------------------------------
; Screen-Display-Data.
;
;  Screen-# : 7
;
;  Screen-Header (Max 40 chl :SCREEN 7
;              Position-On-Screen              Variable-Displayed
;                (1 thru 24)
;
;                       1                              PA-TEST
;  End-of-Screen-Display-Data.
;
```

```
;-----------------------------------------------------------------------
;  Screen-Display-Data.
;  Screen-#  :  8
;  Screen-Header (Max 40 ch) :SCREEN 8
;          Position-On-Screen          Variable-Displayed
                (1 thru 24)
                        1                          PA-TEST
;  End_of_Screen_Display_Data.
;
```

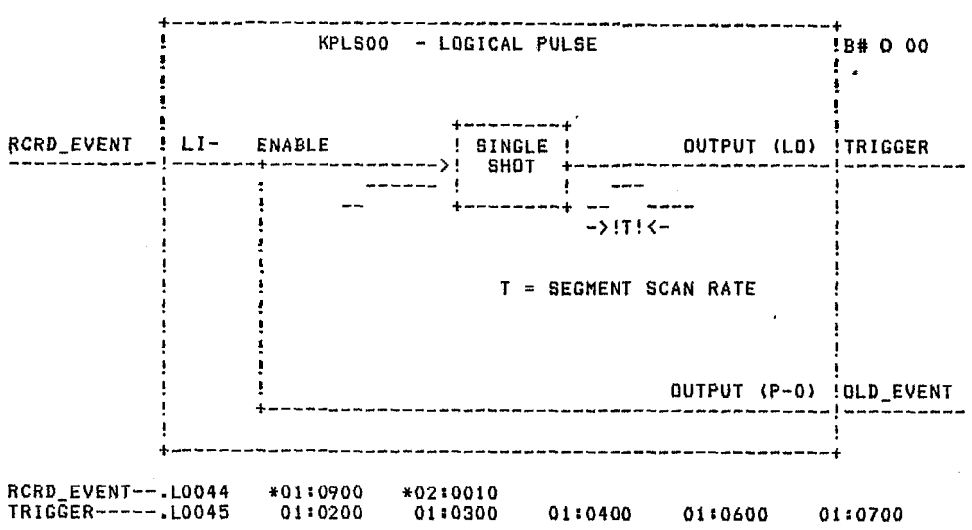
```
;-----------------------------------------------------------------------
;    Screen-Display-Data.
;
;   Screen-#    :   9
;
; Screen-Header (Max 40 ch) :SCREEN 9
;
;              Position-On Screen            Variable-D isplayed
;                 (1 thru 24)
;
;                        1                              PA-TEST
;   End_of_Screen_Display_Data.
;
```

```
;-----------------------------------------------------------------------
;  Screen-Display-Data.
;
;  Screen-#   :   10
;
;  Screen-Header  (Max 40 ch) :SCREEN 10
;     .
;              Position_On_Screen            Variable-Displayed
;                 (1 thru 24)
;
;                       1                          PA-TEST
;  End_of_Screen_Display_Data.
```

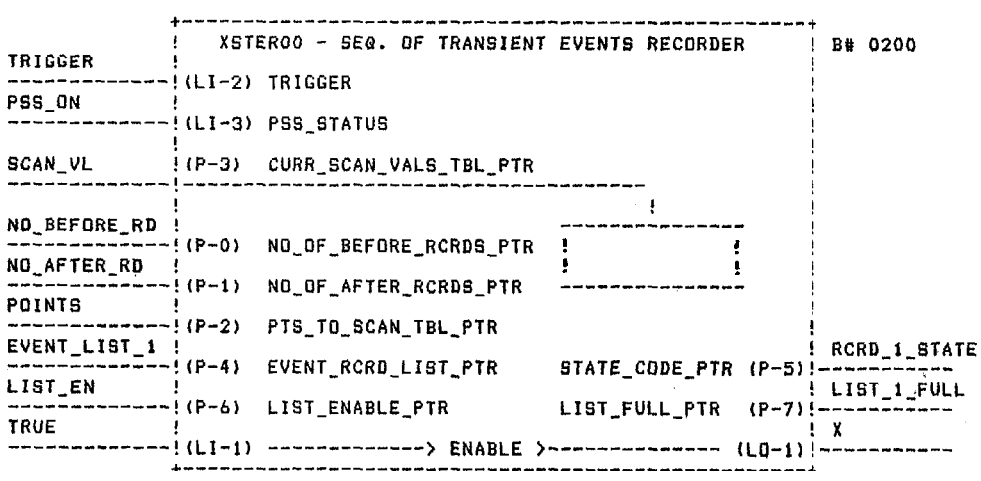
```
..NAME: PSS_EVENT_SEGMENT_0000
..AUTHOR: HOGLE
..COPYRIGHT:    COPYRIGHT        (C)  1986
                BY GENERAL ELECTRIC COMPANY,    U.S.A
..KEYWORDS:
                DISTRIBUTED_MICRO_CONTROLLER,DMC,SEGMENT
..ABSTRACT:
.....DESCRIPTION:

.....DEPENDENCIES:
.....MEMORY_SIZE:
..BUILD_INSTRUCTIONS:
.....PROCESSOR:  DMC_EDIT_R3
.....SEGMENT NAME ..........                    : RCRDER
.....SEGMENT TYPE  (RUN, INTERRUPT, INITIALIZE) : RUN
.....SEGMENT INTERRUPT NUMBER ..........        :
.....SEGMENT SCHEDULING RATE, AND UNIT ..........: 0  MILLISECOND
.....8087 CO-PROCESSOR USED (YES / NO) .......  : YES
.....SEGMENT STACK SIZE (# OF BYTES) ..........  : 500
.....SYMBOLIC NAME SPLIT CHAR (DEFAULT HYPHEN)  :
```

```
*****************************************************************************************************************
*                        *                *   COPYRIGHT (C)1988, BY    i t  SYSTEM:PSS                          *
* DATE: 10/19/88         *  REQUISITION   *   GENERAL ELECTRIC COMPANY *                                         *
* AUTHOR :               * PACIFIC      t   DRIVE SYSTEMS DEPARTMENT  **********************************************
* ©                      *  D.S.D         *   SALEM, V A     USA        it DRAWING : XXXXXX                      *
*                        *  xxxxxx        *                           * NEXT PAGE: 0002                PAGE: 0 0 0 1 *
*****************************************************************************************************************
```



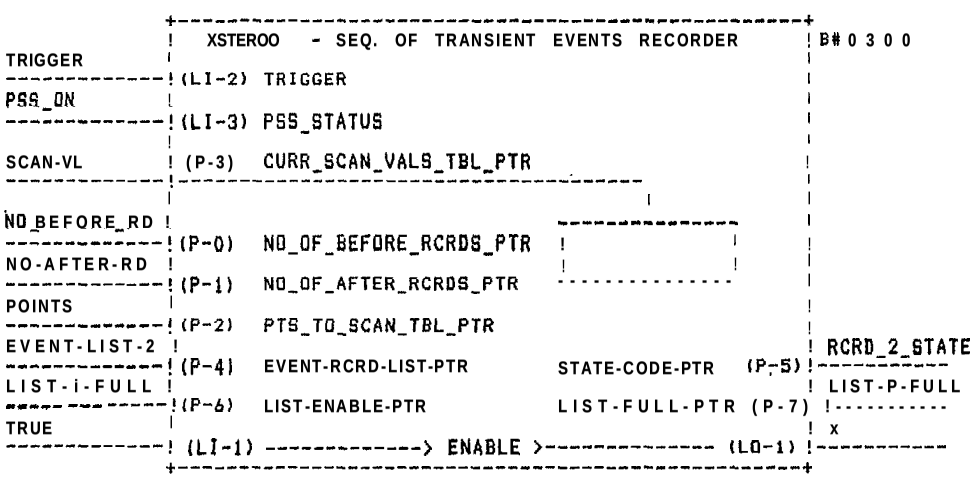

```
====================  ========
SEGMENT:01/BLOCK:0100  PG: 0002
====================  ========
        LD  RCRD_EVENT--.L0044
        BLK KPLS00
        B#  0100
        P-0 OLD_EVENT---.L0046
        STO TRIGGER-----.L0045
        B#  0100
```




```
====================  ========
SEGMENT:01/BLOCK:0200  PG: 0002
====================  ========
        LD  PSS_ON------.D10010
        LD  TRIGGER-----.L0045
        LD  TRUE--------.L0001
        BLK XSTER00
        B#  0200
        P-0 NO_BEFORE_RD-.C01A8C
        P-1 NO_AFTER_RD-.C01AAC
        P-2 POINTS------.C01ACC
        P-3 SCAN_VL-----.V0010C
        P-4 EVENT_LIST_1.C00A0P
        P-5 RCRD_1_STATE.V00E0C
        P-6 LIST_EN-----.C00FEC
        P-7 LIST_1_FULL-.V00A0C
        STO X-----------.L0043
        B#  0200
```

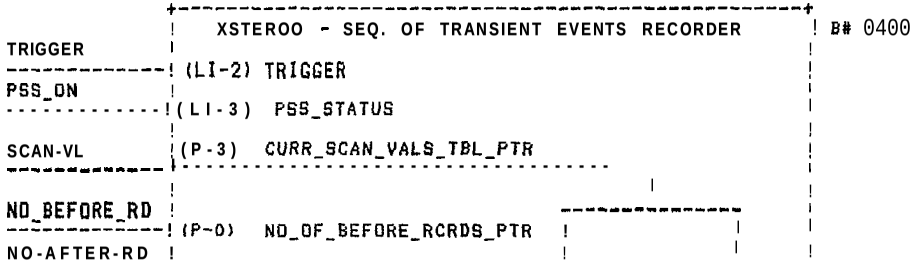
```
PSS_ON------.D10010   01:0300    01:0400    01:0600    01:0700    03:0200
                     *04:0130    04:0210    04:0245    04:0450    05:0220
                      05:0230    05:0240
TRIGGER-----.L0045   *01:0100    01:0300    01:0400    01:0600    01:0700
NO_BEFORE_RD.C01A8C   01:0300    01:0400    01:0600    01:0700    03:0200
NO_AFTER_RD-.C01AAC   01:0300    01:0400    01:0600    01:0700    03:0200
POINTS------.C01ACC   01:0300    01:0400    01:0600    01:0700    02:0010
                      03:0200
SCAN_VL-----.V0010C  *01:0300   *01:0400   *01:0600   *01:0700   *02:0010
                     *03:0200
LIST_EN-----.C00FEC  *01:0800
LIST_1_FULL-.V00A0C   01:0300
X----------.L0043    *01:0300   *01:0400   *01:0500   *01:0600   *01:0700
                     *01:0800   *03:0100   *03:0200
```

```
*************************************************************************************************
*                     *              *   COPYRIGHT (C) 1988,  BY    *  SYSTEM:PSS                *
* DATE: 10/19/88      *  REQUISITION *   GENERAL ELECTRIC COMPANY   *                            *
* AUTHOR :            *  PACIFIC     *   DRIVE SYSTEMS DEPARTMENT   ***************************************
* J                   *  D.S.D       *   SALEM, VA    USA           *  DRAWING : XXXXXX          *
*                     *  XXXXXX      *                              *  NEXT PAGE: 0003  PAGE: 0002  *
*************************************************************************************************
```

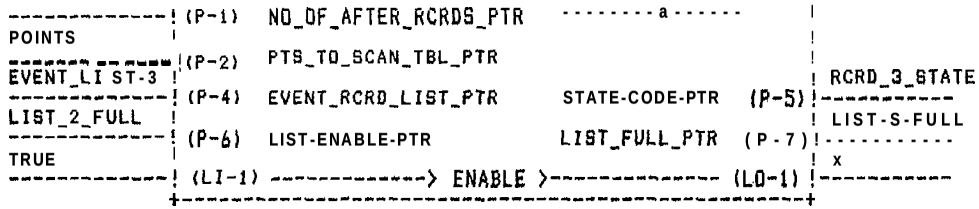


PSS_ON------.010010	01:0200	01 10400	01:0600	01:0700	03:0200
	*04:0130	04:0210	04:0245	04:0450	05:0220
	05:0230	05:0240			
TRIGGER-----.L0045	*01:0100	01:0200	01:0400	01:0600	01:0700
NO_BEFORE_RD.C01A8C	01:0200	01:0400	01:0600	01:0700	03:0200
NO-AFTER-RD-.C01AAC	01:0200	01:0400	01:0600	01:0700	03:0200
POINTS------.C01ACC	01:0200	01:0400	01:0600	01:0700	02:0010
	03:0200				
SCAN_VL-----.V0010C	*01:0200	*01:0400	*01:0600	*01:0700	*02:0010
	*03:0200				
LIST_1_FULL-.V00A0C	*01:0200				
LIST_2_FULL-.V00A2C	01:0400				
X-----------.L0043	*01:0200	*01:0400	*01:0500	*01:0600	*01:0700
	*01:0800	*03:0100	10310200		

```
====================  ========
SEGMENT:01/BLOCK:0300 PC: 0003
====================  ========
        L D  PSS_ON------.010010
        L D  TRIGGER-----.L0045
        LD   TRUE--------.L0001
        BLK XSTEROO
        B# 0 3 0 0
        P-0 NO_BEFORE_RD.C01A8C
        P-1 NO_AFTER_RD-.C01AAC
        P-2  POINTS------.C01ACC
        P-3 SCAN-VL-----.V0010C
        P-4 EVENT_LIST_2.C00A4P
        P-5 RCRD_2_STATE.V00E2C
        P-6 LIST-I-FULL-.V00A0C
        P-7 LIST_2_FULL-.V00A2C
        STO X-----------.L0043
        B# 0 3 0 0
```




```
====================  ========
SEGMENT:01/BLOCK:0400 PC: 0003
====================  ========
        L D  PSS_ON------.010010
        L D  TRIGGER-----.L0045
        LD   TRUE--------.L0001
        BLK XSTEROO
        B# 0 4 0 0
        P-0 NO_BEFORE_RD.C01A8C
        P-1 NO_AFTER_RD-.C01AAC
        P-2  POINTS------.C01ACC
        P-3 SCAN_VL-----.V0010C
        P-4 EVENT_LIST_3.C00A8P
        P-5 RCRD_3_STATE.V00E4C
        P-6 LIST_2_FULL-.V00A2C
        P-7 LIST_3_FULL-.V00A4C
        STO X-----------.L0043
```



B# 0400

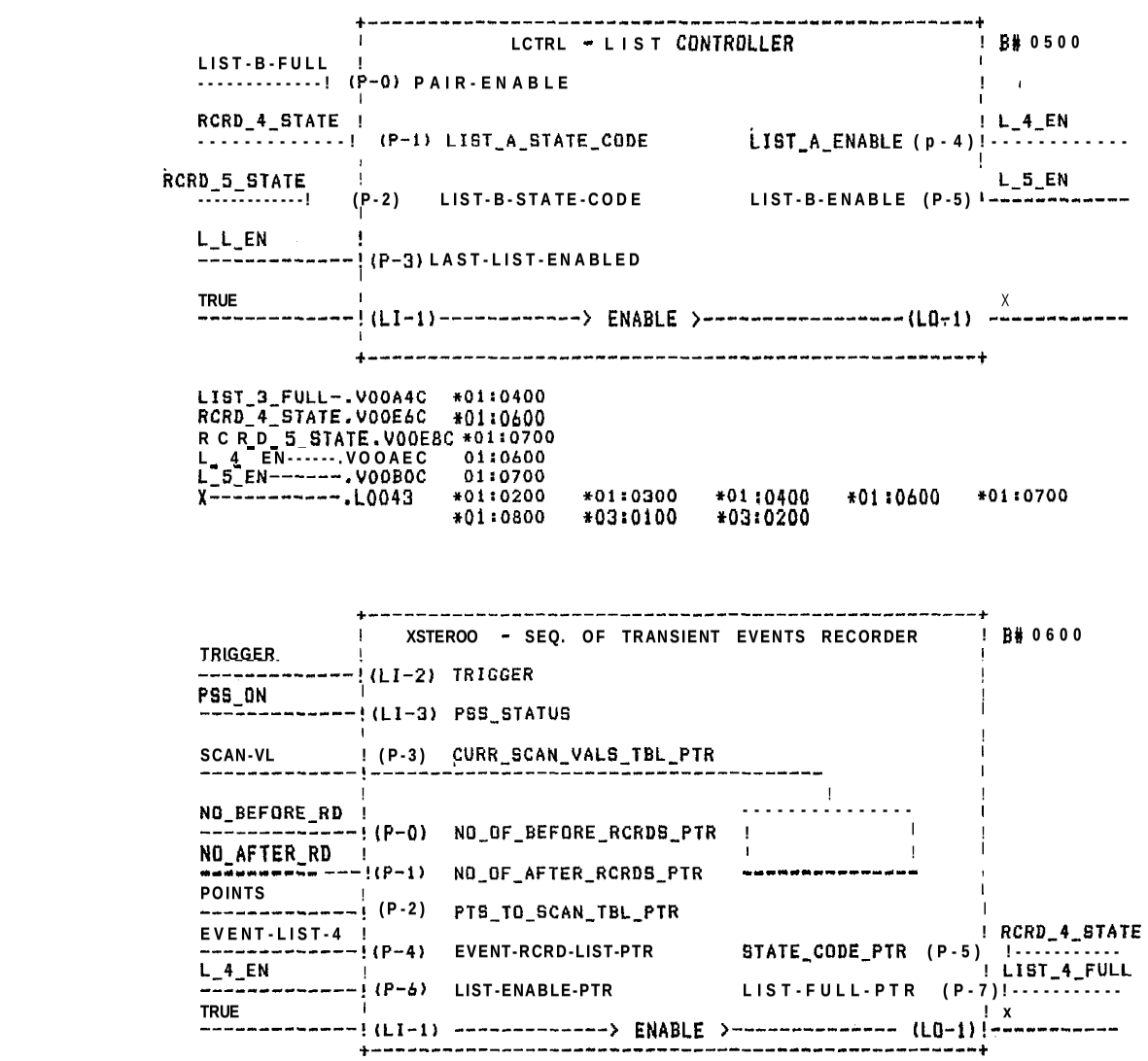
PSS_ON------.010010	01:0200	01 :0300	01:0600	01:0700	03:0200
	*04:0130	04:0210	04:0245	04:0450	05:0220
	05:0230	05:0240			
TRIGGER-----.L0045	*01:0100	01:0200	01:0300	01:0600	01:0700
NO_BEFORE_RD.C01A8C	01:0200	01:0300	01:0600	01:0700	03:0200
NO_AFTER_RD-.C01AAC	01:0200	01:0300	01:0600	01:0700	03:0200
POINTS------.C01ACC	01:0200	01:0300	01:0600	01:0700	02:0010
	03:0200				
SCAN-VL-----.V001OC	*01:0200	*01:0300	*01:0600	*01:0700	*02:0010
	*03:0200				
LIST_2_FULL-.V00A2C	*01:0300				
LIST-S-FULL-. V00A4C	01:0500				
X----------.L0043	*01:0200	*01:0300	*01:0500	*01:0600	*01:0700
	*01:0800	*03:0100	*03:0200		

COPYRIGHT (C) 1988, BY
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPARTMENT
SALEM, VA USA

DATE: 10/19/88
AUTHOR :
J

REQUISITION
PACIFIC
D.S.D
xxxxxx

SYSTEM:PSS
DRAWING : XXXXXX
NEXT PAGE:0004

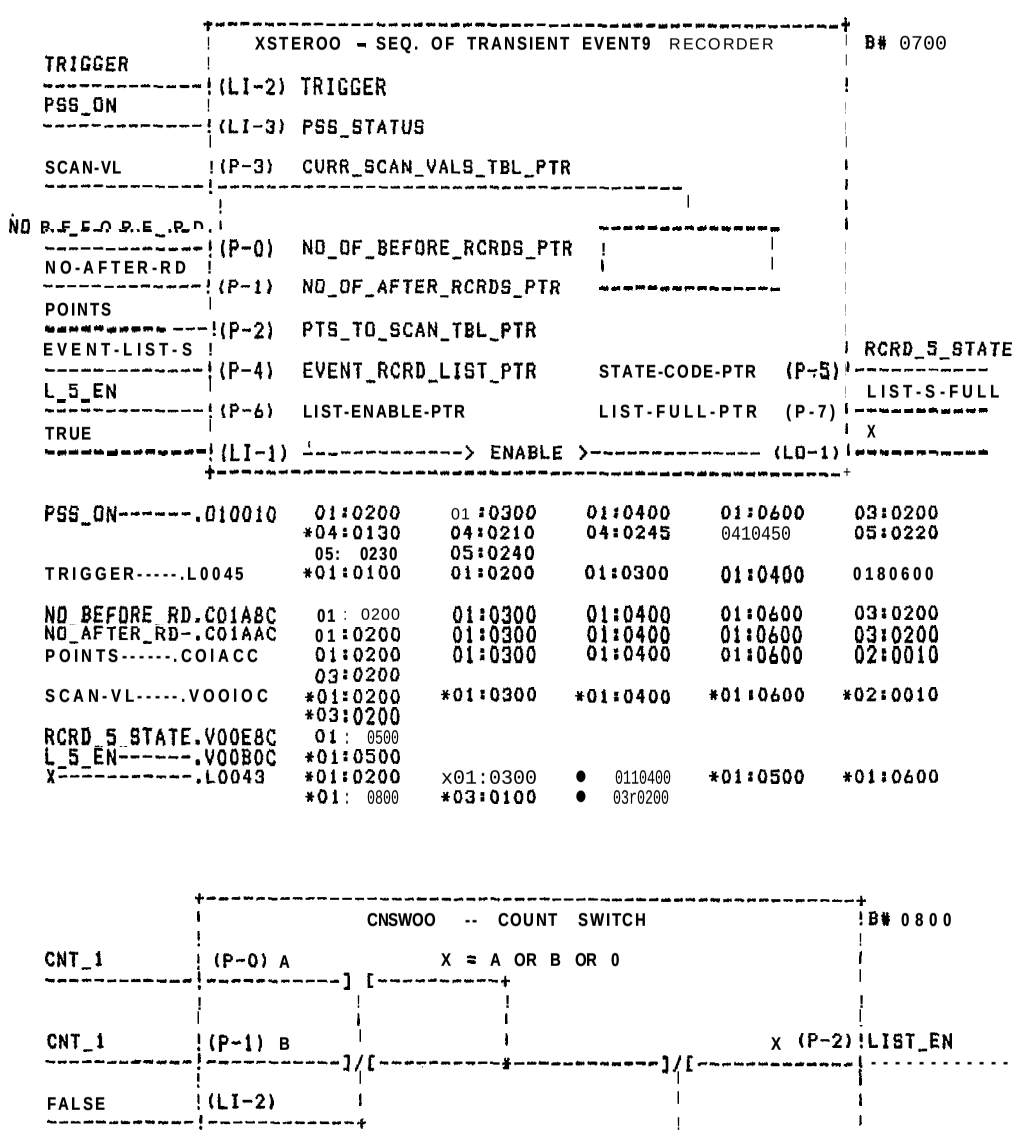



```
====================  ========
SEGMENT:01/BLOCK:0500 PG:0 0 0 4
====================  ========
         LD    TRUE--------.LOOOI
         BLK LCTRLOO
         B# 0 5 0 0
         P-O LIST_3_FULL-.V00A4C
         P-i  RCRD_4_STATE.V00E6C
         P-2 RCRD_5_STATE.V00E8C
         P-3 L_L_EN------.V00ACC
         P-4 L_4_EN------.V00AEC
         P-5 L_5_EN------.V00B0C
         STO  X-----------.LO043
         B# 0 5 0 0
```

```
====================  ========
SEGMENT:01/BLOCK:0600 P C: 0004
====================  ========
         L  D   PSS_ON------,010010
         L  D   TRIGGER-----.L0045
         L  D   TRUE--------,L0001
         BLK XSTEROO
         B# 0 6 0 0
         P-O NO_BEFORE_RD.C01A8C
         P-i NO-AFTER-RD-.C0IAAC
         P-2   POINTS------.C0iACC
         P-3   SCAN-VL-----.V00I0C
         P-4     EVENT-LIST-4,COOACP
         P-S RCRD_4_STATE.V00E6C
         p-6 L_4_EN------,V00AEC
         P-7 LIST_4_FULL-.V00A6C
         STO  X-----------.LO043
         B#0 6 0 0
```

```
PSS_ON------.D10010   01:0200     01:0300     01:0400     01:0700     03:0200
                     *04:0130     04:0210     04:0245     0410450     0510220
                      05:0230     05:0240
TRIGGER-----.L0045   *01:0100     01:0200     01:0300     0110400     01:0700
NO_BEFORE_RD.C01A8C   01:0200     01:0300     01:0400     01:0700     03:0200
NO_AFTER_RD-.C01AAC   01:0200     01:0300     0110400     01:0700     03:0200
POINTS------.C01ACC   01:0200     01:0300     01:0400     01:0700     02:0010
                      03:0200
SCAN-VL-----.V001OC  *01:0200    *01:0300    *01:0400    *01:0700    *02:0010
                     *03:0200
RCRD_4_STATE.V00E6C   01:0500
L_4_EN------.V00AEC  *01:0500
X----------.L0043    *01:0200    *01:0300    *01:0400    *01:0500    *01:0700
                     *01:0800    *03:0100    *03:0200
```

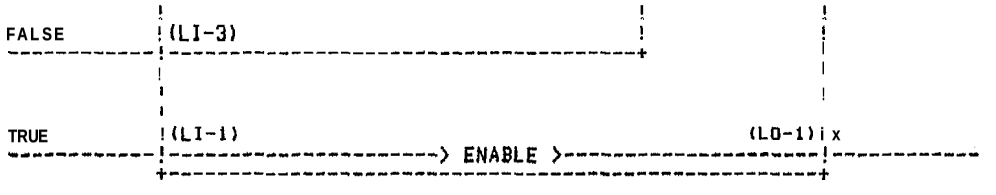
```
*************************************************************************************************************
*                    *                *   COPYRIGHT (C) 1988, BY    *  SYSTEM:PSS                            *
* DATE: 10/19/88     *  REQUISITION   *   GENERAL ELECTRIC COHPANY *                                        *
* AUTHOR :           *  PACIFIC       *   DRIVE SYSTEMS DEPARTMENT  **************************************************
* J                  *  D.S.D         *   SALEM, V A     USA        * DRAWING : XXXXXX                                  *
*                    *  xxxxxx        *                             * NEXT PAGE: 0005                  PAGE: 0 0 0 4 *
*************************************************************************************************************
```




```
=====================  ========
SEGMENT:01/BLOCK:0700 PC: 0005
=====================  ========
        L  D  PSS_ON------.010010
        L  D  TRIGGER-----.L0045
        LD TRUE--------.L000i
        ELK XSTEROO
        B#  0700
        P-0 NO_BEFORE_RD.C01A8C
        P-i NO_AFTER_RD-.C01AAC
        P-2 POINTS------.C01ACC
        P-3 SCAN-VL-----.V00I0C
        P-4 EVENT_LIST_5.C00B0P
        P-5 RCRD_5_STATE.V00E8C
        P-6 L_5_EN------.V00B0C
        P-7 LIST_5_FULL-.V00A8C
        STO X----------.L0043
        B#  6700
```



```
=====================  ========
SEGMENT:01/BLOCK:0800 PC:0005
=====================  ========
        LD   FALSE-------.L0000
        LD   FALSE-------.L0000
        LD    TRUE--------.L000I
        BLK  CNSWOO
        B# 0800
        P-0 CNT_1-------.C0102C
        P-i CNT_1-------.C0102C
        P-2  LIST-EN-----.C00FEC
        STO  X----------.L0043
        B# 0800
```




```
CNT_1--------.C0102C   04:0310    0410350    04:0390    0410570    04:0580
                       04:0590    04:0600    04:0610    04:0620    04:0630
                       04 :0640   04: 0650   04:0660    04:0680
LIST_EN------.C00FEC   01:0200
X-----------.L0043    *01:0200   *01:0300   *01:0400   *01:0500   *01:0600
                       *01:0700   *03:0100   *03:0200
```

```
 FALSE                                                              RCRD_EV
                                                                    ENT
+--] [--+-------+-------+-------+-------+-------+-------+------+--+-------+--(  )-

RCRD_EVENT--.L0044     01:0100    *02:0010
```

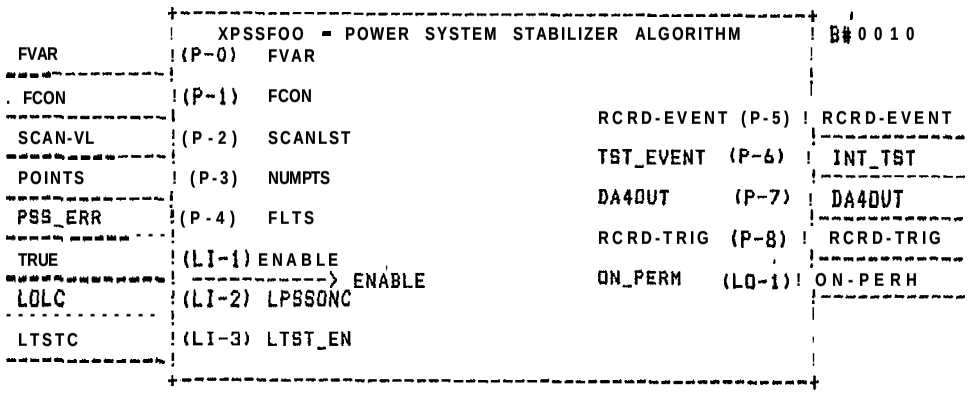
```
====================  ========
SEGMENT:01/BLOCK:0900 PC: 0005
====================  ========
          LD   FALSE-------.L0000
          STO RCRD_EVENT--.L0044
          BY 0900
```

```
*****************************************************************************************************************
*                      *                 *   COPYRIGHT (C) 1988, BY    *  SYSTEM:PS$                              *
* DATE: 10/19/88       *  REQUISITION    *   GENERAL ELECTRIC  COMPANY *                                          *
* AUTHOR :             *  PACIFIC        *   DRIVE SYSTEMS DEPARTMENT  ***************************************
* J                    *  D.S.D          *   SALEM, VA     USA         * DRAWING : XXXXXX                         *
*                      *  XXXXXX         *                             * NEXT PAGE:0006                PAGE:0005 *
*****************************************************************************************************************
```

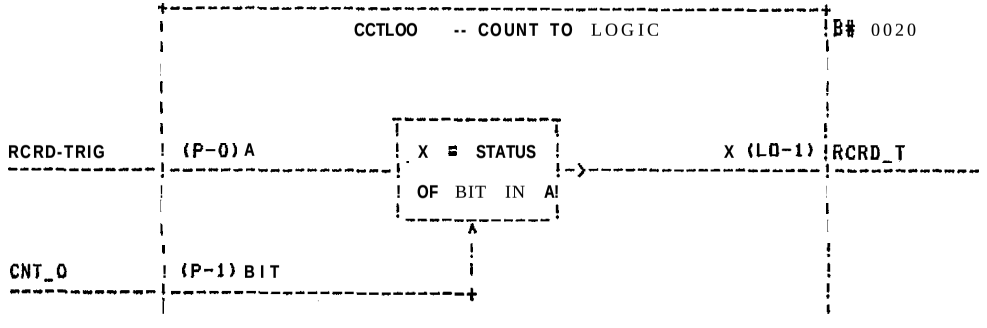
```
..NAME: PSS_FCALC_SEG_0003
..AUTHOR: HANISCH
..COPYRIGHT:    COPYRIGHT       (C)  1986
                BY GENERAL ELECTRIC COMPANY,    U.S.A
..KEYWORDS:
                DISTRIBUTED_MICRO_CONTROLLER,DMC,SEGMENT
..ABSTRACT:
.....DESCRIPTION:

.....DEPENDENCIES:
.....MEMORY_SIZE:
..BUILD_INSTRUCTIONS:
.....PROCESSOR:  DMC_EDIT_R3
.....SEGMENT NAME ..............................: X$PSS
.....SEGMENT TYPE (RUN, INTERRUPT, INITIALIZE) : INTERRUPT
.....SEGMENT INTERRUPT NUMBER ..................: 36
.....SEGMENT SCHEDULING RATE, AND UNIT .........: 0  MILLISECOND
.....8087 CO-PROCESSOR USED (YES / NO) ........ : YES
.....SEGMENT STACK SIZE (# OF BYTES) ..........: 500
.....SYMBOLIC NAME SPLIT CHAR (DEFAULT HYPHEN) :
```

```
*************************************************************************************************
*                     *               *  COPYRIGHT (C) 1988, BY    *  SYSTEM:PSS                 *
* DATE: 10/19/88      * REQUISITION   *  GENERAL ELECTRIC COMPANY *                             *
* AUTHOR :            *  F'ACIFIC     *  DRIVE SYSTEMS DEPARTMENT  ***************************************
* J                   *  D.S.D        *  SALEM, V A    USA         * DRAWING : XXXXXX                    *
*                     *  xxxxxx       *                            * NEXT PAGE: 0007          PAGE: 0006 *
*************************************************************************************************
```

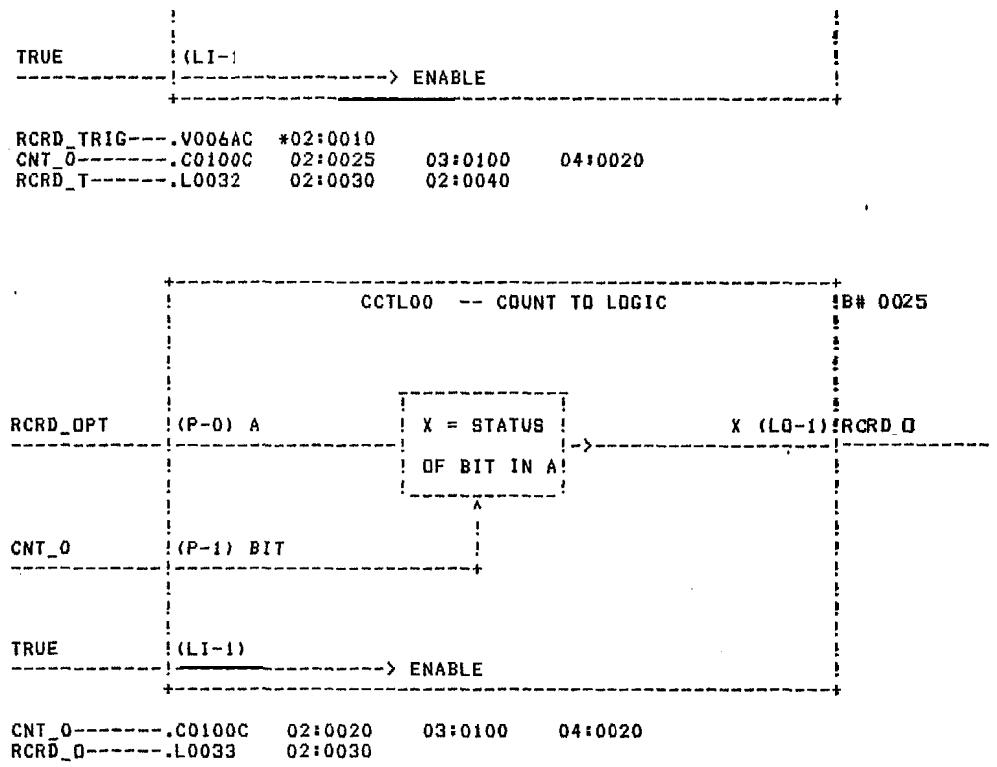



```
LTSTC-------.LOOBS      04:0030   *04:0030    04:0100    04:0110    04:0120
LOLC---------.L0039    *04:0120    04:0130
SCAN-VL-----.VOOIOC     ● 01:0200 *01:0300   *01:0400   *01:0600   *01:0700
                       *03:0200
POINTS------.COIACC     01:0200    01:0300    01:0400    01:0600    01:0700
                        03:0200
PSS_ERR-----.L004E      04:0135     0410144
RCRD_EVENT--.L0044      01:0100   *01:0900
RCRD_TRIG---.V006AC     02:0020
ON_PERM-----.L0036      0410130
```



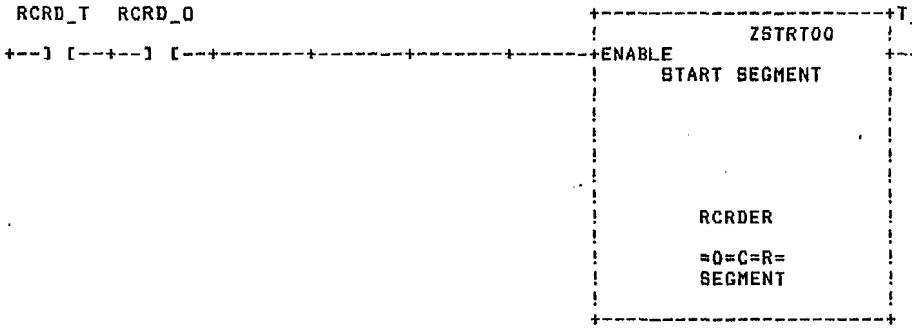

```
=====================  ========
SEGMENT:02/BLOCK:0010 PG:0 0 0 7
=====================  ========
        L D  LTSTC-------.L0035
        LD LOLC---------.L0039
        LD    TRUE--------.LOOOI
        BLK XPSSFOO
        B# 0 0 1 0
        P-O   FVAR--------.VOOFBC
        P-I   FCON--------.COOIOR
        P-2   SCAN-VL-----.VOOIOC
        P-3   POINTS------.COIACC
        P-4 PSS_ERR-----.L004E
        P-5 RCRD_EVENT--.L0044
        P-6 INT_TST-----.L0048
        P-7 DA4OUT------.COOFAP
        P-O RCRD_TRIG---.V006AC
        STO ON_PERM-----.L0036
        B# 0010
```

```
=====================  ========
SEGMENT:02/BLOCK:0020 PG:0 0 0 7
=====================  ========
        LD    TRUE--------.LOOOi
        BLK CCTLOO
        B# 0020
        P-O RCRD_TRIG---.V006AC
        P-I CNT_0-------.C0100C
        STO RCRD_T------.L0032
        B# 0020
```



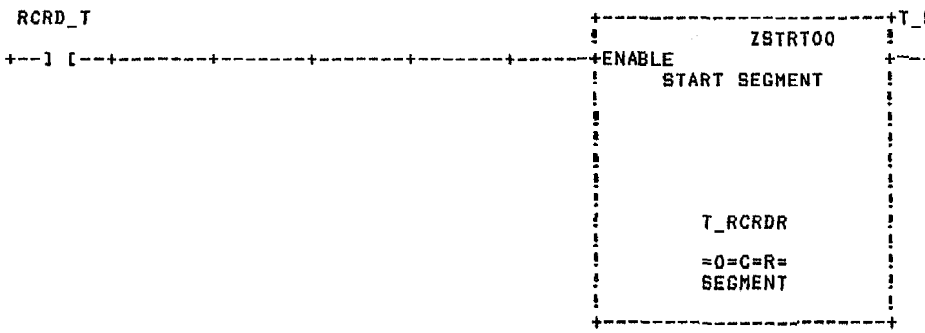

```
====================  ========
SEGMENT:02/BLOCK:0025  PG: 0007
====================  ========
        LD  TRUE--------.L0001
        BLK CCTL00
        B#  0025
        P-0 RCRD_OPT----.C0094C
        P-1 CNT_0-------.C0100C
        STO RCRD_0------.L0033
        B#  0025
```


```
*************************************************************************************************
*                       *                *   COPYRIGHT (C) 1988,  BY    *  SYSTEM:PSB           *
* DATE: 10/19/88        *  REQUISITION   *   GENERAL ELECTRIC COMPANY   *                       *
* AUTHOR :              *  PACIFIC       *   DRIVE SYSTEMS DEPARTMENT   **********************
* J                     *  D.S.D         *   SALEM, VA    USA           *  DRAWING : XXXXXX     *
*                       *  XXXXXX        *                              *  NEXT PAGE: 0008  PAGE: 0007  *
*************************************************************************************************
```




```
RCRD_T------.L0032   *02:0020    02:0040
RCRD_0------.L0033   *02:0025
T_STAT------.L0034   *02:0040
```

```
====================  ========
SEGMENT:02/BLOCK:0030  PG: 0008
====================  ========
         LD  RCRD_T------.L0032
         AN  RCRD_0------.L0033
         BLK ZSTRT00
         B#  0030
         P-0 RCRDER------.C00B8C
         STO T_STAT------.L0034
         B#  0030
```




```
RCRD_T------.L0032   *02:0020    02:0030
T_STAT------.L0034   *02:0030
```

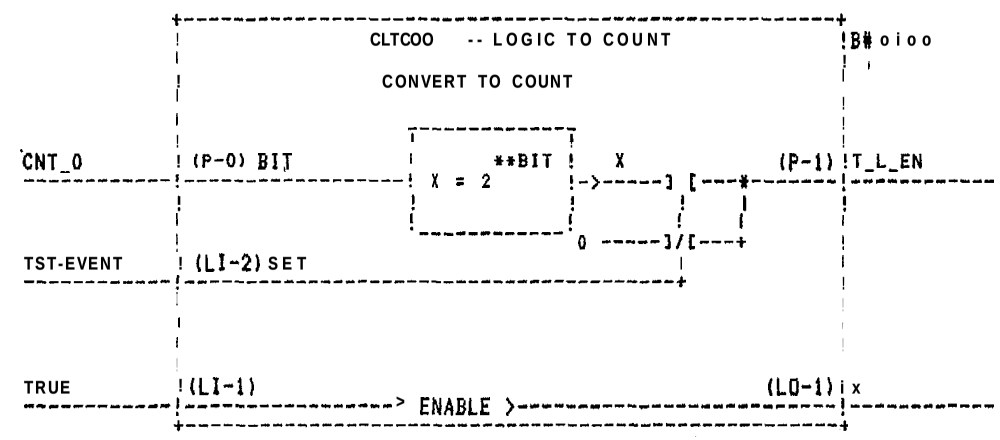
```
====================  ========
SEGMENT:02/BLOCK:0040  PG: 0008
====================  ========
         LD  RCRD_T------.L0032
         BLK ZSTRT00
         B#  0040
         P-0 T_RCRDR-----.C00BAC
         STO T_STAT------.L0034
         B#  0040
```

```
*****************************************************************************************************************
*                       *                 *   COPYRIGHT (C) 1988, BY    *  SYSTEM:PSS                             *
* DATE: 10/19/88        * REQUISITION   *   GENERAL ELECTRIC COMPANY *                                         *
* AUTHOR :              * PACIFIC       *   DRIVE SYSTEMS DEPARTMENT  ***************************************
* J                     * D.6.D         *   SALEM, VA     UBA         * DRAWING : XXXXXX                       *
*                       *  [redacted]   *                             it NEXT PAGE: 0009          PAGE: 0008 *
*****************************************************************************************************************
```

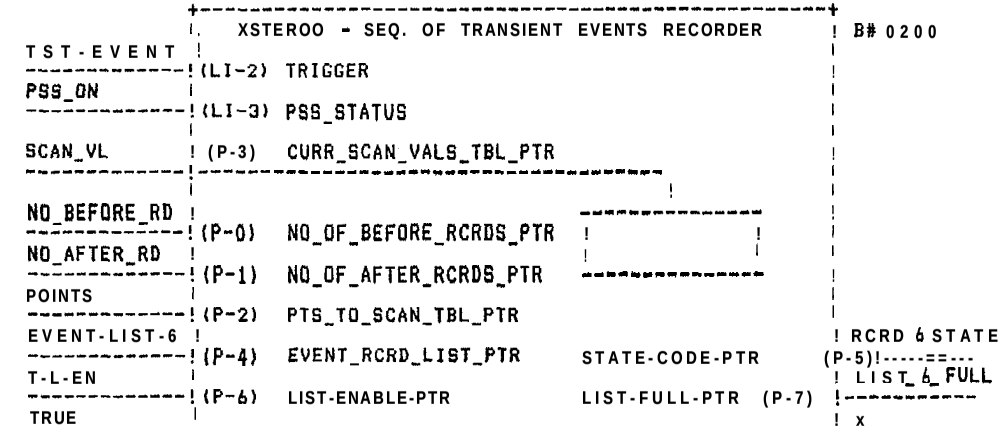
```
..NAME: PSS_TEST_EVENT_SEG_0000
..AUTHOR: HOGLE
..COPYRIGHT:    COPYRIGHT        (C)  1986
                BY GENERAL ELECTRIC COMPANY,     U.S.A
..KEYWORDS:
                DMC,SEGMENT
..ABSTRACT:
.....DESCRIPTION:

.....DEPENDENCIES:
.....MEMORY_SIZE:
..BUILD_INSTRUCTIONS:
.....PROCESSOR:  DMC_EDIT_R3
.....SEGMENT NAME . . . . . ..|.....: T_RCRDR
. . . ..SEGMENT TYPE (RUN, INTERRUPT. INITIALIZE) : RUN
. . . ..SEGMENT INTERRUPT NUMBER . . . . . :
. . . ..SEGMENT SCHEDULING RATE, AND UNIT . . . . . ..| 0 MILLISECONDS
.....8087 CO-PROCESSOR USED (YES / NO) . . . . . : YES
. . . ..SEGMENT STACK SIZE (# OF BYTES) . . . . . : 500
. . . ..SYMBOLIC NAME SPLIT CHAR (DEFAULT HYPHEN) :
```

```
*****************************************************************************************************************
*                          *              *  COPYRIGHT (C) 1988. BY  * SYSTEM:PSS                                  *
* DATE: 10/19/88           * REQUISITION  *  GENERAL ELECTRIC COMPANY *                                           *
* AUTHOR :                 *  PACIFIC     *  DRIVE SYSTEMS DEPARTMENT  ***************************************** *
* J                        * D.9.D        *  SALEM, V A     USA        * DRAWING  : XXXXXX                        *
*                          * XXXXXX       *                            * NEXT PAGE: 0010                PAGE: 0009 *
*****************************************************************************************************************
```




```
TST_EVENT---.L0047     03:0200    *04:0190
CNT_   o-------.coiooc  02:0020     02:0025     04:0020
T_L_EN------.V00B2C    03:0200
X----------.L0043     *01:0200    *01:0300    *01:0400    *01:0500    *01:0600
                      *01:0700    *01:0800    *03:0200
```




```
SEGMENT:03/BLOCK:0100 PC: 0010
        LD  TST_EVENT---.L0047
        LD TRUE--------.LOOOi
        ELK CLTCOO
        BY 0100
        P-O CNT_O-------.C0100C
        P-I T_L_EN------.V00B2C
        STO  X----------.L0043
        B# 0100
```

```
SEGMENT:03/BLOCK:0200 PC: 0010
        L D  PSS_ON------.010010
        L D  TST_EVENT---.L0047
        LD TRUE--------.LOOOI
        BLK XSTEROO
        BY 0200
        P-O NO_BEFORE_RD.C01A8C
        P-I NO_AFTER_RD-.C01AAC
        P-2  POINTS------.COIACC
        P-3  SCAN-VL-----.VOOiOC
        P-4 EVENT_LIST_6.C00B4P
        P-5 RCRD_6_STATE.V00EAC
        P-6 T_L_EN------.V00B2C
        P-7 LIST_6_FULL-.V00AAC
        STO  X----------.LOO43
        B# 0200
```

```
-------------! (LI-1) -------------> ENABLE >-------------- (LO-1) !-----------
             +-------------------------------------------------+

PSS_ON------.D10010   01:0200     01:0300     01 r0400    01:0600     01:0700
                     *04:0130     04:0210     04:0245     04:0450     0510220
                      05:0230     05:0240
TST_EVENT---.L0047    03:0100    *04:0190
NO_BEFORE_RD.C01A8C   01:0200     01:0300     01:0400     01:0600     01:0700

NO_AFTER_RD-.C01AAC   01:0200     01:0300     01:0400     01:0600     01:0700

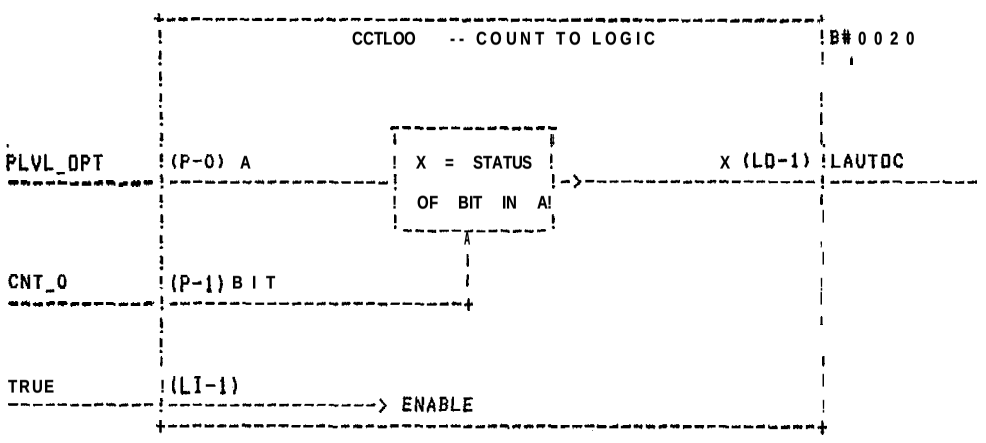
POINTS------.C01ACC   01:0200     01:0300     01:0400     01:0600     01:0700
                      02:0010
SCAN-VL-----.V00I0C  *01:0200    *01:0300    *01 r0400   *01:0600    *01:0700
                     *02:0010
T_L_EN------.V00B2C  *03:0100
X----------.L0043    *01:0200    *01:0300    *01:0400    *01:0500    *01:0600
                     *01:0700    *01:0800    *03:0100
```

```
*************************************************************************************************************
*                     *              *  COPYRIGHT (C) 1988, BY    * SYSTEM:PSS                              *
* DATE: 10/19/88      * REQUISITION *  GENERAL ELECTRIC COMPANY *                                           *
* AUTHDR :            *  PACIFIC     *  DRIVE SYSTEMS DEPARTMENT  ********************************************
* J                   *  D.S.D       *  SALEM. VA     USA         it DRAWING : XXXXXX                       *
*                     *  xxxxxx      *                            * NEXT PAGE: 0011          PAGE: 0010  *
*************************************************************************************************************
```

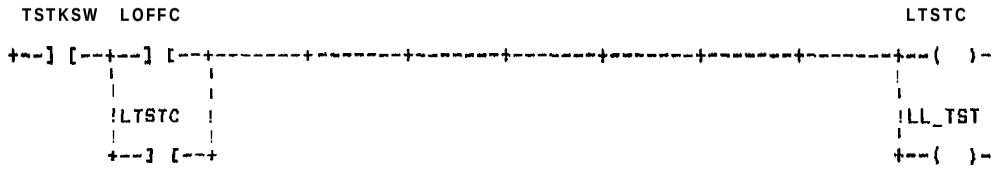
```
..NAME: PGE_IO_SEGMENT_0001
..REVISED_FROM: PGE_IO_SEGMENT_0000
..AUTHOR: GIBBS
..COPYRIGHT:    COPYRIGHT        (C)  1987
                BY  GENERAL  ELECTRIC  COMPANY,     U.S.A
..KEYWORDS:
                DMC,SEGMENT
..ABSTRACT:
. . . ..DESCRIPTION: THIS SEGMENT DDE5 THE NECESSARY
          SEQUENCING TO DETERMINE PSS ON OR PSS OFF
          AND POWER AMPLIFIER UN OR OFF.

....DEPENDENCIES:
.....MEMORY_SIZE:
..BUILD_INSTRUCTIONS:
....PROCESSOR:  DMC_EDIT_R3
. . . ..SEGMENT NAME .....! X$SEQ
. . . ..SEGMENT TYPE (RUN, INTERRUPT, INITIALIZE) : RUN
. . . ..SEGMENT INTERRUPT NUMBER . . ..*..*.....*. :
. . . ..SEGMENT SCHEDULING RATE, AND UNIT . . .....: 100 MILLISECOND
. . ...3037 CO-PROCESSOR USED (YES / NO) ..... : YE%
. . . ..SEGMENT STACK SIZE (# OF BYTES) ..... : 100
. . . ..SYMBOLIC NAME SPLIT CHAR (DEFAULT HYPHEN) :
```

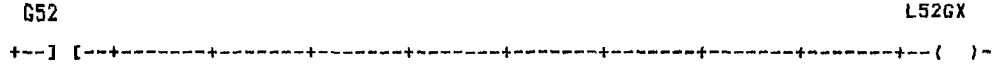
```
*****************************************************************************************************************
*                        *               *   COPYRIGHT (C) 1988, BY     *  SYSTEM:PSS                            *
* DATE: 10/19/88         *  REQUISITION  +   GENERAL ELECTRIC COMPANY *                                        *
*  AUTHOR :              * PACIFIC       *   DRIVE SYSTEMS DEPARTMENT   ************************************************
* J                      *  D.S.D        *   SALEM. VA     USA          * DRAWING : XXXXXX                              *
*                        *  xxxxxx       *                              * NEXT PAGE:0012                  PAGE: 0011 *
*****************************************************************************************************************
```



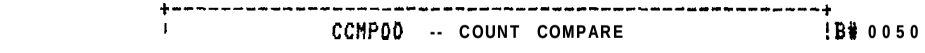

```
CNT_0-------.C0100C    02:0020     02:0025     03~0100
LAUTOC------.LOOBC     04:0100     04:0110
```




```
LOFFC-------.L0037       04:0100    *04:0100     04:0110     04:0120
LTSTC--------.L0035      02:0010     04:0100     04:0110     04:0120
```




```
L52GX--------.L0042     04:0100     04:0110     04:0120
```

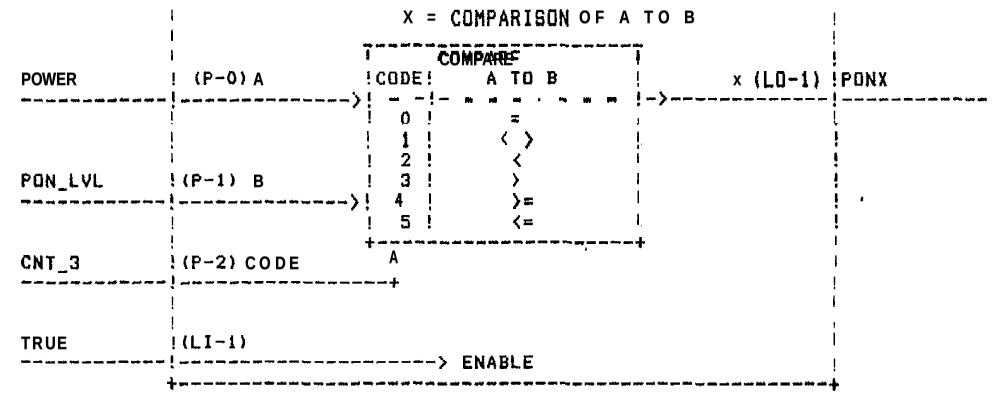



```
SEGMENT:04/BLOCK:0020 PG:0 0 1 2
        LD TRUE--------.LOOOI
        BLK CCTLOO
        B# 0 0 2 0
        P-O PLVL_OPT----.C009AC
        P-i CNT_0-------.C0100C
        STO LAUTOC------.LOOSC
        BY 0020
```

```
SEGMENT:04/BLOCK:0030 PG:0 0 1 2
        LD    TSTKSW------.I10105
        L D   LDFFC--------.L0037
        O R   LTSTC--------.L0035
        ANS
        STO   LTSTC-------.LOO35
        BY 0030
        STO LL_TST------.010103
        BY 0030
```

```
SEGMENT:04/BLOCK:0040 PG:0 0 1 2
        L D   G52---------.I10002
        STO L52GX--------.L0042
        B# 0 0 4 0
```

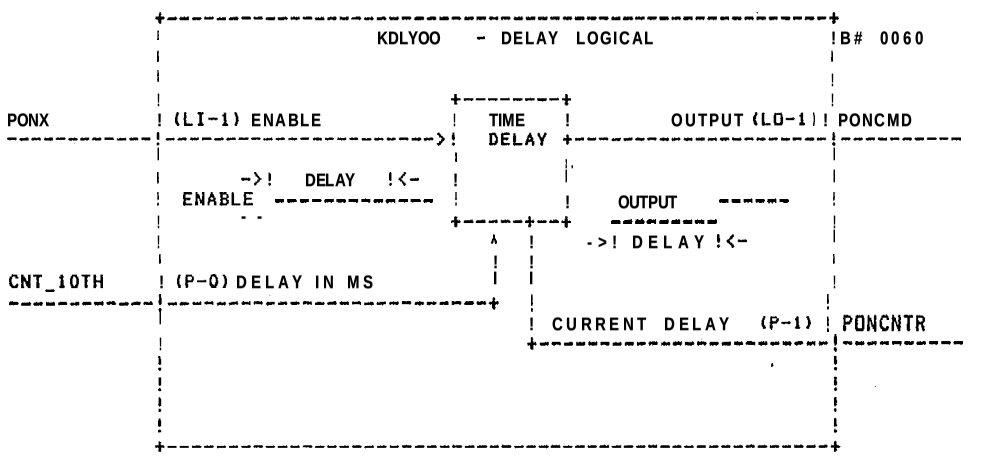
```
SEGMENT:04/BLOCK:0050 PG:0 0 1 2
        LD    TRUE--------.LOOOI
        BLK  CCWPOO
        BY 0050
```




```
P-O  POWER-------.VOOIEC
P-1  PON_LVL-----.C0096C
P-2  CNT_3-------.C0106C
STO  PONX--------.L0040
B# 0050
```

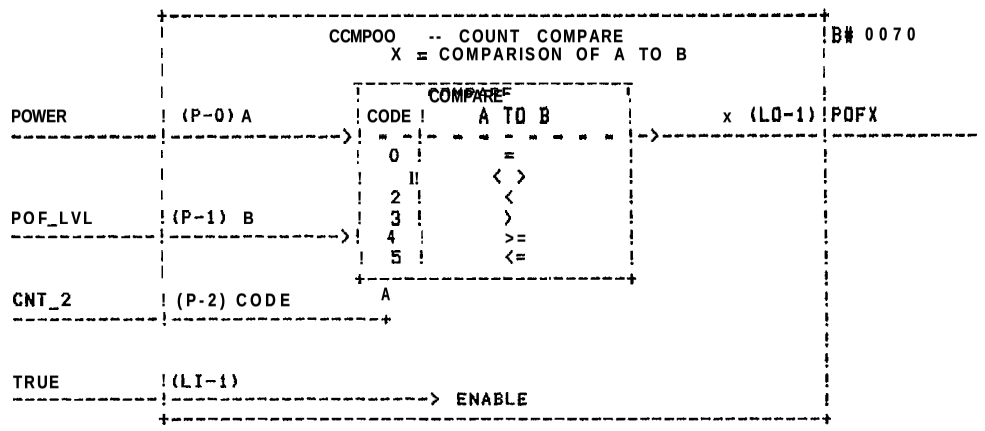
```
POWER-------.VOOIEC    04:0070    04:0640
CNT_3-------.C0106C    04:0340    04:0380    0410420
PONX--------.L0040     0410060
```


```
*****************************************************************************************************************
*                    *                  *  COPYRIGHT (C) 1988, BY     *  SYSTEM:PSS                             *
* DATE: 10/19/88     * REQUISITION      *  GENERAL ELECTRIC COMPANY   it                                        *
* AUTHOR :           * PACIFIC          *  DRIVE SYSTEMS DEPARTMENT   ***************************************
* J                  *  D.S.D           *  SALEM, VA     USA          * DRAWING : XXXXXX                        *
*                    *  XXXXXX          *                             * NEXT PAGE:0013                PAGE:0012 *
*****************************************************************************************************************
```



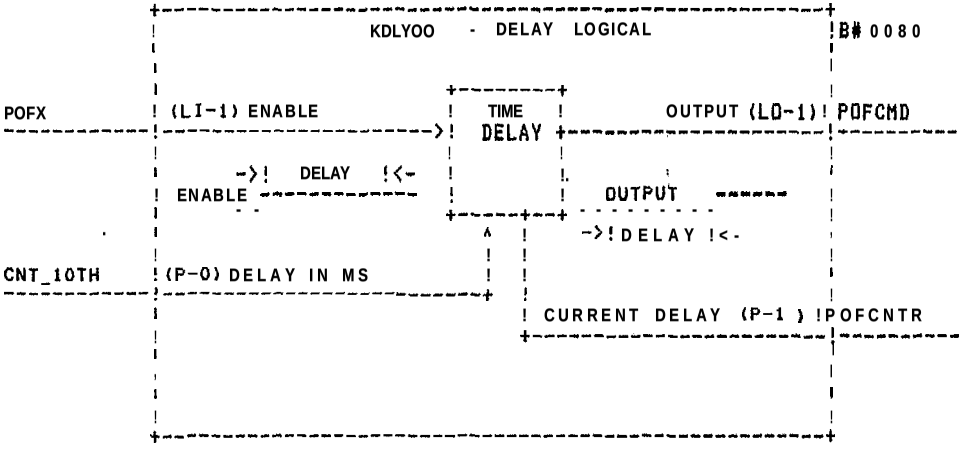

```
PONX--------.L0040    *04:0050
CNT_10TH----.C011AC    04:0080
PONCMD------.L003E     04:0090
```

```
SEGMENT:04/BLOCK:0060 PG:0 0 1 3
          LD PONX--------.L0040
          BLK KDLYOO
          B# 0 0 6 0
          P-O CNT_10TH----.C011AC
          P-I PONCNTR-----.V006CC
          STO PONCMD------.L003E
          B# 0 0 6 0
```




```
POWER-------.V0OIEC     04:0050     04:0640
POFX--------.LOO41      04:0080
```

```
SEGMENT:04/BLOCK:0070 PG:0 0 1 3
          LD    TRUE--------.LOOOI
          BLK  CCMPOO
          B# 0 0 7 0
          P-O POWER--------.V001EC
          P-I POF_LVL-----.C0098C
          P-2 CNT_2--------.C0104C
          STO POFX---------.L0041
          BY 0070
```




```
POFX--------.L0041     *04:0070
CNT_10TH----.C011AC     04:0060
POFCMD------.L003F      04:0090     0410135     04:0144
```

```
SEGMENT:04/BLOCK:0080 PG:0 0 1 3
================================
        LD POFX--------.L0041
        BLK KDLYOO
        B# 0 0 8 0
        P-O CNT_10TH----.C011AC
        P-L POFCNTR-----.V006EC
        STO POFCMD------.L003F
        B# 0 0 8 0
```

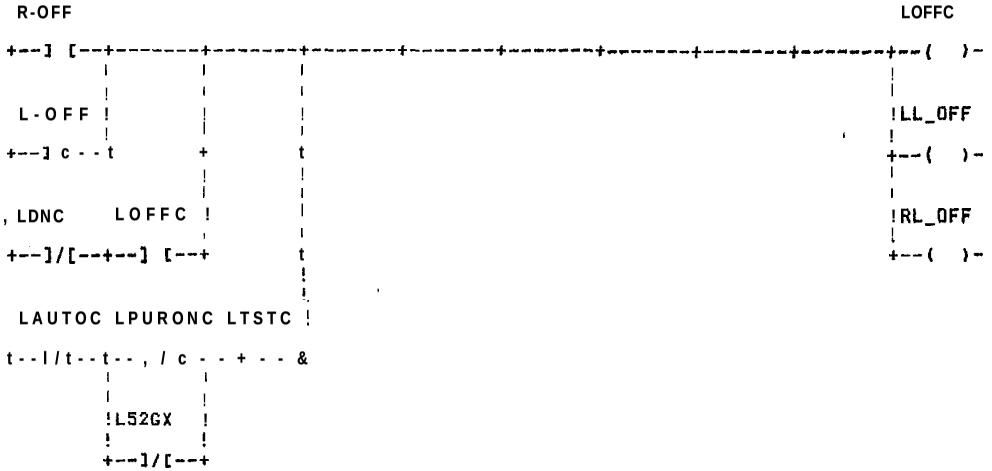
```
 PONCHO  POFCMD                                                          LPWRONC

+--]  [--+--]/[--+-------+-------+-------+-------+-------+-------+-------+--( )-
         |
         |
 LPWRONC |
         |
+--] [--+
```

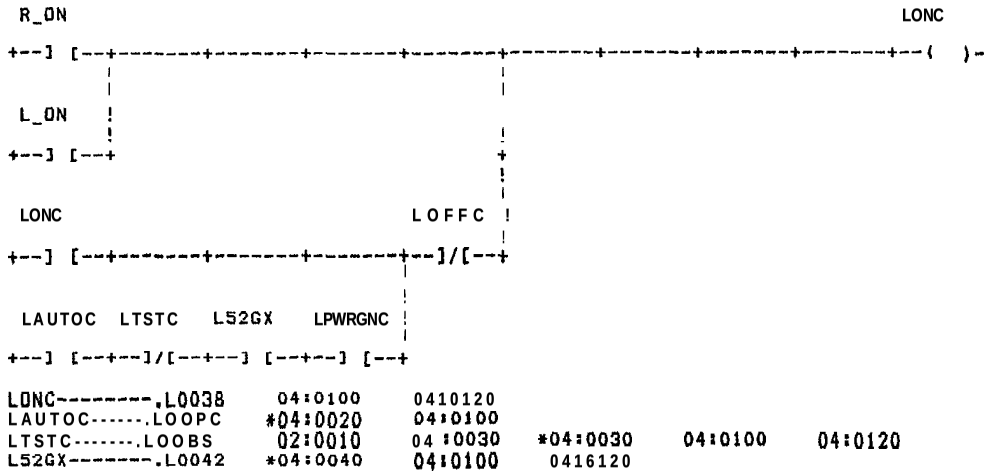
```
PONCMD------.L003E    *04:0060
LPWRONC-----.L003D     04:0100     04:0110     04:0120
POFCMD------.L003F    to4 :0080    04:0135     0410144
```

```
==========================
SEGMENT:04/BLOCK:0090  PG: 0013
==========================
        L D  PONCMD------.L003E
        O R  LPWRONC-----.L003D
        ANF POFCMD------.L003F
        STO LPWRONC-----.L003D
        BY 0090
```

```
*****************************************************************************************************************
*                      *                    *                                                                     *
* DATE: 10/19/88       * REQUISITION      * COPYRIGHT (C) 1988. BY    * SYSTEM:PSS                                *
*  AUTHOR :            * PACIFIC          * GENERAL ELECTRIC COMPANY *                                            *
* J                    *  D.S.D           * DRIVE SYSTEMS DEPARTMENT  ***********************************************
*                      * xxxxxx           * SALEM, VA    USA          * DRAWING : XXXXXX                          *
*                      *                  *                           * NEXT PACE: 0014          PACE: 0013 *
*****************************************************************************************************************
```



LDNC--------.L0038	04:0110	*04:0110	04:0120		
LOFFC-------.L0037	04:0030	0410110	0480120		
LAUTOC------.L003C	*04:0020	04:0110			
LPWRONC-----.L003D	04:0090	*04:0090	04:0110	0410120	
L52GX-------.L0042	*04:0040	04:0110	04:0120		
LTSTC-------.L0035	02:0010	04:0030	*04:0030	04:0110	0480120



LONC--------.L0038	04:0100	0410120				
LAUTOC------.LOOPC	*04:0020	04:0100				
LTSTC-------.LOOBS	02:0010	04:0030	*04:0030	04:0100	04:0120	
L52GX-------.L0042	*04:0040	04:0100	0416120			

```
====================  ========
SEGMENT:04/BLOCK:0100  PG: 0014
====================  ========
        LD  R-OFF--------.I10001
        O R  L_OFF-------.I10100
        LDF    LDNC--------.L003B
        AN    LOFFC-------.L0037
        ORS
        LDF  LAUTOC------.L003C
        LDF  LPWRONC-----.L003D
        ORF  L52GX-------.L0042
        ANS
        ANF  LTSTC-------.L0035
        ORS
        STO  LOFFC-------.L0037
        B#   0100
        STO  LL_OFF------.010100
        B#   0100
        STO  RL_OFF------.010004
        BY   0100
```

```
====================  ========
SEGMENT:04/BLOCK:0110 PG:0 0 1 4
====================  ========
        L  D   R ON--------.I10000
        O R  L_ON--------.I10101
        LD  LONC--------.L0038
        L  D   LAUTOC------.L003C
        ANF LTSTC-------.L0035
        A  N   L52GX-------.L0042
        A  N   LPWRONC-----.L003D
        OR5
        ANF    LOFFC-------.L0037
        ORS
        STO  LONC--------.L0038
        BY  0110
```

```
LPWRONC-----.L003D     04:0090    *04:0090     04:0100     04:0120
LOFFC-------.L0037     04:0030     04:0100    *04:0100     04:0120
```

```
 LTSTC            LONC      LOFFC                                          LOLC

+--] [--+-------+--] [--+--]/[--+-------+-------+-------+-------+-------+--(  )-
        |
        |
 L52GX  LPWRONC!

+--] [--+--] [--+
```

```
LTSTC-------.L0035     02:0010     04:0030    *04:0030     0410100     04:0110

L52GX-------.L0042    *04:0040     04:0100     04:0110
LPWRONC-----.L003D     04:0090    *04:0090     04:0100     04:0110
LONC--------.L0038     04:0100     04:0110    *04:0110
LOFFC-------.L0037     04:0030     04:0100    *04:0100     04:0110
LOLC--------.L0039     02:0010     04:0130
```

```
 ON_PERM LOLC                                                          PSS_ON

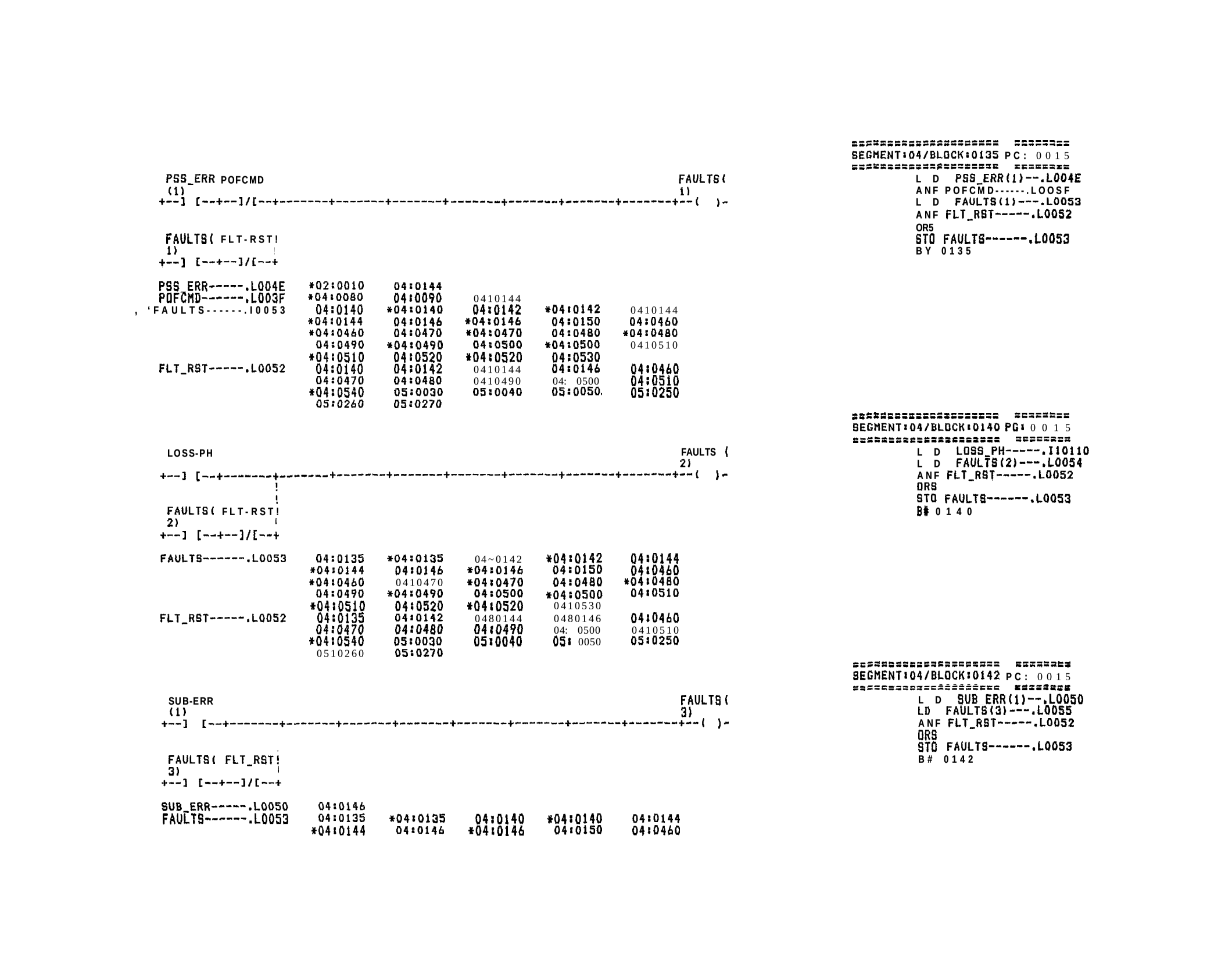
+--] [--+--] [--+-------+-------+-------+-------+-------+-------+-------+--(  )-
                                                                        |
                                                                        |
                                                                        !LL_ON
                                                                        |
                                                                        +--(  )-
                                                                        |
                                                                        |
                                                                        !RL_ON
                                                                        |
                                                                        +--(  )-
```

```
ON_PERM-----.L0036    *02:0010
LOLC--------.L0039     02:0010    *04:0120
PSS_ON------.010010    01:0200     01:0300     01:0400     01:0600     01:0700
                       03:0200     04:0210     0410245     04:0450     0510220
                       05:0230     05:0240
```

```
====================  ========
SEGMENT:04/BLOCK:0120 PG:0014
====================  ========
        L  D  LTSTC-------.L0035
        L  D  L52GX-------.L0042
        A  N  LPWRONC-----.L003D
        OR9
        A  N  LONC--------.L0038
        ANF   LOFFC-------.L0037
        STO LOLC--------.L0039
        B# 0120
```

```
====================  ========
SEGMENT:04/BLOCK:0130 PG:0014
====================  ========
        LD  ON_PERM-----.L0036
        A  N  LOLC--------.L0039
        STO PSS_ON------.010010
        BP 0130
        STO LL_ON-------.010101
        B# 0130
        STO RL_ON-------.010003
        B# 0130
```

```
*************************************************************************************************
*                     *                *  COPYRIGHT (C)1988,BY    *  SYSTEM:PSS                  *
* DATE: 10/19/88      * REQUISITION  *  GENERAL ELECTRIC COMPANY *                              *
* AUTHOR :            * PACIFIC      *  DRIVE SYSTEMS DEPARTMENT  ***************************************
* J                   *  D.S.D       *  SALEM, VA     USA         * DRAWING : XXXXXX                      *
*                     *xxxxxx        *                            * NEXT PAGE  0015          PAGE:0014 *
*************************************************************************************************
```


PSS_ERR POFCMD
(1)
FAULTS(
1)
FAULTS(FLT-RST!
1)
PSS_ERR-----.L004E
POFCMD------.L003F
FAULTS------.I0053
FLT_RST-----.L0052
SEGMENT:04/BLOCK:0135 PC: 0015
L D PSS_ERR(1)--.L004E
ANF POFCMD------.LOOSF
L D FAULTS(1)---.L0053
ANF FLT_RST-----.L0052
OR5
STO FAULTS------.L0053
BY 0135
LOSS-PH
FAULTS (
2)
FAULTS(FLT-RST!
2)
FAULTS------.L0053
FLT_RST-----.L0052
SEGMENT:04/BLOCK:0140 PG: 0015
L D LOSS_PH-----.I10110
L D FAULTS(2)---.L0054
ANF FLT_RST-----.L0052
ORS
STO FAULTS------.L0053
B# 0140
SUB-ERR
FAULTS(
3)
FAULTS(FLT_RST!
3)
SUB_ERR-----.L0050
FAULTS------.L0053
SEGMENT:04/BLOCK:0142 PC: 0015
L D SUB_ERR(1)--.L0050
LD FAULTS(3)---.L0055
ANF FLT_RST-----.L0052
ORS
STO FAULTS------.L0053
B# 0142

```
                    *04:0460    04: 0470    *04:0470    04 :0480    *04:0480
                     04:0490                 0410500    *04:0500     04:0510
                    *04:0510    *04:0520    ● 04:0520   0410530
FLT_RST-----.L0052   04:0135     04:0140     04:0144     04:0146     04:0460
                     04:0470     04:0480     04:0490     04:0500     04:0510
                    *04:0540    05: 0030     05:0040    05: 0050     05:0250
                     05:0260    0510270
```

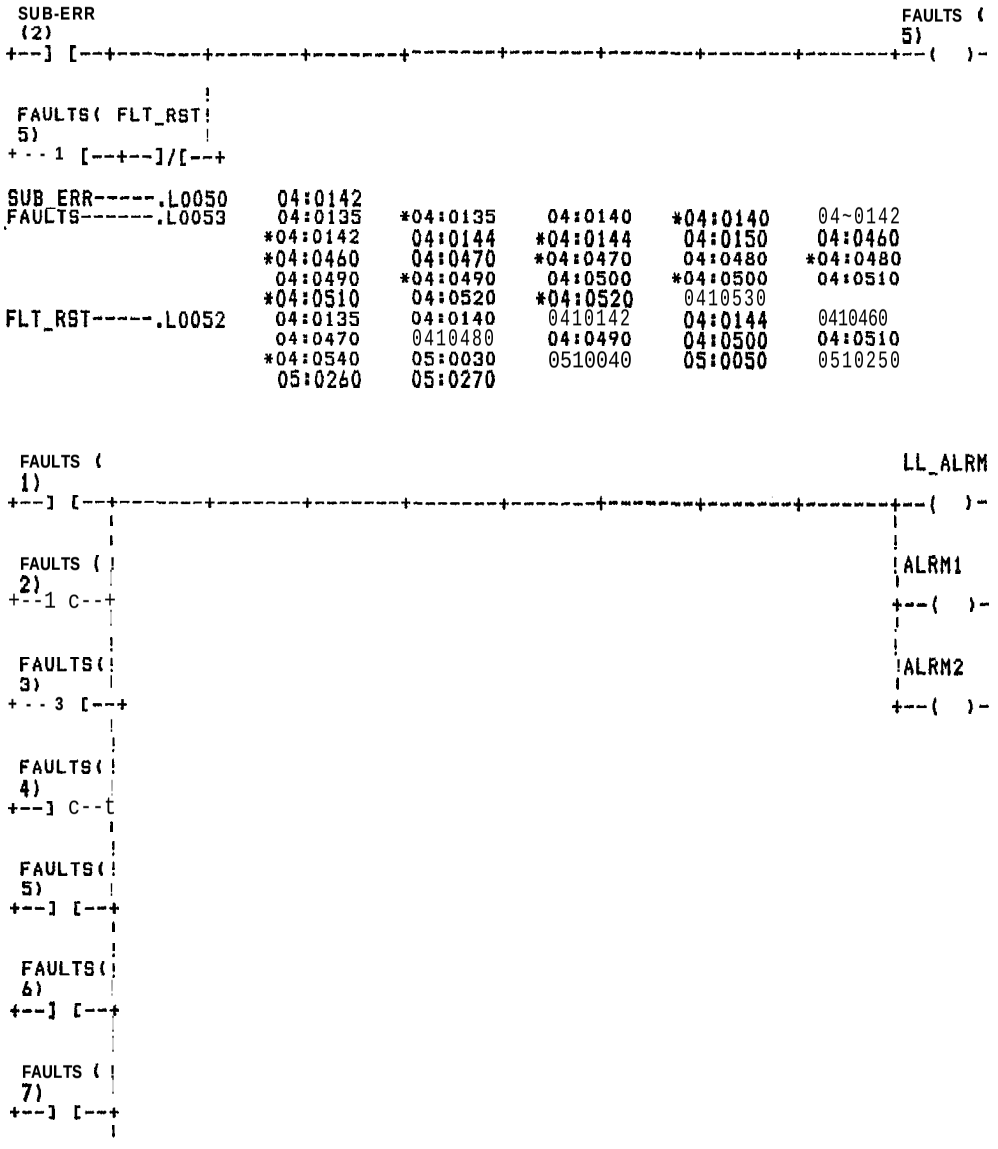
```
PSS_ERR POFCMD                                                        FAULTS(
 (2)                                                                   4)
+--] [--+--]/[--+-------+-------+-------+-------+-------+-------+-------+--(  )-
```

```
FAULTS( FLT_RST!
 4)
+--] [--+--]/[--+
```

```
PSS_ERR-----.L004E  *02:0010    04:0135
POFCMD------.LOOBF  *04:0080    04:0090     04:0135
FAULTS------.L0053   04:0135    *04:0135     04:0140    *04:0140     04:0142
                    *04:0142     04:0146    *04:0146     04:0150     0410460
                    ● 04:0460    04:0470    *04:0470     0410480    *04:0480
                     04:0490    *04:0490     04:0500    *04:0500     04:0510
                    *04:0510     04:0520    *04:0520     04:0530
FLT_RST-----.L0052   04:0135     04:0140     04:0142     04:0146
                     04:0470     0410480     0410490     04:0500     04:0510
                    *04:0540     05:0030     05:0040     05:0050     05:0250
                     05:0260     05:0270
```

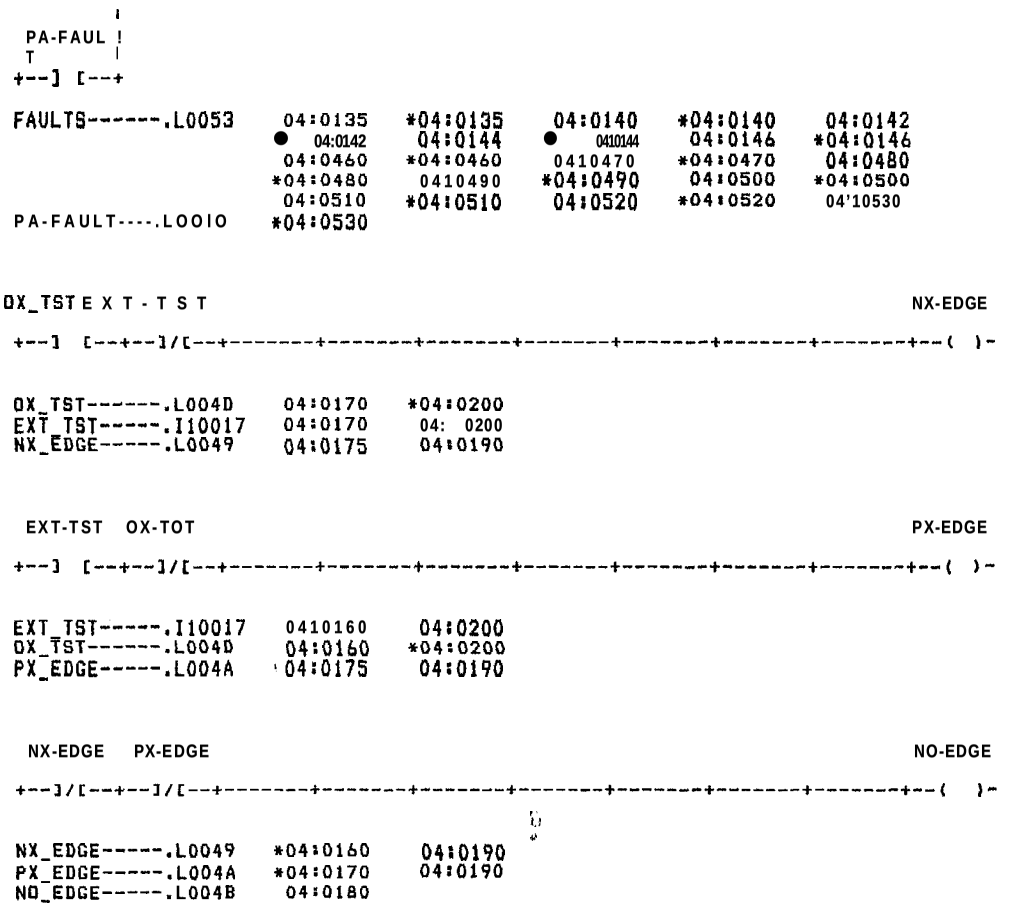
```
====================  ========
SEGMENT:04/BLOCK:0144 PC: 0015
====================  ========
        L D  PSS_ERR(2)--.L004F
        ANF POFCMD------.L003F
        L D  FAULTS(4)---.L0056
        ANF FLT_RST-----.L0052
        ORB
        STO FAULTS------.LOOSB
        BY 0144
```

```
*****************************************************************************************************************
*                        *                *  COPYRIGHT (C) 1988, BY    *  SYSTEM:PSS                             *
* DATE: 10/19/88         * REQUISITION  *  GENERAL ELECTRIC COMPANY *                                          *
* AUTHOR :               * PACIFIC      *  DRIVE SYSTEMS DEPARTMENT  ***************************************************
* J                      *  D.S.D       t  SALEM, VA     USA          * DRAWING : XXXXXX                                *
*                        t xxxxxx       *                            * NEXT PAGE: 0016                PAGE: 0015 *
*****************************************************************************************************************
```




```
=====================  ========
SEGMENT:04/BLOCK:0146  PG: 0016
=====================  ========
         L  D   SUB_ERR(2)--.L0051
         L  D   FAULTS(5)---.L0057
         ANF FLT_RST-----.L0052
         ORS
         STO FAULTS------.L0053
         BY 0146
```

```
=====================  ========
SEGMENT:04/BLOCK:0150 PG:0 0 1 6
=====================  ========
         LD  FAULTS(1)---.L0053
         L  D   FAULTS(2)---.L0054
         ORS
         L  D   FAULTS(3)---.L0055
         ORS
         L  D   FAULTS(4)---.L0056
         ORS
         L  D   FAULTS(5)---.L0057
         ORS
         L  D   FAULTS(6)---.L0058
         ORS
         LD  FAULTS(7)---.L0059
         ORS
         O  R   PA_FAULT----.L0010
         STO LL_ALRM-----.D10102
         BY 0150
         STO ALRM1-------.D10000
         BY 0150
         BTO ALRM2-------.D10001
         B# 0150
```

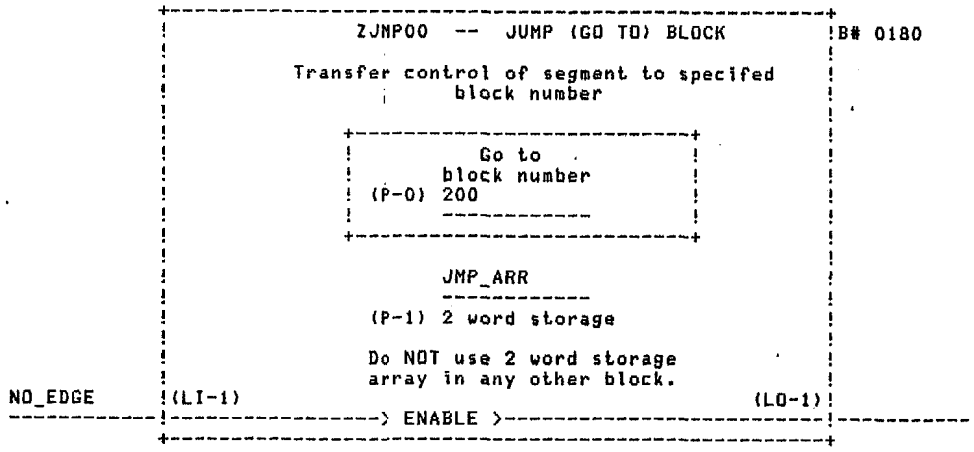



```
SEGMENT:04/BLOCK:0160 PC: 0016
        L D  OX_TST------.L004D
        ANF EXT_TST-----.I10017
        STO NX_EDGE-----.L0049
        B#  0160

SEGMENT:04/BLOCK:0170 PG: 0016
        L D  EXT_TST-----.I10017
        ANF OX_TST------.L004D
        STO PX_EDGE-----.L004A
        BY 0170

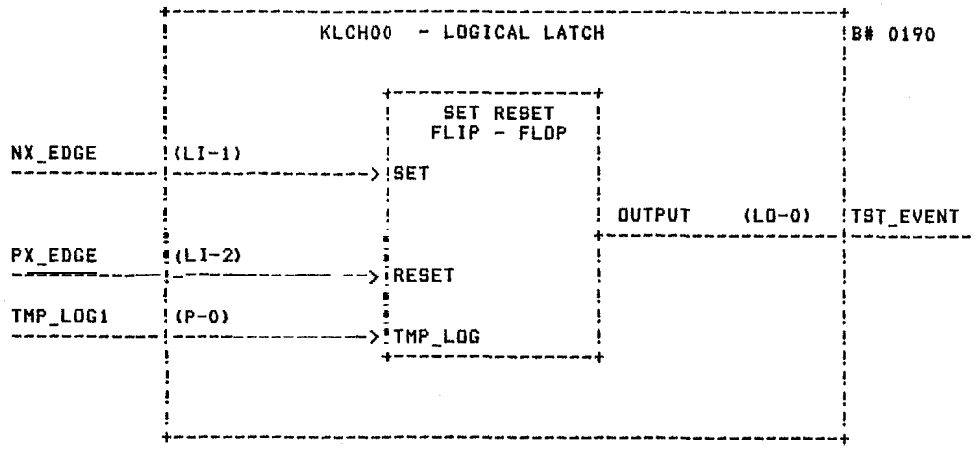
SEGMENT:04/BLOCK:0175 PG: 0016
        LDF NX_EDGE-----.L0049
        ANF PX_EDGE-----.L004A
        STO NO_EDGE-----.L004B
        B# 0175
```

```
*************************************************************************************************
*                        *                *  COPYRIGHT (C) 1988, BY    *  SYSTEM:PSS                *
* DATE: 10/19/88         * REQUISITION  *  GENERAL ELECTRIC COMPANY *                            *
* AUTHOR :               * PACIFIC      *  DRIVE SYSTEMS DEPARTMENT  ************************************************
* J                      * D.S.D        *  SALEM, VA     USA         * DRAWING : XXXXXX          *
*                        * XXXXXX       *                            * NEXT PAGE:0017  PAGE: 0016 *
*************************************************************************************************
```



NO_EDGE-----.L004B *04:0175

```
====================  ========
SEGMENT:04/BLOCK:0180  PG: 0017
====================  ========
        LD  ND_EDGE-----.L004B
        BLK ZJMP00
        B#  0180
        P-0 200---------.C02A0C
        P-1 JMP_ARR-----.V00B4C
```



PX_EDGE-----.L004A *04:0170 04:0175
NX_EDGE-----.L0049 *04:0160 04:0175
TST_EVENT---.L0047 03:0100 03:0200

```
====================  ========
SEGMENT:04/BLOCK:0190  PG: 0017
====================  ========
        LD  PX_EDGE-----.L004A
        LD  NX_EDGE-----.L0049
        BLK KLCH00
        B#  0190
        P-0 TMP_LOG1----.L004C
        STO TST_EVENT---.L0047
        B#  0190
```

```
====================  ========
```

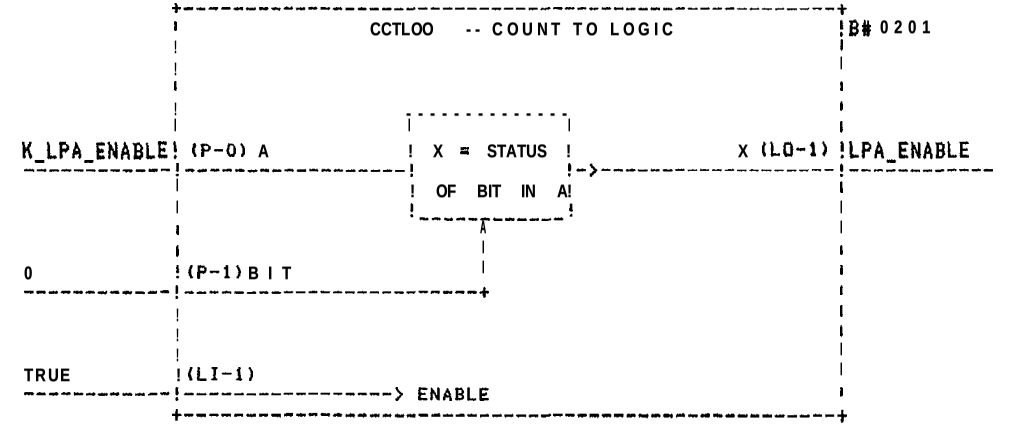
```
SEGMENT:04/BLOCK:0200 P C :  0 0 1 7
=====================  ========
         L  D   EXT_TST------.I10017
         STO  OX_TST-------.L004D
         B# 0 2 0 0


=====================  ========
SEGMENT:04/BLOCK:0201 PG: 0 0 1 7
=====================  ========
         LD     TRUE--------.L000I
         BLK  CCTL00
         BY  0201
         P-0  K_LPA_ENABLE.C00BCC
         P-1  0-----------.C02A2C
         STO  LPA_ENABLE--.L0011
         BY  0201
```

```
 EXT_TST                                                                  OX_TST

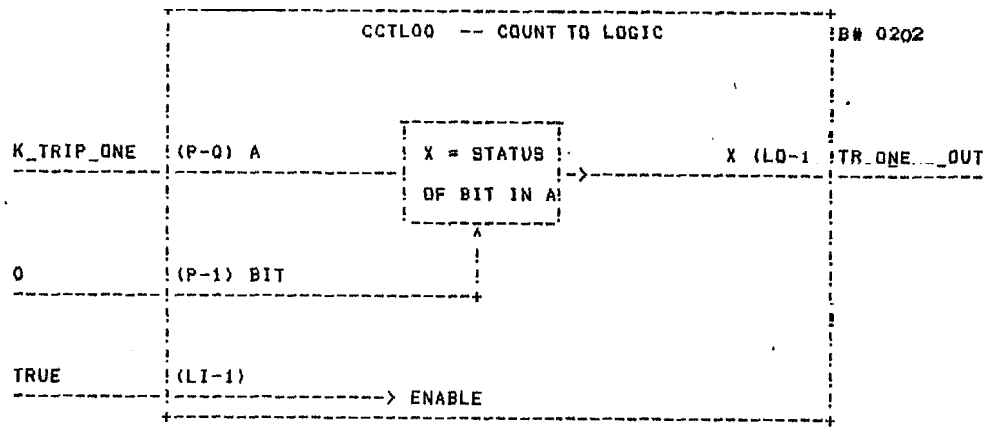
 +--] [--+-------+-------+-------+-------+-------+-------+-------+-------+--(  )-

EXT_TST-----.I10017     04:0160     04:0170
 OX_TST------.L004D     04:0160     04:0170
```



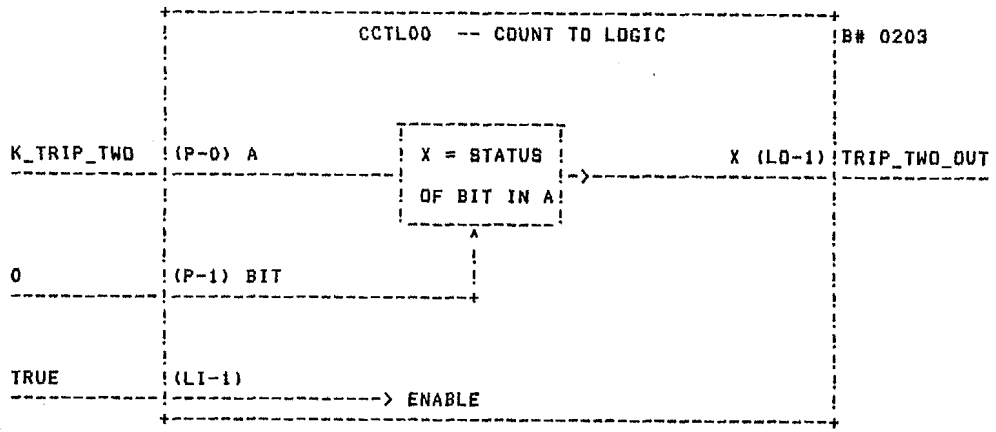

```
LPA_ENABLE--.L0011     04:0215
```

```
*****************************************************************************************************************
*                        *                *  COPYRIGHT (C)1988, BY    *  SYSTEM:PSS                                 *
* DATE: 10/19/88         * REQUISITION  *  GENERAL ELECTRIC COMPANY *                                             *
* AUTHOR :               * PACIFIC      *  DRIVE SYSTEMS DEPARTMENT  ***********************************************
* J                      *  D.S.D       *  SALEM, V A     USA        * DRAWING # XXXXXX                          *
*                        * xxxxxx       *                           * NEXT PAGE # 0018                PAGE: 0 0 1 7 *
*****************************************************************************************************************
```



TRIP_ONE_OUT.L001C 04:0300

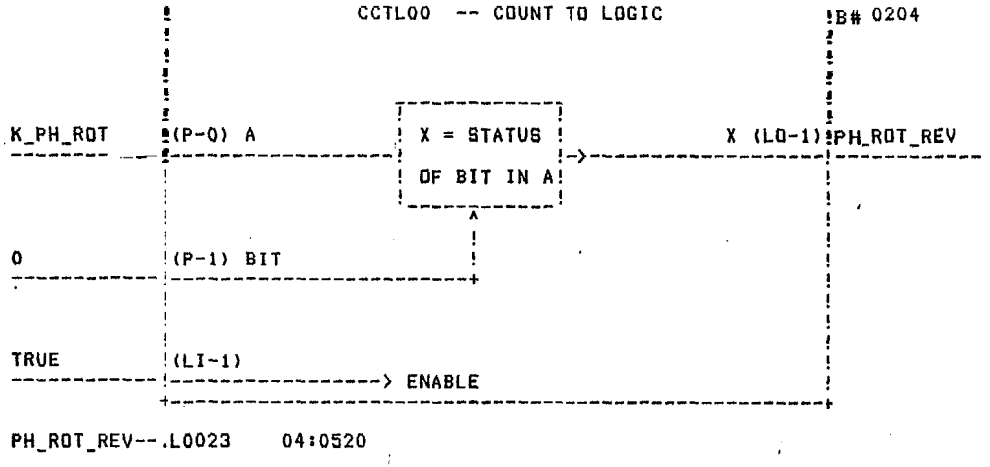
```
====================  ========
SEGMENT:04/BLOCK:0202  PG: 0018
====================  ========
        LD  TRUE--------.L0001
        BLK CCTL00
        B#  0202
        P-0 K_TRIP_ONE--.C00BEC
        P-1 0----------.C02A2C
        STO TRIP_ONE_OUT.L001C
        B#  0202
```



TRIP_TWO_OUT.L001D 04:0300

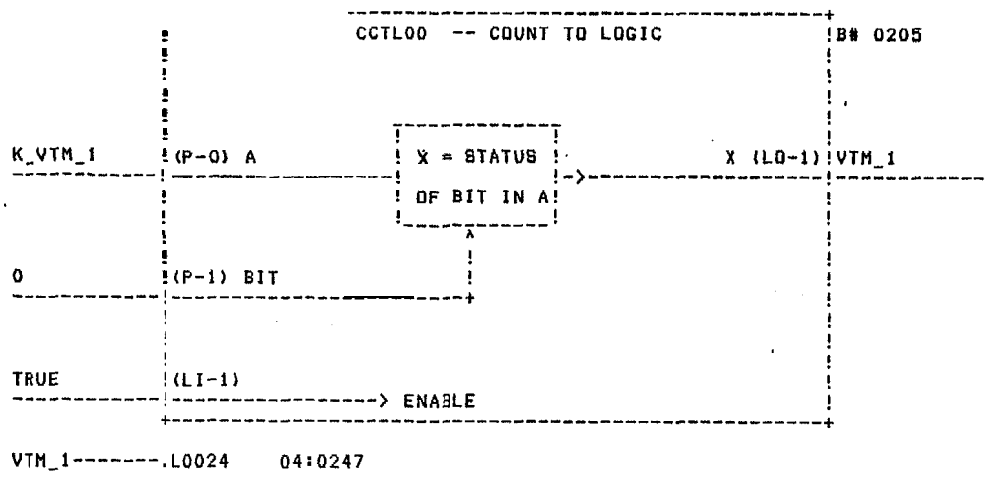
```
====================  ========
SEGMENT:04/BLOCK:0203  PG: 0018
====================  ========
        LD  TRUE--------.L0001
        BLK CCTL00
        B#  0203
        P-0 K_TRIP_TWO--.C00C0C
        P-1 0----------.C02A2C
        STO TRIP_TWO_OUT.L001D
        B#  0203
```

```
====================  ========
SEGMENT:04/BLOCK:0204  PG: 0018
====================  ========
        LD  TRUE--------.L0001
        BLK CCTL00
```

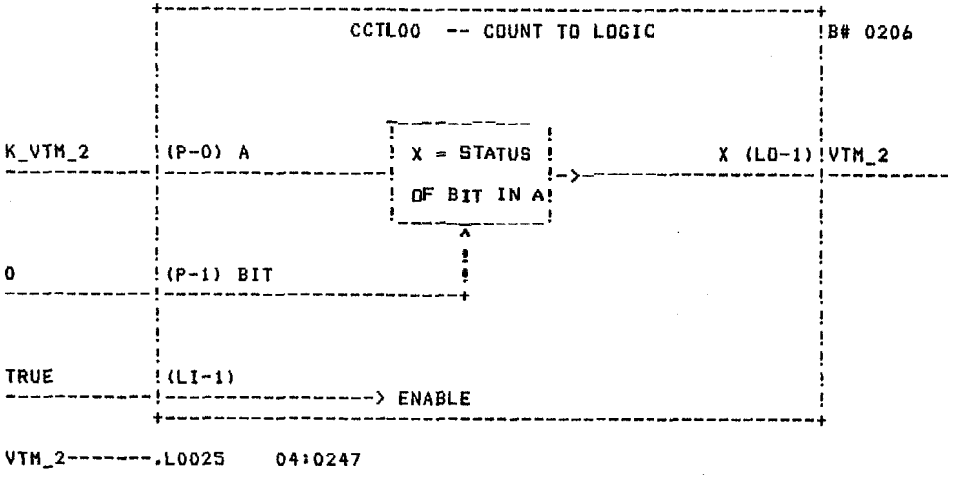



```
B#  0204
P-0 K_PH_ROT----.C00C2C
P-1 0----------.C02A2C
STO PH_ROT_REV--.L0023
B#  0204
```

```
*************************************************************************************************************
*                      *              *   COPYRIGHT (C) 1988,  BY    *  SYSTEM:PSS                           *
* DATE: 10/19/88       *  REQUISITION *   GENERAL ELECTRIC COMPANY   *                                       *
* AUTHOR :             *  PACIFIC     *   DRIVE SYSTEMS DEPARTMENT   ***************************************
* J                    *  D.S.D       *   SALEM, VA    USA           *  DRAWING : XXXXXX                     *
*                      *  XXXXXX      *                              *  NEXT PAGE: 0019          PAGE: 0018  *
*************************************************************************************************************
```

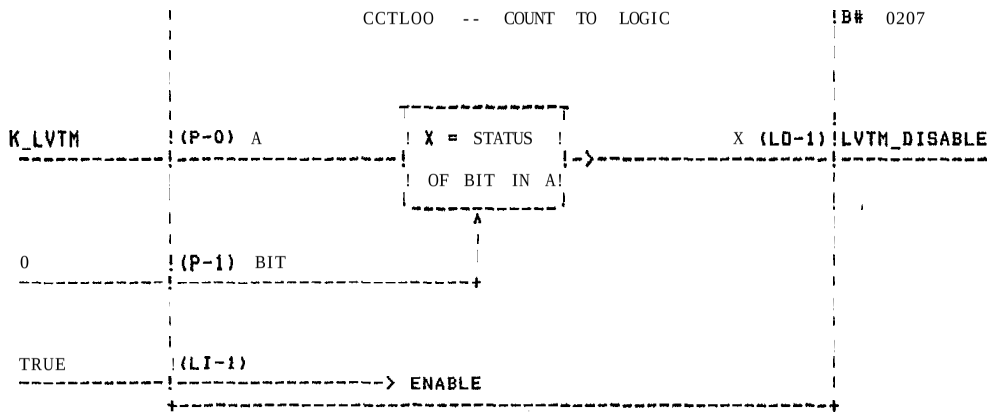



```
====================  ========
SEGMENT:04/BLOCK:0205  PG: 0019
====================  ========
        LD  TRUE---------.L0001
        BLK CCTL00
        B#  0205
        P-0 K_VTM_1------.C00C4C
        P-1 0------------.C02A2C
        STO VTM_1--------.L0024
        B#  0205
```




```
====================  ========
SEGMENT:04/BLOCK:0206  PG: 0019
====================  ========
        LD  TRUE---------.L0001
        BLK CCTL00
        B#  0206
        P-0 K_VTM_2------.C00C6C
        P-1 0------------.C02A2C
        STO VTM_2--------.L0025
        B#  0206
```

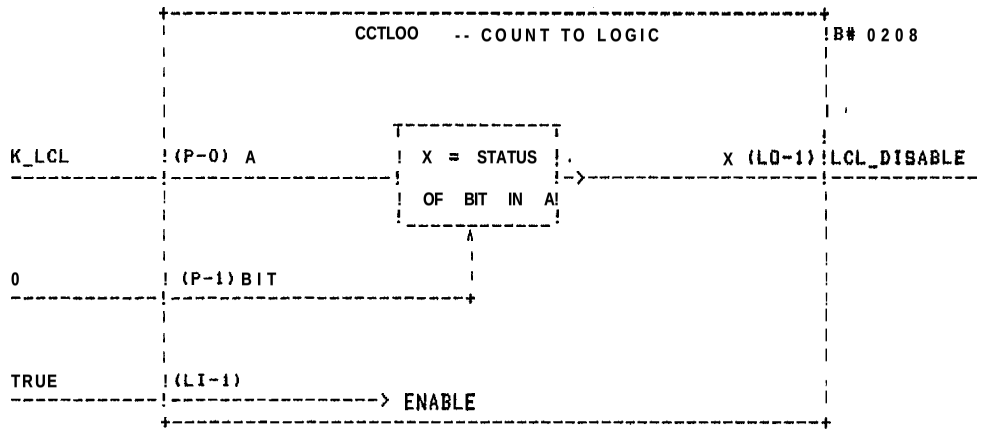
```
====================  ========
SEGMENT:04/BLOCK:0207  PG: 0019
====================  ========
        LD  TRUE---------.L0001
        BLK CCTL00
```



LVTM_DISABLE.L0028 04:0247

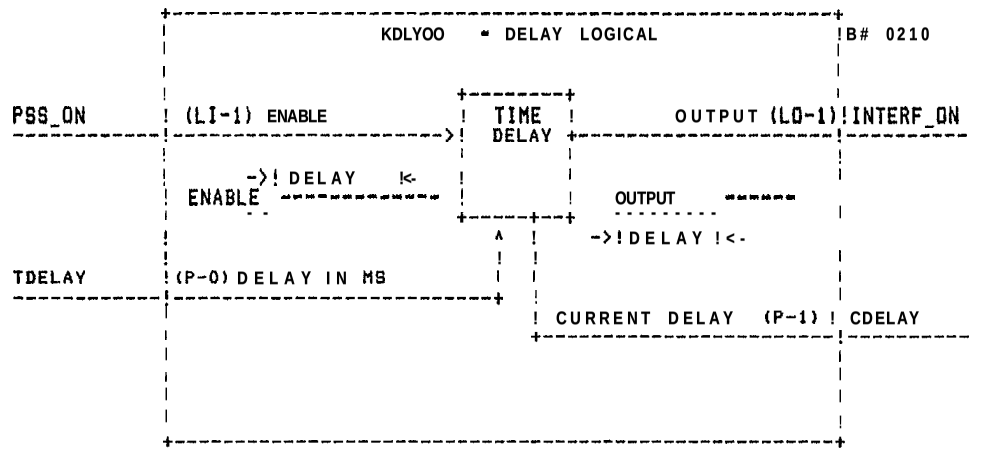
B# 0207
P-0 K_LVTM------.C00CAC
P-1 0----------.C02A2C
STO LVTM_DISABLE.L0028
B# 0207

DATE: 10/19/88	REQUISITION	COPYRIGHT (C) 1988, BY	SYSTEM:PSS	
AUTHOR :	PACIFIC	GENERAL ELECTRIC COMPANY it		
J	D.S.D	DRIVE SYSTEMS DEPARTMENT	DRAWING : XXXXXX	
	xxxxxx	SALEM, VA USA	NEXT PAGE: 0020	PAGE: 0019



LCL_DISABLE-.L0029 04:0245

```
====================  ========
SEGMENT:04/BLOCK:0208 PC: 0020
====================  ========
        LD TRUE--------.L0006
        BLK CCTLOO
        B# 0208
        P-O K_LCL-------.C00CCC
        P-1 0-----------.C02A2C
        STO LCL_DISABLE-.L0029
        B# 0208
```




```
PSS_ON------.B10010   01:0200    01 JO300    01:0400 0 1 :0600 0 1 1 0 7 0 0
                      03: 0200   *04:0130    04:0245     04:0450     05:0220
                      05: 0230    05:0240
INTERF_ON---.L002F    04: 0230    05:0220    05:0230 0 5 1 0 2 4 0
```

```
====================  ========
SEGMENT:04/BLOCK:0210 PC: 0020
====================  ========
        LD  PSS_ON------.B10010
        BLK KDLYOO
        BY 0210
        P-O TDELAY------.C00F6C
        P-I CDELAY------.V0068C
        BTO INTERF_ON---.L002F
        BY 0210
```

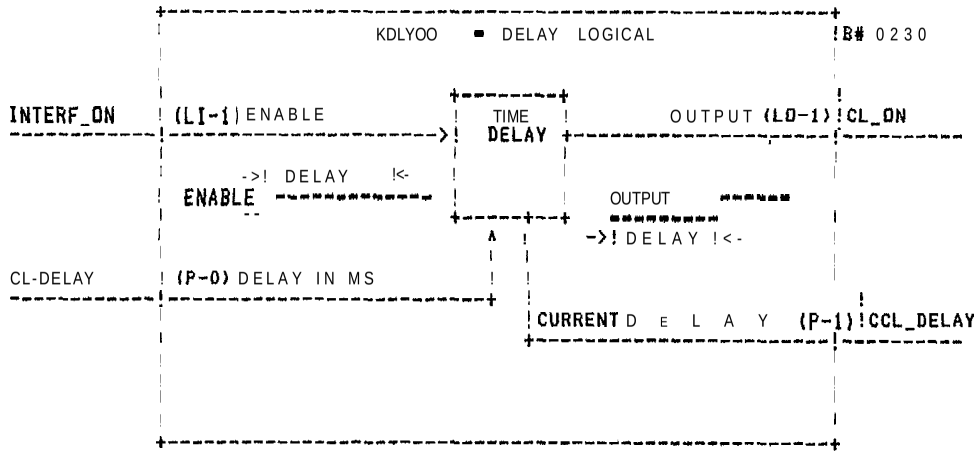
```
====================  ========
```

```
SEGMENT:04/BLOCK:0215 PG:0020
=====================  ========
        L D  LPA_ENABLE--.L0011
        STO PA_ENABLE---.L0012
        B# 0215
```

```
  LPA_ENA                                                                PA-ENAB
  BLE                                                                    LE
 +--] [--+-------+-------+-------+-------+-------+-------+-------+-------+--(  )-

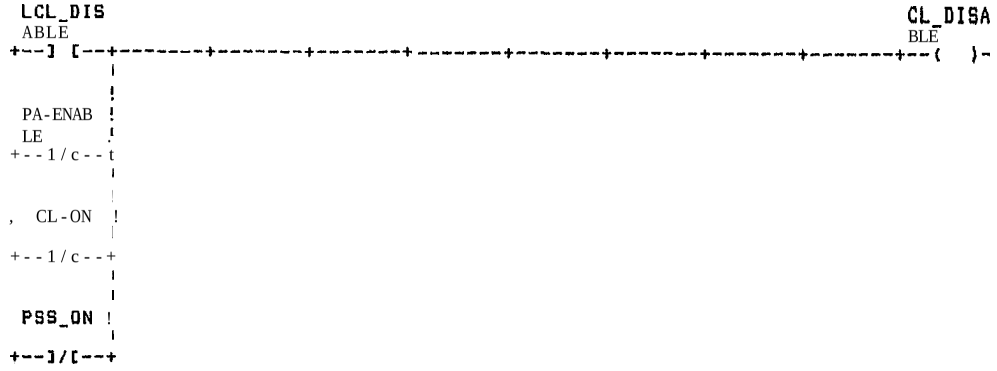
LPA_ENABLE--.L0011     *04:0201
 PA_ENABLE---.L0012     04:0245    04:0247    0410430    0410440    0410450
                        04:0460    04:0470    04:0480    0410490    04:0500
                        04:0510    05:0220    05:0230    05:0240    0510230
                        05:0260    05:0270
```

```
=====================  ========
SEGMENT:04/BLOCK:0230 PG:0020
=====================  ========
        LD  INTERF_ON---.L002F
        BLK KDLYOO
        B# 0230
        P-O CL-DELAY----.COOCEC
        P-I CCL_DELAY---.V0032C
        STO CL-ON-------.L002B
        B# 0230
```

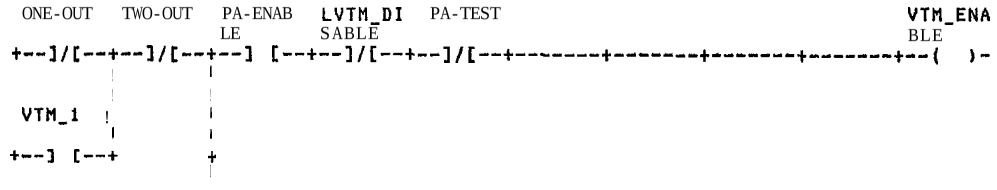



```
INTERF_ON---.L002F    *04:0210    05:0220    05:0230    05:0240
CL-ON-------.L002B     04:0245
```

```
*****************************************************************************************************************************
*                       *                 *   COPYRIGHT (C) 1988, BY     *  SYSTEM:PSS                                        *
* DATE: 10/19/88        *  REQUISITION    *   GENERAL ELECTRIC COMPANY   *                                                    *
* AUTHOR :              *  PACIFIC        *   DRIVE SYSTEMS DEPARTMENT   ***************************************************
*  J                    *   D.S.D         *   SALEM, VA     USA          * DRAWING : XXXXXX                                    *
*                       *xxxxxx           *                              * NEXT PAGE: 0021                  PAGE: 0020 *
*****************************************************************************************************************************
```




```
LCL_DISABLE-.L0029   *04:0208
PA_ENABLE---.L0012   *04:0215    04:0247    0410430    04:0440    04:0450
                      04:0460    0410470   04:0480    04:0490    04:0500
                      04:0510    0510220    05:0230    05:0240    05:0250
                      05: 0260   05:0270
CL_ON-------.L002B   *04:0230
PSS_ON------.010010   01:0200    01:0300    01:0400    01:0600    0110700
                      03:0200   *04:0130    04:0210    04:0450    05:0220
                      05:0230    05:0240
CL_DISABLE--.L002A    04:0460    04:0470    04:0480    05:0030    05:0040
                      05:0050    05:0250    05:0260    05:0270   *06:0010
```



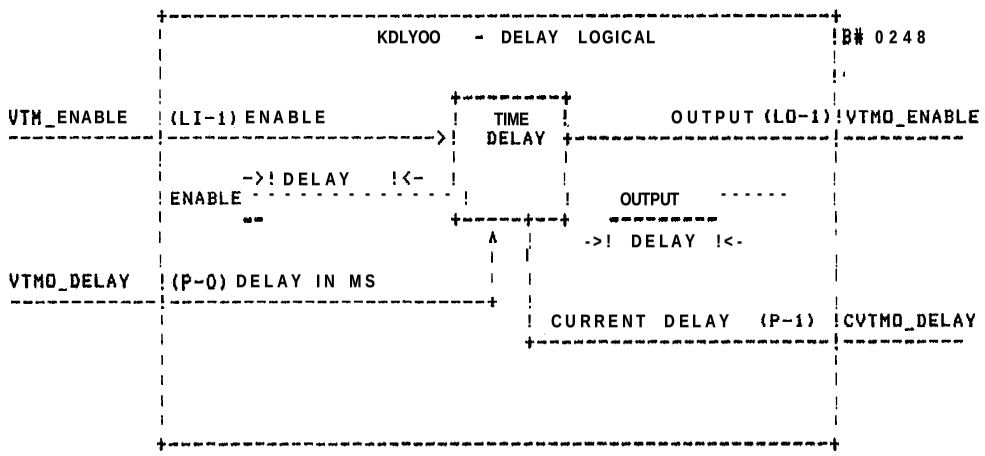

```
ONE-OUT-----.LOOiA   *04:0290    0410300    04:0560
VTM_1-------.L0024   *04:0205
TWO_OUT-----.L001B   • 04:0280    04 10300   04:0560
VTM_2-------.L0025   *04:0206
PA_ENABLE---.L0012   *04:0215    04:0245    04:0430    04:0440    04:0450
                      04:0460    04:0470    04:0480    04:0490    04:0500
                      04:0510    05:0220    05:0230    0510240    05:0250
                      05:0260    05:0270
LVTM_DISABLE.L0028   *04:0207
```

```
=====================  ========
SEGMENT:04/BLOCK:0245  PG: 0021
=====================  ========
          LD   LCL_DISABLE-.L0029
          ORF  PA_ENABLE---.L0012
          ORF  CL_ON-------.L002B
          ORF  PSS_ON------.010010
          STO  CL_DISABLE--.L002A
          B#   0245
```

```
=====================  ========
SEGMENT:04/BLOCK:0247  PG: 0021
=====================  ========
          LDF      ONE-OUT-----.LOO1A
          OR   VTM_1-------.L0024
          ANF      TWO-OUT-----.LOO1B
          OR   VTM_2-------.L0025
          AN   PA_ENABLE---.L0012
          ANF  LVTM_DISABLE.L0028
          ANF      PA-TEST-----.010016
          STO  VTM_ENABLE--.L0026
          B#   0247
```

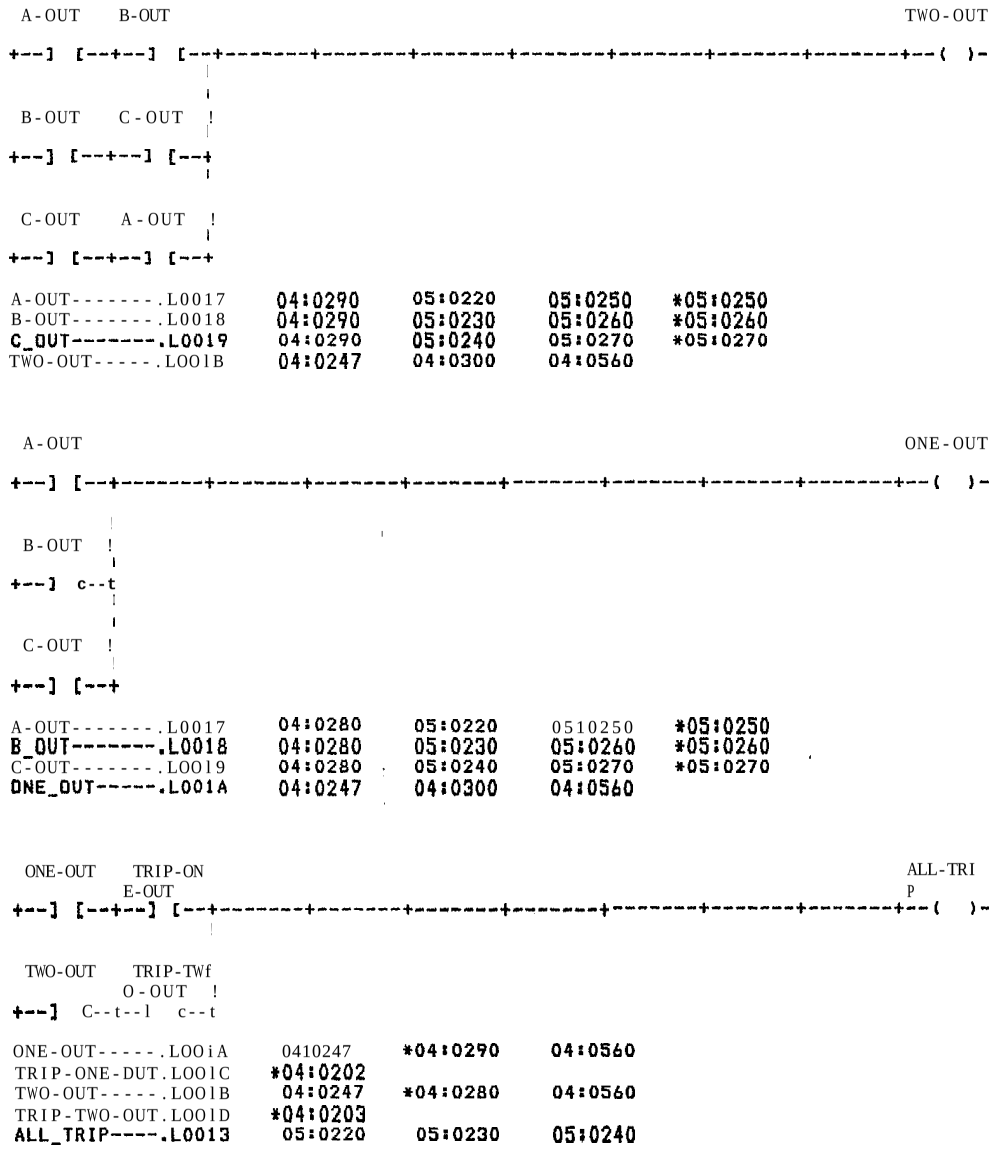
```
PA_TEST-----.D10016  *04:0450    04:0455
VTM_ENABLE--.L0026    04:0248    04:0304
```

```
SEGMENT:04/BLOCK:0248 PG:0021
L D  VTM_ENABLE--.L0026
BLK KDLYOO
BY 0248
P-O VTMD_DELAY--.C00C8C
P-i CVTMD_DELAY-.V0030C
STO VTMD_ENABLE-.L0027
BY 0248
```




```
VTM_ENABLE--.L0026   *04:0247    04:0304
VTMD_ENABLE-.L0027    04:0490    04:0500    04:0510    05:0030    05:0040
                      05:0050    05:0250    05:0260    05:0270
```

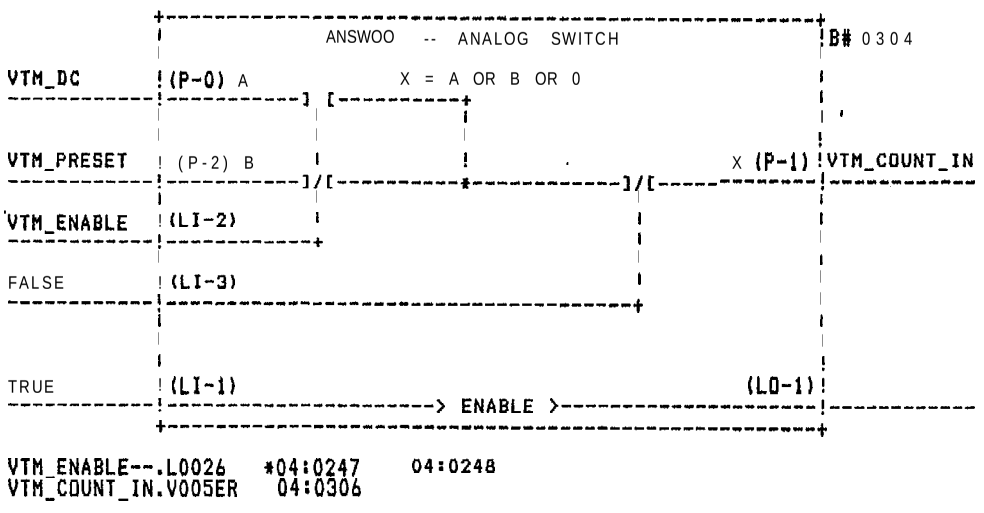
```
*****************************************************************************************************************
*                    *                *   COPYRIGHT (C) 1988, BY     *  SYSTEM:PSS                                 *
* DATE: 10/19/88     *  REQUISITION   *   GENERAL ELECTRIC COMPANY *                                             *
* AUTHOR :           *  PACIFIC       *   DRIVE SYSTEMS DEPARTMENT   ***********************************************
* J                  *  D.S.D         *   SALEM, V A     USA         * DRAWING : XXXXXX                          *
*                    *  XXXXXXX       *                              * NEXT PAGE: 0022                PAGE: 0021 *
*****************************************************************************************************************
```




```
======================  ========
SEGMENT:04/BLOCK:0280  PG:  0022
======================  ========
          LD   A-OUT-------.L0017
          A N  B_OUT-------.L0018
          L D  B_OUT-------.L0018
          A N  C_OUT-------.L0019
          ORS
          LD   C-OUT-------.L0019
          AN   A-OUT-------.L0017
          OR9
          STO  TWO-OUT-----.LOO1B
          BY   0280
```

```
======================  ========
SEGMENT:04/BLOCK:0290  PC:  0022
======================  ========
          LD   A-OUT-------.L0017
          O R  B_OUT-------.L0018
          OR   C-OUT-------.L0019
          STO      ONE-OUT-----.LOO1A
          BP   0290
```

```
======================  ========
SEGMENT:04/BLOCK:0300  PG:  0022
======================  ========
          LD   ONE-OUT-----.LOO1A
          AN   TRIP-ONE-OUT.LOO1C
          LD   TWO-OUT-----.LOO1B
          A N  TRIP_TWO_OUT.L001D
          ORS
          STO  ALL_TRIP----.L0013
          B#   0300
```

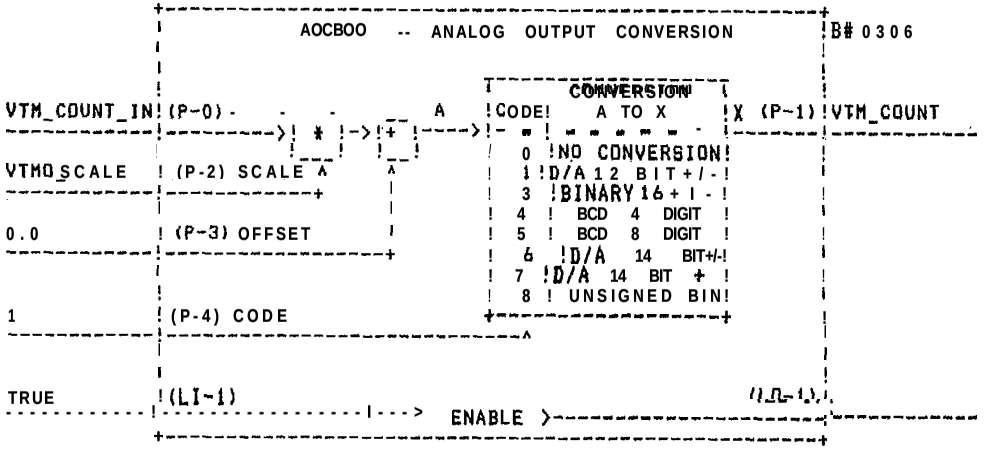


VTM_ENABLE--.L0026 *04:0247 04:0248
VTM_COUNT_IN.V005ER 04:0306

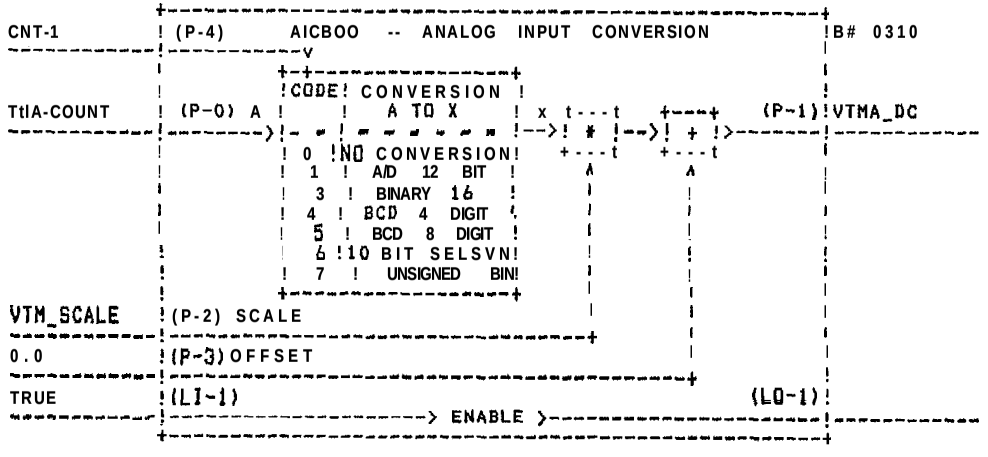
SEGMENT:04/BLOCK:0304 PC: 0022

```
LD   FALSE-------.LOOOO
L D  VTM_ENABLE--.L0026
LD   TRUE--------.LOOOI
BLK  ANSWOO
B# 0304
P-o VTM_DC------.COODAR
P-I VTM_COUNT_IN.V005ER
P-2 VTM_PRESET--.COODER
```


DATE: 10/19/88
AUTHOR : J
REQUISITION * PACIFIC * D.S.D * xxxxxx
COPYRIGHT (C)1988, BY GENERAL ELECTRIC COHPANY
DRIVE SYSTEM5 DEPARTMENT
SALEM, VA USA
SYSTEM:PSa
DRAWING : XXXXXX
NEXT PAGE:0023




```
VTM_COUNT_IN.V005ER  *04:0304
```

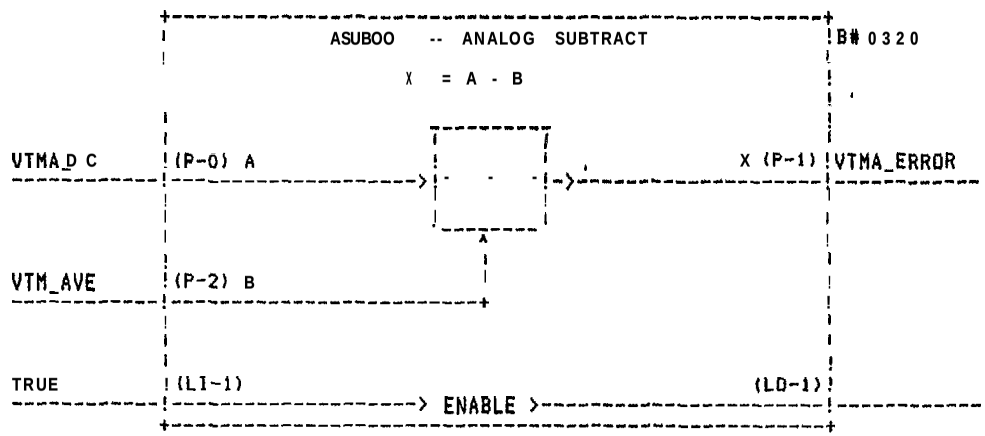



```
VTMA_DC-----.V0034R   04:0320
VTM_SCALE---.C00E2R   04:0350    04:0390
CNT_1-------.C0102C   01:0800    04:0350    04:0390    0410570    04:0580
                      0410590    04:0600    0410610    04:0620    04:0630
```

```
SEGMENT:04/BLOCK:0306 PC: 0023
        L D  TRUE--------.L0001
        BLK AOCB00
        B# 0306
        P-O VTM_COUNT_IN.V005ER
        P-i VTM_COUNT---.V0062C
        P-2   VTMO-SCALE--.C00EbR
        P-3 0.0---------.C02A4R
        P-4 1-----------.C029EC
```

```
SEGMENT:04/BLOCK:0310 PC: 0023
        LD    TRUE--------.L000I
        BLK AICBOO
        B#  0310
        P-O    TMA-COUNT---.V00ZAC
        P-I  VTMA-DC-----.V0034R
        P-2 VTM_SCALE---.C00E2R
        P-3 0.0---------.C02A4R
        P-4 CNT_1-------.C0102C
```

04:0640 04:0650 04:0660 04:0680



VTMA_DC-----.V0034R *04:0310
VTMA_ERROR--.V0040R 04:0330
VTM_AVE-----.C00D0R 04:0360 04:0400

SEGMENT:04/BLOCK:0320 PG:0023

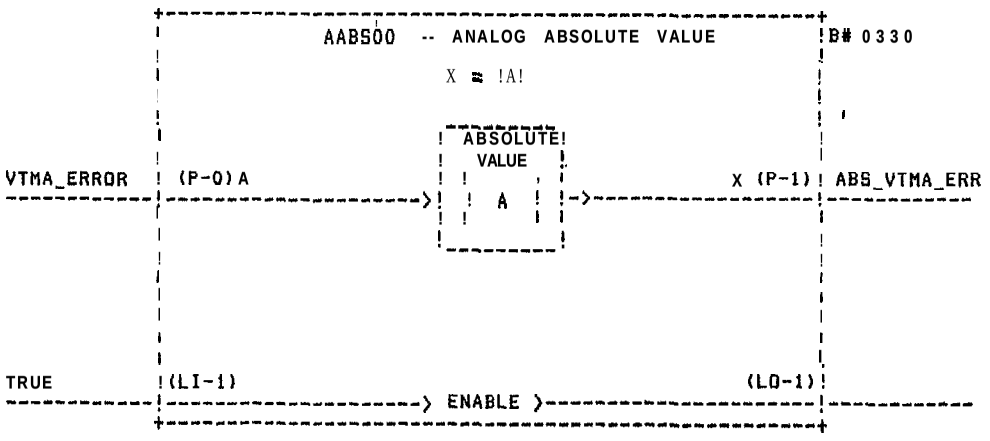
LD TRUE--------.LOOOi
BLK ASUBOO
B# 0320
P-0 VTMA_DC-----.V0034R
P-1 VTMA_ERROR--.V0040R
P-2 VTM_AVE-----.C00D0R

*DATE: 10/19/88
*AUTHOR :
* J
*

* REQUISITION *
* PACIFIC *
* D.9.D *
* xxxxxx *

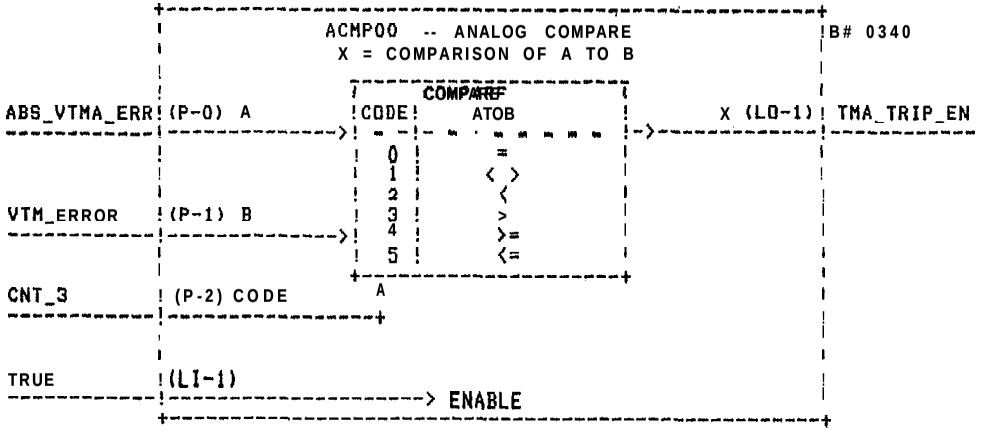
* COPYRIGHT (C) 1988, BY *
* GENERAL ELECTRIC COMPANY *
* DRIVE SYSTEMS DEPARTMENT *
* SALEM, VA USA *

* SYSTEM:PSS *
* DRAWING : XXXXXX *
* NEXT PAGE: 0024 *
PAGE: 0023



VTMA_ERROR--.V0040R *04:0320
ABS_VTMA_ERR.V004CR 04:0340

```
====================  ========
SEGMENT:04/BLOCK:0330 PC: 0024
====================  ========
         LD    TRUE--------.LOOOi
         BLK AAB500
         BY 0330
         P-O VTMA_ERROR--.V0040R
         P-I ABS_VTMA_ERR.V004CR
```

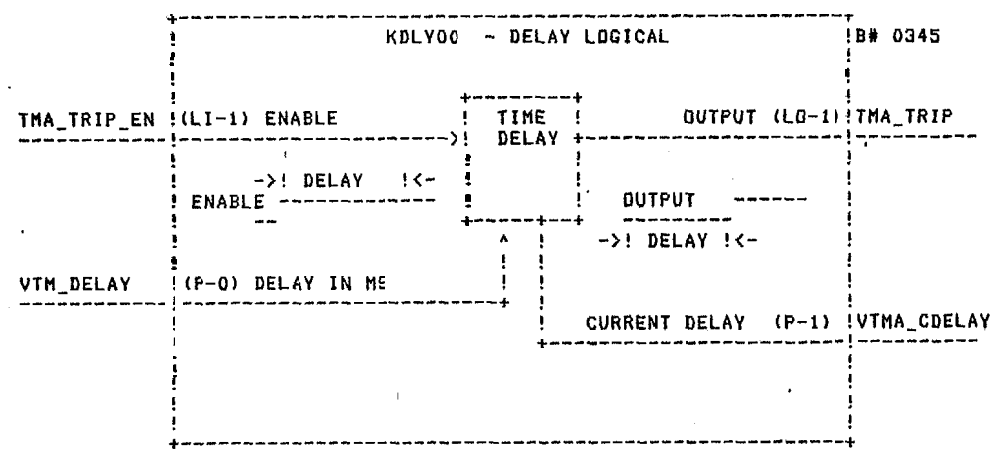


ABS_VTMA_ERR.V004CR *04:0330
VTM_ERROR---.C00D4R 0410380 04:0420
CNT_3-------.C0106C 04:0050 04:0380 04:0420
TMA_TRIP_EN-.L001E 04:0345

```
====================  ========
SEGMENT:04/BLOCK:0340 PG:0024
====================  ========
         LD    TRUE--------.LOOOi
         BLK ACMP00
         BY 0340
         P-O ABS_VTMA_ERR.V004CR
         P-I VTM_ERROR---.C00D4R
         P-Z CNT_3-------.C0106C
         STO TMA_TRIP_EN-.L001E
         BY 0340
```

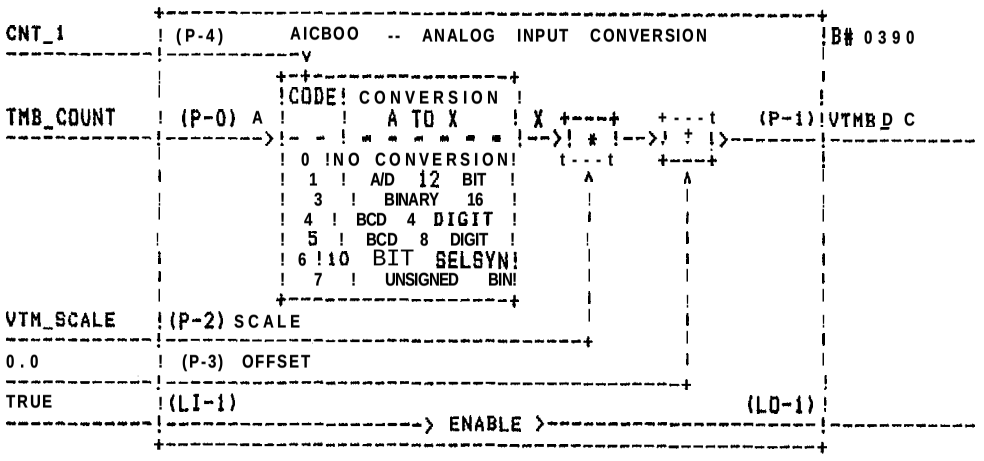
```
====================  ========
```

```
         LD  TMA_TRIP_EN-.L001E
         BLK KDLY00
         B#  0345
         P-0 VTM_DELAY---.C00D8C
         P-1 VTMA_CDELAY-.V0058C
         STO TMA_TRIP----.L0014
         B#  0345
```



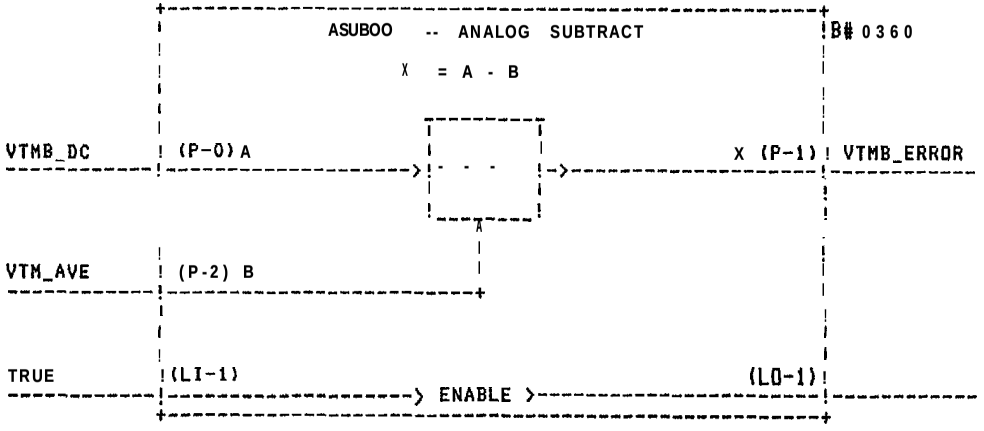

```
TMA_TRIP_EN-.L001E   *04:0340
VTM_DELAY---.C00D8C   04:0385    04:0425
TMA_TRIP----.L0014    04:0490    05:0030    05:0250
```


```
*************************************************************************************************************
*                      *                *   COPYRIGHT (C) 1988,  BY    *  SYSTEM:PSS                        *
* DATE: 10/19/88       *  REQUISITION   *   GENERAL ELECTRIC COMPANY   *                                    *
* AUTHOR :             *  PACIFIC       *   DRIVE SYSTEMS DEPARTMENT   ***************************************
* J                    *  D.S.D         *   SALEM, VA    USA           *  DRAWING : XXXXXX                  *
*                      *  XXXXXX        *                              *  NEXT PAGE: 0025      PAGE: 0024   *
*************************************************************************************************************
```




```
VTMB_DC------.V0038R   04:0360
VTM_SCALE---.C00E2R    04:0310     0410390
CNT_1-------.C0102C    01:0800     0410310     04:0390     04:0570     04:0580
                       04:0590     04:0600     04:0610     0410620     04:0630
                       04:0640     04:0650     04:0660     0410680
```

```
SEGMENT:04/BLOCK:0350 PG:0 0 2 5
        LD    TRUE--------.L0001
        BLK AICBOO
        BY  0350
        P-0 TMB_COUNT---.V002CC
        P-1 VTMB_DC-----.V0038R
        P-2 VTM_SCALE---.C00E2R
        P-3 0.0---------.C02A4R
        P-4 CNT_1-------.C0102C
```

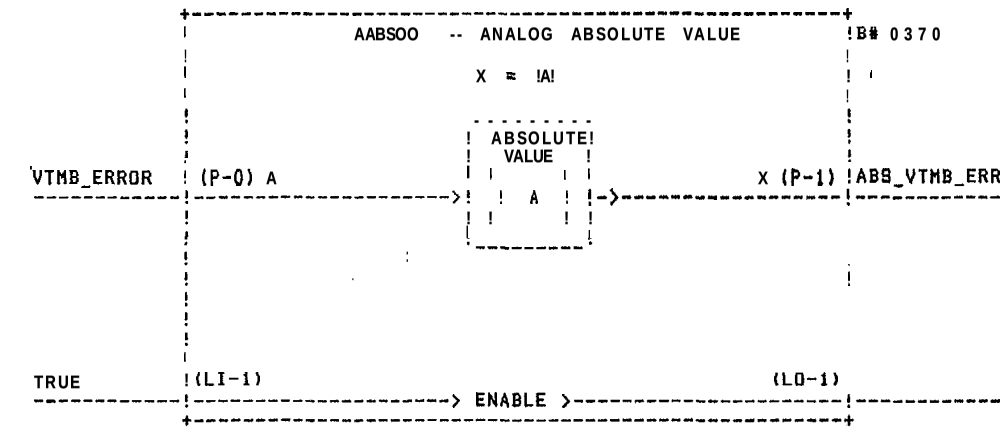



```
VTMB,DC-----.  V0038R  *04:0350
```

```
SEGMENT:04/BLOCK:0360 P C:  0 0 2 5
        LD    TRUE--------.L0001
        BLK ASUBOO
        B# 0 3 6 0
        P-0 VTMB_DC-----.V0038R
        P-1 VTMB_ERROR--.V0044R
        P-2 VTM_AVE-----.C00D0R
```

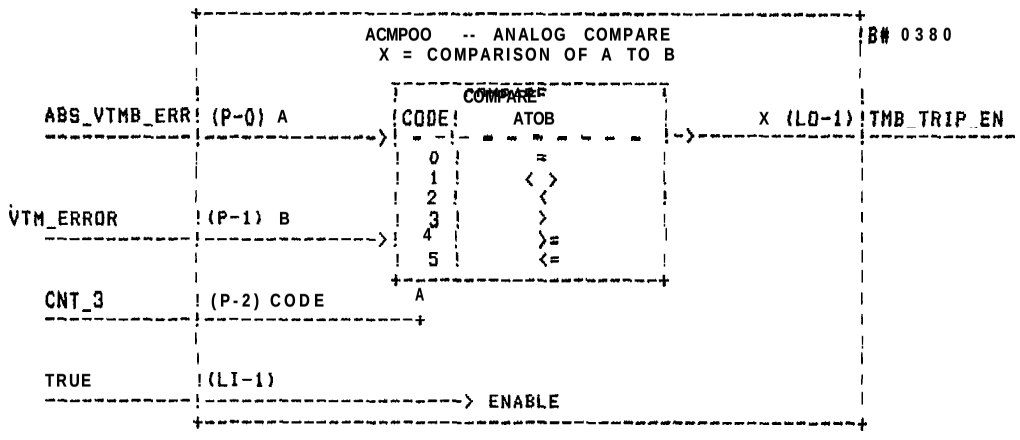
```
VTMB_ERROR--.V0044R    04:0370
VTM_AVE-----.C00D0R    04:0320     04~0400
```

```
====================  ========
SEGMENT:04/BLOCK:0370 PG:0 0 2 5
====================  ========
         LD    TRUE--------.L0001
         BLK AABS00
         B# 0 3 7 0
         P - O VTMB_ERROR--.V0044R
         P - I  ABS_VTMB_ERR.V0050R
```



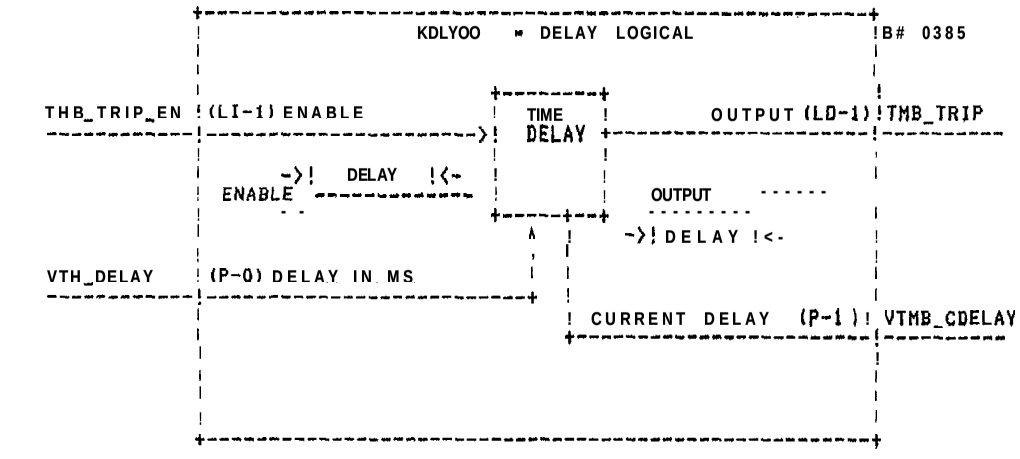

```
VTMB_ERROR--.V0044R  *04:0360
ABS_VTMB_ERR.V0050R   04:0380
```

```
*************************************************************************************************************
*                      *                *  COPYRIGHT (C) 1988, BY    *  SYSTEM:PSS                          *
* DATE: 10/19/88       * REQUISITION  *  GENERAL ELECTRIC COMPANY *                                        .
* AUTHOR :             *  PACIFIC     *  DRIVE SYSTEMS DEPARTMENT  ***************************************************
* J                    *  D.S.D       *  SALEM, V A    USA         * DRAWING : XXXXXX                                *
*                      * XXXXXX       *                            * NEXT PAGE: 0024                 PAGE: 0 0 2 5 *
*************************************************************************************************************
```



ABS_VTMB_ERR.V0050R *04:0370
VTM_ERROR---.C00D4R 04:0340 04:0420
CNT_3-------.C0106C 04:0050 04:0340 04:0420
TMB_TRIP_EN-.L001F 04:0385

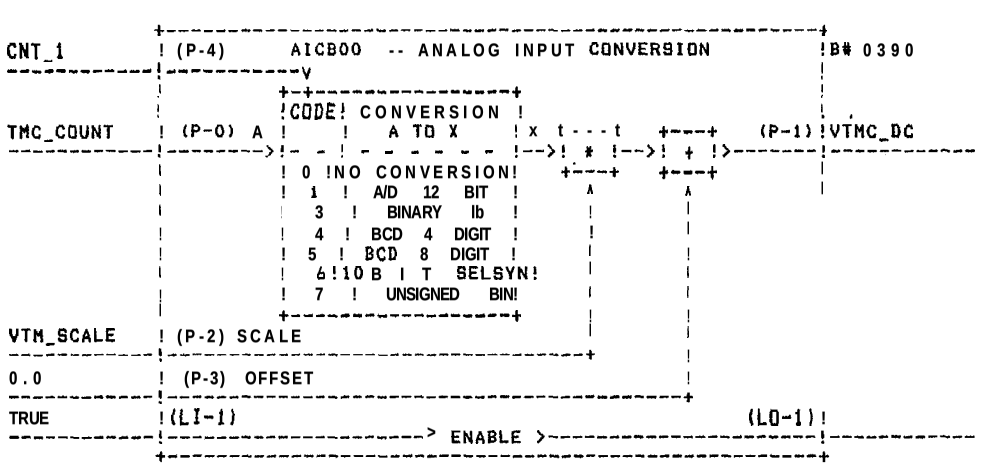
```
====================  ========
SEGMENT:04/BLOCK:0380 PG:0 0 2 6
====================  ========
        LD TRUE--------.L0001
        BLK ACMP00
        BY 0380
        P-0 ABS_VTMB_ERR.V0050R
        P-1 VTM_ERROR---.C00D4R
        P-2 CNT_3-------.C0106C
        STO TMB_TRIP_EN-.L001F
        B#  0380
```



TMB_TRIP_EN-.L001F *04:0380
VTM_DELAY---.C00D8C 04:0345 04:0425

```
====================  ========
SEGMENT:04/BLOCK:0385 PG:0 0 2 6
====================  ========
        L D  TMB_TRIP_EN-.L001F
        BLK KDLY00
        B#  0385
        P-0 VTM_DELAY---.C00D8C
        P-1 VTMB_CDELAY-.V005AC
        STO TMB_TRIP----.L0015
        BY 0385
```

TMB_TRIP----.L0015 04:0500 05:0040 05:0260



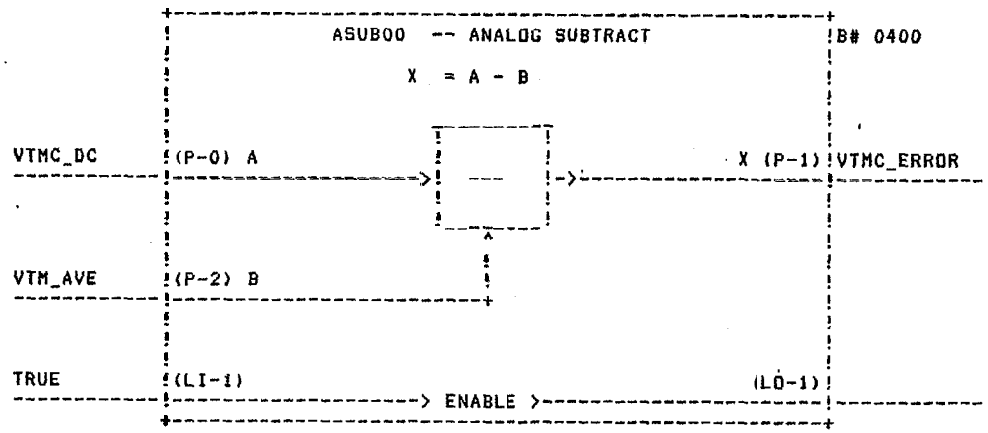
SEGMENT:04/BLOCK:0390 PC: 0026

LD TRUE--------.LOOOI
BLK AICBOO
B# 0390
P-0 TMC_COUNT---.V002EC
P-1 VTMC_DC-----.V003CR
P-2 VTM_SCALE---.C00E2R
P-3 0.0---------.C02A4R
P-4 CNT_1-------.C0102C

VTMC_DC-----.V003CR 04:0400
VTM_SCALE---.C00E2R 04:0310 04:0350
CNT_1-------.C0102C 01:0800 04:0310 04:0350 04:0570 04:0580
04:0590 04:0600 04:0610 04:0620 0410630
04:0640 04:0650 04:0660 04:0680

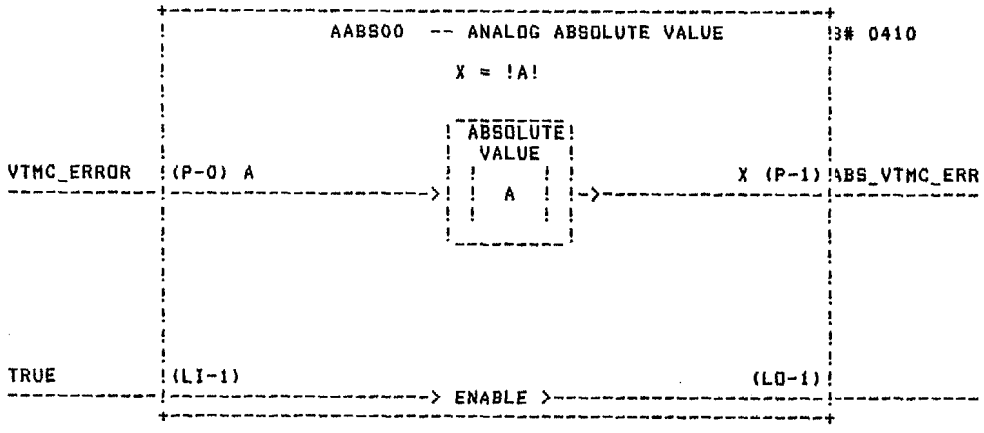
* COPYRIGHT (C) 1988, BY GENERAL ELECTRIC COMPANY
DRIVE SYSTEM9 DEPARTMENT
SALEM, VA USA

* DATE: 10/19/88
* AUTHOR : REQUISITION PACIFIC D.S.D
* SYSTEM:PSS
* DRAWING : XXXXXX
* NEXT PAGE: 0027
PAGE: 0026




```
VTMC_DC-----.V003CR  *04:0390
VTMC_ERROR--.V0048R   04:0410
VTM_AVE-----.C00D0R   04:0320    04:0360
```

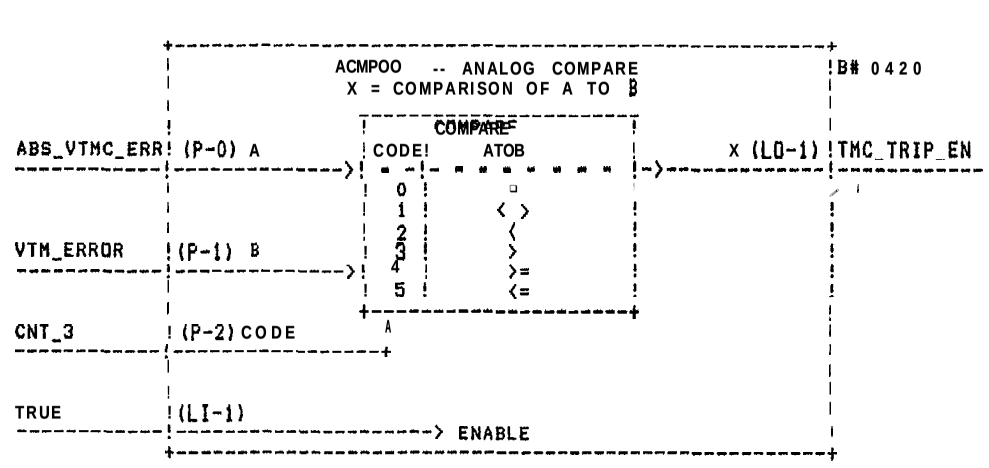
```
====================  ========
SEGMENT:04/BLOCK:0400  PG: 0027
====================  ========
        LD  TRUE--------.L0001
        BLK ASUB00
        B#  0400
        P-0 VTMC_DC-----.V003CR
        P-1 VTMC_ERROR--.V0048R
        P-2 VTM_AVE-----.C00D0R
```




```
VTMC_ERROR--.V0048R  *04:0400
ABS_VTMC_ERR.V0054R   04:0420
```

```
====================  ========
SEGMENT:04/BLOCK:0410  PG: 0027
====================  ========
        LD  TRUE--------.L0001
        BLK AABS00
        B#  0410
        P-0 VTMC_ERROR--.V0048R
        P-1 ABS_VTMC_ERR.V0054R
```

```
====================  ========
SEGMENT:04/BLOCK:0420  PG: 0027
```

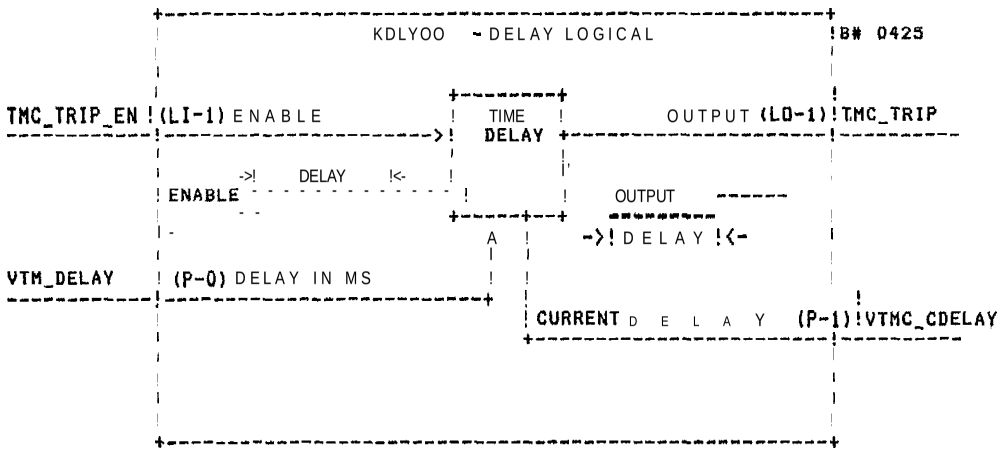



```
LD    TRUE--------.L0001
BLK  ACMPOO
BY 0420
P-0 ABS_VTMC_ERR.V0054R
P-1 VTM_ERROR---.C00D4R
P-2 CNT_3--------.C0106C
STO TMC_TRIP_EN-.L0020
B# 0420
```

```
ABS_VTMC_ERR.V0054R   *04:0410
VTM_ERROR---.C00D4R    04:0340 0 4 1 0 3 8 0
CNT_3--------.C0106C   04:0050     04:0340     04:0380
TMC_TRIP_EN-.L0020     04:0425
```


COPYRIGHT (C) 1988, BY
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPARTMENT
SALEM, VA USA

DATE: 10/19/88	REQUISITION	SYSTEM:PSS	
AUTHOR: J	PACIFIC	DRAWING: XXXXXX	
	D.S.D	NEXT PAGE: 0028	PAGE: 0027
	XXXXXX		




```
TMC_TRIP_EN-.L0020    *04:0420
VTM_DELAY---.C00D8C    04:0345     04:0385
TMC_TRIP----.L0016     04:0510     05:0050     05:0270
```

```
 PA-ENAB                                                                GP_PSSP
 LE                                                                     A
+--] [--+-------+-------+-------+-------+-------+-------+-------+-------+--(  )-

PA_ENABLE---.L0012    *04:0215    04:0245    04:0247    04:0440    04:0450
                       04:0460    04:0470    0410480    04:0490    04:0500
                       04:0510    05:0220    05:0230    05:0240    05:0250
                       05:0260    05:0270
```

```
 PA-ENAB                                                                GP,PSS
 LE
+--]/[--+-------+-------+-------+-------+-------+-------+-------+-------+--(  )-

PA_ENABLE---.L0012    *04:0215    04:0245    0410247    04:0430    04:0450
                       04:0460    04:0470    04:0480    0410490    04:0500
                       04:0510    05:0220    05:0230    0580240    0510250
                       05:0260    05:0270
```

```
 PSS_ON LPA_TES PA-ENAB                                                 PA-TEST
        T       LE
+--]/[--+--] [--+--] [--+-------+-------+-------+-------+-------+-------+--(  )-
```

```
====================  ========
SEGMENT:04/BLOCK:0425 PG:0028
====================  ========
        L D  TMC_TRIP_EN-.L0020
        BLK KDLYOO
        B# 0425
        P-O VTM_DELAY---.C00D8C
        P-I VTMC_CDELAY-.V005CC
        STO TMC_TRIP----.L0016
        B# 0425
```

```
====================  ========
SEGMENT:04/BLOCK:0430 PG:0028
====================  ========
        L D  PA_ENABLE---.L0012
        STO GP_PSSPA----.D10014
        B# 0430
```

```
====================  ========
SEGMENT:04/BLOCK:0440 PG:0028
====================  ========
        LDF PA_ENABLE---.L0012
        STO GP_PSS------.D10015
        B# 0440
```

```
====================  ========
SEGMENT:04/BLOCK:0450 PG:0028
====================  ========
        LDF PSS_ON------.D10010
        A N  LPA_TEST----.L0021
        A N  PA_ENABLE---.L0012
        STO PA_TEST-----.D10016
```

```
                                                                                                  BY  0450

PSS_ON------.010010   01:0200    01:0300    01:0400    01:0600    01:0700
                      03:0200   *04:0130    04:0210    04:0245    0510220
                      05:0230    05:0240
PA_ENABLE---.L0012   *04:0215    04:0245    0410247    04:0430    04:0440
                      04:0460    04:0470    0410480    04:0490    0410500
                      04:0510    05:0220    05:0230    05:0240    05:0250
                      05:0260    05:0270
PA-TEST-----.010016   04:0247    04:0455


                                                                  ======================  ========
                                                                  SEGMENT:04/BLOCK:0455 PC: 0028
                                                                  ======================  ========
                                                                          LDF PA_TEST-----.010016
                                                                          STD VTM_AC------.010017
                                                                          B Y  0455

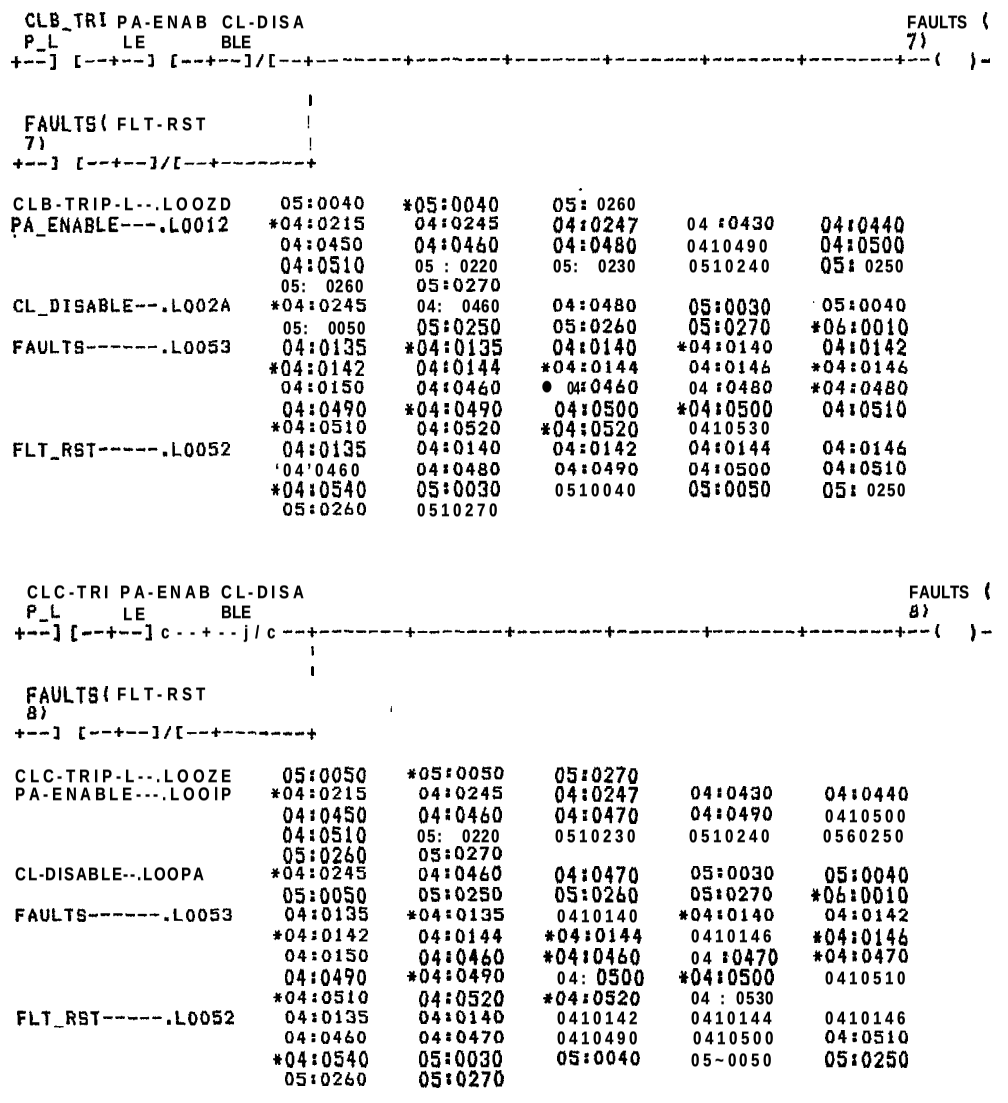
. PA-TEST                                                                                  VTM_AC
+--]/[--+-------+-------+-------+-------+-------+-------+-------+-------+--(  )-

PA-TEST-----.010016   0410247   *04:0450


                                                                  ======================  ========
                                                                  SEGMENT:04/BLOCK:0460 PG:0028
                                                                  ======================  ========
                                                                          LD  CLA_TRIP_L--.L002C
 CLA_TRI PA_ENAB CL_DISA                                                  A N  PA_ENABLE---.L0012
 P_L     LE      BLE                                                      ANF CL_DISABLE--.L002A
+--] [--+--] [--+--]/[--+-------+-------+-------+-------+-------+-------+--( )-  L D  FAULTS(6)---.L0058
                                                                          ANF FLT_RST-----.L0052
                                                                          ORS
 FAULTS ( FLT_RST                                                         STD FAULTS------.L0053
 6)                                                                       B# 0460
+--] [--+--]/[--+-------+

CLA_TRIP_L--.L002C   05:0030   *05:0030    05:0250
PA_ENABLE---.L0012  *04:0215    04:0245    04:0247    04:0430    04:0440
                     04:0450    04:0470    0410480    04:0490    04:0500
                     04:0510    05:0220    05:0230    05:0240    0510250
                     05:0260    05:0270
CL_DISABLE--.L002A  *04:0245    04:0470    04:0480    05:0030    05:0040
                     05:0050    0510250    05:0260    05:0270   *06:0010
FAULTS------.10053   04:0135   *04:0135    04:0140   *04:0140    04:0142
                    *04:0142    04:0144   *04:0144    0480146   *04:0146
                     04:0150    04:0470   *04:0470    04:0480   *04:0480
                     04:0490   *04:0490    0410500   *04:0500    04:0510
                    *04:0510    04:0520   *04:0520    04:0530
FLT_RST-----.L0052   04:0135    04:0140    04:0142    04:0144    04:0146
                     04:0470    04:0480    04:0490    04:0500    04:0510
                    *04:0540    05:0030    05:0040    05:0050    05:0250
                     05:0260    05:0270
```

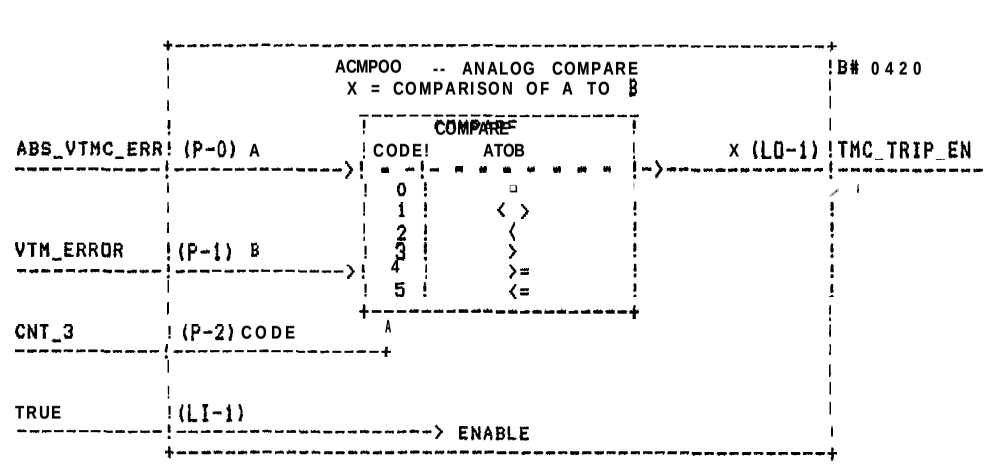
```
*************************************************************************************************************
*                      *                *  COPYRIGHT (C)1988, BY    *  SYSTEM:PSS                             *
* DATE: 10/19/88       * REQUISITION  *  GENERAL ELECTRIC COMPANY *                                         *
*  AUTHOR :            * PACIFIC      *  DRIVE SYSTEMS DEPARTMENT  ***************************************************
* J                    *  D.S.D       *  SALEM, VA    USA          * DRAWING : XXXXXX                                    *
*                      *xxxxxx        *                           * NEXT PACE: 0029                  PAGE: 0028 *
*************************************************************************************************************
```




```
SEGMENT:04/BLOCK:0470 PG:0029
        LD  CLB_TRIP_L--.L002D
        AN  PA-ENABLE---.LOOI?
        ANF CL_DISABLE--.L002A
        LD  FAULTS(7)---.L0059
        ANF FLT_RST-----.L0052
        ORS
        STO FAULTS------.L0053
        BY 0470
```

```
SEGMENT:04/BLOCK:0480 PG:0029
        LD  CLC-TRIP-L--.LOOZE
        AN  PA-ENABLE---.LOOIP
        ANF CL_DISABLE--.L002A
        LD  FAULTS(8)---.L005A
        ANF FLT_RST-----.L0052
        ORS
        STO FAULTS------.LOOSB
        BY 0480
```

```
SEGMENT:04/BLOCK:0490 PG: 0029
```

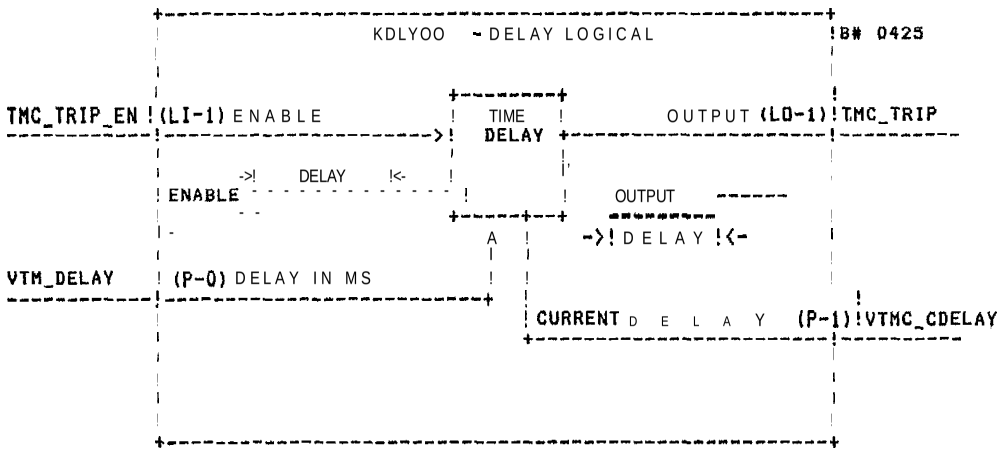



```
LD    TRUE--------.L0001
BLK  ACMPOO
BY 0420
P-0 ABS_VTMC_ERR.V0054R
P-1 VTM_ERROR---.C00D4R
P-2 CNT_3--------.C0106C
STO TMC_TRIP_EN-.L0020
B# 0420
```

```
ABS_VTMC_ERR.V0054R   *04:0410
VTM_ERROR---.C00D4R    04:0340 0 4 1 0 3 8 0
CNT_3--------.C0106C   04:0050     04:0340     04:0380
TMC_TRIP_EN-.L0020     04:0425
```


COPYRIGHT (C) 1988, BY
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPARTMENT
SALEM, VA USA

DATE: 10/19/88	REQUISITION	SYSTEM:PSS	
AUTHOR: J	PACIFIC	DRAWING: XXXXXX	
	D.S.D	NEXT PAGE: 0028	PAGE: 0027
	XXXXXX		




```
TMC_TRIP_EN-.L0020    *04:0420
VTM_DELAY---.C00D8C    04:0345     04:0385
TMC_TRIP----.L0016     04:0510     05:0050     05:0270
```

```
 PA-ENAB                                                                GP_PSSP
 LE                                                                     A
+--] [--+-------+-------+-------+-------+-------+-------+-------+-------+--(  )-

PA_ENABLE---.L0012    *04:0215    04:0245    04:0247    04:0440    04:0450
                       04:0460    04:0470    0410480    04:0490    04:0500
                       04:0510    05:0220    05:0230    05:0240    05:0250
                       05:0260    05:0270
```

```
 PA-ENAB                                                                GP,PSS
 LE
+--]/[--+-------+-------+-------+-------+-------+-------+-------+-------+--(  )-

PA_ENABLE---.L0012    *04:0215    04:0245    0410247    04:0430    04:0450
                       04:0460    04:0470    04:0480    0410490    04:0500
                       04:0510    05:0220    05:0230    0580240    0510250
                       05:0260    05:0270
```

```
 PSS_ON LPA_TES PA-ENAB                                                 PA-TEST
        T       LE
+--]/[--+--] [--+--] [--+-------+-------+-------+-------+-------+-------+--(  )-
```

```
====================  ========
SEGMENT:04/BLOCK:0425 PG:0028
====================  ========
        L D  TMC_TRIP_EN-.L0020
        BLK KDLYOO
        B# 0425
        P-O VTM_DELAY---.C00D8C
        P-I VTMC_CDELAY-.V005CC
        STO TMC_TRIP----.L0016
        B# 0425
```

```
====================  ========
SEGMENT:04/BLOCK:0430 PG:0028
====================  ========
        L D  PA_ENABLE---.L0012
        STO GP_PSSPA----.D10014
        B# 0430
```

```
====================  ========
SEGMENT:04/BLOCK:0440 PG:0028
====================  ========
        LDF PA_ENABLE---.L0012
        STO GP_PSS------.D10015
        B# 0440
```

```
====================  ========
SEGMENT:04/BLOCK:0450 PG:0028
====================  ========
        LDF PSS_ON------.D10010
        A N  LPA_TEST----.L0021
        A N  PA_ENABLE---.L0012
        STO PA_TEST-----.D10016
```

```
                                                                                                  BY  0450

PSS_ON------.010010   01:0200    01:0300    01:0400    01:0600    01:0700
                      03:0200   *04:0130    04:0210    04:0245    0510220
                      05:0230    05:0240
PA_ENABLE---.L0012   *04:0215    04:0245    0410247    04:0430    04:0440
                      04:0460    04:0470    0410480    04:0490    0410500
                      04:0510    05:0220    05:0230    05:0240    05:0250
                      05:0260    05:0270
PA-TEST-----.010016   04:0247    04:0455


                                                                  ======================  ========
                                                                  SEGMENT:04/BLOCK:0455 PC: 0028
                                                                  ======================  ========
                                                                          LDF PA_TEST-----.010016
                                                                          STD VTM_AC------.010017
                                                                          B Y  0455

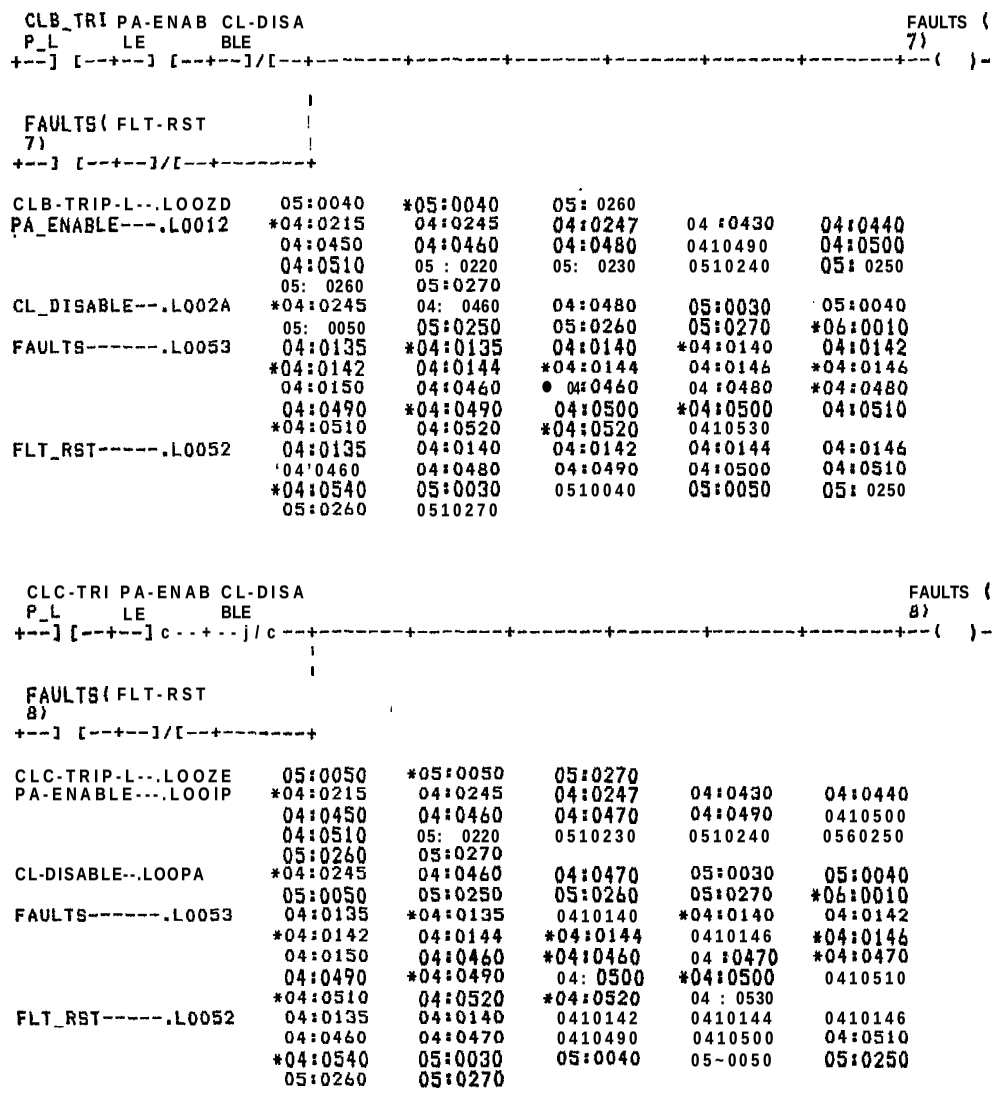
. PA-TEST                                                                                  VTM_AC
+--]/[--+-------+-------+-------+-------+-------+-------+-------+-------+--(  )-

PA-TEST-----.010016   0410247   *04:0450


                                                                  ======================  ========
                                                                  SEGMENT:04/BLOCK:0460 PG:0028
                                                                  ======================  ========
                                                                          LD  CLA_TRIP_L--.L002C
 CLA_TRI PA_ENAB CL_DISA                                                  A N  PA_ENABLE---.L0012
 P_L     LE      BLE                                                      ANF CL_DISABLE--.L002A
+--] [--+--] [--+--]/[--+-------+-------+-------+-------+-------+-------+--( )-  L D  FAULTS(6)---.L0058
                                                                          ANF FLT_RST-----.L0052
                                                                          ORS
 FAULTS ( FLT_RST                                                         STD FAULTS------.L0053
 6)                                                                       B# 0460
+--] [--+--]/[--+-------+

CLA_TRIP_L--.L002C   05:0030   *05:0030    05:0250
PA_ENABLE---.L0012  *04:0215    04:0245    04:0247    04:0430    04:0440
                     04:0450    04:0470    0410480    04:0490    04:0500
                     04:0510    05:0220    05:0230    05:0240    0510250
                     05:0260    05:0270
CL_DISABLE--.L002A  *04:0245    04:0470    04:0480    05:0030    05:0040
                     05:0050    0510250    05:0260    05:0270   *06:0010
FAULTS------.10053   04:0135   *04:0135    04:0140   *04:0140    04:0142
                    *04:0142    04:0144   *04:0144    0480146   *04:0146
                     04:0150    04:0470   *04:0470    04:0480   *04:0480
                     04:0490   *04:0490    0410500   *04:0500    04:0510
                    *04:0510    04:0520   *04:0520    04:0530
FLT_RST-----.L0052   04:0135    04:0140    04:0142    04:0144    04:0146
                     04:0470    04:0480    04:0490    04:0500    04:0510
                    *04:0540    05:0030    05:0040    05:0050    05:0250
                     05:0260    05:0270
```

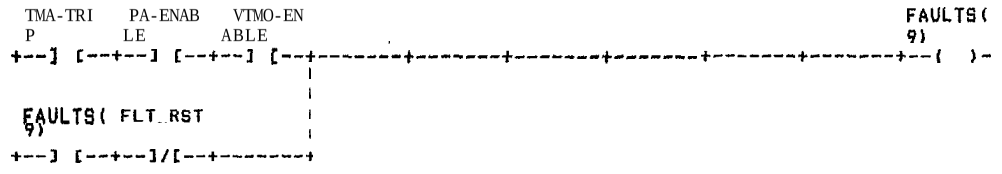
```
*************************************************************************************************************
*                      *                *  COPYRIGHT (C)1988, BY    *  SYSTEM:PSS                             *
* DATE: 10/19/88       * REQUISITION  *  GENERAL ELECTRIC COMPANY *                                         *
*  AUTHOR :            * PACIFIC      *  DRIVE SYSTEMS DEPARTMENT  ***************************************************
* J                    *  D.S.D       *  SALEM, VA    USA          * DRAWING : XXXXXX                                    *
*                      *xxxxxx        *                           * NEXT PACE: 0029                  PAGE: 0028 *
*************************************************************************************************************
```




```
SEGMENT:04/BLOCK:0470 PG:0029
        LD  CLB_TRIP_L--.L002D
        AN  PA-ENABLE---.LOOI?
        ANF CL_DISABLE--.L002A
        LD  FAULTS(7)---.L0059
        ANF FLT_RST-----.L0052
        ORS
        STO FAULTS------.L0053
        BY 0470
```

```
SEGMENT:04/BLOCK:0480 PG:0029
        LD  CLC-TRIP-L--.LOOZE
        AN  PA-ENABLE---.LOOIP
        ANF CL_DISABLE--.L002A
        LD  FAULTS(8)---.L005A
        ANF FLT_RST-----.L0052
        ORS
        STO FAULTS------.LOOSB
        BY 0480
```

```
SEGMENT:04/BLOCK:0490 PG: 0029
```

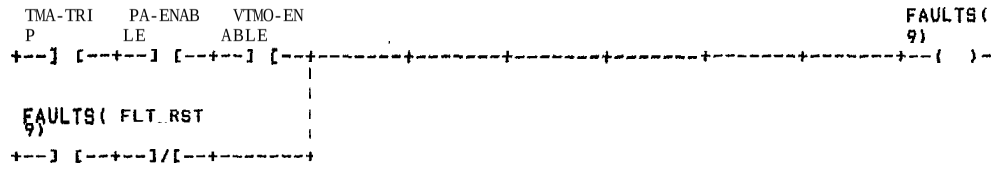



```
LD   TMA_TRIP----.L0014
AN   PA-ENABLE---.LOOI?
A N   VTMO_ENABLE-.L0027
L D  FAULTS(9)---.L005B
ANF  FLT_RST-----.L0052
ORB
STO  FAULTS------.L0053
B#   0490
```

TMA_TRIP----.L0014	*04:0345	05:0030	05:0250			
PA-ENABLE---.LOO1Z	● 04r0215	04:0245	04:0247	04:0430	04:0440	
	04:0450	04:0460	04:0470	0410480	04:0500	
	04:0510	05:0220	05:0230	05:0240	05:0250	
	05:0260	05:0270				
VTMO_ENABLE-.L0027	*04:0248	04:0500	04:0510	05:0030	05:0040	
	05:0050	05:0250	0510260	05:0270		
FAULTS------.L0053	04:0135	*04:0135	0410140	*04:0140	04:0142	
	*04:0142	04:0144	*04:0144	04:0146	*04:0146	
	04:0150	04:0460	*04:0460	0410470	*04:0470	
	04:0480	*04:0480	04:0500	*04:0500	04:0510	
	*04:0510	04:0520	*04:0520	04:0530		
FLT_RST-----.L0052	04:0135	0410140	04:0142	04:0144.	04:0146	
	0410460	0410470	04:0480	04:0500	04:0510	
	*04:0540	05:0030	05:0040	05:0050	0510250	
	05:0260	05:0270				

COPYRIGHT (C) 1988, BY GENERAL ELECTRIC COMPANY
REQUISITION PACIFIC D.S.D
DRIVE SYSTEMS DEPARTMENT
SALEM, VA USA

DATE: 10/19/88
AUTHOR :
SYSTEM:PSS
DRAWING : XXXXXX
NEXT PAGE: 0030

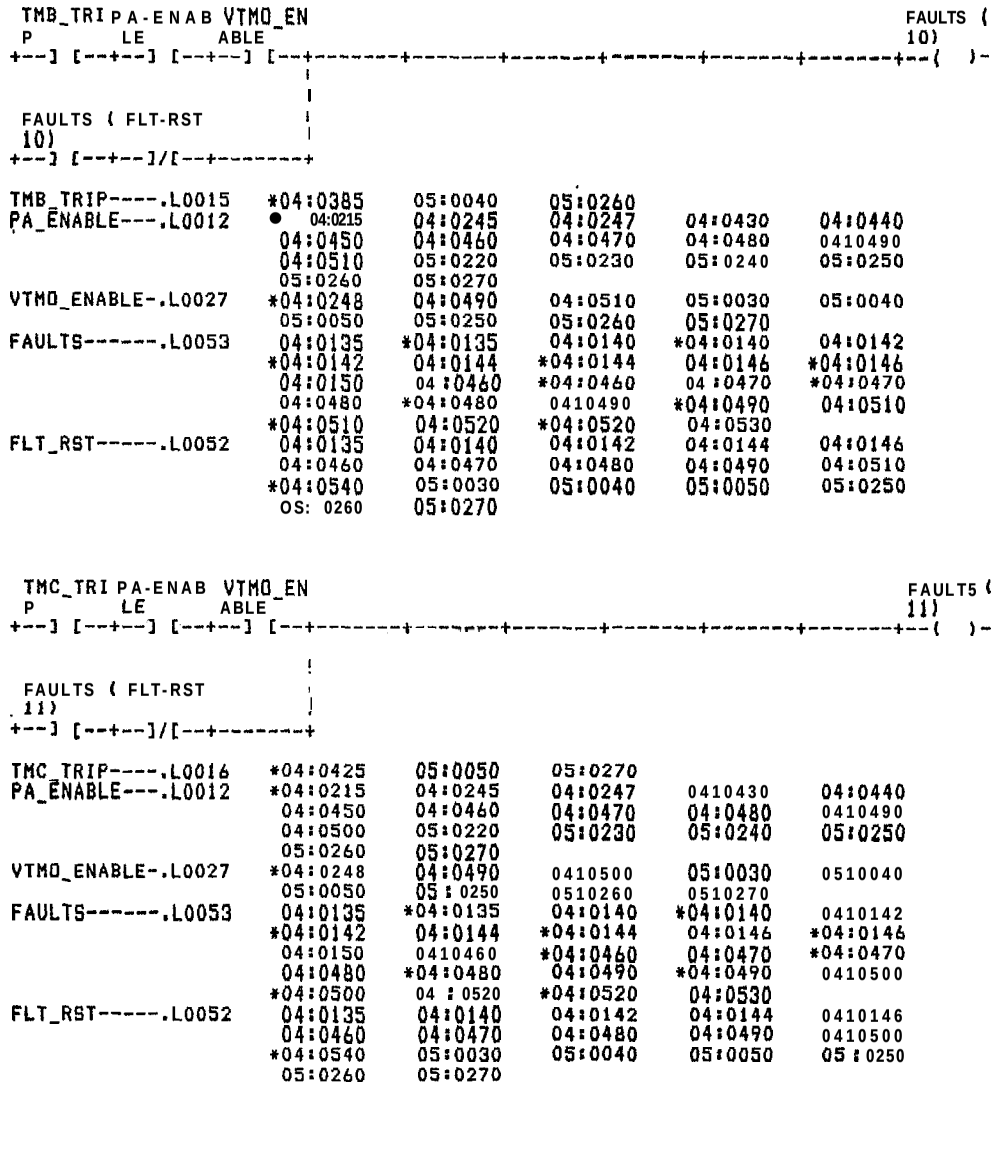



```
LD   TMA_TRIP----.L0014
AN   PA-ENABLE---.LOOI?
A N   VTMO_ENABLE-.L0027
L D  FAULTS(9)---.L005B
ANF  FLT_RST-----.L0052
ORB
STO  FAULTS------.L0053
B#   0490
```

TMA_TRIP----.L0014	*04:0345	05:0030	05:0250			
PA-ENABLE---.LOO1Z	● 04r0215	04:0245	04:0247	04:0430	04:0440	
	04:0450	04:0460	04:0470	0410480	04:0500	
	04:0510	05:0220	05:0230	05:0240	05:0250	
	05:0260	05:0270				
VTMO_ENABLE-.L0027	*04:0248	04:0500	04:0510	05:0030	05:0040	
	05:0050	05:0250	0510260	05:0270		
FAULTS------.L0053	04:0135	*04:0135	0410140	*04:0140	04:0142	
	*04:0142	04:0144	*04:0144	04:0146	*04:0146	
	04:0150	04:0460	*04:0460	0410470	*04:0470	
	04:0480	*04:0480	04:0500	*04:0500	04:0510	
	*04:0510	04:0520	*04:0520	04:0530		
FLT_RST-----.L0052	04:0135	0410140	04:0142	04:0144.	04:0146	
	0410460	0410470	04:0480	04:0500	04:0510	
	*04:0540	05:0030	05:0040	05:0050	0510250	
	05:0260	05:0270				

COPYRIGHT (C) 1988, BY GENERAL ELECTRIC COMPANY
REQUISITION PACIFIC D.S.D
DRIVE SYSTEMS DEPARTMENT
SALEM, VA USA

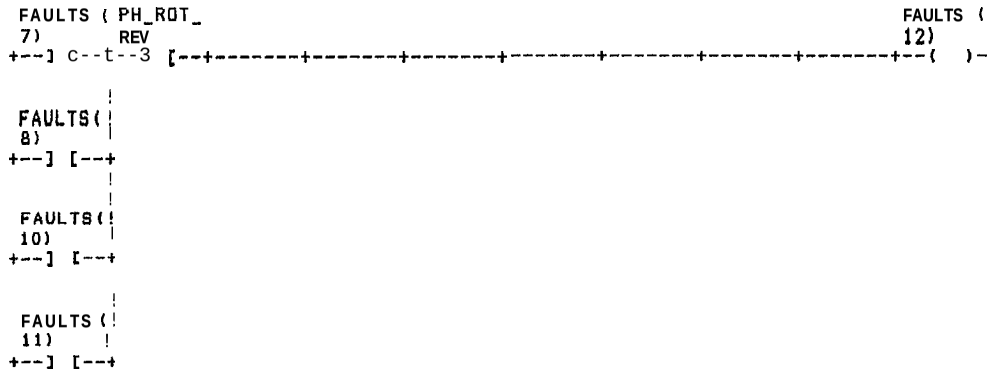
DATE: 10/19/88
AUTHOR :
SYSTEM:PSS
DRAWING : XXXXXX
NEXT PAGE: 0030




```
SEGMENT:04/BLOCK:0500 PG:0030
          L D  TMB_TRIP----.L0015
          A N  PA_ENABLE---.L0012
          A N  VTMO_ENABLE-.L0027
          L D  FAULTS(10)--.L005C
          ANF FLT_RST-----.L0052
          ORS
          STO FAULTS------.L0053
          B# 0500
```

```
SEGMENT:04/BLOCK:0510 PG:0030
          L D  TMC_TRIP----.L0016
          A N  PA_ENABLE---.L0012
          A N  VTMO_ENABLE-.L0027
          L D  FAULTS(11)--.L005D
          ANF FLT_RST-----.L0052
          ORS
          STO FAULTS------.L0053
          BY 0510
```

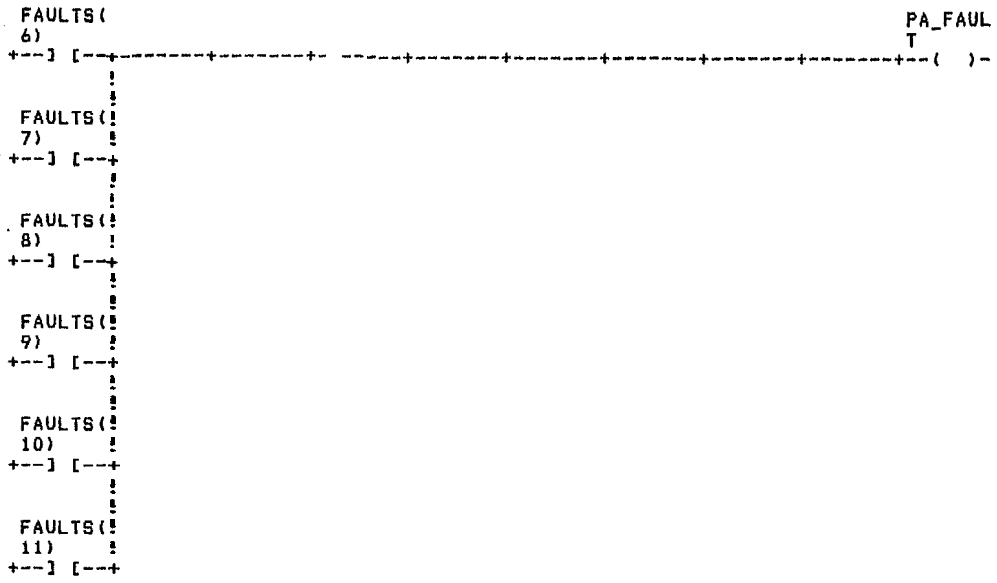
```
SEGMENT:04/BLOCK:0520 PC: 0030
```




```
L  D   FAULTS(7)---.L0059
LD   FAULTS(B)---.LOOSA
ORS
L  D   FAULTS(10)--.L005C
OR8
L  D    FAULTS(11)--.L005D
ORS
A  N   PH_ROT_REV--.L0023
STO  FAULTS------.L0053
B# 0 5 2 0
```

```
FAULTS------.L0053     04:0135   *04:0135    04:0140   *04:0140    04:0142
                      *04:0142    04:0144   *04:0144    04:0146   *04:0146
                       04:0150    04:0460   *04:0460    04:0470   *04:0470
                       04:0480   *04:0480    0410490   *04:0490    0410500
                      •04:0500    04:0510   *04:0510    04:0530
PH_ROT_REV--.L0023    *04:0204
```

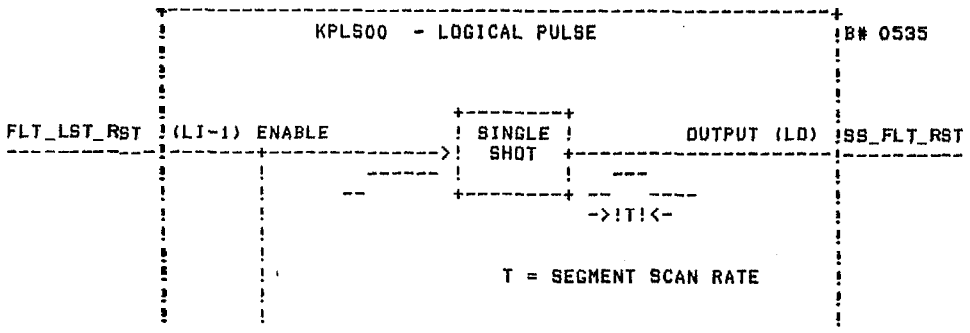
```
*************************************************************************************************************
*                      *                  *  COPYRIGHT (C)1988, BY    *  SYSTEM:PSS                          *
* DATE: 10/19/88       * REQUISITION  *  GENERAL ELECTRIC COMPANY *                                      *
* AUTHOR :             * PACIFIC      *  DRIVE SYSTEMS DEPARTMENT   ************************************************
* J                    *  D.S.D       *  SALEM. VA     USA          * DRAWING : XXXXXX                          *
*                      * XXXXXX       *                             it NEXT PAGE: 0031                PAGE: 0 0 3 0 *
*************************************************************************************************************
```




```
FAULTS------.L0053    04:0135   *04:0135    04:0140   *04:0140    04:0142
                     *04:0142    04:0144   *04:0144    04:0146   *04:0146
                      04:0150   *04:0460   *04:0460    04:0470   *04:0470
                      04:0480   *04:0480    04:0490   *04:0490    04:0500
                     *04:0500    04:0510   *04:0510    04:0520   *04:0520

PA_FAULT----.L0010    04:0150
```

```
======================  ========
SEGMENT:04/BLOCK:0530  PG: 0031
======================  ========
         LD   FAULTS(6)---.L0058
         LD   FAULTS(7)---.L0059
         ORS
         LD   FAULTS(8)---.L005A
         ORS
         LD   FAULTS(9)---.L005B
         ORS
         LD   FAULTS(10)--.L005C
         ORS
         LD   FAULTS(11)--.L005D
         ORS
         STO PA_FAULT----.L0010
         B#   0530
```



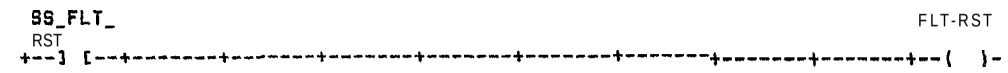

```
======================  ========
SEGMENT:04/BLOCK:0535  PG: 0031
======================  ========
         LD   FLT_LST_RST-.I10003
         BLK KPLS00
         B#   0535
         P-0 TMP_FLT_RST-.L0031
         STO SS_FLT_RST--.L0030
         B#   0535
```



SS_FLT_RST--.L0030 04:0540

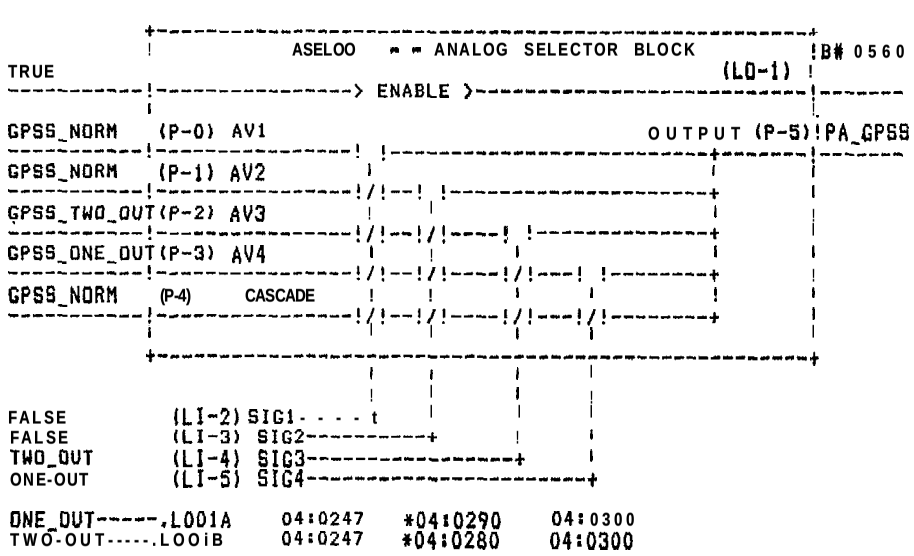
SEGMENT:04/BLOCK:0540 PG:0031

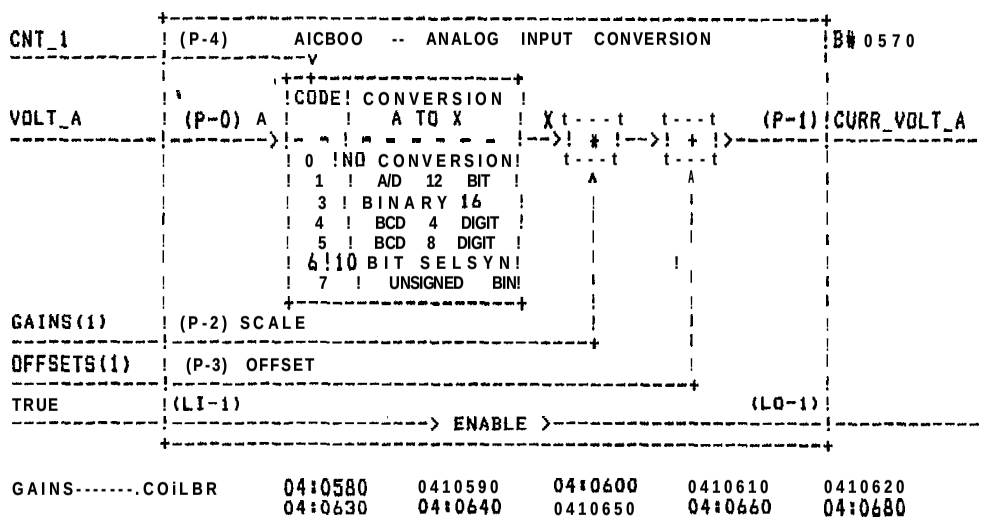
L D SS_FLT_RST--.L0030
STO FLT-RST-----.LOOSP
BY 0540



SS_FLT_RST--.L0030	*04:0535				
FLT_RST-----.L0052	04:0135	04:0140	04:0142	04:0144	0410146
	04:0460	04:0470	04:0480	0410490	04:0500
	04:0510	05:0030	05:0040	05:0050	0580250
	05:0260	0580270			

DATE: 10/19/88	REQUISITION	COPYRIGHT (C) 1988, BY	SYSTEM:PSS	
AUTHOR :	PACIFIC	GENERAL ELECTRIC COMPANY		
J	D.S.D	DRIVE SYSTEMS DEPARTMENT		
	xxxxxx	SALEM, V A USA	DRAWING : XXXXXX	
			NEXT PAGE: 0032	PAGE: 0031



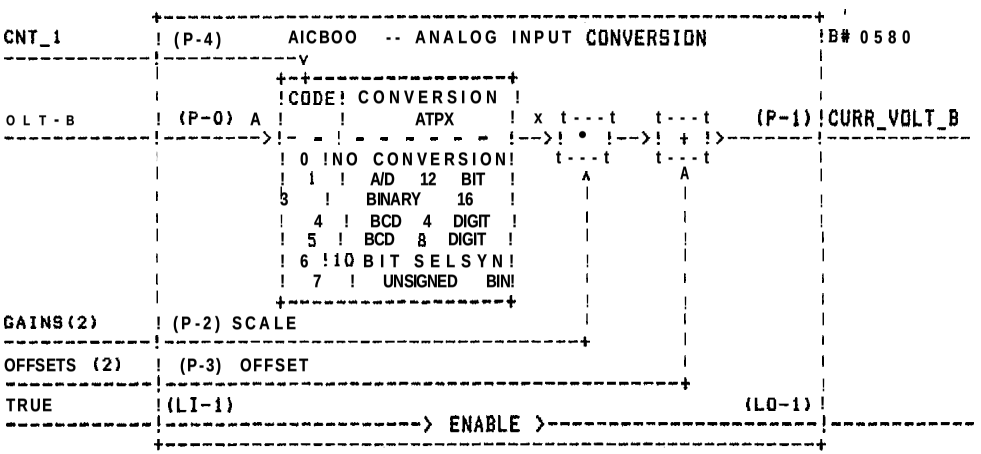



```
====================  ========
SEGMENT:04/BLOCK:0560 PG:0 0 3 2
====================  ========
        LD   ONE-OUT-----.LOOIA
        LD   TWO-OUT-----.LOOiB
        LD   FALSE-------.LOOOO
        LD   FALSE-------.LOOOO
        LD TRUE---------.LOOOI
        BLK  ABEL00
        BY 0560
        P-O GPSS_NORM---.COOEAR
        P-I GPSS_NORM---.COOEAR
        P-2 GPSS_TWO_OUT.COOF2R
        P-3 GPSS_ONE_OUT.COOEER
        P-4 GPSS_NORM---.COOEAR
        P-S PA_GPSS-----.V0064R
```

```
====================  ========
SEGMENT:04/BLOCK:0570 PC: 0032
====================  ========
        LD    TRUE--------.LOOOI
        BLK AICBOO
        BY 0570
        P-O  VOLT-A-------.VOOiOC
        P-I  CURR_VOLT_A-.V0070R
        P-2  GAINS(1)----.C0168R
        P-3  OFFSETS(1)--.C0128R
        P-4  CNT_1-------.C0102C
```

```
OFFSETS-----.C0128R   04:0580   04:0590   04:0600   04:0610   04:0620
                      04:0630   04:0640   04:0650   0410660   04:0680
CNT_1-------.C0102C   01:0800   04:0310   04:0350   04:0390   04 :0580
                      04:0590   04:0600   04:0610   04:0620   04:0630
                      04:0640   04:0650   04:0660   04:0680
```

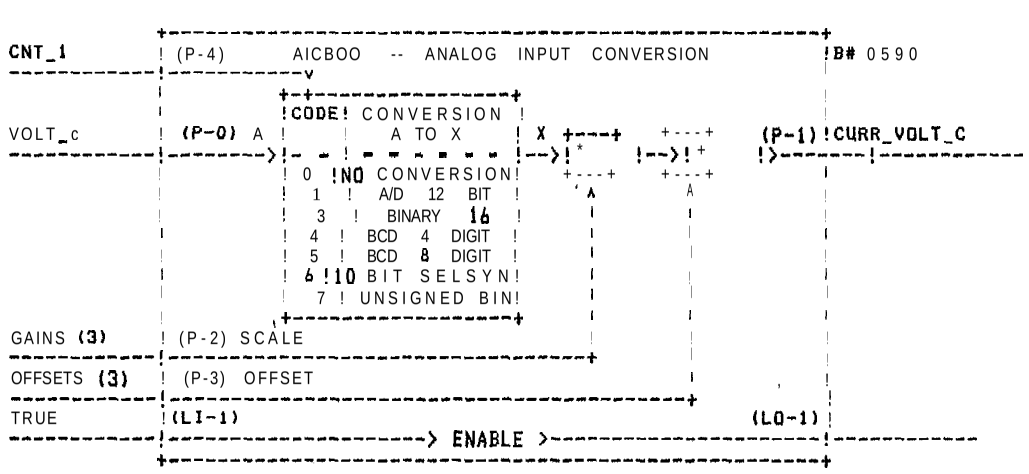
```
SEGMENT:04/BLOCK:0580 PG:0032
LD    TRUE--------.LOOOi
BLK  AICBOO
BY 0580
P-O   VOLT-B------.VOOiZC
P-I  CURR_VOLT_B-.V0074R
P-2  GAINS(2)----.C016CR
P-3  OFFSETS(2)--.C012CR
P-4  CNT_1-------.C0102C
```




```
GAINS-------.C0168R   04:0570   04 :0590   04:0600   04:0610   04:0620
                      04:0630   04:0640    04:0650   0410660   04:0680
OFFSETS-----.C0128R   04:0570   04:0590    04:0600   04:0610   04:0620
                      04:0630   04:0640    04:0650   04:0660   04:0680
CNT_1-------.C0102C   01:0800   04:0310    04:0350   04:0390   04:0570
                      04:0590   04:0600    04:0610   04:0620   0410630
                      04:0640   04:0650    04:0660   04:0680
```

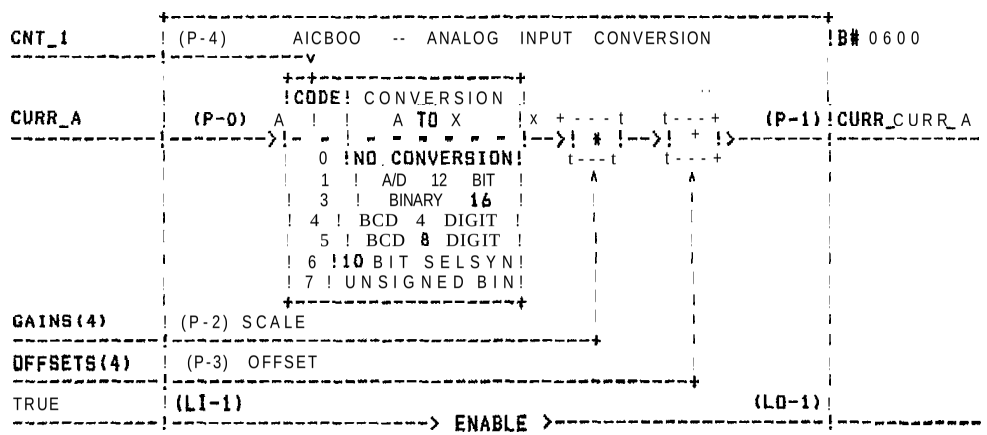

COPYRIGHT (C) 1988, BY
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPARTMENT
SALEM, VA USA


```
it DATE: 10/19/88
* AUTHOR :
* J
*
* REQUISITION
* PACIFIC
*  D.S.D
* XXXXXX
SYSTEM:PSS
DRAWING : XXXXXX
NEXT PAGE:0033
PAGE: 0032
```

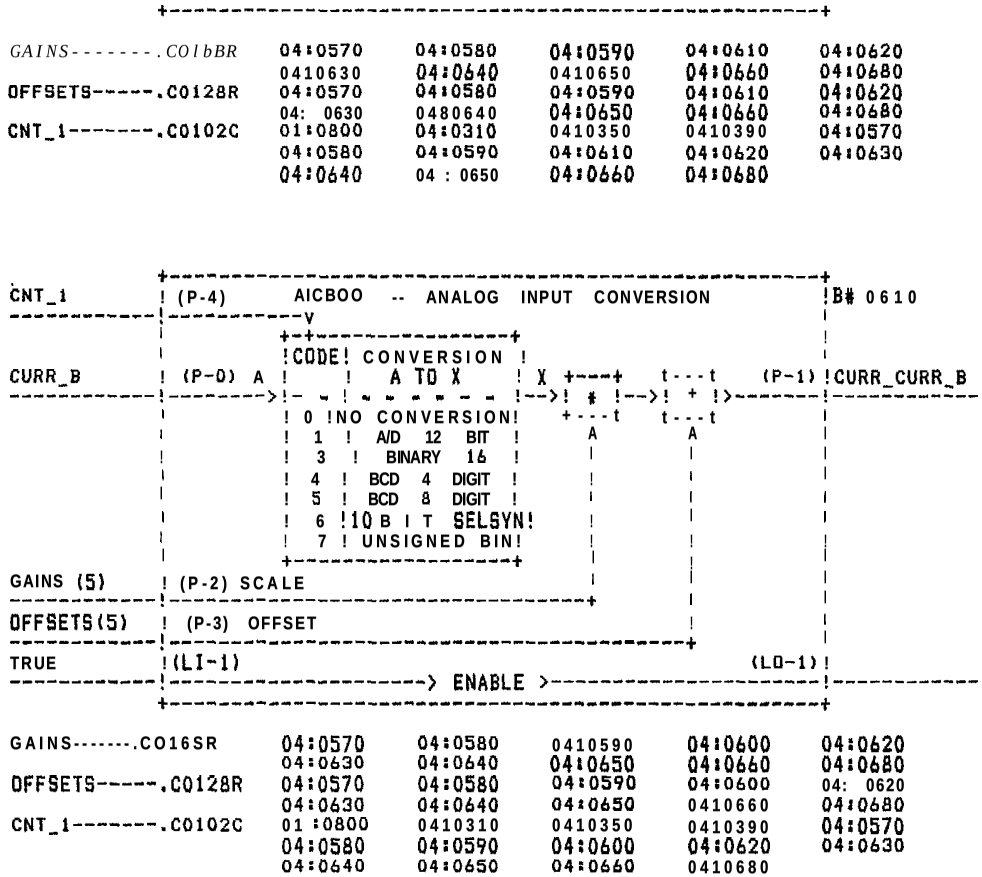



```
GAINS-------.C0168R     04:0570     04:0580     04: 0600     04:0610     04:0620
                        04:0630     04:0640     04:0650      04:0660     04:0680
OFFSETS------.C0128R    04:0570     04:0580     04:0600      04:0610     04:0620
                        04:0630     04:0640     04:0650      04:0660     04:0680
CNT_1-------.C0102C     01:0800     04:0310     04:0350      0410390     04 I 0570
                        04:0580     04:0600     04:0610      04 :0620    04:0630
                        04:0640     04~0650     04:0660      04:0680
```

```
==========================  ========
SEGMENT:04/BLOCK:0590 PC: 0033
==========================  ========
          LD TRUE--------.LOOOI
          BLK AICBOO
          B# 0590
          P-O VOLT_C------.V0014C
          P - I CURR_VOLT_C-.V0078R
          P - 2 GAINS(3)----.C0170R
          P - 3 OFFSETS(3)--.C0130R
          P - 4 CNT_1-------.C0102C
```



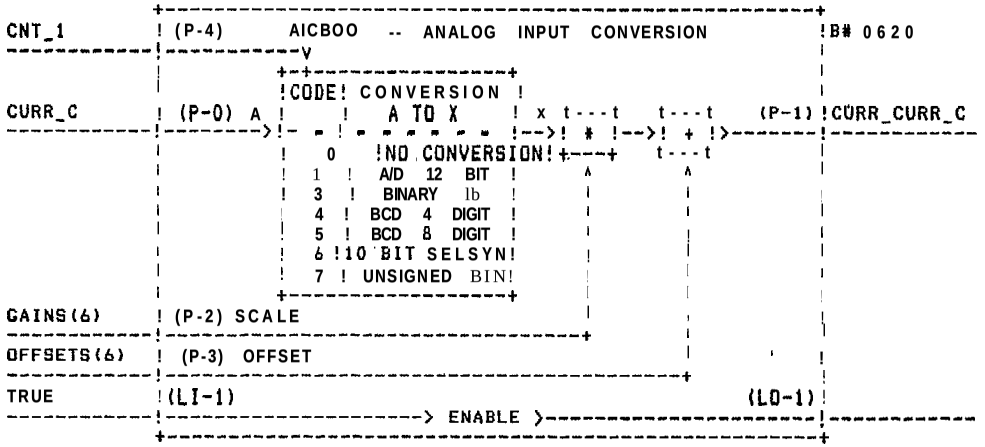

```
==========================  ========
SEGMENT:04/BLOCK:0600 PG:0033
==========================  ========
          LD TRUE--------.LOOOI
          BLK AICBOO
          BY 0600
          P-O CURR_A------.V0016C
          P - I CURR_CURR_A-.V007CR
          P-2 GAINS(4)----.C0174R
          P-3 OFFSETS(4)--.C0134R
          P - 4 CNT_1-------.C0102C
```




```
SEGMENT:04/BLOCK:0610 PG:0033
LD    TRUE--------.LOOOI
BLK AICBOO
R#__0610
P-O CURR_B-------.V0018C
P-I CURR_CURR_B-.V0080R
P-2 GAINS(5)----.C0178R
P-3 OFFSETS(5)--.C0138R
P-4 CNT_1--------.C0102C
```

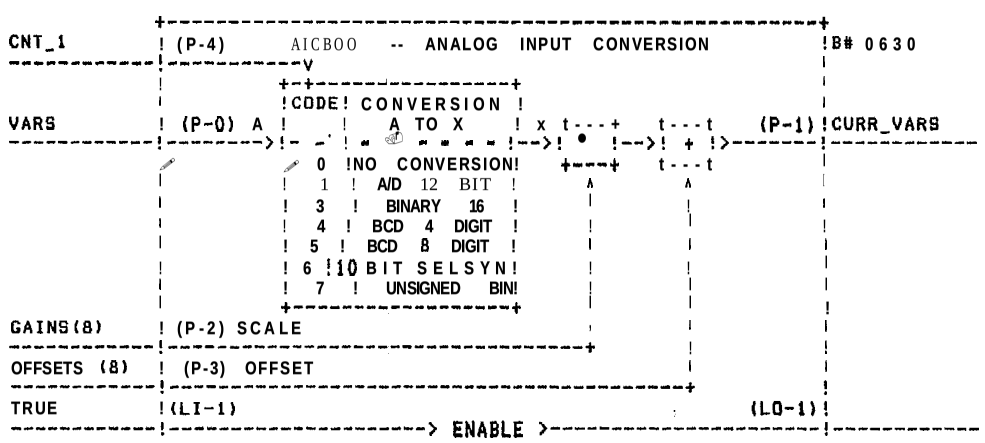

```
*  COPYRIGHT (C) 1988, BY
*  GENERAL ELECTRIC COMPANY
*  DRIVE SYSTEMS DEPARTMENT
*  SALEM, VA  USA
```


```
* DATE: 10/19/88
it AUTHOR :
* J
*
* REQUISITION
* PACIFIC
*  D.S.D
*  xxxxxx
* SYSTEM:PSS
* DRAWING : XXXXXX
* NEXT PAGE:0034
PAGE:0033
```

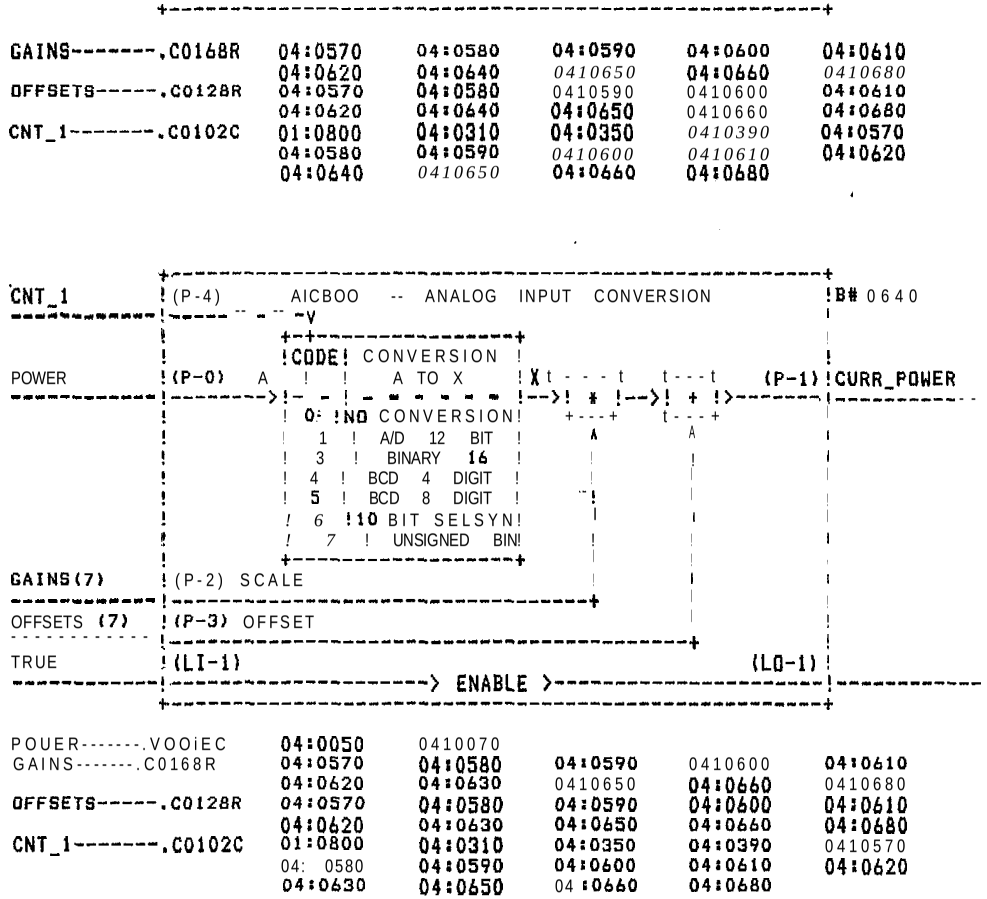



```
GAINS--------.C0168R   04:0570   04:0580   04: 0590   04:0600   04:0610
                       04:0630   04:0640   04:0650    04:0660   04:0680
OFFSETS------.C0128R   04:0570   04:0580   04:0590    0410600   04:0610
                       04:0630   04:0640   04:0650    04:0660   04:0680
CNT_1--------.C0102C   01:0800   04:0310   04 :0350   04:0390   04:0570
                       04:0580   04:0590   04:0600    04:0610   04: 0630
                       04:0640   04:0650   04:0660    04:0680
```

```
====================  ========
SEGMENT:04/BLOCK:0620 PG:0 0 3 4
====================  ========
          LD    TRUE--------.LOOOI
          BLK  AICBOO
          BY 0620
          P-O CURR_C-------.V001AC
          P-1 CURR_CURR_C-.V0084R
          P-2 GAINS(6)----.C017CR
          P-3 OFFSETS(6)--.C013CR
          P-4 CNT_1--------.C0102C
```




```
====================  ========
SEGMENT:04/BLOCK:0630 PC: 0034
====================  ========
          LD    TRUE--------.LOOOI
          BLK  AICBOO
          BY 0630
          P-O   VARS--------.VOOICC
          P-1 CURR_VARS---.V0088R
          P-2 GAINS(8)----.C0184R
          P-3 OFFSETS(8)--.C0144R
          P-4 CNT_1--------.C0102C
```



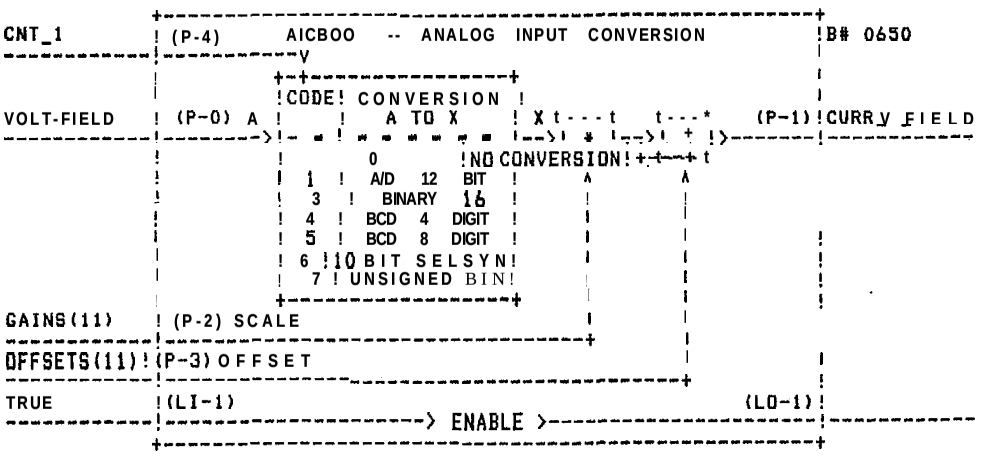

```
SEGMENT:04/BLOCK:0640 PC: 0034
        LD TRUE--------.LOOOI
        BLK AICBOO
        B# 0 6 4 0
        P-O POWER-------.VOOIEC
        P-I CURR_POWER--.V008CR
        P-2 GAINS(7)----.C0180R
        P-3 OFFSETS(7)--.C0140R
        P-4 CNT_1-------.C0102C
```

POUER-------.VOOiEC 04:0050 0410070
GAINS-------.C0168R 04:0570 04:0580 04:0590 0410600 04:0610
04:0620 04:0630 0410650 04:0660 0410680
OFFSETS-----.C0128R 04:0570 04:0580 04:0590 04:0600 04:0610
04:0620 04:0630 04:0650 04:0660 04:0680
CNT_1-------.C0102C 01:0800 04:0310 04:0350 04:0390 0410570
04: 0580 04:0590 04:0600 04:0610 04:0620
04:0630 04:0650 04 :0660 04:0680

COPYRIGHT (C) 1988, BY GENERAL ELECTRIC COMPANY it DRIVE SYSTEMS DEPARTMENT SALEM, V A USA

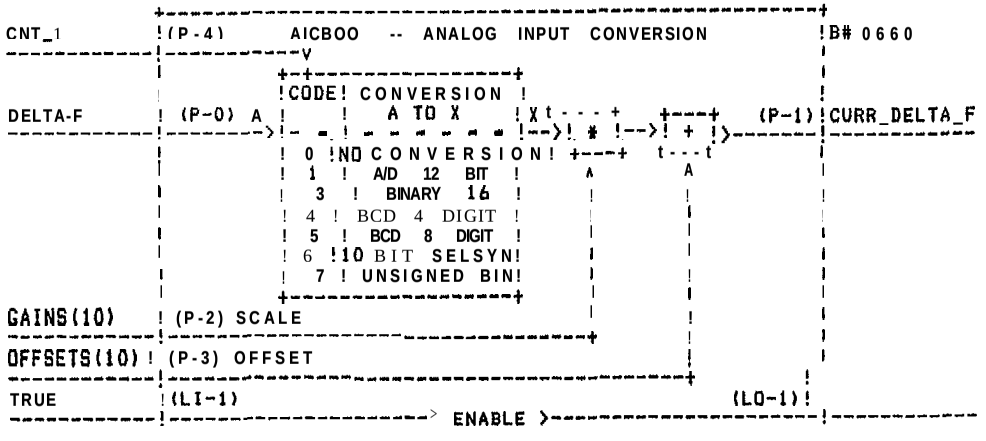
DATE! 10/19/88
AUTHOR !
REQUISITION
PACIFIC
D.8.D
XXXXXX

SYSTEM:PSS
DRAWING ! XXXXXX
NEXT PAGE: 0035

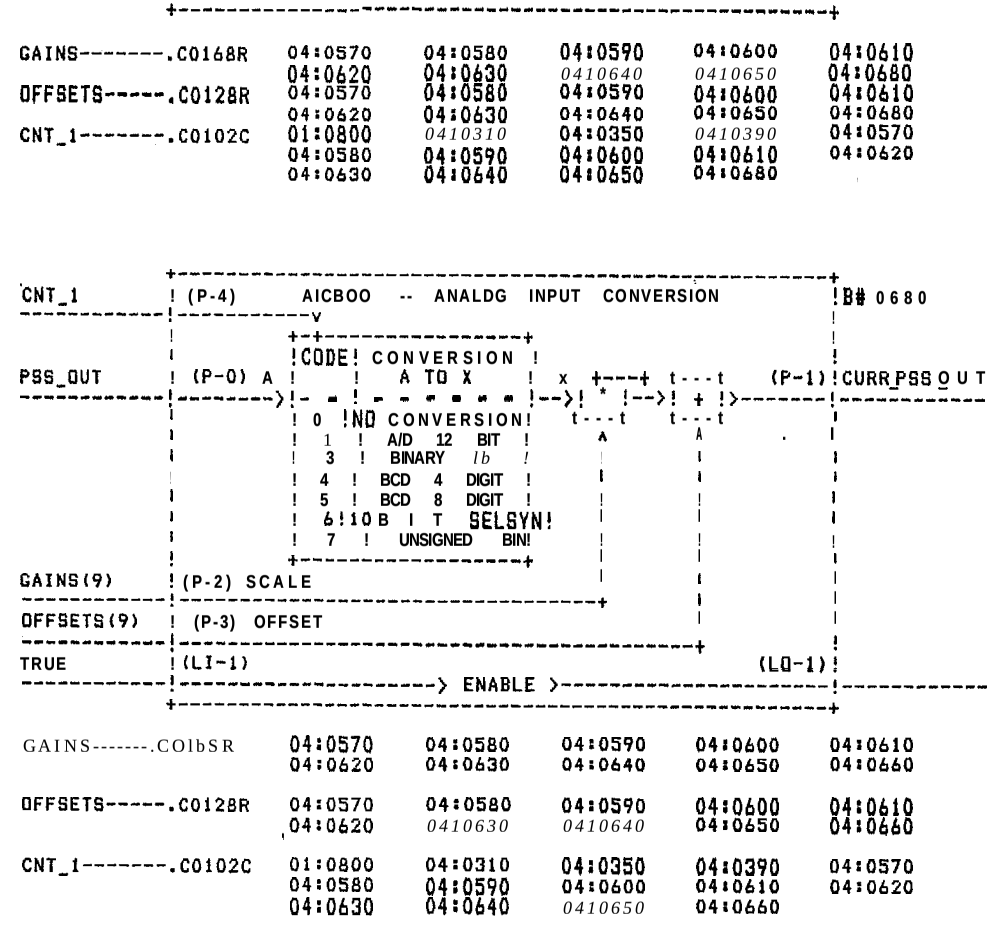


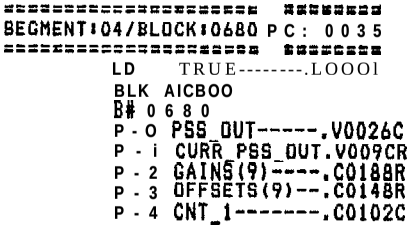

```
GAINS-------.C0168R     04:0570    04:0580    04:0590    04:0600    0410610
                        04:0620    0410630    04:0640    04:0660    04: 0680
OFFSETS------.C0128R    04:0570    04:0580    04:0590    04:0600    04:0610
                        04:0620    04:0630    04:0350    04:0660    04:0680
CNT_1--------.C0102C    01:0800    04:0310    04:0600    04:0390    04r0.570
                        04:0580    04:0590               0410610    04:0620
                        04:0630    04:0640    04:0660    04:0680
```

```
SEGMENT:04/BLOCK:0650 PC: 0035
        LD    TRUE--------.LOOOI
        BLK AICBOO
        B Y 0650
        P-O VOLT_FIELD--.V0020C
        P-I CURR_V_FIELD.V0090R
        P-2 GAINS(11)---.C0190R
        P-3 OFFSETS(11)-.C0150R
        P-4 CNT_1--------.C0102C
```




```
SEGMENT:04/BLOCK:0660 PG:0035
        L D  TRUE---------.L0001
        BLK AICBOO
        B# 0660
        P-O DELTA-F-----.VOOPZC
        P-I CURR_DELTA_F.V0094R
        P-2 GAINS(10)---.C018CR
        P-3 OFFSETS(10)-.C014CR
        P-4 CNT_1--------.C0102C
```



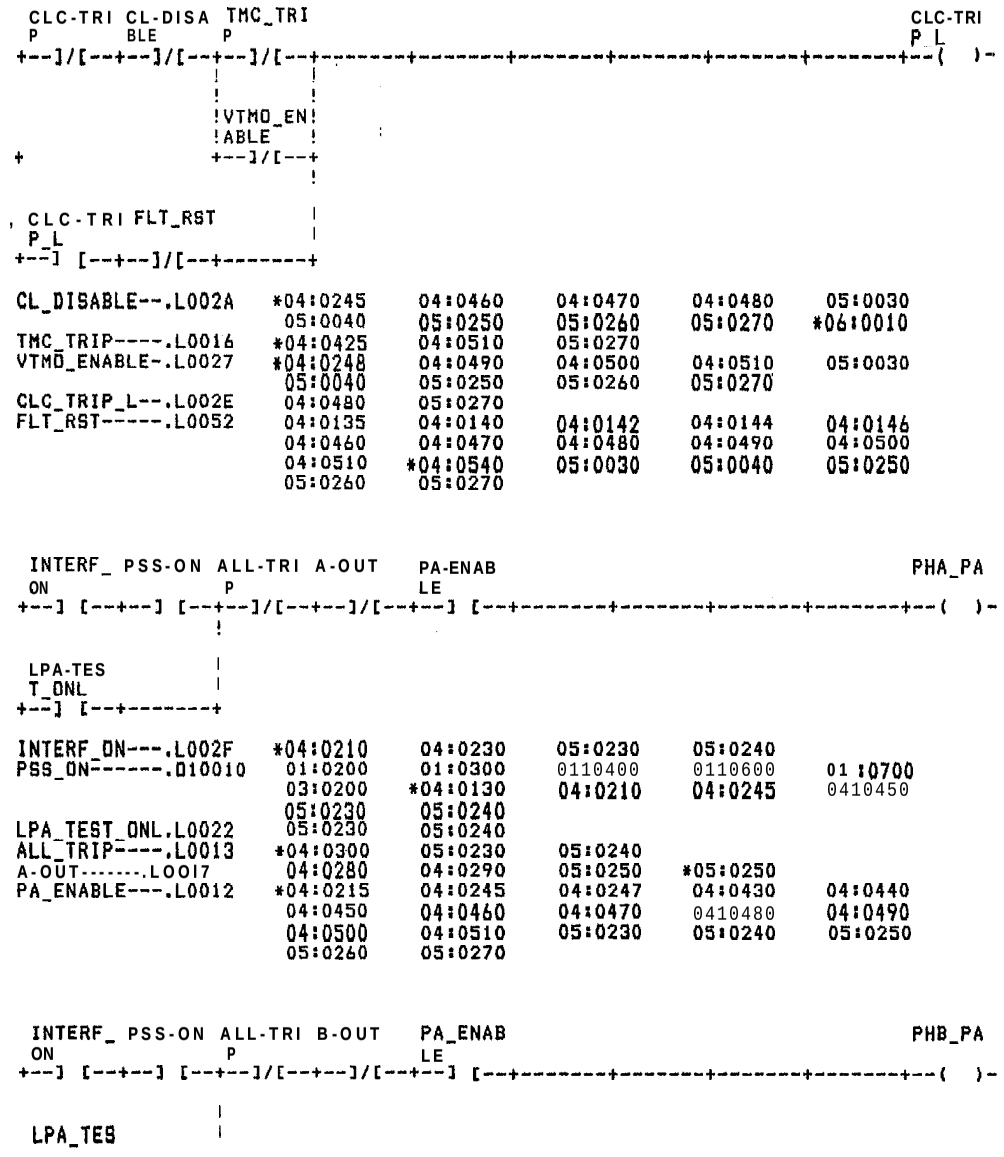


* DATE: 10/19/88 * REQUISITION * COPYRIGHT (C)1988, BY * SYSTEM:PSS *
* AUTHOR : * PACIFIC t GENERAL ELECTRIC COMPANY *
* J * D.S.D * DRIVE SYSTEMS DEPARTMENT *
* * xxxxxx * SALEM, VA USA it DRAWING : XXXXXX *

```
..NAME: PGE_IO_SEG_FAST_0000 ,
..AUTHOR: GIBBS
..COPVRIGHT:     COPYRIGHT        (C)  1987
                 BY GENERAL ELECTRIC COMPANY,     U.S.A
..KEYWORDS:
                 DMC,SEGMENT
..ABSTRACT:
. . . ..DESCRIPTION: THIS SEGMENT READS THE CUSTOMER INPUTS
          AND DETERMINES BASED ON CURRENT LIMIT THE POWER
          AMPLIFIER  STATE.

....DEPENDENCIES:
....MEMORY_SIZE:
..BUILD_INSTRUCTIONS:
:::::PROCESSOR:   DMC_EDIT_R3
. . . .SEGMENT NAME ..............................  X$IOSEG
. . . . .SEGMENT TYPE (RUN, INTERRUPT, INITIALIZE) : RUN
. . . ..SEGMENT INTERRUPT NUMBER ..................  :
. . . ..SEGMENT SCHEDULING RATE, AND UNIT . . . . . ..:  10 MILLI SECOND
. . . ..8087 CO-PROCESSOR USED (YES / NO) . . . . . :  YES
. . . .SEGMENT STACK SIZE (# OF BYTES)  . . . . . :  100
. . . ..SYMBOLIC NAME SPLIT CHAR (DEFAULT HYPHEN) :
```

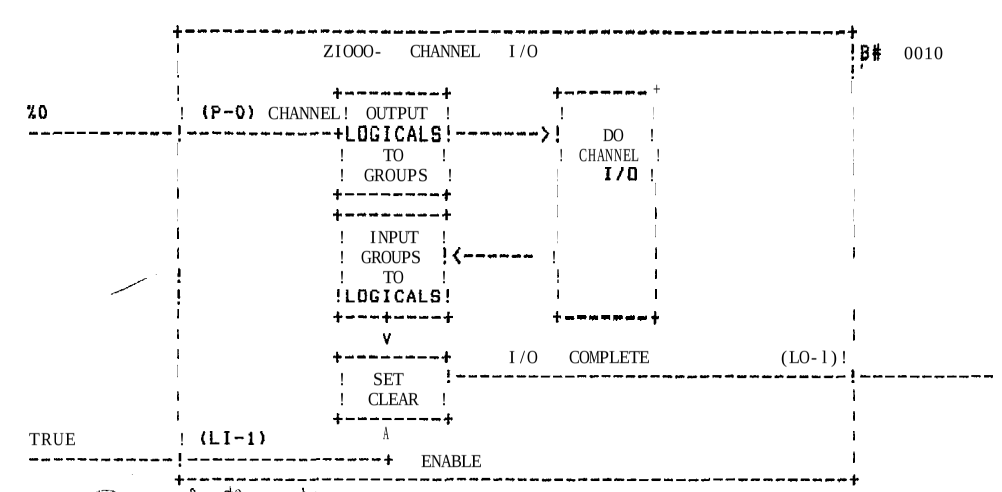
```
*************************************************************************************************************
*                          *              *   COPYRIGHT (C) 1988, BY    *  SYSTEM:PSS                        *
*  DATE: 10/19/88          *   REQUISITION *   GENERAL ELECTRIC COMPANY *                                    *
* AUTHOR :                 *  PACIFIC      *   DRIVE SYSTEMS DEPARTMENT  ************************************************
* J                        *   D.S.D       *   SALEM, VA     USA         * DRAWING : XXXXXX                                 *
*                          *  XXXXXX       *                             * NEXT PAGE: 0037                  PAGE: 0036 *
*************************************************************************************************************
```

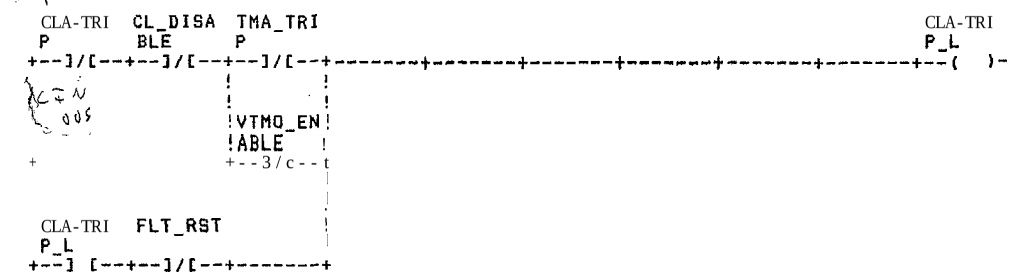



```
====================  ========
SEGMENT:05/BLOCK:0050 PG:0038
====================  ========
        LDF CLC_TRIP----.I10007
        ANF CL_DISABLE--.L002A
        LDF TMC_TRIP----.L0016
        ORF VTMO_ENABLE-.L0027
        ANS
        L D  CLC_TRIP_L--.L002E
        ANF FLT_RST-----.L0052
        ORS
        STO CLC_TRIP_L--.L002E
        BY 0050
```

```
====================  ========
SEGMENT:05/BLOCK:0220 PG:0038
====================  ========
        LD  INTERF_ON---.L002F
        A N  PSS_ON------.D10010
        O R  LPA_TEST_ONL.L0022
        ANF ALL_TRIP----.L0013
        ANF   A-OUT-------.L0017
        A N  PA_ENABLE---.L0012
        STO PHA_PA------.D10005
        B# 0 2 2 0
```

```
====================  ========
SEGMENT:05/BLOCK:0230 PG:0038
====================  ========
        LD  INTERF_ON---.L002F
        A N  PSS_ON------.D10010
        O R  LPA_TEST_ONL.L0022
        ANF ALL_TRIP----.L0013
        ANF   B-OUT-------.L0018
        A N  PA_ENABLE---.L0012
```



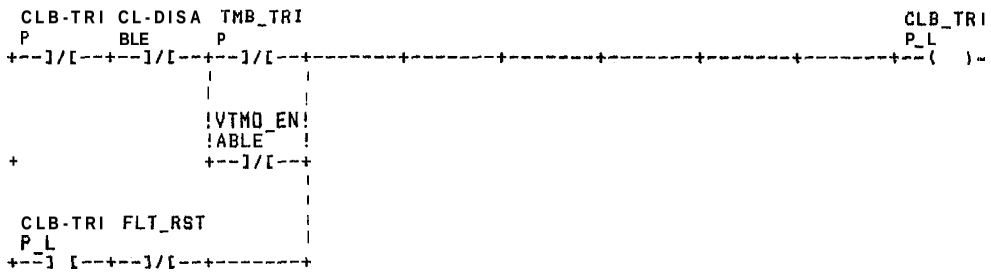



```
CL_DISABLE--.L002A   *04:0245   04:0460   04:0470   04: 0480   05: 0040
                      05:0050   05:0250             05: 0270
TMA_TRIP----.L0014   *04:0345   04 :0490   05:0260
VTMO_ENABLE-.L0027   *04:0248   04:0490   0410500   04:0510   05:0040
                      05:0050   05:0250   05: 0260   05: 0270
CLA_TRIP_L--.L002C    04:0460   05:0250
FLT_RST-----.L0052    04:0135   04:0140   04:0142   04:0144   04:0146
                      04:0460   04:0470   04:0480   04:0490   04:0500
                      04:0510  *04:0540   05:0040   05: 0050   05: 0250
                      05:0260   05:0270
```

```
SEGMENT:05/BLOCK:0010  PG:  0037
          LD   TRUE--------.L0001
          BLK  21000
          B#   0010
          P-0  %0----------.C02AAB
```

```
SEGMENT:05/BLOCK:0030  PC:  0037
          LDF  CLA_TRIP----.I10005
          ANF  CL_DISABLE--.L002A
          LDF  TMA_TRIP----.L0014
          ORF  VTMO_ENABLE-.L0027
          ANS
          L D   CLA_TRIP_L--.L002C
          ANF  FLT_RST-----.L0052
          ORS
          STO  CLA_TRIP_L--.L002C
          B#   0030
```

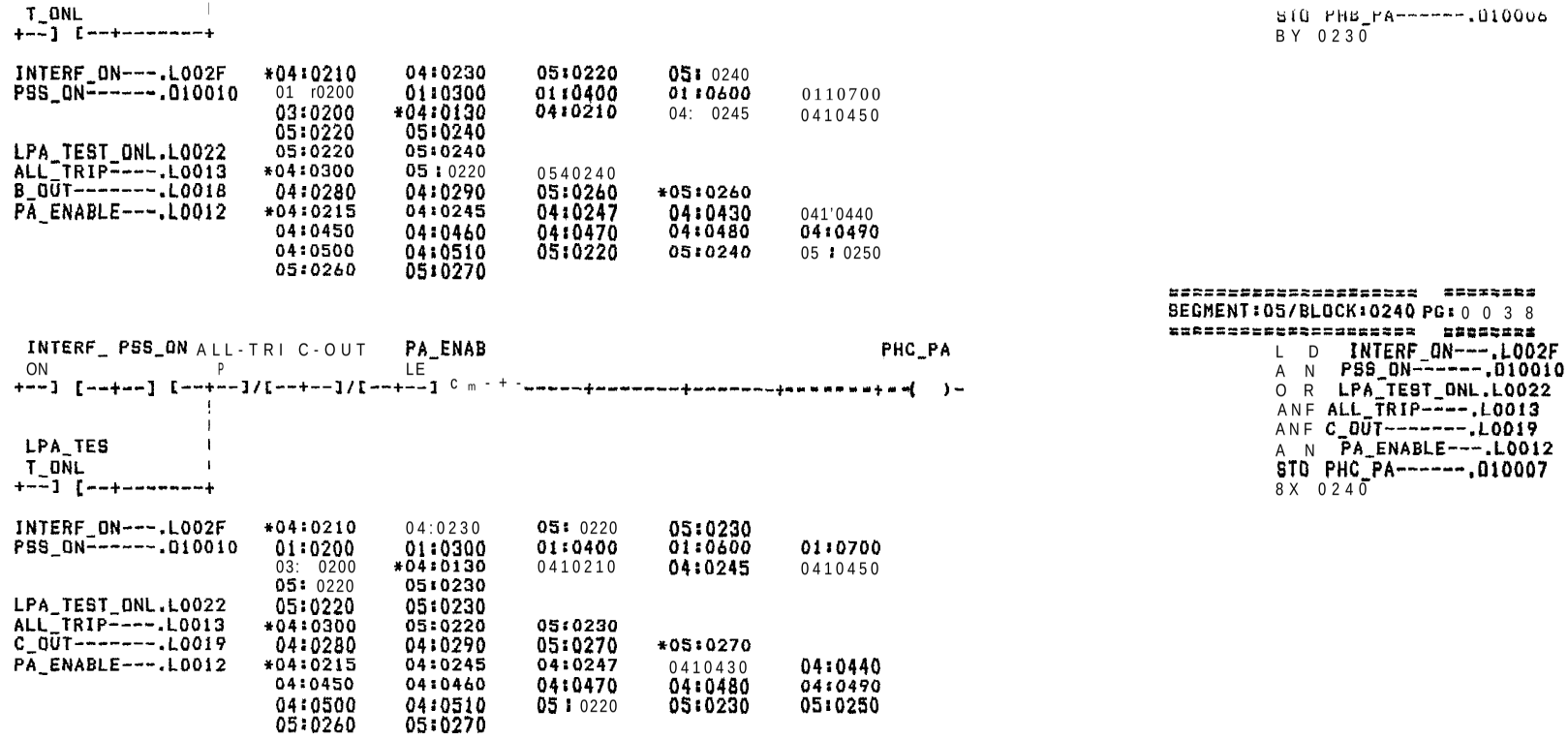
```
SEGMENT:05/BLOCK:0040  PC:  0037
```




```
CL_DISABLE--.L002A    *04:0245    04:0460    04:0470    0410480    05: 0030
                       05:0050    05 :0250   0510260    05:0270   *06:0010
TMB_TRIP----.L0015    *04:0385    04:0500    05:0260
VTMO_ENABLE-.L0027    *04:0248    04:0490    04:0500    04:0510    05:0030
                       05: 0050   05:0250    05: 0260   05:0270
CLB_TRIP_L--.L002D     04:0470    05:0260
FLT_RST-----.L0052     04:0135    04:0140    04:0142    04:0144    04:0146
                       04:0460    0410470    04:0480    04 :0490   a4 :0500
                       04:0510   *04:0540    05:0030    05:0050    05:0250
                       05: 0260   05:0270
```

```
LDF   CLB,TRIP----.I10006
ANF CL_DISABLE--.L002A
LDF TMB_TRIP----.L0015
ORF VTMO_ENABLE-.L0027
ANS
L  D  CLB_TRIP_L--.L002D
ANF FLT_RST-----.L0052
ORS
STO CLB_TRIP_L--.L002D
B# 0040
```

```
*                    *              *  COPYRIGHT (C) 1988, BY    *  SYSTEM:PSS                                  *
* DATE: 10/19/88     * REQUISITION  *  GENERAL ELECTRIC COMPANY *                                              *
* AUTHOR :           * PACIFIC      *  DRIVE SYSTEMS DEPARTMENT  ***********************************************
* J                  *  D.S.D       *  SALEM, V A    USA         * DRAWING : XXXXXX                           *
*                    *  XXXXXXXX    *                            * NEXT PAGE: 0038          PAGE: 0037 *
```



	*04:0210	04:0230	05:0220	05: 0240	
	01 r0200	01:0300	01:0400	01:0600	0110700
	03:0200	*04:0130	04:0210	04: 0245	0410450
	05:0220	05:0240			
	05:0220	05:0240			
	*04:0300	05 : 0220	0540240		
	04:0280	04:0290	05:0260	*05:0260	
	*04:0215	04:0245	04:0247	04:0430	041'0440
	04:0450	04:0460	04:0470	04:0480	04:0490
	04:0500	04:0510	05:0220	05:0240	05 : 0250
	05:0260	05:0270			

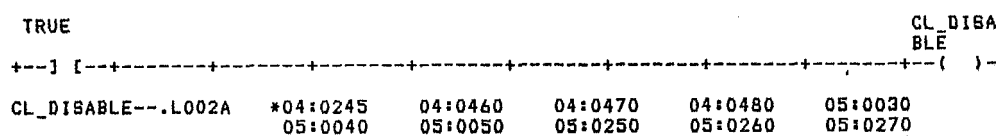
	*04:0210	04:0230	05: 0220	05:0230	
	01:0200	01:0300	01:0400	01:0600	01:0700
	03: 0200	*04:0130	0410210	04:0245	0410450
	05: 0220	05:0230			
	05:0220	05:0230			
	*04:0300	05:0220	05:0230		
	04:0280	04:0290	05:0270	*05:0270	
	*04:0215	04:0245	04:0247	0410430	04:0440
	04:0450	04:0460	04:0470	04:0480	04:0490
	04:0500	04:0510	05 : 0220	05:0230	05:0250
	05:0260	05:0270			

COPYRIGHT (C) 1988, BY
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPARTMENT
SALEM, VA USA

DATE: 10/19/88
AUTHOR : J
REQUISITION
PACIFIC
i t D.S.D

SYSTEM:PSS
DRAWING: X X X X X X
NEXT PAGE: 0039

DATE: 10/19/88	REQUISITION	CDPYRIGHT (C) 1988, BY	SYSTEM:PSS	
AUTHOR I	PACIFIC	GENERAL ELECTRIC COMPANY it		
J	D.S.D	DRIVE SYSTEMS DEPARTMENT		
	xxxxxx	SALEM, V A USA	DRAWING I XXXXXX	
			NEXT PAGE I 0042	PAGE: 0041




```
====================  ========
SEGMENT:06/BLOCK:0010  PG: 0041
====================  ========
         LD  TRUE--------.L0001
         STO CL_DISABLE--.L002A
         B#  0010
```